

Certification Specifications for Flight Simulation Training Devices (CS-FSTD)

Issue 1

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SUBPART A – GENERAL

CS FSTD.GEN.001 Applicability

These certification specifications (CSs) prescribe standards for the initial qualification of all aeroplane and rotorcraft flight simulation training devices (FSTDs) and serve as the qualification basis for subsequent evaluations.

GM1 FSTD.GEN.001 Applicability

GUIDANCE ON THE USE OF THE CERTIFICATION SPECIFICATIONS FOR FLIGHT SIMULATION TRAINING DEVICES

- (a) The structure and suggested use of the CSs for FSTDs (CS-FSTD) are the following.

Based on the training needs identified in compliance with relevant regulations (e.g. Part-FCL (Annex I) of Commission Regulation (EU) No 1178/2011), an applicable FSTD capability signature (FCS) is identified. This FCS is then used to determine the applicable FSTD-related requirements, as explained below:

- (1) for FSTD general requirements, refer to Subpart B;
 - (2) for the necessary design data, validation data and validation data roadmap (VDR) (note that the CSs for simulator data under operational suitability data (OSD) may be applicable), refer to Subpart C;
 - (3) for FSTD objective tests, refer to Subpart D;
 - (4) for FSTD functions and subjective tests, refer to Subpart E.
- (b) Requirements for the application process, the testing of the FSTDs and associated documentation are discussed further in Part-ORA (Annex VII) of Commission Regulation (EU) No 1178/2011.

CS FSTD.GEN.005 Flight simulation training device simulation features

- (a) 'FSTD simulation feature' means a domain of simulation for which general requirements and/or testing requirements have been identified. If combined with their individual feature fidelity levels, these FSTD simulation features create an FCS. The 14 FSTD simulation features can be grouped into three categories and are defined as follows.
- (b) **Aircraft simulation** comprises the following FSTD simulation features.
- (1) **Flight deck layout and structure (FDK).** Defines the level of flight deck enclosure and the physical (or perceived physical) structure, spatial representation and layout of the flight deck environment, instrument layout and presentation, panels and switches, interaction with the flight deck components and flight crew member seating.

Note: This feature excludes the flight controls and aircraft system operation and functionalities.

- (2) **Flight control forces and hardware (CLH).** Defines the physical appearance, travel, tactile feel and force feedback operation of the flight controls and systems affecting aircraft configuration.
- (3) **Flight control systems operation (CLO).** Defines the extent of the flight control system functions and controls affecting aircraft configuration that are to be modelled in the FSTD. These include primary flight control surface displacements (e.g. surface position for aeroplanes, blade angles for helicopters) and the operation modes and logic of the flight control system and systems affecting aircraft configuration, flight and manoeuvre envelope protection functions and flight deck indications and messages.
- (4) **Aircraft systems (SYS).** Defines the extent of the aircraft system simulation required to be modelled in the FSTD. The requirements apply if the ownership is equipped with the appropriate systems. This feature concerns the operation, functionalities and indications of the aircraft systems and the interdependencies of different systems.

Aircraft systems mean, for example, instruments, communications, navigation, autopilot, flight director (FD), flight management, flight guidance, hydraulic, electrical, fuel, pneumatic, pressurisation, air conditioning, power plant, auxiliary power unit (APU), caution and warning, fire protection, ice and rain protection and surveillance systems and also terrain awareness warning systems (TAWs) / helicopter terrain awareness and warning systems (HTAWs), airborne collision avoidance systems (ACASs) and on-board weather radar systems and other systems.

Note: This feature excludes the flight controls. Refer to the 'flight control forces and hardware (CLH)' and 'flight control systems operation (CLO)' FSTD features for their requirements.

Note: Refer to the 'flight deck layout and structure (FDK)' FSTD feature for the requirements for the aircraft system hardware/interface.

Note: Refer to the 'performance and handling on ground (GND)', 'performance and handling in-ground effect (IGE)', performance and handling out-of-ground effect (OGE), 'vibration cueing (VIB)', 'motion cueing (MTN)', 'sound cueing (SND)' and 'visual cueing (VIS)' FSTD features for the requirements for the effects associated with aircraft system operation (e.g. activation of the anti-icing system may result in a change in aircraft flight performance and handling, and vibration, motion, sound and visual cues).

- (5) **Performance and handling on ground (GND).** Defines the mathematical models and associated data to be used to describe the ground-handling characteristics and aerodynamic, propulsion, ground reaction and operating site surface conditions required to be modelled in the FSTD whenever the aircraft is in contact with the surface. These include acceleration, deceleration, turning, braking and the effects of wind (including crosswind) as relevant to the aircraft.

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- (6) **Performance and handling in-ground effect (IGE).** Defines the mathematical models and associated data to be used to describe the aerodynamic, flight dynamic and lift/thrust characteristics required to be modelled in the FSTD in-ground effect. The height of ground effect is the wing span, or main rotor diameter, unless otherwise declared by the original equipment manufacturer (OEM) or data provider (as applicable).
- (7) **Performance and handling out-of-ground effect (OGE).** Defines the mathematical models and associated data to be used to describe the aerodynamic, flight dynamic and lift/thrust system characteristics required to be modelled in the FSTD away from the ground and out-of-ground effect.
- (c) **Cueing simulation** comprises the following FSTD simulation features.
- (1) **Sound cueing (SND).** Defines the sound cues to be modelled. The sound cues are those related to sounds generated both externally and internally in relation to the flight deck. These include but are not limited to the sounds of engines, aerodynamics, propulsion, rotors, operating site elements (e.g. runway rumble), weather effects and aircraft components audible to the flight crew, such as pumps, motors, fans, vents, airflow and wipers.
- Note: Sounds generated by the aircraft systems for pilot awareness (e.g. communication systems, system and avionics audio tones, messages, cautions, warnings or any sound for indication purposes) are covered separately in the aircraft systems (SYS) FSTD feature.
- (2) **Vibration cueing (VIB).** Defines the vibrations and buffets to be modelled in the FSTD that may be felt by the pilots during all phases of flight from effects such as airframe buffet, control surface buffet, engine, propeller/rotor and transmission vibrations, systems operation, weather and ground operations. These vibrations may vary significantly depending on the flight parameters, for example, rotorcraft hover in- or out-of-ground effect, hover into wind, downwind hover, hover taxi, transition to forward flight, translational lift, cruise, turns, steep turns and autorotation, as relevant to the aircraft type.
- Note: This feature does not contain the motion effects (e.g. touchdown cues of the landing gear). Refer to the motion cueing (MTN) FSTD feature for their requirements.
- (3) **Motion cueing (MTN).** Defines the requirements for the motion cues that are generated by aircraft accelerations (due to forces on flight and ground operations) and the effects of certain events or situations.
- Note: This feature does not contain the vibration and buffet cues. Refer to the vibration cueing (VIB) FSTD feature for their requirements.
- (4) **Visual cueing (VIS).** Defines the type of out-of-the-flight-deck-window image optical display (e.g. collimated or non-collimated mirror) and visual field of view (FOV) (horizontal and vertical) that is required to be seen by the pilots using the FSTD from their reference eyepoint(s).

(d) **Environment simulation** comprises the following FSTD simulation features.

- (1) **Navigation (NAV).** Defines the level of simulation for the exchange of navigation, communication and surveillance data and interactions with air traffic services, including associated instructor controls, covering the transmission of signals and data exchange through radio-frequency systems, navigation aids (e.g. ILS, VOR, DME, NDB, GNSS – GPS, SBAS, GBAS), radio communication systems (e.g. VHF, HF, ACARS, CPDLC), surveillance technologies (e.g. ADS-B, ADS-C) and ATC data links.

Note: The aircraft navigation equipment and instruments (e.g. navigation receivers, GNSS units) and their databases (e.g. flight management system (FMS) or GNSS navigation databases) are considered in the ‘flight deck layout and structure (FDK)’ and ‘aircraft systems (SYS)’ FSTD features.

- (2) **Atmosphere and weather (ATM).** Defines the level of complexity of the simulated weather conditions, from ambient temperature and pressure to full thunderstorm modelling etc.
- (3) **Operating sites and terrain (OST).** Defines the complexity and level of detail of the operating sites (e.g. aerodrome, landing areas) and terrain modelling required to be simulated. This includes such items as generic versus customised operating sites (e.g. airport, heliport, vertiport or other landing area), visual scenes and terrain elevation. Since some aircraft types can land at a variety of landing area types, often outside aerodromes and/or controlled airspace, the environment includes these types of landing areas in the visual scene modelling.

CS FSTD.GEN.010 Flight simulation training device simulation feature fidelity levels

(a) ‘FSTD simulation feature fidelity level’ means the level of fidelity assigned to each of the defined FSTD simulation features. The fidelity levels are as follows.

- (1) **Specific (S).** The highest level of fidelity for a given FSTD feature.
- (2) **Representative (R).** The intermediate level of fidelity for a given FSTD feature.
- (3) **Generic (G).** The lowest level of fidelity for a given FSTD feature.
- (4) **None (N).** N means either of the following:
 - (i) the FSTD feature is not installed, functional or available for use in training;
 - (ii) a feature is installed, but is not required and in this case, it should not be distracting or detracting from the other features for the intended use of the device.

(b) The requirements for the fidelity levels for each feature are given in this CS (CS-FSTD).

GM1 FSTD.GEN.010 Flight simulation training device simulation feature fidelity levels

GUIDANCE ON THE FEATURE FIDELITY LEVELS

Fidelity levels for each group of FSTD simulation features are described in a high-level overview in Table 1 below. These broad summaries are provided for guidance only and are not intended to replace the requirements. The detailed requirements are specified in the relevant subparts.

Note that the 'aircraft systems (SYS)' and 'flight deck layout and structure (FDK)' FSTD features allow for only some aircraft systems and their interfaces to be simulated; refer to CS FSTD.QB.104 and CS FSTD.QB.101, respectively.

Table 1. Fidelity levels (excluding 'N') for each feature category

FSTD feature groups	Guidance on the features	Fidelity level	Summary description	Verification and validation
Aircraft simulation • FDK • CLH • CLO • SYS • GND • IGE • OGE	GM1 FSTD.QB.101 GM1 FSTD.QB.102 GM1 FSTD.QB.103 GM1 FSTD.QB.104 GM1 FSTD.QB.105 GM1 FSTD.QB.106 GM1 FSTD.QB.107	S	Replicates the appearance, tactile feel, system operation, performance and handling of the simulated aircraft type and variant. The highest level of simulation model fidelity.	Initial qualification requires the highest level of verification and validation, mainly with numerical tolerances for objective tests, in the context of which the device is compared with data for the particular aircraft type and variant.
		R	Is characteristic of the appearance, tactile feel, system operation, performance and handling of the simulated aircraft type. Simulation may include elements from different variants of the same aircraft type. The intermediate level of simulation model fidelity.	Initial qualification requires an intermediate level of verification and validation, mainly with either numerical tolerances or correct trend and magnitude (CT&M) for objective tests, in the context of which the device is compared with data for the particular aircraft type. The data may originate from different variants of the same aircraft type.
		G	Is characteristic of the appearance, tactile feel, system operation, performance and handling of the simulated group of aircraft. Simulation may include elements from different aircraft of the same group of aircraft. The lowest level of simulation model fidelity.	Initial qualification requires only the lowest level of verification and validation, mainly with tolerances of CT&M for objective tests. The data may originate from different aircraft of the same group of aircraft and from a wider variety of permissible validation data sources.
Cueing simulation • SND • VIB • MTN • VIS	GM1 FSTD.QB.108 GM1 FSTD.QB.109 GM1 FSTD.QB.110 GM1 FSTD.QB.111	S	The highest level of cueing requiring the most realistic simulation.	Initial qualification requires the highest level of verification and validation.
		R	The intermediate level of cueing.	Initial qualification requires an intermediate level of verification and validation.
		G	The lowest level of cueing.	Initial qualification requires only the lowest level of verification and validation.
Environment simulation • NAV • ATM • OST	GM1 FSTD.QB.112 GM1 FSTD.QB.113 GM1 FSTD.QB.114	S	The highest level of environment simulation requiring the most realistic simulation.	Initial qualification requires the highest level of verification and validation, mainly through functions and subjective testing.
		R	The intermediate level of environment simulation.	Initial qualification requires an intermediate level of verification and validation, mainly through functions and subjective testing.
		G	The lowest level of environment simulation.	Initial qualification requires only the lowest level of verification and validation, through functions and subjective testing.

CS FSTD.GEN.015 Terminology

For the purpose of this CS-FSTD, the following definitions and terminology apply:

- (1) 'Acceptable change' means a change to configuration, software, etc. that qualifies as a potential candidate for an alternative approach to validation.
- (2) 'Active force feedback' means, in the context of a flight controls system, a dynamic system that produces FSTD control forces reflecting those of the aircraft in all phases of flight in normal, abnormal and emergency operations.
- (3) 'Additional capabilities' means various mission equipment including but not limited to tactical equipment, air work equipment, search and rescue equipment, external loads and ship landing.
- (4) 'Aft of the pilot seats' means the area of the FSTD behind the pilots' seats or the rear bulkhead that normally contains instructor and observer seating and other non-aircraft-related installations.
- (5) 'Aircraft' means either an aeroplane or a rotorcraft and is used agnostically to refer to either of these.
- (6) 'Aircraft configuration' means the set-up of components, like flaps and landing gear, used to adjust or optimise the aircraft's performance during different phases of flight.
- (7) 'Aircraft data' means data acquired in real aircraft.
- (8) 'Aircraft performance data' means performance data published by the aircraft manufacturer in documents such as the aircraft flight manual (AFM), operations manual, performance engineering manual or equivalent.
- (9) 'Airport' means a defined area on land or water (including any buildings, installations and equipment) intended to be used either wholly or in part for the arrival, departure and surface movement of aircraft. Synonymous with 'aerodrome' in this document.
- (10) 'Airspeed' means calibrated airspeed (in kt) unless otherwise specified.
- (11) 'Alpha/beta envelope plot' means a two-dimensional (2D) plot of FSTD envelopes, with the alpha (α) axis representing the angle of attack and the beta (β) axis representing the angle of side-slip. The type of envelope being plotted varies. For example, a plot may be used to depict the 'FSTD validation envelope'.
- (12) 'Alternate engines/avionics' means an FSTD that includes the simulation of an additional engine/avionics fit.
- (13) 'Alternate FSTD platform' means different combination(s) of flight decks and platforms in the context of a multi-configuration FSTD (e.g. roll-on/roll-off solutions).
- (14) 'Alternative flight test data' means the performance, stability and control, and other necessary test parameters, electronically or manually recorded in an aircraft using simplified and non-invasive techniques, such as flight deck video and data bus recordings, to establish a set of relevant reference parameters with which FSTD parameters can be compared.

- (15) 'Altitude' means pressure altitude (in m or ft) unless otherwise specified.
- (16) 'Augmented reality' (AR) means that the real-world element is enhanced with a computer-generated image overlay component using a see-through head-mounted display (HMD) providing a composite view to the user.
- (17) 'Automatic testing' means using FSTD testing tools to provide the capability to set up test scenarios such that the environment, aircraft configuration, aircraft systems and control settings are exactly replicated. If auto-tests are run, the pilot control inputs are fed from the auto-test system directly into the aircraft and flight models that exactly replicate the pilot demands in magnitude and phase for the most complex sequences. This automated process ensures that exactly the same settings are used each time, removing the possibility of human error, which could creep in if using manual settings. The result is that a test scenario can be repeated exactly for a number of test runs for the case.
- (18) 'Background radio communications' means radiotelephony messages between ATC and other traffic that are heard on the active frequency by the flight crew. The word 'background' refers to the fact that these messages are not intended for the ownship.
- (19) 'Bank' means bank/roll angle (in degrees).
- (20) 'Baseline' means, in the context of alternate engines or avionics validation data, a fully flight-test-validated production aircraft simulation. It may represent a new aircraft type or a major derivative.
- (21) 'Breakout' means the force required at the pilot's primary controls to achieve initial movement of the control position.
- (22) 'Buffet' means the instability caused by turbulent streams generated by the aircraft itself (e.g. airflow separation or shock wave oscillation, or the wake striking the airframe).
- (23) 'Closed loop' means an objective test tuning method in which the test input stimuli are adjusted (e.g. by closed-loop controllers or drivers also known as 'math-pilots') to follow a predefined target response.
- (24) 'Collimated display' means a display system in which light rays from a given point in the image are made parallel, creating the illusion that the image is at a significant distance from the viewer, with parallax error negligible.
- (25) 'Computer-controlled aircraft' (CCA) means an aircraft in which the pilot's inputs to the control surfaces are transferred and augmented via computers.
- (26) 'Confined landing area' means a landing site with restricted dimensions, surrounded by obstacles such as trees, buildings, wires or terrain, requiring specialised approach and departure techniques.
- (27) 'Control sweep' means a movement of the appropriate pilot's control from neutral to an extreme limit in one direction (forwards, aft, right or left), a continuous movement

back through neutral to the opposite extreme position and then a return to the neutral position.

- (28) 'Convertible FSTD' means an FSTD in which hardware and software can be changed so that the FSTD simulates a different aircraft type, variant or group. The same FSTD platform, flight deck shell, motion system, visual system, computers and necessary peripheral equipment can thus be used in more than one simulation.
- (29) 'Correct trend and magnitude' (CT&M) means a method of comparing FSTD objective test results against validation data without the use of numerical tolerances, by using appropriate engineering judgement.
- (30) 'Critical engine parameter(s)' means the primary parameter(s) used to set and measure the engine power/thrust delivered in the aircraft.
- (31) 'Damping – critical' means the minimum damping of a second-order system such that no overshoot occurs in reaching a steady-state value after being displaced from a position of equilibrium and released. This corresponds to a relative damping ratio of 1:0.
- (32) 'Damping – overdamped' means the damping of a second-order system such that it has more damping than is required for critical damping, as described above. This corresponds to a relative damping ratio of more than 1:0.
- (33) 'Damping – underdamped' means the damping of a second-order system such that a displacement from the equilibrium position and free release results in one or more overshoots or oscillations before reaching a steady-state value. This corresponds to a relative damping ratio of less than 1:0.
- (34) 'Daylight visual' means a visual system capable of meeting, as a minimum, system brightness, contrast ratio requirements and performance criteria appropriate for the level of qualification or FCS sought.
- (35) 'Design data' means the set of aircraft and system data and other information used to develop and test the simulation models, flight deck hardware and other simulation aspects.
- (36) 'Driven' means a state in which the objective test input stimulus or variable is 'driven' or deposited by automatic means, generally a computer input. Typical methods are the use of open-loop and closed-loop drivers.
- (37) 'Engineering judgement' means making a determination on a problem or test result using accepted engineering principles and all available relevant information.
- (38) 'Engineering report' (ER) means a document produced by the FSTD manufacturer to describe the data and methods used to design and verify an FSTD against the requirements in this CS-FSTD.
- (39) 'Engineering simulation' means an integrated set of mathematical models representing a particular aircraft configuration, which is typically used by the aircraft manufacturer or data provider for a wide range of engineering analysis tasks including

engineering design, development and certification. It is also used to generate data for checkout, proof-of-match (POM) / validation and other training FSTD data documents.

- (40) 'Engineering simulator' means the aircraft manufacturer's or data provider's simulator, which typically includes a full-scale representation of the simulated aircraft flight deck, operates in real time and can be flown by a pilot to subjectively evaluate the simulation. It contains the engineering simulation models for FSTDs, which are also released by the aircraft manufacturer or data provider to the industry. The engineering simulator may or may not include actual on-board system hardware in lieu of software models.
- (41) 'Engineering simulator/simulation data' means data generated by an engineering simulation or engineering simulator, depending on the aircraft manufacturer's or data provider's processes.
- (42) 'Engineering simulator/simulation validation data' means validation data generated by an engineering simulation or engineering simulator.
- (43) 'Entity' means, in the context of the simulated environment, an aircraft, ground vehicle or other dynamic object.
- (44) 'Entry into service' means the original state of the configuration and systems at the time a new or major derivative aircraft is first put into commercial operation.
- (45) 'Equipment specification list' (ESL) means a list as defined in Article 2 of Regulation (EU) No 1178/2011.
- (46) 'Essential match' (EM) means a comparison of two objective test results for which the differences should be negligible.
- (47) 'Evaluation' is defined in Article 2 of Regulation (EU) No 1178/2011
- (48) 'Extended reality' (XR) is an encompassing term to refer to virtual reality (VR), mixed reality (MR) and AR using an HMD.
- (49) 'FSTD capability signature' (FCS) is defined in Article 2 of Regulation (EU) No 1178/2011.
- (50) 'Field of regard' in relation to an XR headset is defined as the range of vision of an XR headset. It is typically measured in degrees and describes the angular range of the display, both horizontally and vertically, from the perspective of the headset's wearer, including when moving the head around a central reference point.
- (51) 'First segment' means the portion of the take-off profile from lift-off to completion of gear retraction.
- (52) 'Flight deck' means the shell of the cockpit and all associated panels, seats, controls, etc.
- (53) 'Flight deck cut-off angle' means the downward angle from the pilot's reference eyepoint at which the view is blocked by the aircraft structure and flight deck components (e.g. glareshield).

- (54) 'Flight test data' means actual aircraft data obtained by the aircraft manufacturer or other data supplier during an aircraft flight test programme.
- (55) 'Footprint data' means objective test results recorded in an FSTD.
- (56) 'Footprint test' means objective test results recorded in an FSTD that serve as the reference for recurrent testing of the FSTD.
- (57) 'Field of view' (FOV), in relation to a visual system, means the range of displayed imagery that is visible from both pilot eyepoints in the horizontal and vertical directions.
- (58) 'Field of view' (FOV), in relation to an XR headset, is defined as the extent of the observable display of the headset in a static position. It is typically measured in degrees and describes the angular extent of the display, both horizontally and vertically, from the perspective of the HMD's wearer. The observable FOV depends on the geometry of the human skull, most crucially the eye relief distance (distance from the rear lens to the eye).
- (59) 'Free response' means the response of the aircraft after completion of a control input or disturbance in a hands-off manoeuvre, i.e. with no pilot control inputs.
- (60) 'Frozen/locked' means a state in which a variable is held constant over time.
- (61) 'Flight simulation training device' (FSTD) is defined in Article 2 of Regulation (EU) No 1178/2011.
- (62) 'FSTD configuration' means that set of components and software necessary to ensure that the device can provide the necessary training, testing or checking capability for the FCS being sought.
- (63) 'FSTD data' means the various types of data used by the FSTD manufacturer and the applicant to design, manufacture, test and maintain the FSTD.
- (64) 'FSTD operator' means the organisation directly responsible to the competent authority for requesting and maintaining the qualification of an FSTD.
- (65) 'FSTD qualification' is defined in Article 2 of Regulation (EU) No 1178/2011.
- (66) 'FSTD training envelope' means the high and moderate confidence regions of the FSTD validation envelope.
- (67) 'FSTD validation envelope' means an alpha/beta envelope (or equivalent method) depicting the confidence level of the aerodynamic model. The envelope is defined by three regions: (i) flight test validated, (ii) wind tunnel or analytical and (iii) extrapolated (refer to CS FSTD.QB.107.2.STALL for further definitions).
- (68) 'Fuel used' means the quantity of fuel consumed as recorded in the aircraft simulation. This can be represented in various units, such as kilograms, pounds, litres, gallons, percentage of total or any other valid format that provides relevant data to the pilot.
- (69) 'Full stall' or 'post-stall region' means flight conditions at an angle of attack greater than the critical angle of attack.

- (70) 'Full sweep' means the movement of the controller from neutral to a stop, usually the aft or right stop, then to the opposite stop and then to the neutral position.
- (71) 'Functions test' means a quantitative or qualitative assessment of the operation and performance of an FSTD by a suitably qualified evaluator. The test can include verification of the correct operation of controls, instruments and systems of the simulated aircraft systems under normal and non-normal conditions.
- (72) 'General requirements' means the requirements given in CS FSTD.QB.101, CS FSTD.QB.102, CS FSTD.QB.103, CS FSTD.QB.104, CS FSTD.QB.105, CS FSTD.QB.106, CS FSTD.QB.107, CS FSTD.QB.108, CS FSTD.QB.109, CS FSTD.QB.110, CS FSTD.QB.111, CS FSTD.QB.112, CS FSTD.QB.113, CS FSTD.QB.114 and CS FSTD.QB.115.
- (73) 'Ground effect' means the change in aerodynamic characteristics due to the modification of the airflow around the aircraft caused by the presence of the ground at heights above the ground or water equal to the length of the aeroplane's wingspan or rotor diameter of a rotorcraft, unless otherwise declared by the aircraft manufacturer or data provider (as applicable).
- (74) 'Ground reaction' means forces acting on the aircraft due to contact with the ground. These forces include the effects of strut deflections, tyre friction, side forces, structural contact and other appropriate aspects. These forces change appropriately with, for example, weight and speed.
- (75) 'Group of aircraft' means aircraft that have similar handling and operational characteristics. For example, aircraft operated under the privileges of a particular pilot class rating are regarded as a distinct group of aircraft.
- (76) 'Heavy' in the context of aircraft performance means with operational mass at or near the maximum for the specified flight condition.
- (77) 'Height' means the height above ground level (AGL) (in m or ft).
- (78) 'High angle of attack' means flying at an angle of attack higher than in normal operation, beyond the first indication of stall or stall protection systems, whichever occurs first.
- (79) 'Head mounted display', also known as a head-worn display, is a device worn on the head that projects images directly in front of the user's eyes, creating an immersive visual experience.
- (80) 'Icing accountability' means a demonstration of minimum required performance while operating in the maximum and intermittent maximum icing conditions of the applicable airworthiness requirement. It refers to changes from normal (as applicable to the individual aircraft design) in take-off, climb (en route, approach, landing) or landing operating procedures or performance data, in accordance with the AFM, for flight in icing conditions or with ice accumulation on unprotected surfaces.
- (81) 'Initial evaluation' is defined in Article 2 of Regulation (EU) No 1178/2011.

- (82) 'Integrated testing' means the testing of the FSTD such that all aircraft system models are active and contribute appropriately to the results. None of the aircraft system models should be substituted with models or other algorithms intended for only testing purposes. This may be accomplished by using controller displacements as the inputs. These controllers should represent the displacement of the pilot's controls and these controls should have been calibrated.
- (83) 'Intended use' means the training, testing and checking tasks that the device is built to be able to support.
- (84) 'Interpupillary distance' (IPD) is the measurement of the distance between the centres of the pupils of the eyes when the observer focuses on an object at a certain distance. The measurement is required to ensure the proper alignment and fit of an XR headset and thus the optimal visual comfort and performance of the virtual display. Industry standards typically define it as the distance between the centres of the pupils when looking straight ahead, measured in millimetres. The IPD may be expressed either as a single value for the distance between both pupils, or as a value for each eye.
- (85) 'Irreversible control system' means a control system in which the movement of the control surface will not backdrive the pilot's control on the flight deck.
- (86) 'Landing area', in association with the 'operating sites and terrain (OST)' FSTD feature, means any modelled surface and/or structure used for ground operations, take-off, landing or other close-proximity flight operations (e.g. aerodrome, airport, airfield, heliport, vertiport, ship deck).
- (87) 'Landing site', in association with the 'operating sites and terrain feature (OST)' FSTD feature, means a non-airport and non-prepared landing area for rotorcraft use (e.g. accident site or rescue mission destination).
- (88) 'Latency' means the additional time, beyond that of the basic perceivable response time of the aircraft, due to the response time of the FSTD.
- (89) 'Legacy FSTD' is defined in Article 2 of Regulation (EU) No 1178/2011.
- (90) 'Light', in the context of aircraft performance, means with operational mass at or near the minimum for the specified flight condition.
- (91) 'Line-oriented flight training' refers to flight crew training that involves full-mission simulation of situations that are representative of line operations, with specific emphasis on situations that involve communications, management and leadership. It means 'real-time', full-mission training.
- (92) 'Manual testing' means FSTD testing in which the pilot conducts the test without computer inputs except for initial set-up. All modules of the simulation should be active.
- (93) 'Master qualification test guide' (MQTG) means the qualification test guide (QTG) to be used as the reference for future testing and evaluations. The MQTG approval and amendment processes are presented in Part-ORA and Part-ARA (Annexes VI and VII, respectively) of Commission Regulation (EU) No 1178/2011.

- (94) 'Medium', in the context of aircraft performance, means the normal operational weight for the flight segment.
- (95) 'Modification' is defined in Article 2 of Regulation (EU) No 1178/2011.
- (96) 'Mixed reality' (MR) means that the real environment (including camera-perceived) elements and computer-generated elements are merged. Physical and virtual content are presented to the observer on an HMD. All elements may coexist in MR and interact in real time. For example, the image from a head-mounted camera system is integrated into the HMD virtual environment.
- (97) 'Near performance-limited condition' means, if related to approach to stall or stall, a stall event occurring close to the lowest limit of the:
 - (i) maximum certified altitude (structural) or maximum operating envelope;
 - (ii) thrust-limited altitude; and
 - (iii) buffet- or manoeuvre-limited altitude.
- (98) 'Night visual' means a visual system capable of meeting, as a minimum, the visual cueing FSTD feature system brightness, contrast ratio requirements and performance criteria appropriate for the level of qualification sought. The system, if used in training, should provide, as a minimum, all features applicable to the twilight visual scene, as defined below, with the exception of the need to portray reduced ambient intensity that removes ground cues that are not self-illuminating or illuminated by ownship lights (e.g. landing lights).
- (99) 'Nominal' means the normal operational weight, configuration, speed, etc. for the flight segment specified.
- (100) 'Non-normal control' means, in reference to CCA, the state in which one or more of the intended control, augmentation or protection functions are not fully available. Note that terms such as 'alternate', 'direct', 'secondary' and 'backup' may be used to define the actual level of degradation for different aircraft types.
- (101) 'Normal control' means, in reference to CCA, the state in which the intended control, augmentation and protection functions are fully available.
- (102) 'Objective test' or 'objective testing' means a quantitative assessment based on a comparison with data.
- (103) 'Off-axis' means the axis (x, y, z, pitch, roll, yaw) other than the primary axis for an objective test. For example, helicopters typically have cross-couplings leading to a flight control input resulting in an output on the controlled axis (primary axis), but also on another, perpendicular axis (off-axis).
- (104) 'One step' means the degree of changes to an aircraft that would be allowed as an acceptable change, relative to a fully flight-test-validated simulation. The intention of this approach is that changes would be limited to one step rather than a series of steps away from the baseline configuration. It is understood, however, that those changes

that support the primary change (e.g. weight, thrust rating and control system gain changes accompanying a body length change) are considered part of the 'one step'.

- (105) 'Open-loop' means an objective testing method in which the test input or variable is driven or directly replayed. Small explainable offsets/biases are acceptable, but closed-loop followers are not acceptable.
- (106) 'Operating site' means an aerodrome, airport, airfield, heliport, vertiport or other landing area.
- (107) 'Operating site clutter' means the set of ground entities added to a visual scene to create a sense of activity. Clutter may include both static and dynamic models, such as gate infrastructure, baggage carts, ground personnel, ground service vehicles and aircraft parked or undertaking ground movements.
- (108) 'Other data' means non-aircraft data used as validation data that is typically approved for use with features of fidelity levels R and G. Other data may include aircraft performance data, footprint data, aircraft CSs and data from other sources (refer to Subpart C for further information).
- (109) 'Other flight controls' means flight controls that are not classified as primary flight controls and controls that affect aircraft configuration. These include but are not limited to trims, flap levers, speed brake levers and landing gear levers.
- (110) 'Other traffic' means entities other than the ownship in the simulated environment. This traffic will include other aircraft, both airborne and on the ground, and may also include ground vehicles at the operating site.
- (111) 'Ownship' means the user-controlled aircraft simulated within the FSTD.
- (112) 'Phase of flight' means, in the case of aeroplanes, taxiing, take-off run, take-off flight path (including climb), cruise, descent, approach, final approach, missed approach, landing (including the landing roll) and any other phases of flight as required. In the case of rotorcraft, it means taxiing, hovering, take-off, final approach, missed approach, landing and any other phases of flight as required.
- (113) 'Pitch trim' refers to the pitch trim tab, horizontal stabiliser, canard, etc. as applicable to the simulated aircraft.
- (114) 'Platform' means, for example, the:
 - (i) motion system;
 - (ii) visual system;
 - (iii) computers;
 - (iv) necessary additional equipment; or
 - (v) any combination of the above that can be used in more than one configuration.

- (115) 'Proof-of-match' (POM) means a test comparison that shows agreement, within defined tolerances, between model responses and flight test cases at identical test and atmospheric conditions.
- (116) 'Predicted data' means data derived from sources other than flight tests for the particular aircraft.
- (117) 'Primary axis' means the axis (x, y, z, pitch, roll, yaw) under scrutiny during an objective test. For example, the primary axis for a power change dynamic test is the pitch axis.
- (118) 'Primary flight controls' means the controls that are required for immediate control of the aircraft. For a typical aeroplane, these are usually identified as the column, wheel, pedal, engine/thrust control levers (e.g. throttles, propeller levers), tiller and toe brakes. For a typical helicopter, these are usually identified as cyclic, collective and pedals.
- (119) 'Primary reference document' (PRD) is defined in Article 2 of Regulation (EU) No 1178/2011.
- (120) 'Protection functions' means CCA system functions designed to protect an aircraft from exceeding its flight and manoeuvre limitations.
- (121) 'Pulse input' means an abrupt input to a control followed by an immediate return to the initial position.
- (122) 'Qualification test guide' (QTG) is defined in Article 2 of Regulation (EU) No 1178/2011.
- (123) 'Rationale' means the provision of additional information or an explanation of differences, deviations or anomalies to comply with a requirement through the provision of alternative data or the justification of such deviations, differences or anomalies as an acceptable basis on which to fulfil the requirements.
- (124) 'Recurrent evaluation' is defined in Article 2 of Regulation (EU) No 1178/2011.
- (125) 'Re-host' means the use of unmodified and executable aircraft equipment code running on a different computing platform in a simulated environment.
- (126) 'Re-target' means the use of aircraft equipment source code recompiled for a different computing platform in a simulated environment.
- (127) 'Reversible control system' means a partially powered or unpowered control system in which the movement of the control surface will backdrive the pilot's control on the flight deck or affect its feel characteristics.
- (128) 'Robotic test' means a basic performance check of a system's hardware and software components. Exact test conditions are defined to allow for repeatability. The components are tested in their normal operational configuration and may be tested independently of other system components.
- (129) 'Second segment' means the portion of the take-off profile from after gear retraction to the end of climb at take-off safety speed and initial flap/slat retraction.

- (130) 'Simulated air traffic control environment' (SATCE) means a synthetic environment for flight training in which ATC services and other traffic entities are simulated, including radio and data communications between ATC and all traffic.
- (131) 'Simulated aircraft' means the simulated aircraft type and variant, type or group of aircraft used as a reference for an FSTD simulation feature.
- (132) 'Snapshot' means a presentation of one or more variables at a given instant in time.
- (133) 'Special evaluation' is defined in Article 2 of Regulation (EU) No 1178/2011.
- (134) 'Statement of justification' (SOJ) means documentation giving a justification for and information on how each requirement is met or why it is considered not to be applicable.
- (135) 'Step input' means an abrupt input held at a constant value.
- (136) 'Subjective development' means the use of a documented process, prior to the initial evaluation, to supplement other sources of data, resolve issues with the available aircraft data or tailor the FSTD performance and handling qualities to better match the aircraft through the judgement of qualified personnel.
- (137) 'Subjective test' or 'subjective testing' means a qualitative assessment based on established standards as interpreted by a suitably qualified person.
- (138) 'Technically verified engineering simulator/simulation' means an aircraft manufacturer's engineering simulation that has undergone a review by the European Union Aviation Safety Agency (EASA) and been found to be an acceptable source of data to supplement flight test data.
- (139) 'Test condition' means the flight phase, aircraft configuration, flight, environmental, flight manoeuvre or weight / centre-of-gravity condition as required for applicable objective validation tests in the table of validation tests.
- (140) 'Third segment' means the portion of the take-off profile after flap/slat retraction is complete.
- (141) 'Throttle lever angle' (TLA) means the angle of the pilot's primary engine control lever(s) on the flight deck.
- (142) 'Time history' means a presentation of the change of a variable with respect to time.
- (143) 'Tolerance', in the context of objective tests, refers to the maximum acceptable deviation from the validation data. Tolerances may be either numerical or based on an assessment of CT&M.
- (144) 'Transport delay' means the total FSTD system processing time required from an input signal from a pilot primary flight control until the motion system, visual system or instrument response. It is the overall time delay incurred from signal input until output response. It does not include the characteristic delay of the aircraft simulated.
- (145) 'Twilight visual' means a visual system capable of meeting, as a minimum, the system brightness and contrast ratio requirements and performance criteria for dusk and

dawn appropriate for the level of qualification sought. The system, if used in training, should provide, as a minimum, full-colour presentations of reduced ambient intensity (as compared with a daylight visual system) sufficient to conduct a visual approach, landing and airport movement (taxi).

- (146) 'Type of aircraft' means the type of aircraft as defined in point FCL.010 of Part-FCL (Annex I) of Regulation (EU) No 1178/2011.
- (147) 'Upgrade' means the improvement or enhancement of an FSTD for the purpose of achieving a different FCS.
- (148) 'Upset' means an undesired aeroplane state characterised by unintentional deviations from parameters experienced during normal operations. An aeroplane upset may involve pitch or bank angle deviations or inappropriate airspeeds for the given conditions.
- (149) 'User' means the organisation or person requesting training, checking or testing through the use of an FSTD.
- (150) 'Validation' is defined in Article 2 of Regulation (EU) No 1178/2011.
- (151) 'Validation data' is defined in Article 2 of Regulation (EU) No 1178/2011.
- (152) 'Validation flight test data' means performance, stability, control and other necessary test parameters electrically or electronically recorded in an aircraft using a calibrated data acquisition system of sufficient resolution and verified as sufficiently accurate to establish a reference set of relevant parameters with which FSTD parameters can be compared.
- (153) 'Variant', in relation to aircraft, means a particular aircraft model/name within an aircraft type that is operated with a pilot type rating (e.g. the A320-100 and A321-200 are different variants of the A320). An aircraft type or model may have only one variant (e.g. at the time of publishing this CS-FSTD, the A380 aircraft type has only one variant, the A380-800).
- (154) 'Validation data roadmap' (VDR) is defined in Article 2 of Regulation (EU) No 1178/2011.
- (155) 'Verification' is defined in Article 2 of Regulation (EU) No 1178/2011.
- (156) 'Vibration' means periodic or random motion effects or movement created by the aircraft or other environmental effects.
- (157) 'Visual ground segment (VGS) test' means a test designed to assess the accurate portrayal of the visual scene presented at the pilot's eyepoint at decision height (DH) on an ILS approach for a given set of flight conditions, with consideration given to the specific runway approach lighting systems and the geometry of the airframe and cockpit.
- (158) 'Visual system response time' means the interval from an abrupt control input to the completion of the visual display scan of the first video field containing the resulting different information.

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- (159) 'Virtual reality' (VR) means a computer-generated, fully virtual visual representation of a three-dimensional (3D) environment that immerses the user in an artificial world on an HMD.
 - (160) 'Well-understood effect' means an incremental change made to a configuration or system that can be accurately modelled using proven predictive methods based on the known characteristics of the change.
 - (161) 'x-axis' means the axis pointing through the nose of the aircraft.
 - (162) 'y-axis' means the axis pointing to the right of the x-axis (facing in the pilot's direction of view), perpendicular to the x-axis.
 - (163) 'z-axis' means the axis pointing down through the bottom of the aircraft, perpendicular to the x-y plane in accordance with the right-hand rule.

CS FSTD.GEN.020 Abbreviations

The abbreviations used in this CS-FSTD are as follows.

2D	=	two-dimensional (a 2D instrument approach operation means an instrument approach operation using lateral navigation guidance only)
3D	=	three-dimensional (a 3D instrument approach operation means an instrument approach operation using both lateral and vertical navigation guidance)
A	=	aeroplane
ACARS	=	aircraft communication addressing and reporting system
ACAS	=	airborne collision avoidance system
A_d	=	total initial displacement of pilot controller (initial displacement to final resting amplitude)
ADF	=	automatic direction finder
ADS-B	=	automatic dependent surveillance – broadcast
ADS-C	=	automatic dependent surveillance – contract
ADS-R	=	automatic dependent surveillance – rebroadcast
AEO	=	all engines operating
AFM	=	aircraft flight manual
AGL	=	above ground level (m or ft)
A_n	=	sequential amplitude of overshoot after initial x-axis crossing (e.g. A_1 means the first overshoot)
ANSI	=	American National Standards Institute
ANSP	=	air navigation service provider
AOA	=	angle of attack (degrees)
AP	=	autopilot
APSD	=	acceleration power spectral density
APU	=	auxiliary power unit
APV	=	approach procedures with vertical guidance
AR	=	augmented reality
ARINC	=	Aeronautical Radio Incorporated
ARA	=	airborne radar approach
AT	=	autothrust
ATC	=	air traffic control

ATIS	=	automatic terminal information service
ATM	=	atmosphere and weather FSTD feature
ATO	=	approved training organisation
Baro	=	barometric
CAT I	=	category I operations with a decision height not lower than 200 ft and runway visual range in accordance with the operational weather minima
CAT II	=	category II operations with a decision height below 200 ft and down to 100 ft and runway visual range in accordance with the operational weather minima
CAT III	=	category III operations with a decision height below 100 ft or no decision height and runway visual range in accordance with the operational weather minima
CAVOK	=	ceiling and visibility OK (used in aviation weather reports)
CCA	=	computer-controlled aircraft
CCD	=	charge-coupled device
cd/m ²	=	candela per square metre (3.4263 cd/m ² = 1 ft-lambert)
CDFA	=	continuous descent final approach
CL max	=	maximum coefficient of lift
CLH	=	flight control forces and hardware FSTD feature
CLO	=	flight control systems operation FSTD feature
cm	=	centimetre
CG	=	centre of gravity
CPDLC	=	controller–pilot data link communication
CS	=	certification specification
CS-23	=	certification specifications for normal, utility, aerobatic and commuter aeroplanes
CS-25	=	certification specifications and acceptable means of compliance for large aeroplanes
CS-27	=	certification specifications and acceptable means of compliance for small rotorcraft
CS-29	=	certification specifications and acceptable means of compliance for large rotorcraft
CS-FSTD	=	certification specifications for flight simulation training devices
CS-SIMD	=	certification specifications for simulator data

CT&M	=	correct trend and magnitude
daN	=	decanewtons (1 daN = 10 N)
dB	=	decibel
deg	=	degree (°)
DH	=	decision height
DLIC	=	data link initiation capability
DME	=	distance measuring equipment
DOF	=	degree of freedom
EASA	=	European Union Aviation Safety Agency
EFB	=	electronic flight bag
EFVS	=	enhanced flight vision system
EGT	=	exhaust gas temperature
EM	=	essential match
EPR	=	engine pressure ratio
ESL	=	equipment specification list
F&S	=	functions and subjective testing
EVS	=	enhanced vision system
FAA	=	Federal Aviation Administration
FADEC	=	full authority digital engine control
FANS	=	future air navigation system
FATO	=	final approach and take-off area
FCOM	=	flight crew operating manual
FCS	=	flight simulation training device capability signature
FD	=	flight director
FDK	=	flight deck layout and structure FSTD feature
FF	=	engine fuel flow
FFT	=	fast Fourier transform
FIS-B	=	flight information service – broadcast
FL	=	flight level
FMS	=	flight management system
FOR	=	field of regard
FOV	=	field of view

fpm	=	feet per minute
FSTD	=	flight simulation training device
ft	=	foot (1 ft = 0.3048 m)
ft-lambert	=	foot-lambert (1 ft-lambert = 3.4263 cd/m ²)
g	=	acceleration due to gravity (m/s ² or ft/s ²) (1 g = 9.80665 m/s ²)
G	=	generic (as related to FSTD feature fidelity level)
GBAS	=	ground-based augmentation system
GM	=	guidance material
GND	=	performance and handling on ground FSTD feature
GNSS	=	global navigation satellite system
grms	=	mathematical expression for g's root mean
grms ²	=	mathematical expression for g's root mean squared
G/S	=	glideslope
GLS	=	GBAS landing system
H	=	helicopter/rotorcraft
HF	=	high frequency
HMD	=	head-mounted display
HSI	=	horizontal situation indicator
HTAWS	=	helicopter terrain awareness and warning system
HUD	=	head-up display
Hz	=	hertz (unit of frequency) (1 Hz = one cycle per second)
IAS	=	indicated airspeed
IATA	=	International Air Transport Association
ICAO	=	International Civil Aviation Organization
IEC	=	International Electrotechnical Commission
IFR	=	instrument flight rules
IGE	=	performance and handling in-ground effect FSTD feature
ILS	=	instrument landing system
IMC	=	instrument meteorological condition
in	=	inch (1 in = 2.54 cm)
Init	=	initial objective test (in accordance with tolerances)
IOS	=	instructor operating station

IPD	=	interpupillary distance
ITT	=	inter-turbine temperature
JAWS	=	Joint Airport Weather Studies
km	=	kilometre (1 km = 0.62137 sm)
kt	=	knot (calibrated airspeed unless otherwise specified) (1 kt = 0.5148 m/s = 1.6878 ft/s)
lb	=	pound (mass) (1 lb = 0.4536 kg)
lbf	=	pound-force, unit of force (1 lbf = 4.448 N = 0.4448 daN)
LDP	=	landing decision point
LNAV	=	lateral navigation
LOC	=	localiser
LOC-BC	=	localiser back course
LPV	=	localiser performance with vertical guidance
m	=	metre (1 m = 3.28084 ft)
MCC	=	multi-crew cooperation
MDA	=	minimum descent altitude
min	=	minutes
MLS	=	microwave landing system
mm	=	millimetres
M _{MO}	=	maximum operating limit speed (Mach)
MQTG	=	master qualification test guide
MR	=	mixed reality
ms	=	millisecond
MTN	=	motion cueing FSTD feature
MTOW	=	maximum take-off weight
<i>n</i>	=	load factor
<i>n</i>	=	sequential period of a full cycle of oscillation
N	=	none (as related to FSTD feature fidelity level)
n/a	=	not applicable
N1	=	engine low-pressure rotor speed (typically expressed as a percentage of maximum revolutions per minute)
N2	=	engine high-pressure rotor speed (typically expressed as a percentage of maximum revolutions per minute)

NASA	=	National Aeronautics and Space Administration
Nf	=	free turbine speed (typically expressed as percent of maximum RPM), also known as power turbine speed
NAV	=	navigation FSTD feature
NDB	=	non-directional beacon
Ng	=	gas generator speed (typically expressed as a percentage of maximum revolutions per minute)
NM	=	nautical mile (1 NM = 1 852 m)
NOTAR	=	no tail rotor (a technology used in helicopters to eliminate the need for a conventional tail rotor)
NVG	=	night-vision goggle
NR	=	main rotor speed
NVIS	=	night-vision imaging system
N _x	=	load factor in the aircraft x-axis direction
N _y	=	load factor in the aircraft y-axis direction
N _z	=	load factor in the aircraft z-axis direction
NWA	=	nosewheel angle (degrees)
OAT	=	outside air temperature
OEI	=	one engine inoperative
OEM	=	original equipment manufacturer
OGE	=	performance and handling out-of-ground effect FSTD feature
OSD	=	operational suitability data (as in Part 21 (Annex I) of Commission Regulation (EU) No 748/2012)
OST	=	operating sites and terrain FSTD feature
P ₀	=	time from pilot controller release until initial x-axis crossing (x-axis defined by the resting amplitude)
P ₁	=	first full cycle of oscillation after the initial x-axis crossing
P ₂	=	second full cycle of oscillation after the initial x-axis crossing
PFD	=	primary flight display
P _n	=	sequential period of oscillation
POM	=	proof of match
PSD	=	power spectral density
psi	=	pounds per square inch (1 psi = 6 894.76 Pa)

QFE	=	altimeter pressure setting related to a particular reference datum point (e.g. aerodrome elevation)
QNH	=	altimeter pressure setting related to sea level
QTG	=	qualification test guide
QTG(A)	=	qualification test guide for aeroplane flight simulation training device
QTG(H)	=	qualification test guide for rotorcraft flight simulation training device
R	=	representative (as related to FSTD feature fidelity level)
RAeS	=	Royal Aeronautical Society
Rec	=	recurrent objective test (in accordance with tolerances)
RFM	=	rotorcraft flight manual
RMS	=	root mean square
RNP	=	required navigation performance
RNP APCH	=	approach in accordance with the navigation specification 'RNP APCH' of International Civil Aviation Organization Doc 9613 for performance-based navigation (may have minima for lateral navigation, lateral/vertical navigation, localiser performance and localiser performance with vertical guidance)
RNP AR APCH	=	approach in accordance with the navigation specification 'RNP AR APCH' of International Civil Aviation Organization Doc 9613 for performance-based navigation ('AR' stands for 'authorisation required')
RPM	=	revolutions per minute
RVR	=	runway visual range (m or ft)
s	=	second
S	=	specific (as related to FSTD feature fidelity level)
SATCE	=	simulated air traffic control environment
SBAS	=	satellite-based augmentation system
sm	=	statute mile (1 sm = 1 609 m)
SME	=	subject matter expert
SMGCS	=	surface movement guidance and control system
SND	=	sound cueing FSTD feature
SOJ	=	statement of justification
SPL	=	sound pressure level
SYS	=	aircraft systems FSTD feature
T(A _d)	=	tolerance applied to residual amplitude

T(P)	=	tolerance applied to period
TACAN	=	tactical air navigation
TASE	=	training areas of specific emphasis associated with the OSD reports
TAWS	=	terrain awareness warning system
TDM	=	training device manufacturer
TDP	=	take-off decision point
T _i	=	total time from initial throttle movement until a critical engine parameter reaches 10 % of its total response above idle power
TIS-B	=	traffic information service – broadcast
TLA	=	throttle lever angle
TLOF	=	touchdown and lift-off
T _t	=	total time from initial throttle movement until a critical engine parameter reaches 90 % of its total response above idle power
UPRT	=	upset prevention and recovery training
V ₁	=	critical decision speed
V ₂	=	take-off safety speed
VDR	=	validation data roadmap
VFR	=	visual flight rules
VGS	=	visual ground segment
VHF	=	very high frequency
VIB	=	vibration cueing (FSTD feature)
VIS	=	visual cueing (FSTD feature)
VMC	=	visual meteorological condition
V _{MCA}	=	minimum control speed (air)
V _{MCG}	=	minimum control speed (ground)
V _{MCL}	=	minimum control speed (landing)
V _{MO}	=	maximum operating limit speed (airspeed)
V _{MU}	=	minimum unstick speed
VNAV	=	vertical navigation
VOR	=	very-high-frequency omnidirectional range
VR	=	virtual reality
V _R	=	rotation speed

V_S	=	stall speed (or minimum speed in the stall)
V_Y	=	optimum climbing speed / best rate of climb airspeed
WAT	=	weight, altitude, temperature
XR	=	extended reality

SUBPART B – QUALIFICATION BASIS

GENERAL

CS FSTD.QB.001 Qualification standards

- (a) Any FSTD submitted for initial evaluation should be evaluated against all applicable CS-FSTD criteria for the requested FCS.
- (b) The FSTD should be thoroughly tested to demonstrate that it meets all the applicable CS-FSTD criteria for the requested FCS, ensuring the accuracy, functionality and reliability of the simulation.
- (c) The FSTD should be subjected to:
 - (1) objective tests; and
 - (2) functions and subjective tests.
- (d) The qualification test guide (QTG), including all data, supporting material and information, should be submitted in a format that allows for its efficient review and evaluation before the FSTD can be qualified and gain an FCS.
- (e) If applicable, the QTG should be based on the aircraft validation data as defined by the OSD established in accordance with Part 21 (Annex I) of Commission Regulation (EU) No 748/2012.

CS FSTD.QB.005 Minimum flight simulation training device capability signature

- (a) In order to use this CS-FSTD as the qualification basis for any FSTD, the FCS should have at least the features and fidelity levels presented in Table 1 below.

Table 1. Minimum features and fidelity levels for an FCS for an FSTD

FSTD feature	Fidelity level
Flight deck layout and structure	G
Flight control forces and hardware	N
Flight control systems operation	G
Aircraft systems	G (see table note)
Performance and handling on ground	N
Performance and handling in-ground effect	N
Performance and handling out-of-ground effect	G
Sound cueing	N
Vibration cueing	N

Motion cueing	N
Visual cueing	N
Navigation	N
Atmosphere and weather	G
Operating sites and terrain	N

Note: At least the aircraft systems required under CS FSTD.QB.010 should be simulated.

- (b) An aeroplane FSTD that is to be qualified to conduct upset prevention and recovery training (UPRT) should have at least the features and fidelity levels presented in Table 2 below and fulfil the UPRT requirements of this CS-FSTD.

Table 2. Minimum FSTD features and fidelity levels for an FCS for an aeroplane FSTD to be qualified for UPRT

Feature	Fidelity level
Flight deck layout and structure	S
Flight control forces and hardware	S
Flight control systems operation	S
Aircraft systems	S
Performance and handling on ground	S
Performance and handling in-ground effect	S
Performance and handling out-of-ground effect	S
Sound cueing	R
Vibration cueing	R
Motion cueing	S
Visual cueing	R
Navigation	S
Atmosphere and weather	S
Operating sites and terrain	S

- (c) If the FCS of an aeroplane FSTD has at least the features and fidelity levels presented in Table 2 above, then the device should also comply with the UPRT requirements of this CS-FSTD.

GM1 FSTD.QB.005(a) Minimum flight simulation training device capability signature

MINIMUM FLIGHT SIMULATION TRAINING DEVICE CAPABILITY SIGNATURE FOR A FLIGHT SIMULATION TRAINING DEVICE

While some features may be at fidelity level N, it is fundamental that a select number of the features are at level G, R or S to qualify any FSTD in accordance with this CS-FSTD, which potentially enables

the FSTD to be used for accredited training. Rationales for the minimum FCS selection can be summarised as follows.

- (a) **Flight deck layout and structure** – It is fundamental that an FSTD can enable at least some interface with the user.
- (b) **Flight control forces and hardware** – Some aircraft systems training tasks can be performed with no tactile flight controls hardware, so this feature could be at fidelity level N (e.g. a 2D flat panel trainer).
- (c) **Flight control systems operation** – This feature is fundamental, as flight control systems are fundamental, even for automatic flight system management training, and they interface with multiple aircraft systems.
- (d) **Aircraft systems** – It is essential to have some aircraft system (e.g. basic instruments) simulated.
- (e) **Performance and handling on ground** – If the intended training is limited to flying in- and/or out-of-ground effect only and does not cover ground operations, this feature could be at fidelity level N.
- (f) **Performance and handling in-ground effect** – If the intended training is limited to flying out-of-ground effect, this feature could be at fidelity level N.
- (g) **Performance and handling out-of-ground effect** – This feature is fundamental for any flight training tasks.
- (h) **Sound cueing** – This feature could be at level N. Not all training tasks require sound cueing.
- (i) **Vibration cueing** – This feature could be at level N. Not all training tasks require vibration cueing.
- (j) **Motion cueing** – This feature could be at level N. Not all training tasks require motion cueing (e.g. fixed-base FSTDs).
- (k) **Visual cueing** – This feature could be at level N. Not all training tasks require a visual system or visual cueing.
- (l) **Navigation** – This feature could be at level N if the training tasks are limited to visual navigation under visual flight rules (VFR).
- (m) **Atmosphere and weather** – This feature is fundamental for any flight training tasks. The performance and handling models rely on even basic atmospheric parameter simulation.
- (n) **Operating sites and terrain** – This feature could be at level N. Not all training tasks require a visual system or visual cueing.

GM2 FSTD.QB.005(b);(c) Minimum flight simulation training device capability signature

MINIMUM FLIGHT SIMULATION TRAINING DEVICE CAPABILITY SIGNATURE FOR A FLIGHT SIMULATION TRAINING DEVICE TO BE QUALIFIED FOR UPSET PREVENTION AND RECOVERY TRAINING

- (a) The training requirements specify that an FSTD may be used for UPRT only if its overall fidelity meets the required standards. For this reason, Table 2 of CS FSTD.QB.005 specifies the minimum FCS necessary for an FSTD to be qualified for UPRT.
- (b) An FSTD that meets or exceeds the minimum FCS for UPRT, as specified in Table 2 of CS FSTD.QB.005, should comply with the UPRT requirements. This is to maintain continuity and alignment with established standards for the UPRT qualification of legacy FSTDs.
- (c) FSTDs qualified for full-stall training tasks should meet all applicable UPRT requirements and those specifically applicable to full-stall qualification.

CS FSTD.QB.010 Requirements for a flight simulation training device capability signature

- (a) An FSTD comprises a set of features, each assigned a certain fidelity level. Each feature's fidelity level can be N, G, R or S. The FCS for an FSTD details the features, their associated fidelity levels and the simulated aircraft type and variant, type or group of aircraft for each feature. The features are defined in CS.FSTD.GEN.005 and the levels are defined in CS FSTD.GEN.010.
- (b) This CS-FSTD sets criteria for fidelity levels G, R and S for each feature. To gain a certain level for a certain feature, all the requirements in this CS-FSTD applicable to that level should be fulfilled. As an exception, the individual aircraft systems may be at different fidelity levels, which again affects the aircraft systems FSTD feature in the FCS. The documentation of the FSTD should clearly define the fidelity level of each applicable aircraft system. The following apply for the aircraft systems.
 - (1) The FSTD should simulate the following aircraft systems:
 - (i) basic flight instruments (refer to CS FSTD.QB.101, section 'Flight deck panels, instruments and switches')
 - (ii) caution and warning systems (refer to CS FSTD.QB.104, section 'Caution and warning systems') and
 - (iii) power plant systems (refer to CS FSTD.QB.104, section 'Power plant systems').
 - (2) Other aircraft systems may be simulated to support the intended use of the device.
 - (3) The individual aircraft systems may be at different fidelity levels (N/G/R/S). In this case, aircraft system interdependencies should be taken into consideration (refer to paragraph (e) below).

- (c) If the feature or aircraft system is not simulated, it is at fidelity level N. If the feature or aircraft system does not comply with all the requirements for any of level G, R or S, the feature or system is considered to be at level N.
- (d) The FCS, together with the defined fidelity level of each simulated aircraft system, defines the applicable objective tests and functions and subjective tests that are required for the qualification of the FSTD.
- (e) The FCS, together with the defined fidelity level of each simulated aircraft system, should support the intended use of the device and enable integration, validation and verification of the device. Interdependencies within and between features and aircraft systems should be taken into account when determining the FCS.
- (f) All features and the aircraft systems, at their respective fidelity levels, should enable the complete simulation to operate seamlessly, correspond appropriately and function in an integrated manner and in appropriate sequence. The simulation should operate in this way regardless of whether the input or stimuli originate from the flight deck, including pilot actions and flight control inputs, from the cueing or environment simulation or from the instructor station.

GM1 FSTD.QB.010 Requirements for a flight simulation training device capability signature

- (a) The training requirements define what FCS is required for training tasks. The features and fidelity levels in the FCS enable a wide range of possibilities, so a wide variety of FSTDs may be qualified. For example, an FSTD may not have any flight control hardware on the flight deck, but the flight control operation (e.g. system logic, indications, warning) may be at the highest level.
- (b) The requirements for aircraft features are specified as follows: fidelity level S should simulate a particular aircraft type and variant, level R should simulate an aircraft type, and level G may simulate a group of aircraft. These are the minimum requirements, but they may be exceeded (refer to paragraph (c)(2) below).
- (c) To gain a certain level for any feature, all the applicable requirements (i.e. general requirements in Subpart B, engineering and data requirements in Subpart C, objective test requirements in Subpart D and functions and subjective test requirements in Subpart E) should be fulfilled.
 - (1) For example, if the feature otherwise fulfils the general requirements for level S, but the used data fulfils only level R requirements, the feature can be qualified only to level R.
 - (2) For example, an aircraft feature may simulate a certain aircraft type and have data supporting fidelity level R. However, if the simulation model does not meet all the requirements of level R, only satisfying the requirements of level G, the feature can only be qualified to level G.

- (d) The aircraft systems FSTD feature may have individual systems modelled at different fidelity levels, based on training needs. For example, the flight management and auto-flight systems of an FSTD may be at the specific fidelity level, while the pressurisation system is only at the representative level. The aircraft systems cannot always be treated in isolation, since they are often integrated with other systems. This may result in having to alter some of the aircraft systems in order to properly integrate the aircraft systems in the FSTD. As an example, the FMS and FD systems may be integrated in the real aircraft and consequently the integration of these in the FSTD may not be possible to achieve if one of them is at the G fidelity level while the other one is at the S level.
- (e) Individual features and their fidelity levels cannot be treated in isolation. The training device will be used in an integrated manner, and certain features may have a dependency on other features for integrated operation. The features and systems should operate and correspond appropriately, meaning that, for example, ACAS traffic shown on the flight deck display should appear correspondingly in the visual system, if a visual system is fitted. This may result in the planned FCS having to be altered to ensure compliance and correspondence among all the features. Examples of this are as follows.
- (1) It may not be possible to have the performance and handling FSTD features at the specific fidelity level if the flight control forces and hardware FSTD feature is at the generic level because some objective tests (e.g. 2.a.8) may not demonstrate compliance with the requirements. It may be necessary to enhance the flight control forces and hardware FSTD feature to above level G.
 - (2) It may not be possible to have the performance and handling FSTD features at the specific level if the flight control systems operation FSTD feature is at the generic level. This is because the flight control systems operation gives inputs to the flight dynamic model. In addition, the auto-flight control system contributes to the inputs. To be able to demonstrate compliance, it may be necessary to enhance the flight control systems operation FSTD feature to above level G.
 - (3) It may not be possible to have the vibration cueing FSTD feature at the specific fidelity level if the performance and handling FSTD features have lower fidelity levels because the vibration system receives its inputs from the flight model. If the inputs (e.g. triggering of stall buffet for aeroplanes or translational lift for helicopters) are not given under the correct conditions, the resulting vibration/buffet cues may not fulfil the requirements for the S level.
 - (4) A TAWS/HTAWS is considered as an aircraft system, but, if a visual system is installed, the visual image should seamlessly support the TAWS/HTAWS indications (e.g. mountains, obstacles). The lowest level of the operating sites and terrain FSTD feature may not support this, and consequently the level of that feature may have to be enhanced.
- (f) The FSTD should support an open simulation mode, allowing the flight crew full control over the entire ownship, including its systems, without predefined constraints or scenarios. However, for individual and isolated aircraft systems or their elements, the simulation may

utilise predefined scenarios with set starting conditions and/or sequences of events that must be followed. Any such limitations must be documented in the ESL.

GM1 FSTD.QB.010(e);(f) Requirements for a flight simulation training device capability signature

INTEGRATION OF THE FEATURES

- (a) FSTD features encompass aspects of simulation, cueing and environmental modelling. These elements must work together to create a realistic training experience. This means integration of the whole simulation.
- (b) The components and operations of the FSTD can consistently be broken down into individual features. For example, for a landing gear, the following are required.
 - (1) The physical appearance, travel and tactile feel of the landing gear lever and the force required to move it are covered by the flight control forces and hardware FSTD feature.
 - (2) The operation modes and logic of the landing gear system are covered by the flight control systems operation FSTD feature.
 - (3) If the landing gear is operated in flight, the shape of the aircraft changes, affecting its aerodynamic properties. This is covered by the performance and handling FSTD features. These features rely on an atmospheric model, which is covered by the atmosphere and weather FSTD feature.
 - (4) If the landing gear is operated in flight, it may generate buffeting and sound. These are covered by the vibration cueing and sound cueing FSTD features, respectively.
 - (5) If the landing gear is operated in flight, the motion system provides cueing for changes in aircraft attitude due to the changes in airspeed and trajectory. During landing, the motion system provides a touchdown effect. These are covered by the motion cueing FSTD feature.

CS FSTD.QB.015 Flight simulation training device general requirements for feature fidelity levels

- (a) The general requirements specified in CS FSTD.QB.101, CS FSTD.QB.102, CS FSTD.QB.103, CS FSTD.QB.104, CS FSTD.QB.105, CS FSTD.QB.106, CS FSTD.QB.107, CS FSTD.QB.108, CS FSTD.QB.109, CS FSTD.QB.110, CS FSTD.QB.111, CS FSTD.QB.112, CS FSTD.QB.113 and CS FSTD.QB.114 are provided for each FSTD feature and fidelity level for both aeroplane and rotorcraft FSTDs. Certain aircraft types (e.g. tiltrotor aircraft) have characteristics of both aeroplanes and rotorcraft. For such aircraft, the applicable general requirements may be a combination of both aeroplane and rotorcraft requirements and/or special conditions.
- (b) The miscellaneous general requirements applicable to all FSTDs, but not related to any feature fidelity level, are given in CS FSTD.QB.115.

- (c) For the aircraft simulation FSTD features group (refer to CS FSTD.GEN.005), at the specific fidelity level, the FSTD should simulate a certain aircraft type and variant. At the representative fidelity level, the FSTD should simulate a certain aircraft type. At the generic fidelity level, the FSTD should be characteristic of a group of aircraft and the FSTD's performance and handling and system simulation should comply with the certification requirements of real aircraft for that group of aircraft.
- (d) For each FSTD feature, the general requirements are split into individual requirements with unique numbers. Tick marks indicate the fidelity levels (G, R, S) to which the requirements are applicable. For each FSTD feature, all the individual requirements applicable to the fidelity level of this FSTD feature should be met.
- (e) Objective tests should support the demonstration of compliance with the general requirements and consequently establish a reference for recurrent evaluations. In the tables of general requirements, the columns for 'QTG', 'QTG(A)' (the qualification test guide for aeroplane flight simulation training device) and 'QTG(H)' (the qualification test guide for rotorcraft flight simulation training device) contain, if applicable, the reference to the section of objective tests or a reference to a certain objective test. The applicable individual objective tests and requirements are determined in accordance with Subpart D. These objective tests should be included in the MQTG.
- (f) Functions and subjective testing (F&S) is a process requiring pilot expertise to assess the FSTD. In the tables of general requirements, the column 'F&S' contains tick marks to indicate if compliance with the requirement should be ensured and demonstrated by F&S. Details of functions and subjective test requirements can be found in Subpart E. Many tests are for certain piloting tasks or manoeuvres, such as take-off. Such tests are associated with more than one feature and therefore require integrated testing, simultaneously checking all the features and cues in the simulation. The F&S criteria are dictated by the general requirements applicable to the FCS.

GM1 FSTD.QB.015(e) FSTD general requirements for feature fidelity levels

OBJECTIVE TESTS

- (a) The QTG references in the tables of general requirements enable traceability between requirements and verification and validation. The cross references to the objective test sections ensure a minimum level of traceability for the general requirements.
- (b) The columns for objective testing in the general requirement tables are defined as follows:
 - (1) 'QTG(A)' columns refer to the objective tests for aeroplanes;
 - (2) 'QTG(H)' columns refer to the objective tests for rotorcraft;
 - (3) 'QTG' columns are used for the features for which the tests are not assigned to aeroplanes or rotorcraft separately.

(c) Examples of objective test reference notations are as follows:

- (1) '1.a' is a reference to objective test section 1.a;
- (2) '2.c.8.b' is a reference to the individual objective test 2.c.8.b;
- (3) '3.e, 3.f' is a reference to objective test sections 3.e and 3.f;
- (4) '1.e–1.h' is a reference to all the objective test sections from 1.e to 1.h.

GM1 FSTD.QB.015(e);(f) FSTD general requirements for feature fidelity levels

VERIFICATION & VALIDATION

In the context of FSTD assessment, verification and validation are fundamentally interconnected processes that play a crucial role in ensuring the compliance of training devices. While verification consists of testing to confirm that an FSTD meets and maintains the applicable technical qualification requirements, validation focuses on assessing its capabilities for training, testing and checking and its use as a training device. These processes encompass both objective testing and functions and subjective testing. Within this CS-FSTD, the terms 'verification' and 'validation' should be understood as a unified entity, reflecting the comprehensive approach to evaluating the compliance of the FSTD.

CS FSTD.QB.017 Additional functionality

An FSTD may have additional aircraft systems and/or functionality for which this CS-FSTD does not specify requirements. To verify and validate the suitability of those, the organisation requesting the qualification of an FSTD should:

- (a) declare and justify the impact of the additional functionality on all the features;
- (b) give a proposal to the competent authority on how to validate (i.e. through objective testing and/or functions and subjective testing) the additional functionality;
- (c) verify that the aircraft systems and functionality operate as described in the appropriate reference (e.g. AFM or its supplement);
- (d) validate and verify that the additional functionality and associated cueing are suitable for training use;
- (e) include the applicable information in the MQTG and ESL.

GM1 FSTD.QB.017 Additional functionality

- (a) The words 'additional aircraft systems and/or functionality' mean various training or mission scenarios, associated equipment and relevant environmental conditions.
- (b) Training or mission scenarios often include a visual element that depicts the operating site and terrain and/or are typically related to various rotorcraft missions. The following examples provide a non-exhaustive list of typical functionalities for rotorcraft FSTDs: airborne radar approaches, rig landing, ship landing, ditching, sinking ship scenarios, emergency evacuations, pursuit scenarios, cargo hook or hoist operations, firefighting operations, power line or

pipeline inspections, mountain rescue operations, accident site, search and rescue, aerial work and seasonal variations of the operating site.

- (c) The purpose is to assess and evaluate the additional aircraft systems and/or functionality systematically to ensure a positive transfer of training. This assessment should cover all aspects of the simulation, including system hardware and operation, performance and handling and cueing.
- (d) Any additional aircraft system and/or functionality may affect many of the aircraft features. For example, any external equipment – such as load on a helicopter hoist – will affect the performance and handling FSTD features because the load affects weight, drag, moments, dynamics, crash limits, and so forth. The flight deck layout and structure FSTD feature would be affected if there were any additional panels or pilot interfaces. Additionally, the aircraft systems FSTD feature would be affected, as the system simulation would need to account for its operation. The visual image contents would be affected if the load should be visible in the visual image. The sound cueing and effects may be affected if, for example, the load scrapes the ground or crashes. Additionally, the vibration cueing FSTD feature could be impacted if the load generates buffeting. Similar applies to any other feature affected.

CS FSTD.QB.020 Statements of justification

All the general requirements defined in CS FSTD.GEN.015 should be presented in statements of justification (SOJs) by the organisation requesting the qualification of an FSTD. The SOJs should fulfil the following requirements.

- (a) They should follow the numbering of the general requirement tables given in CS FSTD.QB.101, CS FSTD.QB.102, CS FSTD.QB.103, CS FSTD.QB.104, CS FSTD.QB.105, CS FSTD.QB.106, CS FSTD.QB.107, CS FSTD.QB.108, CS FSTD.QB.109, CS FSTD.QB.110, CS FSTD.QB.111, CS FSTD.QB.112, CS FSTD.QB.113, CS FSTD.QB.114 and CS FSTD.QB.115.
- (b) The SOJs should list all the general requirements (including their numbers) that are applicable to the FCS of the FSTD.
- (c) They should give a justification of how each general requirement is met or why it is considered not to be applicable to the FSTD in question. The given justification should be clear and sufficiently detailed to give evidence of and information on how the requirement is fulfilled.

The SOJs should refer to the sources of information and show the compliance rationale to explain how compliance was reached, how the referenced material was used, the applicable mathematical equations and parameter values and the conclusions reached.

For general requirements that are not applicable to the simulated aircraft, the SOJs should state 'not applicable' (n/a) and provide an explanation, if warranted.

- (d) They should be part of the MQTG and/or engineering report (ER).
- (e) The SOJs should be dated and signed by the organisation requesting the qualification of the FSTD.
- (f) They should be revised if modifications are made to the device.

GM1 FSTD.QB.020(b);(c) Statements of justification

- (a) An SOJ should be provided for each general requirement at the applicable fidelity level for each feature. For example, if a feature is at the representative level, the SOJ should list each general requirement for that level.
- (b) The SOJ should then explain how each requirement is met or why it is not applicable to the FSTD in question. Any requirements deemed not applicable to the simulated aircraft should include a sufficient explanation. For instance, if a general requirement applies only to multi-engine aircraft and the FSTD simulates a single-engine aircraft, a simple note indicating this is adequate. For cases that are less obvious, the explanation for any non-applicable requirements should be detailed enough to ensure that the rationale is clear.
- (c) It is unnecessary to duplicate information across multiple SOJs. If the same justification applies to several SOJs, it is appropriate to provide it once and reference it in other relevant SOJs.
- (d) It is only necessary to consider SOJs for the chosen fidelity level for each requirement.
- (e) If a requirement is identified for a particular aircraft category (aeroplane or rotorcraft), it applies only to that category for fidelity levels and features. Requirements without such a qualifier apply to both aeroplanes and rotorcraft, as relevant to the simulated configuration and fidelity level.

GM1 FSTD.QB.020 Statements of justification

- (a) The SOJ should be clear and sufficiently detailed to give evidence of and information on how the requirement is fulfilled. The scope of the information should be appropriate for the requirement for the FSTD in question. For example, an SOJ regarding a complex topic (e.g. a stall event, aircraft stability, icing) is necessarily longer than an SOJ for a simple and straightforward topic (e.g. visual field of view). Complex topics have more aspects to consider in the SOJ.
- (b) The examples below demonstrate the difference in the scope of the information for some fictitious cases.
 - (1) Table 1 provides an example of the SOJ for a feature that uses an OEM data package. The SOJ is simple and straightforward because it is expected that the OEM has all the necessary information to create a detailed data package that supports the accurate replication of the aircraft. If the FSTD were to use a different data source (e.g. reverse engineering), the SOJ should include much more detailed information to demonstrate how the requirements were being met.
 - (2) Table 2 provides an example of the SOJ for a feature at the generic fidelity level. For any SOJ for a feature at the generic level, it is important to provide clear information on what is being simulated and how it is being done.
 - (3) Table 3 provides an example of the SOJ if the general requirement is not applicable to the simulated aircraft.

- (4) Table 4 provides an example of the SOJ for a complex requirement – such as airframe icing – in which the MQTG references an additional SOJ in which a further, detailed description is provided.

Table 1. Example of an SOJ for an FSTD simulating an Airbus A320 (computer-controlled aircraft) and having the flight control systems operation FSTD feature at the specific fidelity level

Number	Requirement	Statement of justification
3.4.1	The flight control system functions for flight and manoeuvre envelope protection functionality should replicate the simulated aircraft type and variant.	The simulation is based on the OEM data package, VDR and validation data. The flight control system operation replicates all aspects of the Airbus sEFCS V06.00.00, including the flight and manoeuvre envelope protection functionalities. The objective tests in MQTG test section 2.h demonstrate that these functionalities accurately replicate the aircraft. Additionally, functions and subjective testing confirm that the functionalities perform correctly under various conditions and configurations.

Table 2. Example of an SOJ for an FSTD simulating a single-engine turbine aeroplane and having the atmosphere and weather FSTD feature at the generic fidelity level

Number	Requirement	Statement of justification
13.2.1	The standard atmosphere should be simulated, and the instructor should have control over key parameters.	The mathematical model in the FSTD for the standard atmosphere adheres to the International Standard Atmosphere (ISA) as published by the International Organization for Standardization (ISO) in ISO 2533:1975. The model includes the distribution of pressure, temperature, density and viscosity as functions of altitude from 1 500 ft below mean sea level up to the tropopause (11 km). The simulated aircraft does not have sufficient performance to climb above the tropopause, so the simulation of the atmosphere above this level is not applicable to the device. The atmospheric variables are calculated using formulas via floating-point calculations, ensuring that there are no discontinuities. The instructor has the controls to affect the following atmospheric parameters: — altimeter pressure setting related to sea level (1 hPa resolution); — temperature at mean sea level; — temperature lapse rate. In addition, the instructor has the controls for the following weather settings: — wind speed and direction at various altitudes; — wind gusts; — turbulence level; — cloud layers.

Table 3. Example of an SOJ for a helicopter FSTD having the aircraft systems FSTD feature at the specific fidelity level

Number	Requirement	Statement of justification
4.9.2	Anti-icing systems replicating those installed in the simulated aircraft type and variant should operate with appropriate effects upon ice formation on airframe, engines and instrument sensors.	This requirement is not applicable to this FSTD. The simulated configuration of the aircraft does not include anti-icing systems.

Table 4. Example of an SOJ for a complex specific fidelity level requirement in which the MQTG references an additional SOJ in which a further, detailed description is provided

Number	Requirement	Statement of justification
7.7.1, 7.7.2	7.7.1: The flight model should include the effects of icing, if applicable, on the airframe, the lift/thrust system aerodynamics and the engine(s). 7.7.2: Icing models should simulate the aerodynamic degradation effects of ice accretion on the lift/thrust system's lifting surfaces, including loss of efficiency, effect on power setting, change in pitching, rolling moments, decrease in control effectiveness and changes in control forces, in addition to any overall increase in drag or aircraft fuselage gross mass, with resulting effects on power. [Note: 7.7.2 is not quoted in full here, as this is only a guidance example.]	The simulation is based on the OEM data package, VDR and validation data. The performance model is based on OEM engineering simulation data in the form of incremental drag, lift, side force, pitching, rolling and yawing moment coefficients. The objective tests in Section 2.i demonstrate the objective performance of the model. Additionally, functional and subjective testing confirm that the functionalities perform correctly under various conditions and configurations. The effects of the airframe and engine icing model are described in further detail in the ER section 'Ice simulation model'.

CS FSTD.QB.030 Qualification test guide

- (a) The QTG resulting from the initial evaluation process becomes the MQTG (refer to Part-ORA and Part-ARA of Commission Regulation (EU) No 1178/2011). The MQTG may be later revised as applicable (e.g. due to modifications).

- (b) The QTG delivered to the competent authority for the purpose of becoming the MQTG should contain all the test results, SOJs and other information to demonstrate that the FSTD meets the requirements and test criteria. It should include:
- (1) a title page;
 - (2) an FSTD information page (for each FSTD configuration in the case of a convertible FSTD) providing:
 - (i) the FSTD identification number used by the organisation requesting the qualification of the FSTD,
 - (ii) the simulated aircraft for each feature (i.e. group, type or type and variant) in the aircraft simulation FSTD feature group (refer to CS FSTD.GEN.005),
 - (iii) a general description of the components of each feature,
 - (iv) references to aerodynamic design data or sources for the aerodynamic model,
 - (v) references to engine design data or sources for the engine model,
 - (vi) references to flight control design data or sources for the flight control model (which should include a declaration of the simulated primary flight control system, such as CCA, reversible or irreversible flight control system),
 - (vii) avionics equipment system identification and references to simulated avionics source data, including a list of the avionics systems that are simulated, how these systems are simulated and which OEM documents (pilot guide, flight crew operating manual (FCOM), AFM, etc.) can be used as the references for the system,
 - (viii) the FSTD model and manufacturer,
 - (ix) the month and year of FSTD delivery,
 - (x) the FSTD computer identification,
 - (xi) the visual display system type and manufacturer (if fitted),
 - (xii) the visual image generation system type and manufacturer (if fitted),
 - (xiii) the motion system type and manufacturer (if fitted);
 - (3) the table of contents;
 - (4) a list of effective pages;
 - (5) a revision history;
 - (6) a list of objective tests (including test number, run date, time, test revision and software load);
 - (7) a glossary of terms, acronyms and symbols used;
 - (8) the SOJs;
 - (9) the VDR;

- (10) the engineering report (ER);
- (11) the recording procedures and required equipment for the objective tests;
- (12) all the applicable objective tests and results;
- (13) all the applicable functions and subjective tests and results;
- (14) a record of the review, analysis and acceptance by the organisation requesting the qualification for each objective test, which may be provided either by signing each test individually or by indicating acceptance in a signed list of tests;
- (15) a record of the review and acceptance by the organisation requesting the qualification for the functions and subjective test results, with the results signed by the people accountable for the testing;
- (16) the means to demonstrate evidence of the approval of the MQTG and its revisions and also a section for signatures from the organisation and the competent authority.

GM1 FSTD.QB.030(b)(2)(vii) Qualification test guide

AVIONICS EQUIPMENT SYSTEMS

This guidance material (GM) gives guidance on what is expected for the different documents listing the avionics equipment.

- (a) FSTD information page — expectations on the avionics equipment

The FSTD information page of the QTG (refer to CS FSTD.QB.030, paragraph (b)(2)) should provide an overview of the simulated avionics systems, including their identifying information (e.g. block point number, standard, version or other unique identifier). It should reference the ER section that contains the detailed references to the OEM source documents (or equivalent, including the design basis, modelling techniques and data sources, as required by CS FSTD.QB.104, section 'Aircraft system simulation basis').

- (b) Engineering report — expectations on the avionics equipment

The ER should serve as the main description of the simulated avionics systems (refer to CS FSTD.ENG.020, paragraph (c)):

- (1) summarising the extent of the aircraft system simulation (full or partial functionality);
- (2) describing the simulation techniques used (e.g. own modelling, re-hosting, OEM software);
- (3) referencing OEM documents (e.g. pilot guide, FCOM, AFM) with sufficient detail to define the aircraft system's standard or version;
- (4) providing traceability through version information.

If multiple types of data documents are referenced for simulating a system (e.g. supplemental type certificates, system descriptions, aircraft manuals), the ER should explicitly identify and detail these sources.

Note that the information expected in the ER may overlap with the SOJs. Refer to GM1 FSTD.ENG.020(c) for more information.

(c) Equipment specification list — expectations on the avionics equipment

The ESL should align with the FSTD information page by summarising the avionics system versions and fidelity levels. This ensures consistency between the information page, ESL and ER without unnecessary duplication. ESL requirements are given in Part-ORA (Annex VII) of Commission Regulation (EU) No 1178/2011.

FLIGHT DECK LAYOUT AND STRUCTURE

CS FSTD.QB.101 Flight simulation training device general requirements for flight deck layout and structure

1: Flight deck layout and structure
Defines the level of flight deck enclosure and the physical (or perceived physical) structure, spatial representation and layout of the flight deck environment, instrument layout and presentation, panels and switches, interaction with the flight deck components and flight crew member seating. Note: This feature excludes the flight controls. Refer to the flight control forces and hardware FSTD feature in CS FSTD.QB.102 for their requirements. Note: This feature does not cover the aircraft system operation and functionalities (e.g. logic of the systems). Refer to the aircraft systems FSTD feature in CS FSTD.QB.104 for their requirements.

The general requirements for the flight deck layout and structure FSTD feature are presented in the table below.

1: Flight deck layout and structure								
Feature general requirements				Feature fidelity level			Verification and validation	
				G	R	S	QTG(A)	QTG(H) F&S
1.1	Enclosure requirements							
1.1.1	The flight deck, including any fitted instructor or observer seat, should be in a distraction-free environment. The flight deck, for FSTD purposes, consists of all that space forward of a cross section of the fuselage at the most extreme aft setting of the pilot seats.			✓	✓	✓		✓

1: Flight deck layout and structure							
Feature general requirements		Feature fidelity level			Verification and validation		
		G	R	S	QTG(A)	QTG(H)	F&S
1.1.2	The flight deck should be a physical full-scale replica of the simulated aircraft type and variant, including structure, shell, bulkheads and panels. The shell may be open to the rear.			✓			✓
1.1.3	Additional required flight crew member duty stations are also considered part of the flight deck and should replicate the aircraft.			✓			✓
1.1.4	Bulkheads aft of the pilots' seats containing items such as switches, and supplementary radio panels (to which the flight crew may require access during any event after pre-flight flight deck preparation is complete) are also considered part of the flight deck and should replicate the aircraft. Note: Bulkheads containing only additional equipment (e.g. landing gear pin storage compartments, fire axes or extinguishers, spare light bulbs, aircraft document pouches) may be omitted.			✓			✓
1.1.5	The flight deck should be characteristic of the simulated aircraft type. The enclosure may be limited to the flight deck panels and instruments (without the flight deck structure) or may be perceived as a flight deck shell.		✓				✓
1.1.6	A flight deck area should be characteristic of the simulated group of aircraft. The enclosure may be limited to the flight deck panels and instruments (without the flight deck structure).	✓					✓
1.1.7	For convertible FSTDs: A flight deck that is characteristic of the simulated group of aircraft may have physical dimensions such that the enclosure may be acceptable to simulate more than one group of aircraft in a convertible FSTD. Each FSTD conversion should be characteristic of the group of aircraft being simulated. Some switches, instruments, panels, masking, etc. may therefore need to be changed for conversions.	✓					✓

1: Flight deck layout and structure								
Feature general requirements				Feature fidelity level			Verification and validation	
		G	R	S	QTG(A)	QTG(H)	F&S	
1.2	Flight deck windows							
1.2.1	Windows, eyebrow and chin windows and such should be replicated at the correct sizes and positions. The glazing should provide the reflection effects for the simulated aircraft, in cases this is applicable (e.g. instrument reflections in night flight conditions, mirroring effects). As applicable, equipment for the operation of the flight deck windows should be included, but the actual windows need not be operable.			✓				✓
1.2.2	Any window accessories and other aircraft parts visible within the visual FOV that obstruct the view or have a functional use (e.g. wipers, demisting wiring, pitot tubes, handles, wire cutters, external mirrors, ice detectors) should be fitted, as applicable to the aircraft. A non-functional replica, silhouette or visual representation is acceptable.			✓				✓
1.3	Flight crew member seating							
1.3.1	Flight crew member seats, seat belts/restraints and the movement mechanism should replicate those in the aircraft being simulated. Note: Aircraft flight deck observer seats are not considered to be additional flight crew member duty stations and may be omitted.			✓				✓
1.3.2	Flight crew member seats should be characteristic of those in the aircraft being simulated. They should afford the occupants the ability to achieve the flight deck design eye reference position established for the simulated aircraft. Note: Aircraft flight deck observer seats are not considered to be additional flight crew member duty stations and may be omitted. Note: Seats that are characteristic need not be actual aircraft seats.		✓					✓

1: Flight deck layout and structure							
Feature general requirements		Feature fidelity level			Verification and validation		
		G	R	S	QTG(A)	QTG(H)	F&S
1.3.3	Flight crew member seats should have sufficient adjustment to allow the occupant to achieve proper posture and see and interact with the simulated instruments, panels, switches and flight controls as appropriate for the group of aircraft. Note: Non-aircraft seats are permitted. Any suitable adjustable seating may be used, provided that it enables the occupant to adopt proper posture and effectively interact with the simulated controls.	✓					✓
1.3.4	For an FSTD with a motion cueing system, any on-board flight deck seat(s) should be adequately secured and fitted with positive restraint devices of sufficient integrity to safely restrain the occupant during any known or predicted motion system excursion.	✓	✓	✓			✓
1.4	Flight deck panels, instruments and switches Note: As presented in CS FSTD.QB.104, the operation of individual aircraft systems may be at different fidelity levels (N/G/R/S). If an aircraft system is not simulated, the associated panel(s), switches and hardware need not be simulated, or they may be representations that do not support interaction. Refer to CS FSTD.QB.101.1.						
1.4.1	Basic flight instruments, as applicable to the simulated aircraft, should be fitted.	✓	✓	✓			✓
1.4.2	The flight deck instruments, panels and switches should support all the aircraft systems that are simulated. Simultaneous interaction of them should be possible.	✓	✓	✓			✓
1.4.3	The flight deck instruments, panels and switches should be a complete full-scale replica of the aircraft type and variant. Note: Any visible hardware and associated controls and switches not required to support flight crew training (e.g. flight deck panels or functions to be used only by aircraft maintenance personnel) do not need to be functional but should be fitted.			✓			✓

1: Flight deck layout and structure								
Feature general requirements		Feature fidelity level			Verification and validation			
		G	R	S	QTG(A)	QTG(H)	F&S	
1.4.4	The flight deck instruments, panels and switches for the aircraft systems that are simulated should be represented in the flight deck in a spatially characteristic layout for the simulated aircraft type (i.e. the distance between instruments and switches and the location relative to the pilot seats should be such that any deviations in the locations and sizes of panels, instruments and switches should not distract the pilots in the flow of their tasks or in operating the panels, instruments and switches).		✓					✓
1.4.5	Flight deck instruments, panels and switches should be characteristic of the simulated group of aircraft (as applicable).	✓						✓
1.4.6	The appearance of the simulated instrument, if viewed from the principal flight crew member's angle, should replicate that of the actual aircraft instrument.		✓	✓				✓
1.4.7	Any instrument reading inaccuracy due to viewing angle and parallax present in the actual aircraft instrument should be replicated in the simulated instrument display image. Any non-realistic induced viewing angle error and parallax should be minimised on shared instruments, such as engine displays and standby indicators.			✓				✓

1: Flight deck layout and structure								
Feature general requirements		Feature fidelity level			Verification and validation			
		G	R	S	QTG(A)	QTG(H)	F&S	
1.4.8	Instrument panels should have the following characteristics. (1) All instruments and instrument panel layouts should be dimensionally correct, with differences, if any, being imperceptible to the pilot. (2) Instruments should replicate those of the aircraft, including full instrument functionality and embedded logic, if any. (3) Instruments displayed should be free of quantisation (stepping), aliasing and artefacts. (4) Instrument display characteristics should replicate those of the aircraft, including resolution, colours, luminance, fonts, fill patterns, line styles and symbology. The display brightness should correspond to the appropriate lighting control setting; the maximum brightness should be tuned to the simulator lighting conditions. (5) As applicable, instruments should have faceplates, bezels and bugs that replicate those in the aircraft. (6) For 3D instruments, such as an electromechanical instrument, the display image should appear to have the same perceived 3D depth as that of the replicated instrument. The appearance of the simulated instrument, if viewed from any angle, should replicate that of the actual aircraft instrument. Any instrument reading inaccuracy due to viewing angle and parallax present in the actual aircraft instrument should be duplicated in the simulated instrument display image. (7) Typical vibrations as those of the instrument in the aircraft should be included. Note: Electronically displayed images with physical overlay or masking for FSTD instruments are acceptable (e.g. simulated touchscreen with hardware keypad control display unit).		✓				✓	
1.5	Flight deck haptic requirements							

1: Flight deck layout and structure								
Feature general requirements		Feature fidelity level			Verification and validation			
		G	R	S	QTG(A)	QTG(H)	F&S	
1.5.1	Flight deck instruments, panels and switches should replicate those of the aircraft type and variant in shape, operation and tactile feel, including 3D geometry, range of travel, force and other general haptic characteristics.			✓			✓	
1.5.2	If tactile hardware is fitted in the FSTD, flight deck instruments, panels and switches should be similar to those of the aircraft type in shape, operation and tactile feel, including 3D geometry, range of travel, force and other mechanical characteristics.		✓				✓	
1.5.3	If tactile hardware is fitted in the FSTD, flight deck instruments, panels and switches should be aircraft-like, consistent with the characteristics of the simulated group of aircraft, in shape, operation and tactile feel.	✓					✓	
1.5.4	If touchscreen representations of 3D switches, panels or instruments are fitted in the FSTD, flight deck instruments, panels and switches should be similar to those of the aircraft type in 3D appearance and operation, including the range of travel and type of action. Note: Multi-step actions should be represented, but need not be replicated. Note: Visually overlaid elements, such as pop-ups or graphical highlights, may be used to support the usability of multi-function switches and knobs (e.g. a multi-barrel VHF tuning knob), provided that the original functionality and logic of the control are preserved.		✓				✓	
1.5.5	If touchscreen representations of 3D switches, panels or instruments are fitted in the FSTD, they should support multitouch operations for multi-crew operations.		✓				✓	
1.5.6	If touchscreen representations of 3D switches, panels or instruments are fitted in the FSTD, flight deck instruments, panels and switches should be aircraft-like, as per group of aircraft, in appearance.	✓					✓	
1.5.7	If touchscreen representations of 3D controls are fitted in the FSTD, the operation of any touchscreen controls should be efficient and not distracting to the crew members.	✓	✓				✓	

1: Flight deck layout and structure								
Feature general requirements		Feature fidelity level			Verification and validation			
		G	R	S	QTG(A)	QTG(H)	F&S	
1.6	Circuit breaker panels							
1.6.1	All flight deck circuit breakers should replicate those in and be located as in the aircraft. Only those circuit breakers used in a normal, abnormal or emergency procedure need to be presented, simulated in an aircraft-like form and functionally accurate. The tactile feel and the technique, effort and direction required to manipulate the circuit breakers should replicate those in the aircraft.			✓				✓
1.6.2	A representative circuit breaker panel should be presented (photographic, touchscreen or electronic reproductions are acceptable). Only those circuit breakers used in a normal, abnormal or emergency procedure need to be simulated and functionally correct. Note: Circuit breaker panels in the area forward, surrounding or overhead of the pilot seats should be located in a spatially representative location. Circuit breaker panels aft of the pilot seats may be represented on the instructor station.		✓					✓
1.6.3	Only those circuit breakers used in a normal, abnormal or emergency procedure need to be presented, simulated in an aeroplane-like form and functionally correct.	✓						✓
1.7	Flight deck equipment							
1.7.1	Functional flight deck equipment (which is not a fitted panel or instrument, such as headsets, oxygen masks and electronic flight bags (EFBs)) that is accessed by the flight crew during normal, abnormal, emergency and non-normal operations, as applicable, should provide the functionality necessary to replicate its use in the aircraft. Note: Use of actual aircraft parts is not required, provided that the simulated flight deck equipment performs in a manner equivalent to the aircraft equipment.			✓				✓
1.7.2	If fitted, functional flight deck equipment (which is not a fitted panel or instrument, such as headsets, oxygen masks and EFBs) that is accessed by the flight crew during normal, abnormal, emergency and non-normal operations should be representative of the aircraft.		✓					✓

1: Flight deck layout and structure								
Feature general requirements		Feature fidelity level			Verification and validation			
		G	R	S	QTG(A)	QTG(H)	F&S	
1.7.3	Additional equipment required to be accessed by the flight crew, such as landing gear pin storage compartments, fire axes or extinguishers, spare light bulbs and aircraft document pouches, or reasonable facsimile, should be available but may be relocated to a suitable location as near as practical to the original position. Fire axes and any similar-purpose equipment may be represented in silhouette or by a photograph or a similar technique.			✓			✓	
1.7.4	If fitted, additional equipment required to be accessed by the flight crew, such as landing gear pin storage compartments, fire axes or extinguishers, spare light bulbs and aircraft document pouches, may be represented by a photograph or a similar technique.		✓				✓	
1.7.5	Flight deck escape hatches, ropes and similar emergency equipment required to be accessed by the flight crew should be represented in the flight deck, but need not be fully functional.			✓			✓	
1.8	Instrument lighting requirements							
1.8.1	Instrument lighting should replicate that of the aircraft; it should be operated from the same control (e.g. lighting panel) as in the aircraft; the logic of operation (e.g. the type of illumination dimmed with one particular button) should be as per the aircraft.			✓			✓	
1.8.2	Instrument lighting for panels and instruments should be sufficient for the operation of the flight deck instruments. Backlighting for panels and instruments may be installed, but is not required.	✓	✓				✓	
1.9	Flight deck lighting							
1.9.1	Flight deck lighting should replicate that in the aircraft and should not cause unrealistic reflections on flight deck systems (e.g. night-vision goggles (NVGs), unrealistic effects on instruments caused by flight deck lighting).			✓			✓	

1: Flight deck layout and structure							
Feature general requirements		Feature fidelity level			Verification and validation		
		G	R	S	QTG(A)	QTG(H)	F&S
1.9.2	The lighting environment for panels and instruments should be sufficient for their operation.	✓	✓				✓
1.10	Ambient lighting						
1.10.1	The ambient light level should ensure the readability of the instruments, panel nomenclature, placards, charts, etc. at all times.	✓	✓	✓			✓
1.10.2	Any simulated ambient lighting, if fitted, should be dynamically consistent (e.g. the light level is to be correlated with the environmental brightness) with the visual display. The ambient lighting should provide an even level of illumination that is not distracting to the flight crew member(s).			✓			✓
1.10.3	Any simulated ambient lighting, if fitted, should provide an even level of illumination that is not distracting to the flight crew member(s).		✓				✓
1.11	XR systems: virtual flight deck requirements Provides requirements for flight decks fully or partially perceived through an HMD. Note: Requirements marked with ✓ ¹ indicate that an objective test section for these is not available. The tests should be defined in the special conditions.						
1.11.1	Type- and variant-specific FSTDs require a perceived full-scale physical flight deck replica of the aircraft represented with all elements in hardware for pilot interaction. This includes other physical structural elements (e.g. frame, other seats, doors) of the flight deck structure that impact the pilot's movement (as applicable to training). The entire flight deck and relevant aircraft parts should be visually perceivable by the pilot. Haptic feedback should be specific to the aircraft simulated.			✓			✓

1: Flight deck layout and structure								
Feature general requirements		Feature fidelity level			Verification and validation			
		G	R	S	QTG(A)	QTG(H)	F&S	
1.11.2	A perceived full-scale flight deck should be representative of the aircraft simulated, with all elements in hardware for pilot interaction. Other structural elements (e.g. frame, other seats, doors) should be implemented to the extent relevant for training. The entire flight deck and relevant aircraft parts should be visually perceivable by the pilot. Touch panels are acceptable to represent flight deck elements if training is not adversely affected.		✓				✓	
1.11.3	Flight deck elements may be operable by alternate means, such as gesture control. The methods used should not adversely affect training. The entire flight deck and relevant aircraft parts should be visually perceivable by the pilot.	✓					✓	
1.11.4	Acceptable means for using checklists and maps should be provided in the virtual environment.	✓	✓	✓			✓	
1.11.5	The flight deck representation should be aligned with implemented physical structures with respect to eyepoint, view and geometry. If a motion system is fitted, the tracking should be robust enough to compensate for the effects of motion.	✓	✓	✓	✓ ¹	✓ ¹	✓	
1.11.6	The flight deck states of implemented hardware and the visual representation of these elements (e.g. switches, levers, guards) should be consistent with the simulation.	✓	✓	✓			✓	
1.11.7	The virtual flight deck should represent the simulated aircraft. The dimensions, colours, lighting and behaviour of flight deck elements, controls and equipment should be accurate. Special attention should be paid to the correct representation of optical indicators (e.g. warning lights). A list of instrument/indicator colours with physical colour references and representation target values for the FSTD should be provided.		✓	✓	✓ ¹	✓ ¹	✓	
1.11.8	The effects of lighting (e.g. shadows and reflections) in the flight deck should be realistic and accurately represent real-world lighting conditions as they would appear in the actual aircraft.		✓	✓			✓	

1: Flight deck layout and structure								
Feature general requirements		Feature fidelity level			Verification and validation			
		G	R	S	QTG(A)	QTG(H)	F&S	
1.11.9	The representation of the flight deck, colours and simulated light effects (e.g. shadows, reflections) should remain visually consistent and should not be susceptible to interference from light sources outside the simulation.	✓	✓	✓			✓	
1.11.10	Relevant flight deck colours should replicate the aircraft simulated.			✓			✓	
1.11.11	If a direct view is not possible (e.g. if using VR systems), the pilot's own body pose (e.g. hands) needs to be represented in the virtual environment (as required by the training tasks). If there are multiple crew members, they should be able to see each other's body pose in the virtual environment. For multi-crew training, the specifications of the crew member visualisations within the virtual environment should be described in the special conditions.		✓	✓			✓	
1.11.12	All relevant systems for flight deck perception (visual system, flight deck hardware, body pose tracking) should be aligned and stable during operation.	✓	✓	✓			✓	
1.12	XR systems: head-tracking system Note: Requirements marked with ✓ ¹ indicate that an objective test section for these is not available. The tests should be defined in the special conditions.							
1.12.1	The XR system should include an HMD with accurate head tracking.	✓	✓	✓			✓	
1.12.2	The tracking of the HMD should be fast and robust. Tracking noise, glitches and other disruptive effects should not be present during operation.	✓	✓	✓			✓	
1.12.3	If a motion system is fitted, effects of motion should not noticeably affect the head tracking or inaccurately impact the pilot's view.	✓	✓	✓	✓ ¹	✓ ¹	✓	
1.12.4	The visual (virtual) flight deck in the HMD should always be aligned with the flight deck hardware.		✓	✓			✓	

1: Flight deck layout and structure								
Feature general requirements		Feature fidelity level			Verification and validation			
		G	R	S	QTG(A)	QTG(H)	F&S	
1.12.5	The head tracking should have six degrees of freedom (DOFs) and accurately change the viewpoint in the flight deck.	✓	✓	✓			✓	

GM1 FSTD.QB.101 Flight simulation training device general requirements for flight deck layout and structure

Summaries of the fidelity levels for the flight deck layout and structure FSTD feature are presented in the table below. These summaries are broad overviews provided for guidance only. They do not encompass all aspects and requirements and are not intended to replace CS FSTD.QB.101.

1: Flight deck layout and structure

Specific:

A physical full-scale replica of the flight deck of the aircraft (type and variant) being simulated is required. (Note: This feature excludes the flight controls.)

The flight deck panels, instruments, switches and structures are properly located, replicating those in the simulated aircraft type and variant.

Switches, panels and instruments are simulated by using tactile hardware replicating those of the aircraft.

Representative:

A flight deck that is characteristic of the simulated aircraft type. (Note: This feature excludes the flight controls.) The enclosure may be limited to the flight deck panels and instruments (without the flight deck structure) or may be a perceived flight deck shell.

The flight deck panels, instruments and switches for the simulated aircraft systems are represented in the flight deck in a spatially characteristic layout for the simulated aircraft type.

Switches, panels or instruments may be simulated by using either tactile hardware or touchscreen representations.

Generic:

A flight deck area that is characteristic of the simulated group of aircraft. (Note: This feature excludes the flight controls.) The enclosure may be limited to the flight deck panels and instruments (without the flight deck structure).

The flight deck panels, instruments and switches are characteristic of the simulated group of aircraft.

Switches, panels or instruments may be simulated by using either tactile hardware or touchscreen representations.

GM2 FSTD.QB.101 Flight simulation training device general requirements for flight deck layout and structure

FLIGHT INSTRUMENTS

Each FSTD should have the basic flight instruments, as applicable to the simulated aircraft. These mean the primary flight instruments, such as attitude indicators, altimeters, airspeed indicators, vertical speed indicators, turn coordinators and heading indicators (directional gyros) or other representations of these, such as primary flight displays (PFDs), navigation displays or electronic flight instrument systems, as applicable to the simulated aircraft.

At the specific fidelity level, these should replicate those of the simulated aircraft type and variant. At the representative fidelity level, these should be characteristic of the simulated aircraft type (i.e. simulate an instrument arrangement that can be installed in a real aircraft of this type). At the generic fidelity level, these should be characteristic of those of the simulated group of aircraft. The markings (e.g. limits) should be characteristic of those of the simulated aircraft and should correspond to those of the flight dynamic model. For example, the airspeed indicator's marking for optimum climbing speed / best rate of climb airspeed (V_Y) should match the airspeed that gives the best rate of climb in the FSTD in nominal conditions. Similar applies to minimum control speed and so on.

For this feature, the basic flight instruments are considered as an entity in which all the instruments are at the same fidelity level to ensure a positive transfer of training. If the fitted instruments are simulated at different fidelity levels, the entity should be considered to be only at the level of the lowest simulated instrument.

GM3 FSTD.QB.101 Flight simulation training device general requirements for flight deck layout and structure

FLIGHT DECK LAYOUT AND STRUCTURE USING EXTENDED REALITY SYSTEMS

This GM gives guidance on the flight deck representation or interfaces (e.g. panels, switches, buttons) that are simulated through the use of XR systems.

(a) Guidance on the flight deck requirements

CS FSTD.QB.101 outlines the requirements for the flight deck layout and structure FSTD feature, which may be (partially) represented using an XR system with an HMD.

If an XR system is used instead of a fully physical flight deck representation, certain aspects of CS FSTD.QB.101 may require alternative means of compliance. In such cases, compliance should be demonstrated through standard SOJs and in addition special conditions, which define how the XR-based implementation meets or exceeds the intended FSTD requirements and training objectives.

The special conditions should focus on how the XR-based flight deck meets the applicable requirements, addressing elements such as the accuracy of control placement, interaction methods, visual and spatial representation and any compensatory design features used to ensure effective training outcomes. Any limitations of the XR implementation and their potential impact on training should also be documented within the special conditions to ensure transparency in the qualification.

(b) Guidance on the provisions for XR systems

In addition to meeting the general flight deck layout and structure FSTD feature requirements, an FSTD using XR systems should comply with the XR-specific provisions outlined in CS FSTD.QB.101 section 'XR systems: virtual flight deck requirements'. These provisions address the representation of the flight deck within the virtual environment and the functionality of its components.

(c) Guidance on the head-tracking requirements

XR systems should meet the technical requirements for head-tracking accuracy and reliability as specified in CS FSTD.QB.101 section 'XR systems: head-tracking system'. This ensures precise and consistent interaction with the virtual flight deck.

(d) Guidance on the XR systems using AR and MR visual display systems

While the provisions above primarily address virtual reality (VR) systems, the same principles apply to other XR configurations, such as augmented reality (AR) and mixed reality (MR), if HMDs are used to present or overlay simulated flight deck elements.

Any differences introduced by the use of AR or MR should be addressed through device-specific special conditions. These special conditions should demonstrate how the intended validation, verification, training fidelity, usability and safety levels are achieved and describe any compensatory design features used. In the cases of AR and MR, particular attention should be given to the alignment between virtual and physical elements, the stability and latency of visual overlays and the impact of any limitations on spatial orientation or control accessibility.

CS FSTD.QB.101.1 Flight simulation training device requirements for mixed-fidelity flight deck panels, instruments and switches

This CS defines the requirements for the use of mixed fidelity levels in the flight deck layout and structure FSTD feature.

- (a) For the specific fidelity level, all flight deck panels, instruments and switches should meet the requirements for the specific fidelity level. If an aircraft system is not simulated, the associated panels, switches and hardware need not be simulated (i.e. level N), or they may be representations that do not support interaction.
- (b) For the representative and generic fidelity levels, if the flight deck is equipped with panels, instruments and switches that are simulated at different fidelity levels (G, R or S), the flight deck should be considered to be only at the level of the lowest simulated panel, instrument or switch.
- (c) All flight deck panels, instruments and switches, at their individual fidelity levels, should enable the simulation to operate seamlessly and function in an integrated manner.
- (d) The equipment specification list (ESL) should document system interfaces in accordance with the requirements of Part-ORA (Annex VII) of Commission Regulation (EU) No 1178/2011.

GM1 FSTD.QB.101.1 Flight simulation training device requirements for mixed-fidelity flight deck panels, instruments and switches

This GM clarifies how to qualify the flight deck layout and structure FSTD feature in cases if mixed fidelity levels are used across the flight deck. This may be in combination with aircraft systems at different fidelity levels, which leads to the flight deck representation or interfaces (e.g. panels, switches, buttons) covering different fidelity levels.

- (a) General considerations for mixed-fidelity flight decks are as follows.

CS FSTD.QB.101 outlines the requirements for the flight deck layout and structure FSTD feature, covering the entire flight deck (e.g. enclosure, windows, lighting, seating) and aircraft system interfaces (e.g. panels, switches, buttons).

CS FSTD.QB.010 and CS FSTD.QB.104 allow individual aircraft systems to be simulated at different fidelity levels (N/G/R/S). This means that not all aircraft systems need to be simulated within the FSTD, which directly impacts the types of training tasks that the device can support.

- (b) Qualification of the flight deck layout and structure FSTD feature if certain aircraft systems are not simulated should proceed as follows.

CS FSTD.QB.101 specifies that, if an aircraft system is not simulated, the corresponding flight deck interface (e.g. panels, switches, hardware) is not required; it may be either omitted or represented as a static, non-interactive element.

For example, an FSTD may simulate all aircraft systems at the S fidelity level, except for the cargo compartment fire protection system, which is not simulated (level N). In this case:

- (1) the cargo compartment fire protection panel, switches and push-buttons for system interfaces are not required (they may be represented statically);
- (2) the flight deck layout and structure FSTD feature can still achieve the specific fidelity level, as the missing interface corresponds to a system that is not simulated.

- (c) Qualification of mixed-fidelity interfaces for the flight deck layout and structure FSTD feature at the specific fidelity level should proceed as follows.

At the specific fidelity level, it is assumed that all installed panels, instrument, switches and peripherals are at the same level (i.e. level S), as part of a complete full-scale replica. There may be some optional equipment that is not installed or not simulated (i.e. level N).

- (d) Qualification of mixed-fidelity interfaces for flight deck layout and structure FSTD feature at the representative or generic fidelity level should proceed as follows.

CS FSTD.QB.010 states that an FSTD feature is only qualified to a given fidelity level if all associated requirements for that level are met or exceeded. Therefore, for a flight deck at the representative fidelity level, it is assumed that the minimum fidelity level of all installed panels, instruments, switches and peripherals is representative or greater.

If aircraft system interfaces within the flight deck vary in fidelity, the lowest fidelity level among them determines the overall qualification level for the flight deck layout and structure FSTD feature. This ensures consistency in the qualification and alignment with the intended training use of the FSTD.

For example, say an FSTD has most flight deck interfaces (e.g. panels, switches, push-buttons) at the specific fidelity level. However, certain interfaces, such as the radio communication panel and navigation radio panel, are at the representative fidelity level. In this case:

- (1) the flight deck layout and structure FSTD feature can only be qualified to the representative fidelity level, as the lowest fidelity level among the interfaces dictates the qualification level;
 - (2) the differences in fidelity across cockpit panels, instruments and switches for the flight deck layout and structure FSTD feature is documented in the ESL.
- (e) Consistency for mixed-fidelity configurations at the representative or generic fidelity level should be ensured as follows.

While mixed fidelity levels are possible, generally, maintaining a consistent fidelity level across flight deck interfaces is preferred to ensure training effectiveness and usability in multiple situations. As noted above, certain aircraft system interfaces may be omitted if the corresponding aircraft system is not simulated. If mixed fidelity levels are used, the intended use of the device should be taken into consideration in the design and acceptance of the device for training. A level of consistency across the flight deck should be ensured (e.g. to facilitate the use of the device for training on abnormal system procedures or for instrument navigation tasks).

- (f) The ESL should clearly document any omitted system interfaces, or any non-interactive representations used in place of functional elements used, and any fidelity differences across cockpit panels and consoles. This documentation ensures transparency in the qualification and use of the device and helps to align the FSTD with its intended training use.

FLIGHT CONTROL FORCES AND HARDWARE

CS FSTD.QB.102 Flight simulation training device general requirements for flight control forces and hardware

2: Flight control forces and hardware
Defines the physical appearance, travel, tactile feel and force feedback operation of the flight controls and systems affecting aircraft configuration. Note: Primary flight controls are the controls that are required for immediate control of the aircraft. For a typical aeroplane, these are usually identified as the column, wheel, pedal, engine/thrust control levers (e.g. throttles, propeller levers), tiller and toe brakes. For a typical helicopter, these are usually identified as cyclic, collective and pedals. Note: Other flight controls are the flight controls that are not classified as primary flight controls and controls that affect aircraft configuration. These include but are not limited to trims, flap levers, speed brake levers and landing gear levers. Note: The individual flight controls may be at different fidelity levels (N/G/R/S).

The general requirements for the flight control forces and hardware FSTD feature are presented in the table below (applicable if hardware is fitted).

2: Flight control forces and hardware								
Feature general requirements						Feature fidelity level		
						Verification and validation		
						G	R	S
						QTG(A)	QTG(H)	F&S
2.1	Primary flight control forces and travel							

2: Flight control forces and hardware								
Feature general requirements		Feature fidelity level			Verification and validation			
		G	R	S	QTG(A)	QTG(H)	F&S	
2.1.1	Primary flight controls should be fitted that replicate the appearance and tactile feel of the simulated aircraft type and variant. Primary flight control forces and travel should react in the same manner as in the simulated aircraft type and variant under the same flight and system conditions throughout the flight envelope and, if the FSTD includes a motion platform, throughout the motion platform displacement envelope.			✓	2.a	2.a	✓	
2.1.2	For aeroplanes: Applicable only to aeroplane FSTDs that are to be qualified for full-stall training tasks. For aeroplanes equipped with a stick pusher system, the control forces, displacement and consequent surface position should correspond to those of the aeroplane being simulated. The SOJ should verify that the stick pusher system has been simulated and validated using the aeroplane manufacturer's design data or data from another acceptable source. The SOJ should address, at a minimum, the stick pusher activation and cancellation logic and the system dynamics, control displacement and forces as a result of the stick pusher activation.			✓	2.a.10		✓	
2.1.3	For aeroplanes: For aeroplanes equipped with longitudinal control augmentation functionality (e.g. control force modifier), the control forces, displacement and surface position should correspond to those of the aeroplane being simulated. The SOJ should verify that the relevant system/functionality has been simulated and validated using the aeroplane manufacturer's design data or data from another acceptable source. The SOJ must address, at a minimum, the system/functionality activation and cancellation logic and the system dynamics, control displacement and forces as a result of the activation.		✓	✓			✓	

2: Flight control forces and hardware								
Feature general requirements		Feature fidelity level			Verification and validation			
		G	R	S	QTG(A)	QTG(H)	F&S	
2.1.4	Primary flight controls should be fitted that are characteristic of the simulated aircraft type in appearance and tactile feel. Primary flight control forces and travel should react in the same manner as in the simulated aircraft type under the same flight and system conditions throughout the flight envelope.		✓		2.a	2.a	✓	
2.1.5	Primary flight controls should be fitted that are characteristic of the simulated group of aircraft in appearance and tactile feel. Primary flight control forces and travel should be characteristic of the simulated group. Primary flight control force gradients do not have to change as flight conditions, such as configuration or airspeed, change.	✓			2.a	2.a	✓	
2.2	Integration with aircraft systems							
2.2.1	Integration of the flight controls hardware with the aircraft systems: If an aircraft system affects the primary flight control forces, the primary flight control forces should be characteristic of the simulated aircraft type (e.g. hydraulic system status affecting the control forces, force required to overcome the autopilot).		✓	✓			✓	
2.2.2	Integration of the flight controls hardware with the aircraft systems: If an aircraft system affects the flight control forces, the flight control forces should be characteristic of the simulated group of aircraft system operation (e.g. hydraulic system status affecting the control forces, force required to overcome the autopilot).	✓					✓	
2.3	Control feel dynamics							
2.3.1	Primary flight control dynamics should replicate the simulated aircraft type and variant.			✓	2.b	2.a		
2.3.2	Primary flight control dynamics should be characteristic of the simulated aircraft type.		✓		2.b	2.a		

2: Flight control forces and hardware							
Feature general requirements		Feature fidelity level			Verification and validation		
		G	R	S	QTG(A)	QTG(H)	F&S
2.4	Other flight controls and controls that affect aircraft configuration Note: This section concerns controls such as trims (pitch, roll, yaw), flap levers, speed brake levers and landing gear levers.						
2.4.1	The flight control hardware should replicate that of the simulated aircraft type and variant. The forces, tactile feel and travel should replicate those of the simulated aircraft type and variant.			✓	2.a	2.a	✓
2.4.2	The flight control hardware should be characteristic of the simulated aircraft type. The forces, tactile feel and travel should be characteristic of the simulated aircraft type.		✓		2.a	2.a	✓
2.4.3	If physical controls are fitted, the flight control hardware should be characteristic of the simulated group of aircraft. The forces, tactile feel and travel should be characteristic of the simulated group of aircraft.	✓			2.a	2.a	✓
2.4.4	If touchscreen representations of the flight controls are fitted, they should be aircraft-like, as per group of aircraft, in appearance and operation. The operation of any touch controls should be efficient and not distracting to the crew members.	✓					✓
2.5	XR systems: virtual flight control requirements Provides requirements for flight controls fully or partially perceived through an HMD. Note: Requirements marked with $\sqrt{1}$ indicate that an objective test section for these is not available. The tests should be defined in the special conditions.						
2.5.1	The virtual representation of the flight controls should correspond to the physical flight controls and their movement.		✓	✓	$\sqrt{1}$	$\sqrt{1}$	✓
2.5.2	Primary flight controls should be physically implemented in addition to being represented virtually. Other elements may be operable by alternate means, such as gesture control. The methods used should not adversely affect training.	✓					✓

GM1 FSTD.QB.102 Flight simulation training device general requirements for flight control forces and hardware

Summaries of the fidelity levels for the flight control forces and hardware FSTD feature are presented in the table below. These summaries are broad overviews provided for guidance only. They do not encompass all aspects and requirements and are not intended to replace CS FSTD.QB.102.

2: Flight control forces and hardware
Specific: The appearance, tactile feel, forces and travel of the flight controls and configuration controls replicate those of the simulated aircraft type and variant throughout the flight envelope.
Representative: The appearance, tactile feel, forces and travel of the flight controls and configuration controls are characteristic of the simulated aircraft type throughout the flight envelope.
Generic: The appearance, tactile feel, forces and travel of the flight controls and configuration controls are characteristic of the simulated group of aircraft. Flight control forces do not have to change as flight conditions, such as configuration or airspeed, change.

GM2 FSTD.QB.102 Flight simulation training device general requirements for flight control forces and hardware

- (a) The following should be considered for flight control forces.

For many aircraft, the primary flight control forces or travel vary throughout the flight envelope. For example, if airspeed changes, the flight control forces change. A typical method of simulating this phenomenon is to use an active force feedback system, in which a computer-driven system continuously adjusts the simulated flight controls so that the pilot experiences appropriate control forces. It may also be possible to achieve this by using a passive system in which the control forces are created only by a mechanical system (e.g. springs, dampers), but building and validating such a system could face difficulties.

- (b) The following should be considered for control augmentation.

The words 'longitudinal control augmentation functionality' mean aeroplane functions that modify or enhance the stability and control characteristics. Examples include functions that modify control forces to enhance stick force gradients during speed or angle-of-attack changes or stall identification. There are separate aeroplane FSTD requirements for stick pusher systems and for longitudinal control augmentation functionality.

GM3 FSTD.QB.102 Flight simulation training device general requirements for flight control forces and hardware

FLIGHT CONTROLS AT DIFFERENT FIDELITY LEVELS

This GM gives guidance on the flight controls if the controls are at different fidelity levels (N/G/R/S).

As CS FSTD.QB.102 states, the individual flight controls may be at different fidelity levels. Each flight control should be evaluated against the CS FSTD.QB.102 general requirements on an individual basis.

The requirements outlined in CS FSTD.QB.102 distinguish between primary flight controls and other flight controls. According to CS FSTD.QB.102, for a primary flight control to qualify as level G, R or S, it must be simulated using tactile hardware. Consequently, a primary flight control would be considered to be at level N if represented solely by a touchscreen interface. Note that CS FSTD.QB.102 allows for the use of touchscreen interfaces at the generic fidelity level for the other flight controls.

It is possible for some primary flight controls to be at level N. For example, an FSTD may feature a tactile hardware control column and wheel but may not include any pedals (i.e. pedals at level N). While such an FSTD could be utilised for various training tasks, it would not be suitable for many others.

If flight controls operate at different fidelity levels, utilise touchscreen representations or are not simulated at all (i.e. are at level N), this can affect the training outcomes. Therefore, the ESL should detail the flight controls and their individual fidelity levels. The training requirements will provide further information on how these variations in fidelity impact training.

FLIGHT CONTROL SYSTEMS OPERATION

CS FSTD.QB.103 Flight simulation training device general requirements for flight control systems operation

3: Flight control systems operations
Defines the extent of the flight control system functions and controls affecting aircraft configuration that are to be modelled in the FSTD. These include primary flight control surface displacements (e.g. surface position for aeroplanes, blade angles for helicopters) and the operation modes and logic of the flight control system and systems affecting aircraft configuration, flight and manoeuvre envelope protection functions and flight deck indications and messages.

The general requirements for the flight control systems operation FSTD feature are presented in the table below.

3: Flight control systems operations							
Feature general requirements		Feature fidelity level			Verification and validation		
		G	R	S	QTG(A)	QTG(H)	F&S
3.1	Primary flight control systems operation						
3.1.1	Primary flight control systems should replicate the simulated aircraft type and variant operation in normal and non-normal modes, including backup systems, and should reflect the failures of associated systems. Appropriate flight deck indications and messages should be replicated.			✓			✓
3.1.2	Primary flight control systems should be characteristic of the simulated aircraft type operation in normal and non-normal modes, including backup systems, and should reflect the failures of associated systems. Appropriate flight deck indications and messages should be replicated.		✓				✓
3.1.3	Primary flight control systems should be characteristic of the simulated group of aircraft and allow aircraft operation with appropriate flight deck indications.	✓					✓

3: Flight control systems operations									
Feature general requirements				Feature fidelity level			Verification and validation		
				G	R	S	QTG(A)	QTG(H)	F&S
3.2	Primary flight control operation								
3.2.1	Primary flight control displacements should generate flight control surface positions (for aeroplanes) or blade angles (or equivalent, for rotorcraft) that replicate those of the simulated aircraft type and variant.					✓	2.a	2.a	✓
3.2.2	Primary flight control displacements should generate flight control surface positions (for aeroplanes) or blade angles (or equivalent, for rotorcraft) that are characteristic of the simulated aircraft type.				✓		2.a	2.a	✓
3.2.3	Primary flight control displacements should have control authority that is characteristic of the simulated group of aircraft.			✓					✓
3.2.4	For CCA: A SOJ should explain and demonstrate how the primary flight control authority changes as the aircraft configuration changes.			✓	✓	✓			✓
3.3	Operation of other flight controls and controls that affect aircraft configuration Note: This section concerns controls such as trims (pitch, roll, yaw), flap levers, speed brake levers and landing gear levers.								
3.3.1	The operation of other flight controls and controls that affect aircraft configuration should replicate that of the simulated aircraft type and variant.					✓			✓
3.3.2	The operation of other flight controls and controls that affect aircraft configuration should be characteristic of the simulated aircraft type.				✓				✓
3.3.3	The operation of other flight controls and controls that affect aircraft configuration should be characteristic of the simulated group of aircraft.			✓					✓

3: Flight control systems operations								
Feature general requirements				Feature fidelity level			Verification and validation	
		G	R	S	QTG(A)	QTG(H)	F&S	
3.4	Flight and manoeuvre envelope protection functions Note: Objective tests are applicable only to CCA. For non-CCA, the relevant flight and manoeuvre envelope protection functions are validated through functions and subjective testing only.							
3.4.1	The flight control system functions for flight and manoeuvre envelope protection functionality should replicate those of the simulated aircraft type and variant.			✓	2.h		✓	
3.4.2	The flight control system functions for flight and manoeuvre envelope protection functionality should be characteristic of the simulated aircraft type.		✓		2.h		✓	
3.4.3	The flight control system functions for flight and manoeuvre envelope protection functionality should be characteristic of the simulated group of aircraft.	✓					✓	
3.5	Flight control integration with aircraft systems							
3.5.1	For aeroplanes: For aeroplanes equipped with a longitudinal control augmentation functionality (e.g. control force modifier), the surface positions should correspond to those of the aeroplane being simulated. The SOJ should verify that the relevant system/functionality has been modelled, programmed and validated using the aeroplane manufacturer's design data or data from another acceptable source. The SOJ must address, at a minimum, the system/functionality activation and cancellation logic and the system dynamics, control displacement and forces as a result of the activation.		✓	✓			✓	

3: Flight control systems operations								
Feature general requirements		Feature fidelity level			Verification and validation			
		G	R	S	QTG(A)	QTG(H)	F&S	
3.5.2	Integration of the primary flight control system operation with the aircraft systems: If an aircraft system (e.g. auto-flight control system) affects the primary flight control system, the primary flight control system operation should correspond to the simulated aircraft system operation (e.g. autopilot may move the flight controls, automatically trim the flight controls or move the actuators). The SOJ should describe whether or not the primary flight control system is simulating a CCA and whether or not the effects of reversible flight controls are modelled (and, if modelled, in which axes).		✓	✓			✓	
3.5.3	Integration of the primary flight control system operation with the aircraft systems: If an aircraft system (e.g. auto-flight control system) affects the primary flight control system, the primary flight control system operation should be characteristic of the simulated group of aircraft system operation (e.g. autopilot may move the flight controls, automatically trim the flight controls or move the actuators). The SOJ should describe which aircraft systems affect the primary flight controls and identify which autopilot system is being simulated. This should include whether the system is modelling a CCA and whether the effects of reversible flight controls are modelled (and, if modelled, in which axes).	✓					✓	

GM1 FSTD.QB.103 Flight simulation training device general requirements for flight control systems operation

Summaries of the fidelity levels for the flight control systems operation FSTD feature are presented in the table below. These summaries are broad overviews provided for guidance only. They do not encompass all aspects and requirements and are not intended to replace CS FSTD.QB.103.

3: Flight control systems operations
Specific: The primary flight control displacements generate primary flight control surface positions (for aeroplanes) or blade angles (for rotorcraft) as in the simulated aircraft type and variant under the same flight and system conditions. The operation and logic of the flight control and configuration control systems replicate those of the simulated aircraft type and variant, including system interdependencies, in both normal and non-normal modes. These include backup systems and reflect the failures of associated systems of the aircraft type and variant. Appropriate flight control system logic, flight deck indications and messages are replicated for the simulated aircraft type and variant.
Representative: The primary flight control displacements generate primary flight control surface positions (for aeroplanes) or blade angles (for rotorcraft) that are similar to those of the aircraft type being simulated under the same flight and system conditions. The operation and logic of the flight control and configuration control systems match those of the simulated aircraft type, including system interdependencies, in both normal and non-normal modes. These include backup systems and reflect the failures of associated systems of the aircraft type. Appropriate flight deck indications and messages are represented.
Generic: Primary flight control displacements have control authority that is characteristic of the simulated group of aircraft. The operation and logic of the flight control and configuration control systems are characteristic of the simulated group of aircraft. Appropriate flight deck indications to allow aircraft operation.

GM2 FSTD.QB.103 Flight simulation training device general requirements for flight control systems operation

GUIDANCE ON PRIMARY FLIGHT CONTROLS OPERATION

This GM gives guidance on primary flight controls operation.

- (a) As CS FSTD.QB.103 requires, at the specific and representative fidelity levels, the objective tests in test section 2.a should validate flight control surface positions (for aeroplanes) or blade angles (or equivalent, for rotorcraft). For aeroplane FSTDs, this is straightforward and self-explanatory. For rotorcraft FSTDs, the simulation may be based on different models and

may not even use the exact blade angle as part of the flight dynamic model (e.g. a disc model may be used). However, successful model development typically requires that the flight control position in relation to the rotor position is measured during the data-gathering process. The data gathering may record the flight control position (e.g. cyclic) versus blade angles, actuator positions and/or swashplate attitudes. It is acknowledged that the FSTD model may not perfectly match the measured rotor blade angle (or equivalent) in 2.a objective tests because of the complexity of the rotorcraft flight dynamic and flight control models. For rotorcraft, the resulting aircraft response versus flight control input is of greater importance and is validated using a multitude of objective tests. The 2.a objective tests should still demonstrate the rotor blade angle (or equivalent) versus the flight control input for the above-mentioned reasons and to demonstrate the repeatability of the flight control system for recurrent test purposes.

- (b) As CS FSTD.QB.103 requires, at the generic fidelity level, the primary flight control displacements should have control authority that is characteristic of the simulated group of aircraft. At this fidelity level, the simulation may be simplified and may not consider flight control surface positions for aeroplanes or blade angles (or equivalent) for rotorcraft. Because of this, the authority of the primary flight controls is to be characteristic of the simulated group of aircraft. The word 'authority' should be understood to mean the surface position or blade angle (or equivalent) similar to that at the representative and specific fidelity levels or the aircraft response (e.g. pitch rate v pitch flight control input) as applicable to the simulation method used. The SOJ should explain the simulation of the primary flight control system and how the resulting aircraft response is calculated.
- (c) For rotorcraft, the flight control system may use multiple actuators with different rates and logic by using multiple inputs. The simulation of the whole system should be characteristic of the system of the simulated rotorcraft group. Due to non-linearities and possible actuator saturation, the flight controls of rotorcraft/helicopters may have varying levels of authority depending on the flight state.

GM3 FSTD.QB.103 Flight simulation training device general requirements for flight control systems operation

FLIGHT CONTROL SYSTEMS OPERATION AT THE GENERIC FIDELITY LEVEL

This GM gives guidance on the flight control systems operation FSTD feature at the generic fidelity level.

- (a) Different aircraft use different primary flight control systems. The system may, for example, be reversible or irreversible, include stabilisation augmentation, include flight envelope protection systems or include automatic trim functions; the aircraft may be a fly-by-wire CCA. If the flight control systems operation FSTD feature is at the generic fidelity level and the simulated group of aircraft has complex flight control systems, the FSTD's flight control operation is required to have realistic characteristics and should be similar to that of a real aircraft within the simulated group of aircraft. This should be demonstrated in the SOJ. For example, for a fast fly-by-wire jet aeroplane equipped with a stabiliser, the pitch controller

input may lead to a deflection of the elevator, followed by an auto-trim function on the stabiliser. The logic and gains of these functions should be appropriate for the group of aircraft.

- (b) The simulation of CCA or flight envelope protection systems should integrate with those of other features. The associated secondary flight controls, flight deck panels, annunciations and indications should correlate with the flight control systems operation FSTD feature. The rate and sequence of any automatic trim functions should be realistic.
- (c) It is possible that, for computer-controlled aeroplanes, the same flight control input will result in different authority (i.e. different magnitude of surface deflections or level of authority) for different configurations. This means that objective tests such as 2.a.1, 2.a.2 and 2.a.3 may have different slopes and results if the tests are performed for different flap settings. This affects the feel or sensitivity of the aircraft. The FSTD should simulate these aspects as appropriate. The SOJ should explain how aircraft configuration affects flight control authority.
- (d) The simulation of the failures of complex flight control systems (e.g. loss of stabilisation system, reversion mode of CCA) should have an effect on the performance and handling characteristics. The features should integrate and they should provide a positive transfer of training.
- (e) While this feature, at the generic fidelity level, does not have to simulate a certain aircraft, it should be characteristic of the simulated group of aircraft and include associated phenomena and effects. For the generic fidelity level, it should be demonstrated that there is an aircraft within the simulated group of aircraft with similar flight control system operation as the FSTD. It should be noted that the simulation of complex aircraft is complicated even at the generic fidelity level.

GM4 FSTD.QB.103 Flight simulation training device general requirements for flight control systems operation

INTEGRATION OF THE FLIGHT CONTROL SYSTEM WITH THE AUTO-FLIGHT CONTROL SYSTEM

This GM gives guidance on the integration of the flight control system with the auto-flight control system.

- (a) CS FSTD.QB.010 notes that the individual aircraft systems may be at different fidelity levels. GM1 FSTD.QB.010 explains that the individual aircraft systems cannot always be treated in isolation, since they often integrate with other systems. The aircraft system requirements are presented in CS FSTD.QB.104.
- (b) For a real aircraft, the auto-flight control system may be an integrated and inseparable part of the flight control system. Many helicopters and larger aeroplanes have such integrated systems. In addition to the auto-flight modes or upper modes (e.g. for heading hold, altitude hold, indicated airspeed hold, and navigation mode), such integrated systems may provide stabilisation (e.g. different levels of stability augmentation, stability augmentation systems) and protections. For such an integrated system, based on the references above, the FSTD's auto-flight control system simulation and the flight control systems operation FSTD feature

should be at the same fidelity level (G/R/S) as or at a higher fidelity level (G/R/S) than the flight control system. The SOJ should identify what auto-flight control system is being simulated.

GM5 FSTD.QB.103 Flight simulation training device general requirements for flight control systems operation

FLIGHT AND MANOEUVRE ENVELOPE PROTECTION FUNCTIONS

This GM gives guidance on the flight and manoeuvre envelope protection functions to be simulated in FSTDs. Note that the effect of flight and manoeuvre envelope protection functions should be considered, regardless of the system that controls them.

- (a) CS FSTD.QB.103 specifies the requirements for flight and manoeuvre envelope protection functions, meaning any protection or safety functions associated with aircraft flight control systems, including but not limited to the following.
 - (1) Computer-controlled aircraft (CCA) may incorporate protection functions for overspeed, minimum speed, attitude and other flight parameters. These functions automatically limit pilot control authority and adjust flight control surfaces and/or engine thrust to protect against undesired flight states.
 - (2) Non-CCA may be equipped with various protection systems, such as stall protection, overspeed protection and/or pitch attitude protection. These systems typically provide limited protection by resisting pilot input without fully taking control, increasing the force required to push further into a potentially hazardous range.
 - (3) In certain flight conditions, conventional aircraft may physically limit or obstruct the range of flight control movement to prevent excessive pitch attitudes that could lead to a stall.
 - (4) Some conventional aircraft are equipped with a stick pusher system that automatically moves the pitch control forward when an impending stall is detected, reducing the angle of attack to prevent a full stall.
 - (5) Auto-flight control systems in helicopters may incorporate various protection functions, such as airspeed limitation, attitude limitation, automatic level-off or fly-up when in close proximity to the ground and unusual attitude recovery.
- (b) Protection functions may or may not be integrated with the auto-flight control system, which can make it challenging to determine whether a protection function is part of the flight control systems operation or aircraft systems FSTD feature. This highlights the importance of the integration of these features. Refer to GM4 FSTD.QB.103, CS FSTD.QB.010 and GM1 FSTD.QB.010.
- (c) Flight control force requirements are specified in CS FSTD.QB.102, which also addresses stick pusher systems. System operation modes and logic within flight control system operation, including flight and manoeuvre protection functions, are covered in CS FSTD.QB.103.
- (d) CS FSTD.QB.103 specifies the requirements for the verification and validation of these functions. Note that the objective test requirements differ between CCA and conventional

aircraft, as well as among feature fidelity levels. If objective tests are not applicable, these functions should still undergo verification and validation through functions and subjective testing.

AIRCRAFT SYSTEMS

CS FSTD.QB.104 Flight simulation training device general requirements for aircraft systems

4: Aircraft systems

Defines the extent of the aircraft system simulation required to be modelled in the FSTD. The requirements apply if the ownship is equipped with the appropriate systems. This feature concerns the operation, functionalities and indications of the aircraft systems and the interdependencies of different systems.

Note: Aircraft systems mean, for example, instrument, communication, navigation, autopilot, flight director, flight management, flight guidance, hydraulic, electrical, fuel, pneumatic, pressurisation, air conditioning, power plant, auxiliary power unit, caution and warning, fire protection, ice and rain protection and surveillance systems and also TAWS/HTAWS, ACAS and on-board weather radar systems.

Note: Refer to CS FSTD.QB.010 for the required aircraft systems and their individual fidelity levels.

Note: The individual aircraft systems may be at different fidelity levels (N/G/R/S).

Note: This feature excludes the flight controls. Refer to the flight control forces and hardware FSTD feature in CS FSTD.QB.102 and the flight controls system operation FSTD feature in CS FSTD.QB.103 for their requirements.

Note: Refer to the flight deck layout and structure FSTD feature in CS FSTD.QB.101 for the aircraft system hardware/interface requirements.

Note: Refer to the performance and handling FSTD features in CS FSTD.QB.105 through FSTD.QB.107, the vibration cueing FSTD feature in CS FSTD.QB.109, the motion cueing FSTD feature in CS FSTD.QB.110, the sound cueing FSTD feature in CS FSTD.QB.108 and the visual cueing FSTD feature in CS FSTD.QB.111 for the requirements for the effects associated with aircraft system operation (e.g. activation of the anti-icing system may result in a change in aircraft flight performance and handling, and vibration, motion, sound and visual cues).

The general requirements for the aircraft systems FSTD feature are presented in the table below.

4: Aircraft systems						
Feature general requirements				Feature fidelity level		Verification and validation
		G	R	S	QTG	F&S
4.1	Aircraft system operation					
4.1.1	Aircraft systems should replicate those of the simulated aircraft type and variant with functionality for flight crew operation, including system interdependencies. Once activated, correct normal system operation should result from system management by the flight crew and not require input from the instructor's controls. Systems should be operative to the extent that all normal, abnormal and emergency operating procedures can be accomplished.			✓		✓
4.1.2	Aircraft systems should be representative of those of the simulated aircraft type with sufficient functionality for flight crew operation, including system interdependencies. Once activated, correct normal system operation should result from system management by the flight crew and not require input from the instructor's controls. Systems should be operative to the extent that appropriate normal, abnormal and emergency operating procedures can be accomplished.		✓			✓
4.1.3	Aircraft systems should be characteristic of those of the simulated group of aircraft and should have sufficient functionality for flight crew operation, including system interdependencies. Once activated, appropriate normal system operation should result from system management by the flight crew and not require input from the instructor's controls. System functionality should enable sufficient normal and, as applicable, emergency and abnormal operating procedures to be accomplished.	✓				✓
4.1.4	Circuit breakers for the simulated aircraft systems that are required for system operation should be functional. Note: Refer to CS FSTD.QB.101.	✓	✓	✓		✓
4.1.5	The aircraft systems are to include enhanced vision systems and NVGs, if required by the aircraft type and variant certification.			✓		✓

4: Aircraft systems						
Feature general requirements		Feature fidelity level			Verification and validation	
		G	R	S	QTG	F&S
4.1.6	Integration of the aircraft systems and the FSTD: Aircraft systems should be correctly integrated with each other and with all other FSTD features. (Refer to CS FSTD.QB.010.)	✓	✓	✓		✓
4.2	Aircraft system simulation basis					
4.2.1	The design basis – including data references, as applicable – for the simulation of the relevant aircraft systems by the training device manufacturer (TDM) should be summarised in the ER in terms of the: (1) extent of aircraft system simulation (full or partial functionality/capability with respect to accomplishing all normal, abnormal and/or emergency procedures); (2) aircraft system simulation technique, covering: (a) the TDM's own modelling, (b) aircraft-OEM-provided modelling (executable, binary or loadable software aircraft part), (c) aircraft hardware or line-replaceable unit, (d) re-hosting or re-targeting of aircraft OEM or avionics vendor software, (e) reverse engineering, (f) other modelling approaches as defined by the TDM.	✓	✓	✓		
4.3	Aircraft system databases					
4.3.1	If the aircraft system uses a database (e.g. FMS or GNSS navigation database, chart view, TAWS terrain or obstacle data, magnetic variation database, performance database), the database should be valid and appropriate to support the system operation at the applicable fidelity level (G/R/S) of the system. For systems with navigation-related functions (e.g. FMS or GNSS), the database should include the areas, regions and operating sites that are represented in the simulation of the navigation FSTD feature. The coverage of the database should be consistent with the areas and systems represented in the FSTD.	✓	✓	✓		✓

4: Aircraft systems						
Feature general requirements		Feature fidelity level			Verification and validation	
		G	R	S	QTG	F&S
4.4	Instrument indications					
4.4.1	All relevant instrument indications involved in the simulation of the aircraft type and variant should automatically respond to attitude changes of the aircraft, pilot inputs or changes in system status and also to any external stimuli relevant to the instrument or indication being simulated. The instrument simulation should replicate the real instruments, including inherent errors such as those caused by atmospheric conditions, icing effects, barometric altimeter errors due to outside air temperature, magnetic compass turn and acceleration errors, latency in the barometric altimeter and inaccuracies in radio navigation instruments. Numerical values should be presented in the units used in the simulated aircraft type and variant.			✓		✓
4.4.2	All relevant instrument indications involved in the simulation of the aircraft type should automatically respond to attitude changes of the aircraft, pilot inputs or changes in system status and also to any external stimulus relevant to the instrument or indication being simulated. The instrument simulation should be characteristic of the real instruments, including inherent errors such as those caused by atmospheric conditions, icing effects, barometric altimeter errors due to outside air temperature, magnetic compass turn and acceleration errors, latency in the barometric altimeter and inaccuracies in radio navigation instruments. Numerical values should be presented in the units used in the simulated aircraft type.		✓			✓
4.4.3	All relevant instrument indications involved in the simulation of the aircraft should automatically respond to attitude changes of the aircraft, pilot inputs or changes in system status. The instrument simulation should be characteristic of the simulated group of aircraft and include inherent errors such as those caused by atmospheric conditions, icing effects, barometric altimeter errors due to outside air temperature, magnetic compass turn and acceleration errors, latency in the barometric altimeter and inaccuracies in radionavigation instruments.	✓				✓
4.5	Caution and warning systems					

4: Aircraft systems						
Feature general requirements		Feature fidelity level			Verification and validation	
		G	R	S	QTG	F&S
4.5.1	The caution and warning systems, including audio warnings, aural cues and flight deck speakers associated with aircraft system alerts (e.g. system malfunctions or emergency conditions), should operate within the prescribed operational tolerances and replicate the simulated aircraft type and variant functionalities and characteristics.			✓		✓
4.5.2	The caution and warning systems, including audio warnings, aural cues and flight deck speakers associated with aircraft system alerts (e.g. system malfunctions or emergency conditions), should operate within the prescribed operational tolerances and be representative of the simulated aircraft type functionalities and characteristics.		✓			✓
4.5.3	The caution and warning systems, including audio warnings, aural cues and flight deck speakers associated with aircraft system alerts (e.g. system malfunctions or emergency conditions), should be characteristic of the simulated group of aircraft.	✓				✓
4.6	Communication and surveillance systems Applies if the appropriate systems are simulated.					
4.6.1	The communication systems, including avionics sounds (e.g. intercom, radio communications) and flight deck speakers, should replicate the simulated aircraft type and variant functionalities and characteristics. These systems should operate in accordance with the operational tolerances prescribed for the applicable aircraft type and variant.			✓		✓
4.6.2	The communication systems, including avionics sounds (e.g. intercom, radio communications) and flight deck speakers, should be representative of the functionalities and characteristics of the installed systems in the simulated aircraft type. These systems should operate in accordance with the operational tolerances and aircraft-specific characteristics prescribed for the applicable aircraft type.		✓			✓

4: Aircraft systems						
Feature general requirements		Feature fidelity level			Verification and validation	
		G	R	S	QTG	F&S
4.6.3	The communication systems, including avionics sounds (e.g. intercom, radio communications) and flight deck speakers, should be characteristic of the functionalities and characteristics of the installed systems in the simulated group of aircraft. These systems should operate in accordance with the operational characteristics.	✓				✓
4.6.4	In addition to communications, if the ownship is fitted with headsets, then the appropriate navigation, caution and warning equipment sounds and avionics sounds should be audible in the headsets as in the aircraft.		✓	✓		✓
4.6.5	In addition to communications, if headsets are fitted, the appropriate navigation, caution and warning equipment sounds and avionics sounds should be audible in the headsets as in the aircraft. If simulated sounds are played in the headsets, the volume of the simulated sounds should be controllable independently of that of the aircraft system sounds.	✓				✓
4.6.6	The ownship communication system should support human-provided manual role play of ATC services using radio communications with the ownship.	✓	✓	✓		✓
4.6.7	Radio transmissions from the instructor to the ownship should simulate the auditory and operational characteristics of the relevant communication system (e.g. VHF, HF), including effects such as signal degradation, background static and reduced intelligibility, as appropriate (e.g. HF).	✓	✓	✓		✓
4.6.8	The data link or surveillance systems (e.g. ACARS, CPDLC, ADS-B, ADS-C, future air navigation system (FANS)) should function in a manner that accurately reflects the actual aircraft, enabling realistic interaction with the associated services, as applicable.		✓	✓		✓
4.7	Navigation systems Applies if the appropriate systems are simulated.					

4: Aircraft systems						
Feature general requirements		Feature fidelity level			Verification and validation	
		G	R	S	QTG	F&S
4.7.1	The navigation systems, including avionics sounds related to navigation operations (e.g. system alerts and automatic call-outs), should replicate the simulated aircraft type and variant functionalities and characteristics.			✓		✓
4.7.2	The navigation systems, including avionics sounds related to navigation operations (e.g. system alerts and automatic call-outs), should be representative of the functionalities and characteristics of the installed systems in the simulated aircraft type.		✓			✓
4.7.3	The navigation systems, including avionics sounds related to navigation operations (e.g. system alerts and automatic call-outs), should be characteristic of the functionalities of the installed systems in the simulated group of aircraft.	✓				✓
4.8	ACAS Applies if the appropriate systems are simulated.					
4.8.1	The ACAS should replicate the simulated aircraft type and variant functionalities and characteristics. The system should also support failure scenarios.			✓		✓
4.8.2	The ACAS should be representative of the functionalities and characteristics of the installed systems in the simulated aircraft type.		✓			✓
4.8.3	The ACAS should be characteristic of the functionalities and characteristics of the installed systems in the simulated group of aircraft.	✓				✓
4.8.4	If other traffic is simulated and their equipment includes a transponder, appropriate ACAS traffic advisories and resolutions should be triggered. If an ACAS traffic advisory or resolution advisory is triggered, other traffic is not required to follow correct ACAS protocols.	✓	✓	✓		✓
4.8.5	The performance of other traffic should be realistic, and its behaviour should respond appropriately to the ACAS advisory scenario.	✓	✓	✓		✓

4: Aircraft systems						
Feature general requirements		Feature fidelity level			Verification and validation	
		G	R	S	QTG	F&S
4.8.6	Instructor controls: Several different aircraft traffic scenarios should be provided through the instructor operating station (IOS) to generate different ACAS warning advisories, including Mode C and Mode S transponder functionality, as appropriate.	✓	✓	✓		✓
4.9	Icing effects and anti-icing systems Refer to CS FSTD.QB.107.1.ICING.					
4.9.1	Icing should affect aircraft systems, if applicable (e.g. pitot–static systems, temperature sensors).	✓	✓	✓		✓
4.9.2	Anti-icing systems replicating those installed in the simulated aircraft type and variant should operate with appropriate effects on ice formation on the airframe, engines and instrument sensors.			✓		✓
4.9.3	Anti-icing systems corresponding to those typically installed in the simulated aircraft type should be operative.		✓			✓
4.9.4	Simplified airframe and engine systems, corresponding to those typically installed on the simulated group of aircraft, should be in place.	✓				✓
4.9.5	The simulation models should include the effects of the icing of the pitot–static systems, including the effects of anti-icing systems, as applicable to the aircraft.	✓	✓	✓		✓
4.10	Special equipment					
4.10.1	If the FSTD includes a head-up display (HUD), it should comply with the requirements specified in CS FSTD.QB.104.HUD.		✓	✓		✓
4.10.2	If the FSTD includes an enhanced flight vision system (EFVS), it should comply with the requirements specified in CS FSTD.QB.104.EFVS.		✓	✓		✓
4.10.3	An FSTD designed to support operations with a night-vision imaging system (NVIS) should comply with the requirements specified in CS FSTD.QB.104.NVIS.		✓	✓		✓

4: Aircraft systems						
Feature general requirements		Feature fidelity level			Verification and validation	
		G	R	S	QTG	F&S
4.11	Power plant systems					
4.11.1	Power plant systems should replicate the simulated aircraft type and variant with functionality for flight crew operation in normal, abnormal and emergency operations.			✓		✓
4.11.2	Power plant system functionality should enable appropriate normal, abnormal and emergency operating procedures for the simulated aircraft type to be accomplished.		✓			✓
4.11.3	Power plant systems functionality should enable sufficient operating procedures for the simulated group of aircraft to be accomplished, including the effects of malfunctions.	✓				✓
4.12	Instructor controls					
4.12.1	Instructor controls should be provided to control all appropriate system variables and insert all abnormal or emergency conditions into the simulated aircraft systems. Simulated malfunctions should result in integrated system responses with system interdependencies and allow for the exercise of the procedures in relevant operating manuals.			✓		✓
4.12.2	Instructor controls should be provided to control all appropriate system variables and insert appropriate abnormal or emergency conditions into the simulated aircraft systems. Simulated malfunctions should allow for the exercise of the procedures in relevant operating manuals.		✓			✓
4.12.3	Instructor controls should be provided to control all appropriate system variables and insert sufficient abnormal or emergency conditions into the simulated aircraft systems. Simulated malfunctions should result in integrated system responses with system interdependencies characteristic of the simulated group of aircraft.	✓				✓
4.13	Multi-crew cooperation (MCC)					

4: Aircraft systems						
Feature general requirements				Feature fidelity level		Verification and validation
		G	R	S	QTG	F&S
4.13.1	If the FSTD is to be qualified for MCC, the device and its aircraft systems should comply with the requirements specified in CS FSTD.QB.120.	✓	✓	✓		✓

GM1 FSTD.QB.104 Flight simulation training device general requirements for aircraft systems

Summaries of the fidelity levels for the aircraft systems FSTD feature are presented in the table below. These summaries are broad overviews provided for guidance only. They do not encompass all aspects and requirements and are not intended to replace CS FSTD.QB.104.

4: Aircraft systems
Specific: Simulated aircraft systems replicate the aircraft type and variant system operation with appropriate system interdependencies. Simulated systems are operative to the extent that all normal, abnormal and emergency operating procedures can be accomplished.
Representative: Simulated aircraft system operation and logic represent the simulated aircraft type, including system interdependencies. Simulated systems are operative to the extent that appropriate normal, abnormal and emergency operating procedures can be accomplished.
Generic: Simulated aircraft systems are characteristic of the systems used in the simulated group of aircraft. Simulated aircraft systems have the functionality to enable sufficient normal and, as applicable, emergency and abnormal operating procedures to be accomplished.

GM2 FSTD.QB.104 Flight simulation training device general requirements for aircraft systems

FLIGHT INSTRUMENTS

- (a) The basic flight instruments are defined in GM2 FSTD.QB.101. For the aircraft systems FSTD feature, they should be considered as an entity in which all the instruments are at the same fidelity level to ensure a positive transfer of training. If the fitted flight instruments are simulated at different fidelity levels, the entity should be considered to be only at the fidelity level of the lowest simulated instrument. This is because, in order to achieve any fidelity level, all the requirements applicable to that level must be fulfilled (refer to CS FSTD.QB.010).
- (b) At the generic fidelity level, the flight instruments should be characteristic of the simulated group of aircraft. This means that the instruments are required to have realistic characteristics and should be similar to those of a real aircraft within the simulated group of aircraft. In addition, the instrument power sources should comply with the requirements for those of real aircraft. For example, the airworthiness directives for real aircraft may stipulate that the attitude indicator and turn coordinator must not share the same power source due to redundancy requirements. As a result, it is customary for one of these instruments to be powered by electrical means while the other utilises vacuum power. This should be simulated in an FSTD with instruments at the generic fidelity level. In the FSTD, in the case of the failure

of one of these power sources, only the instrument(s) powered by the failed source should be affected.

GM3 FSTD.QB.104 Flight simulation training device general requirements for aircraft systems

This GM gives guidance on what is expected of the aircraft system(s) at the different fidelity levels (N/G/R/S).

As CS FSTD.QB.104 states, the individual aircraft systems may be at different fidelity levels. Each aircraft system should be evaluated against the CS FSTD.QB.104 general requirements on an individual basis.

The differences between the fidelity levels are based on the following aspects.

- (a) For the specific fidelity level, the general requirements in CS FSTD.QB.104 require that the system replicate the simulated aircraft type and variant with correct functionality. It is expected that, for the specific level, all the system simulation aspects function as the system does in the aircraft (i.e. replicate its operation).
- (b) For the representative fidelity level, the system is required to be representative of the simulated aircraft type with sufficient functionality. The following aspects give guidance on the representative fidelity level.
 - (1) The word 'representative' here means that the system logic, sequences, indications and operation should be correct, but the latencies (i.e. timings, delays of events) and rates may differ from those of the simulated aircraft. For example, adjusting the pressurisation system manually should result in the correct sequence of valves and pack operation, but the timing of those and of the resulting cabin altitude rate need not replicate those in the simulated aircraft; they may be only characteristic.
 - (2) The words 'sufficient functionality' mean that the system may not simulate all the system functionalities or capabilities of the simulated aircraft system. For example, the FMS may not support 3D approaches, while the real aircraft system would be capable of them. Such differences should be declared in the documentation (e.g. ESL; refer to Part-ORA (Annex VII) of Commission Regulation (EU) No 1178/2011).
 - (3) The data requirements for the specific and representative fidelity levels are different, as presented in CS FSTD.QB.104 and Subpart C. If the data requirements for the specific fidelity level cannot be demonstrated, the aircraft system in question may still comply with the data requirements of the representative level.
- (c) For the generic fidelity level, the aircraft system(s) do not need to simulate any particular real aircraft system but should be characteristic of the systems that can be installed in real aircraft. As such, the simulated aircraft systems at the generic level should operate in accordance with the certification standards for real aircraft (e.g. certification specifications for normal, utility, aerobatic and commuter aeroplanes (CS-23), certification specifications and acceptable means of compliance for large aeroplanes (CS-25), certification specifications and acceptable means of compliance for small rotorcraft (CS-27), certification specifications and acceptable

means of compliance for large rotorcraft (CS-29) and European technical standard orders (CS-ETSO)).

- (1) For example, an autopilot system at the generic level does not need to simulate a particular real aircraft autopilot system, but its modes, functions, operations and interdependencies should resemble those typically experienced by pilots in real aircraft of the same group. This means that the autopilot's modes should have typical names, and its mode laws, reversions, gains, limits, indications and logic should be characteristic of the simulated group of aircraft.
- (2) Refer to GM2 FSTD.QB.104 for additional examples and information on flight instruments at the generic level.

CS FSTD.QB.104.HUD Head-up display requirements for flight simulation training devices

- (a) Applicability requirements are as follows.
 - (1) This CS applies to all FSTDs with a HUD installation at the specific and representative fidelity levels. While the requirements in this CS concern many features, they are addressed here as a single entity.
 - (2) For the purposes of this CS, 'HUD' will be used as a general term for any alternative aircraft instrument system that displays information to a pilot through a combiner glass in the normal out-of-the-window view.
 - (3) Objective tests for new, modified or upgraded FSTDs incorporating a HUD system should contain an SOJ. The SOJ should be confirmation that the HUD hardware and software, including associated displays, function in the same way as those of the HUD installed in the aircraft. A block diagram detailing the input and output signal flow and comparing it with the aircraft configuration should support this SOJ.
- (b) FSTD HUD standards are as follows.
 - (1) For a HUD system at the specific fidelity level, the display equipment should replicate the functionalities and characteristics of the real equipment and be provided as fitted in the aircraft flight deck.
 - (2) For a HUD system at the representative fidelity level, the display equipment can be provided as fitted in the aircraft flight deck, or the data may be displayed on the out-of-the-window display. If the data is displayed outside the flight deck (e.g. overlaid on the visual image), the HUD symbology should be clipped in line with the HUD viewing box, ensuring that it aligns with the visible area as seen through the pilot's HUD. The data image should remain controllable by the pilot in the same manner as in the aircraft. This functionality is only applicable to single-pilot operations unless a baffle is installed to prevent the HUD symbology from being visible to the other flight crew members, allowing control of the display to remain with the pilot operating the HUD system.

- (3) For non-collimated systems, only one display can be used by the pilot flying due to alignment with the out-of-the-window displays.
 - (4) Whether the HUD system is an actual aircraft system or is software simulated, the system should perform its intended function for each operation and phase of flight.
 - (5) A repeater display of all parameters displayed on the pilot's combiner should be located on the IOS or at another location approved by the competent authority. The repeater display format should replicate that of the combiner and should display only the HUD symbology parameters, excluding the out-of-the-window view.
- (c) Objective testing requirements are as follows.
- (1) The HUD attitude should align with the ownship attitude. Static calibration tests should be included for HUD attitude alignment in the QTG. These tests may be combined with the alignment tests for the FSTD visual system. Refer to CS FSTD.QTG.A.405 or CS FSTD.QTG.H.405 for objective tests.
 - (2) HUD systems that are software simulated (i.e. all components, including image generation, are simulated; not being an actual aircraft system) should include transport delay tests in all three axes. Refer to CS FSTD.QTG.A.600 or CS FSTD.QTG.H.600 for system integration tests.
- (d) Subjective testing requirements are as follows.
- (1) The ground and flight tests that should be conducted for the qualification of HUD systems are listed below and may be combined with subjective manoeuvres not dedicated to HUD testing. Only those phases of flight for which the particular HUD system is authorised should be tested and detailed in the QTG. The testing should be conducted using daylight, twilight and night conditions as available or applicable. The tests are as follows:
 - (i) pre-flight inspection of the HUD system;
 - (ii) taxi:
 - (A) HUD taxi guidance,
 - (B) combiner horizon's match against the visual horizon within the manufacturer's tolerance.
 - (iii) take-off testing should cover:
 - (A) normal take-off in visual meteorological conditions (VMCs) (centre line guidance if available);
 - (B) instrument take-off using the lowest runway visual range (RVR) authorised for the particular HUD;
 - (C) engine-out take-off;
 - (D) maximum demonstrated crosswind take-off;
 - (E) wind shear during take-off.

- (iv) in-flight testing should cover:
 - (A) climb;
 - (B) turns;
 - (C) cruise;
 - (D) descent.
 - (v) approach testing should cover:
 - (A) normal approach in VMCS;
 - (B) ILS approach with a crosswind:
 - (a) flight path vector represents the inertial path of the aircraft,
 - (b) course indication matches the track over the ground,
 - (c) HUD combiner does not excessively degrade the approach lights;
 - (C) engine-out approach and landing;
 - (D) non-precision approach;
 - (E) circling approach, if applicable;
 - (F) missed approach (normal and engine out);
 - (G) maximum demonstrated crosswind approach and landing;
 - (H) wind shear on approach.
 - (vi) malfunction testing should cover:
 - (A) malfunctions causing abnormal pre-flight tests;
 - (B) malfunctions logically associated with training during take-off and approach;
 - (C) malfunctions associated with any approved flight manual abnormal procedures that are not included above.
- (b) If the real HUD system is certified without an emergency power backup, the simulated HUD should also blank out and reboot following any temporary power loss. The power supply and expected behaviour should be based on the manufacturer's data and documented in the SOJ.

CS FSTD.QB.104.EFVS Enhanced flight vision system requirements for flight simulation training devices

- (a) Applicability requirements are as follows.
 - (1) This CS applies to all FSTDs with EFVS installations at the specific and representative fidelity levels and applies in addition to the HUD requirements detailed in CS

FSTD.QB.104.HUD. While the requirements in this CS concern many features, they are addressed here as a single entity.

- (2) For the purposes of this CS, 'EFVS' will be used as a general term for any alternative aircraft visual enhancement aid using imaging sensors, such as an infrared radiometer or a radar, that displays information to a pilot through a HUD combiner glass in the normal out-of-the-window view.
 - (3) Objective tests for FSTDs incorporating an EFVS should contain an SOJ. The SOJ should be confirmation that the EFVS hardware and software, including associated displays and annunciation, function in the same way as or in an equivalent way to those of the EFVS(s) installed in the aircraft. A block diagram detailing the input and output signal flow and comparing it with the aircraft configuration should support this SOJ.
- (b) FSTD EFVS standards are as follows.
- (1) Whether the EFVS is an actual aircraft system or is software simulated, the system should be shown to perform its intended function for each operation and phase of flight.
 - (2) The minimum FSTD requirements for qualifying an EFVS in an FSTD at the specific or representative fidelity level are as follows.
 - (i) The EFVS FSTD hardware and software, including associated flight deck displays and annunciation, should function in the same way as or in an equivalent way to those of the EFVS installed in the aircraft.
 - (ii) A repeater display of the pilot's combiner should be located on the IOS or at another location approved by the competent authority. It should include a duplicate display of the EFVS and HUD scene, as seen through the pilot's HUD combiner glass or on the flight deck flight displays.
 - (iii) For aircraft in which the EFVS image presentation is combined with the HUD, the repeated HUD image on the IOS should also include the EFVS image with correct registration.
 - (iv) IOS weather presets should be provided for EFVS minima.
 - (v) A minimum of one airport / operating site should be modelled for EFVS operation. That model should have an ILS and a non-precision approach (with vertical navigation (VNAV) if required by the AFM for the simulated aircraft type) available. In addition to modelling the EFVS, the airport model should meet the applicable visual system requirements in the CS FSTD.QB.114 general requirements for the operating sites and terrain FSTD feature.
 - (vi) The FVS should demonstrate thermal crossover effects during the day-to-night transition. The scene should correctly represent the thermal characteristics of the scene during a day-to-night transition.
 - (vii) For an EFVS at the representative fidelity level, the display equipment can be provided as fitted in the aircraft flight deck, or the data may be displayed on

the out-of-the-window display. If the data is displayed outside the flight deck (e.g. overlaid on the visual image), the EFVS symbology should be clipped in line with the EFVS viewing box, ensuring that it aligns with the visible area as seen through the pilot's EFVS. The data image should remain controllable by the pilot in the same manner as in the aircraft. This functionality is only applicable to single-pilot operations unless a baffle is installed to prevent the EFVS symbology from being visible to the other flight crew members, allowing control of the display to remain with the pilot operating the EFVS.

- (c) For objective testing requirements:
 - (1) refer to CS FSTD.QTG.A.410 or CS FSTD.QTG.H.410 for objective tests;
 - (2) refer to CS FSTD.QTG.A.600 or CS FSTD.QTG.H.600 for system integration tests.
- (d) Subjective testing requirements are as follows.
 - (1) Handling qualities, performance and FSTD system operation, while using the EFVS, should be subjectively assessed.
 - (2) The ground and flight tests and other checks required for the EFVS are listed below. They include manoeuvres and procedures to ensure that the EFVS functions and performs appropriately for use in pilot training and checking in the manoeuvres and procedures, including any OSD special emphasis items or items delineated in the relevant Joint Operations Evaluation Board (Joint Aviation Authorities) or EASA Operations Evaluation Board reports. The testing should be conducted using daylight, twilight and night conditions and cover the following aspects.
 - (i) For the IOS, check to ensure that the IOS has preset selections that match the training programme.
 - (ii) For pre-flight tests, carry out normal pre-flight procedures and checks, including warnings and annunciations.
 - (iii) For taxi:
 - (A) observe parallax caused by sensor position;
 - (B) observe ground hazards, especially other aircraft and nearby terrain;
 - (C) note that, due to the way the infrared sensor works in EFVS, signs may appear as a block (unreadable) due to the absence of temperature variation between the letters and the background.
 - (iv) For take-off:
 - (A) to test normal take-off in night VMCs, observe the terrain and surrounding visual scene;
 - (B) to test instrument take-off using visual RVR settings of 180 m, ensure that the RVR perceived through the EFVS is better than the visual RVR (i.e. at least 750 m).

- (v) For in-flight operations:
 - (A) adjust the scene to VMCs and see if the image horizon conforms with the visual horizon and the combiner horizon;
 - (B) using a VMC night or twilight scene, select a thunderstorm at a distance of at least 20 nm (37 km) and see if the imager detects the clouds.
- (vi) The following approaches should be tested.
 - (A) Normal approach in night VMCs.
 - (B) ILS approach requirements are as follows.
 - (a) Select the preset that allows the pilot to see the EFVS image at approximately 500 ft. This will preset the EFVS visibility to approximately 2 300 m and the visual RVR to 750 m.
 - (b) Fly or reposition the aircraft to 500 ft AGL on the ILS. Freeze position. The pilot flying should be able to see the image of the runway approach lights. The pilot not flying should not be able to see any lights. (Some very slight bleed-through of strobes is acceptable but not of steady lights.)
 - (c) Continue the approach and freeze position at 200 ft AGL. The pilot flying should be able to see approximately 1 NM down the runway, and the pilot not flying should be able to visually acquire the approach lights and runway end identifier lights.
 - (d) Continue the approach and landing. Observe the blooming effect of the airport lights.
 - (C) Non-precision approach.
 - (D) Missed approach.
 - (E) Note that emphasis should be placed on the FSTD's capability to demonstrate that the EFVS is able to display the visual cues needed for the pilot to identify the visual references required to descend below the published decision altitude if conducting instrument approaches with vertical guidance. The EFVS should continue to provide glide path and alignment information between DH and touchdown. During landing roll-out, visual alignment information should be available to the pilot.
- (vii) The visual segment and landing should be tested:
 - (A) from the non-precision approach;
 - (B) from the precision approach.

- (viii) Abnormal procedure tests should cover:
 - (A) EFVS malfunctions on the ground;
 - (B) EFVS malfunctions in the air.
- (e) Due to the uniqueness of this system and the normal FSTD environmental visual selections, the IOS should have preset weather conditions for EFVS operations. Recommended settings are such that the EFVS 'visual' reference can be attained at approximately 500 ft (150 m) AGL, for category I operations with a DH not lower than 200 ft and RVR in accordance with the operational weather minima (CAT I) and EFVS authorised minima, and below the minima to force a go-around.

CS FSTD.QB.104.NVIS Night-vision imaging system requirements for flight simulation training devices

- (a) Applicability requirements are as follows.
 - (1) This CS applies to all FSTDs with a night vision imaging system (NVIS), such as NVGs, at specific or representative fidelity levels. While the requirements in this CS concern many features, they are addressed here as a single entity.
 - (2) For the purposes of this CS, 'NVIS' will be used as a general term for an NVIS defined as a direct-view (image viewed directly in line with the image intensification process), head-mounted, binocular, light intensification appliance that enhances the ability to maintain visual surface reference at night.
 - (3) Objective tests for new, updated or upgraded FSTDs incorporating an NVIS should contain an SOJ. The SOJ should be confirmation that the NVIS hardware and software, including associated displays and annunciation, function in the same way as or in an equivalent way to those of the system(s) installed in the aircraft. A block diagram detailing the input and output signal flow and comparing it with the aircraft configuration should support this SOJ.
- (b) FSTD NVIS standards should meet the following requirements.
 - (1) Whether the NVIS is an actual aircraft system or is software simulated, the system should be shown to perform its intended function for each operation and phase of flight.
 - (2) The minimum FSTD requirements for qualifying an NVIS in an FSTD at the specific or representative fidelity level are as follows.
 - (i) The NVIS FSTD hardware and software, including associated flight deck displays and annunciation, should function in the same or in an equivalent way to those of the NVIS installed in the aircraft.
 - (ii) If the NVIS information is displayed to a pilot through a HUD combiner, a repeater display of the pilot's combiner should be located on the IOS or at another location approved by the competent authority. It should include a

duplicate display of the NVIS, as seen through the pilot's HUD combiner glass or on the flight deck flight displays.

- (iii) The FSTD environment must be suitable for NVIS training.
 - (A) Windshield panes are required for any FSTD used for NVIS training tasks. The effects of the windshield on NVIS visuals and non-NVIS visuals should be as similar as possible to those in the aircraft.
 - (B) Displays or lighting outside the flight deck area, if installed, should not interfere with or distract from NVIS training.
 - (C) Rotorcraft operational steerable (visible-light, non-infrared) searchlight or aeroplane landing/taxi lights, as appropriate for the aircraft configuration, used for the identification of wires, poles and other obstructions during landing, take-off and en route should be simulated. Use of the searchlight should realistically aid in the detection of poles, wires and obstacles.
 - (D) Suitable seat adjustment for proper NVIS training should be in place.
 - (E) Proper flight deck lighting, suitable for NVIS training, should be in place.
 - (F) There should be no distracting light leaks, glare or unrealistic halo effects.
 - (G) External lighting should not interfere with or distract from NVIS training.
- (c) For objective testing, refer to CS FSTD.QTG.A.415 or CS FSTD.QTG.H.415
- (d) Subjective testing should take place as follows.
 - (1) Handling qualities, performance and FSTD system operation, while using the NVIS, should be subjectively assessed.
 - (2) The ground and flight tests and other checks required for the qualification of the NVIS are listed below. They include manoeuvres and procedures to ensure that the NVIS functions and performs appropriately for use in pilot training and checking in the manoeuvres and procedures and/or special areas of emphasis. The testing should be conducted using night conditions. The visual scene should simulate the actual night environment as viewed both aided and unaided at moon illumination levels between 0 % and 100 %. If moon elevation angles other than 90° are simulated, then shadows computed as a function of the moon angle should be displayed. Required testing is as follows.
 - (i) For IOS tests, check to ensure that the IOS has preset selections that match the training programme.

- (ii) Pre-flight tests should:
 - (A) inspect and adjust NVIS appliances in accordance with the operator's approved (OEM-recommended) procedures;
 - (B) evaluate the flight deck environment for the suitability of a head-mounted NVIS and determine the ability to view flight deck instrumentation, switches and controls below NVIS FOV;
 - (C) carry out normal pre-flight procedures and checks, including warnings and annunciations.
- (iii) Taxi/take-off tests should evaluate the use of NVIS for focus, depth perception, brightness, contrast and FOV while performing the following tasks:
 - (A) taxiing on wheels (rotorcraft and aeroplane) and/or while hovering (rotorcraft),
 - (B) normal take-off from hover (rotorcraft) or normal rolling take-off (aeroplane),
 - (C) take-off with abnormal occurrence (engine failure, category A/B).
- (iv) In-flight operations should:
 - (A) check for adequate ground reference while in straight and level flight;
 - (B) check for adequate ground reference while manoeuvring (turns, climbs and descents);
 - (C) evaluate night scenes with a moon for suitability with NVIS;
 - (D) navigate to a designated area via ground reference;
 - (E) check navigation by instruments;
 - (F) check flight deck lighting (instruments, panel illumination, ability to read checklists, etc.).
- (v) Approach testing should:
 - (A) check visual approach to a confined area or unimproved/unlit runway without reference to ground lighting;
 - (B) check engine-out approach for multi-engine aircraft;
 - (C) respond to a change from VMCs to instrument meteorological conditions (IMCs).
- (vi) Abnormal procedure tests:
 - (A) should be undertaken during hover, take-off, cruise, approach and landing;

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- (B) may select malfunctions from the abnormal procedures section of the operator's approved flight manual.
 - (vii) Note that emphasis should be placed on the FSTD's capability to demonstrate compatibility with use in NVIS training and the ability to display adequate visual scenes such that the pilot can identify the required visual references for night VFR flight.

PERFORMANCE AND HANDLING ON GROUND

CS FSTD.QB.105 Flight simulation training device general requirements for performance and handling on ground

5: Performance and handling on ground

Defines the mathematical models and associated data to be used to describe the ground-handling characteristics and aerodynamic, propulsion, ground reaction and operating site surface conditions required to be modelled in the FSTD whenever the aircraft is in contact with the surface. These include acceleration, deceleration, turning, braking and the effects of wind (including crosswind) as relevant to the aircraft.

The general requirements for the performance and handling on ground FSTD feature are presented in the table below.

5: Performance and handling on ground								
Feature general requirements				Feature fidelity level			Verification and validation	
		G	R	S	QTG(A)	QTG(H)	F&S	
5.1	Ground reaction Covers the reaction of the aircraft upon contact with the ground (also including ship deck, sea and ditching, as applicable) or ground-based infrastructure (e.g. platform, rig, mooring mast) through its dedicated system during departure and arrival phases. The equipment of the aircraft system for contacting the ground should be considered accordingly (landing gear, skids, bumpers, other).							
5.1.1	A ground reaction model that replicates the simulated aircraft type and variant upon contact with the operating site (e.g. ground or ground-based infrastructure) is to include but not be limited to strut/skid deflections, tyre or skid friction, side forces, oleo deflections, effects of each main landing gear and other appropriate parameters, such as weight and speed, necessary to identify the flight condition and aircraft configuration.			✓				✓

5: Performance and handling on ground								
Feature general requirements		Feature fidelity level			Verification and validation			
		G	R	S	QTG(A)	QTG(H)	F&S	
5.1.2	A ground reaction model that is characteristic of the simulated aircraft type upon contact with the operating site (e.g. ground or ground-based infrastructure) is to include but not be limited to strut/skid deflections, tyre or skid friction, side forces, effects of each main landing gear and other appropriate parameters, such as weight and speed, necessary to identify the flight condition and aircraft configuration.		✓					✓
5.1.3	A ground reaction model that is characteristic of the simulated group of aircraft upon contact with the operating site (e.g. ground or ground-based infrastructure) is to include but not be limited to strut/skid deflections, tyre or skid friction, side forces and other appropriate parameters, such as weight and speed, necessary to identify the flight condition and configuration.	✓						✓
5.1.4	For rotorcraft: A ground resonance model (if applicable to the simulated aircraft) that is characteristic of the simulated aircraft type is to include but not be limited to the effect of the flight control position, the rotorcraft weight and balance, the main rotor speed and the stiffness of the undercarriage.		✓	✓				✓
5.2	Ground movement characteristics Ground movement characteristics – during taxi, take-off and landing movement – include the behaviour of contact between the ground and aircraft elements (e.g. brakes, turning radius, rate of turn) and the elements of the atmosphere that may be encountered in ground operations (e.g. crosswinds).							
5.2.1	The handling characteristics and effect of control inputs (to include braking, deceleration and turning radius) should replicate those of the simulated aircraft type and variant.			✓	1.a	1.b		✓
5.2.2	The handling characteristics and effect of control inputs (to include braking, deceleration and turning radius) should be characteristic of the simulated aircraft type.		✓		1.a	1.b		✓
5.2.3	The handling characteristics and effect of control inputs should be characteristic of the simulated group of aircraft.	✓						✓

5: Performance and handling on ground							
Feature general requirements		Feature fidelity level			Verification and validation		
		G	R	S	QTG(A)	QTG(H)	F&S
5.2.4	The effect of control inputs to counteract atmospheric effects (to include crosswind, if applicable) should replicate those of the simulated aircraft type and variant.			✓	1.b, 2.e	1.b, 1.j	✓
5.2.5	The effect of control inputs to counteract atmospheric effects (to include crosswind, if applicable) should be characteristic of the simulated aircraft type.		✓		1.b, 2.e	1.b, 1.j	✓
5.2.6	The effect of control inputs to counteract atmospheric effects should be characteristic of the simulated group of aircraft.	✓			1.b, 2.e	1.b, 1.j	✓
5.3	Surface and operating site conditions Describes aircraft performance on ground for nominal or contaminated conditions (wet, icy, etc.) on ground or particular infrastructure.						
5.3.1	Stopping and directional performance and handling characteristics for at least the following landing and take-off surface conditions should be based on aircraft-related data for a landing or running landing and should replicate those of the simulated aircraft type and variant (as appropriate for the aircraft area of operations and aircraft configuration): (1) dry; (2) wet (for rotorcraft, soft and hard surface); (3) icy; (4) patchy wet; (5) patchy ice; (6) wet on rubber residue in touchdown zone (for aeroplane); and (7) slope landings (for rotorcraft).			✓	1.e		✓

5: Performance and handling on ground								
Feature general requirements		Feature fidelity level			Verification and validation			
		G	R	S	QTG(A)	QTG(H)	F&S	
5.3.2	Stopping and directional performance and handling characteristics for at least the following landing surface conditions should be based on aircraft-related data for a landing or running landing (if appropriate for the aircraft area of operations and aircraft configuration) and should be characteristic of the simulated aircraft type: (1) dry; and (2) wet.		✓		1.e		✓	
5.3.3	Stopping and directional performance and handling characteristics for dry landing surface conditions for a landing or running landing (if appropriate for the aircraft area of operations and aircraft configuration) should be characteristic of the simulated group of aircraft.	✓					✓	
5.4	Brake and tyre failures							
5.4.1	Brake and tyre failure dynamics (including anti-skid) and the effects of temperature on brake efficiency should replicate those of the simulated aircraft type and variant (e.g. reduced brake efficiency in the case of steel brakes). Stopping and directional forces should be characteristic for all environmental landing conditions.		✓	✓			✓	
5.5	Strikes							
5.5.1	The effects of the aircraft striking the ground – such as tail strikes, pod strikes, wing tip strikes, rotor strikes and impacts involving wire cutters or slung loads – or striking objects, whether artificial or natural, should be simulated as a function of the aircraft geometry.		✓	✓			✓	
5.6	Aerodynamics							
5.6.1	An aerodynamic and flight dynamic model that includes ground effects should replicate the simulated aircraft type and variant.			✓	1.a, 1.b, 2.e	1.b, 1.j	✓	

5: Performance and handling on ground								
Feature general requirements		Feature fidelity level			Verification and validation			
		G	R	S	QTG(A)	QTG(H)	F&S	
5.6.2	Ground reaction and aerodynamic modelling should be included for the effects of crosswind and side-slip on directional control.			✓	1.a, 1.b, 2.e	1.b, 1.j	✓	
5.6.3	For aeroplanes: Aerodynamic modelling that includes the normal and reverse dynamic thrust effect on control surfaces, aeroelastic effect and representations of non-linearities due to side-slip should replicate the simulated aircraft type and variant.			✓	1.a, 1.b, 2.e		✓	
5.6.4	For aeroplanes: Aerodynamic and ground reaction modelling for the effects of reverse thrust on directional control should be included.		✓		2.e		✓	
5.6.5	An acceptable simulation of ground effect includes modelling of lift/thrust, airframe drag, pitching moment, trim and power while on the ground. This should include the effects on lateral–directional characteristics, such as roll damping.			✓	1.a, 1.b, 2.e	1.b, 1.j	✓	
5.6.6	Aerodynamic, flight dynamic and lift/thrust system modelling should be characteristic of the simulated aircraft type on the ground.		✓		1.a, 1.b, 2.e	1.b, 1.j	✓	
5.6.7	Aerodynamic, flight dynamic and lift/thrust system modelling should be characteristic of the simulated group of aircraft on the ground.	✓			1.b, 2.e	1.b, 1.j	✓	
5.7	Ground dynamics and handling							
5.7.1	The performance and handling should replicate those of the simulated aircraft type and variant during the on-ground segment of take-off and landing.			✓	1.a, 1.b, 2.e	1.b, 1.c, 1.j	✓	
5.7.2	The performance and handling should be characteristic of the simulated aircraft type during the on-ground segment of take-off and landing.		✓		1.a, 1.b, 2.e	1.b, 1.c, 1.j	✓	

5: Performance and handling on ground							
Feature general requirements		Feature fidelity level			Verification and validation		
		G	R	S	QTG(A)	QTG(H)	F&S
5.7.3	The performance and handling should be characteristic of the simulated group of aircraft during the on-ground segment of take-off and landing.	✓			1.a, 1.b, 2.e	1.b, 1.c, 1.j	✓
5.8	Flight dynamic model						
5.8.1	A flight dynamics model should replicate the simulated aircraft type and variant on the ground, including aerodynamic and lift/thrust system modelling that accounts for all combinations of drag and thrust and all combinations of airspeed and power normally encountered in flight, including the effect of change in aircraft attitude, aerodynamic forces and moments, forces and moments of rotors, aerodynamic interference, altitude, temperature, mass, centre-of-gravity location and configuration.			✓	1.a, 1.b, 2.e	1.b, 1.c, 1.j	✓
5.8.2	A flight dynamics model should be characteristic of the simulated aircraft type on the ground, including aerodynamic and lift/thrust system modelling that accounts for various combinations of drag and thrust normally encountered in flight, including the effect of change in aircraft attitude, aerodynamic forces and moments, forces and moments of rotors, altitude, temperature, mass, centre-of-gravity location and configuration.		✓		1.a, 1.b, 2.e	1.b, 1.c, 1.j	✓
5.8.3	A flight dynamics model should be characteristic of the simulated group of aircraft on the ground, including aerodynamic and lift/thrust system modelling that accounts for various combinations of drag and thrust normally encountered in flight, including the effect of change in aircraft attitude, altitude, temperature, mass, centre-of-gravity location and configuration.	✓					✓
5.9	Mass properties						
5.9.1	An aircraft-type-specific mass properties model, including mass or weight, centre of gravity and moments of inertia as functions of payload, fuel or any consumable loading, should be implemented.			✓			✓
5.9.2	A representative, aircraft-like, mass properties model, including mass, centre of gravity and moments of inertia as functions of payload, fuel or any consumable loading, should be implemented.		✓				✓

5: Performance and handling on ground								
Feature general requirements		Feature fidelity level			Verification and validation			
		G	R	S	QTG(A)	QTG(H)	F&S	
5.9.3	A generic, aircraft-like, mass properties model, including mass, centre of gravity and moments of inertia as functions of payload, fuel or any consumable loading, should be implemented.	✓						✓
5.9.4	The effects of pitch attitude and fuel slosh on the aircraft centre of gravity should be simulated.		✓	✓				✓
5.10	Aerodynamic flight control handling response							
5.10.1	The flight control input should result in the aircraft response that replicates that of the simulated aircraft type and variant.			✓	1.a, 1.b, 1.e, 2.e	1.b, 1.j		✓
5.10.2	The flight control input should result in the aircraft response that is characteristic of the simulated aircraft type.		✓		1.a, 1.b, 1.e, 2.e	1.b, 1.j		✓
5.10.3	The flight control input should result in the aircraft response that is characteristic of the simulated group of aircraft.	✓			1.b, 1.e, 2.e	1.b, 1.j		✓
5.11	Atmospheric effects Defines the effect of atmospheric disturbances on performance and handling.							
5.11.1	The performance and handling characteristics of the simulated aircraft should be consistent with the atmospheric model (refer to CS FSTD.QB.113). The performance and handling characteristics should respond to atmospheric effects to the extent that these are covered by the atmospheric model, at its fidelity level.	✓	✓	✓				✓
5.11.2	For aeroplanes: The performance and handling characteristics should support the effects of specific atmospheric phenomena, such as crosswind, wind shear and microburst (as applicable), consistent with the fidelity level and content of the atmospheric model (refer to CS FSTD.QB.113). These effects should replicate the effect on the simulated aircraft type and variant.			✓				✓

5: Performance and handling on ground							
Feature general requirements		Feature fidelity level			Verification and validation		
		G	R	S	QTG(A)	QTG(H)	F&S
5.11.3	For aeroplanes: The performance and handling characteristics should support the effects of atmospheric phenomena such as crosswind, wind shear and microburst (as applicable) in a way that is characteristic of the effect on the simulated aircraft type.		✓				✓
5.11.4	The performance and handling characteristics should be consistent and correlate with any additional atmospheric and weather models implemented (refer to CS FSTD.QB.113) and should reflect their effects on the simulated aircraft type and variant.			✓			✓
5.12	Icing effects Defines the effects of icing on the aircraft (e.g. the airframe, the lift/thrust system, aerodynamics, flight dynamics, surfaces and engines, as applicable). Refer to CS FSTD.QB.107.1.ICING.						
5.12.1	The flight model should include the effects of icing, if applicable, on the airframe, the lift/thrust system aerodynamics and the engine(s). Refer to CS FSTD.QB.107.1.ICING, paragraph (a).			✓			✓
5.12.2	For aeroplanes certified for flight into icing conditions: Icing models should simulate the aerodynamic degradation effects of ice accretion on the lift/thrust system's lifting surfaces, including loss of efficiency, effect on power setting, change in pitching, rolling moments, decrease in control effectiveness and changes in control forces, in addition to any overall increase in drag or aircraft fuselage gross mass, with resulting effects on power. Refer to CS FSTD.QB.107.1.ICING, paragraph (a).			✓			✓
5.12.3	For rotorcraft certified for flight into icing conditions: The icing model should simulate the aerodynamic degradation effects of airframe and rotor icing and performance degradation due to icing. Refer to CS FSTD.QB.107.1.ICING, paragraph (b).			✓			✓

5: Performance and handling on ground							
Feature general requirements		Feature fidelity level			Verification and validation		
		G	R	S	QTG(A)	QTG(H)	F&S
5.12.4	For aeroplanes and rotorcraft not certified for flight into icing conditions: Modelling should include the effects of icing, as appropriate, on the airframe, aerodynamics, weight, balance and engine(s). Icing models with performance degradation due to icing should be provided. Refer to CS FSTD.QB.107.1.ICING, paragraph (c).			✓			✓
5.12.5	For aeroplanes and rotorcraft certified for flight into icing conditions: Modelling that includes the effects of icing, as appropriate, on the airframe, aerodynamics, weight, balance and engine(s) should include icing models with the associated performance degradation. Refer to CS FSTD.QB.107.1.ICING, paragraph (d).	✓	✓				✓
5.12.6	For aeroplanes certified for flight into icing conditions: The engine icing model should simulate the effects of icing on the power plant, including performance and efficiency, and the effects of the build-up and balance of the engine (e.g. roughness and vibration). Refer to CS FSTD.QB.107.1.ICING, paragraph (a).			✓			✓
5.13	Engine performance						
5.13.1	The engine performance should replicate that of the simulated aircraft type and variant.			✓			✓
5.13.2	The engine performance should be characteristic of the simulated aircraft type.		✓				✓
5.13.3	The engine performance should be characteristic of the simulated group of aircraft.	✓					✓
5.14	Integration						
5.14.1	Performance and handling on the ground should correlate with the environmental conditions and aircraft system simulation.		✓	✓			✓
5.14.2	Performance and handling on the ground should be affected by the environmental conditions and aircraft system simulation, as applicable.	✓					✓

5: Performance and handling on ground								
Feature general requirements					Feature fidelity level		Verification and validation	
		G	R	S	QTG(A)	QTG(H)	F&S	
5.15	Performance and handling transition							
5.15.1	The transition between flight regimes on-ground and in-ground effect should be smooth.	✓	✓	✓			✓	

GM1 FSTD.QB.105 Flight simulation training device general requirements for performance and handling on ground

Summaries of the fidelity levels for the performance and handling on ground FSTD feature are presented in the table below. These summaries are broad overviews provided for guidance only. They do not encompass all aspects and requirements and are not intended to replace CS FSTD.QB.105.

5: Performance and handling on ground

Specific:

The aircraft's reaction upon ground contact and its aerodynamics and lift/thrust system while on the ground replicate the behaviour of the actual aircraft. The handling characteristics and effect of control inputs (e.g. braking, turning, engine thrust, effect of the rudder) replicate those of the simulated aircraft type and variant. Crosswind effects, brake and tire failures, oleo deflections and performance and handling on various runway contaminants are simulated realistically.

Representative:

The aircraft's reaction upon ground contact and its aerodynamics and lift/thrust system while on the ground are characteristic of the simulated aircraft type. The handling characteristics and effect of control inputs (e.g. braking, turning, engine thrust, effect of the rudder) are characteristic of the simulated aircraft type. Performance and handling on dry and wet runways are simulated.

Generic:

The aircraft's reaction upon ground contact and its aerodynamics and lift/thrust system while on the ground are characteristic of the simulated group of aircraft. The handling characteristics and effect of control inputs (e.g. braking, turning, engine thrust, effect of the rudder) are similarly generic and characteristic of the simulated group of aircraft. Performance and handling only on dry runways is required.

PERFORMANCE AND HANDLING IN-GROUND EFFECT

CS FSTD.QB.106 Flight simulation training device general requirements for performance and handling in-ground effect

6: Performance and handling in-ground effect

Defines the mathematical models and associated data to be used to describe the aerodynamic, flight dynamic and lift/thrust characteristics required to be modelled in the FSTD in-ground effect. The height of ground effect is the wing span, or main rotor diameter, unless otherwise declared by the OEM or data provider (as applicable).

The general requirements for the performance and handling in-ground effect FSTD feature are presented in the table below.

6: Performance and handling in-ground effect								
Feature general requirements				Feature fidelity level			Verification and validation	
		G	R	S	QTG(A)	QTG(H)	F&S	
6.1	Ground effect							
6.1.1	Aerodynamics and lift/thrust system modelling should replicate the simulated aircraft type and variant in-ground effect.			✓	1.b, 2.e, 2.f	1.c, 1.d, 1.j	✓	
6.1.2	Aerodynamics and lift/thrust system modelling should be characteristic of the simulated aircraft type in-ground effect.		✓		1.b, 2.e, 2.f	1.c, 1.d, 1.j	✓	
6.1.3	Aerodynamics and lift/thrust system modelling should be characteristic of the simulated group of aircraft in-ground effect to permit ground–air transitions.	✓			1.b, 2.e	1.c, 1.d, 1.j	✓	

6.1.4	An acceptable simulation of ground effect includes modelling of lift/thrust system efficiency, airframe drag, pitching moment and trim and power/thrust while in-ground effect and also the transition effect from in-ground-effect to out-of-ground-effect conditions and vice versa. This should include the effects on the lateral-directional characteristics from ground effect, such as roll damping, as applicable to the aircraft.			✓	1.b, 2.e, 2.f	1.c, 1.d, 1.j	✓
6.2	Ground effect dynamics and handling						
6.2.1	The performance and handling characteristics during take-off, final approach and landing should replicate those of the simulated aircraft type and variant. These should include the transition to/from ground effect.			✓	1.b, 2.e, 2.f	1.c, 1.j	✓
6.2.2	The performance and handling characteristics during take-off, final approach and landing should be characteristic of the simulated aircraft type. These should include the transition to/from ground effect.		✓		1.b, 2.e, 2.f	1.c, 1.j	✓
6.2.3	The performance and handling characteristics during take-off, final approach and landing should be characteristic of the simulated group of aircraft. These should include the transition to/from ground effect.	✓			1.b, 2.e	1.c, 1.j	✓
6.2.4	For aeroplanes: Aerodynamic modelling should include the normal thrust effect on control surfaces, the aeroelastic effect and representations of non-linearities due to side-slip.			✓	1.b, 2.e, 2.f		✓
6.2.5	The flight dynamic responses and stability in-ground effect should replicate those of the simulated aircraft type and variant.			✓	1.b, 2.e, 2.f	1.c, 1.j, 2.b	✓
6.2.6	The flight dynamic responses and stability in-ground effect should be characteristic of the simulated aircraft type.		✓		1.b, 2.e, 2.f	1.c, 1.j, 2.b,	✓
6.2.7	The flight dynamic responses and stability in-ground effect should be characteristic of the simulated group of aircraft.	✓			1.b, 2.e	1.c, 1.j, 2.b,	✓
6.2.8	For rotorcraft: The performance and handling characteristics in hover in-ground effect and through translational lift should replicate those of the simulated aircraft type and variant.			✓		1.d, 2.b	✓

6.2.9	For rotorcraft: The performance and handling characteristics in hover in-ground effect and through translational lift should be characteristic of the simulated aircraft type.		✓			1.d, 2.b	✓
6.2.10	For rotorcraft: The performance and handling characteristics in hover in-ground effect and through translational lift should be characteristic of the simulated group of aircraft.	✓				1.d, 2.b	✓
6.3	Aerodynamic flight control handling response						
6.3.1	The flight control input should result in the aircraft response that replicates that of the simulated aircraft type and variant.			✓	1.b–1.f, 2.c–2.h	1.c, 1.j, 2.b	✓
6.3.2	The flight control input should result in the aircraft response that is characteristic of the simulated aircraft type.		✓		1.b–1.f, 2.c–2.h	1.c, 1.j, 2.b	✓
6.3.3	The flight control input should result in the aircraft response that is characteristic of the simulated group of aircraft.	✓			1.b–1.f, 2.c–2.e	1.c, 1.j, 2.b	✓
6.4	Flight dynamic model						
6.4.1	A flight dynamics model should replicate the simulated aircraft type and variant in-ground effect, including aerodynamic and lift/thrust system modelling that accounts for all combinations of drag and thrust and all combinations of airspeed and power normally encountered in flight, including the effect of change in aircraft attitude, aerodynamic forces and moments, forces and moments of rotors, aerodynamic interference, altitude, temperature, mass, centre-of-gravity location and configuration.			✓	1.b–1.f, 2.c–2.h	1.c, 1.j, 2.b,	✓
6.4.2	A flight dynamics model should be characteristic of the simulated aircraft type in-ground effect, including aerodynamic and lift/thrust system modelling that accounts for various combinations of drag and thrust normally encountered in flight, including the effect of change in aircraft attitude, aerodynamic forces and moments, forces and moments of rotors, altitude, temperature, mass, centre-of-gravity location and configuration.		✓		1.b–1.f, 2.c–2.h	1.c, 1.j, 2.b,	✓

6.4.3	A flight dynamics model should be characteristic of the simulated group of aircraft in-ground effect, including aerodynamic and lift/thrust system modelling that accounts for various combinations of drag and thrust normally encountered in flight, including the effect of change in aircraft attitude, altitude, temperature, mass, centre-of-gravity location and configuration.	✓					✓
6.5	Mass properties						
6.5.1	An aircraft-type-specific mass properties model, including mass or weight, centre of gravity and moments of inertia as functions of payload, fuel or any consumable loading, should be implemented.			✓			✓
6.5.2	A representative, aircraft-like, mass properties model, including mass, centre of gravity and moments of inertia as functions of payload, fuel or any consumable loading, should be implemented.		✓				✓
6.5.3	A generic, aircraft-like, mass properties model, including mass, centre of gravity and moments of inertia as functions of payload, fuel or any consumable loading, should be implemented.	✓					✓
6.5.4	The effects of pitch attitude and fuel slosh on the aircraft centre of gravity should be simulated.		✓	✓			✓
6.6	Atmospheric effects Defines the effect of atmospheric disturbances on performance and handling.						
6.6.1	The performance and handling characteristics of the simulated aircraft should be consistent with the atmospheric model (refer to CS FSTD.QB.113). The performance and handling characteristics should respond to atmospheric effects to the extent that these are covered by the atmospheric model, at its fidelity level.	✓	✓	✓			✓
6.6.2	For aeroplanes: The performance and handling characteristics should support the effects of specific atmospheric phenomena, such as crosswind, wind shear and microburst (as applicable), consistent with the fidelity level and content of the atmospheric model (refer to CS FSTD.QB.113). These effects should replicate the effect on the simulated aircraft type and variant.			✓	2.g		✓
6.6.3	For aeroplanes: The performance and handling characteristics should support the effects of atmospheric phenomena such as crosswind, wind shear and microburst (as applicable) in a way that is characteristic of the effect on the simulated aircraft type.		✓		2.g		✓

6.6.4	The performance and handling characteristics should be consistent and correlate with any additional atmospheric and weather models implemented (refer to CS FSTD.QB.113) and should reflect their effects on the simulated aircraft type and variant.			✓			✓
6.7	Icing effects Defines the effects of icing on the aircraft (e.g. the airframe, the lift/thrust system, aerodynamics, flight dynamics, surfaces and engines, as applicable). Refer to CS FSTD.QB.107.1.ICING.						
6.7.1	The flight model should include the effects of icing, if applicable, on the airframe, the lift/thrust system aerodynamics and the engine(s). Refer to CS FSTD.QB.107.1.ICING, paragraph (a).			✓			✓
6.7.2	For aeroplanes certified for flight into icing conditions: Icing models should simulate the aerodynamic degradation effects of ice accretion on the lift/thrust system lifting surfaces, including loss of efficiency, effect on power setting, change in pitching, rolling moments, decrease in control effectiveness and changes in control forces, in addition to any overall increase in drag or aircraft fuselage gross mass, with resulting effects on power. Refer to CS FSTD.QB.107.1.ICING, paragraph (a).			✓			✓
6.7.3	For rotorcraft certified for flight into icing conditions: The icing model should simulate the aerodynamic degradation effects of airframe and rotor icing and performance degradation due to icing. Refer to CS FSTD.QB.107.1.ICING, paragraph (b).			✓			✓
6.7.4	For aeroplanes and rotorcraft not certified for flight into icing conditions: Modelling should include the effects of icing, as appropriate, on the airframe, aerodynamics, weight, balance and engine(s). Icing models with performance degradation due to icing should be provided. Refer to CS FSTD.QB.107.1.ICING, paragraph (c).			✓			✓
6.7.5	For aeroplanes and rotorcraft certified for flight into icing conditions: Modelling that includes the effects of icing, as appropriate, on the airframe, aerodynamics, weight, balance and engine(s) should include icing models with the associated performance degradation. Refer to CS FSTD.QB.107.1.ICING, paragraph (d).	✓	✓				✓

6.7.6	For aeroplanes certified for flight into icing conditions: The engine icing model should simulate the effects of icing on the power plant, including performance and efficiency, and the effects of the build-up and balance of the engine (e.g. roughness and vibration). Refer to CS FSTD.QB.107.1.ICING, paragraph (a).			✓			✓
6.8	Engine performance						
6.8.1	The engine performance should replicate that of the simulated aircraft type and variant.			✓			✓
6.8.2	The engine performance should be characteristic of the simulated aircraft type.		✓				✓
6.8.3	The engine performance should be characteristic of the simulated group of aircraft.	✓					✓
6.9	Integration						
6.9.1	Performance and handling in-ground effect should correlate with the environmental conditions and aircraft systems simulation.		✓	✓			✓
6.9.2	Performance and handling in-ground effect should be affected by the environmental conditions and aircraft systems simulation, as applicable.	✓					✓
6.10	Performance and handling transition						
6.10.1	The transition between the flight regimes on ground and in-ground effect and between in-ground effect and out-of-ground effect should be smooth.	✓	✓	✓			✓

GM1 FSTD.QB.106 Flight simulation training device general requirements for performance and handling in-ground effect

Summaries of the fidelity levels for the performance and handling in-ground effect FSTD feature are presented in the table below. These summaries are broad overviews provided for guidance only. They do not encompass all aspects and requirements and are not intended to replace CS FSTD.QB.106.

6: Performance and handling in-ground effect
Specific: Aerodynamic, lift/thrust system and any other relevant modelling close to the ground or in-ground effect replicates the aircraft type and variant. This includes the effect of change in aircraft attitude, airspeed, thrust and other applicable variables.
Representative: Aerodynamic and lift/thrust system modelling close to the ground or in-ground effect is characteristic of the aircraft type. This includes the effect of change in aircraft attitude, airspeed, thrust and other applicable variables.
Generic: Aerodynamic and lift/thrust system modelling close to the ground or in-ground effect is characteristic of the simulated group of aircraft. This includes the effect of change in aircraft attitude, airspeed, thrust and other applicable variables.

PERFORMANCE AND HANDLING OUT-OF-GROUND EFFECT

CS FSTD.QB.107 Flight simulation training device general requirements for performance and handling out-of-ground effect

7: Performance and handling out-of-ground effect

Defines the mathematical models and associated data to be used to describe the aerodynamic, flight dynamic and lift/thrust system characteristics required to be modelled in the FSTD away from the ground and out-of-ground effect.

The general requirements for the performance and handling out-of-ground effect FSTD feature are presented in the table below.

7: Performance and handling out-of-ground effect							
Feature general requirements		Feature fidelity level			Verification and validation		
		G	R	S	QTG(A)	QTG(H)	F&S
7.1	Aerodynamic flight control handling response						
7.1.1	The flight control input should result in the aircraft response that replicates that of the simulated aircraft type and variant.			✓	1.b-1.d, 2.c-2.i	1.c-1.j, 2.b-2.d	✓
7.1.2	The flight control input should result in the aircraft response that is characteristic of the simulated aircraft type.		✓		1.b-1.d, 2.c-2.i	1.c-1.j, 2.b-2.d	✓
7.1.3	The flight control input should result in the aircraft response that is characteristic of the simulated group of aircraft.	✓			1.b-1.d, 2.c-2.i	1.c-1.j, 2.b-2.d	✓
7.2	Flight dynamic model						

7: Performance and handling out-of-ground effect								
Feature general requirements		Feature fidelity level			Verification and validation			
		G	R	S	QTG(A)	QTG(H)	F&S	
7.2.1	A flight dynamics model should replicate the simulated aircraft type and variant out-of-ground effect, including aerodynamic and lift/thrust system modelling that accounts for all combinations of drag and thrust of all combinations of airspeed and power normally encountered in flight, including the effect of change in aircraft attitude, aerodynamic forces and moments, rotor forces and moments, aerodynamic interference, altitude, temperature, mass, centre-of-gravity location and configuration.			✓	1.b–1.f, 2.c–2.h	1.e–1.h, 2.b–2.d	✓	
7.2.2	A flight dynamics model should be characteristic of the simulated aircraft type out-of-ground effect, including aerodynamic and lift/thrust system modelling that accounts for various combinations of drag and thrust normally encountered in flight, including the effect of change in aircraft attitude, aerodynamic forces and moments, rotor forces and moments, altitude, temperature, mass, centre-of-gravity location and configuration.		✓		1.b–1.f, 2.c–2.h	1.e–1.h, 2.b–2.d	✓	
7.2.3	A flight dynamics model should be characteristic of the simulated group of aircraft out-of-ground effect, including aerodynamic and lift/thrust system modelling that accounts for various combinations of drag and thrust normally encountered in flight, including the effect of change in aircraft attitude, altitude, temperature, mass, centre-of-gravity location and configuration.	✓			1.b–1.f, 2.c–2.e	1.e–1.h, 2.b–2.d	✓	
7.2.4	For aeroplanes: Aerodynamic modelling should include Mach effect, normal dynamic thrust effect on control surfaces, aeroelastic effect and representations of non-linearities due to side-slip.			✓	1.b–1.f, 2.c–2.h		✓	
7.2.5	For rotorcraft: The performance and handling characteristics in hover out-of-ground effect and through translational lift and transition to forward flight should replicate the simulated aircraft type and variant.			✓		1.d, 2.b–2.d	✓	
7.2.6	For rotorcraft: The performance and handling characteristics in hover out-of-ground effect and through translational lift and transition to forward flight should be characteristic of the simulated aircraft type.		✓			1.d, 2.b–2.d	✓	

7: Performance and handling out-of-ground effect							
Feature general requirements		Feature fidelity level			Verification and validation		
		G	R	S	QTG(A)	QTG(H)	F&S
7.2.7	For rotorcraft: The performance and handling characteristics in hover out-of-ground effect and through translational lift and transition to forward flight should be characteristic of the simulated group of aircraft.	✓				1.d, 2.b–2.d	✓
7.2.8	The flight dynamics model for performance and handling should replicate that of the simulated aircraft type and variant. This should include the effect of change in parameters for attitude, side-slip, thrust, drag, altitude, temperature, gross weight, moments of inertia, centre-of-gravity location and configuration.			✓	1.b–1.f, 2.c–2.h	1.e–1.h, 2.b–2.d	✓
7.2.9	The flight dynamics model for performance and handling should be characteristic of the simulated aircraft type. This should include the effect of change in parameters for attitude, side-slip, thrust, drag, altitude, temperature, gross weight, moments of inertia, centre-of-gravity location and configuration.		✓		1.b–1.f, 2.c–2.h	1.e–1.h, 2.b–2.d	✓
7.2.10	The flight dynamics model for performance and handling should be characteristic of the simulated group of aircraft. This should include the effect of change in parameters for attitude, side-slip, thrust, drag, altitude, temperature, gross weight, moments of inertia, centre-of-gravity location and configuration.	✓			1.b–1.f, 2.c–2.e	1.e–1.h, 2.b–2.d	✓
7.2.11	The aircraft dynamic handling and stability characteristics should replicate those of the simulated aircraft type and variant.			✓	2.c, 2.d	2.c, 2.d	✓
7.2.12	The aircraft dynamic handling and stability characteristics should be characteristic of the simulated aircraft type.		✓		2.c, 2.d	2.c, 2.d	✓
7.2.13	The aircraft dynamic handling and stability characteristics should be characteristic of the simulated group of aircraft.	✓			2.c, 2.d	2.c, 2.d	✓
7.2.14	The flight dynamics model for performance and handling during approach should replicate the simulated aircraft type and variant. This should include the transition into ground effect.			✓	2.e	1.j	✓

7: Performance and handling out-of-ground effect								
Feature general requirements		Feature fidelity level			Verification and validation			
		G	R	S	QTG(A)	QTG(H)	F&S	
7.2.15	The flight dynamics model for performance and handling during approach should be characteristic of the simulated aircraft type. This should include the transition into ground effect.		✓		2.e	1.j	✓	
7.2.16	The flight dynamics model for performance and handling during approach should be characteristic of the simulated group of aircraft. This should include the transition into ground effect.	✓			2.e	1.j	✓	
7.2.17	The performance and handling characteristics during climb, level flight and descent should replicate those of the simulated aircraft type and variant.			✓	1.c, 1.d, 2.c, 2.d	1.e, 1.f, 1.h, 1.j	✓	
7.2.18	The performance and handling characteristics during climb, level flight and descent should be characteristic of the simulated aircraft type.		✓		1.c, 1.d, 2.c, 2.d	1.e, 1.f, 1.h, 1.j	✓	
7.2.19	The performance and handling characteristics during climb, level flight and descent should be characteristic of the simulated group of aircraft.	✓			1.c, 1.d, 2.c, 2.d	1.e, 1.f, 1.h, 1.j	✓	
7.2.20	The Mach effects in the aerodynamic model should replicate those of the simulated aircraft type and variant, as applicable.			✓			✓	
7.2.21	The Mach effects in the aerodynamic model should be characteristic of the simulated aircraft type, as applicable.		✓				✓	
7.2.22	The Mach effects in the aerodynamic model should be characteristic of the simulated group of aircraft, as applicable.	✓					✓	
7.2.23	The aerodynamic model should replicate the aeroelastic effects.			✓			✓	
7.3	Special effects for rotorcraft Defines the aerodynamic special effects that are applicable to rotorcraft, such as vortex ring and retreating blade stall.							

7: Performance and handling out-of-ground effect							
Feature general requirements		Feature fidelity level			Verification and validation		
		G	R	S	QTG(A)	QTG(H)	F&S
7.3.1	The flight model should include (as applicable to aircraft type and variant) the effects of the: (1) aerodynamic interference effects between the rotor wake and fuselage; (2) influence of the rotor on control and stabilisation systems; (3) representations of non-linearities due to side-slip; (4) vortex ring (settling with power); (5) retreating blade stall; (6) aeroelastic effects; (7) Mach effects on rotors; (8) unanticipated yaw.			✓			✓
7.3.2	The flight model should include (as applicable to aircraft type) the effects of the: (1) aerodynamic interference effects between the rotor wake and fuselage; (2) influence of the rotor on control and stabilisation systems; (3) representations of non-linearities due to side-slip; (4) vortex ring (settling with power) and retreating blade stall; (5) unanticipated yaw.		✓				✓
7.3.3	The flight model should include (as applicable to the simulated group of aircraft) the effects of the: (1) aerodynamic interference effects between the rotor wake and fuselage; (2) influence of the rotor on control and stabilisation systems; (3) representations of non-linearities due to side-slip.	✓					✓
7.3.4	The performance and handling characteristics in hover should replicate those of the simulated aircraft type and variant.			✓		1.d	✓
7.3.5	The performance and handling characteristics in hover should be characteristic of the simulated aircraft type.		✓				✓

7: Performance and handling out-of-ground effect							
Feature general requirements		Feature fidelity level			Verification and validation		
		G	R	S	QTG(A)	QTG(H)	F&S
7.3.6	The performance and handling characteristics in hover should be characteristic of the simulated group of aircraft.	✓					✓
7.4	Special effects for aeroplanes Applies only to aeroplane FSTDs that are to be qualified for UPRT.						
7.4.1	The aerodynamic model has to incorporate data representing the aeroplane's characteristics covering an angle-of-attack and side-slip range to support the training tasks.			✓			✓
7.4.2	The aerodynamic modelling has to support UPRT tasks in the following upset recovery manoeuvres: (1) a nose-high, wings-level aeroplane upset; (2) a nose-low aeroplane upset; and (3) a high-bank-angle aeroplane upset. Other upset prevention and recovery scenarios may be developed by the organisation requesting the qualification of an FSTD. All the upset prevention and recovery manoeuvres and scenarios should be selectable and controllable by the instructor. Refer to CS FSTD.QB.115, section 'UPRT'.			✓			✓

7: Performance and handling out-of-ground effect							
Feature general requirements		Feature fidelity level			Verification and validation		
		G	R	S	QTG(A)	QTG(H)	F&S
7.4.3	Applicable only to aeroplane FSTDs that are to be qualified for full-stall training tasks: The aerodynamic modelling has to support stall-recovery training tasks in the following flight conditions: (1) stall entry at wings level (1 g); (2) stall entry into turning flight of at least 25° bank angle (accelerated stall); (3) stall entry into a power-on condition (required only for propeller-driven aeroplanes); and (4) aeroplane configurations of second-segment climb, high-altitude cruise (near performance-limited condition) and approach or landing. The SOJ should describe the aerodynamic modelling methods, validation and check of the stall characteristics of the FSTD and should include verification that the FSTD has been evaluated by a subject matter expert (SME) pilot acceptable to the competent authority. Refer to CS FSTD.QB.107.1.STALL and CS FSTD.QB.107.2.STALL. Refer to CS FSTD.GEN.015 for a definition of ‘near performance-limited condition’.			✓	2.c.8.a		✓
7.4.4	Applicable to those FSTDs that are not to be qualified for full-stall training tasks. The aerodynamic modelling should support stall event training tasks in the following flight conditions: (1) approach to stall during second-segment climb; (2) approach to stall at high-altitude cruise (near performance-limited condition); and (3) approach to stall during either approach or landing.			✓	2.c.8.b		✓
7.5	Mass properties						
7.5.1	An aircraft-type-specific mass properties model, including mass or weight, centre of gravity and moments of inertia as functions of payload, fuel or any consumable loading, should be implemented. The SOJ should include a range of tabulated target values to enable a demonstration of the mass properties model to be conducted from the instructor’s station.			✓			✓

7: Performance and handling out-of-ground effect							
Feature general requirements		Feature fidelity level			Verification and validation		
		G	R	S	QTG(A)	QTG(H)	F&S
7.5.2	A representative, aircraft-like, mass properties model including mass, centre of gravity and moments of inertia as functions of payload, fuel or any consumable loading, should be implemented.		✓				✓
7.5.3	A generic, aircraft-like, mass properties model, including mass, centre of gravity and moments of inertia as functions of payload, fuel or any consumable loading, should be implemented.	✓					✓
7.5.4	The effects of pitch attitude and fuel slosh on the aircraft centre of gravity should be simulated.		✓	✓			✓
7.6	Atmospheric effects Defines the effect of atmospheric disturbances on performance and handling.						
7.6.1	The performance and handling characteristics of the simulated aircraft should be consistent with the atmospheric model (refer to CS FSTD.QB.113). The performance and handling characteristics should respond to atmospheric effects to the extent that these are covered by the atmospheric model, at its fidelity level.			✓			✓
7.6.2	For aeroplanes: The performance and handling characteristics should support the effects of specific atmospheric phenomena, such as crosswind, wind shear and microburst (as applicable), consistent with the fidelity level and content of the atmospheric model (refer to CS FSTD.QB.113). These effects should replicate the effect on the simulated aircraft type and variant.			✓	1.b, 2.g		✓
7.6.3	For aeroplanes: The performance and handling characteristics should support the effects of atmospheric phenomena such as crosswind, wind shear and microburst (as applicable) in a way that is characteristic of the effect on the simulated aircraft type.		✓		1.b		✓

7: Performance and handling out-of-ground effect							
Feature general requirements		Feature fidelity level			Verification and validation		
		G	R	S	QTG(A)	QTG(H)	F&S
7.6.4	The performance and handling characteristics should be consistent and correlate with any additional atmospheric and weather models implemented (refer to CS FSTD.QB.113) and should reflect their effects on the simulated aircraft type and variant.			✓			✓
7.7	Icing effects Defines the effects of icing on the aircraft (e.g. the airframe, the lift/thrust system, aerodynamics, flight dynamics, surfaces and engines, as applicable). Refer to CS FSTD.QB.107.1.ICING.						
7.7.1	The flight model should include the effects of icing, if applicable, on the airframe, the lift/thrust system aerodynamics and the engine(s). Refer to CS FSTD.QB.107.1.ICING, paragraph (a).			✓			✓
7.7.2	For aeroplanes certified for flight into icing conditions: Icing models should simulate the aerodynamic degradation effects of ice accretion on the lift/thrust system's lifting surfaces, including loss of efficiency, effect on power setting, change in pitching, rolling moments, decrease in control effectiveness and changes in control forces, in addition to any overall increase in drag or aircraft fuselage gross mass, with resulting effects on power. Aeroplane OEM data or other acceptable analytical methods must be used to represent icing effects. Acceptable methods may include wind tunnel analysis or engineering analysis of the aerodynamic effects of icing on the aeroplane's lifting surfaces coupled with tuning and supplemental subjective assessment by a SME pilot knowledgeable in the handling characteristics of the aeroplane affected by ice. The SOJ should describe the effects that provide training in the specific skills for the recognition of icing phenomena and the execution of recovery. The SOJ should detail the source data and any analytical methods used to develop ice accretion models, including verification that these effects have been tested. Refer to CS FSTD.QB.107.1.ICING, paragraph (a).			✓	2.i		✓

7: Performance and handling out-of-ground effect							
Feature general requirements		Feature fidelity level			Verification and validation		
		G	R	S	QTG(A)	QTG(H)	F&S
7.7.3	For rotorcraft certified for flight into icing conditions: The icing model should simulate the aerodynamic degradation effects of airframe and rotor icing and performance degradation due to icing. Refer to CS FSTD.QB.107.1.ICING, paragraph (b).			✓			✓
7.7.4	For aeroplanes and rotorcraft not certified for flight into icing conditions: Modelling should include the effects of icing, as appropriate, on the airframe, aerodynamics, weight, balance and engine(s). Icing models with performance degradation due to icing should be provided. Refer to CS FSTD.QB.107.1.ICING, paragraph (c).			✓			✓
7.7.5	For aeroplanes and rotorcraft certified for flight into icing conditions: Modelling that includes the effects of icing, as appropriate, on the airframe, aerodynamics, weight, balance and engine(s) should include icing models with the associated performance degradation. Refer to CS FSTD.QB.107.1.ICING, paragraph (d).	✓	✓				✓
7.7.6	For aeroplanes certified for flight into icing conditions: The engine icing model should simulate the effects of icing on the power plant, including performance and efficiency, and the effects of the build-up and balance of the engine (e.g. roughness and vibration). Refer to CS FSTD.QB.107.1.ICING, paragraph (a).			✓	2.i		✓
7.8	Engine performance						
7.8.1	The engine performance should replicate that of the simulated aircraft type and variant.			✓	1.f	1.a	✓
7.8.2	The engine performance should be characteristic of the simulated aircraft type.		✓		1.f	1.a	✓
7.8.3	The engine performance should be characteristic of the simulated group of aircraft.	✓			1.f	1.a	✓
7.9	Integration						

7: Performance and handling out-of-ground effect							
Feature general requirements		Feature fidelity level			Verification and validation		
		G	R	S	QTG(A)	QTG(H)	F&S
7.9.1	Performance and handling out-of-ground effect should correlate with the environmental conditions and aircraft systems simulation.		✓	✓			✓
7.9.2	Performance and handling out-of-ground effect should be affected by the environmental conditions and aircraft systems simulation, as applicable.	✓					✓
7.10	Performance and handling transition						
7.10.1	The transition between the flight regimes in-ground effect and out-of-ground effect should be smooth.	✓	✓	✓			✓

GM1 FSTD.QB.107 Flight simulation training device general requirements for performance and handling out-of-ground effect

Summaries of the fidelity levels for the performance and handling out-of-ground effect FSTD feature are presented in the table below. These summaries are broad overviews provided for guidance only. They do not encompass all aspects and requirements and are not intended to replace CS FSTD.QB.107.

7: Performance and handling out-of-ground effect
Specific: Aerodynamic, lift/thrust system and any other relevant modelling out-of-ground effect replicates the aircraft type and variant, accounting for all combinations of drag and thrust (or airspeed and power) normally encountered in flight, including effects of changes in aircraft attitude, aerodynamic forces and moments, rotor forces and moments, aerodynamic interference, altitude, temperature, mass, centre-of-gravity location and configuration.
Representative: Aerodynamic and lift/thrust system modelling is characteristic of the simulated aircraft type, accounting for various combinations of drag and thrust normally encountered in flight, including effects of changes in aircraft attitude, aerodynamic forces and moments, rotor forces and moments, altitude, temperature, mass, centre-of-gravity location and configuration.
Generic: Aerodynamic and lift/thrust system modelling is characteristic of a group of aircraft, accounting for various combinations of drag and thrust normally encountered in flight, including effects of changes in aircraft attitude, altitude, temperature, mass, centre-of-gravity location and configuration.

GM2 FSTD.QB.107 Flight simulation training device general requirements for performance and handling out-of-ground effect

GENERIC FIDELITY LEVEL OF PERFORMANCE AND HANDLING OUT-OF-GROUND EFFECT

This GM gives guidance on the generic fidelity level of performance and handling out-of-ground effect FSTD feature.

As Subpart C states, data for generic performance and handling may also originate from the real aircraft CSs (e.g. CS-23, CS-25, CS-27 or CS-29). The requirements in the above-mentioned CSs may require, for example, that the aircraft have positive damping on dynamic stability. For generic performance and handling, this does not mean that the simulation may be too stable. Generic performance and handling still requires that the flight characteristics are similar to those of a real aircraft that exists within the simulated group of aircraft. This should be demonstrated in the ER. In summary, the real aircraft CSs may not be useful alone but should be considered on a case-by-case basis and may need to be complemented by other data. Refer to Subpart C for more information.

As required by CS FSTD.QB.107, the simulation should include the effects of change in various parameters. For example, it is expected that the stability of any aircraft should be lower at high altitudes than at the sea level. This is caused by the air density and reduced aerodynamic damping at high altitudes. It is important to note that the aerodynamic effects are more complex for aircraft that

fly at higher altitudes or at higher Mach numbers. In addition, the effects are more complex for aircraft that have couplings, such as helicopters or a swept-wing aircraft. Furthermore, the effects are influenced by the aircraft systems. Any stabilisation system (e.g. stability augmentation system, auto-trim, yaw damper) or CCA would have an important effect on performance and handling, associated phenomena and realism. For a positive transfer of training, these effects and phenomena should be simulated and realistic, even at the generic fidelity level. For the generic Level, the ER should demonstrate that there is an aircraft within the simulated group with similar characteristics to that of the FSTD.

It should be noted that the simulation of complex aircraft is complicated even at the generic fidelity level.

GM3 FSTD.QB.107 Flight simulation training device general requirements for performance and handling out-of-ground effect

ENGINE OR POWER PLANT FAILURE

CS FSTD.QB.105, CS FSTD.QB.106 and CS FSTD.QB.107 outline the requirements for performance and handling, including which parameter changes must be accounted for. Thrust and drag are among those parameters. The requirements address any changes in engine or power plant parameters or failure of the engine or power plant. This means that the failure effects (e.g. loss of thrust, increase in drag) must be accounted for and must result in consequent changes in other parameters (e.g. dynamics, changes in performance, attitude, side-slip, airspeed, necessary flight control inputs and trimming).

CS FSTD.QB.107.1.ICING Flight simulation training device requirements for engine and airframe icing

This CS defines the requirements for engine and airframe icing. This CS applies to the following FSTD features: performance and handling on ground, performance and handling in-ground effect, and performance and handling out-of-ground effect.

- (a) For the specific fidelity level for aeroplanes certified for flight into icing conditions, the following elements should be included in the development of the ice accretion models for use in FSTD training.
 - (1) Icing models should enable training on the specific skills required for the recognition of ice accumulation and for generating the required response.
 - (2) Icing models should contain aircraft-specific recognition cues as determined through data supplied by an aircraft OEM or gathered through other suitable analytical methods.
 - (3) At least one icing model should be objectively tested (refer to CS FSTD.QTG.A.240) to demonstrate that it has been implemented correctly and that it generates the correct cues necessary for training.
 - (4) Icing models should simulate engine-specific icing effects, including changes in engine performance, potential engine surges, flameouts, vibration and other adverse effects caused by ice formation on engine components.

-
- (5) Icing model should support recognition cues specific to engine icing, such as unusual fluctuations in engine parameters, audible indications and cockpit alerts, as applicable to the simulated aircraft type and variant.
 - (6) The SOJ required in CS FSTD.QB.107, section 'Icing effects', for aeroplanes should contain the following information to support FSTD qualification of aircraft-specific icing models.
 - (i) There should be a description of expected aircraft-specific recognition cues and degradation effects due to a typical in-flight icing encounter. Typical cues may include loss of lift, decrease in stall angle of attack, changes in pitching moment, a decrease in control effectiveness and changes in control forces, in addition to any overall increase in drag. This description should be based on relevant data sources, such as aeroplane OEM-supplied data, accident/incident data or data from other acceptable sources. If a particular airframe has demonstrated vulnerabilities to a specific type of ice accretion (due to accident/incident history) that requires specific training (e.g. supercooled large-droplet icing or tailplane icing), ice accretion models should be developed that address those training provisions.
 - (ii) There should be a description of the data sources used to develop the ice accretion models. Acceptable data sources may be but are not limited to flight tests, aeroplane certifications, aircraft OEM engineering simulations or other analytical methods based on established engineering principles.
 - (iii) There should be a description of the expected engine recognition cues and degradation effects due to the build-up of ice (on ground and/or in flight). Typical cues may include fluctuation in engine parameters (e.g. compressor speed, exhaust gas temperature (EGT), fuel flow, vibration, thrust output). If an aircraft requires specific training (including procedures) for engine ice accretion management (e.g. engine power adjustments, engine anti-icing system use, recovery actions following icing-induced engine anomalies), the engine icing models should be developed to address those training provisions.
 - (b) For the specific fidelity level for rotorcraft certified for flight into icing conditions:
 - (1) icing models should enable training on the specific skills required for the recognition of ice accumulation and for generating the required response;
 - (2) icing models should contain aircraft-specific recognition cues (e.g. vibration);
 - (3) aircraft systems (e.g. the anti-icing/de-icing systems) should respond properly and have realistic effects.
 - (c) For the specific fidelity level for aeroplanes and rotorcraft not certified for flight into icing conditions, the following should apply.
 - (1) An icing model that is realistic for the aircraft type and supports recognition cues and degradation effects is required.

- (2) Icing models should simulate the effects of ice accretion on lifting surfaces and engine(s), including aerodynamic degradation and engine performance impacts, with cues sufficient for recognition. These cues should reflect the effects of ice on aircraft performance and handling characteristics, such as (as applicable to the simulated aircraft) a loss of lift, a decrease in stall angle of attack, decreases in control effectiveness and forces and an increase in drag. Aeroplane systems, including stall protection and auto-flight systems, should respond to ice accretion in a manner consistent with the simulated aircraft's design and behaviour.
- (d) For the generic and representative fidelity levels for aeroplanes and rotorcraft certified for flight into icing conditions:
 - (1) an icing model is required that is characteristic of the simulated aircraft type or group and supports recognition cues and degradation effects;
 - (2) icing models should simulate the effects of ice accretion on lifting surfaces and engine(s), including (as applicable) loss of lift, reduced stall angle of attack, decreased control effectiveness, increased drag and any engine performance impacts;
 - (3) aircraft systems (e.g. the anti-icing/de-icing systems) should respond properly and have realistic effects;
 - (4) the SOJ should include a description of the simulated recognition cues and degradation effects due to a typical in-flight icing encounter.
- (e) For the generic and representative fidelity levels for aircraft not certified for flight into icing conditions, if the FSTD includes icing simulation, the requirements outlined in paragraph (d) above apply.

GM1 FSTD.QB.107.1.ICING Flight simulation training device requirements for engine and airframe icing

This GM provides additional guidance on the requirements for icing modelling in FSTDs.

- (a) For the specific fidelity level for aeroplanes certified for flight into icing conditions, the following apply.
 - (1) The icing models should ensure that the implemented models accurately reflect the real-world effects of icing on engine and aircraft performance and handling. CS FSTD.QB.107.1.ICING specifies considerations for the SOJ, data sourcing and implementation of icing models. The requirements address the effects of ice accumulation on various engine and airframe components, as applicable to the simulated aircraft type and variant.
 - (2) To align with training objectives, the icing models should:
 - (i) enable recognition and interpretation of icing cues from aircraft systems and handling characteristics;
 - (ii) represent aircraft-specific vulnerabilities and limitations related to icing;

- (iii) support training scenarios involving the use of anti-icing and de-icing systems;
 - (iv) allow for upset prevention and recovery exercises in icing conditions, as applicable to training objectives.
 - (3) The icing models should incorporate the following aspects, as applicable to the simulated aircraft type and variant:
 - (i) Icing models should simulate the effects of ice accumulation dynamically, taking into account variations in temperature, humidity, and airspeed over time.
 - (ii) Recognition cues should include changes in control response, increased drag, altered stall and buffet characteristics, and additional aerodynamic penalties.
 - (iii) Ice accumulation should be considered for wings, control surfaces, stabilisers, fuselage and propellers affecting lift, drag and overall controllability.
 - (iv) Effects on aircraft systems, such as pitot-static instruments and control mechanisms, should be incorporated to ensure accurate simulation of degraded performance.
 - (4) Engine icing considerations are as follows.
 - (i) Engine icing should be incorporated distinctly from airframe icing effects.
 - (ii) Engine icing models should capture the effects of ice accumulation on the following: compressor efficiency, including the risk of surge and potential flameout; thrust loss, particularly in engines without active anti-icing systems; and ice accretion on engine inlets, fan blades and internal components, leading to degraded engine performance.
 - (iii) Recognition cues should include flight deck indication changes, vibration, abnormal engine parameter fluctuations and power losses, helping trainees identify and respond to icing-related engine issues.
- (b) For the specific fidelity level for rotorcraft certified for flight into icing conditions, the following apply.
 - (1) To align with training objectives, the icing models should comply with paragraphs (a)(2)(i) through (a)(2)(iii) above.
 - (2) The SOJ should explain how the icing models affect the aerodynamic model. It should also justify their appropriateness for the simulated aircraft type and variant.
 - (3) The icing models should incorporate the following aspects, as applicable to the simulated aircraft type and variant:
 - (i) Icing effects on rotor blades should include changes in lift, thrust, and control response, as well as dynamic impacts such as ice shedding, balance issues and increased vibrations.

- (ii) The effects of ice accumulation on tail rotors and anti-torque control should be modelled to reflect handling difficulties in icing conditions.
- (c) For the specific fidelity level for aeroplanes and rotorcraft not certified for flight into icing conditions, the following apply.
 - (1) The FSTD should simulate icing accumulation and associated effects even for aircraft types not certified for operation in icing conditions. The effects of icing should align with expectations for the aircraft to ensure realistic effects that support the recognition of ice accumulation.
 - (2) The SOJ should explain how the icing models affect the aerodynamic and engine models and justify their appropriateness for the simulated aircraft type and variant.
- (d) For the generic and representative fidelity levels for aeroplanes and rotorcraft certified for flight into icing conditions, the following apply.
 - (1) To align with training objectives, the icing models should:
 - (i) enable the recognition and interpretation of icing cues from aircraft systems and handling characteristics;
 - (ii) represent icing-related vulnerabilities and limitations;
 - (iii) support training scenarios involving the use of anti-icing and de-icing systems.
 - (2) The SOJ should explain how the icing models affect the aerodynamic and engine models. It should also justify their appropriateness for the simulated aircraft.
 - (3) In terms of typical cues:
 - (i) typical airframe icing recognition cues and degradation effects may include a loss of lift, a decrease in stall angle of attack, changes in pitching moment, a decrease in control effectiveness and changes in control forces, in addition to any overall increase in drag;
 - (ii) typical engine icing recognition cues may include flight deck indication changes, vibration, abnormal engine parameter fluctuations and power loss.

CS FSTD.QB.107.1.STALL Flight simulation training device requirements for stall modelling

This CS is applicable only to aeroplane FSTDs that are to be qualified for full-stall training tasks. This CS specifies standards related to stall modelling.

- (a) For the angle-of-attack and side-slip range of the aerodynamic model:
 - (1) for continuity purposes, the model should remain useable beyond the FSTD training envelope to the extent that it allows completion of the recovery training; and
 - (2) if known limitations exist in the aerodynamic model for particular stall event manoeuvres (e.g. aeroplane configuration, approach-to-stall entry methods and

limited range for continuity of the modelling), these limitations should be declared in the required SOJ and the ESL.

- (b) Characteristics of stall modelling include:
- (1) for aerodynamic stall modelling, degradation of the static/dynamic lateral–directional stability;
 - (2) degradation in control response (pitch, roll, yaw);
 - (3) uncommanded roll response or roll-off requiring significant control deflection to counter;
 - (4) apparent randomness or non-repeatability;
 - (5) changes in pitch stability;
 - (6) Mach effects;
 - (7) stall buffet, as appropriate to the aeroplane type;
- (c) The model should be capable of capturing the variations seen in the stall characteristics of the aeroplane (e.g. the presence or absence of a pitch break, deterrent buffet or other indications of a stall if present on the aeroplane).
- (d) If known limitations exist in the aerodynamic model for particular stall manoeuvres (e.g. aeroplane configuration and stall entry methods), these limitations should be declared in the required SOJ and the ESL.
- (e) Specific guidance should be available to the instructor, clearly communicating the flight configurations and stall manoeuvres that have been evaluated in the FSTD for use in training.
- (f) FSTDs that are to be qualified for full-stall training tasks should also meet the IOS requirements for UPRT tasks as described in CS FSTD.QB.115.

CS FSTD.QB.107.2.STALL Flight simulation training device requirements for stall manoeuvres

This CS is applicable only to aeroplane FSTDs that are to be qualified for full-stall training tasks.

This CS specifies the requirements for aeroplane FSTDs that are used to satisfy training requirements for stall manoeuvres conducted at angles of attack beyond the first indication of the stall, up to stall break and recovery.

(a) **General provisions**

The provisions for high-angle-of-attack modelling should be applied to evaluate the recognition cues and performance and handling qualities of a developing stall through the stall identification angle of attack and stall recovery. Strict time-history-based evaluations, involving comparison against flight test data, may not adequately validate the aerodynamic model in an unsteady and potentially unstable flight regime, such as stalled flight. As a result, the objective tests of Subpart D do not contain strict tolerances for any parameter at angles of attack beyond the stall identification angle of attack. In lieu of mandating such objective

tolerances, an SOJ should define the source data and methods used to develop the aerodynamic stall model.

(b) Fidelity provisions

The provisions for the evaluation of full-stall training manoeuvres should include the following elements.

- (1) Aeroplane-type-specific recognition cues of the first indication of the stall (e.g. the stall warning system or aerodynamic stall buffet) should be included.
- (2) Aeroplane-type-specific recognition cues of an impending aerodynamic stall should be included.
- (3) Recognition cues and handling qualities from stall break through recovery that are sufficiently 'characteristic' of the aeroplane being simulated to allow successful completion of the stall-recovery training tasks should be included.

For the purposes of stall manoeuvre evaluation, the term 'characteristic' is defined as a level of fidelity that is type-specific for the simulated aeroplane to the extent that the training objectives can be satisfactorily accomplished. Therefore, the term 'characteristic' in this CS is specifically limited to the elements of the aerodynamic model in the post-stall region.

(c) Statement of Justification (aerodynamic model)

At a minimum, the following should be addressed in the SOJ.

(1) Source data and modelling methods

The SOJ should identify the sources of data used to develop the aerodynamic model. These data sources may be the aeroplane OEM, the original FSTD manufacturer / data provider or other data providers acceptable to the competent authority. Of particular interest is a mapping of test points in the form of an alpha/beta envelope plot for a minimum of flaps-up and flaps-down aeroplane configurations. For the flight test data, a list of the types of manoeuvres used to define the aerodynamic model for angle-of-attack ranges greater than the first indication of stall should be provided per flap setting. Flight test reports, if available, describing the stall characteristics of the aeroplane type being modelled, issued by the OEM or flight test pilot, can be referred to. In cases in which it is impractical to develop and validate a stall model with flight test data (e.g. due to safety concerns involving the collection of flight test data past a certain angle of attack), the data provider is expected to make a reasonable attempt to develop a stall model through the required angle-of-attack range using analytical methods and empirical data (e.g. wind tunnel data).

(2) Validity range

The organisation requesting the qualification of an FSTD should declare the range of angle of attack and side-slip in which the aerodynamic model remains valid for training. Satisfactory aerodynamic model fidelity should be shown through stall-recovery training tasks. For the purposes of determining the validity range, the stall

identification angle of attack is defined as the angle of attack at which the pilot is given a clear and distinctive indication to cease any further increase in the angle of attack if one or more effects occur out of:

- (i) no further increase in pitch when the pitch control is held at the full aft stop for two seconds, leading to an inability to arrest the descent rate;
- (ii) an uncommanded nose-down pitch that cannot be readily arrested, which may be accompanied by an uncommanded rolling motion;
- (iii) buffeting of a magnitude and severity that is a strong and effective deterrent to a further increase in the angle of attack;
- (iv) the activation of a stick pusher.

For the validity range, the modelling continuity should allow for an angle-of-attack range that is adequate to allow for the completion of stall recovery; for pusher-equipped aeroplanes, this should be adequate to capture any inappropriate action during the recovery procedure.

For aeroplanes equipped with a stall envelope protection system, the model should allow training with the protection systems disabled or otherwise degraded (e.g. a degraded flight control mode as a result of a pitot–static system failure).

(3) Model characteristics

Within the declared model validity range, the SOJ should address and the aerodynamic model should incorporate certain stall characteristics, as applicable to the aeroplane type, namely:

- (i) degradation of the static/dynamic lateral–directional stability;
- (ii) degradation in control response (pitch, roll, yaw);
- (iii) uncommanded roll acceleration or roll-off requiring significant control deflection to counter;
- (iv) apparent randomness or non-repeatability;
- (v) changes in pitch stability;
- (vi) stall hysteresis;
- (vii) Mach effects;
- (viii) stall buffet; and
- (ix) angle-of-attack rate effects.

An overview of the methodology used to address these features should be provided.

(d) Statement of justification, subject matter expert (SME) pilot’s evaluation

The FSTD operator should provide an SOJ confirming that the simulation stall model has been subjectively evaluated by an SME pilot knowledgeable of the aeroplane’s stall characteristics (refer to (c)(1) above).

The FSTD operator is also required to provide an SOJ to state that the simulation stall model, as defined above, has been implemented and verify that the aerodynamic stall training tasks can be accomplished on the FSTD.

The purpose is to ensure that the stall model has been sufficiently evaluated using those general aeroplane configurations and stall entry methods that are likely to be conducted in training.

The SOJ will only be required at the time the FSTD is initially qualified for stall training tasks, as long as the FSTD's stall model remains unmodified (i.e. what was originally evaluated and qualified). If an FSTD shares common aerodynamic and flight control models with those of an engineering or development simulator, the competent authority will accept an SOJ from the aeroplane manufacturer or data provider confirming that the stall characteristics have been subjectively assessed by an SME pilot on the engineering/development simulator (refer to CS FSTD.ENG.040).

In this context, a 'development simulator' means a data provider's simulator that serves as a platform for the development of alternative engineering simulation data and models. This could be based on a specific aeroplane or a representation of its type. It would typically operate in real time and, if necessary, could be flown by a pilot to subjectively evaluate the simulation.

(e) SME pilot requirements

In order to qualify as an acceptable SME to evaluate the stall model characteristics, the SME should meet the following criteria.

- (1) Has held or currently holds a type rating / qualification in the aeroplane being simulated.
- (2) Has direct experience in conducting stall manoeuvres in an aeroplane that shares the same type rating as the make, model and series of the simulated aeroplane. This stall experience should include hands-on manipulation of the controls at angles of attack sufficient to identify the stall (e.g. deterrent buffet, stick pusher activation) through recovery to stable flight.
- (3) If the SME's stall experience is in an aeroplane of a different make, model and series within the same type rating, differences in aeroplane-specific stall recognition cues and handling characteristics should be addressed using available documentation. This documentation may include aeroplane operating manuals, aeroplane manufacturer flight test reports or other documentation that describes the stall characteristics of the aeroplane.
- (4) Be familiar with the intended stall training manoeuvres to be conducted in the FSTD (e.g. general aeroplane configurations, stall entry methods) and the cues necessary to accomplish the required training objectives.

(f) **Deviation from the SME pilot requirements**

The organisation requesting the qualification of an FSTD may submit a request to the competent authority for approval of a deviation from the SME pilot's experience provisions under this paragraph. This request for deviation should include:

- (1) an assessment of pilot availability, demonstrating that an SME pilot, meeting the experience described in this CS, is not available;
- (2) alternative methods to subjectively evaluate the FSTD's capability to provide the stall recognition cues and handling characteristics needed to accomplish the training objectives.

(g) **Statement of justification (SOJ) — subjective tests**

The necessity of subjective tests arises from the need to confirm that the simulation model has been integrated correctly and performs as declared under paragraph (d) above. It is vital to examine, for example, whether the simulation validity range provides modelling continuity that is adequate to allow for the completion of stall recovery.

Considerations for aeroplane certification flight tests are as follows.

In aeroplane certification flight tests, there is no requirement to go beyond the maximum coefficient of lift (CL max), and the aeroplane is not to be held indefinitely in a full-stall condition, so this provision should be applied in the same way during the simulator's subjective evaluation.

The subjective tests of the simulation model should assess modelling continuity when slightly increasing the angle of attack beyond the validity range CL max defined in subparagraph (c)(2) of this CS.

The increase in the angle of attack beyond the validity range CL max should be limited to a value no greater than the maximum angle achieved two seconds after stall recognition, which is sufficient to allow for a proper recovery manoeuvre.

Stall recognition is defined by the occurrence of one or more of the following effects, as applicable to the simulated aeroplane type:

- (1) no further increase in pitch occurs when the pitch control is held at the full aft stop for two seconds, leading to an inability to arrest the descent rate;
- (2) an uncommanded nose-down pitch that cannot be readily arrested, which may be accompanied by an uncommanded rolling motion;
- (3) buffeting of a magnitude and severity that is a strong and effective deterrent to a further increase in the angle of attack;
- (4) activation of a stick pusher.

If known limitations exist in the aerodynamic model for particular stall event manoeuvres (e.g. aeroplane configuration, approach-to-stall entry methods and limited range for continuity of the modelling), these limitations should be declared in the required SOJ.

SOUND CUEING

CS FSTD.QB.108 Flight simulation training device general requirements for sound cueing

8: Sound cueing
Defines the sound cues to be modelled. The sound cues are those related to sounds generated both externally and internally in relation to the flight deck. These include but are not limited to the sounds of engines, aerodynamics, propulsion, rotors, operating site elements (e.g. runway rumble), weather effects and aircraft components audible to the flight crew, such as pumps, motors, fans, vents, airflow and wipers. Note: Sounds generated by the aircraft systems for pilot awareness (e.g. communication systems, system and avionics audio tones, messages, cautions, warnings or any sound for indication purposes) are covered separately in the aircraft systems FSTD feature in CS FSTD.QB.104.

The general requirements for the sound cueing FSTD feature are presented in the table below.

8: Sound cueing						
Feature general requirements				Feature fidelity level		Verification and validation
		G	R	S	QTG	F&S
8.1	Sound system					
8.1.1	Significant sounds perceptible to the flight crew during flight operations should replicate those of the simulated aircraft type and variant.			✓		✓
8.1.2	Significant sounds perceptible to the flight crew during flight operations should be characteristic of the simulated aircraft type.		✓			✓

8: Sound cueing						
Feature general requirements		Feature fidelity level			Verification and validation	
		G	R	S	QTG	F&S
8.1.3	Significant sounds perceptible to the flight crew during flight operations should be characteristic of the simulated group of aircraft, including basic engine, propeller, rotor, airframe and aircraft system sounds.	✓				✓
8.2	Sound contents and qualities					
8.2.1	Sound frequencies and amplitudes should replicate those of the simulated aircraft type and variant. Sounds should include engine, propeller, transmission, rotor, airframe, aircraft system and environmental sounds, as well as those generated by pilot or instructor actions. Sounds should be a function of flight conditions – for example, variable IAS, aircraft configuration, variable engine noise, rotor speed and blade slap, as appropriate to the aircraft. Sounds should correlate with aircraft systems, the simulation of flight control systems and the motion/vibration platform (if installed).			✓	5.a, 5.b, 5.c	✓
8.2.2	Sounds should include engine, propeller, transmission, rotor, airframe, aircraft system and environmental sounds, as well as those generated by pilot or instructor actions. Sounds should be a function of flight conditions – for example, variable IAS, aircraft configuration, variable engine noise, rotor speed and blade slap, as appropriate to the aircraft. Sounds should correlate with aircraft systems, the simulation of flight control systems and the motion/vibration platform (if installed).		✓		5.a, 5.b, 5.c	✓
8.2.3	Sounds should include engine, rotor and aircraft system sounds, as well as those generated by pilot or instructor actions. Sounds should be a function of flight conditions – for example, variable IAS, aircraft configuration, variable engine noise, rotor speed and blade slap, as appropriate to the aircraft.	✓			5.a, 5.b, 5.c	✓
8.2.4	The background noise level of the FSTD should be below the set tolerances.	✓	✓	✓	5.d	

8: Sound cueing							
Feature general requirements		Feature fidelity level			Verification and validation		
		G	R	S	QTG	F&S	
8.2.5	The frequency response of the speakers should be within the set tolerances.		✓	✓	5.e		
8.3	Crash and strike sounds						
8.3.1	Sounds should include a sound effect for a crash when the simulated aircraft exceeds limitations, such as but not limited to structural or flight envelope exceedances.	✓	✓	✓			✓
8.3.2	Sounds should include a sound effect for a crash when impacting ground structures or another aircraft.	✓	✓	✓			✓
8.3.3	Sounds should include sound effects for aircraft striking the ground or objects (e.g. tail strike, pod strike, wing tip strike, rotor strike, wire cutter, slung load).			✓			✓
8.4	Environmental sounds						
8.4.1	Significant environmental sounds should be consistent with the simulated weather.		✓	✓			✓
8.4.2	Environmental sounds are not required. If present, they should be consistent with the simulated weather.	✓					
8.5	Sound volume						
8.5.1	If the IOS has a volume control, the following applies. Full volume should correspond to actual volume levels in the validation data meeting all the qualification requirements. If full volume is not selected, the IOS should have an indication that is visible at all times.			✓			✓
8.5.2	If the IOS has a volume control, the following applies. Full volume should correspond to the volume level agreed as initially qualified. If full volume is not selected, the IOS should have an indication of the volume level that is visible at all times.	✓	✓				✓

8: Sound cueing							
Feature general requirements				Feature fidelity level			Verification and validation
		G	R	S	QTG	F&S	
8.6	Sound directionality						
8.6.1	Sounds should directionally replicate those of the simulated aircraft type and variant. Specific sounds that are directional (e.g. skid impacts during an asymmetric touchdown, windshield wipers, asymmetric engine sounds, directional failure sounds) should be directionally differentiated.			✓			✓
8.6.2	Sounds should be directionally characteristic of the simulated aircraft type.		✓				✓
8.6.3	Engine, propeller and rotor sounds should be directionally characteristic if they are asymmetric in the simulated group of aircraft for normal, abnormal or emergency situations (e.g. a multi-engine aeroplane with wing-mounted engines is expected to have asymmetric engine sounds when only one engine is operating).	✓					✓

GM1 FSTD.QB.108 Flight simulation training device general requirements for sound cueing

Summaries of the fidelity levels for the sound cueing FSTD feature are presented in the table below. These summaries are broad overviews provided for guidance only. They do not encompass all aspects and requirements and are not intended to replace CS FSTD.QB.108.

8: Sound cueing
Specific Directional sounds replicate those of the simulated aircraft type and variant, and include engine, propeller, transmission, rotor, airframe, aircraft system and environmental sounds, as well as those generated by pilot or instructor actions. Sounds are a function of flight conditions. Significant environmental sounds are consistent with the simulated weather. Sound frequencies and amplitudes replicate those of the simulated aircraft type and variant.
Representative: Directional sounds are characteristic of the simulated aircraft type, and include engine, propeller, transmission, rotor, airframe, aircraft system and environmental sounds, as well as those generated by pilot or instructor actions. Sounds are a function of flight conditions. Significant environmental sounds are consistent with the simulated weather.
Generic: Significant sounds are characteristic of the simulated group of aircraft, and include engine, rotor and aircraft system sounds, as well as those generated by pilot or instructor actions. Engine, propeller and rotor sounds are directional if they are asymmetric. Environmental sounds are not required.

GM2 FSTD.QB.108 Flight simulation training device general requirements for sound cueing

The sound cueing FSTD feature covers the aircraft sounds external to the flight deck environment: engine sounds, aerodynamic sounds and sounds that are simulation effects, such as those related to the weather, and sound effects, such as crash sounds, that provide cues to pilots.

It is recognised that the sound cues for the specific fidelity level may not necessarily be a perfect replication of those of the simulated type and variant, but they will be validated objectively against aircraft type data in addition to being validated subjectively.

VIBRATION CUEING

CS FSTD.QB.109 Flight simulation training device general requirements for vibration cueing

9: Vibration cueing

Defines the vibrations and buffets to be modelled in the FSTD that may be felt by the pilots during all phases of flight from effects such as airframe buffet, control surface buffet, engine, propeller/rotor and transmission vibrations, systems operation, weather and ground operations. These vibrations may vary significantly depending on the flight parameters, for example, rotorcraft hover in or out-of-ground effect, hover into wind, downwind hover, hover taxi, transition to forward flight, translational lift, cruise, turns, steep turns and autorotation, as relevant to the aircraft type.

Note: This feature does not contain the motion effects (e.g. touchdown cues of the landing gear). Refer to the motion cueing FSTD feature in CS FSTD.QB.110 for their requirements.

Note: Refer to the definitions of ‘vibration’ and ‘buffet’ in CS FSTD.GEN.015.

The general requirements for the vibration cueing FSTD feature are presented in the table below.

9: Vibration cueing							
Feature general requirements				Feature fidelity level		Verification and validation	
				G	R	S	
				QTG		F&S	
9.1	Vibration cues general						

9: Vibration cueing						
Feature general requirements		Feature fidelity level			Verification and validation	
		G	R	S	QTG	F&S
9.1.1	Tactile vibrations and buffets that result from the operation of the aircraft should replicate those of the simulated aircraft type and variant in so far as the vibration or buffet marks an event or aircraft state that can be physically perceived in the flight deck. The vibrations and buffets should be simulated on the following flight deck elements, as applicable to the aircraft: (1) floor and seats; (2) glareshield; (3) instrument panel; (4) overhead panel; and (5) pedestal/aisle stand.			✓		✓
9.1.2	Tactile vibrations and buffets that result from the operation of the aircraft should be characteristic of the simulated aircraft type in so far as the vibration or buffet marks an event or aircraft state that can be physically perceived in the flight deck. At a minimum, vibrations and buffets should be simulated on the following flight deck elements, as applicable to the aircraft: (1) for aeroplanes: seats; (2) for rotorcraft: flight deck structure and seats.		✓			✓
9.1.3	Tactile vibrations and buffets that result from the operation of the aircraft should be characteristic of the simulated group of aircraft in so far as the vibration or buffet marks an event or aircraft state that can be physically perceived by the pilot. At a minimum, the tactile vibrations and buffets should be perceivable at the pilot station (e.g. through the flight deck structure or seat(s)).	✓				✓

9: Vibration cueing						
Feature general requirements				Feature fidelity level		Verification and validation
		G	R	S	QTG	F&S
9.2	Vibration and buffet cues					
9.2.1	Objective vibration tests are required and should include recorded results that allow the comparison of relative amplitudes versus frequency: (1) for relevant frequencies up to at least 20 Hz for aeroplanes, and up to the main rotor n/rev frequency for rotorcraft; (2) for at least three axes – x, y and z – to represent the effects as experienced in the aircraft.		✓	✓	3.g	
9.2.2	The FSTD should be programmed and instrumented using an accelerometer in such a manner that the characteristic vibration/buffet modes can be measured by the objective tests.			✓		
9.2.3	For aeroplanes: The FSTD should provide vibration and buffet effects, including, as applicable to the simulated aircraft type and variant: (1) cues on the ground due to thrust effect with brakes set; (2) cues on the ground due to spoiler / speed brake extension and thrust reversal; (3) cues due to the extension and retraction of landing gear; (4) cues in the air due to flap and spoiler / speed brake extension; (5) cues due to atmospheric disturbances; (6) cues due to Mach and manoeuvre buffet; (7) cues due to approach-to-stall buffet and stall buffet. For (7), if there are known flight conditions in which buffet is the first indication of the stall or if no stall buffet occurs, this characteristic should be included in the model. An objective test for stall buffet is required only for aeroplane FSTDs that are to be qualified for full-stall training tasks or for those aircraft that exhibit stall buffet before the activation of the stall warning system.			✓	3.g	✓

9: Vibration cueing						
Feature general requirements		Feature fidelity level			Verification and validation	
		G	R	S	QTG	F&S
9.2.4	For aeroplanes: The FSTD should provide vibration and buffet effects, including, as applicable to the simulated aircraft type: (1) cues on the ground due to the thrust effect with brakes set; (2) cues on the ground due to spoiler / speed brake extension and thrust reversal; (3) cues due to the extension and retraction of landing gear; (4) cues in the air due to flap and spoiler / speed brake extension; (5) cues due to atmospheric disturbances; (6) cues due to Mach and manoeuvre buffet; (7) cues due to the approach-to-stall buffet (included in the model if there are known flight conditions in which buffet is the first indication of the stall).		✓		3.g	✓
9.2.5	For aeroplanes, the FSTD should provide vibration and buffet effects, including, as applicable to the simulated group of aircraft: (1) cues due to configuration changes (e.g. flaps, landing gear, spoilers, speed brakes); (2) cues due to thrust; (3) cues due to aerodynamic regimes (e.g. overspeed, approach to stall).	✓				✓

9.2.6	For rotorcraft: The FSTD should provide vibration and buffet effects, including, as applicable to the simulated aircraft type and variant: (1) cues due to lift/thrust system effects, including main and tail rotor speed and disk loading vibration effects and cues due to main rotor and anti-torque system blade-tracking and balance effects; (2) cues due to translational lift and the effect of speed changes (acceleration and deceleration); (3) cues due to primary anti-torque system (e.g. tail rotor, fenestron, no tail rotor (NOTAR) system); (4) cues due to transverse flow effects; (5) cues due to the extension and retraction of landing gear (if applicable to the simulated aircraft); (6) cues due to autorotation; (7) cues due to retreating blade stall; (8) cues due to vortex ring (setting with power); (9) cues due to high-speed rotor vibration; (10) cues due to basic anti-torque system failure; (11) cues due to operation in the low-relative-speed environment (e.g. relative wind from all directions); (12) cues due to icing effects; (13) cues due to atmospheric disturbances; (14) cues due to hydraulic system failures; (15) cues due to detailed malfunction effects (to include those associated with the main rotor damper, vibration suppression systems, and anti-torque bearing, coupling, gearbox or driveshaft); (16) if the simulated rotorcraft type is equipped with a vibration reduction system, vibration cues simulating the situation in which the vibration reduction system is inoperative.			✓	3.g	✓
9.2.7	For rotorcraft: The FSTD should provide vibration and buffet effects, including, as applicable to the simulated aircraft type:		✓		3.g	✓

9: Vibration cueing						
Feature general requirements		Feature fidelity level			Verification and validation	
		G	R	S	QTG	F&S
	(1) cues due to lift/thrust system vibration effects, including main rotor and tail rotor speed and disk loading vibration effects and cues due to main rotor and anti-torque system blade-tracking and balance effects; (2) cues due to translational lift and the effect of speed changes (acceleration and deceleration); (3) cues due to primary anti-torque system (e.g. tail rotor, fenestron, NOTAR system); (4) cues due to transverse flow effects; (5) cues due to the extension and retraction of landing gear (if applicable to the simulated aircraft); (6) cues due to autorotation; (7) cues due to retreating blade stall; (8) cues due to vortex ring (setting with power); (9) cues due to high-speed rotor vibration; (10) cues due to basic anti-torque system failure; (11) cues due to operation in the low-relative-speed environment (e.g. relative wind from all directions); (12) cues due to icing effects; (13) cues due to atmospheric disturbances; (14) if the simulated rotorcraft type is equipped with a vibration reduction system, vibration cues simulating the situation in which the vibration reduction system is inoperative.					
9.2.8	For rotorcraft: The FSTD should provide vibration and buffet effects, including, as applicable to the simulated group of aircraft: (1) the primary main rotor speed and disk loading vibration effects; (2) the buffet due to translational lift and the effect of speed changes.	✓				✓

9: Vibration cueing						
Feature general requirements				Feature fidelity level		Verification and validation
		G	R	S	QTG	F&S
9.3	Integration					
9.3.1	Vibrations and buffets should correlate with aircraft systems and flight control systems simulation, including any effects due to abnormal system conditions.		✓	✓		✓
9.3.2	Vibrations and buffets should be affected by the aircraft systems and flight control systems, as applicable, including any significant cues due to abnormal system conditions.	✓				✓

GM1 FSTD.QB.109 Flight simulation training device general requirements for vibration cueing

Summaries of the fidelity levels for the vibration cueing FSTD feature are presented in the table below. These summaries are broad overviews provided for guidance only. They do not encompass all aspects and requirements and are not intended to replace CS FSTD.QB.109.

9: Vibration cueing
Specific: Vibrations and buffets that result from the operation of the aircraft replicate those of the simulated aircraft type and variant in so far as the vibration or buffet marks an event or aircraft state that can be sensed in the flight deck. The vibrations and buffets are simulated at least on the floor and seats, glareshield, instrument panel, overhead panel and pedestal / aisle stand.
Representative: Vibrations and buffets that result from the operation of the aircraft are characteristic of the simulated aircraft type in so far as the vibration or buffet marks an event or aircraft state that can be sensed in the flight deck. The vibrations and buffets are simulated at least on the seats for aeroplanes, and on the flight deck structure and seats for rotorcraft.
Generic: Vibrations and buffets that result from the operation of the aircraft are characteristic of the simulated group of aircraft in so far as the vibration or buffet marks an event or aircraft state that can be sensed in the flight deck. The vibrations and buffets are perceivable at the pilot station (e.g. through the flight deck structure or seat(s)).

GM2 FSTD.QB.109 Flight simulation training device general requirements for vibration cueing

The vibration cueing FSTD feature covers the aircraft vibrations and buffet cues that are experienced in the aircraft flight deck. Some cues are intended to replicate cues in the aircraft and are therefore also objectively verified. Other cues are simulation effects to provide general cues of the effects that pilots would experience, but they may not necessarily replicate what the aircraft would do.

It is recognised that the vibration cues for the specific fidelity level may not necessarily be a perfect replication of those of the simulated type and variant, but they will be validated objectively against aircraft type data in addition to being validated subjectively. Refer to the objective test requirements in Subpart D.

Vibration or buffet cueing may be triggered by any factor related to the operation of the aircraft. The words 'operation of the aircraft' should be understood to mean manoeuvring the aircraft (e.g. changes in attitude, altitude, angle of attack, load factor, airspeed and thrust), changes in the airframe shape

(e.g. usage of slats or flaps, extension of speed brakes, extension of a ram air turbine, opening of a door, use of tilting external lights) or usage of the aircraft systems that may affect vibrations or buffets.

At the generic fidelity level, the frequency of the vibration effects should be characteristic of the simulated group of aircraft. It is recognised that the amplitude of the hardware vibration system for the generic fidelity level may be limited.

GM3 FSTD.QB.109 Flight simulation training device general requirements for vibration cueing

VIBRATION SYSTEMS AT THE SPECIFIC OR REPRESENTATIVE FIDELITY LEVEL FOR ROTORCRAFT

- (a) The role of vibrations in pilot cueing is as follows.
 - (1) Motion cues in rotary wing aircraft have a wide bandwidth of frequencies and amplitudes, consisting of cues ranging from large sustained accelerations up to high-frequency vibrations generated by the rotor harmonics. Vibrations in helicopters, in addition to creating a harsh operating environment, provide pilots with rotor dynamic feedback critical to their ability to control the aircraft. Normal and abnormal flying conditions are therefore sensed by the pilots through the vibration levels and amplitudes and are integral to helicopter flying. Rotor malfunctions or conditions such as icing or damage are rapidly identified subjectively by sensing the increased vibration levels and change in characteristics.
 - (2) The FSTD training environment should subject the pilot to realistic and sufficiently intense vibration levels to support an effective transfer of training. Vibrations, if accurately simulated and harmonised with visual and sound system cues, ensure that the pilot develops proper control strategies while experiencing representative workloads.
 - (3) Three characteristics of the vibrations must be accurately reproduced to create an authentic flying environment and provide pilots with representative aircraft vibrations: the trends, the axes and the levels of vibrations. For example, the vibration trends will inform the pilot that the helicopter has entered a transition stage between hover and low-speed level flight. Helicopter vibrations are multidimensional and are perceived as occurring in more than one DOF at a time. Simulating combinations of vibrations in the x-, y- and z-axes has been demonstrated to be significant for pilot training. Accurate reproduction of vibration levels provides subjective information on the stresses that certain manoeuvres exert on the helicopter.
- (b) Limitations of using a six degrees of freedom motion system to reproduce vibrations are as follows.
 - (1) The simulation of vibration cues for rotorcraft as produced by a conventional six-DOF motion system is limited. While most motion systems are capable of reproducing vibrations, the dynamic range of helicopter vibration amplitudes and frequencies (typically 3–50 Hz) exceeds the limited bandwidth capability of synergistic motion

systems (typically up to 10 Hz in the vertical axis and less than 10 Hz in the longitudinal and lateral axes).

- (2) Moreover, the application of representative vibrations to the entire FSTD structure may adversely impact the lifespan of some FSTD components, such as the visual system.
- (c) Advantages of a dedicated three-DOF vibration system are as follows.
- (1) To augment the performance of a six-DOF motion system and achieve an accurate reproduction of vibrations while minimising stresses on the FSTD structure, the motion cueing frequency bandwidth may be separated into two. Dedicated cueing devices may then be assigned to reproduce each specific frequency range. The lower frequency range is used to drive the motion system, and the higher frequency range, with the majority of the vibration information, is used to drive a vibration system.
 - (2) Two solutions may be used for simulating the vibration with three DOFs:
 - (i) A vibration system consisting of a three-DOF system tailored for vibrations and installed under the flight deck can be used as shown in Figure 1. This system combines a high bandwidth, independent driving axes (to avoid cross-coupling effects) and high stiffness.
 - (ii) A vibration system consisting of a three-DOF system can be used to make the seats, controls and main instrument board vibrate independently from the flight deck. This solution decreases the moving mass relatively to the payload and therefore minimises the risk of resonance.

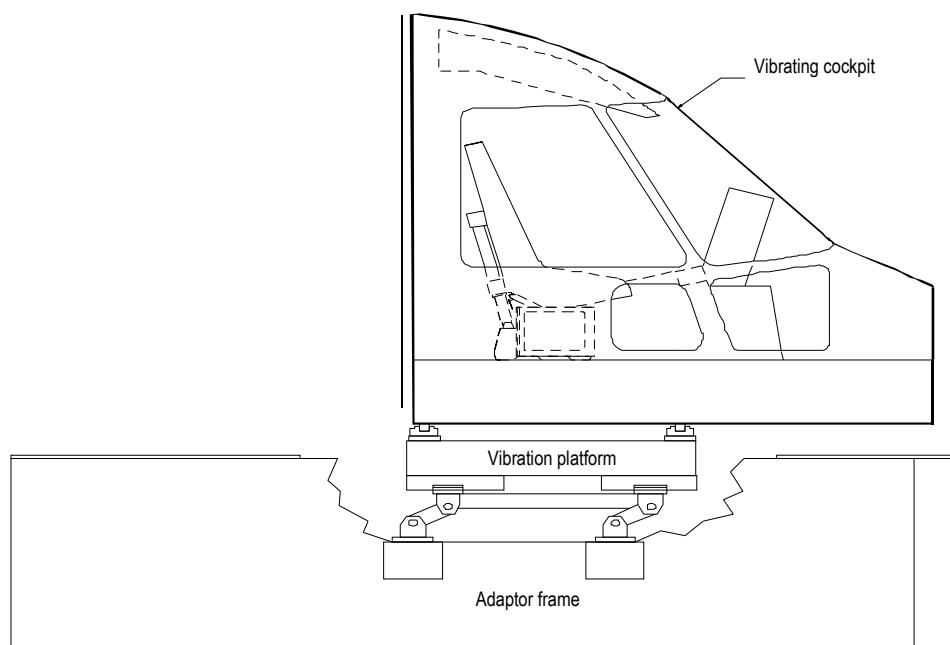


Figure 1. An example of a three-DOF flight deck vibration system

MOTION CUEING

CS FSTD.QB.110 Flight simulation training device general requirements for motion cueing

10: Motion cueing

Defines the requirements for the motion cues that are generated by aircraft accelerations (due to forces on flight and ground operations) and the effects of certain events or situations.

Note: This feature does not contain the vibration and buffet cues. Refer to the vibration cueing FSTD feature in CS FSTD.QB.109 for their requirements.

The general requirements for the motion cueing FSTD feature are presented in the table below.

10: Motion cueing								
Feature general requirements				Feature fidelity level		Verification and validation		
				G	R	S	QTG	F&S
10.1	Motion cues general							
10.1.1	A motion system with a minimum of six degrees of freedom (DOF) should be fitted. The pilot receives an effective motion cue and stimulus, which provide the appropriate sensations of the acceleration of the aircraft's six DOFs. The motion system should be capable of generating sustained acceleration cues and generating the key acceleration and angular cues that are characteristic of the aircraft dynamics in all axes for all the applicable dynamic flight conditions, manoeuvres and events (e.g. taxiing, take-off, rotation, climb, turns, acceleration, deceleration, descent, approach, flare, landing, engine failures, emergencies). The motion system should provide both transient responses and quasi-static platform attitudes that support the perception of aircraft motion and orientation.					✓		✓

10: Motion cueing						
Feature general requirements		Feature fidelity level			Verification and validation	
		G	R	S	QTG	F&S
	The FSTD motion cueing system should have a high tilt coordination gain, high rotational gain and high correlation with respect to the aircraft simulation model. The cues replicate aircraft accelerations, angular rates and motion effects (e.g. landing bumps) to the maximum extent possible within the physical limitations of the motion system. The motion cueing performance and repeatability related to aircraft accelerations and manoeuvres should be demonstrated for the applicable manoeuvres. The SOJ should explain in detail the selected motion technology and envelopes (both the envelope reached physically and the envelope that the cueing uses). The SOJ should explain the physical magnitude of the motion movements and washouts used by the cueing for the applicable objective cueing performance signature tests.					
10.1.2	A motion system with a minimum of three DOFs for pitch, roll and heave should be fitted. The pilot receives a motion cue and stimulus, which provide the appropriate sensations of acceleration. The motion system should be capable of generating sustained acceleration cues and generating the key acceleration and angular cues that are characteristic of the aircraft dynamics in the appropriate axes for all the applicable dynamic flight conditions, manoeuvres and events (e.g. taxiing, take-off, rotation, climb, turns, acceleration, deceleration, descent, approach, flare, landing, engine failures, emergencies). The motion system should provide both transient responses and quasi-static platform attitudes, within its available DOFs, that support the perception of aircraft motion and orientation. The SOJ should explain in detail the selected motion technology, envelopes (both the envelope reached physically and the envelope that the cueing uses) and the physical magnitudes and washouts used by the cueing for the applicable manoeuvres.		✓			✓
10.1.3	A cueing system providing motion effects should be fitted. Cues of the aircraft accelerations or attitudes are not required at this fidelity level. If such capabilities are simulated, they should comply with level R requirements.	✓				✓

10: Motion cueing						
Feature general requirements		Feature fidelity level			Verification and validation	
		G	R	S	QTG	F&S
	The SOJ should explain in detail what motion effects are simulated and how.					
10.1.4	For aeroplane FSTDs to be qualified for UPRT: For devices with a motion cueing FSTD feature fidelity at level S, special consideration is given to the motion system response during upset prevention and recovery and approach-to-stall or stall-recovery manoeuvres. Notwithstanding the limitations of the motion system, the operator should place particular emphasis on tuning out incorrect cues in the motion system responses, if possible.			✓		✓
10.1.5	The cross-coupling effects of the motion movements should be compensated for so that they are not discernible by a human during flight. Motion cues should always provide a correct sensation and provide positive cues for training purposes. Inappropriate cues that could contribute to a negative transfer of training are not acceptable. The short-term cueing and long-term cueing should be coordinated and blended together.		✓	✓		✓
10.1.6	The washout should be below the human-perceivable threshold during flight and not exhibit any adverse motion cueing effects.		✓	✓		✓
10.2	Motion system performance					
10.2.1	The motion system performance (e.g. frequency response) should comply with the objective test requirements.		✓	✓	3.a, 3.b, 3.c, 3.d	
10.3	Motion effects for aeroplane FSTDs Describes the motion special effects for simulated aircraft cueing.					
10.3.1	Motion effects for an aeroplane FSTD should give cues that are realistic for the simulated aircraft dynamics to indicate the threshold at which a flight crew member recognises an event or situation. Cues should be given on all the appropriate axes (six DOFs) as applicable to the event or situation for:			✓	3.e, 3.f	✓

10: Motion cueing						
Feature general requirements		Feature fidelity level			Verification and validation	
		G	R	S	QTG	F&S
	(1) effects of runway and taxiway rumble, oleo deflections, ground speed, uneven runway, runway contamination with the associated anti-skid characteristics, running over runway centre line lights and taxiway surface roughness; (2) bumps associated with landing gear mechanism movement; (3) touchdown cues for main and nose gear; (4) nose wheel scuffing; (5) tyre failure; (6) engine damage (as applicable to the simulated failure, aircraft type and engine model); (7) tail, pod and wing strike.					
10.3.2	Motion effects for an aeroplane FSTD should give cues that are characteristic of the simulated aircraft dynamics to indicate the threshold at which a flight crew member recognises an event or situation. Cues should be given at least on the heave axis as applicable to the event or situation for: (1) effects of runway and taxiway rumble, oleo deflections, ground speed, uneven runway, runway contamination with the associated anti-skid characteristics, and running over centre line lights; (2) bumps associated with landing gear mechanism movement; (3) touchdown cues for main and nose gear; (4) nose wheel scuffing; (5) tyre failure; (6) engine damage (as applicable to the simulated failure, aircraft type and engine model); (7) tail, pod and wing strike.		✓		3.e	✓
10.3.3	Touchdown cues (relative to each landing gear) should reflect the effects of lateral and directional cues resulting from crab or crosswind landings and from the rate of descent.		✓	✓	3.e, 3.f	✓

10: Motion cueing						
Feature general requirements		Feature fidelity level			Verification and validation	
		G	R	S	QTG	F&S
10.3.4	Motion effects for an aeroplane FSTD should give cues that are characteristic of the simulated aircraft dynamics to indicate the threshold at which a flight crew member recognises an event or situation. Cues should be given at least on the heave axis as applicable to the event or situation for: (1) effects of runway and taxiway rumble, oleo deflections, uneven runway, runway contamination with the associated anti-skid characteristics, and running over runway centre line light characteristics (with such effects as functions of ground speed); (2) touchdown cues for main and nose gear; (3) nose wheel scuffing; (4) engine damage (as applicable to the simulated failure, aircraft type and engine model); (5) airframe ground strike (e.g. tail strike).	✓				✓
10.4	Motion effects for rotorcraft FSTDs Describes the motion special effects for the simulated aircraft cueing.					
10.4.1	Motion effects for a rotorcraft FSTD should give cues that are realistic for the simulated aircraft dynamics to indicate the threshold at which a flight crew member recognises an event or situation. Cues should be given on all the appropriate axis (six DOFs) as applicable to the event or situation for: (1) effects of runway and taxiway rumble, oleo deflections, ground speed, uneven runway, runway contamination, running over runway centre line lights and taxiway characteristics; (2) touchdown cues for main and nose gear or for skids, based on their geometry; (3) ground resonance; (4) tyre failure; (5) engine, transmission or rotor damage (as applicable to the simulated failure and aircraft type); (6) airframe ground strike (e.g. tail strike).			✓	3.e, 3.f	✓
10.4.2	Motion effects for a rotorcraft FSTD should give cues that are characteristic of the simulated aircraft to indicate the threshold at which a flight crew member recognises an event or situation.		✓		3.e	✓

10: Motion cueing						
Feature general requirements		Feature fidelity level			Verification and validation	
		G	R	S	QTG	F&S
	Cues should be given at least on the heave axis as applicable to the event or situation for: (1) touchdown cues for main and nose gear or for skids, based on their geometry; (2) ground resonance; (3) tyre failure; (4) engine, transmission or rotor damage (as applicable to the simulated failure and aircraft type); (5) airframe ground strike (e.g. tail strike).					
10.4.3	Motion effects for a rotorcraft FSTD should give cues that are characteristic of the simulated group of aircraft to indicate the threshold at which a flight crew member recognises an event or situation. Cues should be given at least on the heave axis as applicable to the event or situation for: (1) touchdown cues for main and nose gear or for skids; (2) tyre failure; (3) engine, transmission or rotor damage (as applicable to the simulated failure and aircraft type); (4) airframe ground strike (e.g. tail strike).	✓				✓
10.4.4	Touchdown cues (relative to each landing gear) should reflect the effects of lateral and directional cues resulting from asymmetry or side-slip at touchdown and from the rate of descent.		✓	✓		✓
10.5	Motion system safety					
10.5.1	The motion system should have a safety mechanism to monitor and ensure that the motion performs only the commanded movements and stops in cases of uncommanded movements.	✓	✓	✓		
10.5.2	The motion system should have easily accessible emergency stop buttons available to the instructor and flight crew. The button for the flight crew should be located in a way as to prevent confusion with those of the operated flight deck.	✓	✓	✓		

GM1 FSTD.QB.110 Flight simulation training device general requirements for motion cueing

Summaries of the fidelity levels for the motion cueing FSTD feature are presented in the table below. These summaries are broad overviews provided for guidance only. They do not encompass all aspects and requirements and are not intended to replace CS FSTD.QB.110.

10: Motion cueing
Specific: The pilot receives effective motion cues and stimuli, which provide the appropriate sensations of accelerations in all axes. Motion cues have a high correlation with the aircraft movements.
Representative: The pilot receives motion cues and stimuli, which provide appropriate sensations of accelerations. However, the cues are not provided in all axes, and both the magnitude of the cues and the realism of the experience are limited compared to those of actual flight.
Generic: The pilot receives motion effects, such as touchdown cues, engine damage and runway rumble. Cues of the aircraft accelerations or attitudes are not required at this fidelity level.

GM2 FSTD.QB.110 Flight simulation training device general requirements for motion cueing

- (a) The motion cueing FSTD feature covers the aircraft motion and acceleration cues that are experienced in the aircraft flight deck to enhance the simulation experience and training effectiveness. In addition, this feature covers motion effects (i.e. dynamic cues provided for isolated events during flight). Examples of motion effects and motion cueing are given below.
- (1) An example of a motion effect is a tyre failure during taxiing. The motion effect should reflect the sensation of the tyre bursting.
 - (2) Examples of motion cues include key acceleration and angular cues associated with aircraft manoeuvres such as taxi turns, braking, take-off run, rotation, turns and flare. The performance and handling model provides the corresponding acceleration and angular rate data to the motion system, which moves the FSTD to deliver the appropriate cues to the pilot.
- (b) Cueing may consist of a combination of motion effects, including discrete motion impulses, longer-term motion cues and vibrations or buffeting. For example, a simulated bird strike resulting in engine damage may include a sharp, momentary lateral motion cue (e.g. a 'kick'), accompanied by appropriate visual and aural effects, instrument indications and other cues. The subsequent engine damage may result in asymmetric thrust, leading to aircraft yaw and roll. The motion system should provide cues corresponding to these movements. In addition,

vibrations and/or buffeting effects may be present as a consequence of in-flight thrust loss (refer to CS FSTD.QB.109).

- (c) A motion system cannot provide or sustain the same accelerations as the pilots feel in real aircraft because of the limited size and envelope of the motion system. Some cues are intended to replicate the aircraft motions and are therefore also objectively verified. Other cues are simulation effects (which may be artificial) to provide general cues of the effects that pilots would experience, but they may not necessarily replicate what the aircraft would do.

It is recognised that the physical space and mechanics of the motion system may limit the range and fidelity of motion cueing that can be realistically reproduced. Motion systems often exhibit cross-couplings between axes, necessitating careful consideration of cue strength to avoid misleading or incorrect cues. To minimise the impact of incorrect cues, it may be necessary to downscale the motion envelope. However, this approach can significantly affect the perception of correct cues as well. For more information, refer to GM1 FSTD.QTG.A.300 and GM1 FSTD.QTG.H.300.

VISUAL CUEING

CS FSTD.QB.111 Flight simulation training device general requirements for visual cueing

11: Visual cueing
Defines the type of out-of-the-flight-deck-window image optical display (e.g. collimated or non-collimated mirror) and visual field of view (horizontal and vertical) that is required to be seen by the pilots using the FSTD from their reference eyepoint(s). Note: This feature concerns the qualities and capabilities of the visual image. For requirements for the contents of the visual image (e.g. runway, lights, terrain, buildings), refer to FSTD features operating sites and terrain in CS FSTD.QB.114 and atmosphere and weather in CS FSTD.QB.113.

The general requirements for the visual cueing FSTD feature are presented in the table below.

11: Visual cueing							
Feature general requirements				Feature fidelity level			Verification and validation
				G	R	S	QTG F&S
11.1	Display system						
11.1.1	The system should be free from optical discontinuities and artefacts that create non-realistic cues. Artefacts may include but are not limited to colour balance, colour correctness, incorrect image movement, aliasing or edge-blending issues. If the system consists of multiple channels, they should have the same image quality (within the minimum required FOV) and the image should continue smoothly between channels (i.e. edge blending) for day, twilight and night. Visible joins or edges (i.e. sharp contours) in the image from adjoining display surfaces may be acceptable if necessary, but should be minimised.			✓	✓	✓	✓
11.1.2	The computational resources for the visual system should be sufficient to ensure a stable, seamless and realistic visual display.			✓	✓	✓	✓

11: Visual cueing						
Feature general requirements				Feature fidelity level		Verification and validation
		G	R	S	QTG	F&S
11.2	Display geometry					
11.2.1	The geometry should be presented on the visual display correctly. The SOJ should explain the geometry of the installation.		✓	✓		✓
11.2.2	The geometry of the image should be free from distracting discontinuities.	✓	✓	✓		✓
11.3.1	Display FOV					
11.3.1	The FOV in the training device replicates the aircraft FOV without obstruction.			✓		✓
11.3.2	The visual representation of flight deck cut-off angle at the pilot reference eyepoint should match the simulated aircraft.	✓	✓	✓	4.e.1	✓
11.4	Display FOV for aeroplanes					
11.4.1	Continuous collimated FOV of at least 200° horizontally and 40° vertically should be ensured for each pilot simultaneously. The display should be sufficient to provide peripheral vision cues.			✓	4.a.1	
11.4.2	Continuous FOV of at least 200° horizontally and 40° vertically should be ensured for each pilot simultaneously.		✓		4.a.1	
11.4.3	A FOV of a minimum of 45° horizontally and 30° vertically, unless restricted by the type of aircraft, should be ensured for each pilot simultaneously. The display may consist of one or more flat screens.	✓			4.a.1	
11.5	Display FOV for rotorcraft					
11.5.1	Continuous FOV of at least 200° horizontally and 60° vertically FOV should be ensured for each pilot simultaneously. The display should be sufficient to provide peripheral vision cues. The visual system need not be collimated.			✓	4.a.1	

11: Visual cueing						
Feature general requirements		Feature fidelity level			Verification and validation	
		G	R	S	QTG	F&S
11.5.2	Continuous FOV of at least 200° horizontally and 60° vertically should be ensured for each pilot simultaneously. The visual system need not be collimated.		✓		4.a.1	
11.5.3	A FOV of a minimum of 150° horizontally and 40° vertically, unless restricted by the type of aircraft, should be ensured for each pilot simultaneously. The display may consist of one or more flat screens. The visual system need not be collimated.	✓			4.a.1	
11.5.4	The horizontal FOV should be no less than 75° measured either side of the centre of the design eyepoint, unless the simulated rotorcraft justifies an alternative FOV distribution.		✓	✓	4.a.1	
11.5.5	The horizontal FOV should be no less than 60° measured either side of the centre of the design eyepoint, unless the simulated rotorcraft justifies an alternative FOV distribution.	✓			4.a.1	
11.5.6	The vertical FOV should be distributed such that 20° is above the horizon and 40° is below, unless the simulated rotorcraft justifies an alternative FOV distribution.		✓	✓	4.a.1	
11.5.7	The vertical FOV should be distributed such that 15° is above the horizon and 25° is below, unless the simulated rotorcraft justifies an alternative FOV distribution.	✓			4.a.1	
11.6	Geometry					
11.6.1	For collimated displays: The parallax effects should be minimised. The centre of the image should be aligned with the flight deck centre line and the horizon. From each eyepoint position, the centre of the image is to be between 0° and 2° inboard in the horizontal plane and within ±0.25° vertically. The difference between the left and right eyepoint horizontal angles should not exceed 1°.		✓	✓	4.a.2	✓

11: Visual cueing						
Feature general requirements		Feature fidelity level			Verification and validation	
		G	R	S	QTG	F&S
11.6.2	For aeroplanes using non-collimated displays: The parallax effects should be minimised. The parallax error should not exceed 10° in the horizontal plane and $\pm 0.25^\circ$ in the vertical plane simultaneously for each pilot, if aligned for the point midway between the left- and right-seat eyepoints.		✓		4.a.2	✓
11.6.3	For rotorcraft using non-collimated displays: The parallax effects should be minimised. The parallax error should not exceed 10° in the horizontal plane and $\pm 0.25^\circ$ in the vertical plane simultaneously for each pilot, if aligned for the point midway between the left- and right-seat eyepoints.		✓	✓	4.a.2	✓
11.6.4	The parallax effects should be minimised.	✓	✓	✓		✓
11.6.5	For aeroplanes using non-collimated displays: The visual system should be capable of aligning at any time to each of the pilot eye positions and to the flight deck centre line. If the image is aligned to the pilot flying, the misalignment observed by the non-flying pilot should not exceed 20°.	✓	✓			✓
11.6.6	For rotorcraft using non-collimated displays: The visual system should be capable of aligning at any time to each of the pilot eye positions and to the flight deck centre line. If the image is aligned to the pilot flying, the misalignment observed by the non-flying pilot should not exceed 20°.	✓	✓	✓		✓
11.6.7	The minimum distance from the pilot's eye position to the surface of a direct-view display may not be less than the distance to any front panel instrument.	✓				
11.6.8	The image should not be distorted.	✓	✓	✓	4.a.2	✓
11.7	Display surface resolution					

11: Visual cueing						
Feature general requirements		Feature fidelity level			Verification and validation	
		G	R	S	QTG	F&S
11.7.1	The display surface resolution should be demonstrated using a test pattern for each channel that ensures that objects in the visual display, when viewed from the pilot's eyepoint, are resolved within an appropriate visual angle suitable for realistic scene representation. This should take into account the characteristics of the display system, including collimation effects, projection method, blending zones and viewing geometry, to ensure realistic scene representation across the entire FOV.	✓	✓	✓	4.a.3	
11.8	Light point size					
11.8.1	The light point size should be no greater than 5 arc minutes.			✓	4.a.4	
11.8.2	The light point size should be no greater than 8 arc minutes.	✓	✓		4.a.4	
11.8.3	The light point size in the visual system should be demonstrated using a test pattern for each channel that accurately represents the lights used for airport lighting when viewed from the pilot's eyepoint.	✓	✓	✓	4.a.4	
11.9	Surface contrast ratio Describes the measurable difference between the brightest and darkest areas of the display.					
11.9.1	For collimated displays, the display contrast ratio should be adequate to clearly identify image visual elements and not less than 8:1.			✓	4.a.5	
11.9.2	For collimated displays, the display contrast ratio should be adequate to clearly identify image visual elements and not less than 5:1.		✓		4.a.5	
11.9.3	For non-collimated displays, the display contrast ratio should be adequate to clearly identify image visual elements and not less than 5:1.		✓	✓	4.a.5	
11.9.4	The display contrast ratio should be demonstrated using a test pattern for each channel to ensure that the contrast is clearly distinguishable.		✓	✓	4.a.5	

11: Visual cueing						
Feature general requirements		Feature fidelity level			Verification and validation	
		G	R	S	QTG	F&S
11.9.5	The display contrast ratio should be sufficient to clearly differentiate the brightest and darkest areas of the display.	✓	✓	✓		✓
11.10	Light point contrast ratio					
11.10.1	The light point contrast ratio should be sufficient to clearly identify illumination sources and lights and not less than 25:1.		✓	✓	4.a.6	
11.10.2	The light point contrast ratio should be demonstrated using a test pattern for each channel to ensure that illumination sources and lights are clearly identifiable. The contrast ratio should be sufficient to allow for the clear detection of relevant lighting visual elements.		✓	✓	4.a.6	
11.10.3	The light point contrast ratio should be sufficient to be able to clearly differentiate light points.	✓	✓	✓		✓
11.11	Light point brightness					
11.11.1	The light point brightness should be no less than 20 cd/m ² (5.8 ft-lamberts).			✓	4.a.7	
11.11.2	The light point brightness should be no less than 17 cd/m ² (5.0 ft-lamberts).		✓		4.a.7	
11.11.3	The light point brightness should be demonstrated using a test pattern for each channel to ensure clear visibility of illumination sources in the visual display. The brightness level of individual light points should be adequate to allow for the detection and realistic representation of relevant lighting visual elements.		✓	✓	4.a.7	
11.11.4	The light point brightness should be sufficient to be able to clearly identify visual elements in the visual scene.	✓	✓	✓		✓
11.12	Surface brightness					

11: Visual cueing						
Feature general requirements		Feature fidelity level			Verification and validation	
		G	R	S	QTG	F&S
11.12.1	The surface brightness should not be less than 20 cd/m ² (5.8 ft-lamberts) at any point across the entire FOV.			✓	4.a.8	
11.12.2	The surface brightness should not be less than 14 cd/m ² (4.1 ft-lamberts) at any point across the entire FOV.		✓		4.a.8	
11.12.3	The overall surface brightness should be demonstrated using a test pattern for each channel to ensure a realistic scene environment. The brightness level of the display should support the clear visibility of all visual elements, including background and illumination sources.		✓	✓	4.a.8	
11.12.4	The surface brightness should be sufficient to be able to clearly identify visual elements in the visual scene.	✓	✓	✓		✓
11.13	Black level and luminance levels (as applicable to the visual display technology used on the FSTD)					
11.13.1	Tests are required, for each channel, to demonstrate that the black level is sufficient to support operations in daylight, twilight and night conditions.		✓	✓	4.a.9	
11.13.2	The black level should support operations in daylight, twilight and night conditions.	✓	✓	✓		✓
11.13.3	Tests are required, for each channel, to demonstrate that the luminance levels are sufficient to support operations in daylight, twilight and night conditions.		✓	✓	4.a.9	
11.13.4	The luminance levels should support operations in daylight, twilight and night conditions.	✓	✓	✓		✓
11.14	Motion blur (as applicable to the visual display technology used on the FSTD)					
11.14.1	A test is required for each channel to demonstrate that the motion blur is below a distracting level.		✓	✓	4.a.10	✓

11: Visual cueing						
Feature general requirements		Feature fidelity level			Verification and validation	
		G	R	S	QTG	F&S
11.14.2	The motion blur should be below a distracting level.	✓	✓	✓		✓
11.15	Speckle test (as applicable to the visual display technology used on the FSTD)					
11.15.1	A speckle test for each channel is required to demonstrate that the speckle is below a distracting level.		✓	✓	4.a.11	✓
11.15.2	The speckle should be below a distracting level.	✓	✓	✓		✓
11.16	VGS					
11.16.1	The ground segment visible to the pilot when conducting a landing manoeuvre in day, twilight and night low-visibility conditions conforms to that of the aircraft.	✓	✓	✓	4.e.1	✓
11.17	Crash effects					
11.17.1	The visual system should have an effect for a crash, for example when impacting with ground structures or another aircraft or exceeding structural or flight envelope limitations.	✓	✓	✓		✓
11.18	Image generator capability					
11.18.1	Scene content should be capable of producing an adequate number of polygons to generate realistic, detailed scene content, including all airport light points relevant to aircraft operation. The scene content should be realistically affected by the viewing distance.		✓	✓		✓
11.18.2	Scene content should be capable of producing an adequate number of polygons to generate realistic, detailed scene content, including airport light points for the active runway. These should be realistically affected by the viewing distance.	✓				✓

11: Visual cueing						
Feature general requirements		Feature fidelity level			Verification and validation	
		G	R	S	QTG	F&S
11.18.3	The visual system should render scenes with a level of detail sufficient to support accurate perception and recognition of relevant environmental visual elements, objects and terrain in all operational scenarios. The system should provide clear and discernible visual details of buildings, apron markings and ground support equipment in airport environments. Curved and complex shapes must appear smooth and realistic, without noticeable angularity or coarse polygonal artefacts. The imagery must ensure smooth transitions and a realistic representation free of unrealistic distractions.	✓	✓	✓		✓
11.18.4	The system should have sufficient capacity to ensure that all moving objects are free from jitter.	✓	✓	✓		✓
11.18.5	The system should ensure that any moving operating site clutter (as defined in CS FSTD.GEN.015) avoids conflicts with the ownship.	✓	✓	✓		✓
11.18.6	The visual system should be capable of displaying static and moving aircraft traffic, both on the ground and in air.		✓	✓		✓
11.18.7	The visual system should be capable of displaying static and moving ground vehicles.		✓	✓		✓
11.18.8	Moving aircraft and ground vehicle traffic should be capable of conflicting with the ownship.		✓	✓		✓
11.19	Visual scene content					
11.19.1	Continuous and sufficient horizontal and vertical FOV specific to the aircraft being simulated should be ensured from the handling pilot's normal eyepoint, with detailed close-up perspective and textured representation of all ambient conditions for each pilot.			✓		✓
11.19.2	Visual cues to assess the rate of change of height, height AGL and translational displacements and rates during take-off, low-altitude/low-air-speed manoeuvring, hover and landing should be ensured. These should include: (1) surface on runways, taxiways and ramps; (2) terrain visual elements.			✓		✓

11: Visual cueing						
Feature general requirements		Feature fidelity level			Verification and validation	
		G	R	S	QTG	F&S
11.19.3	Visual cues to assess the rate of change of height, height AGL and translational displacements and rates during take-off and landing should be ensured.		✓			✓
11.19.4	In terms of visual fidelity, the visual scene should be capable of displaying an identifiable horizon under day, twilight and night conditions with unlimited visibility for determining the aircraft attitude in the pitch, roll and yaw axes. If the visibility is reduced, the horizon should become indistinct or obscured.	✓	✓	✓		✓
11.19.5	The system should be capable of displaying contoured terrain at the resolution demanded by the environment.			✓		✓
11.19.6	Landing area(s) provided should be visible from a distance sufficient to allow the pilot to establish, stabilise and maintain approach angles from the intended 3D final approach position to land in accordance with aircraft performance.		✓	✓		✓
11.19.7	Landing area(s) provided should be visible from a distance sufficient to allow the pilot to establish, stabilise and maintain a shallow approach angle (less than 5°) from 300 m (1 000 ft) AGL to the surface unless obstructed by the flight deck structure.	✓				✓
11.19.8	Focus and clarity of terrain and ground objects should be sharp.		✓	✓		✓
11.19.9	Surface shading effects should be consistent with the simulated sun position. Shadows of 3D objects and lighting effects are appropriate for the directionality of light (e.g. sunshine).			✓		✓
11.19.10	The FSTD should support daylight, twilight and night visual capabilities.	✓	✓	✓		✓
11.19.11	The FSTD should have operational aircraft landing lights, which should be visible in the visual scene in twilight and night conditions.	✓	✓	✓		✓

11: Visual cueing						
Feature general requirements		Feature fidelity level			Verification and validation	
		G	R	S	QTG	F&S
11.19.12	Integration of the aircraft systems and the visual image: If an aircraft system provides information on other traffic or terrain (e.g. ACAS or TAWS/HTAWS), the visual image should correspond to the information provided by the system (e.g. bearing, distance and altitude of the traffic for ACAS or terrain for TAWS/HTAWS).		✓	✓		✓
11.19.13	For aircraft capable of landing on water (e.g. seaplanes, rotorcraft equipped with fixed and/or emergency floats), the following effects should be capable of being displayed (as applicable): (1) sea states; (2) effects of wind lanes on water surfaces.		✓	✓		✓
11.19.14	The visual system should display the ownship parts that are visible within the FOV (e.g. wings, nacelles, propellers, spinners, rotors). The sizes and shapes of those parts should replicate those of the parts of the simulated aircraft type and variant. Any movement (e.g. ailerons, propellers, rotors) should be displayed and should replicate that of the simulated aircraft type and variant.			✓		✓
11.19.15	The visual system should display the ownship parts that are visible within the FOV (e.g. wings, nacelles, propellers, spinners, rotors). The sizes and shapes of those parts should be characteristic of those of the parts of the simulated aircraft type. Any movement (e.g. ailerons, propellers, rotors) should be displayed and should be characteristic of the simulated aircraft type.		✓			✓

11: Visual cueing						
Feature general requirements		Feature fidelity level			Verification and validation	
		G	R	S	QTG	F&S
11.19.16	The visual system should display the ownship parts that are visible within the FOV (e.g. wings, nacelles, propellers, spinners, rotors). The size and shape of those parts should be characteristic of those of the parts of the simulated group. Any movement (e.g. ailerons, propellers, rotors) should be displayed and should be characteristic of the simulated group and should be consistent with that of the other aircraft visual elements (e.g. propellers turning in the same direction as in the flight dynamic model).	✓				✓
11.19.17	The visual system should display the icing effects on the aircraft parts that are visible within the FOV (e.g. wing leading edges, spinners). Note: This does not apply to accessories or parts outside the cockpit that have a functional use (e.g. wipers, demisting wiring, pitot tubes, handles, wire cutters, external mirrors, ice detectors).	✓	✓	✓		✓
11.19.18	Visual cues to assess sink rate and enable depth perception during take-off and landing should be provided.		✓	✓		✓
11.19.19	The visual scene should be free of any negative impact due to any flight deck ambient lighting, IOS or room lighting (e.g. reduction of contrast).	✓	✓	✓		✓
11.20	NIGHT VISION IMAGING SYSTEMS (NVIS) (as applicable) Refer to CS FSTD.QB.104.NVIS.					
11.20.1	The visual image should support NVIS for the black level, surface brightness, surface contrast ratio, lunar illumination, image colour and surface resolution.			✓		✓
11.20.2	Visible black level: The display brightness for a low-visibility night scene, if configured for NVIS operations, should be no more than 0.01 cd/m ² .			✓	4.d.1	
11.20.3	NVIS surface brightness: If configured for NVIS operation, the surface brightness should not be less than 3.4 cd/m ² (1 ft-lambert).			✓	4.d.2	

11: Visual cueing						
Feature general requirements		Feature fidelity level			Verification and validation	
		G	R	S	QTG	F&S
11.20.4	The NVIS surface contrast ratio should be a minimum of 4:1.			✓	4.d.3	
11.20.5	Lunar illumination: The luminance of a fully white Lambertian surface under 100 % lunar illumination at 90° elevation (moon directly overhead) with clear atmospheric conditions should be between 0.075 cd/m ² and 1.025 cd/m ² .			✓	4.d.4	
11.20.6	NVIS image colour: There should be no distracting colour shift from white light in the unaided view. Scenes that exhibit an unrealistic colour tint are not acceptable.			✓		
11.20.7	NVIS surface resolution: Surface resolution should be no greater than 5 arc minutes when viewed through the NVGs under 100 % moon luminescence.			✓	4.d.5	
11.20.8	Surface shading effects should be consistent with the simulated moon position. Shadows of 3D objects and lighting effects should be appropriate for the directionality of light (e.g. moonlight). Surface shading effects should be consistent with the simulated moon position and should include shadows from objects, consistent with the simulated moon position.			✓		✓
11.21	XR systems: HMD Refer to CS FSTD.QB.130.					
11.21.1	FSTDs fitted with an HMD for the visual system that uses VR should meet the requirements for VR displays.	✓	✓	✓		
11.22	XR systems: VR display – headset system requirements Note: Requirements marked with ✓ ¹ indicate that an objective test section for these is not available. The tests should be defined in the special conditions.					
11.22.1	The visible scene should provide an immersive and clear view, with the FOV and field of regard contributing to a seamless and consistent visual experience.		✓	✓	✓ ¹	✓

11: Visual cueing						
Feature general requirements		Feature fidelity level			Verification and validation	
		G	R	S	QTG	F&S
11.22.2	A binocular FOV for the VR display headset should be of a minimum of 80° horizontally (combined) and 65° vertically.		✓	✓	✓ ¹	✓
11.22.3	A binocular FOV for the VR display headset should be of a minimum of 80° horizontally (combined) and 45° vertically.	✓			✓ ¹	✓
11.22.4	The geometry of the visual representation should be correct and free of noticeable distortion. The system should be measured with respect to absolute and relative geometrical characteristics (deg and deg/deg) with a 2D grid of a minimum of 1° over the entire FOV.		✓	✓	✓ ¹	✓
11.22.5	The computational resources for the visual system should be sufficient to ensure a stable frame rate. The frame rate should be above the minimum required to maintain retina persistence and reduce flicker.	✓	✓	✓	✓ ¹	✓
11.22.6	If night scene training is applicable to the FSTD, the visual system should provide sufficient contrast and black level.		✓	✓	✓ ¹	✓
11.22.7	The resolution of the systems used should be sufficient to ensure the readability of the relevant flight deck elements. Resolution characteristics should be tested over the entire FOV. The operator should provide procedures to ensure the correct visual set-up and performance before training.		✓	✓	✓ ¹	✓
11.22.8	Relevant flight deck colours should be those specific to the aircraft simulated. The relevant colour correctness should be objectively tested. The colours should be evenly represented across the display.			✓	✓ ¹	✓
11.22.9	Relevant flight deck colours should be representative of the aircraft simulated and be consistent during simulation. The representativeness of relevant colours should be objectively tested with respect to the degradation of relevant systems. The colours should be evenly represented across the display.	✓	✓		✓ ¹	✓

11: Visual cueing						
Feature general requirements		Feature fidelity level			Verification and validation	
		G	R	S	QTG	F&S
11.22.10	Depth perception in the VR display should be objectively demonstrated and should be subjectively perceived as realistic.		✓	✓	✓ ¹	✓
11.22.11	The IPD setting of the headset should be verified.		✓	✓	✓ ¹	✓
11.22.12	Any HMD-specific capability used that may impact the visual representation should be assessed against the risks of malfunction and degradation (e.g. automatic IPD, foveated rendering, HMD tracking, colour calibration, automatic exposure or white balancing). Means to monitor system integrity during operation should be implemented.	✓	✓	✓		✓
11.23	XR systems: MR display – headset system requirements					
11.23.1	Reserved for future use.					
11.24	XR systems: AR display – headset system requirements					
11.24.1	Reserved for future use.					

GM1 FSTD.QB.111 Flight simulation training device general requirements for visual cueing

Summaries of the fidelity levels for the visual cueing FSTD feature are presented in the table below. These summaries are broad overviews provided for guidance only. They do not encompass all aspects and requirements and are not intended to replace CS FSTD.QB.111.

11: Visual cueing
Specific: Continuous sufficient horizontal and vertical FOV specific to the aircraft being simulated are provided from the handling pilot's reference eyepoint, with detailed close-up perspective and textured representation of all ambient conditions for each pilot. Visual image qualities are at the highest level. Visual cues to assess the rate of change of height, height AGL and translational displacements and rates during take-off, low-altitude/low-air-speed manoeuvring, hover and landing are ensured. The system is capable of displaying contoured terrain at the resolution demanded by the environment. Landing area(s) are visible from a distance sufficient to allow the pilot to establish, stabilise and maintain approach angles from the intended 3D final approach position to land in accordance with aircraft performance.
Representative: Continuous horizontal and vertical FOV are ensured to provide peripheral vision cues. Visual image qualities are at the intermediate level. Visual cues to assess the rate of change of height, height AGL and translational displacements and rates during take-off and landing are ensured. Landing area(s) are visible from a distance sufficient to allow the pilot to establish, stabilise and maintain approach angles from the intended 3D final approach position to land in accordance with aircraft performance.
Generic: The display may consist of one or more flat screens. Visual image qualities are at the lowest level. If a landing area is provided, it is visible from a distance sufficient to allow the pilot to establish, stabilise and maintain a shallow approach angle from 1 000 ft AGL to the surface unless obstructed by the flight deck structure.

GM2 FSTD.QB.111 Flight simulation training device general requirements for visual cueing

The choice of the display system and the FOV requirements should fully consider the intended use of the FSTD. The balance between training and testing/checking may influence the choice and FOV of the display system. In addition, the diverse operational requirements should be addressed.

Guidance on display types is provided in GM3 FSTD.QB.111.

Within the FSTD FOV, the visible image should be as per the aircraft, including any relevant visual obstructions from the aircraft. It should not be obstructed by any non-aircraft components, structure or casted shadows.

As appropriate for the training task(s) and aircraft type, an extended FOV, including chin windows, should be provided. For rotorcraft, these tasks may include take-offs, departures, approaches and landings in confined areas, on sloping ground, in snow/sand or on a rig/ship.

Note that, if a visual display system is fitted, even if it is not considered to be at fidelity level G, R or S, it should be assessed to ensure that it does not adversely affect the qualification of the FSTD.

GM3 FSTD.QB.111 Flight simulation training device general requirements for visual cueing

VISUAL DISPLAY SYSTEMS

This GM provides guidance on the assessment of visual display systems.

- (a) The following is an introduction to visual display systems.

When selecting a visual system configuration, there are many compromises to be made depending on the flight deck geometry, crew complement and intended use of the training device. Some of these compromises and choices regarding display systems are discussed here.

- (b) Basic principles of an FSTD collimated display are as follows.

- (1) The essential feature of a collimated display is that the light rays coming from a given point in a picture are parallel. There are two main implications of the parallel rays. First, the viewer's eyes focus at infinity and have zero convergence, thus providing a cue that the object is distant. Second, the angle to any given point in the picture does not change if viewed from a different position, and thus the object behaves geometrically as though it were located at a significant distance from the viewer. These cues are self-consistent and are appropriate for any object that has been modelled as being at a significant distance from the viewer.
- (2) In an ideal situation, the rays are perfectly parallel, but most implementations provide only an approximation of the ideal. Typically, an FSTD display provides an image located no closer than about 6–10 m from the viewer, with the distance varying over the FOV. A schematic representation of a collimated display is provided in Figure 1 below.

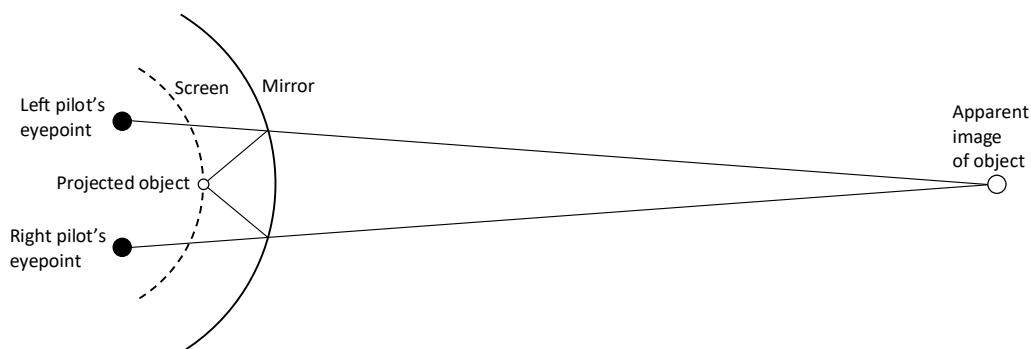


Figure 1. Collimated display

- (3) Collimated displays are well suited to many simulation applications, as the area of interest is relatively distant from the observer, and so the angles to objects should remain independent of the viewing position. Consider the view of the runway seen by the flight crew lined up on an approach. In the real world, the runway is distant, and therefore the light rays from the runway to the eyes are parallel. The runway therefore appears to be straight ahead to both crew members. This situation is well simulated by a collimated display and is presented in Figure 2. Note that the distance to the runway has been shortened for clarity. If drawn to scale, the runway would be farther away and the rays from the two seats would be closer to being parallel.
- (4) While the horizontal FOV of a collimated display can be extended to approximately 210–220°, the vertical FOV is normally limited to about 40–45°. These limitations result from trade-offs in optical quality and interference between the display components and flight deck structures, but the displays were nonetheless sufficient to meet FSTD regulatory approval for aeroplane and rotorcraft FSTDs. Recently, new designs have been introduced with vertical FOVs of up to 60° for some aircraft (e.g. helicopter) applications.

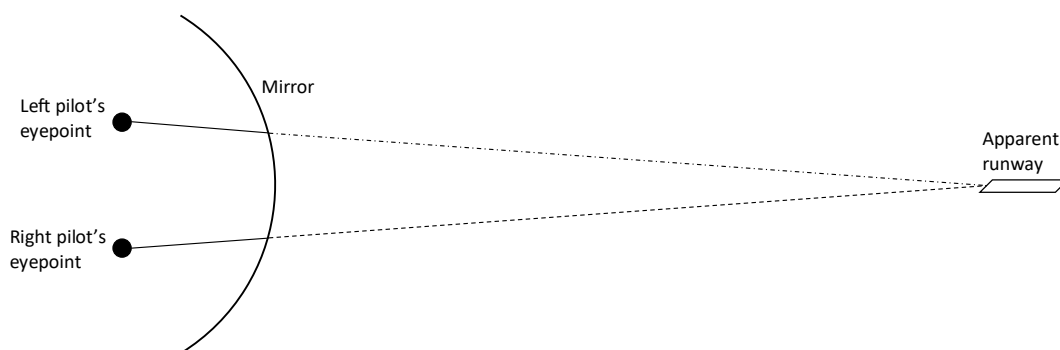


Figure 2. Runway view in a collimated display

- (c) Basic principles of an FSTD dome display are as follows.
- (1) The situation in a dome display is shown in Figure 3. As the angles can be correct for only one eyepoint at a time, the visual system has been calibrated for the right-seat viewer's eyepoint position – the runway appears to this viewer to be straight ahead of the aircraft. To the left-seat viewer, however, the runway appears to be somewhat to the right of the aircraft. As the aircraft is still moving towards the runway, the

perceived velocity vector should be directed towards the runway and this should be interpreted as the aircraft having some yaw offset.

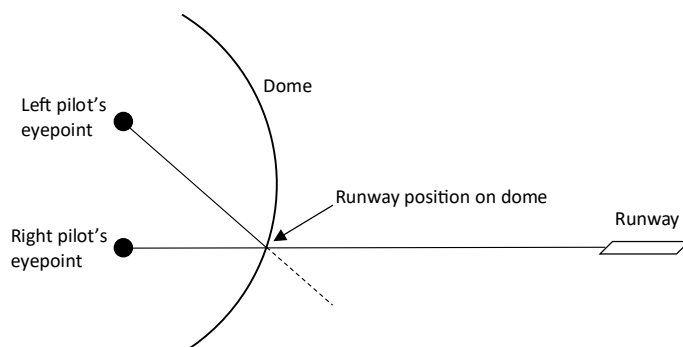


Figure 3. Runway view in a dome display

- (2) The situation is substantially different for near-field objects such as those that are encountered in some aircraft type (e.g. helicopter) operations close to the ground. Here, objects that should be interpreted as being close to the viewer will be misinterpreted as being distant in a collimated display. The errors can actually be reduced in a dome display, as shown in Figures 4 and 5.

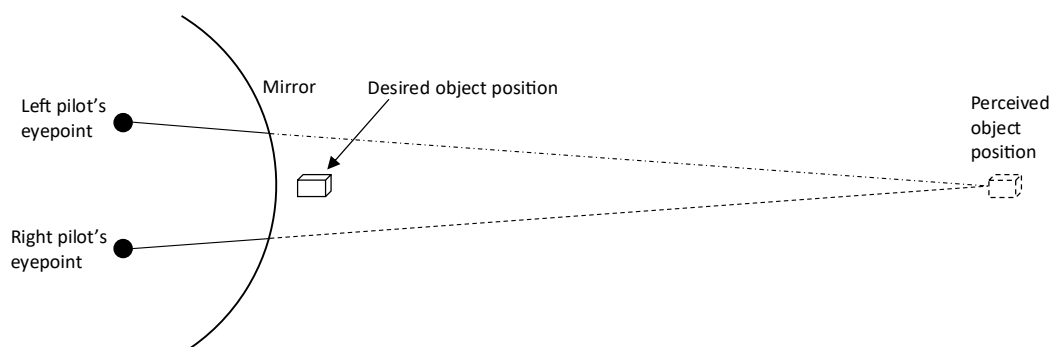


Figure 4. Near-field object in a collimated display

- (3) The FOV possible with a dome display can be larger than that of a collimated display. Depending on the configuration, a FOV of 240° by 90° is possible and can even be exceeded.

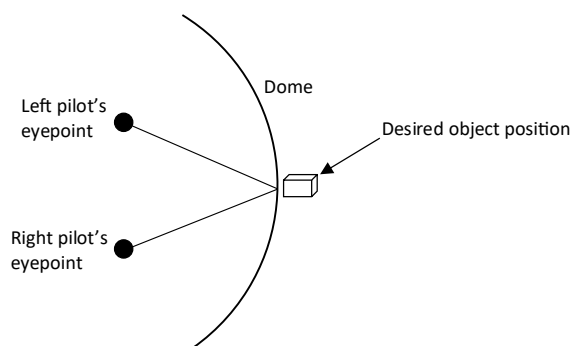


Figure 5. Near-field object in a dome display

(d) Additional display considerations are as follows.

- (1) While the situations described above are for discrete viewing positions, the same arguments can be extended to moving eyepoints, such as those that are produced by the viewer moving their head. In the real world, the parallax effects resulting from head movement provide distance cues. The effect is particularly strong for the relative movement of the flight deck structure in the near field and modelled objects in the distance. Collimated displays provide accurate parallax cues for distant objects, but increasingly inaccurate cues for near-field objects. The situation is reversed for dome displays.
- (2) Stereopsis cues resulting from the different images presented to each eye for objects relatively close to the viewer also provide depth cues. Once again, the collimated and dome displays provide more or less accurate cues depending on the modelled distance of the objects being viewed.

(e) Training implications are as follows.

In view of the basic principles described above, it is clear that neither display approach provides a completely accurate image for all possible object distances. It is therefore important when configuring an FSTD display system to consider the training role of the FSTD. Depending on the training role, one or other display may be the optimum choice. Factors that should be considered when selecting a design approach include the relative importance of training tasks at low altitudes, the role of the two crew members in the flying tasks and the FOV required for particular training tasks.

GM4 FSTD.QB.111 Flight simulation training device general requirements for visual cueing

GUIDANCE ON VISUAL IMAGE DETAILS AND POLYGON COUNTS

The level of detail and realism in the visual scene of an FSTD is significantly influenced by the number of polygons used in the visual system. Higher polygon counts allow for greater detail, contributing to a more immersive and realistic training environment.

For example, a simple ATC tower may require thousands of polygons. In contrast, a more detailed terminal building may consist of hundreds of thousands of polygons. Scaling up further, a complete aerodrome model would require millions of polygons.

The need for detail depends on the training tasks planned for the FSTD. Devices not used for tasks involving proximity to visual objects, such as taxiing to a gate or heliport operations, may require fewer polygons. For scenarios involving close interaction with objects, the visual system must provide sufficient detail to ensure realism, avoiding angularity or coarse polygonal artefacts.

If viewed from close distances, coarse or low-polygon models can negatively impact realism. Therefore, it is critical that curved shapes appear smooth, without visible angular edges or simplified geometry, and that complex shapes appear realistic. This requires an adequate number of polygons.

In addition to polygon count, other factors, such as the visual system's capability to manage rendering performance, texture resolution and lighting effects, must be considered. These elements contribute to the overall perceptual accuracy of the visual cueing.

While this CS-FSTD does not mandate a specific polygon count, the visual system should support a level of detail appropriate for the intended training tasks, while also ensuring adequate system performance.

GM5 FSTD.QB.111 Flight simulation training device general requirements for visual cueing

GUIDANCE ON EXTENDED REALITY SYSTEMS

- (a) XR system capabilities and limitations are as follows.

Special conditions should identify the capabilities and limitations of XR solutions at an early stage of FSTD development to ensure compliance demonstration for qualification. The organisation must understand and describe all relevant aspects of their XR solution within the special conditions, including any novel considerations affecting safety, human performance or the training environment. These aspects should be addressed to ensure an equivalent level of safety to that in the applicable CSs.

- (b) The following applies to AR and MR display systems.

For the purposes of FSTDs using an AR or MR display system, the requirements defined for XR systems (i.e. VR display systems) described in the applicable sections of CS FSTD.QB.111 should be used as the basis for the associated special conditions that are used for the qualification of the FSTD. Refer also to GM1 FSTD.QB.130.

NAVIGATION

CS FSTD.QB.112 Flight simulation training device general requirements for navigation

12: Navigation

Defines the level of simulation for the exchange of navigation, communication and surveillance data and interactions with air traffic services, including associated instructor controls, covering the transmission of signals and data exchange through radio-frequency systems, navigation aids (e.g. ILS, VOR, DME, NDB, GNSS – GPS, SBAS, GBAS), radio communication systems (e.g. VHF, HF, ACARS, CPDLC), surveillance technologies (e.g. ADS-B, ADS-C) and ATC data links.

Note: The aircraft navigation equipment and instruments (e.g. navigation receivers, GNSS units) and their databases (e.g. FMS or GNSS navigation databases) are considered in the FSTD features flight deck layout and structure in CS FSTD.QB.101 and aircraft systems in CS FSTD.QB.104.

The general requirements for the navigation FSTD feature are presented in the table below.

12: Navigation							
Feature general requirements				Feature fidelity level			Verification and validation
				G	R	S	QTG F&S
12.1	Navigation database						
12.1.1	Navigational data should be provided to support operations and simulated aircraft systems for real-world operations at a minimum of five airports / operating sites and to support operational flight scenarios between two operating sites. Earth's magnetic field, including geographic magnetic declination (i.e. magnetic variation), should be simulated within these areas. Navigational data for the airports / operating sites should be complete with corresponding 2D and 3D approach procedures.				✓	✓	✓

12: Navigation						
Feature general requirements		Feature fidelity level			Verification and validation	
		G	R	S	QTG	F&S
12.1.2	Navigational data should be provided to support operations and simulated aircraft systems at a minimum of five airports / operating sites and to support operational flight scenarios between two operating sites. Earth's magnetic field, including geographic magnetic declination (i.e. magnetic variation), should be simulated within these areas. The airports / operating sites and their navigational data may be fictional, in which case associated fictional charts should be provided. The fictional charts should replicate the layout of and contain the same types of information as real aviation charts to ensure that all fictional procedures provide adequate and accurate information for proper use. Navigational data for at least one airport / operating site should be complete with corresponding 2D and/or 3D approach procedures.	✓				✓
12.1.3	Navigational data should be maintained (i.e. kept up to date) as per the aircraft update cycle.			✓		
12.1.4	Navigational data should be capable of being modified or updated. Note: For the requirements for updates to the navigation data (e.g. ground station data, magnetic declination), refer to the requirements in Part-ORA (Annex VII) of Commission Regulation (EU) No 1178/2011.	✓	✓	✓		
12.2	Instructor controls and malfunctions					
12.2.1	The IOS should have controls for navigational aids (e.g. being able to activate/deactivate navigation aids such as ground stations).	✓	✓	✓		✓
12.3	Arrival/departure features					
12.3.1	Navigational data should support the standard arrival and departure procedures at the simulated airports / operating sites.	✓	✓	✓		✓

12: Navigation						
Feature general requirements		Feature fidelity level			Verification and validation	
		G	R	S	QTG	F&S
12.4	Navigation aids					
12.4.1	Navigation aids should be usable within range or line of sight (i.e. should be unusable when outside range or line of sight), as applicable to the geographic area, and dependent on the height of the aircraft above the horizon.		✓	✓		✓
12.4.2	Navigation aids should be usable within range or line of sight (i.e. should be unusable when outside range or line of sight).	✓				✓
12.4.3	Navigational aids should exhibit realistic characteristics (e.g. level of accuracy, VOR cone of confusion).	✓	✓	✓		✓
12.4.4	Navigational aids should be located and aligned appropriately (e.g. with variation at each simulated airport / operating site)	✓	✓	✓		✓
12.5	Radio communication aids					
12.5.1	Radio communication aids should be usable within range or line of sight without restriction, as applicable to the geographic area, and dependent on the height of the aircraft above the horizon (i.e. should be unusable when outside range or line of sight).			✓		✓
12.5.2	Radio communication aids should be usable within range or line of sight.	✓	✓			✓
12.6	GNSS navigation aids					
12.6.1	GNSS navigational aids should be capable of being disrupted.	✓	✓	✓		✓
12.6.2	Malfunctions and/or controls for the instructor should be provided to reduce the GNSS accuracy or availability (e.g. by repeatedly reducing the number of received GNSS satellites one at a time down to zero, GNSS interference) below the normal operational limits.		✓	✓		✓

12: Navigation						
Feature general requirements				Feature fidelity level		Verification and validation
		G	R	S	QTG	F&S
12.7	Automated weather reporting Describes the simulation of fully automatic ATC services delivering reported weather information via radio and data communications. This section is applicable if the ownship is equipped with an associated communication system (e.g. VHF radio or data link) used to receive the reports. Note: If the FSTD to be qualified includes a SATCE system, then refer to CS FSTD.QB.160 for further requirements for integrated automated weather reporting.					
12.7.1	Flight crew access: Automated weather reports (e.g. through the automatic terminal information service (ATIS)) should be available to the flight crew via flight deck communication avionics interfaces, as per the simulated aircraft equipment.	✓	✓	✓		✓
12.7.2	Flight crew access (information): The method and details (e.g. radio frequencies, ACARS) used to access automated weather reports should match the information available to the flight crew (e.g. via charts and documentation, EFB applications and the FMS).	✓	✓	✓		✓
12.7.3	Real-world location reports: Automated weather reports should be available to the flight crew for all stations or operating sites that are modelled to match real-world locations.		✓	✓		✓
12.7.4	Reported weather conditions (single): A minimum of one set of automated reported weather conditions should be available to the flight crew if all weather-related parameters are the same for all stations or operating sites that may be used during a scenario.	✓				✓
12.7.5	Message content: The automated reported weather message should include, at a minimum, station or operating site, reference take-off and landing area, temperature, wind, altimeter setting, clouds, visibility, take-off and landing area conditions and also any predefined conditions if these are required elements of the weather-reporting station or operating site.	✓	✓	✓		✓

12: Navigation						
Feature general requirements		Feature fidelity level			Verification and validation	
		G	R	S	QTG	F&S
12.7.6	Message format (International Civil Aviation Organization (ICAO): The format and content of automated reported weather messages should adhere to standard ICAO specifications in accordance with ICAO <i>Annex 11 – Air traffic services</i> .	✓	✓	✓		✓
12.7.7	Reported weather conditions: The reported weather conditions should correspond to those set on the IOS.	✓	✓	✓		✓
12.8	Surveillance and data link communication systems This section is applicable if the ownship is equipped with the associated aircraft systems (e.g. ADS-B, ADS-C, ACARS, CPDLC, FANS; refer to CS FSTD.QB.104). Note: If the FSTD to be qualified includes a SATCE system, then refer to CS FSTD.QB.160 for further requirements for automated data link communications.					
12.8.1	If the ownship is equipped with ADS-B in capability and surrounding traffic is equipped with ADS-B out capability, the FSTD should simulate the reception of ADS-B messages. The simulation should reflect realistic transmission range, update rates and message content, making traffic surveillance data available to systems that process ADS-B inputs.	✓	✓	✓		✓
12.8.2	If the ownship is equipped with data link communication capabilities, such as an ACARS, CPDLC or FANS, the FSTD should simulate the exchange of messages between the ownship and external systems. The simulation should reflect realistic message formatting, transmission paths, communication latency and message delivery timing in accordance with operational use.	✓	✓	✓		✓

GM1 FSTD.QB.112 Flight simulation training device general requirements for navigation

Summaries of the fidelity levels for the navigation FSTD feature are presented in the table below. These summaries are broad overviews provided for guidance only. They do not encompass all aspects and requirements and are not intended to replace CS FSTD.QB.112.

12: Navigation
Specific: Navigational data, including the corresponding departure, en route and arrival/approach/landing facilities for a selection of operating sites, is provided. Navigation and communication aids are usable within range or line of sight without restriction, as applicable to the geographic area, and dependent on the height of the aircraft above the horizon (i.e. should be unusable when outside range or line of sight). Automated weather reporting (e.g. ATIS) is available if a communication system is installed. The transmission and exchange of surveillance and data link communications reflect realistic characteristics (e.g. range, message content, latency), if associated aircraft systems are installed.
Representative: Navigational data, including the corresponding departure, en route and arrival/approach/landing facilities for a selection of operating sites, is provided. Navigation and communication aids are usable within range or line of sight. Automated weather reporting (e.g. ATIS) is available if a communication system is installed. The transmission and exchange of surveillance and data link communications reflect realistic characteristics (e.g. range, message content, latency), if associated aircraft systems are installed.
Generic: Navigational data, including the corresponding departure, en route and arrival/approach/landing facilities for a selection of operating sites, is provided. Navigation and communication aids are usable within range or line of sight. Navigational data may be fictional, in which case associated fictional charts are provided. Automated weather reporting (e.g. ATIS) is available if a communication system is installed. The transmission and exchange of surveillance and data link communications reflect realistic characteristics (e.g. range, message content, latency), if associated aircraft systems are installed.

GM2 FSTD.QB.112 Flight simulation training device general requirements for navigation

CS FSTD.QB.112 requires that instructor controls be provided for the navigation environment. The IOS should enable the activation of navigation-related failure cases. The following examples indicate

typical controls that may be included, but these are not exhaustive and the list does not imply that all must be provided. The instructor controls can include:

- (a) simulating the failure of individual ground stations (e.g. ILS, VOR, DME, LOC, NDB, markers);
- (b) failing the identification morse code of the ground station;
- (c) adjusting the quality of signals from navigation aids to simulate signal degradation;
- (d) adjusting beam alignment (e.g. to simulate localiser misalignment);
- (e) activating false glideslope signals to simulate signal reflections or harmonics of the main signal;
- (f) losing visible GNSS satellites;
- (g) GNSS jamming;
- (h) GNSS spoofing.

GM3 FSTD.QB.112 Flight simulation training device general requirements for navigation

SURVEILLANCE AND DATA LINK COMMUNICATION SYSTEMS

Real-world aircraft operating rules define when, and in which airspace, ADS-B, ADS-C, or air traffic control (ATC) data link communication capabilities are required. The intended use of the FSTD may include training tasks that require the simulation of these capabilities. This GM provides basic information on these technologies.

Automatic Dependent Surveillance – Broadcast (ADS-B) is a surveillance technology that automatically transmits an aircraft's position and other data to nearby aircraft and ground stations. It supports real-time situational awareness and is used by ATC for surveillance purposes.

Automatic Dependent Surveillance – Contract (ADS-C) is a surveillance technology that sends position and intent data from the aircraft to ATC at defined intervals or in response to specific events. It is typically used to track aircraft in oceanic and remote areas where radar coverage is unavailable.

Data link communications refer to the exchange of non-voice messages between the ownship and ground-based systems—for example, ATC services via CPDLC or company operations via ACARS. These communications may be implemented as part of the FANS (Future Air Navigation System). Data link messages may be transmitted through various technologies, such as satellite links or radio systems.

The navigation feature concerns the simulation of signal transmission and data exchange, including those related to navigation, communication, and surveillance functions, as affected by environmental aspects such as distance and line of sight. For the requirements on how the associated aircraft systems should function, refer to CS FSTD.QB.104. If the FSTD to be qualified includes a SATCE system, then refer to CS FSTD.QB.160.

ATMOSPHERE AND WEATHER

CS FSTD.QB.113 Flight simulation training device general requirements for atmosphere and weather

13: Atmosphere and weather

Defines the level of complexity of the simulated weather conditions, from ambient temperature and pressure to full thunderstorm modelling, etc.

Note: Requirements of this feature concerning the visual scene are applicable only if the visual cueing FSTD feature is at level G, R or S.

Note: The requirements should be read in conjunction with the CS FSTD.FST.020 and CS FSTD.FST.030 visual cueing requirements to fully understand the details to be provided.

The general requirements for the atmosphere and weather FSTD feature are presented in the table below.

13: Atmosphere and weather								
Feature general requirements				Feature fidelity level			Verification and validation	
		G	R	S	QTG(A)	QTG(H)	F&S	
13.1	Environment simulation							
13.1.1	Fully integrated dynamic environment simulation with weather effects (on ground / in air) should be provided. The environment should be integrated with the entire simulation.		✓	✓				✓
13.1.2	A basic atmospheric model, including pressure, temperature, visibility, cloud base and winds, should be provided. The environment should be sufficient to permit accurate system operation and indications and be synchronised with appropriate features to provide integrity.	✓						✓
13.2	Atmosphere simulation							
13.2.1	The standard atmosphere should be simulated, and the instructor should have control over key parameters.	✓	✓	✓				✓

13: Atmosphere and weather								
Feature general requirements		Feature fidelity level			Verification and validation			
		G	R	S	QTG(A)	QTG(H)	F&S	
13.2.2	The atmosphere simulation should include: (1) wind speed, direction and gusts; (2) air temperature and its lapse rate; (3) air pressure; (4) turbulence levels.	✓	✓	✓				✓
13.2.3	Turbulence effects should be simulated by using applicable cues on all axes, including: (1) effects on instruments; (2) effects on aerodynamics and handling as a result of the atmospheric disturbances; (3) effects on the controls, as applicable (e.g. reversible flight controls); (4) motion and/or vibration effects, if installed.	✓	✓	✓				✓
13.3	Visual weather representation Note: This section is applicable only if a visual cueing FSTD feature is simulated and is at fidelity level G, R or S.							
13.3.1	The visual scene should include: (1) various visibility and RVR conditions, including fog and patchy fog effects; (2) effects on operating site lighting (including variable intensity and fog effects).		✓	✓				✓
13.3.2	The visual scene should include various visibility and RVR conditions.	✓						✓
13.3.3	The visual scene should permit gradual visibility changes entering and breaking out of the cloud for all times of day, including: (1) a realistic transition to IMCs; (2) a realistic transition to visual contact from IMCs.	✓	✓	✓				✓
13.3.4	The FSTD should employ cloud layers representing few, scattered, broken and overcast cloud conditions, giving partial or complete obstruction of the ground scene.	✓	✓	✓				✓

13: Atmosphere and weather							
Feature general requirements		Feature fidelity level			Verification and validation		
		G	R	S	QTG(A)	QTG(H)	F&S
13.3.5	For at least one cloud layer, it should be possible to perceive and assess broken and scattered cloud thickness if looking at the clouds from the outside. It should be possible to enter broken and scattered clouds from the sides, with appropriate transitions provided.			✓			✓
13.3.6	The clouds should provide an in-cloud airspeed effect in the visual scene.			✓			✓
13.3.7	The clouds, fog and weather effects should affect the ownship external lights and those of other aircraft within visual range (e.g. haze, reflection) in the visual scene.		✓	✓			✓
13.4	Weather effects Note: Visual representations of these are applicable only if a visual cueing FSTD feature is simulated and is at fidelity level G, R or S.						
13.4.1	The environment simulation should include the following, with visual representations provided as applicable: (1) rain; (2) hail; (3) storm cells (i.e. localised storm area, rain cell); (4) thunderstorms (i.e. including thunder and lightning effects); (5) falling snow; (6) blowing snow; and (7) blowing sand. The precipitation and blowing snow or sand should be affected by the wind direction and velocity.			✓			✓
13.4.2	The FSTD should provide special weather representations of light, medium and heavy precipitation near a thunderstorm on take-off and during approach and landing. Representations need only be presented at and below an altitude of 2 000 ft (610 m) above the operating site surface and within 10 miles (16 km) of the operating site.			✓			✓

13: Atmosphere and weather							
Feature general requirements		Feature fidelity level			Verification and validation		
		G	R	S	QTG(A)	QTG(H)	F&S
13.5	For aeroplanes: wind shear and microburst						
13.5.1	The FSTD should employ wind shear models that provide training on the recognition of and necessary corrective pilot actions for the following critical phases of flight: (1) prior to take-off rotation; (2) at lift-off; (3) during initial climb; and (4) short final approach, below 500 ft AGL.		✓	✓	2.g		✓
13.5.2	The SOJ should reference the wind shear implementation method(s) used. Wind shear models should be representative of measured or accident-derived winds, but may include simplifications that ensure repeatable encounters. For example, models may consist of independent variable winds in multiple simultaneous components. Wind models from the United States Federal Aviation Administration (FAA) <i>Aviation Weather Handbook</i> (Ref. FAA-H-9083-28), the Royal Aircraft Establishment’s <i>Wind Shear and Vortex Wake Research in UK – 1982</i> (Ref. 19830025568), the United States joint airport weather studies project publication <i>The Downburst, Microburst and Macrobust</i> (Ref. PB-148880) or other recognised sources may be implemented and should be supported and properly referenced in the SOJ. Wind models from alternate sources may also be used if they are supported by aeroplane-related data and such data is also properly supported and referenced. Use of alternate data should be coordinated with the competent authority prior to submitting the QTG for approval.		✓	✓	2.g		
13.5.3	The FSTD should employ microburst models.			✓			✓
13.6	For rotorcraft: wind shear and microburst						
13.6.1	The FSTD should employ wind shear models that provide training on the recognition of and necessary corrective pilot actions for wind shear phenomena.			✓			✓

13: Atmosphere and weather							
Feature general requirements				Feature fidelity level		Verification and validation	
		G	R	S	QTG(A)	QTG(H)	F&S
13.6.2	The FSTD should employ microburst models.			✓			✓
13.7	IOS controls – environment						
13.7.1	The instructor controls should be provided for environment and weather effects and/or conditions, as applicable to the aircraft type or group.	✓	✓	✓			✓
13.7.2	The instructor controls should be provided for: (1) wind speed, direction and gusts; (2) air temperature and its lapse rate; (3) air pressure; (4) turbulence levels; (5) visibility and RVR, including fog and patchy fog effect (units used should correspond to those in use at the operating site or in the region); (6) cloud layers with adjustable cloud types, bases, tops, sky coverage and scud effect; (7) rain levels; (8) storm cells (i.e. localised storm area, rain cell); (9) thunderstorms (i.e. including thunder and lightning effects); (10) weather effects.		✓	✓			✓
13.7.3	If simulated, the instructor controls should be provided for: (1) wind speed, direction and gusts; (2) air temperature and its lapse rate; (3) air pressure; (4) turbulence levels; (5) visibility (in km or NM) and RVR (in m), if relevant; (6) cloud layers with adjustable cloud types, bases, tops and sky coverage.	✓					✓

13: Atmosphere and weather							
Feature general requirements		Feature fidelity level			Verification and validation		
		G	R	S	QTG(A)	QTG(H)	F&S
13.7.4	The instructor controls should be provided for: (1) wind shear models with varying degrees of severity; and (2) microbursts. If a wind shear or microburst model is not survivable, it should be labelled on the IOS as unsurvivable.			✓			✓
13.7.5	For aeroplanes to be qualified for UPRT: If wake turbulence scenarios are simulated, they should be assessed in accordance with CS FSTD.QB.017.			✓			✓
13.7.6	For aeroplanes: The instructor controls should be provided for wind shear models with varying degrees of severity. If a wind shear is not survivable, it should be labelled on the IOS as unsurvivable.		✓				✓
13.7.7	The instructor controls should be capable of: (1) configuring multiple cloud layers of different cloud types and coverage; (2) specifying different wind speeds and directions at various altitudes, from ground to high cruise altitudes. These cloud layers and wind conditions should seamlessly blend to create a smooth and continuous atmospheric transition.	✓	✓	✓			✓
13.7.8	The SOJ should include a description of any additional atmospheric and weather effects that are simulated in excess of the required ones. Refer to CS FSTD.QB.017.	✓	✓	✓			

GM1 FSTD.QB.113 Flight simulation training device general requirements for atmosphere and weather

Summaries of the fidelity levels for the atmosphere and weather FSTD feature are presented in the table below. These summaries are broad overviews provided for guidance only. They do not encompass all aspects and requirements and are not intended to replace CS FSTD.QB.113.

13: Atmosphere and weather
Specific: A fully integrated dynamic environment simulation with weather effects is provided. The environment is synchronised with other features. The visual weather presentation provides realistic cues for all weather conditions and effects, including precipitation, storms, wind shears and microbursts.
Representative: A fully integrated dynamic environment simulation with weather effects is provided. The environment is synchronised with other features. The visual weather presentation provides realistic cues for visibility/RVR effects. Wind shear simulation is provided for aeroplanes.
Generic: A basic atmospheric model is provided. The environment is sufficient to permit accurate system operation and indications and is synchronised with appropriate features to provide integrity. The visual weather presentation provides transitions for entering and breaking out of clouds/IMCs and for visibility/RVR conditions.

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ADDITIONAL ATMOSPHERIC AND WEATHER EFFECTS

The FSTD may employ other atmospheric and weather effects in addition to the required ones. The SOJ should include a description of any additional effects. Refer to CS FSTD.QB.017. Such effects may include but are not limited to:

- (a) arctic sea smoke;
- (b) katabatic winds;
- (c) mountain waves and effects (rotors, demarcation lines, etc.);
- (d) oil rig exhaust turbulence;
- (e) wind masking due to buildings, trees, etc.;
- (f) turbulence and wind effects caused by topography and structures.

OPERATING SITES AND TERRAIN

CS FSTD.QB.114 Flight simulation training device general requirements for operating sites and terrain

14: Operating sites and terrain
Defines the complexity and level of detail of the operating sites (e.g. aerodrome, landing areas) and terrain modelling required to be simulated. This includes such items as generic versus customised operating sites (e.g. airport, heliport, vertiport or other landing area), visual scenes and terrain elevation. Since some aircraft types can land at a variety of landing area types, often outside aerodromes and/or controlled airspace, the environment includes these types of landing areas in the visual scene modelling. Note: The requirements should be read in conjunction with the CS FSTD.FST.020 and CS FSTD.FST.030 visual cueing requirements to fully understand the details to be provided. Note: Requirements of this feature concerning the visual image are applicable only if the visual cueing FSTD feature is at fidelity level G, R or S.

The general requirements for the operating sites and terrain FSTD feature are presented in the table below.

14: Operating sites and terrain						
Feature general requirements				Feature fidelity level		Verification and validation
		G	R	S	QTG	F&S
14.1	Aerodrome area scenes The following is the minimum aerodrome area model content requirement to satisfy the requirements of visual capability tests and provide suitable visual cues to allow for the completion of all functions and subjective tests described in Subpart E.					
14.1.1	There should be multiple tier I (refer to CS FSTD.QB.114.1), location-specific, real-world 3D airport or heliport models to demonstrate the capabilities of the FSTD. They should be among the operating sites simulated in accordance with the navigation FSTD feature (refer to CS FSTD.QB.112).			✓		✓

14: Operating sites and terrain						
Feature general requirements		Feature fidelity level			Verification and validation	
		G	R	S	QTG	F&S
14.1.2	The airport, heliport or aircraft landing area(s) may be either fictional or a real-world area. They should be among the operating sites simulated in accordance with the navigation FSTD feature (refer to CS FSTD.QB.112).		✓			✓
14.1.3	For aeroplanes: Multiple fully customised 3D airport models should be provided. They should be among the operating sites simulated in accordance with the navigation FSTD feature (refer to CS FSTD.QB.112).		✓			✓
14.1.4	For rotorcraft: Multiple operating sites, including at least one fully customised 3D airport/heliport model and a minimum of one additional airport or heliport, should be provided. They should be among the operating sites simulated in accordance with the navigation FSTD feature (refer to CS FSTD.QB.112).		✓			✓
14.1.5	The system should include an airport model (e.g. a generic or real-world model) with an aircraft landing area available that is operable at the same operating sites as the airports which are simulated for the navigation FSTD feature (refer to CS FSTD.QB.112).	✓				✓
14.1.6	Airport, heliport and/or helipad models should have realistic terrain modelling, markings and lighting and correct runway orientation, dimensions and taxiways. The visual model should be correctly aligned (e.g. navigation aids, runways, gates, terrain, buildings).			✓		✓
14.1.7	For an airport, helipad and/or heliport, the scene should provide: (1) correct runway or landing area orientation, markings and lighting; and (2) representative terrain modelling, dimensions and taxiways.		✓			✓
14.1.8	The operating site visual scene should be representative of a typical airport/airfield or landing area, including terrain surface (grass, tarmac) and characteristics (soft, hard, slippery), but not necessarily geo-specific.		✓			✓

14: Operating sites and terrain						
Feature general requirements		Feature fidelity level			Verification and validation	
		G	R	S	QTG	F&S
14.1.9	The fidelity of the visual scene should be sufficient for the aircrew to visually identify the airport and/or rotorcraft landing area; determine the position of the simulated aircraft within the visual scene; successfully accomplish take-offs, approaches and landings; and manoeuvre around the airport on the ground or in hover taxi, as necessary.		✓			✓
14.1.10	The operating site visual scene should be representative of a typical airport/airfield or landing site, including terrain surface (grass, tarmac) and characteristics (soft, hard, slippery), but not necessarily geo-specific. The scene should include highly detailed and accurate surface depiction of the terrain surface within an approximate area from 400 m (0.25 sm) before to 400 m (0.25 sm) beyond the runway approach end, with a total width of approximately 400 m (0.25 sm) including the width of the runway.		✓			✓
14.1.11	Generic airport/heliport model(s) with terrain topographical elements should: (1) have a visible horizon; and (2) include runway and taxiway markings and lighting.	✓				✓
14.1.12	The aerodrome visual scene should include the terrain, obstacles and natural and cultural elements within 46 km (25 NM) of the aerodrome. Note: This refers to natural and cultural elements that are typically used for pilot orientation in flight.			✓		✓
14.1.13	The aerodrome visual scene should include representative depiction of terrain, obstacles and natural and cultural elements within 46 km (25 NM) of the aerodrome.		✓			✓
14.1.14	The visual operating sites should support operations in day, twilight and night conditions.	✓	✓	✓		✓
14.1.15	To support flight from one operating site to another, the visual image should provide a smooth transition to new operating scenes or sites.	✓	✓	✓		✓
14.2	Airport / landing area					

14: Operating sites and terrain						
Feature general requirements		Feature fidelity level			Verification and validation	
		G	R	S	QTG	F&S
14.2.1	The runway or landing area should be identifiable from the downwind leg or approach direction at all times of day. Airport, heliport or landing site lights should be representative of those of the airport/heliport/site modelled. For example, runway edge lights should be bidirectional or omnidirectional to enable visual approaches and visual circuits.		✓	✓		✓
14.2.2	The runway or landing area should have runway markings, lighting and signage, including a windsock that responds to wind cues.		✓	✓		✓
14.2.3	As appropriate to the real-world airport: (1) airport models should be able to display multiple runways simultaneously (e.g. two parallel runways and one crossing runway); (2) at least two approach and runway lighting systems should be capable of being lit simultaneously.			✓		✓
14.2.4	The airport should have appropriate approach lighting systems – including VFR circuits and landings, non-precision approaches and landings and precision approaches and landings – with appropriate colours, directionality and spacing. The lighting systems should be appropriate for the operational needs of the simulated aircraft type.		✓	✓		✓
14.2.5	The airport should have runway and approach lighting systems with appropriate colours, directionality and spacing.	✓				✓
14.2.6	If an aircraft can land at a variety of landing area types, including those outside aerodromes and/or controlled airspace, the environment should include these types of landing areas in the visual scene modelling.		✓	✓		✓
14.3	Rotorcraft landing areas and sites					

14: Operating sites and terrain						
Feature general requirements		Feature fidelity level			Verification and validation	
		G	R	S	QTG	F&S
14.3.1	For rotorcraft: There must be at least three non-airport landing areas, including: (1) at least one aircraft landing area situated on a substantially elevated surface with respect to the surrounding structures or terrain (e.g. building top, offshore oil rig); (2) at least one aircraft landing area that meets the definition of a 'confined landing area'; and (3) at least one aircraft landing area on a sloped surface if the slope is at least 2.5°.		✓	✓		✓
14.3.2	For rotorcraft: There should be a sloped landing area that supports different slopes up to the maximum permitted slope.		✓	✓		✓
14.3.3	For rotorcraft FSTD used for ship operations training: At least one surface ship with a landing deck, operating within different sea states.		✓	✓		✓
14.3.4	For rotorcraft: A generic slope landing area should be provided.	✓				✓
14.3.5	For rotorcraft: The airport or heliport and the aircraft landing area may be contained within the same model. If this option is selected, the approach path to the airport runway(s) and the approach path to the aircraft landing area must be different.		✓			✓
14.4	Airport / landing areas – ground models					
14.4.1	Multiple airport / landing area scenes should provide a ground model that supports ground manoeuvring (e.g. typical gate/apron/parking positions for the aircraft type). This should include elements relevant to taxi routes, with marker boards, stop bars, runway guard lights, approach and runway lighting, centre line lights, lead-in and lead-off and taxiway lighting.			✓		✓

14: Operating sites and terrain						
Feature general requirements		Feature fidelity level			Verification and validation	
		G	R	S	QTG	F&S
14.4.2	The displayed scene should correspond to the appropriate surface contaminants and include runway lighting reflections for wet, partially obscured lights for snow or suitable alternative effects. For wet and patchy wet airport contaminants, some other, non-lighting object reflections should also be included, such as but not necessarily or limited to terminal buildings, lamp posts, jetways, aircraft and vehicles.			✓		✓
14.4.3	Ground objects should be presented in sufficient quantity to permit orientation and alignment for ground reference manoeuvring (i.e. traffic pattern and/or high/low reconnaissance), hovering and taxiing. All ground objects must be presented with proper size and perspective, presenting no distraction to the pilot.		✓	✓		✓
14.4.4	Ground vehicles should be available at multiple operating sites undertaking predefined manoeuvres, with instructor control over whether a conflict with the ownship occurs, including on take-off and in landing areas (e.g. a runway incursion threat).		✓	✓		✓
14.4.5	Aircraft traffic should be available at multiple operating sites undertaking predefined manoeuvres, with instructor control over whether a conflict with the ownship occurs. Aircraft need only exhibit basic or characteristic performance appropriate to the aircraft type.		✓	✓		✓
14.5	Aerodrome (e.g. airport, heliport) currency					
14.5.1	It should be possible for specific – tier I and II – aerodromes, airports or heliports or landing areas modelled in the system that represent real-world locations to be updated and maintained to incorporate permanent changes. Note: For requirements on updates to the modelled operating sites, refer to the requirements in Part-ORA (Annex VII) of Commission Regulation (EU) No 1178/2011.			✓		✓
14.6	Geographic area					

14: Operating sites and terrain						
Feature general requirements		Feature fidelity level			Verification and validation	
		G	R	S	QTG	F&S
14.6.1	Terrain should have visual texturing to give sufficient cues to allow pilots to determine speed, altitude, attitude and position information.	✓	✓	✓		✓
14.6.2	Geo-specific scene content and terrain: Colours and textures should represent a particular geographic area environment as appropriate for the location of the operating site (e.g. vegetation, landscape, desert, arctic, jungle).		✓	✓		✓
14.6.3	For rotorcraft: The following effects should be displayed as a function of wind and downwash: (1) movement of trees in confined area landing sites; and (2) movement of vegetation (e.g. grass, trees).			✓		✓
14.6.4	Integration of the aircraft systems and the visual terrain model: If an aircraft system provides information on the terrain (e.g. radio altimeter, TAWS/HTAWS, synthetic vision system, radar, forward-looking infrared camera), the visual image should correspond to the information provided by the system (e.g. topography, obstacles).		✓	✓		✓
14.6.5	Geo-specific scene content and terrain is to include topographical information, such as buildings, trees and obstacles (and other elements, such as unprepared landing sites, confined areas, ships and rigs, as applicable to the aircraft), that allows for VFR navigation training.		✓	✓		✓
14.6.6	Generic scene content and terrain, texture and ground objects are acceptable. Terrain does not need to represent any geo-specific location.	✓				✓
14.6.7	The FSTD should provide operational visual scenes that portray physical relationships known to cause landing illusions for pilots – for example, short runways, landing approaches over water, uphill or downhill runways, rising terrain on the approach path and unique topographical elements.		✓	✓		✓

14: Operating sites and terrain						
Feature general requirements		Feature fidelity level			Verification and validation	
		G	R	S	QTG	F&S
14.6.8	Daylight visual scenes should have sufficient scene content to allow users to recognise the airport, the terrain and major landmarks around the airport. Landmarks include elements such as roads, rivers and fields and also significant identifiable buildings. If 3D models are included, these should reflect the real-world appearance. The scene content must allow a pilot to successfully accomplish a visual landing.		✓	✓		✓
14.6.9	For rotorcraft: Daylight, twilight and night visual scenes will be capable of displaying sufficient scene content to allow users to recognise generic aerodromes, heliports, terrain and landmarks around the final approach and take-off area (FATO) and to accomplish low-airspeed/low-altitude manoeuvres, including lift-off, hover, translation to and from hover into forward flight, landing and touchdown adequate for instrument training.	✓				✓
14.6.10	Twilight visual scenes should have sufficient scene content to allow users to recognise the airport, the terrain and major landmarks around the airport. The scene content must allow a pilot to successfully accomplish a visual landing. Twilight scenes, as a minimum, must provide full-colour presentations of reduced ambient intensity and sufficient surfaces with appropriate textual cues that include self-illuminated objects, such as road networks, ramp lighting and airport signage, to conduct a visual approach, landing and airport movement (taxi).		✓	✓		✓
14.6.11	Night visual scenes should have sufficient content to allow users to recognise the airport, the terrain and major landmarks around the airport. The scene content must allow a pilot to successfully accomplish a visual landing.		✓	✓		✓
14.6.12	Night scene should include stars, ground lights and sufficient texture to clearly identify the aircraft attitude and differentiate between the terrain and sky.	✓	✓	✓		✓
14.7	Visual scene for rotorcraft devices with the VIS feature at the G fidelity level					

14: Operating sites and terrain						
Feature general requirements		Feature fidelity level			Verification and validation	
		G	R	S	QTG	F&S
14.7.1	The system should provide a visual scene with sufficient scene content to allow a pilot to successfully accomplish a visual landing. Scenes should include a definable horizon and typical terrain characteristics such as fields, roads, bodies of water and surfaces illuminated by rotorcraft landing lights. Note: The airport model may be generic.	✓				✓
14.8	Visual feature recognition					
14.8.1	The following visual scene elements should be recognisable at an appropriate distance depending on the time of day (day, twilight, night): (1) surface resolution and runway lighting; (2) taxiways (day) and taxiway lighting (twilight/night); (3) approach and airport/airfield lighting.	✓	✓	✓		✓
14.8.2	Scenes should include a definable horizon and typical terrain characteristics, such as fields, roads and bodies of water. If provided, the directional light seen at the horizon must have correct orientation and be consistent with surface shading effects.			✓		✓
14.8.3	Surfaces on the ground that are in range of the aircraft or external lighting should be appropriately illuminated.			✓		✓
14.8.4	All ground objects must be presented with the proper size and perspective, presenting no distraction to the pilot.			✓		✓
14.8.5	There should be a highly detailed and accurate surface depiction of the terrain surface within an area that is large enough to achieve cross-country flying under VFR conditions for a flight of at least 30 min.			✓		✓

14: Operating sites and terrain						
Feature general requirements		Feature fidelity level			Verification and validation	
		G	R	S	QTG	F&S
14.8.6	There should be sufficient scene and terrain detail within an area that is large enough to achieve cross-country flying under VFR conditions for a flight of at least 30 min.	✓	✓			✓
14.8.7	In good visibility, the horizon should be sufficiently visible to determine the aircraft attitude. The day or night horizon representation can be obscured by multiple ragged or solid cloud layers, with adjustable bottoms and tops of cloud layers, corresponding to the aircraft altitude.		✓	✓		✓
14.8.8	Terrain surface references and topographical elements should be sufficient to support VFR navigation in accordance with appropriate charts (typically 1:500 000, 1:250 000 or 1:100 000 scale mapping).		✓	✓		✓
14.8.9	In good visibility, the horizon should be sufficiently visible to determine the aircraft attitude. The day or night horizon representation can be obscured by a solid cloud layer, with adjustable bottom and top of the cloud layer, corresponding to the aircraft altitude.	✓				✓
14.8.10	Daylight scenes should include sky and terrain suitably coloured and textured to clearly identify aircraft attitude and height.	✓				✓
14.9	Terrain					
14.9.1	A detailed 3D terrain contour model should be provided and support aircraft system simulation (e.g. TAWS) as applicable. The visual representation of the terrain should match the terrain contour model. The terrain model should cover at least the areas simulated in accordance with the navigation feature (refer to CS FSTD.QB.112) in the vicinity of at least one operating site.		✓	✓		✓
14.9.2	A basic terrain model should be provided to support aircraft systems simulation (e.g. TAWS/HTAWS) and should be correlated with the visual model as applicable.	✓				✓
14.10	Sea state					

14: Operating sites and terrain						
Feature general requirements		Feature fidelity level			Verification and validation	
		G	R	S	QTG	F&S
14.10.1	For aeroplanes, if required for training, 3D sea states that can be controlled by the instructor should be included.		✓	✓		✓
14.10.2	For rotorcraft, if required for training: The following effects should be possible to display (as applicable): (1) sea states corresponding to the sea swell conditions selected on the IOS, including the resulting movement of ships used for deck landings; (2) effects of wind conditions on wind-driven waves selected on the IOS; (3) effects of wind lanes on the water surface; (4) marine vessel movement that at least conforms to the sea state in terms of pitch, roll and heave.		✓	✓		✓
14.11	Downwash effects for rotorcraft					
14.11.1	The effects of ownship downwash on various surfaces, such as snow, sand, water, grass, trees and vegetation, should be simulated, including associated effects of reduced visibility due to recirculation (e.g. brownout due to sand, white-out due to snow).		✓	✓		✓
14.12	Low-visibility operations					
14.12.1	The system should include multiple airport scenes with low-visibility details and functionality. The system should include gate/apron/parking positions typical for the aircraft type being simulated, leading to a low-visibility taxi route with marker boards, stop bars, runway guard lights, the required approach and runway lighting, daylight-visible taxi lighting and taxiway lighting, ensuring consistency with real-world low-visibility procedures.			✓		✓

14: Operating sites and terrain						
Feature general requirements		Feature fidelity level			Verification and validation	
		G	R	S	QTG	F&S
14.12.2	There should be at least one airport scene with functionality to support the approach to the lowest minima applicable to the simulated aircraft type and as stated in the ESL. This airport scene should support, for example, a gate/apron/parking position typical of the aircraft type being simulated, leading to low-visibility taxi routes with marker boards, stop bars, runway guard lights and the required approach, runway and taxiway lighting.		✓			✓
14.13	Low-visibility special effects					
14.13.1	If the surface movement guidance and control system (SMGCS) functionality is modelled on the FSTD, real-world operating sites should support low-visibility operations including, as applicable to the real-world operating sites, the SMGCS functionality for each runway to allow low-visibility taxiing consistent with real-world procedures to/from the gate / parking positions and the active runway.			✓		✓
14.13.2	The system will be capable of displaying moving follow-me trucks to guide the ownship to the desired position, as applicable to the real-world procedures.			✓		✓
14.13.3	The system should provide visual effects for: (1) light poles; (2) raised runway edge lights as appropriate; (3) the glow associated with approach lights in low visibility before physical lights are seen; and (4) the glow associated with airport illumination (lamp posts and other items) in low visibility.			✓		✓
14.14	Instructor controls – operating sites					

14: Operating sites and terrain						
Feature general requirements		Feature fidelity level			Verification and validation	
		G	R	S	QTG	F&S
14.14.1	The FSTD should have instructor controls for (as applicable to the models): (1) airport selection; (2) airport lighting (separate controls for each type of light); (3) stop bars; (4) visual aids for navigation; (5) lighting for obstacle clearance (as applicable); (6) runway conditions (e.g. friction or braking action); and (7) surface contaminants for the runways, taxiways and apron or operating site. The default settings and operations should be in accordance with ICAO <i>Annex 14 – Aerodromes – Volume I</i>.	✓	✓	✓		✓
14.14.2	The instructor should be able to control the operating site runway, approach and taxiway lighting with use of five intensity levels and an off setting.			✓		✓
14.14.3	The instructor should be able to control the operating site runway, approach and taxiway lighting intensity.	✓	✓			✓
14.14.4	The instructor station should have controls for activating and deactivating the ground and airborne conflicting traffic.		✓	✓		✓

GM1 FSTD.QB.114 Flight simulation training device general requirements for operating sites and terrain

Summaries of the fidelity levels for the operating sites and terrain FSTD feature are presented in the table below. These summaries are broad overviews provided for guidance only. They do not encompass all aspects and requirements and are not intended to replace CS FSTD.QB.114.

14: Operating sites and terrain
Specific: A selection of specific, real-world 3D airport or heliport models with accurate terrain modelling, runway markings and lighting are provided. The visual scene colours and textures represent a particular geographic area. Major landmarks around the airport are present, and ground objects are realistically sized. There is a highly detailed and accurate surface depiction of the terrain surface within an area sufficient to achieve cross-country flying under VFR conditions. For rotorcraft, the FSTD simulates the effects of ownship downwash on various surfaces. The visual scene corresponds to actual surface conditions.
Representative: At least one fully customised 3D airport model, which may be either fictional or a real-world airport, is provided. The visual scene colours and textures represent a particular geographic area. Major landmarks around the airport are present. Terrain surface references and topographical elements are sufficient to support VFR navigation in accordance with VFR charts. For rotorcraft, the FSTD simulates the effects of ownship downwash on various surfaces.
Generic: Fictional or real-world operating site model(s) with generic scene content, topographical elements, orientation, markings, lighting, dimensions and taxi routes for aircraft ground operations, including take-off and landing. Terrain does not need to represent any geo-specific location and may be flat. Visual texturing gives sufficient cues to permit pilots to determine speed, altitude and position information.

GM2 FSTD.QB.114 Flight simulation training device general requirements for operating sites and terrain

An FSTD may simulate training scenarios for which this CS-FSTD does not specify requirements. Such scenarios are typically related to various rotorcraft missions and/or often include a visual element that depicts the operating site and terrain. For such scenarios, refer to the requirements in CS FSTD.QB.017 and guidance in GM1 FSTD.QB.017.

CS FSTD.QB.114.1 Visual scene operating sites

(a) Applicability of this CS

This CS applies to all operating sites that are aerodromes – including airports and landing areas for rotorcraft – that are used in FSTD visual models for FSTDs with the operating sites and terrain FSTD feature at the specific fidelity level. All such aerodrome operating sites should be classified in accordance with paragraph (b) below. The FSTD operator should classify the operating sites as:

- (1) aerodrome (airport or heliport) models (refer to paragraph (c) below);
- (2) other operating sites and operating areas.

(b) The FSTD operator should have a list of all the operating sites that indicates the classification tier and capabilities of and the requirements that are met for each model. This should be provided to the competent authority prior to the initial evaluation.

(c) The minimum aerodrome model classifications for the S fidelity level are defined as follows.

For tier I, the detailed models accurately represent real-world aerodromes or locations. Whether modelling real-world airports or landing areas for rotorcraft, these operating sites are those that are used to demonstrate compliance with all the requirements of the operating sites and terrain FSTD feature's general requirements in CS FSTD.QB.114.

Not all of the elements of 'low-visibility special effects' described in CS FSTD.QB.114 need to be found in all tier I models.

For tier II, the detailed models of real-world operating sites may not include all of the visual elements and detail required at the specific fidelity level, but they should meet or exceed the requirements in CS FSTD.QB.114 at the representative fidelity level.

GM1 FSTD.QB.114.1 Visual scene operating sites

CLASSIFICATION OF VISUAL SCENE OPERATING SITES

This guidance provides further details of the classification of the visual scenes for the operating sites, which include aerodromes, airports, heliports, landing sites and other modelled operating areas.

(a) The list of operating sites – including aerodromes, landing sites and modelled operating areas – should include sufficient information to provide the user of the device with an understanding of the level of detail that each site offers and whether or not it is updated.

Note that a list of 'generic' (refer to paragraph (c) below) aerodrome models is not required.

- (1) A list of geographic operating areas should include a classification of the level of detail that each area offers for training (e.g. low-level rotorcraft operations, VFR cross-country flights).
- (2) A list of aerodromes should be included. As a non-exhaustive example, the aerodrome list may contain the identification of the visual scene details, for each aerodrome and

site, necessary for operation, namely:

- (i) classification tier;
- (ii) identification of the runways and helispots in use;
- (iii) identification of gates and ramp areas modelled;
- (iv) identification of the level of low-visibility operations (e.g. SMGCS, taxi) supported;
- (v) identification of the special approaches that require particular model details to support training (e.g. RNP AR APCH, point-in-space approach, tactical air navigation as applicable) supported;
- (vi) identification of any winter scenes.

(b) Classification tiers are as follows.

(1) Tier I models are aerodrome (e.g. airport, heliport) models that demonstrate the full requirements of CS FSTD.QB.114. These are highly detailed airport models, providing realistic and immersive visual environments. They replicate the actual airport's physical and operational characteristics to the greatest extent possible. These models are expected to be kept up to date to reflect real-world changes. These models should include but are not limited to:

- (i) high level of detail replicating the airport environment (which should include full 3D renderings of permanent airport structures, including terminals, control towers, hangars and ground support equipment within the airport perimeter that appear on the airport chart;
- (ii) accurate runway and taxiway configurations, including markings, lighting and signage;
- (iii) detailed surrounding landmarks, terrain and obstacles (e.g. hills, buildings or water bodies);
- (iv) visual cues such as approach lighting systems, precision approach path indicators and runway edge lighting systems;
- (v) dynamic elements, such as moving vehicles or other aircraft, to enhance realism.

(2) Tier II models are real-world operational aerodromes that provide sufficient detail for particular training scenarios but may not include all of the visual elements and detail seen in tier I models. These models are acceptable for most training environments. They are expected to be updated to reflect any significant changes in the real-world (e.g. new runways or terminals). The essential visual elements of these models include but are not limited to:

- (i) moderate to high level of detail, focusing on essential visual and operational elements (typically including 3D renderings of significant airport structures, including terminals, control towers, hangars and ground support equipment;

- (ii) runway and taxiway layouts, including markings, lighting and signage;
 - (iii) approach lighting systems, precision approach path indicators and runway edge lighting systems.
- (3) Tier III models are fictional aerodromes that have particular purposes above those of a ‘generic’ model (refer to subparagraph (4), and Table 1 in paragraph (c) below). These are simplified operating site models designed for particular purposes for which full visual accuracy is not critical, but the training objective can still be achieved effectively. Examples include:
- (i) low-visibility operations in which detailed visual elements are not discernible;
 - (ii) instrument approaches in which reliance on instruments rather than external visual cues is the primary focus;
 - (iii) helicopter landing areas where operational procedures are more significant than precise visual replication;
 - (iv) basic terrain and obstructions sufficient for procedural training.

Their use is conditional upon operational restrictions that are communicated to users to prevent trainees from encountering the model’s inaccuracies and also a detailed justification of how the model meets the training objectives despite its limitations. Limits that may be applied to tier III models regarding their usage include, for example ‘valid for use only in visibility conditions less than 1 km or RVR 1 000 m’ or ‘valid for use only for approaches to Runway 22L and 22R’.

- (4) Generic models are airport models that represent operating sites corresponding to real-world locations, without requiring geo-specific terrain, buildings or textures. The visual models should provide scene content, including lakes, coastlines and surrounding terrain, appropriate runway orientation and numbering, markings, lighting, dimensions and generic basic taxi routes for aircraft ground operations, including take-off and landing.

A typical generic airport model may include a single runway, a taxiway on one side and simple structures such as ATC towers and aprons to support basic ground operations.

Visual texturing and scene cues should provide sufficient detail to allow the pilot to assess speed, altitude and positional awareness during ground and airborne operations.

- (c) In the context of the operating sites and terrain FSTD feature for the FSTD, Table 1 below gives guidance on different categories of operating sites.

Table 1. Guidance on different categories of operating sites

	Applicability	
	Aeroplane	Rotorcraft

Aerodromes – tier I	Real-world aerodromes (e.g. airports)	Real-world aerodromes (e.g. airports, heliports, helipads, oil rigs)
Aerodromes – tier II	Real-world aerodromes (e.g. airports)	Real-world aerodromes (e.g. airports, heliports, helipads, oil rigs)
Aerodromes – tier III	Fictional aerodromes	Fictional aerodromes
Aerodromes – generic	Generic aerodromes	Generic aerodromes
Operating areas	Geographic areas used for training (e.g. VFR cross-country flights)	Geographic areas used for training (e.g. low-level flying), which may include landing sites, confined areas and sloping sites

Note: The term ‘generic aerodrome’ above is not related to the fidelity level of the operating sites and terrain FSTD feature, but to the commonly used term for visual models that do not represent real aerodrome buildings or layouts. Refer to (b)(4) above.

MISCELLANEOUS

CS FSTD.QB.115 Miscellaneous flight simulation training device general requirements

15: Miscellaneous

The requirements contained within this section are applicable to all FSTDs and are not related to any feature fidelity levels.

Miscellaneous general requirements are presented in the table below.

15: Miscellaneous				
General requirements		Verification and validation		
		QTG(A)	QTG(H)	F&S
15.1	IOS Describes the overall requirements for the FSTD’s IOS.			
15.1.1	The IOS should be positioned on board or in close proximity to the flight deck, allowing a clear and unobstructed direct view of the pilots, their instruments and the external visual scene.			✓
15.1.2	The IOS should support the use of the FSTD to meet all the other requirements.			✓
15.1.3	An IOS with a seat should be provided.			
15.1.4	The IOS should not distract the instructor from observing the activities of the flight crew.			✓
15.1.5	If motion or control loading is installed, the FSTD should have emergency stop controls for these systems, accessible from both pilot and instructor seats.			

15: Miscellaneous				
General requirements		Verification and validation		
		QTG(A)	QTG(H)	F&S
15.1.6	The functions available on the IOS should include: (1) a map displaying operating sites, runways, navigation aids, ownship position and scale information; (2) current ownship airspeed, heading and altitude; (3) an approach plot showing the ownship's vertical and horizontal position relative to the runway track and glidepath.			✓
15.1.7	If the ownship is equipped with radio communication systems, the IOS interface should enable human ATC role play and communication with the flight crew.			✓
15.1.8	If the ownship is equipped with data link systems (e.g. ACARS, CPDLC), the IOS interface should enable observation and control of data link messages.			✓
15.2	Instructor controls Describes the controls that are required for the instructor during the use of the FSTD.			
15.2.1	Instructor controls should be provided to allow for the easy set-up of an aircraft state, all relevant system variables, repositioning, freezing, resetting and the insertion of malfunctions to simulate abnormal and emergency conditions.			✓
15.3	Simulated malfunctions			
15.3.2	The malfunctions available on the FSTD should be relevant for training and should include a description of the effects and impact for the instructor.			✓
15.3.3	An SOJ should describe how all the simulated malfunctions have been tested and verified.			
15.4	UPRT Applies only to aeroplane FSTDs that are to be qualified for UPRT.			

15: Miscellaneous				
General requirements		Verification and validation		
		QTG(A)	QTG(H)	F&S
15.4.1	The FSTD must have a real-time feedback tool that provides visual information to the instructor/evaluator whenever the FSTD training envelope or aeroplane operating limits have been exceeded. The feedback tool should include the following. (1) FSTD validation envelope. This must be in the form of an alpha/beta envelope (or equivalent method) depicting the confidence level of the aerodynamic model. This confidence level depends on the degree of flight validation or the source of predictive methods. There must be a minimum of flaps-up and flaps-down envelopes available. (2) Flight control inputs. These must enable the instructor/evaluator to assess the pilot's flight control displacements and forces (including fly-by-wire, as appropriate). (3) Aeroplane operational limits. This must display the aeroplane's operational limits during the manoeuvre, as applicable to the configuration of the aeroplane. An SOJ should define the source data used to construct the FSTD validation envelope. Refer to CS FSTD.QB.125.			✓
15.4.2	The FSTD should have a recording mechanism that may be utilised from and is triggered and displayed on the instructor station to display the FSTD's performance.			✓
15.4.3	The FSTD should have upset scenarios in which dynamic upset scenarios and aeroplane system malfunctions are used to drive the FSTD into an upset condition.			✓

15: Miscellaneous				
General requirements		Verification and validation		
		QTG(A)	QTG(H)	F&S
15.4.4	Upset scenarios: If the IOS is equipped with selectable aeroplane upsets, the IOS should provide guidance on the method used to drive the FSTD into an upset condition, including any malfunction or degradation of the FSTD's functionality required to initiate the upset. In such scenarios: (1) the unrealistic degradation of simulator functionality (e.g. degrading flight control effectiveness) to drive an aeroplane upset is generally not acceptable unless used purely as a tool for repositioning the FSTD with the pilot out of the loop; (2) consideration should be given to flight-envelope-protected aeroplanes, as artificially positioning the aeroplane in a specified attitude may incorrectly initialise flight control laws.			✓
15.4.5	Each upset prevention and recovery feature programmed on the IOS and the associated training manoeuvre should be evaluated by a suitably qualified pilot. (1) A suitably qualified pilot should: (a) hold a type rating qualification for the aeroplane being simulated; and (b) be familiar with the upset scenarios, associated recovery methods and the cues necessary to accomplish the required training objectives for the type and variant simulated. (2) The SOJ should also confirm that, for each upset scenario, the recovery manoeuvre can be performed such that the FSTD does not exceed the FSTD training envelope. If the envelope is exceeded, the SOJ should confirm that the FSTD is within the realms of confidence in the simulation accuracy.			✓
15.5	Instructor and observer seating Describes the controls that are required for the instructor during the use of the FSTD.			
15.5.1	The instructor's and observer's seats need not represent those found in the aircraft. In addition to the seat for the instructor, at least two seats for inspectors/observers should also be provided. All three seats should be of adequate comfort for the occupant to remain seated for typical training session durations.			✓

15: Miscellaneous				
General requirements		Verification and validation		
		QTG(A)	QTG(H)	F&S
15.5.2	Both observer seats should have adequate lighting to permit note-taking. Both observers should be able to monitor flight crew / instructor communications, if the ownship is equipped with communication systems.			✓
15.5.3	The seating set-up should ensure that the first observer has a direct, clear and sufficient view of the flight deck (i.e. to monitor flight crew actions) and out of the window without causing obstruction to or the distraction of the instructor.			✓
15.5.4	For an FSTD with a motion cueing system, any on-board instructor and observer seat(s) should be adequately secured and fitted with positive restraint devices of sufficient integrity to safely restrain the occupant during any known or predicted motion system excursion. The seats should be safe and secured for all FSTD operations.			✓
15.6	Computer capacity			
15.6.1	An FSTD should provide continuous cueing and simulation, with hardware and software modules running at frame rates sufficient to ensure seamless and realistic perception by the user.			
15.7	Self-diagnostic testing			
15.7.1	Self-diagnostic testing of the FSTD should be available to determine the integrity of hardware and software operation and to provide a means for quickly and effectively conducting daily testing of the FSTD software and hardware.			
15.8	Automatic testing facilities			
15.8.1	The FSTD should provide automated objective testing capabilities to verify compliance with objective test requirements and facilitate recurrent testing in accordance with objective test requirements (refer to Subpart D).			
15.9	Modifications to FSTD hardware and software			
15.9.1	The FSTD architecture should support updates of the FSTD hardware and software and enable the usage of different software loads (i.e. the separation of current accepted load used for training, test or development load and old loads).			
15.10	System integration			

15: Miscellaneous				
General requirements		Verification and validation		
		QTG(A)	QTG(H)	F&S
15.10.1	The relative responses of the visual system (if fitted), flight deck instruments and initial motion system should be coupled closely to provide integrated sensory cues.	6.a	6.a	✓
15.10.2	The motion response should occur before the visual response.	6.a	6.a	✓
15.10.3	If HUD systems / EFVSs are installed that are not actual aircraft systems, additional transport delay tests should be provided.	6.a	6.a	✓
15.10.4	If a visual system's mode of operation (daylight, twilight and night) can affect performance, additional tests are required. In such cases, the SOJ should provide details of the tests and how results justify the use of this mode.	6.a	6.a	✓
15.11	XR systems: system integration Note: Requirements marked with √ ¹ indicate that an objective test section for these is not available. The tests should be defined in the special conditions.			
15.11.1	The tracking delay should be minimised. Note: Tracking delay refers to the entire process (i.e. movement – image generation – graphic processing unit – HMD).	√ ¹	√ ¹	✓
15.12	XR systems: VR display – instructor system requirements			
15.12.1	The view of the crew member HMD image(s) should be presented on the IOS at all times. The instructor should also be provided with: (1) a fixed out-of-the-window scene to ensure proper situational awareness; and (2) a full view of the pilot to enable assessment of the pilot's behaviour.			
15.12.2	It should be possible for the instructor to see (representations of) the crew members' body positions (i.e. the body pose) in the virtual environment.			

15: Miscellaneous					
General requirements			Verification and validation		
			QTG(A)	QTG(H)	F&S
15.12.3	The instructor station should provide the means to observe the states of flight deck elements (e.g. switches, panels, displays, instruments) in the virtual environment.				
15.13	XR systems: climate requirements				
15.13.1	For cyber sickness prevention and training comfort, the room climate should be controlled. Sufficient fresh air, an appropriate room temperature and an appropriate carbon dioxide (CO ₂) level should be ensured.				

CS FSTD.QB.120 Multi-crew cooperation requirements for flight simulation training devices

For MCC training and operation, the minimum systems, instrumentation and indicators should be installed and simulated as applicable to the simulated aircraft at the levels of FSTD features flight deck layout and structure and aircraft systems.

- (a) For aeroplane FSTDs, these are:
- (1) turbojet or turboprop engines;
 - (2) performance reserves, for cases of engine failure, to be in accordance with aircraft certification requirements (e.g. CS-25);
 - (3) retractable landing gear;
 - (4) a pressurisation system;
 - (5) de-icing systems;
 - (6) a fire detection/suppression system;
 - (7) dual controls;
 - (8) autopilot with automatic approach mode;
 - (9) two VHF transceivers, , accessible to each pilot station;
 - (10) a crew intercom system, supporting communication between flight crew members
 - (11) oxygen masks at each pilot station, fitted with microphones compatible with the intercom and communication systems;
 - (12) two VHF NAV receivers (VOR, ILS, DME);
 - (13) one GNSS;
 - (14) one automatic direction finder (ADF) receiver (as applicable to the aircraft simulated);
 - (15) one marker receiver;
 - (16) one transponder;
 - (17) the following indicators in the same positions on the instrument panels of both pilots (as applicable to the simulated aircraft):
 - (i) airspeed,
 - (ii) flight attitude with FD guidance,
 - (iii) altimeter,
 - (iv) horizontal situation indicator (HSI) or equivalent navigation display,
 - (v) vertical speed,
 - (vi) ADF (as applicable to the aircraft simulated),
 - (vii) VOR, ILS and DME,

- (viii) marker indications (as appropriate),
 - (ix) stopwatch.
- (b) For rotorcraft FSTDs, these are:
 - (1) multi-engine and multi-pilot aircraft;
 - (2) performance reserves, for cases of engine failure, to be in accordance with category A criteria;
 - (3) anti-icing or de-icing systems;
 - (4) a fire detection/suppression system;
 - (5) dual controls;
 - (6) autopilot with upper modes;
 - (7) two VHF transceivers;
 - (8) two VHF NAV receivers (VOR, ILS, DME);
 - (9) one GNSS;
 - (10) one ADF receiver (as applicable to the aircraft simulated);
 - (11) one marker receiver;
 - (12) one transponder;
 - (13) weather radar;
 - (14) the following indicators in the same positions on the instrument panels of both pilots (as applicable to the simulated aircraft):
 - (i) airspeed,
 - (ii) flight attitude,
 - (iii) altimeter and radio altimeter,
 - (iv) horizontal situation indicator or equivalent navigation display,
 - (v) vertical speed,
 - (vi) ADF (as applicable to the aircraft simulated),
 - (vii) VOR, ILS and DME,
 - (viii) marker indications (as appropriate),
 - (ix) stopwatch.

CS FSTD.QB.125 Flight simulation training device standards to support upset prevention and recovery training feedback tools

- (a) Applicability requirements are as follows.

This CS is applicable only to aeroplane FSTDs that are to be qualified for UPRT. For the purposes of this CS, an aeroplane upset is an undesired aeroplane state characterised by unintentional deviations from parameters experienced during normal operations. An aeroplane upset may involve pitch or bank angle deviations and inappropriate airspeeds for the given conditions.

- (b) FSTD standards are as follows.

- (1) The following FSTD standard provisions apply.

- (i) The provisions of this CS define three basic elements that are required for qualifying an FSTD for UPRT manoeuvres.

- (A) **FSTD training envelope:** refer to the definition in CS FSTD.GEN.015.
- (B) **Instructor feedback:** provides the instructor/evaluator with a minimum set of feedback tools to properly evaluate the trainee's performance in accomplishing a UPRT task.
- (C) **Upset scenarios:** if dynamic upset scenarios or aeroplane system malfunctions are used to drive the FSTD into an aeroplane upset condition, specific guidance should be available to the instructor (e.g. on the IOS or in the manual) that describes how the upset scenario is driven along with any malfunction or degradation in FSTD functionality required to stimulate the upset.

- (ii) The FSTD validation envelope is defined by the following three subdivisions.

- (A) **The flight-test-validated region.** This is the region of the flight envelope that has been validated with flight test data, typically by comparing the performance of the FSTD against the flight test data through tests incorporated into the QTG and other flight test data utilised to further extend the model beyond the minimum provisions. Within this region, there is high confidence that the FSTD will respond similarly to the aeroplane. Note that this region is not strictly limited to what has been tested in the QTG. As long as it is ensured that the aerodynamic mathematical model conforms to the flight test results, that portion of the mathematical model is considered to be within the flight-test-validated region.
- (B) **The wind tunnel or analytical region.** This is the region of the flight envelope for which wind tunnel testing or other reliable predictive methods have been used (typically by the aeroplane manufacturer) to define the aerodynamic model. Any extensions to the aerodynamic model that have been evaluated in accordance with the definition of

a 'characteristic' stall model (as described in CS FSTD.QB.107.2.STALL) should be clearly indicated. Within this region, there is moderate confidence that the FSTD will respond in a similar way as the aeroplane.

- (C) **The extrapolated region.** This is the region extrapolated beyond the flight-test-validated and wind tunnel / analytical regions. The extrapolation may be a linear one, a holding of the last value before the extrapolation began or some other set of values. Whether this extrapolated data is provided by the aeroplane or FSTD manufacturer, it is a 'best estimation' only. Within this region, there is low confidence that the FSTD will respond in a similar way as the aeroplane.

(c) IOS feedback mechanism requirements are as follows.

- (1) For the instructor/evaluator to provide feedback to the student during the upset prevention and recovery manoeuvre training, additional information should be accessible that indicates the fidelity of the simulation, the magnitude of the trainee's flight control inputs and the aeroplane operational limits that could potentially affect the successful completion of the manoeuvre(s). At a minimum, the following should be available to the instructor/evaluator.

- (i) The FSTD validation envelope should be provided.
- (ii) Flight control inputs should be provided.
- (iii) Aeroplane operational limits should be provided.

The FSTD should employ a method to provide the instructor/evaluator with real-time information concerning the aeroplane operational limits. The simulated aeroplane's parameters should be displayed dynamically in real time and provided in a time history or equivalent format. At a minimum, the parameters that should be available to the instructor/evaluator should include:

- (A) airspeed and airspeed limits, including the stall speed (V_s) and maximum operating limit airspeed (V_{MO}) / maximum operating limit speed (Mach) (M_{MO});
- (B) load factor and operational load factor limits; and
- (C) angle of attack and stall identification angle of attack (refer to CS FSTD.QB.107.2.STALL for additional information on the definition of the stall identification angle of attack), which may be displayed in conjunction with the FSTD validation envelope.

- (2) A feedback mechanism is available to the instructor/evaluator.

GM1 FSTD.QB.125 Flight simulation training device standards to support upset prevention and recovery training feedback tools

- (a) The following is an introduction to UPRT feedback tool standards.

An SOJ should provide information pertaining to the aeroplane's parameters as described in CS FSTD.QB.115 section 'UPRT'. This GM details some provisions for these.

The objective of the IOS feedback during UPRT exercises is to provide the instructor with the ability to assess the timely and proper control action, including sequence, to complete the recovery in a safe manner.

- (b) Background information on feedback is provided below.

IOS feedback, which may also be provided via a separate mobile device, is used to monitor and debrief the crew regarding UPRT exercises in order to verify that proper control activity was executed. The instructor should have the necessary information to clearly establish whether the recovery was completed within the FSTD training envelope (refer to CS FSTD.QB.125) and take any necessary action to complete the training.

The FSTD should include tools for the instructor to be able to immediately debrief the pilot(s) after the training event. All data recorded for use in the UPRT debrief should be easily, permanently deleted after the UPRT training event.

- (c) In terms of IOS parameters, the tool should normally display the following.

- (1) Pilot-induced control inputs should include the following:

- (i) pitch,
- (ii) roll,
- (iii) rudder pedal,
- (iv) throttles,
- (v) flaps, and
- (vi) speed brake / spoilers.

A time history of control inputs, including flight deck control forces and flight control laws (fly-by-wire aeroplanes), as applicable, should be provided.

In order to ascertain that the control inputs are applied in a correct, timely and smooth manner, the display should indicate these at a sampling frequency rate that is sufficiently high to prevent the user from missing possible abrupt pilot action. This may be limited to the debrief mode following the execution of the exercise or an individual manoeuvre.

- (2) A display of the primary flight parameters should be provided. If applicable, it should display a pseudo-PFD; if a pseudo-PFD is displayed, then the parameters must be the same as the ones displayed on the aeroplane PFD, including the following:

- (i) pitch attitude,

- (ii) roll attitude,
 - (iii) turn/side-slip,
 - (iv) IAS,
 - (v) stall warning speed / stall buffet speed,
 - (vi) V_{MO}/M_{MO} ,
 - (vii) altitude,
 - (viii) rate of climb,
 - (ix) autopilot status, and
 - (x) auto-throttle status.
- (3) Angle of attack should be provided.
 - (4) Angle of side-slip should be provided.
 - (5) G-loading should be provided.
 - (6) Limitations of angle of attack, angle of side-slip and g-loading should be provided.
- (d) In terms of the FSTD validation envelope, the following should be displayed.

- (1) Angle of attack and angle of side-slip should be provided.

One method of showing the limitations of the angle of attack and angle of side-slip is through the simultaneous depiction of the angle of attack versus the angle of side-slip and the corresponding FSTD validation envelope.

- (2) The g-loading should be given.

A V - n diagram is a method of illustrating the aeroplane load factor (n) during a flight manoeuvre in relation to its operational load factor and airspeed limits (V). The diagram boundaries indicate the aeroplane operational flight envelope, and the diagram presents the load factor with the current airspeed and flight configuration. Refer to Figure 1. The lower airspeed limit line represents the values for V_S governed by the aerodynamic characteristics. V_{S1} means the clean stall speed at 1g. The maximum airspeed limit line represents the operational limit airspeeds, such as maximum operating airspeed (V_{MO}), maximum operating Mach (M_{MO}), never exceed speed (V_{NE}), maximum flap operating airspeed (V_{FE}), or maximum gear operating airspeed (V_{LO}). The upper and lower lines represent the operational load factor limits. The displayed data should be in accordance with the aircraft OEM operating recommendations and limitations (i.e. AFM limits). The shape and limits of the V - n diagram depend on the aircraft, its configuration and the environmental and flight conditions.

The V - n diagram's shape also varies as a function of the Mach number and other parameters. For example, for a transport category jet aircraft, the lower airspeed limits may be very different for stall events at nominal cruise altitudes and low

altitudes in cruise configuration. If the $V-n$ diagram does not consider such aspects, the IOS should show a distinct label to indicate this.

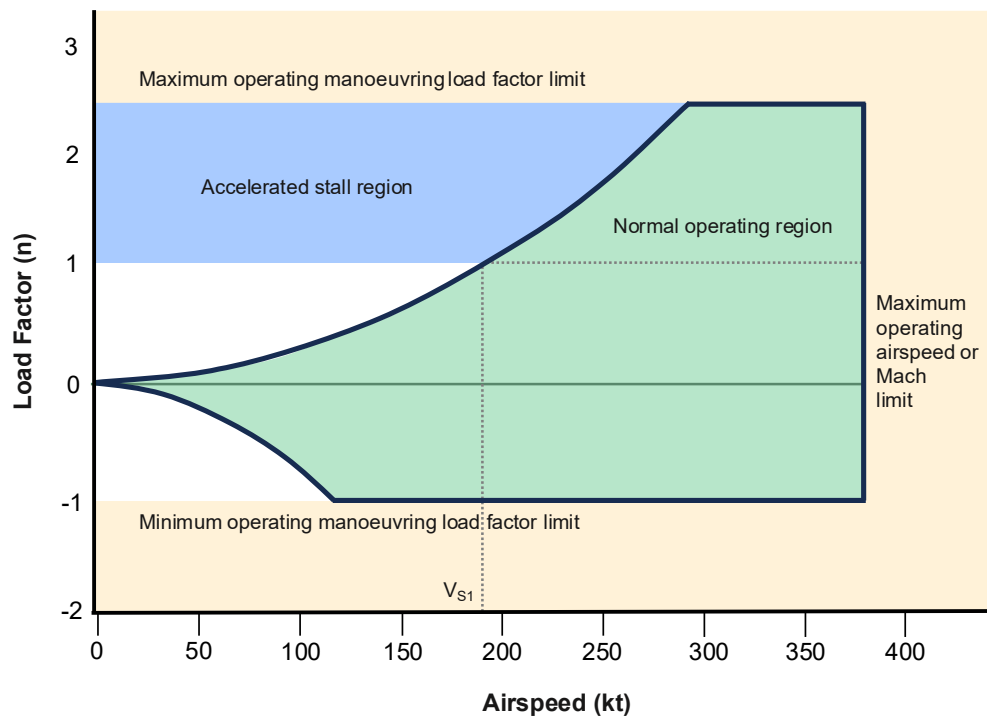


Figure 1. $V-n$ diagram example

- (e) The instructor feedback mechanism should provide the following.

A feedback system should be available to display the real-time performance of the FSTD during UPRT exercises. This real-time feedback enables the instructor to monitor the FSTD's behaviour and ensure that it aligns with the training objectives.

Additionally, the FSTD may include an optional recording mechanism to capture detailed performance data and feedback using the IOS display(s) (e.g. in the form of time history plots). This allows the instructor or evaluator to review the exercises during debriefing on board, facilitating a deeper analysis of the crew's actions and decisions during the training scenarios. A recording mechanism can enhance the overall training value by supporting post-exercise discussions and evaluations.

CS FSTD.QB.130 Qualification of a flight simulation training device with an extended reality system

- (a) Applicability requirements are as follows.

- (1) FSTDs may use an HMD as part of the FSTD in order to use XR. XR-based environments include the use of a VR display, an AR display or other MR displays. If an XR system is used, special conditions may be used to define adequate or appropriate standards for the FSTD as part of the qualification basis.
- (2) XR is a term used to refer to VR, MR and AR.

- (i) VR means that the environment is completely represented virtually on an HMD.
 - (ii) AR means that the real-world element is enhanced with a virtual overlay component using a see-through HMD.
 - (iii) MR means that the real environment (including camera-perceived) elements and computer-generated elements are merged. Physical and virtual content are presented to the observer on an HMD. All elements may coexist in MR and interact in real time. For example, the image from a head-mounted camera system is integrated into the HMD virtual environment.
- (b) FSTD standards are as follows. The special conditions of the qualification basis should take into account the general requirements for XR systems included in Subpart B of this CS-FSTD for:
 - (1) flight deck layout and structure (CS FSTD.QB.101);
 - (2) flight control forces and hardware (CS FSTD.QB.102);
 - (3) visual cueing (CS FSTD.QB.111); and
 - (4) miscellaneous (CS FSTD.QB.115).
- (c) For the flight deck layout and structure FSTD feature, an XR system with an HMD may fully or partially represent the flight deck layout and structure, provided it meets the intended requirements in CS FSTD.QB.101. Special conditions should describe how the general requirements in CS FSTD.QB.101 are addressed, including control placement, interaction, visual and spatial representation and compensatory design features. The XR system should provide accurate, reliable head tracking for proper alignment and interaction.
- (d) For the flight control forces and hardware FSTD feature, primary flight controls should be physically implemented.
- (e) For the visual cueing FSTD feature, HMDs may use one of the XR technologies (AR, MR, VR), the requirements of which will differ depending on the technology used. The general requirements for XR systems using an HMD are described in Subpart B. If a technology is not described, special conditions should be used to describe it.
- (f) Motion cueing considerations are as follows. For cyber sickness prevention and avoidance of a negative transfer of training, a six-DOF motion platform may be used in combination with the HMD. A motion platform with a reduced envelope is considered to be acceptable.
- (g) For objective testing, the objective tests associated with the validation and verification of the performance for XR-display-based FSTDs (e.g. FSTDs with a VR display system) should be set out in the special conditions as part of the qualification basis.
- (h) For the system set-up, an SOJ is required to explain the system set-up for the XR system to describe the overall concept with regard to flight deck and environment representation, interaction with the flight deck and the technology used, including the HMD.

GM1 FSTD.QB.130 Qualification of a flight simulation training device with an extended reality system

GUIDANCE FOR FLIGHT SIMULATION TRAINING DEVICES FITTED WITH EXTENDED REALITY SYSTEMS – AUGMENTED REALITY AND MIXED REALITY DISPLAY SYSTEMS

- (a) For FSTDs, an XR system may employ an HMD or ‘goggles’ as the primary display system. This system may use one of several virtual environment display technologies: VR, AR or MR.
- (b) The XR requirements in CS FSTD.QB.101 for the flight deck layout and structure FSTD feature, and CS FSTD.QB.111 for the visual cueing FSTD feature are designed for VR systems, in which the HMD provides an immersive, computer-generated 3D simulation. The associated special conditions that form the basis for the qualification should justify the application of these general requirements and detail the objective testing methods used for system verification and validation.
- (c) While specific requirements for AR and MR display systems are not explicitly included in this CS-FSTD, they may be addressed using special conditions. If an HMD is used, the general requirements for VR systems (e.g. CS FSTD.QB.101, CS FSTD.QB.111 and CS FSTD.QB.115) should serve as the foundations and be adapted or expanded upon within the special conditions.
- (d) For example, in an FSTD with an AR or MR display system intended for the specific fidelity level of the flight deck layout and structure FSTD feature:
 - (1) a physical full-scale flight deck – as described in CS FSTD.QB.101, section ‘XR systems: virtual flight deck requirements’ – should be provided, along with ensuring compliance with other general requirements for level S;
 - (2) the relationship between the physical flight deck and the AR or MR elements should be explicitly described in the special conditions, including how these elements are aligned and interact with one another.

CS FSTD.QB.140 Multi-configuration devices and their master qualification test guide requirements

- (a) Multi-configuration devices are FSTDs that host multiple configurations of aircraft, aircraft systems, avionics and engine types/ratings and may or may not utilise different flight deck and platform combinations. For devices with major interchangeable assemblies, platforms contain the motion and visual aspects of the device into which various flight decks can be inserted.
- (b) Multi-configuration devices should have the regular accompanying SOJs, objective tests and other documentation. Multi-configuration devices should have the regular accompanying SOJs, objective tests and other documentation. If any of these documents apply only to a particular configuration, they should be provided as separate sets for each configuration. The MQTG(s) should be appropriately arranged so that it is clear how each configuration impacts the device, applicable tests and applicable requirements. The impacts should be specifically addressed through the use of rationales and SOJs addressing which requirements and tests

are applicable and common between configurations, which are independent of the configuration and why this is the case.

- (c) Convertible FSTDs may have objective tests and functions and subjective tests that are applicable only for one FSTD configuration.

GM1 FSTD.QB.140 Multi-configuration devices and their master qualification test guide requirements

The following general principles should be followed and are applicable to all FCSs.

- (a) Simulation software principles are as follows.
- (1) Every FSTD configuration is affected by the software architecture. Convertible FSTD platforms may use different software modules or software loads for each FSTD configuration. The configuration control system should cover convertible FSTD platforms and their unique features.
 - (2) If the software changes between FSTD configurations are known to affect objective tests (e.g. affect handling characteristics or transport delay), then separate objective test cases should be prepared for each FSTD configuration. The MQTG should describe how the software loads and modules are structured for each FSTD configuration. This will indicate whether separate objective tests are needed.
- (b) Visual system principles are as follows.
- Visual objective tests particular to each configuration – such as VGS, system geometry and qualification airfield tests for each configuration, if applicable (e.g. devices with interchangeable flight decks) – should be performed with each configuration. An SOJ should be given to indicate which tests are independent of the configuration.
- (c) Transport delay objective test principles are as follows.
- Consideration should be given to transport delay tests, since different configurations may use different software modules, software loads, hardware and avionics that may have an effect on the results. If this is not considered applicable by the applicant (e.g. because the software loads differ only so slightly that the difference does not affect transport delay), an SOJ should be provided.
- (d) Convertible avionics principles are as follows.
- (1) Convertible avionics are different avionics suites, optional flight management and guidance computers, selectable avionics options and different flight augmentation computer standards or versions.
 - (2) Convertible avionics should be segmented into those systems or components that can significantly affect the objective test results and those that cannot. This analysis should be done by the organisation requesting the qualification of an FSTD, supported by the aircraft manufacturer, avionics manufacturer, data provider or FSTD manufacturer (or a combination of them) on a case-by-case basis as appropriate. (Refer to CS FSTD.ENG.030.)

- (3) If a convertible avionics configuration can affect the objective test results, additional objective test cases should be included for all appropriate configurations.
- (e) For alternate system or equipment:
 - (1) typical examples of alternate systems or equipment are different brake systems (e.g. steel and carbon brakes) and system configurations for aircraft performance purposes (e.g. symmetric aileron deflection for take-off);
 - (2) if alternate systems or equipment affect the objective test results, additional objective test cases should be included for all appropriate configurations.
- (f) Principles for alternate engine type(s) and rating(s) are as follows.

Objective tests should validate engine type and rating characteristics for each device configuration. The CS FSTD.ENG.030 requirements for alternate engine types and ratings can be used across device configurations with appropriate additional objective tests, SOJs and rationales.
- (g) Principles for different groups of aircraft, types and variants are as follows.
 - (1) If multiple aircraft types are simulated, each should have a complete and separate MQTG, except for those tests that are independent of the configuration.
 - (2) If a single aircraft type with multiple variants is simulated, the configurations can affect the objective test results, so additional objective test cases should be included for all appropriate configurations to validate the variants. The additional objective tests may be included as an appendix to the MQTG.
 - (3) If an FSTD simulates aircraft in multiple groups with one or more features of the aircraft simulation FSTD features group at the generic fidelity level, each configuration should have a complete and separate MQTG, except for those tests that are independent of the configuration.
- (h) Principles for devices with major interchangeable assemblies are as follows.
 - (1) Devices using major interchangeable assemblies (flight decks and platforms), for which noticeable hardware entities or modules are changed in order to change the configuration, can often affect objective tests.
 - (2) Devices with major interchangeable assemblies can use various architectures and methods. Not all possible set-ups can be considered in this GM due to their number.
 - (3) Typical examples of devices with major interchangeable assemblies are the roll-on/roll-off solutions, in which different flight decks can be inserted into different platforms. In addition, there could be further roll-on/roll-off configurations for alternate engines, avionics, etc. Refer to other subparagraphs of this GM regarding those configurations.

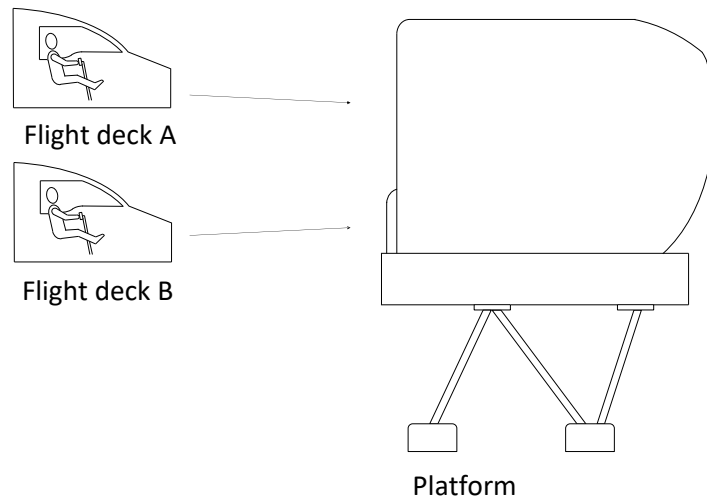


Figure 1. Example of a device with two interchangeable flight decks

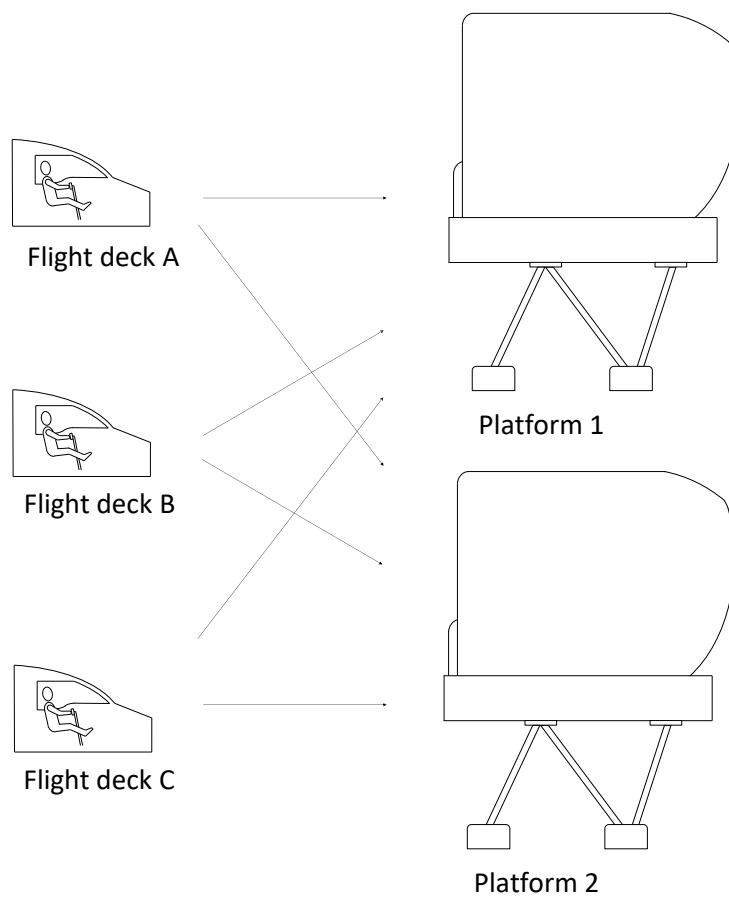


Figure 2. Example of two platforms with three interchangeable flight decks

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- (4) If each flight deck will be inserted into only one dedicated platform (refer to Figure 1), then all of the following apply.
- (i) The MQTG(s) should address all requirements for each combination of flight decks and platforms as appropriate. These may be in the form of SOJs, rationales and/or objective tests.
 - (ii) There should be a complete and separate MQTG for each flight deck.
 - (iii) The MQTG should identify the flight deck and platform. The individual objective tests should also identify these.
- (5) If each flight deck will be inserted into multiple platforms (refer to Figure 2), then all of the following apply.
- (i) The MQTG(s) should address all requirements for each combination of flight decks and platforms as appropriate. These may be in the form of SOJs, rationales and/or objective tests.
 - (ii) Each flight deck should be appointed a platform that is the primary combination for objective tests. This combination of flight deck and platform should have a complete MQTG.
 - (iii) The MQTG should identify the flight deck and platform. The individual objective tests should also identify these.
 - (iv) Since the flight deck will be inserted into multiple platforms, the MQTG should be amended for each combination. The host computer and the software modules may be associated with the platform or the flight deck. This determines the extent to which the software is common between flight deck and platform combinations. Hence, it will define which objective tests may be affected by swaps. For example, if the flight deck will be inserted into two different platforms, then there should be one complete MQTG (i.e. for the primary combination) and an appendix for the tests related to the alternate platform.

SIMULATED AIR TRAFFIC CONTROL ENVIRONMENT

CS FSTD.QB.160 Simulated air traffic control environment requirements for flight simulation training devices

If the FSTD to be qualified includes a SATCE system, the following requirements should be fulfilled at fidelity level R or S. Note that level G requirements are reserved for future use.

Refer to CS FSTD.FST.040 for functions and subjective test standards for SATCE.

ATC and traffic
Simulated air traffic control environment (SATCE) is an automated flight training technology in which ATC services and other traffic entities are simulated as part of the synthetic environment provided by the FSTD. The simulation of services and associated communications other than those of ATC, such as those of companies, operations, ground service providers or cabin crew, are considered outside the scope. ‘Scenarios’ in association with SATCE means real-time exercises — developed specifically for training, testing or checking — that reflect flight operations (e.g. line-oriented simulation and line-oriented flight training). If SATCE is simulated on the FSTD, the ESL should document this capability in accordance with the requirements in Part-ORA (Annex VII) of Commission Regulation (EU) No 1178/2011. The simulation of ATC services and other traffic need only encompass those geographic areas that flight crews are expected to encounter during scenarios, with the scope limited to support planned training objectives while still allowing for various possible outcomes. SATCE requires that a communication system is simulated (refer to CS FSTD.QB.104) at level G, R or S.

The requirements for SATCE are presented in the table below.

SATCE						
Requirements			Feature fidelity level			Verification and validation
			G	R	S	QTG F&S
SATCE.1	Automated weather reporting Describes the simulation of fully automatic ATC services delivering reported weather information via radio and data communications. ATIS is the most common automated weather-reporting system. Other reported weather broadcast systems need only be simulated for regions or locations if they are operational. Refer to CS FSTD.QB.112, section 'Automated weather reporting', for the associated general requirements.					
SATCE.1.1	Message format (ICAO or regional): The format and content of automated reported weather messages should adhere to either standard ICAO specifications in accordance with ICAO <i>Annex 11 – Air traffic services</i> or regional characteristics. Regional messages may have characteristics that are non-standard (in relation to the ICAO format and content). Differences may include message content order, units and language.			✓		✓
SATCE.1.2	Reported weather conditions (multiple): More than one set of automated reported weather conditions should be available to the flight crew if multiple stations or operating sites may be used during a scenario.		✓	✓		✓
SATCE.1.3	Reported weather conditions (assignment): The instructor should be able to associate a particular set of automated reported weather conditions with one or more stations or operating sites, if multiple sets of reported weather conditions are available.		✓	✓		✓
SATCE.1.4	Message format (ICAO): Refer to CS FSTD.QB.112, section 'Automated weather reporting', for the associated general requirements.		✓			✓

SATCE						
Requirements		Feature fidelity level			Verification and validation	
		G	R	S	QTG	F&S
SATCE.1.5	Instructor control (configuration): Reported weather conditions should be configurable via the IOS or lesson-planning tools before a scenario. The instructor should have the ability to override weather-related parameters via the IOS during the scenario.		✓	✓		✓
SATCE.1.6	Instructor control: If the instructor changes the automated reported weather messages by overriding them, these inputs should influence the message(s) for associated stations or operating sites. Changes to automated reported weather messages should not necessarily affect the weather conditions experienced. Similarly, if the instructor changes the weather conditions experienced, the automated reported weather messages should not necessarily be affected.		✓	✓		✓
SATCE.1.7	ATC correlation with reported weather: All weather-related communications from ATC should correlate with automated reported weather message(s), including any changes made by the instructor.		✓	✓		✓
SATCE.1.8	Message content: Refer to CS FSTD.QB.112, section 'Automated weather reporting', for the associated general requirements.		✓	✓		✓
SATCE.1.9	Flight crew access: Refer to CS FSTD.QB.112, section 'Automated weather reporting', for the associated general requirements		✓	✓		✓
SATCE.1.10	Flight crew access (information): Refer to CS FSTD.QB.112, section 'Automated weather reporting', for the associated general requirements.		✓	✓		✓

SATCE						
Requirements				Feature fidelity level		Verification and validation
		G	R	S	QTG	F&S
SATCE.2	Other traffic Describes the simulation of entities other than the ownship in the operational environment. Other traffic may be generated and controlled by FSTD systems other than SATCE, such as applications used for ACAS training, and the visual display system. Note: Other traffic simulation may be limited to the traffic required during a scenario. Note: Requirements concerning the visual image are applicable only if a visual cueing FSTD feature is at fidelity level G, R or S. Refer to CS FSTD.QB.114 and CS FSTD.QB.111 for the requirements for the display of other traffic. Note: 'Normal conditions' describes the operational environment in which no accidents, incidents or similar events occur, resulting in uninterrupted traffic flow. ATC and other traffic can be expected to follow normal procedures, and other traffic adhere to all ATC clearances and instructions. Note: 'Non-normal conditions' describes the operational environment in which events such as accidents and incidents or adverse weather occur, resulting in interrupted traffic flow. ATC and other traffic may not follow normal procedures and may introduce errors that may cause a threat to the ownship. The ownship may also cause non-normal conditions. 'Emergency conditions' describes the operational environment in which the safety of an aircraft or the people on board or on the ground is endangered. Normal procedures and traffic flow are likely to be disrupted, if simulated, emergency conditions may be caused by ATC and/or other traffic and/or the ownship.					
SATCE.2.1	Ground vehicles: For tier I operating sites (refer to CS FSTD.QB.114.1), ground vehicles undertaking manoeuvres under ATC should be present in the simulated operational environment. Ground vehicle simulation may be limited to those that require clearance to operate under ATC and would typically have a radio presence on ATC frequencies.			✓		✓

SATCE						
Requirements		Feature fidelity level			Verification and validation	
		G	R	S	QTG	F&S
SATCE.2.2	Ground vehicle performance: For tier I operating sites, ground vehicles should exhibit typical performance during all manoeuvres. Typical ground vehicle performance may include speed, acceleration, turning and braking/deceleration performance appropriate for the vehicle type.			✓		✓
SATCE.2.3	Ground vehicle behaviour: For tier I operating sites, ground vehicles should follow operating site rules and procedures, including typical speed restrictions and runway holding point procedures, under normal conditions.			✓		✓
SATCE.2.4	Ground vehicle behaviour (dynamic): Ground vehicles should exhibit dynamic interactions with ATC, the ownship and other traffic under normal conditions. 'Dynamic interactions' means that ATC, the ownship and other traffic can influence the timing and manoeuvre of a traffic entity during a real-time simulation. Note: This may not apply for non-normal and emergency conditions.			✓		✓
SATCE.2.5	Other aircraft: Other aircraft traffic undertaking manoeuvres under ATC should be present in the simulated environment, both airborne and on the ground. Other airborne traffic may perform actions such as changes in level, heading and speed and may fly published routes, including standard instrument departure routes, standard terminal arrival routes, approaches and holding patterns. Other aircraft traffic on the ground may perform manoeuvres such as pushback, taxi-out, take-off, landing, taxi-in and parking.		✓	✓		✓

SATCE						
Requirements		Feature fidelity level			Verification and validation	
		G	R	S	QTG	F&S
SATCE.2.6	Other aircraft performance: Other aircraft traffic should exhibit typical performance during all flight phases. Typical aircraft airborne performance may include speeds, turn rates, climb/descent rates and cruise altitudes appropriate for the aircraft type. Typical performance for aircraft on the ground may include taxiing speed, acceleration, cornering and braking/deceleration performance appropriate for the aircraft type.		✓	✓		✓
SATCE.2.7	Other aircraft behaviour (dynamic): Other aircraft traffic should exhibit dynamic interactions with ATC, the ownship and other traffic under normal conditions. 'Dynamic interactions' means that ATC, the ownship and other traffic can influence the timing and manoeuvre of a traffic entity during a real-time simulation. Note: This may not apply for non-normal and emergency conditions.		✓	✓		✓
SATCE.2.8	Other aircraft behaviour (routing): Other aircraft traffic should follow appropriate routing, both in the air and on the ground, under normal conditions that matches routing information available to the ownship flight crew.		✓	✓		✓
SATCE.2.9	Other aircraft behaviour (movements): Other aircraft traffic should depart from and land to take-off and landing areas designated for the appropriate aircraft category and weight class.		✓	✓		✓
SATCE.2.10	Other aircraft behaviour (taxiing): Other aircraft traffic should follow taxi / air taxi routes that are appropriate for the aircraft category and weight class. Other fixed-wing aircraft traffic should not transit over non-supporting surfaces under normal conditions.		✓	✓		✓

SATCE						
Requirements		Feature fidelity level			Verification and validation	
		G	R	S	QTG	F&S
SATCE.2.11	Other aircraft behaviour (parking): Other aircraft traffic should park at gates, stands and other designated parking locations suitable for the relevant aircraft category and weight class.		✓	✓		✓
SATCE.2.12	Other traffic transponder state: If other aircraft traffic is simulated as equipped with a transponder, the transponder state should also be simulated. Transponder states should include Mode A (no altitude reporting), Mode C/S (altitude reporting) and off (not powered) or in standby. If the simulated aircraft is equipped with ACAS capability, the ACAS system should appropriately display other traffic transponder status on flight deck instruments.			✓		✓
SATCE.2.13	Other traffic transponder mode of operation: If other aircraft traffic is simulated as equipped with a transponder, the transponder mode setting should be appropriate for that aircraft's phase of flight and position.			✓		✓
SATCE.2.14	Other aircraft types and liveries: Other aircraft traffic should have aircraft types and liveries typical of their operator at the operating site location. Other aircraft types are not required to be current.			✓		✓
SATCE.2.15	Other aircraft visual effects: Other aircraft traffic animation and visual effects should be simulated. Visual effects should include the representation of aircraft lighting, landing gear and control surfaces. The scope and fidelity of the visual effects may be limited to the extent required during a scenario.			✓		✓
SATCE.2.16	Other traffic automation: Other aircraft and/or ground vehicle movements should be fully automated, so that instructor management of other traffic movements during a scenario is not required.		✓	✓		✓

SATCE						
Requirements		Feature fidelity level			Verification and validation	
		G	R	S	QTG	F&S
SATCE.2.17	Other traffic correlation with ATC: The behaviours of other aircraft and/or ground vehicles should correlate with ATC radio communications under normal conditions (i.e. other traffic should acknowledge and manoeuvre appropriately in response to ATC clearances and instructions).		✓	✓		✓
SATCE.2.18	Other traffic separation: Other aircraft traffic minimum separation criteria should be typical of ATC procedures under normal conditions. This includes ownship separation from other traffic and separation among other traffic.		✓	✓		✓
SATCE.2.19	Other aircraft call signs and liveries: Other aircraft traffic should have allocated call signs that match their liveries observed in the visual scene. Other aircraft liveries are not required to be up to date (i.e. in current use in real-world operations).		✓	✓		✓
SATCE.2.20	Ground vehicle conflicts: Ground vehicles should be capable of creating a conflict with the ownship, including on take-off and in landing areas (e.g. a runway incursion threat). Ground conflict threats should be configurable before the start of a scenario or initiated via the IOS during a scenario.			✓		✓
SATCE.2.21	Other aircraft ground conflicts: Other aircraft should be capable of creating a ground conflict with the ownship, including on take-off and in landing areas (e.g. a runway incursion threat). Ground conflict threats should be configurable before the start of a scenario or initiated via the IOS during a scenario.		✓	✓		✓
SATCE.3	Background radio communications Describes the simulation of radio transmissions between ATC and other traffic, not intended for the ownship. Note: 'SATCE system' means the overall system delivering the simulated ATC environment as part of the FSTD. This comprises hardware and software and includes interfaces, controls and information made available to the instructor and flight crew.					

SATCE						
Requirements		Feature fidelity level			Verification and validation	
		G	R	S	QTG	F&S
SATCE.3.1	Presence: Background radio transmissions between ATC and other aircraft and/or ground vehicle traffic should be available to the ownship flight crew, if other traffic is present in the scenario and would have a radio presence in real-world operations.		✓	✓		✓
SATCE.3.2	ATC services and other traffic operations: Background radio transmissions between ATC and other traffic should correlate with the ATC service and other traffic operational context, such as phases of flight, positions and manoeuvres. Note: In general, the number of radio transmissions between ATC and other traffic should reflect the amount of other traffic manoeuvring in the simulated environment.		✓	✓		✓
SATCE.3.3	Correctness: Background radio transmissions between ATC and other traffic should be correct and complete (i.e. transmissions should not contain any significant erroneous information or omissions) under normal conditions.		✓	✓		✓
SATCE.3.4	Radio discipline (overstepping): ATC and other traffic radio transmissions should follow correct radio discipline by allowing a current transmission to finish, including current ownship transmissions, before starting a new transmission, under normal conditions (i.e. new transmissions should not step over current transmissions). Note: This may not apply under non-normal and emergency conditions.		✓	✓		✓

SATCE						
Requirements		Feature fidelity level			Verification and validation	
		G	R	S	QTG	F&S
SATCE.3.5	Radio discipline (interrupting): ATC and other traffic radio transmissions should follow correct radio discipline by allowing a current conversation (set of transmissions) to finish, including current ownship conversations, before starting a new conversation, under normal conditions (i.e. current conversations should not be interrupted, such as transmitting before the readback of a previous clearance). Note: This may not apply under non-normal and emergency conditions.		✓	✓		✓
SATCE.3.6	Radio discipline (ownship – overstepping): The SATCE system should detect and record (log) if the ownship flight crew step over radio transmissions from other radio users (e.g. ATC or other traffic).		✓	✓		✓
SATCE.3.7	Radio discipline (ownship – interrupting): The SATCE system should detect and record (log) if the ownship flight crew interrupt other currently active radio conversations, under normal conditions.		✓	✓		✓
SATCE.3.8	Ownship radio support: The SATCE system should support concurrent radio communications on all radios available to the ownship flight crew (e.g. on a multi-crew flight deck, the pilot flying may be monitoring VHF 1 while at the same time the pilot monitoring may be transmitting and receiving on VHF 2).		✓	✓		✓
SATCE.4	ATC services Describes the simulation of various distinct air traffic management roles, often allocated to different frequencies, accessible by the ownship flight crew. Note: ATC services simulation may be limited to those services required during a scenario.					
SATCE.4.1	ATC procedures (location-specific): Regional and/or location-specific published ATC procedures and associated radio communications should be simulated and apply to the ownship and other traffic.			✓		✓

SATCE						
Requirements		Feature fidelity level			Verification and validation	
		G	R	S	QTG	F&S
SATCE.4.2	Radio ranging: ATC radio transmissions should be received by the ownship within realistic and typical reception range or line of sight, as applicable to the geographic area and dependent on the height of the aircraft above the horizon. The fidelity of the radio ranging simulation may be limited to the extent required during a scenario.		✓	✓		✓
SATCE.4.3	ATC service provision: ATC services should be simulated, managing the ownship and other traffic within controlled ground and airspace. ATC services should be fully automated so that instructor management of ATC services during a scenario is not required.		✓	✓		✓
SATCE.4.4	Roles and frequency allocation: ATC service roles and allocated frequencies should correlate with each other and with the information available to the ownship flight crew.		✓	✓		✓
SATCE.4.5	ATC procedures: Standard ATC procedures and associated radio communications should be simulated and apply to the ownship and other traffic. ATC procedures should be used as published by ICAO and/or the national aviation authority and/or the air navigation service provider (ANSP).		✓	✓		✓
SATCE.4.6	ATC procedures (correlation): ATC procedures should correlate with the published information available to the ownship flight crew (e.g. via charts and documentation, EFB applications and the FMS). Refer to CS FSTD.QB.112.		✓	✓		✓
SATCE.4.7	ATC service continuity: ATC service provision to other traffic should maintain continuity across ATC sector boundaries within controlled airspace. Other traffic may be handed off by one controller and accepted by the next, as appropriate within the airspace/ground environment.		✓	✓		✓

SATCE						
Requirements				Feature fidelity level		Verification and validation
		G	R	S	QTG	F&S
SATCE.5	Language and phraseology Describes the method of simulated ATC communications and the particular set of fixed expressions used.					
SATCE.5.1	Standard phraseology (regional): Radio communications between ATC and other traffic and from ATC to the ownship should include published regional and/or location-specific phraseology.			✓		✓
SATCE.5.2	Standard phraseology: Radio transmissions between ATC and other traffic and from ATC to the ownship should comply, if possible, with the phraseologies detailed in ICAO Document 4444 (<i>Procedures for Air Navigation Services – Air traffic management</i>) and ICAO Annex 10 – <i>Aeronautical telecommunications: Volume II – Communication procedures</i> , including those with procedure for air navigation service status, supported by ICAO Document 9432 (<i>Manual of Radiotelephony</i>). If the ICAO documentation does not include standard phraseology for particular operations, phraseology standards or guidance published by the national aviation authority and/or the ANSP may be used.		✓	✓		✓
SATCE.5.3	Language: Radio communications between ATC and other traffic and from ATC to the ownship should be in English as per ICAO Document 4444 (<i>Procedures for Air Navigation Services – Air traffic management</i>).		✓	✓		✓
SATCE.6	Voice characteristics Describes the characteristics and qualities of simulated speech used for radio communications. If this is achieved automatically using synthetic speech generation, the focus should be on achieving realistic voice audio from ATC services over that from other traffic.					

SATCE						
Requirements		Feature fidelity level			Verification and validation	
		G	R	S	QTG	F&S
SATCE.6.1	ATC voices (multiple): If more than one frequency and/or ATC service is simulated, multiple distinct voices should be assigned to the ATC function, as would be experienced in real-world operations. The number of voices should allow differentiation between ATC services.			✓		✓
SATCE.6.2	Other traffic voices (multiple): If more than one other traffic item is simulated, multiple distinct voices should be assigned to other traffic. The number of voices should allow differentiation between other traffic items.			✓		✓
SATCE.6.3	ATC voices: Distinct voices should be assigned to the ATC function that are diverse enough to distinguish between different ATC services. A voice used for ATC may be assigned to more than one ATC service.		✓	✓		✓
SATCE.6.4	ATC voice assignment: Radio transmissions to the ownship should occur using the same ATC voice or voices used to simulate ATC transmissions to other traffic.		✓	✓		✓
SATCE.6.5	Dedicated ATC voices: ATC voices should be dedicated to the ATC function for the duration of a scenario. A voice used to simulate an ATC function should not be used to simulate another function (e.g. a pilot) within the same scenario. Note: It is recommended that ATC voices remain dedicated to the ATC function if a training, testing or checking session involves more than one scenario.		✓	✓		✓

SATCE						
Requirements		Feature fidelity level			Verification and validation	
		G	R	S	QTG	F&S
SATCE.6.6	Other traffic voices: Distinct voices should be assigned to other traffic that are diverse enough to distinguish between different items of other traffic. A voice used for other traffic may be assigned to more than one other traffic item.		✓	✓		✓
SATCE.7	Operating site and airspace modelling Describes the scope of data and functionality required for the simulation of ATC services to the ownship and other traffic on controlled ground and in controlled airspace. Modelling requires data that may include ATC-related airspace, ATC procedures and operating site data. This section also includes SATCE system control and the correlation of ground lighting at operating sites. Note: Requirements concerning the visual image are applicable only if the visual cueing FSTD feature is at fidelity level G, R or S.					
SATCE.7.1	Operating sites (location-specific): A simulated ATC environment typical of the location should be available at operating sites, within the same areas simulated by the navigation FSTD feature (refer to CS FSTD.QB.112). Operating site procedures should be modelled from published sources, if available, such as the site operator and/or the national aviation authority and/or ICAO. The number of operating sites at which SATCE is available may be limited to those required during a scenario.			✓		✓
SATCE.7.2	Controlled airspace (location-specific): Simulation of terminal and en route controlled airspace typical of the location should be supported, within the same areas simulated by the navigation FSTD feature (refer to CS FSTD.QB.112). Controlled airspace should be modelled on real-world data, if available, from the location and/or the ANSP and/or the national aviation authority. Controlled airspace may be limited to those routes and sectors required during a scenario.			✓		✓

SATCE						
Requirements		Feature fidelity level			Verification and validation	
		G	R	S	QTG	F&S
SATCE.7.3	Multiple connected ground movement areas: Ownship and other traffic movements should be supported for multiple ground movement areas, including take-off and landing areas, taxi routes and parking locations if this reflects real-world operations at the operating site. The number and complexity of connected ground movement areas should be extensive enough to support the scenario.			✓		✓
SATCE.7.4	Dual-direction runway movements: For conventional take-off and landing aircraft, ownship and other traffic movements should be supported in either direction on a single physical runway surface if this reflects real-world operations at the operating site.			✓		✓
SATCE.7.5	Multiple take-off and landing areas: Ownship and other traffic movements on more than one physical take-off and landing area surface should be supported, if the real-world operating site has multiple areas and if required during a scenario. Location-specific operational limitations may be reflected in traffic movements (e.g. a closed take-off and landing area or limitations imposed by noise abatement or terrain).			✓		✓
SATCE.7.6	Runway operation modes: For conventional take-off and landing aircraft, runway operation modes should include take-off only, landing only or both take-off and landing (mixed-mode use) if this reflects real-world operations.			✓		✓
SATCE.7.7	Take-off and landing area lighting: The SATCE system should control the ground lighting state for take-off and landing areas, including lighting configuration and intensity, and correlate lighting changes with routine ATC traffic movement procedures for site operations, time of day and reported weather. Lighting control and correlation need only extend to those take-off and landing areas observable by the flight crew during a scenario.		✓	✓		✓

SATCE						
Requirements		Feature fidelity level			Verification and validation	
		G	R	S	QTG	F&S
SATCE.7.8	Holding point lighting: The SATCE system should control the ground lighting state of holding point stop bars, including those associated with take-off and landing area entry points, and correlate lighting changes with ATC clearances to the ownship and other traffic. Lighting control and correlation need only extend to those holding points observable by the flight crew during a scenario.		✓	✓		✓
SATCE.7.9	Taxi route lighting: The SATCE system should control the ground lighting state of taxi routeways, including taxiway centre lights and lead-on/lead-off lights, and correlate lighting changes with ATC clearances to the ownship and other traffic. Lighting control and correlation need only extend to those taxi routes observable by the flight crew during a scenario.		✓	✓		✓
SATCE.7.10	Operating sites: A simulated ATC environment characteristic of the region should be available at operating sites, within the same areas simulated by the navigation FSTD feature (refer to CS FSTD.QB.112). Operating site procedures should be modelled from published sources, if available, such as operators of similar sites and/or the national aviation authority and/or ICAO. The number of operating sites at which SATCE is available may be limited to those required during a scenario.		✓			✓
SATCE.7.11	Controlled airspace: Simulation of terminal and en route controlled airspace characteristic of the region should be supported, within the same areas simulated by the navigation FSTD feature (refer to CS FSTD.QB.112). Controlled airspace should be modelled on real-world data, if available, from the region and/or the ANSP and/or the national aviation authority. Controlled airspace may be limited to those routes and sectors required during a scenario.		✓			

SATCE						
Requirements		Feature fidelity level			Verification and validation	
		G	R	S	QTG	F&S
SATCE.7.12	Minimum connected ground movement areas: Ownship and other traffic movements should be supported for ground movement areas that include, at a minimum, one take-off and landing area, one taxi route and one parking location that are connected. Ground movement areas should be extensive enough to support the scenario.		✓			✓
SATCE.7.13	Single-direction runway movements: For conventional take-off and landing aircraft, ownship and other traffic movements should be supported in at least one direction on a single physical runway surface.		✓			✓
SATCE.8	Weather Describes the simulation of ATC procedures and other traffic routing and behaviours influenced by meteorological conditions.					
SATCE.8.1	Site operations and reported weather: Traffic departure and arrival routing, active take-off and landing areas and the direction of take-off and landing should be correlated with the reported weather for the operating site location, during normal operations.		✓	✓		✓
SATCE.8.2	ATC procedures and reported weather: ATC services should implement appropriate procedures for the reported weather for the operating site location, under normal conditions. Weather conditions may affect procedures that determine traffic flow, sequencing and separation.		✓	✓		✓
SATCE.9	Voice communications Describes ATC service provision and radio communications between ATC and the ownship.					

SATCE						
Requirements		Feature fidelity level			Verification and validation	
		G	R	S	QTG	F&S
SATCE.9.1	Ownship non-normal conditions: ATC services should support radio communications to and from the ownship using standard phraseology (if defined) during the simulation of ownship non-normal conditions. The scope and fidelity of ATC support under ownship non-normal conditions need only be to the extent required during a scenario.			✓		✓
SATCE.9.2	Ownship emergency conditions: ATC services should support radio communications to and from the ownship using standard phraseology (if defined) during the simulation of ownship emergency conditions. At a minimum, ATC services should recognise and respond to distress ('MAYDAY') or urgency ('PAN PAN') ownship transmissions. The scope and fidelity of ATC support under ownship emergency conditions need only be to the extent required during a scenario.			✓		✓
SATCE.9.3	Ownship routing (taxi routing and parking): ATC services should provide clearances and instructions to the ownship to follow taxi routes and park at gates, stands and other designated parking locations appropriate to the ownship aircraft category and weight class.		✓	✓		✓
SATCE.9.4	Voice continuity: ATC radio communications to the ownship should use the same voices used to simulate radio communications from ATC to other traffic.		✓	✓		✓
SATCE.9.5	Time of day: ATC radio communications to the ownship and other traffic, including automated reported weather messages, should be correlated with the simulation time.		✓	✓		✓

SATCE						
Requirements		Feature fidelity level			Verification and validation	
		G	R	S	QTG	F&S
SATCE.9.6	Communication initiation: The SATCE system should support both ATC- and flight-crew-initiated radio communications between ATC and the ownship.		✓	✓		✓
SATCE.9.7	ATC services and ownship operations: ATC radio communications to the ownship should be correlated with the ATC service and the ownship flight plan (intent), phase of flight, position and manoeuvres.		✓	✓		✓
SATCE.9.8	ATC service continuity: ATC service provision to the ownship should maintain continuity across ATC sector boundaries within controlled airspace, ensuring that there is no loss or interruption of services under normal conditions.		✓	✓		✓
SATCE.9.9	Standby: The SATCE system should support standby responses from either ATC or the flight crew. If the flight crew communicates a standby response to ATC, then ATC should respect this request. If ATC communicates a standby response to the flight crew, then after a short period, ATC should initiate a subsequent conversation.		✓	✓		✓
SATCE.9.10	Say again: The SATCE system should support requests for information to be repeated from either ATC or the flight crew. ATC should repeat a transmission or part of a transmission after receiving a repeat request from the flight crew, such as 'SAY AGAIN'. ATC should issue a request for a transmission or part of a transmission to be repeated if the flight crew transmission is not understood or unintelligible.		✓	✓		✓

SATCE						
Requirements		Feature fidelity level			Verification and validation	
		G	R	S	QTG	F&S
SATCE.9.11	Content errors and omissions: ATC should be capable of responding to or correcting content errors and omissions in radio transmissions from the flight crew. At a minimum, this should include the capability for ATC to correct common ownship readback and acknowledgement errors, including missing required transmissions, and erroneous or omitted required parameters (including the ownship call sign).		✓	✓		✓
SATCE.9.12	Incorrect frequency transmissions: ATC should respond in an appropriate manner to a radio transmission by the flight crew on an incorrect or inappropriate frequency monitored by ATC. ATC need not provide a response to an ownship transmission on a frequency that is not monitored.		✓	✓		✓
SATCE.9.13	Clearance deviations: If the ownship does not follow an ATC clearance or instruction, ATC should respond with an appropriate radio transmission. Ownship deviations may include not respecting the assigned speed, heading or altitude/flight level when airborne or deviating from a cleared route when on the ground (e.g. failing to hold short of runway).		✓	✓		✓
SATCE.9.14	Ownship routing (flight plan): ATC services should provide clearances and instructions to the ownship to follow routing in accordance with the ownship flight plan. If required during a scenario, ATC should support changes to the ownship flight plan.		✓	✓		✓
SATCE.9.15	Ownship routing (published routes): ATC services should provide clearances and instructions to the ownship to follow published routes, if available, although routes are not required to be current. Route clearances from ATC should match the information available to the flight crew.		✓	✓		✓

SATCE						
Requirements		Feature fidelity level			Verification and validation	
		G	R	S	QTG	F&S
SATCE.9.16	Ownship routing (take-off and landing areas): ATC services should provide the ownship with landing and departure clearances to and from take-off and landing areas designated for the ownship aircraft category and weight class.		✓	✓		✓
SATCE.10	ATC data link communications Describes the simulation of certain non-voice messages between ATC services and the ownship. Data link communications unrelated to ATC (e.g. company communications, email services) are considered outside the scope of a SATCE system. Note: Data link communications need only be simulated as applicable or relevant to the simulated aircraft type and operation.					
SATCE.10.1	Message sequence: Data link messages should follow a correct and coherent sequence of transmissions.			✓		✓
SATCE.10.2	Message indications: Data link messages, contracts and connections should result in correct flight deck visual and/or audio indications.			✓		✓
SATCE.10.3	Message presentation: Data link messages should be presented to the flight crew with the correct indications and in a format that replicates that of the simulated aircraft equipment. Refer to CS FSTD.QB.104 for general requirements for the aircraft systems FSTD feature at fidelity level S.			✓		✓
SATCE.10.4	Timing delays: Delays in message timing due to equipment, network, ATC service response time and other delays should be typical of real-world operations. An excessive delay may indicate failed delivery.			✓		✓

SATCE						
Requirements		Feature fidelity level			Verification and validation	
		G	R	S	QTG	F&S
SATCE.10.5	Message content: Data link message content should adhere to published standards. Published standards may include those from ICAO and/or the national aviation authority and/or other industry organisations, such as the Radio Technical Commission for Aeronautics and Aeronautical Radio Incorporated (ARINC).			✓		✓
SATCE.10.6	Air traffic service clearances: Air traffic service clearance messages should be consistent with published routes, waypoints, flight information regions and real-world ATC centres. The routes, waypoints, flight information regions and real-world ATC centres should match those in the information available to the flight crew.			✓		✓
SATCE.10.7	Data link weather: Data link weather messages should support text- and/or graphic-based weather services and correlate with reported weather conditions. Text reports should be consistent with the information from other automated weather-reporting services, if available.			✓		✓
SATCE.10.8	Data link initiation capability (DLIC): The simulation of DLIC should allow the flight crew to establish a connection with a CPDLC service provider that corresponds to a real-world facility.			✓		✓
SATCE.10.9	Connection management: The SATCE system should simulate the transfer between an active data authority and the next data authority at the appropriate time and/or distance from a control boundary and support the termination of a data link connection by the flight crew or the ATC service provider. CPDLC and ADS-C connections should support transfers and terminations.			✓		✓
SATCE.10.10	CPDLC: The SATCE system should support the flight crew in sending and receiving CPDLC messages. The CPDLC message set need only contain those messages required to support the scenario.			✓		✓

SATCE						
Requirements		Feature fidelity level			Verification and validation	
		G	R	S	QTG	F&S
SATCE.10.11	CPDLC (regional protocols): CPDLC messages should be consistent with regional protocols and use the published message set available for the corresponding regional data authority. The CPDLC message set need only contain those messages required to support the scenario.			✓		✓
SATCE.10.12	ADS-C: ADS-C messages should be available through simulations of real-world data authorities that support ADS-C messaging. The ADS-C information displayed should reflect the ownship position and context. The ADS-C message set need only contain those messages required to support the scenario.			✓		✓
SATCE.10.13	Flight information service – broadcast (FIS-B): The SATCE system should support FIS-B messages. Messages, including weather messages, and graphics should correlate with those of other reporting systems available to the flight crew. The FIS-B message set need only contain those messages required to support the scenario.			✓		✓
SATCE.10.14	Service failures: The SATCE system should support data link service failures and recovery, including CPDLC service failures. A data link service failure should be configurable before the start of a scenario or initiated via the IOS during a scenario.			✓		✓
SATCE.11	System correlation Describes the characteristics necessary for a SATCE system to correlate with various FSTD systems, so that all information concerning the simulated ATC environment available to the flight crew and instructor, including navigation data, visual and audio cues, is in harmony. Note: Requirements in this section concerning the visual scene are applicable only if a visual cueing FSTD feature is at fidelity level G, R or S.					

SATCE						
Requirements		Feature fidelity level			Verification and validation	
		G	R	S	QTG	F&S
SATCE.11.1	ACAS (standard procedures): Standard procedures and associated pilot and ATC radio communications to an ownship ACAS event should be supported in accordance with ICAO Document 4444 (<i>Procedures for Air Navigation Services – Air traffic management</i>), including ATC responses to the flight crew reporting that an ACAS manoeuvre has been completed and any subsequent altitude/flight level clearances and instructions.			✓		✓
SATCE.11.2	Traffic on the visual system: Other aircraft and/or ground vehicle traffic within visual range of the ownship should be visible to the flight crew in the visual scene if the weather conditions and line of sight permit such visibility.		✓	✓		✓
SATCE.11.3	Traffic on the visual system (alignment): Other aircraft and/or ground vehicle traffic positions and routing should be aligned with the FSTD visual display system operating site model, including other aircraft taxi routes, parking locations and take-off and landing areas. Other traffic should appear at the ground level or AGL and avoid collisions with buildings and/or other traffic under normal conditions.		✓	✓		✓
SATCE.11.4	Navigation data alignment (other aircraft): Other aircraft traffic positions and routing should be aligned with the FSTD navigation data available to the flight crew.		✓	✓		✓
SATCE.11.5	Navigation data alignment (airspace): Airspace volumes and sectorisations for terminal and en route controlled airspace should be aligned with the FSTD navigation data available to the flight crew.		✓	✓		✓

SATCE						
Requirements		Feature fidelity level			Verification and validation	
		G	R	S	QTG	F&S
SATCE.11.6	Traffic on flight deck displays: If the simulated aircraft is equipped with traffic display capability, other traffic within range should be shown on the flight deck instrument panel displays. Refer to CS FSTD.QB.104 for the general requirements for the aircraft systems FSTD feature at fidelity level S.		✓	✓		✓
SATCE.11.7	ADS-B traffic: If the simulated aircraft is equipped with ADS-B in capability and other traffic is equipped with ADS-B out capability, other traffic ADS-B information should be available to the flight crew. Automatic dependent surveillance – rebroadcast and traffic information services – broadcast stations may also use and/or provide other traffic ADS-B information if they are simulated by the FSTD or the SATCE system.		✓	✓		✓
SATCE.12	IOS interfaces and controls Describes the functions and capabilities necessary for users to obtain information and manage the SATCE system. The SATCE system should reduce instructor workload for certain tasks, including the need to provide ATC services to the ownship.					
SATCE.12.1	Access to data link communications: If a data link is simulated, the SATCE system should provide downlink and uplink data link messages via the IOS during a scenario, for monitoring and intervention purposes.			✓		✓
SATCE.12.2	Other traffic context and information: The SATCE system should provide visibility of the wider traffic context and provide other aircraft and/or ground vehicle information via the IOS during a scenario. Other aircraft information may include position, call sign, aircraft type and flight plan details. Other traffic information need only be provided to the extent required to support the scenario.		✓	✓		✓

SATCE						
Requirements		Feature fidelity level			Verification and validation	
		G	R	S	QTG	F&S
SATCE.12.3	Access to radio communications: The SATCE system should provide flight crew and ATC radio communications via the IOS during a scenario, for monitoring and intervention purposes.		✓	✓		✓
SATCE.12.4	Simulation functions (common): The SATCE system should support commonly used simulation control and scenario set-up functions with minimal training disruption and instructor management. At a minimum, these FSTD functions should include total/flight freeze, resets and repositions.		✓	✓		✓
SATCE.12.5	Disable SATCE: The SATCE system should support the disabling of the SATCE functionality via the IOS during a training scenario. Disabling the SATCE functionality, including synthetic ATC services and other traffic, should cause minimal training disruption and require minimal instructor management.		✓	✓		✓
SATCE.12.6	Mute and restore (background radio communications): The SATCE system should support muting background radio communications between ATC and other traffic and then restoring the audio via the IOS during a scenario. Radio communications from ATC to the ownship should remain unaffected by muting and restoring background radio communications. While mute is enabled, other traffic simulation should continue so that the impact on the flight crew's situational awareness is minimised.		✓	✓		✓
SATCE.12.7	Other traffic (configurable presence): The SATCE system should support the configurable presence of other aircraft and/or ground vehicles in the simulated environment via the IOS or lesson-planning tools before the start of a scenario (i.e. without other traffic present, ATC services may be provided to the ownship only). Note: The absence of other traffic should not affect the capability of the FSTD visual system to display other traffic (refer to CS FSTD QB.111).		✓	✓		✓

SATCE						
Requirements		Feature fidelity level			Verification and validation	
		G	R	S	QTG	F&S
SATCE.12.8	Other traffic (controllable presence): The SATCE system should support the controllable presence of other aircraft and/or ground vehicles in the simulated environment via the IOS during a scenario (i.e. other traffic may be dynamically added or removed during a scenario).		✓	✓		✓
SATCE.12.9	Other traffic (configurable flow): The SATCE system should support configuring the number of other aircraft and/or ground vehicles manoeuvring in the simulated environment via the IOS or lesson-planning tools before the start of a scenario.		✓	✓		✓

GM1 FSTD.QB.160 Simulated air traffic control environment requirements for flight simulation training devices

Summaries of the fidelity levels for SATCE are presented in the table below. These summaries are broad overviews provided for guidance only. They do not encompass all aspects and requirements and are not intended to replace CS FSTD.QB.160.

The subheadings in CS FSTD.QB.160 are based on ARINC Specification 439B, *Simulated air traffic control environments in flight simulation training devices*, to enable the traceability of the headings to that industry standard and guidance document.

16: SATCE
Specific: Fully automated ATC services (e.g. clearances, instructions and automated weather reporting) that are location-specific are provided to both the ownship and other aircraft and ground vehicle traffic. Both ATC- and flight-crew-initiated radio and data link communications (as applicable) are supported. ATC is capable of responding to or correcting errors by the ownship flight crew (e.g. regarding flight path, ground routing or radio transmissions). Radio transmissions between ATC and the ownship and ATC and other traffic (if present) use both standard and regional and/or location-specific phraseology. ATC services at the operating site support the movement of the ownship, other aircraft and ground vehicles on/in multiple connected runways, taxiways and parking areas. Simulated ATC and traffic are fully integrated with the visual scene and aircraft systems (e.g. ADS-B, ACAS, radio and data link communications). The IOS provides contextual information and controls for SATCE. It can manage ATC and other traffic (e.g. presence, flow, threats) or disable SATCE entirely.
Representative: Fully automated ATC services (e.g. clearances, instructions and automated weather reporting) that are characteristic of the location are provided to both the ownship and other aircraft traffic. Both ATC- and flight-crew-initiated radio communications are supported. ATC is capable of responding to or correcting errors by the ownship flight crew (e.g. regarding flight path, ground routing or radio transmissions). Radio transmissions between ATC and the ownship and ATC and other traffic (if present) use standard phraseology. ATC services at the operating site support the movement of the ownship and other aircraft on/in at least one connected runway, taxiway and parking area. Simulated ATC and traffic are fully integrated with the visual scene and aircraft systems (e.g. ADS-B, ACAS, radio communications). The IOS provides contextual information and controls for SATCE. It can manage ATC and other traffic (e.g. presence, flow, threats) or disable SATCE entirely.

Generic:

Reserved for future use.

SUBPART C – ENGINEERING JUSTIFICATION

CS FSTD.ENG.001 General

- (a) Subpart C establishes the types of data and associated documentation required to validate the aircraft and cueing features during an initial evaluation.
- (b) The FCS defines how closely the FSTD simulates the aircraft, cueing and environment. The word ‘aircraft’ here means any aircraft used as a reference for the FSTD. It can be an aircraft type and variant (feature fidelity level S), an aircraft type (feature fidelity level R) or a group of aircraft (feature fidelity level G). For example, if the FSTD features for performance and handling are at the specific fidelity level, the FSTD should be within certain tolerances to the real aircraft type and variant and should hence fly and handle very similarly to the real aircraft type and variant. Data from the real aircraft type and variant is therefore needed to build the simulation and to show that objective tests are within tolerances. At lower levels, the requirements are correspondingly lower.
- (c) Two types of data are required for the development and qualification of an FSTD.
 - (1) Design data should be used to develop simulation models and flight deck hardware.
 - (2) Validation data should be used to objectively confirm that the simulation models reflect the static and dynamic performance characteristics of the aircraft and that the cueing features (if required to be objectively validated) conform to those of the aircraft. Acceptable sources of validation data are described in this subpart.
- (d) An ER should be provided as part of the MQTG for any FSTD, regardless of its FCS. The ER should establish and justify the sources of design data and validation data.
- (e) A VDR should be provided as part of the MQTG for any FSTD, regardless of its FCS. The VDR should define the validation data to be used for each objective test. Multiple VDRs may be provided as appropriate, such as in cases in which the validation data is from multiple data providers or aircraft variants or if the different aircraft features are based on different VDRs. The VDR(s) should be acceptable to the competent authority.
- (f) The VDR(s) should be submitted to the competent authority as early as possible in the planning stages for any FSTD for which qualification to the standards contained in this CS-FSTD is planned.

GM1 FSTD.ENG.001 General

While this subpart is primarily concerned with design and validation data to support the aircraft simulation features, the cueing features and transport delay tests should also be taken into consideration. For example, the various data reports may include information related to the position of the pilot’s eye and the flight deck cut-off angle (visual ground segment test), information related to the computation of the transport delay tests and vibration and sound validation data, etc.

CS FSTD.ENG.010 Design data

- (a) Design data comprises the set of aircraft and systems data and other information used to develop and test the simulation models, flight deck hardware and other simulation aspects. Design data may be collected from multiple sources.
- (b) The ER should include a clear description and justification of all design data sources.

GM1 FSTD.ENG.010 Design data

TYPES OF DESIGN DATA

- (a) The design data provided may be from multiple sources, in accordance with the requirements for each feature fidelity level. Table 1 provides a non-exhaustive list of examples of applicable primary design data sources for each feature fidelity level. Other, secondary design data sources may also be used if appropriately justified for the given feature fidelity level.

Table 1. Examples of design data sources applicable to each fidelity level

Design data source	G	R	S	Comments
Aircraft manufacturer data	✓	✓	✓	
Avionics and systems manufacturer data	✓	✓	✓	Such as systems descriptions and wiring diagrams
Supplemental type certificate holder	✓	✓	✓	
Genuine aircraft software/hardware	✓	✓	✓	Such as stimulated or re-hosted avionics and systems
Genuine simulation software package	✓	✓	✓	Such as aircraft manufacturer simulation software
In-aircraft data acquisition	✓	✓	✓	For example, by third-party data providers
FAA-published 'Alternative objective data for FTD level 5'	✓	✓		Refer to Title 14 of the Code of Federal Regulations, Part 60, Appendix B
Aircraft manuals	✓	✓		
System descriptions	✓	✓		
Public domain data	✓			

- (b) In order to demonstrate compliance with the requirements for the aircraft systems FSTD feature at the specific or representative fidelity level, the aircraft system simulation should be based on and be traceable to the design data source for the simulated aircraft. This traceability is important due to the diversity of aircraft systems and the various simulation methods and data acquisition techniques available.

GM2 FSTD.ENG.010 Design data

DESIGN DATA JUSTIFICATION

The engineering report should contain descriptions of any design data used in the development of the FSTD and simulation models.

The quality, fidelity and scope of the design data should be adequate to ensure that the FSTD and the simulation models can be built to comply with the requested FCS and support a positive transfer of training in the applicable flight envelope and that the FSTD aligns with the simulated aircraft.

Design data is especially important for modelling the aircraft flight characteristics, including flight control system operation, aircraft systems and performance-and-handling-related features.

Design data is needed for the flight deck layout and structure to ensure proper touch and feel. Examples include flight deck dimensions, hardware, materials and button and knob movement ranges.

The simulation of aircraft system operation (e.g. logic, indications, delays, system interdependencies) requires data. The source of this data could be OEM design data or data from aircraft testing by third-party data providers. System models developed by reference to aircraft test data instead of OEM design data should use very systematic and comprehensive methods to acquire the necessary data. Due to this complexity, the ER should demonstrate that the data-gathering and model development process was appropriate for the fidelity level.

The competent authority may require further information on the design data whenever necessary.

CS FSTD.ENG.015 Validation data

- (a) Validation data comprises the set of aircraft and other data to be compared against FSTD objective test results in the MQTG to demonstrate that the simulation and simulation models are within the set tolerances. The source of the validation data for each objective test should be defined by the VDR, as described in CS FSTD.ENG.025. Validation data not derived from aircraft flight tests should be substantiated in the ER.
- (b) Regardless of the source and fidelity level, the validation data should comply with the objective test requirements in Subpart D. The validation data should have the appropriate configuration, flight path, trim state, control inputs, etc., as specified in the objective test requirements. The data should include all parameters necessary for objective comparison against the FSTD. Comparison between validation data and the FSTD is objective and meaningful only if the quality of the validation data is adequate.
- (c) Validation data variables, along with sign conventions, should be defined in a nomenclature list. This list should be included at an appropriate location in the MQTG.
- (d) Since individual FSTD features can be at different levels, the validation data requirements should be considered separately for each FSTD feature.
- (e) Validation data requirements vary based on feature fidelity levels.
 - (1) If the fidelity level is S for the aircraft simulation and cueing features, the initial evaluation should be based on objective evaluation against validation data for the particular aircraft type and variant.

Validation data from the same aircraft type and/or variant may be acceptable provided that the differences and effects are understood and explainable (e.g. in

terms of engines, avionics, body length). The VDR or ER should contain the rationale for using the data, including the expected impact on the FSTD's performance and handling characteristics.

Flight test data is the primary validation data source. Validation data may include FSTD results (footprint data) in isolated cases if specifically allowed by this CS-FSTD.

Submissions of data other than flight test data should include a rationale or an explanation of validity with respect to the available flight test information.

For FSTDs simulating aircraft for which the application for aircraft type certification was issued after 17 February 2014, if the fidelity level is S for the aircraft simulation features, the source data should be the data as defined by the OSD established in accordance with Part 21 (Annex I) of Commission Regulation (EU) No 748/2012.

- (2) If the fidelity level is R for the aircraft simulation and cueing features, the initial evaluation should be based on objective evaluation against validation data for the aircraft type, complemented, if necessary, by subjective development.

Data from multiple variants within the same type may be used to establish the set of validation data, provided the differences between variants are properly accounted for.

Validation data other than flight test data should be substantiated in the ER to demonstrate that the proposed validation data for the feature in question is characteristic of the aircraft type.

The ER should include detailed information on any subjective development and justification of the FSTD results. This approach should be used if agreed with the competent authority.

- (3) If the fidelity level is G for the aircraft simulation and cueing features, the initial evaluation should be based on objective evaluation against validation data for the group of aircraft, which may be complemented by subjective development.

Aircraft data (as defined in CS FSTD.GEN.015) is not required for each individual objective test but should be provided, at a minimum, for each objective test section.

Validation data sources other than flight test data should be substantiated in the ER to demonstrate that the proposed validation data for the feature in question is characteristic of the group of aircraft.

The ER should include detailed information on any subjective development and justification of the FSTD results.

- (f) If validation data has been acquired from multiple sources, the highest-quality data sources should be prioritised and used.

GM1 FSTD.ENG.015 Validation data

TYPES OF VALIDATION DATA

(a) Validation data may be provided from multiple sources, in accordance with the requirements for each feature fidelity level. This section provides a non-exhaustive list of examples and usage. The ER should include a clear description of all data sources, including the basis for derived parameters. These descriptions may be listed in a flight test report, if provided, or another document and referenced in the ER.

(b) Examples of validation data are as follows.

(1) 'Flight test data' is defined under CS FSTD.GEN.015 and refers to aircraft data that is used to objectively confirm that the FSTD reflects the static and dynamic handling and performance characteristics of the simulated aircraft and its relevant systems. Flight test data should be collected using a calibrated data acquisition system of sufficient resolution and verified as accurate.

(2) 'Published performance data' refers to the data used by pilots for flight-planning purposes. Performance data may include cruise speeds, climb rates, stopping distances, stall speed values and hover ceilings, and other performance data published by the OEM. If using performance data to support validation for fidelity levels S and R, the ER should provide a rationale for any tolerance parameters not included in the performance data. For example, cruise performance data would typically not include stabiliser trim positions.

Note that the AFM data may be normalised to represent a conservative, least-performing case for the aircraft fleet.

(3) Engineering simulator/simulation validation data is described in CS FSTD.ENG.040.

(4) Predictive and theoretical data includes the results of commonly accepted simulations or predictive models and also recognised theoretical results. Such data should be described and justified in the ER. Examples of such data include formulas applicable to well-isolated and well-understood physical phenomena, methods and models documented in academic books or professional literature, verified open-source tools developed by renowned research institutions and verified commercial products from well-established industry sources.

(5) Footprint data from an FSTD flown manually by a suitably qualified pilot may be used. The footprint data, the qualifications of the pilot and the use of the data should be properly justified in the ER. For example, safety limitations in AFMs may prohibit certain manoeuvres or test cases and may hence be one justification for using footprint data.

Footprint data can be recorded in the FSTD itself. In the event of an approved change to a qualified FSTD that affects objective test results, the competent authority may require that the test result be regenerated under the new conditions to gather new footprint data.

Footprint data from another qualified FSTD can be used as validation data for a new FSTD. Footprint data from another qualified FSTD is generally limited to the following use cases, if at least one of the conditions below is met.

- (i) The feature fidelity level of the qualified FSTD exceeds that of the new FSTD. For example, data for a particular feature may be sourced from an FSTD which implements the feature at the specific fidelity level, if the FSTD being qualified implements this feature at the generic or representative fidelity level.
 - (ii) The qualified FSTD simulates the same aircraft type as the new FSTD, both devices have the associated feature(s) at the same fidelity level and an essential match can be demonstrated between the footprint data and the new FSTD.
- (6) 'Alternative flight test data' is defined under CS FSTD.GEN.015 and refers to aircraft data collected on the ground and in flight, using cost-effective methodologies, to support fidelity levels R and G. Such alternative methodologies, if properly executed, do not require external instrumentation or other equipment that would impact the aircraft's airworthiness.

Examples of alternative methodologies include flight deck video recordings, recordings from an electronic data bus or aircraft data recorder (avionics data card or data bus output, health and usage monitoring systems (HUMS), flight data recorder (FDR), etc.) and manually recorded kneeboard data.

Using alternative flight test methodologies, some parameters may be derived rather than measured directly. For example, the angle of attack (AOA) can be sufficiently derived if the flight test programme ensures the collection of acceptable-level, unaccelerated, trimmed flight data. The AOA may be validated by conducting the three basic 'fly-by' trim tests. The FSTD time history tests should begin in level, unaccelerated and trimmed flight and the results should be compared with the flight test pitch angle. Similarly, control surface positions can be derived from in-flight measurements of flight deck control force/position and on-ground measurements of cable stretch and gearing, accounting for any non-linearities.

Flight deck video should be accurately synchronised with other data sources and should provide sufficient resolution to allow for the magnification of the display to make appropriate measurements and comparisons. The detail provided by the video should provide sufficient clarity and accuracy to measure the necessary parameter(s) to at least half of the tolerance authorised for the particular test being conducted and allow for the integration of the parameter(s) in question to obtain a rate of change. Video data may not provide the time resolution for 'fast' aircraft responses, such as short-period response or control dynamics.

A typical flight test case using alternative methodologies might acquire data using an inertial measurement system and a synchronised video of the calibrated flight deck instruments, recordings of the force/position measurements of flight deck controls and a clear visual directional reference for a known magnetic bearing (e.g. a runway centre line). The ground track and wind-corrected heading could be used for the side-slip angle.

- (7) FAA alternative objective data is published in Title 14 of the Code of Federal Regulations, Part 60, Appendix B, for different categories of fixed-wing aircraft.
- (8) Data in the public domain may include academic publications, National Aeronautics and Space Administration (NASA) technical reports and other sources known to employ appropriate engineering methodology.
- (9) Aircraft CSs may be used to substantiate FSTD performance and handling qualities trends for some tests – for example, positively damped phugoid and Dutch roll modes. If the aircraft CSs only identify trends, it is important to also justify magnitudes as part of the ER. For example, a critically damped FSTD phugoid would be justified in trend (positively damped) but not in magnitude (damping ratio too high).
- (10) Subjective development may be used to supplement other sources of data, resolve issues with the available aircraft data or tailor the FSTD performance and handling qualities to better match the aircraft, within the constraints of CT&M and based on the judgements of qualified personnel. The subjective development process should be documented in the ER.

Subjective development may be used to fill gaps in existing data packages for fidelity levels R and G, as described in CS FSTD.ENG.015.

Subjective development may result in using the FSTD to record a footprint test result.

- (c) Table 1 provides a summary of acceptable validation data sources for each feature fidelity level. This list is not comprehensive; other sources may be proposed and justified in the ER. The list is divided into four groups. These groups are arranged in descending order based on the expected data quality of the sources within each group. Sources in group A are generally considered to provide the highest quality of data, followed by sources in group B, then group C and finally group D, which is expected to offer the lowest quality among the groups.

Sources marked (✓) should be used only if agreed with the competent authority, as they may lack the necessary fidelity or level of detail for assessing tolerances.

If a tick mark is not listed, the type of validation data may still be used as a secondary source to support the ER justification.

Exceptions to this for an interim FSTD qualification are described in CS FSTD.ENG.060.

Table 1. Summary of validation data sources applicable to each fidelity level

Validation data source	G	R	S	Comments
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A	Flight test data	✓	✓	✓	Conventional flight test programme
	Published performance data	✓	✓	✓	AFM and rotorcraft flight manual charts, OEM performance calculators, etc.
B	Engineering simulator/simulation data	✓	✓	(✓)	Refer to CS FSTD.ENG.040
C	Predictive and theoretical data	✓	✓	(✓)	Permissible for level S in isolated cases, with proper rationale
	FSTD footprint data	✓	(✓)	(✓)	Validation data may be supported by subjective development and FSTD footprint data. Permissible for S in isolated cases, with proper rationale.
	Alternative flight test data	✓	✓		Limited-scope flight test programme
D	FAA-published 'Alternative objective data for FTD level 5'	✓	(✓)		Refer to Title 14 of the Code of Federal Regulations, Part 60, Appendix B
	Public domain data	✓	(✓)		
	Aircraft CSs	✓	(✓)		

(d) In general, data should fulfil the following.

- (1) Data should include all parameters necessary for objective comparison against the FSTD (if the aircraft includes unique characteristics or has a configuration that does not align with the assumptions of this CS-FSTD, the ER and MQTG should propose alternative parameters and methodologies to meet the intent of the objective tests).

For example, if the aircraft uses an elevator trim tab instead of stabiliser trim, it would not be acceptable to disregard stabiliser trim tolerances. Instead, the validation data should include trim tab data.
- (2) Data should include both inputs and outputs, with sufficient resolution to repeat the manoeuvre in the FSTD.
- (3) Data should include stable trim conditions for several seconds prior to initiating each manoeuvre.
- (4) Data should avoid time shifts between parameters.

It should be noted that the scope of the design and validation data required to support the FSTD development and verification process generally scales with the aircraft performance envelope and complexity. For example, an FSTD simulating a fly-by-wire jet operating at high altitude and high Mach would require substantially more data than an FSTD simulating a single-engine piston aeroplane.

Refer to ARINC Specification 450-1, *Flight Simulator Design and Performance Data Requirements*, as amended, for additional guidance.

- (e) The validation data requirements are considered separately for each feature. For example, if the FSTD feature performance and handling on ground is at the generic fidelity level, then the validation data for this feature needs to meet only the requirements for the generic level. If the FSTD feature performance and handling out-of-ground effect of the same device is at the specific level, then the validation data for this feature needs to meet the requirements for the specific level.
- (f) The following guidance on validation data requirements applies, dependent on the feature fidelity level.
 - (1) If the fidelity level is R for the aircraft simulation and cueing features and validation data is aggregated from multiple variants within the same type, differences between variants should support a coherent FSTD model. Otherwise, data may need to be limited to a single variant. For example, two helicopter variants may share a common type certificate despite having fundamentally different rotor hub designs. In such a case, all runs within a given test (e.g. lateral control responses in different directions and at different airspeeds) should use validation data from a single variant so that the proper trends are evident.
 - (2) If the fidelity level is G for the aircraft simulation and cueing features, flight test validation data may not be available for each test. For example, alternative flight test data could be used to verify the roll response rate during cruise. The same control input could be repeated in the FSTD in a landing configuration without requiring a second flight test case, demonstrating reduced aileron effectiveness at the lower airspeed.

CS FSTD.ENG.020 Engineering report

- (a) This CS provides standards for the ER, which describes the data and methods used to design and verify an FSTD in line with the requirements in this CS-FSTD. The ER is required for all FSTDs, regardless of fidelity level, and should be part of the MQTG (refer to CS FSTD.QB.030).
- (b) The scope of the ER should comprise the following FSTD features:
 - (1) Flight deck layout and structure;
 - (2) Flight control forces and hardware;
 - (3) Flight control systems operation;
 - (4) Aircraft systems;
 - (5) Performance and handling on ground;
 - (6) Performance and handling in-ground effect;
 - (7) Performance and handling out-of-ground effect;
 - (8) Sound cueing; and

- (9) Vibration cueing.
- (c) The ER should contain the following information.
 - (1) **Design data sources.** A clear description of all design data sources, including their scope and the methods used for their acquisition, should be provided. Refer to CS FSTD.ENG.001 and CS FSTD.ENG.010. For aircraft systems, also refer to CS FSTD.QB.104, section 'Aircraft system simulation basis'.
 - (2) **Validation data sources.** A list of all validation data sources, including their acquisition methods and substantiation, should be provided. Refer to CS FSTD.ENG.015.
 - (i) For flight test validation data, the description may be a reference to the VDR. For other sources of validation data, particularly alternative flight test data for fidelity levels R and G, the ER should include a detailed description of the flight test programme or a reference to the validation data provider's flight test report. Any such references should be included as attachments to the MQTG.
 - (ii) If applicable, this section should also include a description of any subjective development, including the objective tests affected, the process used and the qualifications of the pilots providing subjective feedback.
 - (3) **Simulation models.** A description of the simulation models and their associated hardware solutions should be provided, addressing the following aspects.
 - (i) **Scope.** This should be a description that details the coverage of the simulation models and hardware solutions, including the aspects of aircraft behaviour and operating conditions that they simulate.
 - (ii) **Implementation.** This should cover the structure, integration and execution of the models, including computational methods and, as applicable, system interactions.
 - (iii) **Validity.** This should cover the verification and validation processes used to ensure the accuracy and consistency of simulation across the entire operational envelope. While the objective tests demonstrate compliance under particular conditions, aircraft configurations and flight states, the description must extend beyond these sample points and provide a high-level description of how consistency and continuity with the aircraft's flight characteristics are ensured across the entire operational envelope.
 - (4) **Objective test verification process.** Information on how each objective test should be verified against validation data should be provided, if not included as part of the objective test in the QTG. The information in the ER may be grouped for the tests that use the same validation data source and provider.
 - (5) **Compliance of objective tests for the initial qualification.** For objective test results making use of CT&M tolerances, for footprint tests or if a test assessment against the validation data is not possible due to the type of data used (e.g. if alternative flight

test data and other data is used), a justification of the objective test result should be given in the ER.

The justification should include the following information, as appropriate for each objective test.

- (i) It should include a presentation of the validation data, which may include plots, tables and/or text descriptions.
 - (ii) It should include a written explanation of why the FSTD's objective test result is considered compliant and acceptable, including how it demonstrates the tolerances, and using engineering judgement, as applicable. If there is no aircraft data (as defined in CS FSTD.GEN.015), other data sources may be used (refer to CS FSTD.ENG.015).
- (d) As part of the MQTG, the ER is considered to be a 'living document', changing with the FSTD and simulated aircraft. Therefore, the ER may be later revised to accommodate changes to the FSTD or simulated aircraft. If a modification to the FSTD requires a corresponding change to the validation data, such as to incorporate a performance update following a change to the aircraft, the ER should be updated.

GM1 FSTD.ENG.020(c) Engineering report

ENGINEERING REPORT AND STATEMENTS OF JUSTIFICATION

CS FSTD.ENG.020 specifies the requirements for the ER. Some of the information expected in the ER may overlap with the SOJs required by CS FSTD.QB.020. While both the ER and SOJs should be submitted in a format that allows efficient review and evaluation, wide duplication of information across documents is not intended.

For example, SOJs associated with CS FSTD.QB.102 should justify the primary flight control forces and travel. If the applied tolerances for these forces, as required by Subpart D, are CT&M, the ER should provide a written justification of the test results for 2.a objective tests (refer to CS FSTD.ENG.020, paragraph (c)(5)(ii)). In such a case, the justification in the ER may overlap with the relevant SOJ. Instead of duplicating the information, it is acceptable to reference the corresponding document.

GM2 FSTD.ENG.020(c) Engineering report

COMPLIANCE OF OBJECTIVE TESTS FOR THE INITIAL QUALIFICATION

This GM gives guidance on the contents of the ER by providing fictional examples to justify objective tests for different cases and feature fidelity levels. Note that if the features are at different fidelity levels, the ER should provide sufficient information to account for this and justify the objective test results of the integrated FSTD.

- (a) Example 1 presents an example of an aeroplane FSTD that simulates all aircraft features at the representative fidelity level, based on alternative flight test data. This example is for a single test. It is not intended to unnecessarily repeat the same information across multiple objective tests. Therefore, some of the information in this example (e.g. details on the design data) may be better presented elsewhere in the ER or could be grouped for multiple tests. However, this

example is provided to demonstrate the overall logic and key principles of the justification in the ER.

- (b) Example 2 presents an example of an aeroplane FSTD that simulates all aircraft features at the generic fidelity level. The example is for a single test. There is no flight test data, but other sources are used. The test uses the footprint method.

Example 1. Engineering report example of an aeroplane FSTD with all aircraft features at level R

Engineering report / 2.c.4: Gear change dynamics – gear extension

Test scope and method

The purpose of this test is to verify that the FSTD's ability to simulate the dynamics of gear extension, including changes in aircraft pitch, drag and vibration effects during the deployment of landing gear, is representative of those of the real aircraft. This test ensures that the FSTD matches the behaviour observed in the real aircraft within CT&M.

Design and validation data

The flight test data is 'alternative flight test data' (GM1 FSTD.ENG.015) if a high-resolution and high-frame-rate flight deck video was recorded and later analysed for data points with a resolution of 2 Hz. Observed flight parameters were manually quantified and converted into a tabular dataset. This method has a level of inaccuracy, but it is precise enough for this test purpose.

String potentiometers were used to acquire the positions of the primary flight controls at 20 Hz, while the positions of the secondary flight controls were assessed from the video recording. The process for gathering design data and the data quality are discussed in Section 2 of this ER, with details of the conducted flight test cases given in Appendix B.

The design data for the landing gear and its effect on the flight dynamics consists of the flight manual system description for the landing gear, the required validation data cases and three additional test points at different airspeeds (see Appendix B).

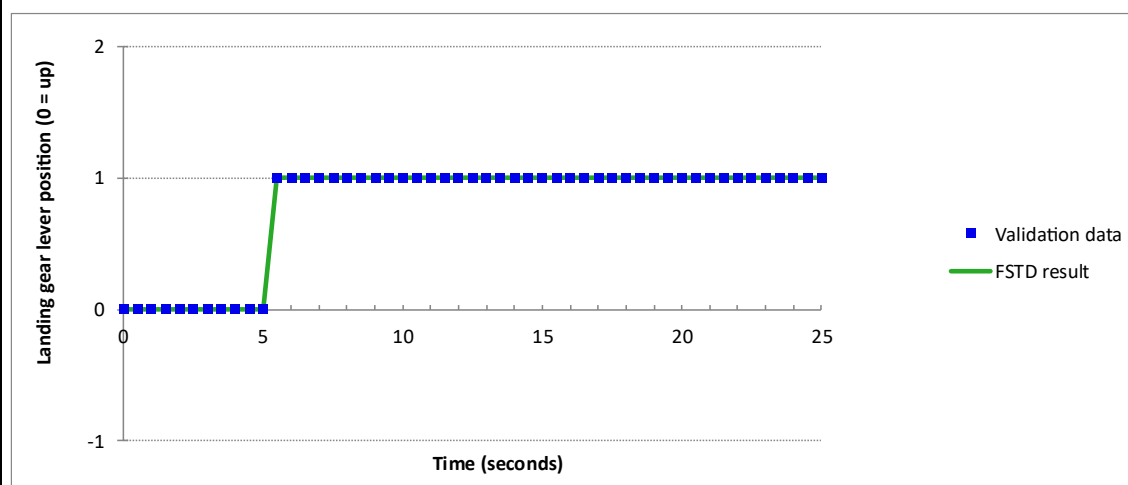
The flight tests were accomplished using the -200 version, while the FSTD simulates the -300 version of the aircraft. The differences for this test are considered negligible (see Appendix E for the differences between 200 and 300) and such an arrangement is acceptable for the representative level.

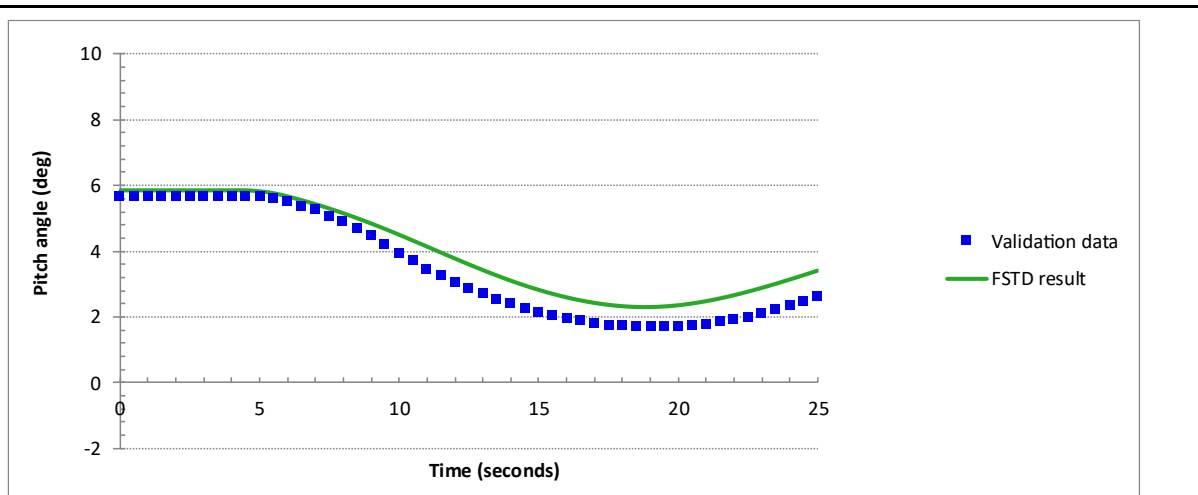
The scope and quality of the acquired data were found to be adequate to support the design and validation of this FSTD at the representative fidelity level.

Objective test analysis

The aircraft was stabilised at level flight at 130 kt. At five seconds, the landing gear lever was moved down. Free response was recorded; no flight controls were touched during the test. Initial conditions and the time history plots are given below.

Parameter	Flight test	FSTD
Gross weight (kg)	4 130	4 130
Centre of gravity (%)	24.3	24.3





[Note: This fictional example omits other parameters of the initial conditions and test graphs. A real ER is expected to include all relevant data and graphs.]

Objective test analysis and justification

The FSTD is characteristic of the real aircraft's pitch dynamics. All the tolerance parameters match the validation data for the trend and magnitude throughout the time history. The other parameters also show a representative match for the validation data. The FSTD's gear extension cycle duration was 7.7 seconds, which aligns closely with the real aircraft's 8.1 second cycle.

Due to the resolution of the video-based data (0.5 second intervals), minor dynamic effects such as transient oscillations cannot be captured accurately. However, for a stable aircraft for this manoeuvre, there are no such rapid dynamic effects. The quality and resolution of the data are found to be adequate. The objective test result demonstrates that the FSTD matches the validation data within CT&M.

Example 2. Engineering report example of an aeroplane FSTD with all aircraft features at level G

Engineering report / 2.d.2: Roll response (rate) – cruise

Test scope

The purpose of this test is to verify that the FSTD roll response is characteristic of the group of aircraft during steady cruise conditions. Specifically, the test ensures that the simulated roll rate meets the requirements of CS-25, AMC 25.147(d), which governs controllability during rolling manoeuvres.

Design data

The data used to develop and test the simulation models is as follows.

1. The CS-25 requirement for real aircraft of this group is set out in CS 25.147, which states the following regarding lateral control.
Lateral control; all engines operating. With the engines operating, roll response must allow normal manoeuvres (such as recovery from upsets produced by gusts and the initiation of evasive manoeuvres). There must be enough excess lateral control in sideslips (up to sideslip angles that might be required in normal operation), to allow a limited amount of manoeuvring and to correct for gusts. Lateral control must be enough at any speed up to V_{FC}/M_{FC} to provide a peak roll rate necessary for safety, without excessive control forces or travel.
AMC 25.147(f) further explains this for the cruise configuration.
It should be possible in the conditions specified below to roll the aeroplane from a steady 30° banked turn through an angle of 60° so as to reverse the direction of the turn in not more than 7 seconds. In these demonstrations the rudder may be used to the extent necessary to minimise sideslip. The demonstrations should be made rolling the aeroplane in either direction, and the manoeuvres may be unchecked.
2. In Title 14 of the FAA's Code of Federal Regulations (Part 60, Appendix B, 'Alternative objective data for FTD level 5'), Table B2E ('Multi-engine turbo-propeller airplane') states that the FSTD must have a roll rate of 4–25°/s if the aileron is deflected one third of its maximum travel.
3. For subjective development, the flight model of the FSTD was tuned to have the roll rate in the limits prescribed by CS-25 and Part 60 of Title 14 of the Code of Federal Regulations, as stated above. The build-up of the roll rate and the roll overshoot characteristics required subjective testing. These characteristics were refined with the assistance of an expert pilot (experienced type rating instructor; see name and credentials in Appendix D). The pilot performed multiple roll-handling and roll response tests in the FSTD under different conditions and all configurations and provided detailed feedback to fine-tune the simulated dynamics until they matched the expectations based on the real-world experience.
4. NASA report *Flight investigation of the roll requirements for transport airplanes in the landing approach* (NASA TN D-7062) was used to further verify the reasonableness of the FSTD's results.

Use of the above sources is in accordance with CS-FSTD.

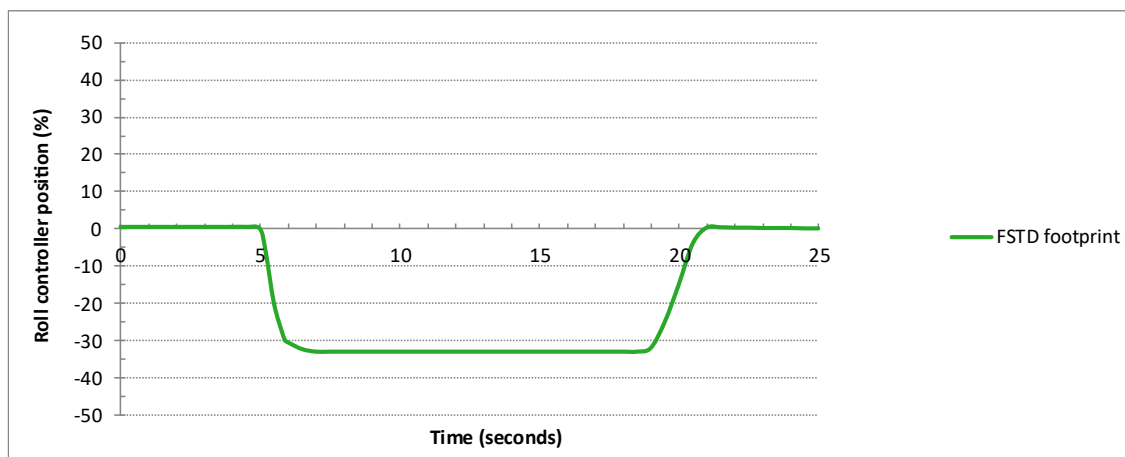
Objective test

The FSTD is set at cruise conditions (215 kt at flight level 150) in a right turn with a steady 30° bank angle. From a steady-state turning flight, the roll controller is abruptly deflected one third of its travel to the left. The aircraft experiences a roll rate to the left. Roll is continued until reaching a 30° left bank angle.

The time between 20° right bank and 20° left bank angle is measured, and an average roll rate is calculated for this region. Note that the timing begins only at 20° right bank when the roll rate has fully developed.

Initial conditions and the time history plots are given below.

Parameter	FSTD
Gross weight (kg)	21 000
Centre of gravity (%)	25.0



[Note: This fictional example omits other parameters of the initial conditions and test graphs. A real ER is expected to include all relevant data and graphs.]

Justification

The time measured in the FSTD for the roll rate from 20° right bank to 20° left bank with one-third left-wheel input (-33 %) is 4.2 s. This means an average roll rate of $(20^\circ + 20^\circ)/4.2 \text{ s} = 9.5^\circ/\text{s}$.

The FAA alternative objective data requirement for this test is to have a roll rate of 4 to 25°/s if the aileron is deflected one third of its maximum travel. The FSTD test result demonstrates that the FSTD roll rate is within this range.

A roll rate of 9.5°/s allows for a 60° roll to be completed in 6.3 s, meeting the CS-25 requirement for real aeroplanes described above.

The results are aligned with the NASA TN D-7062 for a satisfactory roll rate.

The build-up of the roll rate due to the roll controller input is logical: not instantaneous, but not experiencing any spurious lags either. As a result of subjective testing, the resulting roll dynamics and overall roll handling are found to be acceptable and in compliance with CS 25.147. The test result demonstrates that the FSTD roll rate for cruise conditions is characteristic of the group of aircraft. The used data is in compliance with CS-FSTD. The footprint of the FSTD can be accepted and taken as a reference for subsequent recurrent testing.

GM1 FSTD.ENG.020(c)(5) Engineering report

CORRECT TREND AND MAGNITUDE

- (a) CT&M is a method of comparing FSTD objective test results against validation data without the use of numerical tolerances. Results should demonstrate positive correlation between the FSTD and validation data for the inputs and outputs, with values and variations characteristic of the validation data. For example, a flap change should cause the FSTD to pitch in the same direction as the simulated aircraft would, with characteristic magnitude. The rationale for any significant differences should be provided as part of the ER.

Validation data should be overplotted with FSTD results to aid in the assessment of CT&M, unless the test is a snapshot test. Overplotting could occur as part of the objective test or as supplemental plots provided in the ER. Overplotting is most important for dynamic manoeuvres such as landings, take-offs and control responses. It is recognised that not all validation data can be overplotted; for example, some data may be provided in tabular form

or at a low sample rate. Refer to CS FSTD.ENG.015 and GM1 FSTD.ENG.015 for the applicable requirements and further guidance.

- (b) The assessment of objective test results involving CT&M should be based on sound engineering judgement, with a focus on simulation behaviour that may be perceptible to the flight crew / pilot and on replication of the manoeuvre as shown in the validation data.

For objective tests with a CT&M requirement, all the parameters should demonstrate CT&M. The main parameters under scrutiny during the test are the primary axis input and output parameters. Good engineering judgement may be applied for the off-axis parameters.

Generic and representative FSTD features should still clearly resemble the characteristics of the simulated aircraft. The ‘sensitivity’ or ‘feel’ of flying is largely created by the stability of the aircraft. For a positive transfer of training, the FSTD should have a similar trend and magnitude as the simulated aircraft; it is not sufficient to provide the correct trend while ignoring magnitude.

For example, the flight dynamics model of an FSTD should not be too stable or too unstable. Excessive stability can be experienced, for example, in phugoid, Dutch roll and spiral stability tests as a very quick return to the original flight state. Excessive instability can be seen in, for example, exaggerated pitch changes during short-period dynamic tests.

If the FSTD with the FSTD features for performance and handling at the generic fidelity level exhibits exceptionally stable flying characteristics (e.g. no overshoots during a phugoid test), the ER should justify this behaviour by providing clear and objective information to show that some aircraft in this group of aircraft has similar stability. If the FSTD’s handling qualities are too stable relative to the group of aircraft, the FSTD is at risk of providing a negative transfer of training.

- (c) ‘Trend’ is typically defined as the first-time derivative of a given parameter, but the first-order dynamic characteristics (frequency and damping) may also need to be evaluated. Examples of trends include:

- (1) the shape and direction of a control response;
- (2) correlation and phase relationships between aircraft DOFs following a control input;
- (3) oscillations increasing or decreasing over time;
- (4) changes in performance with weight, density and other dependencies;
- (5) changes in control force with airspeed, g-loading and other dependencies;
- (6) tendency to converge or diverge.

If two numbers (e.g. a change in a flight parameter) have the same order of magnitude, they are about the same size. Magnitude is necessarily subjective and may require justification. Differences in magnitude should be reasonable with respect to the aircraft envelope and scale/precision of the flight deck instruments, and they should remain within the same portion of the envelope unless the trend shows divergence.

- (d) Objective tests have different purposes and they measure different aspects and phenomena. Certain parameters require a closer match than others to be considered CT&M. For example, 4 degrees in pitch angle may not be within CT&M, while 4 kt in airspeed is more likely to still be acceptable for CT&M. It is impossible to give all-inclusive guidance or criteria to cover all the different objective tests and cases. The following information helps in interpreting and judging whether an objective test demonstrates CT&M.

If an FSTD feature is at the representative or generic fidelity level, there is a wide variety of permissible validation data sources. For example, if the validation data for an objective test is subjective development, the FSTD's footprint is justified in the ER. The justification should carefully explain what is happening in the test (e.g. the flight state and all the movements through the time history) and why the footprint can be considered to be characteristic of the simulated group of aircraft. Justification of the footprint should always be supported by other sources (e.g. quotation from academic literature that confirms that the footprint is characteristic of the simulated group of aircraft). Justification for a simple aircraft (e.g. single-engine piston aeroplane) is naturally easier than for more complex aircraft (e.g. multi-engine turbojet aircraft with swept-wings and non-linearities due to Mach effects, or helicopter with cross-couplings and a stabilisation system). The footprint and the justification together should demonstrate the CT&M.

Figures 1 to 4 below illustrate examples of objective tests for an aeroplane FSTD with the performance and handling FSTD features at the representative fidelity level. For each example, guidance is provided on whether the test outcome demonstrates CT&M. In all of these examples, the validation data consists of flight test data from the simulated aircraft variant. The examples show plots for only one parameter of the test. Figure 1 below shows examples of acceptable CT&M for gear change dynamics. The validation data is flight test data that is gathered by first trimming the aircraft for a level flight. Data is recorded for five seconds before the landing gear is selected down. The test shows the free response of the aircraft, without moving flight controls during the test. The data is recorded at least 15 seconds after the landing gear has fully extended. The landing gear extension creates drag and causes disturbances to the airflow. This results in a pitching-down moment. The airspeed changes and the pitch trim causes a pitch-up tendency to move back towards the original airspeed.

The figure also shows two examples of FSTD objective test results. Objective test example A begins from a pitch attitude that differs a little from that in the validation data. Such a difference may be caused by initial trimming or differences in initial conditions. The graph shows good timing and the trend is correct. The change in pitch angle is a little less than in the validation data, but the magnitude can still be considered to be adequate for this test and parameter. Note that other parameters of this test should still be assessed.

Example B shows that the pitching begins a little later than in the validation data. This could be due to the simulated extension time of the landing gear. The rationale could provide more information on that. The initial trend is correct. The magnitude of the change in pitch angle differs from that of the validation data but can still be considered acceptable for CT&M. The trend of the pitch-up movement at the end of the test is lower than that in the validation data, but is still in the correct direction and can still be considered acceptable for CT&M.

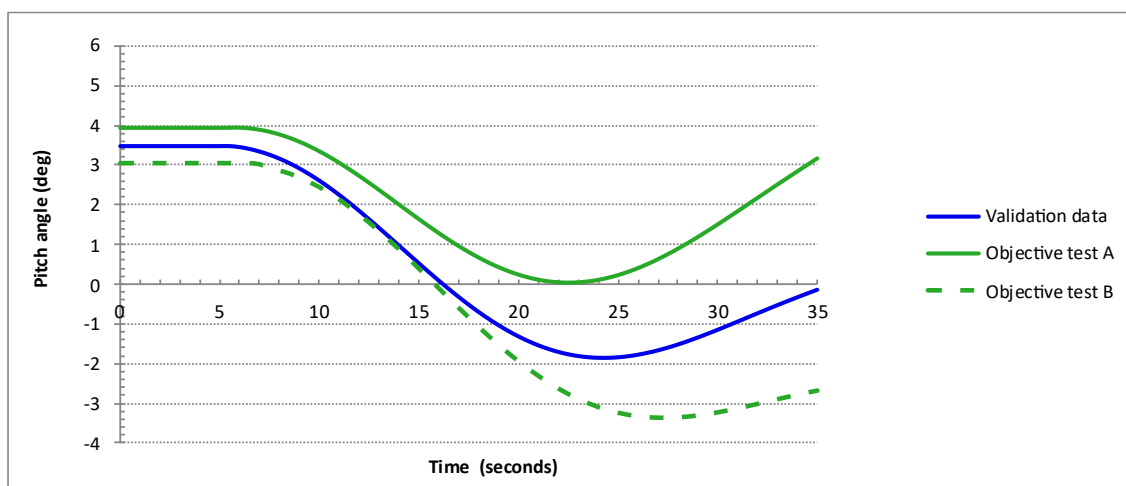


Figure 1. Examples of objective test results that demonstrate CT&M

Figure 2 below shows examples of results that do not demonstrate CT&M. The test and validation data is the same as in Figure 1. Example C shows pitch movement in the wrong direction, so example C does not demonstrate CT&M. Example D has the trend in the correct direction, but the trend is clearly lower, and the magnitude is significantly less than that in the validation data, so example D does not demonstrate CT&M. Example E has the initial trend in the correct direction, but the magnitude of the change in pitch angle is clearly more than that in the validation data. Example E does not show a trend of pitching up during the last 10 seconds of the test. These are differences that could be noted by the pilots: the FSTD in example E would require more pilot input than the real aircraft would when the landing gear is extended. Example E does not demonstrate CT&M.

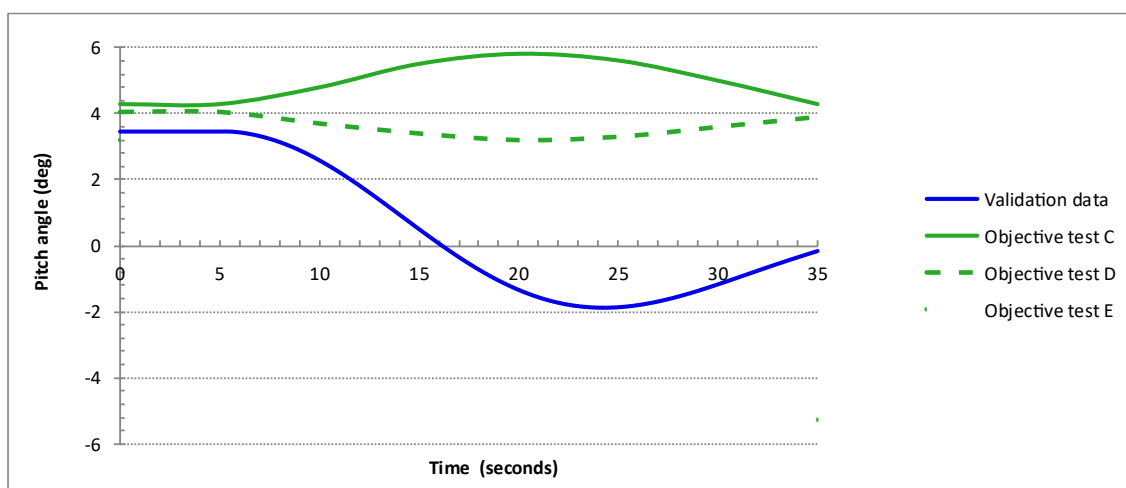


Figure 2. Examples of objective test results that do not demonstrate CT&M

Figure 3 below shows more examples of results that do not demonstrate CT&M. The test and validation data is the same as in Figure 1. Example F shows movement in the correct direction with the correct pitch rate, but the movement happens after a significant delay. The delay could be caused by test timing or the slowness of the simulated gear extension or by issues regarding the flight dynamics model. A rationale could be given, but the timing still affects the trend. Example F does not demonstrate CT&M. Example G has a good trend and magnitude

after five seconds (i.e. after the landing gear lever is moved). However, the first five seconds of the test show that the pitch angle is changing. This means that the FSTD is not trimmed for level flight. The flight state is not correct. The initial test state affects the result (i.e. output) of the test. Because of this, the test results must be deemed unacceptable. CT&M cannot be determined if the initialisation (i.e. input) is not appropriate. So, example G does not demonstrate CT&M.

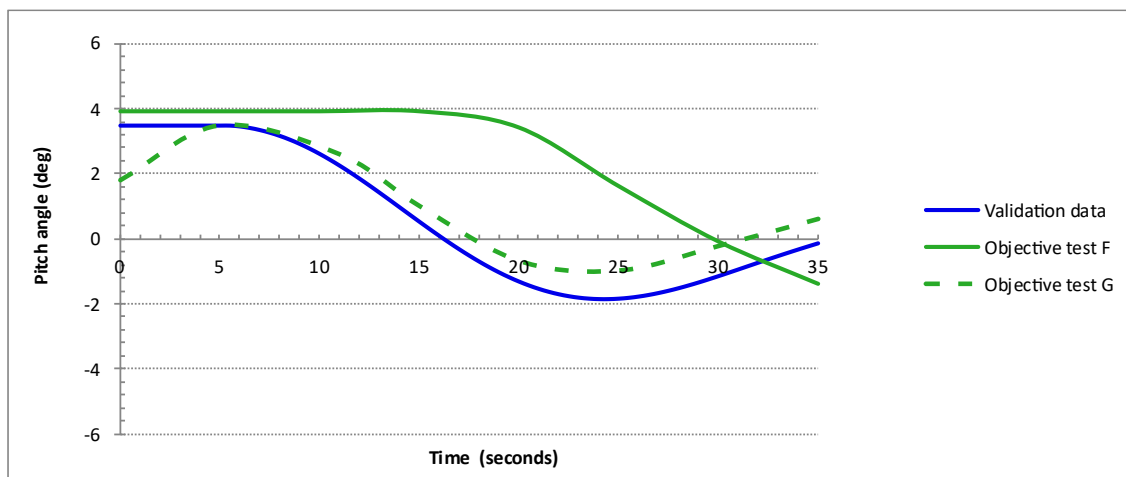


Figure 3. Further examples of objective test results that do not demonstrate CT&M

Figure 4 below shows the final examples of results that do not demonstrate CT&M. The test and validation data is the same as in Figure 1. Example H shows an FSTD result that has an unexplained oscillation. The pitch angle of the validation data is changing smoothly, so the FSTD's oscillating response to landing gear extension does not show CT&M and is not acceptable. This result could be caused by a closed-loop controller that is moving the pitch controller in an attempt to follow the pitch angle. Such a controller should not be used if the test purpose is to measure the free response. Example I shows that the FSTD movement happens in the correct direction, but the change happens much quicker, so the correctness of the trend is questionable. In addition, the magnitude of the change is significantly higher than that in the validation data. Flying such an FSTD would 'feel' different from flying the real aircraft (i.e. it would clearly require more controlling than the real aircraft when the landing gear were to be extended). Example I does not demonstrate CT&M.

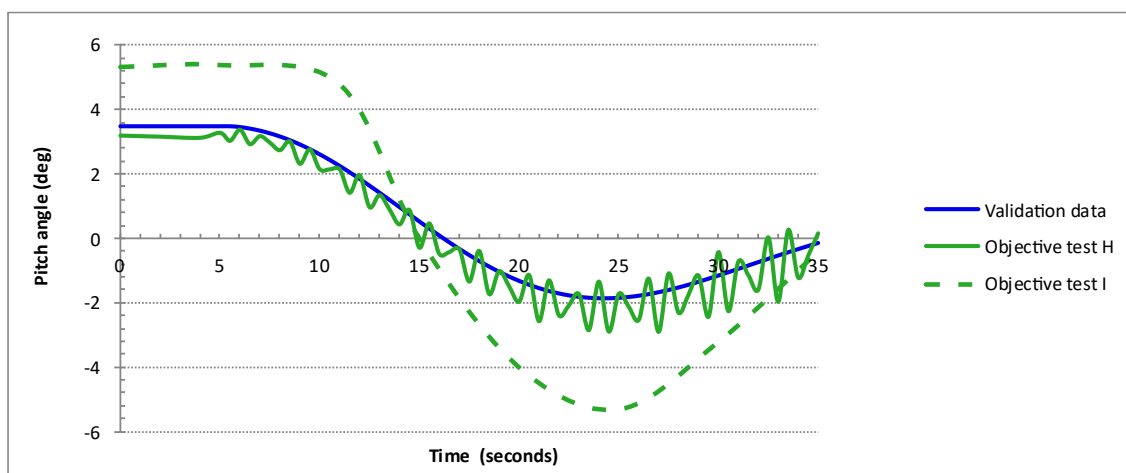


Figure 4. Final examples of objective test results that do not demonstrate CT&M

- (e) Examples of test results that do not demonstrate CT&M are:
- (1) dynamics that are overdamped relative to the simulated aircraft (e.g. phugoid dynamics);
 - (2) a clearly higher level of stability than that in the simulated aircraft (e.g. spiral stability);
 - (3) the opposite variation of parameters (e.g. pitching up instead of pitching down) following inputs or configuration changes (e.g. speed brake or flap extension);
 - (4) insufficient or exaggerated trim change in relation to airspeed;
 - (5) control force or travel that clearly differs from that of the simulated aircraft;
 - (6) failure to account for aircraft-specific issues, such as weight-on-wheels dependencies;
 - (7) a lack of roll–yaw coupling in swept-wing Dutch roll;
 - (8) a lack of cross-coupling in helicopter manoeuvring;
 - (9) misapplied group of aircraft data (e.g. sidestick forces applied to a yoke);
 - (10) data including only a few parameters (e.g. IAS, pitch and altitude) and omitting other key parameters (e.g. roll, yaw, configuration, trim and engine settings) (refer to GM1 FSTD.QTG.015, paragraph (a)).

GM1 FSTD.ENG.020 Engineering report

INDUSTRY STANDARDS

When showing compliance, the organisation requesting the qualification of an FSTD should account for industry standards and best practices, such as but not limited to:

- (a) ARINC Specification 450-1, *Flight Simulator Design and Performance Data Requirements*, as amended and as appropriate to the FCS;
- (b) ARINC Report 610C, *Guidance for design of aircraft equipment and software for use in training devices*, as amended (refer also to GM1 FSTD.FST.001).

CS FSTD.ENG.025 Validation data roadmap

- (a) This CS provides standards for the VDR, which defines the source(s) of aircraft data that could be used for validating each objective test result. A VDR document contains the guidance material from the aircraft validation data supplier recommending the best possible sources of data to be used as validation data in the QTG.
- (b) The VDR is required for all FSTDs, regardless of fidelity level, should be produced by the validation data provider and should be part of the MQTG.
- (c) The VDR should clearly identify (in matrix format) the sources of data for all required tests. It should also provide guidance regarding the validity of this data for a particular engine type and thrust/torque rating configuration and the revision levels of all avionics affecting aircraft handling qualities and performance. The document should include rationales or explanations if, for example, data or parameters are missing, engineering simulator/simulation data is to be used, flight test methods require explanation or the cause and effect of any deviation from data requirements need to be clarified. Additionally, the document should make reference to other appropriate sources of validation data (e.g. sound and vibration data documents).

GM1 FSTD.ENG.025 Validation data roadmap

- (a) A VDR should provide clear and explicit information on the validation data sources for each objective test. If the FSTD is based on multiple VDRs (e.g. separate VDRs for different features), the source of validation data for each objective test must be clearly defined. It may be necessary to merge the information from multiple VDRs into one document. The method should be agreed with the competent authority. If multiple VDRs are merged into one document, the original VDRs from the data providers should also be attached to the MQTG to ensure traceability.
- (b) A VDR should be submitted to the competent authority as early as possible in the planning stages for any FSTD planned for which qualification to the standards contained in this CS-FSTD is planned.
- (c) Table 1 depicts an example VDR matrix identifying sources of validation data for a typical validation data package to support fidelity level S. Only the first page of the full matrix is shown, and some test conditions were deleted for brevity.

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- (d) Relevant regulatory material should be consulted, and all applicable tests addressed in the actual VDR document should be submitted. The validation sources, validation data documents and comments provided in the example in Table 1 are for reference only. The actual data sources and documents will depend on the particular airframe/engine combination under consideration. The following set of guidelines should be used when this example is applied to a particular VDR document.
- (1) The CCA mode column should be included if applicable.
 - (2) Columns for each validation source (e.g. each flight test airframe/engine combination and the simulation configuration) should be included.
 - (3) A column should be included for each document being referenced as a source of validation data. The term 'integrated' in the document title indicates that the test conditions contained in these documents conform to the definition of 'integrated testing' in the terminology section (CS FSTD.GEN.015).
 - (4) The columns for validation data sources should be arranged in order, starting from the highest-quality data source.
- (e) Tables 2 and 3 provide examples of another presentation of VDR matrices identifying sources of validation data for an abbreviated list of tests, along with detailed information for a typical test case. In this example, certain data elements have been substituted with series of Xs. A complete matrix should address all test conditions. A complete set of detailed information pages for tests quoted in the matrix would be provided with this presentation.
- (f) Tables 4 and 5 provide excerpts of fictional example VDR matrices to support fidelity levels R and G, respectively.

Table 1. VDR – example 1

ICAO/ IATA#	Test description	CCA mode N — Normal law, D — Direct law	Validation source				Validation document						Validation source category	Rationales R — Rationale page attached	Comments (This VDR is for aeroplane 2 with DEF-74 engines)
			Aeroplane 1 flight test data	Aeroplane 2 flight test data	Engineering simulation data (Aeroplane 2 with DEF-74 engines)	Aerodynamics POM	Flight controls POM	Ground handling POM	Propulsion POM	Integrated POM	Aeroplane 2 Integrated validation data	Aeroplane 2 Validation data roadmap (integrated)			
1.a.1	Minimum radius turn.	N		2				NE					FT		
1.a.2	Rate of turn versus nosewheel steering angle (2 speeds).	N			3						D73		ES		
1.b.1	Ground acceleration time and distance.	N			3							D74	ES		Data is included in normal take-off (1.b.4).
1.b.2	Minimum control speed, ground (V_{mcg}).	N		1		D74				D74			FT		
1.b.3	Minimum unstick speed (V_{mu}).	N			3							D74	ES		
1.b.4	Normal take-off.	N	2		3	C78						D74	ES	R	
1.b.5	Critical engine failure on take-off.	N		1		D74							FT		
1.b.6	Crosswind take-off.	N		1		D74							FT		
1.b.7	Rejected take-off.	N		1	3	D74						D74	FT/ES	R	Test procedure anomaly; see rationale.
1.b.8	Dynamic engine failure after take-off.	N		1		D74				D74			FT		
1.c.1	Normal climb all engines operating.	N,D	2		3	D73						D74	ES	R	FT data flown in direct mode; see rationale.
1.c.2	One-engine-inoperative second segment climb.	N		1		D74							FT		AFM data available for reference.
1.c.3	One-engine-inoperative en-route climb.	N			3							D74	ES		
1.c.4	One-engine-inoperative approach climb.	N			3							D74	ES		Run with and without icing accountability.
1.d.1	Level flight acceleration.	N	2		3	C78						D74	FT/ES	R	FSTD manufacturer to evaluate use of FT in QTG.
1.d.2	Level flight deceleration.	N	2		3	C78						D74	FT/ES	R	FSTD manufacturer to evaluate use of FT in QTG.
1.d.3	Cruise performance.	N			3							D74	ES		
1.d.4	Idle descent.	N			3						D73		ES		

1.d.5	Emergency descent.	N			3						D73		ES		
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Table 2. VDR – example 2

QTG	Test description A = Engine 1: xx kN. B = Engine 2: xx kN. D: <i>Direct law.</i> N: <i>Normal law.</i> Alt: <i>Alternate law or system alternate conditions (e.g. hydraulics off).</i>	CCA mode	Validation source				QTG — Natural/computer controlled aeroplane Doc. xxxx Common tests	QTG — Natural/computer controlled aeroplane Doc. xxxx A3xx-xxx — Engine specific tests	Performance test cases for FSTD Doc. xxxx	Autopilot tests Doc. xxxx	Engine tests for FSTD qualification Doc. xxxx	Static control and dynamic control checks Doc. xxxx	Brake static and dynamic control checks Doc. xxxx	Common sound QTG: xxxx	Engine specific sound QTG: xxxx	Common vibration QTG: xxxx	Engine specific vibration QTG: xxxx	Long range flight controls latency tests Doc. xxxx
			Aeroplane flight test	Proof of match (POM)	Engineering simulator test	AFM data												
1.a.1	Minimum radius turn.	N	B				X											
1.a.2.1	Rate of turn versus nosewheel steering angle — Speed 1.	D	A	A			X											
1.a.2.2	Rate of turn versus nosewheel steering angle — Speed 2.	D	A	A			X											
1.b.1	Ground acceleration time and distance.	N	B	B				X										
1.b.2	Minimum control speed, ground (V_{mcg}).	D			B			X										
1.b.3	Minimum unstick speed (V_{mu}).	D	A	B			X											

Table 3. VDR – example 3

1. PERFORMANCE	1.a TAXI	1.a.2 Rate of turn versus nosewheel steering angle (NWA)	Conditions: Ground
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A – Requirements

Document:	ICAO Doc 9625 – <i>Manual of Criteria for the Qualification of Flight Simulation Training Devices</i> , Volume I – <i>Aeroplanes</i> , Fourth Edition.						
Tolerance:	±2°/s or ±10 % of turn rate.						
Flight Condition:	Ground.						
Comments:	Plot a minimum of two speeds, greater than minimum turning radius speed, with a spread of at least 5 kt ground speed.						
Type:	I	II	III	IV	V	VI	VII
					✓		✓

B – Data Package

Configuration:	#	Avionics 1	FCSC	FADEC	BSCU	Flight test validation data	Engineering simulation validation data	Proof of match
	1	Std xx	Std xx	Std xx	Std xx	XXXXXX Engine		
	2	Std xx	Std xx	Std xx	Std xx			XXXXXX Engine
	3							
	4							
	5							
	6							

Rationales:	#	
	1	Rationale 1.
	2	Rationale 2.
	3	
	4	
	5	
	6	

Table 4. Fictional VDR excerpt for an FSTD with the FSTD features flight control forces and hardware and flight control systems operation simulating a Boeing B747 at the representative fidelity levels

Test No	Test name	Validation data source				Additional information on the validation data source Detailed justification is presented in the ER
		Flight test data	AFM	Public domain data	Subjective development	
2.a.1	Pitch controller position versus force and surface position calibration	✓		✓		Data is from NASA Document ID 19730001300 which shows a measurement from a real B747 aircraft.
2.a.2	Roll controller position versus force and surface position calibration	✓		✓		Data is from NASA Document ID 19730001300 which shows a measurement from a real B747 aircraft.
2.a.3	Rudder pedal position versus force and surface position calibration	✓		✓		Data is from NASA Document ID 19730001300 which shows a measurement from a real B747 aircraft.
2.a.4	Nose wheel steering controller force and position calibration	✓				FSTD manufacturer's measurement in real B747-400 aircraft (registration OH-YES).
2.a.5	Rudder pedal steering calibration		✓		✓	Data for maximum deflection is from the Boeing B747-400 FCOM (Document Number D6-30151-400, revision 21), Section 14.10.3; intermediate positions are validated subjectively
2.a.8	Alignment of flight deck throttle lever versus selected engine parameter		✓	✓	✓	Engine N1 for go-around and maximum climb are from the Boeing B747-400 FCOM (Document Number D6-30151-400, revision 21), Section PI.10.11 Engine idle parameters (N1, N2, FF) are from digital flight data recorder data from Japan Transport Safety Board Report AI2011-5, Figure 2 Other throttle positions are interpolated between these and are validated subjectively

Table 5. Fictional VDR excerpts for an FSTD with the FSTD feature performance and handling out-of-ground effect simulating a single-engine piston aeroplane at the generic fidelity level

Test No	Test name	Validation data source					Additional information on the validation data source Detailed justification is presented in the ER
		Flight test data	AFM	Public domain data	CS-23	Subjective development	
2.c.8.b	Approach-to-stall characteristics for 2nd segment climb (normal climb and power-on stall event), high-altitude cruise (straight and turning flight) and landing	✓	✓				FSTD manufacturer's flight test in real aircraft (Cessna 172, registration PH-ISH) for time-history data for all test cases. Stall speed data also validated by Cessna 172 AFM data.
2.c.9	Phugoid dynamics			✓			Time-history data from NASA Document ID 19660030615, Figure 19b
2.c.10	Short-period dynamics				✓	✓	Footprint method is used Subjective development was performed to fulfil the requirements in CS 23.2145 for stability
2.d.1	Minimum control speed (air) (V_{MCA})						Not applicable to a single-engine aircraft See rationale #2.
2.d.2	Roll response (rate)				✓	✓	Footprint method is used. Subjective development was performed to fulfil the requirements in CS 23.157 for roll rate

CS FSTD.ENG.030 Alternate engine or avionics validation data

This CS defines the standards for additional/alternate engine or avionics validation data.

(a) Background information on these standards is provided below.

- (1) For a new aircraft type, the majority of the flight validation data is collected on the first aircraft configuration with a 'baseline' engine fit and a 'baseline' avionics configuration, which forms the basis of the models and the data package. This dataset is then used to validate all FSTDs simulating that aircraft type.
- (2) In the case of FSTDs simulating an aircraft with a different engine fit from the baseline or with a revised avionics configuration or more than one avionics configuration, additional flight test validation data may be needed.
- (3) If a FSTD with multiple engine fits is to be qualified, the MQTG should contain test validation data for selected cases in which engine differences are expected to be significant.
- (4) If an FSTD with alternate avionics configurations is to be qualified, the MQTG should contain test validation data for selected cases in which the avionics configuration differences are expected to be significant as defined in paragraph (b)(2) of this CS.
- (5) The nature of the complementary validation data required (e.g. flight test data, engineering simulator/simulation data) should be in accordance with the guidelines

prescribed in paragraph (b) below, except if other data is specifically allowed (refer to CS FSTD.ENG.040).

- (b) Validation data requirements for alternate engine fits and alternate avionics configurations are as follows.
 - (1) For tests that are affected by a difference in engine type or thrust rating, the use of flight test data is preferred to validate that particular aircraft engine configuration or the alternate thrust rating.
 - (2) Tests that are significantly affected by a change to the avionics configuration should be supported by flight test data.
 - (3) A matrix or VDR should be provided with the MQTG indicating the appropriate validation data source for each test (refer to CS FSTD.ENG.025). The organisation requesting the qualification should coordinate the FSTD data requirements pertaining to alternate engine or avionics configurations in advance with the competent authority.

GM1 FSTD.ENG.030 Alternate engines or avionics validation data

OBJECTIVE TEST STANDARDS FOR THE QUALIFICATION OF ALTERNATE ENGINES

- (a) The following guidance applies to FSTDs equipped with multiple engine types or thrust ratings. Examples are provided for aeroplane FSTDs, but similar principles can be applied to helicopter FSTDs.
- (b) To validate additional engine types or thrust ratings in an FSTD, a subset of the QTG should be provided. For each test number in Table 1 below, all test conditions as required by the relevant section of Subpart D (refer to the CS FSTD.QTG.A sections) should be provided.
- (c) If the additional engine fit is a different type, information on all the tests under the additional engine type column in Table 1 of this GM should be provided in the QTG.
- (d) If the additional engine type is the same but the thrust rating exceeds that of the baseline engine fit by 5 % or more or is significantly less than the baseline engine rating (a decrease of 15 % or more), all the tests in the additional engine rating column should be provided in the QTG. Otherwise, it might be acceptable to only provide the throttle calibration data (i.e. commanded power setting parameter versus throttle lever angle) and the engine acceleration and deceleration cases. However, if a validation data provider, as defined under the guidelines of CS FSTD.ENG.040, shows that a thrust increase of greater than 5 % will not significantly change the aircraft's flight characteristics, then flight validation data is not needed.
- (e) Regarding sound system and alternate engine fits, for FSTDs with multiple propulsion configurations, any condition listed in the table of objective tests that is identified by the data provider as significantly different due to a change in the engine model should be presented for evaluation as part of the QTG.
- (f) If the certification of the flight characteristics of the aircraft with a new thrust rating (regardless of the percentage change in thrust rating) requires certification flight testing with a comprehensive stability and control flight instrumentation package, the VDR should provide

a list of objective tests supported by flight test data and presented in the QTG (along with the additional tests listed in Table 1 of this GM, for which other sources of validation data are acceptable).

Table 1. Minimum recommended list of objective tests for an additional engine configuration

Test		Additional engine type	Additional engine rating (exceeds change from baseline of +5 % to -15 %)	Additional engine rating (change from baseline within +5 % to -15 %)
1.b.1	Ground acceleration time and distance	X		
1.b.2	Minimum control speed (ground) (V_{MCG})	X	X	
1.b.4	Normal take-off	X		
1.b.5	Critical engine failure on take-off	X		
1.b.7	Rejected take-off	X		
1.b.8	Dynamic engine failure after take-off	X		
1.c.1	Normal climb, all engines operating	X	X	
1.c.2	One-engine-inoperative second-segment climb	X	X	
1.d.1	Level flight acceleration	X		
1.d.2	Level flight deceleration	X		
1.d.3	Cruise performance	X		
1.f.1	Engine acceleration	X	X	X
1.f.2	Engine deceleration	X	X	X
2.a.8	Alignment of flight deck throttle lever versus selected engine parameter	X	X	X
2.c.1	Power change dynamics	X	X	
2.d.1	Minimum control speed (air) (V_{MCA})	X	X	
2.d.5	Engine-inoperative trim	X		
2.e.4	One-engine-inoperative landing	X	X	
2.e.6	All-engine autopilot go-around	X	X	
2.e.7	One-engine-inoperative go-around	X	X	
2.e.8	Directional control (rudder effectiveness) with symmetric reverse thrust	X		
2.e.9	Directional control (rudder effectiveness) with asymmetric reverse thrust	X		
3.g.1	Thrust effects with brakes set	X		
5.a.3	All engines at maximum allowable thrust with brakes set	X		

GM2 FSTD.ENG.030 Alternate engine or avionics validation data**OBJECTIVE TEST STANDARDS FOR THE QUALIFICATION OF ALTERNATE AVIONICS**

- (a) The following guidance applies to FSTDs simulating aircraft with a revised avionics configuration or more than one avionics configuration.
- (b) The aircraft avionics can be segmented into those systems or components that can significantly affect the objective test results and those that cannot. The following avionics systems or components are examples of those for which hardware design changes or software revision updates may lead to significant differences relative to the baseline avionics configuration: flight control computers, controllers for engines, autopilot, braking systems, nose wheel steering systems, high-lift systems and landing gear systems. Related avionics such as stall warning and stability augmentation systems should also be considered. The data provider should identify the objective tests and effects associated with each avionics system change.
- (c) For changes to an avionics system or component that could affect an objective test but if that test is not affected by this particular change (e.g. the avionics change is an update to built-in test equipment or a modification affecting a different flight phase), the objective test can be based on validation data from the previously validated avionics configuration. The organisation requesting qualification of an FSTD should provide a statement from the aircraft manufacturer or data provider clearly stating that this avionics change does not affect the test.
- (d) For an avionics change that affects some tests in the QTG but if no new functionality is added and the impact of the avionics change on the aircraft response is a small and well-understood effect, the QTG may be based on validation data from the previously validated avionics configuration. This should be supplemented with avionics-specific validation data from the aircraft manufacturer's or data provider's engineering simulation generated with the revised avionics configuration. In such cases, the organisation requesting qualification should provide a rationale from the aircraft manufacturer or data provider explaining the nature of the change and its effect on the aircraft response.
- (e) For an avionics change that significantly affects some tests in the QTG, especially if a new functionality is added, the QTG should be based on validation data from the previously validated avionics configuration and supplemental avionics-specific test data necessary to validate the avionics revision. However, additional flight validation data may not be needed if the avionics changes were certified without the need for testing with a comprehensive flight instrumentation package. In this situation, the organisation operating the FSTDs should coordinate the FSTD data requirements in advance with the aircraft manufacturer or data provider and then with the competent authority.
- (f) For changes to an avionics system or component that are non-contributory to the objective test response, the test can be based on validation data from the previously validated avionics configuration. For such changes, it is not necessary to include a rationale explaining that this avionics change does not affect the test.

CS FSTD.ENG.035 Standards for existing flight simulation training device data packages

This CS provides standards applicable if a feature is to be qualified at the specific fidelity level, and the data package used has been developed and employed in a previously qualified FSTD prior to the applicability date of this CS-FSTD.

- (a) Except if specifically stated otherwise in the CS FSTD.QTG.A or CS FSTD.QTG.H sections, validation data for QTG objective tests is expected to be derived from aircraft flight testing.
- (b) Ideally, data packages for all new FSTDs should fully comply with this CS-FSTD for qualifying FSTDs. For aircraft certified prior to the applicability date of this CS-FSTD, due to unavailability of the test aircraft or the necessary instrumentation, it may not always be possible to provide the required validation data for all objective test cases. This scenario may in particular occur for objective test cases whose requirements in this CS-FSTD differ from the previous primary reference documents, or for test cases that are newly introduced. The provision of acceptable data to support the FSTD qualification process is a matter of planning and regulatory agreement.
- (c) Notwithstanding the above paragraphs, except if other types of data are already acceptable, the preferred source of validation data is flight testing. It is expected that data providers will endeavour to provide the required flight test data to the best of their abilities. If any flight test data exists (flown during the certification or any other flight test campaigns) that addresses the requirement, this test data should be provided. If there is any possibility of conducting this flight test during the occasion of a new flight test campaign, this should be done and information provided in the data package at the next issue. If this flight test data is genuinely not available, alternative sources of data may be acceptable using the following hierarchy of preferences.
 - (1) First, sources should be used as defined in flight testing at an alternate but near equivalent condition/configuration.
 - (2) Data from an engineering simulator/simulation as defined in CS FSTD.GEN.015 from an acceptable source (e.g. the data meets the guidelines laid out in CS FSTD.ENG.040) or as used for aircraft certification may be used.
 - (3) Aircraft performance data as defined in CS FSTD.GEN.015 or data from other approved published sources (e.g. production flight test schedule) may be used for the following tests.
 - (i) For aeroplanes, the tests are:
 - (A) 1.c.1, 'Normal climb, all engines operating';
 - (B) 1.c.2, 'One-engine-inoperative second-segment climb';
 - (C) 1.c.3, 'One-engine-inoperative en route climb';
 - (D) 1.c.4, 'One-engine-inoperative approach climb' (for aeroplanes with icing accountability);

- (E) 1.e.3, 'Stopping distance, wheel brakes, wet runway'; and
 - (F) 1.e.4, 'Stopping distance, wheel brakes, icy runway'.
- (ii) For rotorcraft, the tests are:
 - (A) 1.d, 'Hover performance' (in-ground effect, out-of-ground effect); and
 - (B) 1.g, 'Climb performance and trimmed flight control position' (all engines operating, one engine inoperative).
- (4) If no other data is available, the following sources may be acceptable, subject to a case-by-case review with the competent authorities concerned, taking into consideration the FCS sought for the FSTD. For fidelity level S, these sources should be used in exceptional circumstances only.
 - (i) Unpublished but acceptable sources may be used (e.g. calculations, simulations, videos or other simple means of flight test analysis or recording).
 - (ii) Footprint data from the actual FSTD requiring qualification, validated by subjective assessment by a pilot acceptable to the competent authority, may be used. Only if all other possible alternative sources of data have been thoroughly sought without success may footprint data be acceptable, subject to a case-by-case review with the competent authority concerned, taking into consideration the level of qualification or FCS sought for the FSTD.

Wherever practical, FSTD footprint data should be compared against objectively validated test cases. For example, if an existing data package contains flight test data for normal take-off but does not contain crosswind take-off data, the organisation requesting the qualification could run a crosswind take-off test in the FSTD and apply CT&M methodology to compare the data against the validation flight test data for normal take-off, providing a justification for the differences in control inputs induced by the crosswind.
- (d) In certain cases, it may make good engineering sense to provide more than one test to support a particular objective test requirement. An example is a minimum control speed (ground) (V_{MCG}) test for multi-engine aeroplanes, in which the flight test engine and thrust profile do not match the simulated engine. The V_{MCG} test could be run twice: once with the flight test thrust profile as an input and a second time with a fully integrated response to a fuel cut on the simulated engine.
- (e) For aircraft certified prior to the date of issue of this CS-FSTD, an operator may, after reasonable attempts to obtain suitable flight test data have failed, indicate in the QTG or VDR if flight test data is unavailable or unsuitable for a particular test. For each case, if the preferred data is unavailable, a rationale should be provided laying out the reasons for the non-compliance and justifying the alternate data and or test(s).
- (f) New flight test data should be gathered if the available data is insufficient.

CS FSTD.ENG.040 Standards for engineering simulator/simulation validation data

This CS provides the standards applicable to engineering simulator/simulation validation data.

- (a) If a fully flight-test-validated simulation is modified as a result of changes to the simulated aircraft configuration, a data provider may choose, with the prior agreement of the competent authority, to supply validation data from an engineering simulator/simulation to selectively supplement flight test data.

This arrangement is confined to changes that are incremental in nature and that are both easily understood and well defined.

- (b) To supply engineering simulator/simulation validation data, a data provider should:
- (1) have a proven track record of developing successful data packages used in the qualification of FSTDs;
 - (2) have demonstrated high-quality prediction methods through comparisons of predicted data and flight-test-validated data;
 - (3) have demonstrated that the engineering simulator/simulation represents the aircraft characteristics throughout the certified flight envelope;
 - (4) have an engineering simulator/simulation that:
 - (i) has simulation models that run in an integrated manner;
 - (ii) uses the same simulation models as those released to the training community (which are also used to produce stand-alone POM and checkout documents);
 - (iii) has a rigorous configuration control system; and
 - (iv) has been found to be a high-fidelity simulation of the aircraft by the pilots of the aircraft manufacturer or data provider, operators and the competent authority;
 - (5) use the engineering simulator/simulation to produce an adequate set of integrated POM cases;
 - (6) have an effective configuration control system in place covering the engineering simulator/simulation.
- (c) Data providers seeking to take advantage of this arrangement should contact the competent authority at the earliest opportunity. The use of engineering simulator/simulation validation data is subject to a technical verification through review by and the agreement of the competent authority.
- (d) Validation data from a technically verified engineering simulator/simulation may be used for changes that are incremental in nature and that are both easily understood and well defined.

- (e) To be admissible as a source of validation data, an engineering simulator should:
- (1) have to exist as a physical entity, complete with a flight deck characteristic of the simulated aircraft, with controls sufficient for manual flight;
 - (2) have a visual system and preferably also a motion system;
 - (3) if appropriate, have actual avionics boxes interchangeable with the equivalent software simulations, to support the validation of released software;
 - (4) have a rigorous configuration control system covering hardware and software;
 - (5) have been found to be a high-fidelity simulator of the aircraft by the pilots of the aircraft manufacturers or data provider, operators and the competent authority.

GM1 FSTD.ENG.040(b) Standards for engineering simulator/simulation validation data

ENGINEERING SIMULATOR/SIMULATION VALIDATION DATA TO SUPPORT VALIDATION

- (a) Background information on the models is provided below.
- (1) In the case of fully flight-test-validated simulation models of a new or major derivative aircraft, it is likely that the simulation models will become progressively less closely aligned with the simulated aircraft as the aircraft configuration is revised.
 - (2) Traditionally, as the aircraft configuration has been revised, the simulation models have been revised to reflect changes. In the case of aerodynamic, engine, flight control and ground-handling models, this revision process normally results in the collection of additional flight test data and the subsequent release of new models and validation data.
 - (3) The quality of the prediction of simulation models has advanced to the point that differences between the predicted and flight test validation models are often quite small.
 - (4) Some data providers utilise the same simulation models in their engineering simulations as those released to the training community. These simulations range from physical engineering simulators with and without aircraft hardware to non-real-time workstation-based simulations.
- (b) Guidance on using engineering simulator/simulation validation data is provided below.
- (1) For the specific fidelity level, flight test data should be the primary data source, and engineering simulator/simulation validation data should be used only if necessary and justified by a rationale in the VDR.
 - (2) For the representative and generic fidelity levels, engineering simulator/simulation validation data can be used if substantiated in the ER (refer to CS FSTD.ENG.015).
 - (3) If a fully flight-test-validated simulation is modified as a result of changes to the simulated aircraft configuration, a data provider may choose, with the prior

agreement of the competent authority, to supply validation data from an engineering simulator/simulation to selectively supplement flight test data.

- (4) If data from an engineering simulator/simulation is used, the engineering simulation process should be verified by the competent authority. Refer to CS-SIMD Simulator Data.
- (5) In all cases, a data package verified to current standards against flight testing should be developed for the aircraft entry-into-service configuration of the baseline aircraft.
- (6) In cases in which the use of engineering simulator/simulation validation data is envisaged, a complete proposal should be presented to the competent authority. The proposal should contain evidence of the data provider's past achievements in high-fidelity modelling.
- (7) During the technical verification, data providers may demonstrate that the engineering simulator/simulation represents the aircraft characteristics throughout the certified flight envelope by providing evidence of use in aircraft design, development and certification.
- (8) A configuration management process should be maintained, including an audit trail that clearly defines the simulation model changes step by step away from a fully flight-test-validated simulation, so that it is possible to reverse the changes and restore the baseline (flight-test-validated) version.
- (9) The competent authority should conduct technical reviews of the proposed plan and the subsequent validation data to establish the acceptability of the proposal.
- (10) The implementation of the engineering simulator/simulation used to produce only system data may not need all the above elements.
- (11) The precise procedure followed to use engineering simulator/simulation validation data will vary from case to case between data providers and based on the type of change. The process should be applicable to one step away from a fully flight-test-validated simulation. Refer to CS FSTD.GEN.015 for the definition of 'one step'.
- (12) The type of modifications covered by this alternative procedure will be restricted to those with well-understood effects, such as:
 - (i) software (e.g. flight control computer, autopilot);
 - (ii) simple (in aerodynamic terms) geometric revisions (e.g. body length);
 - (iii) engines, limited to non-propeller-driven aircraft;
 - (iv) control system gearing/rigging/deflection limits; and
 - (v) brake, tyre and steering revisions.
- (13) The organisation requesting qualification of an FSTD, with the assistance of the data provider that wishes to take advantage of this alternative procedure, is expected to demonstrate a sound engineering basis for the proposed approach. Such a basis should include an analysis showing that the predicted effects of the change(s) were

incremental in nature and were both easily understood and well defined, confirming that additional flight test data was not required. In the event that the predicted effects are not deemed to be sufficiently accurate, it might be necessary to collect a limited set of flight test data to validate the predicted increments.

- (14) Any applications for this procedure should be reviewed by the competent authority.

CS FSTD.ENG.050 Computer-controlled aircraft

This CS defines the requirements concerning CCA FSTDs for which the associated aircraft features are at the specific fidelity levels.

- (a) For the objective testing of CCA FSTDs, flight test data is required for both the normal and non-normal control states, as applicable to the simulated aircraft and as indicated in the objective test requirements. Tests in the non-normal state should always include the least-augmented state attainable by the aircraft. Tests for other levels of control state degradation may be required, as detailed by the competent authority at the time of defining the set of aircraft tests for FSTD data.
- (b) All the objective tests defined in the tables of objective tests set out in all CS FSTD.QTG.A or CS FSTD.QTG.H sections require test results in the normal control state unless specifically noted otherwise following the CCA designation. However, if the test results are independent of the control state, non-normal control data may be used.
- (c) If non-normal control states are required, test data should be provided for one or more non-normal control states, including the least-augmented state attainable by the aircraft.
- (d) If normal, non-normal or other degraded control states are not applicable to the aircraft being simulated, appropriate rationales should be included in the VDR.

CS FSTD.ENG.060 Data for interim flight simulation training device qualification

In the case of the introduction of new aircraft programmes, when compliance with the requirements established in this subpart for FSTD qualification is not possible, the competent authority may consider interim qualification in accordance with Part-ARA (Annex VI) of Commission Regulation (EU) No 1178/2011, that is:

- (a) Design and validation data for an interim FSTD qualification applies only to the following FSTD features:
 - (1) Flight deck layout and structure;
 - (2) Flight control forces and hardware;
 - (3) Flight control systems operation;
 - (4) Aircraft systems;
 - (5) Performance and handling on ground;
 - (6) Performance and handling in-ground effect;

- (7) Performance and handling out-of-ground effect;
 - (8) Sound cueing; and
 - (9) Vibration cueing.
- (b) The interim qualification is only applicable to the representative and specific fidelity levels.
- (c) The design and validation data for an interim FSTD qualification should be part of the new process for gathering aircraft data.
- (d) The FSTD features sound cueing and vibration cueing are the only features for which the validation data used may be solely based on subjective development and footprint data for an interim R or interim S level.
- (e) The justification for the interim qualification, including the description of the associated validation data, should be set out in an FSTD interim qualification plan for the competent authority.

GM1 FSTD.ENG.060 Data for interim flight simulation training device qualification

- (a) Refer to Part-ARA (Annex VI) of Commission Regulation (EU) No 1178/2011 for standards of interim flight simulation training device qualification.
- (b) The approval by the Agency of the definition of the scope of the aircraft validation source data to support the objective qualification as part of the OSD can also be an approval of a VDR for an interim qualification in the case of preliminary data. The preliminary data to be used should be based on this approval.
- (c) There should be assurance that the preliminary data is the aircraft manufacturer's or data provider's best representation of the aircraft and reasonable certainty that final data will not deviate to a large degree from these preliminary, but refined, estimates. First of all, there should be an interim acceptance of the data in the type certification process under Part 21 of Commission Regulation (EU) No 748/2012. Furthermore, the data derived from these predictive or preliminary techniques should be validated by available sources, including, at least, the following sources.
 - (1) A report should explain the predictive method used and detail past successes of the method in similar projects. For example, the aircraft manufacturer or data provider could show the application of the method to an earlier aircraft model or predict the characteristics of an earlier model and compare the results with the final data for that model.
 - (2) Early flight test results will often be derived from aircraft certification tests and should be used to maximum advantage for early FSTD validation. Certain critical tests, which would normally be done early in the aircraft certification programme, should be included to validate essential pilot training and certification manoeuvres. These include cases in which a pilot is expected to cope with an aircraft failure mode, including engine failures. The early flight test data available will depend on the aircraft

manufacturer's or data provider's flight test programme design and may not be the same in each case. However, it is expected that the flight test programme of the aircraft manufacturer or data provider will include provisions for the generation of very early flight test results for FSTD validation.

- (d) The following is an example of the design data and sources that might be used in the development of an interim qualification plan.
 - (1) The plan should consist of the development of a QTG based on a mix of flight test and engineering simulator/simulation data. For data collected from specific aircraft flight tests or other flights, the required designed model and data changes necessary to support an acceptable POM should be generated by the aircraft manufacturer or data provider.
 - (2) The details of the interim qualification data are recorded in the ER in accordance with CS FSTD.ENG.020.
 - (3) In order for the two sets of data to be properly validated, the aircraft manufacturer or data provider should compare their simulation model responses against the flight test data, if driven by the same control inputs and subjected to the same atmospheric conditions as those recorded in the flight test. The model responses should result from a simulation in which the following systems are run in an integrated fashion and are consistent with the design data released to the FSTD manufacturer:
 - (i) propulsion;
 - (ii) aerodynamics;
 - (iii) mass properties;
 - (iv) flight controls;
 - (v) stability augmentation; and
 - (vi) brakes and landing gear.
- (e) The FSTD interim qualification plan submitted to the competent authority should set out the timeline and methods by which the interim validation data will be completed and superseded by the final data.

SUBPART D – OBJECTIVE TESTS

GENERAL

CS FSTD.QTG.001 General

- (a) The FSTD features should be validated using objective tests for the requested FCS.
 - (1) Each objective test should compare the performance and system operation of the FSTD against the validation data as described in Subpart C for the particular test, unless specifically noted otherwise (e.g. certain motion, visual and sound objective tests);
 - (2) Each objective test should match the validation data within the limits of the prescribed tolerances of the test.
- (b) Each objective test should establish the FSTD performance and system operation to be used by the FSTD operator as the ongoing recurrent objective testing validation basis for maintaining the FSTD qualification.

CS FSTD.QTG.005 Definition of applicable objective tests for flight simulation training devices

- (a) The tables of objective tests defined in all CS FSTD.QTG.A or CS FSTD.QTG.H sections of this subpart should be used to determine the applicable tests to be included in the QTG and the tolerances applicable to each test. This determination is based on the fidelity level of each feature in the requested FCS.
- (b) The tables of objective tests define, for each test, the FSTD feature(s) that are primarily validated by each tolerance parameter. If the FSTD feature(s) primarily validated by any of the tolerance parameters for a test are simulated on the FSTD at a fidelity level for which a tolerance is specified in the 'Tolerances' columns of the table, the test is applicable to the FSTD and should be included in the QTG.
 - (1) The tolerances applicable to the test are those defined for the requested FCS fidelity level of the FSTD's primary feature(s) validated for each tolerance parameter.
 - (2) If a tolerance parameter is validated by more than one primary feature, the tolerances applicable should be as follows:
 - (i) for the performance and handling features (GND, IGE, OGE), those defined for the requested FCS fidelity level should be used for the portion of the test that spans each feature;
 - (ii) in all other cases, refer to the relevant tables of additional requirements.

- (3) If a tolerance parameter is defined as applicable to the test but cannot be demonstrated due to the simulated aircraft, the reasons for this should be described in an SOJ.
- (c) The primary feature(s) validated by a test as defined in the tables of objective tests is dependent on other FSTD features being modelled to a sufficient level of simulation detail to support the validation of the primary feature(s) fidelity level.
 - (1) If the FSTD features that interact and contribute to an objective test result are simulated at the same fidelity level, the test should be within the applicable tolerances.
 - (2) If the FSTD features that interact and contribute to an objective test result are simulated at lower fidelity levels, the simulation detail may not be sufficient to achieve an objective test result within the applicable tolerances. In such cases, the simulation of those supporting features should be enhanced (e.g. through the use of higher-fidelity design data or enhanced simulation) to improve the objective test result to within the applicable tolerances. Note that the enhancement of an FSTD feature will not affect the FCS unless all requirements for the higher fidelity level of the enhanced feature are met.
- (d) An individual test should be provided in the QTG for each of the test conditions defined in the tables of objective tests for applicable tests. This means that if a test is required but the test cannot be conducted on the FSTD due to FSTD features that are needed to meet other requirements, such as test conditions, not being simulated (fidelity level N), an SOJ should be provided to explain the non-applicability.
- (e) Additional requirements for individual objective tests or a section of objective tests (e.g. take-offs), if applicable, are defined in tables in each CS FSTD.QTG.A or CS FSTD.QTG.H section of the objective tests. These should be reviewed and demonstrated in the relevant individual objective tests.
- (f) The ESL and SOJs for the general requirements (refer to Subpart B) should be reviewed against the simulated aircraft and FSTD configurations, avionics, engine type/ratings, etc. to ensure that the appropriate objective tests, consistent with the content of the ESL and SOJs, are provided in the QTG to validate the FSTD.
- (g) Each objective test should use the applicable validation data defined in the VDR (refer to Subpart C).
- (h) This subpart defines the objective tests for aeroplanes and rotorcraft. Certain aircraft categories (e.g. tiltrotor aircraft) have characteristics of both aeroplanes and rotorcraft. Therefore, as applicable, the QTG should include the applicable objective tests for aeroplanes or rotorcraft.
- (i) Certain aircraft types (e.g. fly-by-wire helicopters) may have new technologies or characteristics that are not directly considered in this CS-FSTD. In such cases, the list of applicable objective tests may differ from that in this CS-FSTD and could include additional

objective tests to validate aspects not directly considered in this CS-FSTD. This should be agreed with the competent authority as part of the VDR.

GM1 FSTD.QTG.005 Definition of applicable objective tests for flight simulation training devices

- (a) To determine whether an objective test is applicable to an FSTD, the tables of objective tests defined in all CS FSTD.QTG.A or CS FSTD.QTG.H sections should be reviewed using the following guidance for each test:
 - (1) refer to the 'Tolerance parameter' column of the test — for each tolerance parameter stated in the test, note the FSTD feature(s) defined in the 'Primary feature(s) validated' column;
 - (2) note the fidelity level of the same FSTD feature(s) in the requested FCS for the FSTD;
 - (3) refer to the 'Tolerances' columns of the test:
 - (i) if a tolerance is specified for the fidelity level of the feature(s) in the requested FCS, the test is applicable to the FSTD and should be included in the QTG,
 - (ii) the tolerances applicable to the test are those specified for the fidelity level of the FSTD feature(s) primarily validated.
- (b) The following examples provide guidance on the use of the tables of objective tests to define test and tolerance applicability for an FSTD.
 - (1) The following apply to a rotorcraft FSTD with feature OGE at fidelity level S.
 - (i) As per CS FSTD.QTG.H.120, test 1.e is applicable to the FSTD and must be included in the QTG, as tolerances are specified in the 'Tolerances' columns for vertical velocity, directional control position or blade angle and collective control position or blade angle.
 - (ii) Vertical velocity, directional control position or blade angle and collective control position or blade angle must have the relevant tolerances defined for fidelity level S applied to each parameter.
 - (iii) The relevant numerical tolerances defined for an initial objective test (Init) must be applied to each parameter for the initial qualification. An essential match of each parameter as defined for a recurrent objective test (Rec) is expected for recurrent testing.
 - (2) The following apply to an aeroplane FSTD with feature GND at fidelity level R and feature IGE at fidelity level S and simulating an aircraft with irreversible flight controls.
 - (i) As per CS FSTD.QTG.A.105, test 1.b.4 is applicable to the FSTD and must be included in the QTG, as tolerances are specified in the 'Tolerances' columns for airspeed, pitch angle, angle of attack, height and column force.
 - (ii) Airspeed, pitch angle, angle of attack and height must have the relevant tolerances defined for fidelity level S applied to each parameter.

- (iii) The column force tolerance is not applicable, as the aircraft simulated does not have reversible flight controls. This should be described in an SOJ.
 - (iv) The CT&M tolerance defined for fidelity level R Init must be applied to airspeed, pitch angle, angle of attack and height for the portion of the test conducted on ground. The relevant numerical tolerances defined for fidelity level S Init must be applied to each parameter for the portion of the test conducted in-ground effect.
 - (v) An essential match of each parameter as defined for fidelity levels R and S Rec is expected for recurrent testing.
- (3) The following apply to an aeroplane FSTD with features CLO and CLH at fidelity level R and feature GND not simulated (fidelity level N).
 - (i) As per CS FSTD.QTG.A.200, test 2.a.9 is applicable to the FSTD and must be included in the MQTG, as tolerances are specified in the 'Tolerances' columns for force and brake system pressure.
 - (ii) The test cannot be conducted on the FSTD, as the lack of the GND feature would mean that the requirement for conducting the test in ground test conditions cannot be met.
 - (iii) This should be explained in an SOJ.
- (c) If a test spans several of the performance and handling FSTD features (GND, IGE, OGE), but no tolerance is applicable to the portion of the test conducted for one or two of the features due to the requested FCS fidelity level(s) of those features, the test should still be conducted to demonstrate compliance with all requirements defined for the test.

An example is provided here. For an aeroplane FSTD with feature GND at fidelity level G and features IGE and OGE at fidelity level R:

 - (1) As per CS FSTD.QTG.A.220, test 2.e.3 is applicable, should be conducted in accordance with the requirement 'Test from a minimum of 61 m (200 ft) AGL to a 50 % decrease in main landing gear touchdown speed' and should not stop at the point where the test transitions from the IGE feature to the OGE feature.
 - (2) All the tolerance parameters are applicable to the test, but the tolerances are only applicable to the portion of the test conducted in-ground effect (IGE feature) and out-of-ground effect (OGE feature).
- (d) If an objective test is required but is not applicable to the simulated aircraft, the test should still be included in the list of applicable objective tests and an SOJ provided in QTG to explain why the test cannot be conducted on the FSTD. For example:
 - (1) An FSTD simulating a single-engine aeroplane cannot meet the requirements for OEI objective tests.
 - (2) An autopilot landing test is applicable only to FSTDs with appropriate auto-land capability.

- (e) This subpart defines the objective tests for aeroplanes and rotorcraft. Certain aircraft categories (e.g. tiltrotor aircraft) have characteristics of both aeroplanes and rotorcraft. Therefore, as applicable, the QTG should include the relevant objective tests for aeroplanes or rotorcraft.
- (f) Certain aircraft types (e.g. fly-by-wire helicopters) may have new technologies or characteristics that are not directly considered in this CS-FSTD. In such cases, the list of applicable objective tests may differ from those in this CS-FSTD and could include additional objective tests to validate aspects not directly considered in this CS-FSTD. This should be agreed with the competent authority as part of the VDR (refer to Subpart C).

CS FSTD.QTG.010 Objective test content

- (a) Each objective test in the QTG should contain sufficient information for the FSTD operator to demonstrate compliance with the applicable requirements of this CS for the test.
- (b) The following information should be provided for each objective test.
 - (1) Test number: this should provide a clear link to the numbering defined in the relevant CS FSTD.QTG.A or CS FSTD.QTG.H section.
 - (2) Test title: this should be short and definitive, and based on the test title referred to in the relevant CS FSTD.QTG.A or CS FSTD.QTG.H section.
 - (3) Test objective: this should be a brief summary of what the test is intended to demonstrate and how the objective is to be met.
 - (4) References: these are the validation data source documents referred to in the VDR, including both the document number and the page or condition number and other supporting data or information, if applicable.
 - (5) Initial conditions: a full and comprehensive list of the FSTD test initial conditions is required. For performance and handling objective tests (CS FSTD.QTG.A.100 through CS FSTD.QTG.A.240, and CS FSTD.QTG.H.100 through CS FSTD.QTG.H.215), these conditions should include the following, as applicable to and available for the aircraft simulated, test condition and fidelity level of each feature, as a minimum (for all other objective tests, include the relevant initial condition parameters necessary to confirm the correct test conditions):
 - (i) gross weight;
 - (ii) centre of gravity;
 - (iii) moments of inertia (for aeroplanes);
 - (iv) pressure altitude;
 - (v) radar altitude or mean gear height above ground*;
 - (vi) airspeed (calibrated, indicated or ground speed if airspeed is not valid, but as specified in the validation data) and vertical speed;
 - (vii) for aeroplanes, Mach number (for cruise/high-altitude condition);

- (viii) pitch, roll and yaw angles (Euler angles);
- (ix) AOA and side-slip angle;
- (x) linear and rotational velocities and accelerations / load factors (each axis);
- (xi) outside air temperature;
- (xii) wind speed and direction*;
- (xiii) engine power rating selection, fuel flow and bleed condition;
- (xiv) key engine, propeller and transmission parameters, as applicable to the simulated aircraft, including parameters related to thrust and power output (e.g. manifold pressure, propeller revolutions per minute (RPM), engine low-pressure rotor speed (N1), engine pressure ratio (EPR) and torque for aeroplanes; torque and main rotor speed for rotorcraft), engine temperature parameters (e.g. EGT, inter-turbine temperature, cylinder head temperature) and other core engine parameters (e.g. engine fuel flow, engine high-pressure rotor speed (N2), gas generator speed (Ng), free turbine speed (Nf));
- (xv) stability augmentation status (each axis), including the flight control law mode for fly-by-wire aircraft and autopilot state or mode, if applicable;
- (xvi) primary flight control positions (pitch, roll and yaw) for aeroplanes and rotor blade angles or equivalent for rotorcraft;
- (xvii) for aeroplanes, forces on the primary flight controls (pitch, roll and yaw);
- (xviii) for aeroplanes, trim settings (pitch, roll and yaw);
- (xix) landing gear positions; and
- (xx) for aeroplanes, secondary flight controls such as leading- and trailing-edge flap/slat, speed brake and spoiler positions.

Note: The ‘*’ symbol denotes extra parameters required for tests performed on or near the ground.

- (6) Manual test procedures: they should clearly describe how the FSTD will be set up and operated for each test if flown manually by a person. Procedures should be sufficient to enable the test to be flown using only reference to flight deck instrumentation.
- (7) Automatic test procedures: they should describe how to execute the test using an automatic testing system. They should cover the execution of performance and handling tests and also motion, visual and sound tests if the automatic testing system may need manual steps to be conducted by the FSTD operator (e.g. assessment of a visual test pattern) in the process of executing a test.
- (8) Information should be provided on how each test is constructed and how the automatic test system is controlling the test (e.g. which parameters are driven through the application of either open- or closed-loop drivers or are locked at a constant fixed value).

- (9) Evaluation criteria: specify the applicable tolerances and main parameter(s) under scrutiny during the test and how the tolerances are applied in the initial qualification and/or recurrent testing processes.
- (10) Any rationales necessary to justify out-of-tolerance results or to support the use of engineering judgement in the conduct of an objective test should be provided.
- (11) Validation data should be provided.
 - (i) A copy of the validation data should be clearly marked with the document, page number, data provider and the test number and title. If necessary, a further definition of the point at which the information was extracted from the validation data source should be provided.
 - (ii) If the primary feature(s) validated by the objective test are at fidelity level R or G, it is acceptable to refer to the ER.
- (12) Test revision number: this should define the revision of the objective test script that conducts the test and generates the test output, and it should serve as the means to demonstrate configuration control of the objective test script.
- (13) Test result and comparison of results: FSTD test result obtained when running the test on the FSTD. The result should:
 - (i) indicate the FSTD on which the test was run;
 - (ii) indicate the software load/version used and other information necessary to identify the configuration of the FSTD;
 - (iii) indicate whether the test was run manually or automatically.
 - (iv) indicate the date and time of the test run;
 - (v) have each page reflect the test page number and the total number of pages in the test result;
 - (vi) indicate the assessment of the results against evaluation criteria.
 - (vii) for recurrent tests, cover all items listed in paragraph (b)(13)(i)–(vi) and additionally:
 - (A) the date and time of the MQTG test result;
 - (B) the test revision number of the MQTG result.

GM1 FSTD.QTG.010 Objective test content

Some test content information may not be applicable to certain types of objective tests, particularly in the areas of motion, sound and visual. As an example, manual test procedures (refer to CS FSTD.QTG.010, paragraph (b)(6)) and information on test construction (refer to CS FSTD.QTG.010, paragraph (b)(8)) are unlikely to be needed for these types of tests. In these cases, the information may be omitted from the test.

CS FSTD.QTG.015 Objective test result presentation

- (a) For the initial qualification:
- (1) each objective test should be presented in a form and manner to allow for a comparison of the FSTD result against the validation data;
 - (2) each objective test should be presented in a form and manner to allow for an assessment of the FSTD result against the applicable tolerances;
 - (3) the initial conditions for each test should be provided, including the initial conditions for the FSTD result and the validation data;
 - (4) to facilitate objective assessment of the FSTD performance and system operation, a recording of the relevant FSTD parameters should be made for each objective test and presented as time history plots, directly overplotted with the same parameters from the validation data, if available;
 - (5) the following time history plots should be provided for each objective test:
 - (i) all tolerance parameters;
 - (ii) all driven parameters (i.e. the FSTD parameters that are 'driven' with corresponding parameters from the validation data in order to execute the test, such as primary and secondary flight control inputs);
 - (iii) relevant primary axis parameters to facilitate a review of the FSTD performance, including evidence of end-to-end testing, for the primary axis of interest throughout the test;
 - (iv) sufficient off-axis parameters to confirm the overall FSTD performance, including evidence of end-to-end testing, throughout the test;
 - (6) If an objective test may be provided as a snapshot:
 - (i) initial conditions should be provided;
 - (ii) it should be possible to compare the tolerance FSTD snapshot parameters against the corresponding validation data parameters;
 - (7) the size and scaling of time history plots should provide the resolution necessary for the efficient evaluation and identification of changes in parameter values.
- (b) For recurrent test presentation:
- (1) If the recurrent test tolerance is specified as 'EM' (Essential Match):
 - (i) each objective test should provide a method of directly comparing the current FSTD result against the MQTG result for recurrent testing to confirm that there are no changes in performance from the MQTG and that an essential match with the MQTG can be achieved for all test parameters;
 - (ii) each objective test should provide a method of directly comparing the current FSTD results against the validation data;

- (iii) to facilitate the objective assessment of an essential match between the current FSTD result and the MQTG result and to also compare the FSTD result against the validation data, the FSTD result, the MQTG result and the validation data should all be directly overplotted and the initial conditions for the FSTD result, MQTG result and validation data should be provided in a manner that allows for easy comparison;
- (2) If the recurrent test tolerance is specified as 'same as Init' (i.e. same as for the initial qualification), the requirements of CS FSTD.QTG.015, paragraph (a), apply;
- (3) If the recurrent test tolerance is specified as a numerical tolerance:
 - (i) each objective test should be presented in a form and manner that allows for a comparison of the FSTD result against the MQTG result;
 - (ii) each objective test should be presented in a form and manner that allows for an assessment of the FSTD result against the MQTG result using the applicable tolerance;
- (4) If an objective test may be provided as a snapshot, it should be possible to compare the snapshot parameters from the current FSTD result against the MQTG result and the validation data.

GM1 FSTD.QTG.015 Objective test result presentation

- (a) Plotted parameters should cover the following.
 - (1) As guidance, certain groups of parameters should be considered for inclusion in each performance and handling objective test, dependent on flight regime and manoeuvre. The parameters should support the analysis of primary axis and off-axis FSTD inputs and resulting aircraft performance, if not already covered due to being either tolerance or driven parameters. The parameter groups are:
 - (i) aircraft state parameters, such as:
 - (A) airspeed / Mach / ground speed, altitude and radio height above ground,
 - (B) pitch, roll and yaw angles (Euler angles),
 - (C) angular/Euler rates;
 - (ii) flight controls (e.g. control positions and blade angles, surface positions);
 - (iii) secondary flight controls (e.g. control positions, system outputs);
 - (iv) configuration parameters (e.g. flaps, gear);
 - (v) engine key parameters (e.g. engine control positions, output parameters, rotor speed);
 - (vi) environment parameters (e.g. wind speed and direction);
 - (vii) system parameters (e.g. envelope protection functions, alerts, stick shaker).

- (2) It is preferable for plotted parameters to be ordered axis by axis, such as:
- (i) main flight parameters (speed, altitude, rate of climb, etc.);
 - (ii) pitch/longitudinal axis (flight control, angular rates, attitude, etc.);
 - (iii) roll/lateral axis (flight control, angular rates, attitude, etc.);
 - (iv) yaw/directional axis (flight control, angular rates, attitude, etc.);
 - (v) vertical axis (collective, main rotor speed, torque, etc.);
 - (vi) other parameters added to each section, as needed for the particular test (e.g. side-slip, load factor, angle of attack).
- (b) The following applies to tolerance bands.
- Tolerance bands may be used in time history plots to indicate the upper and lower limits of the applicable tolerances. These should support the initial qualification process of analysing the FSTD result against the validation data.
- (c) The following apply to the presentation of validation data.
- (1) Aircraft data documents included in the QTG may be photographically reduced only if such reduction will not cause distortions or difficulties in scale interpretation or resolution.
 - (2) Validation data variables, along with sign conventions, should be defined in a nomenclature list. This list should be included at some appropriate location in the QTG.
- (d) All the tests with plotted parameters should include a legend identifying each plot and any associated tolerance bands.
- (e) In showing compliance with CS FSTD.QB.001, paragraph (d), account should be taken, as appropriate, of ARINC Report 436-1, *Guidelines for Electronic Qualification Test Guide*, as amended.

CS FSTD.QTG.020 Conducting objective tests

- (a) For the initial qualification, the following apply.
- (1) It should be possible to conduct each performance and handling objective test automatically using an automatic test driver programme by following the automatic test procedure.

Motion, visual, sound and system integration objective tests should be conducted automatically to the maximum extent possible. All steps required to be executed manually to complete the test and obtain a test result should be defined within the automatic test procedure.
 - (2) Test inputs, including those that are locked at particular values, used to set up the initial conditions for each test should show an essential match to the validation data, with engineering judgement applied if differences are present.

-
- (3) Performance and handling tests for the stabilisation of snapshots and execution of time histories should be driven automatically through the application of open- or closed-loop drivers to the relevant pilot control inputs or by locking certain pilot control inputs to a fixed constant value as appropriate:
- (i) if capturing snapshots for test initial conditions or snapshots, the FSTD should be in a steady trimmed condition as per the validation data at the point at which the snapshot is captured;
 - (ii) open-loop drivers:
an open-loop driver should not apply any deviation to the shape, timing and magnitude of the corresponding pilot control input validation data parameter, unless justified by a supporting rationale;
 - (iii) closed-loop drivers:
use of a closed-loop driver on a pilot control input should only be applied if the corresponding validation data parameter exhibits frequent varying inputs throughout the time history that may integrate throughout the duration of the test into larger errors in terms of differences between the FSTD result and validation data than if an open loop driver were used;
 - (iv) locked parameters:
some pilot control inputs that do not change throughout the duration of the time history may be locked at the constant fixed value for the duration of the test.
- (b) For recurrent testing:
- (1) All test inputs used to set up the initial conditions for a recurrent test should be made using the same inputs that were used for the corresponding MQTG result.
 - (2) All pilot control inputs for a recurrent test should be made using the same open- and closed-loop drivers that were used for the corresponding MQTG result.
- (c) For manual testing, the following apply.
- (1) It should be possible to run each performance and handling objective test manually by following the manual test procedure. Other FSTD features (e.g. VIS), if present on the FSTD, should provide appropriate cues to support running the test.
 - (2) Performance and handling objective tests should be capable of being conducted manually from all pilot seats if both sets of primary flight controls are not mechanically linked in the simulated aircraft.
- (d) For integrated and end-to-end testing:
- (1) If executing an objective test, the FSTD test should be running in the same integrated manner using all FSTD features as it would be if the FSTD is being used in training.

- (2) Unless otherwise justified, all pilot control inputs (e.g. primary and other flight controls) needed to execute each performance and handling objective test, for both snapshot and time history tests, should be made through the relevant pilot controls either manually or automatically with an open- or closed-loop driver by the test driver programme.
- (e) Tests run on a computer that is independent of the FSTD are not acceptable.

GM1 FSTD.QTG.020 Conducting objective tests

- (a) For the initial qualification, the following apply.
 - (1) The application of open- and closed-loop drivers and locked parameters should be as follows.
 - (i) For open-loop drivers:

Open-loop drivers are the preferred type of driver to be used, as they ensure that the test input is the same as that used in the validation data. This will lead to a reliable assessment of the test outputs.
 - (ii) For closed-loop drivers:
 - (A) Typically, closed-loop drivers should only be applied to highly dynamic tests, such as take-offs and landings. They would not be expected to be applied to the pilot control input in the primary axis for tests assessing aircraft free response (e.g. aeroplane longitudinal free response, tests 2.c.1, 2.c.2, 2.c.3 and 2.c.4) and tests in which the FSTD performance under a fixed control input is assessed (e.g. aeroplane step input of flight deck roll controller (or roll overshoot) (test 2.d.3)).
 - (a) An example of acceptable use is in rotorcraft take-off and landing tests. In these tests, both flight control inputs and aircraft performance and handling parameters (test outputs) are subject to tolerances. A closed-loop driver may be used.
 - (b) Another example of acceptable use is a closed-loop driver on the wheel in an aeroplane longitudinal test to ensure that the bank angle does not adversely impact the longitudinal performance. In this case, the closed-loop driver operates on a secondary axis.
 - (B) In all applications of a closed-loop driver, the resultant modified flight control input should still be consistent with the patterns a pilot would produce during the same manoeuvre and should align with the validation data.

- (iii) For locked parameters:

a pilot control input should only be locked for the duration of a test if it is necessary to ensure that there is no change throughout the test, such as ensuring no change in the gear or flap handle.
- (b) For recurrent testing:

If the MQTG test result is a 'footprint' result, the recurrent test may use the MQTG result pilot control outputs as open- or closed-loop driver inputs that are 'driven' into the simulation for the purposes of executing the recurrent test automatically.
- (c) For manual testing:
 - (1) If executing a test manually, it should be possible to achieve a result that matches the validation data within the limits of the prescribed tolerances of the test.
 - (2) Reference to the validation data and/or automatic test results is encouraged for complex tests in order to help guide the manual control inputs.
 - (3) The objective test system should not inhibit pilot input of any flight control (primary or other) if performing a test manually.
- (d) For integrated and end-to-end testing:
 - (1) The requirement for objective tests to be executed through the relevant pilot control inputs (known as 'end-to-end testing') with the FSTD running in an integrated manner (known as 'integrated testing') was created during the development of ICAO Document 9625, *Manual of criteria for the qualification of flight simulators*, 1st edition, by a Royal Aeronautical Society (RAeS) working group. The following text was inserted in the document.

It is not the intent, nor is it acceptable, to test each simulator subsystem independently. Over-all integrated testing of the simulator must be accomplished to assure that the total simulator system meets the prescribed standards.
 - (2) Objective tests may be executed without the motion system active only in tests in which the motion system is not integral to the objective of the test or does not affect any of the tolerance parameters. For example, motion typically does not affect performance and handling tests but may be needed for a stall test to demonstrate buffets.

CS FSTD.QTG.025 Assessing objective tests

- (a) For the initial qualification, the following apply:
 - (1) Each objective test should be assessed to ensure that the FSTD result matches the validation data within the limits of the prescribed tolerances for initial conditions, snapshots and the duration of time histories, as applicable to the test.

- (i) Some objective tests (e.g. take-offs and landings) span several flight regimes (e.g. on ground, in-ground effect and out-of-ground effect). All tolerance parameters should remain within the limits of the prescribed tolerances of that feature for the duration of the test.
 - (ii) Some validation data parameters may exhibit rapid time history variations that require engineering judgement when making assessments of the FSTD result compared with the validation data. Such judgement should not be linked to a single parameter. All relevant parameters related to a given manoeuvre or test condition should be assessed for overall interpretation.
 - (iii) If it is difficult or impossible to match a validation data parameter in a snapshot or throughout a time history within the limits of the prescribed tolerance, a rationale should be provided in the QTG detailing the reason for the deviation and why the test result is to be considered acceptable.
 - (iv) All parameters that are not subject to tolerances should demonstrate that they match the validation data, with any variations being comparable to the variations expected during the manoeuvres of the objective test. Significant differences should be justified in the QTG by a supporting rationale.
- (2) If the tolerance applicable to the test is due to the primary feature(s) validated being modelled at the specific level of fidelity and engineering simulator/simulation data or other non-flight test data is used as an allowable form of validation data, the FSTD result should match the validation data within 40 % of the applicable tolerances defined in the relevant CS FSTD.QTG.A or CS FSTD.QTG.H section.
- If the tolerance applicable to the test is due to the primary feature(s) validated being modelled at the representative (R) or generic (G) fidelity level, the FSTD result should match the validation data within the applicable tolerances defined in the relevant CS FSTD.QTG.A or CS FSTD.QTG.H section.
- (3) If an objective test result is provided as a 'footprint', justification of the footprint should be given as part of the ER (refer to Subpart C).
- (b) For recurrent test assessment:
- (1) If the recurrent test tolerance is specified as 'EM' (essential match), each objective test should be assessed to ensure that an essential match with the MQTG test result has been achieved for all initial condition parameters, time history plots and snapshots, if applicable;
 - (2) If the recurrent test tolerance is specified as 'same as Init' (i.e. same as for the initial qualification), the requirements of CS FSTD.QTG.025, paragraph (a), apply;
 - (3) If the recurrent test tolerance is specified as a numerical tolerance, the tolerance should be applied in relation to the FSTD result and the MQTG result.

GM1 FSTD.QTG.025 Assessing objective tests

- (a) The principles of test inputs affecting test outputs are as follows.
 - (1) For any objective test, demonstration that the test inputs used for setting up the initial conditions and any further snapshots or for driving pilot controls either open or closed loop throughout a time history, showing an essential match to the validation data is fundamental in being able to reliably assess the FSTD outputs.
 - (2) Test inputs required and FSTD outputs assessed can vary based on the type of test. Examples of inputs could be aircraft initial configuration, atmospheric conditions and/or primary and other flight control positions for performance and handling tests; particular flight model rates and accelerations for some motion tests; and commands to display a visual test pattern for some visual tests. In each of these examples, the outputs – which would be the response of the simulated aircraft, the physical movement of the motion system or the measured result from the visual test pattern – would be assessed against the evaluation criteria for the tests.
- (b) For the initial qualification, the following apply.
 - (1) The interpretation of test tolerances should be as follows.
 - (i) If two tolerance values are given for a parameter, the less restrictive one may be used, unless otherwise indicated.
 - (ii) If tolerances are expressed as a percentage, the following apply.
 - (A) For parameters that have units of per cent or parameters normally displayed in the flight deck in units of per cent (e.g. N1, N2, engine torque or power), a percentage tolerance should be interpreted as an absolute tolerance unless otherwise specified. That is, for an observation of 50 % N1 and a tolerance of 5 %, the acceptable range should be from 45 % to 55 %.
 - (B) For parameters not displayed in units of per cent, a tolerance expressed only as a percentage should be interpreted as the percentage of the current validation data value of that parameter during the test. However, for parameters varying around a zero value, a minimum absolute value should be agreed with the competent authority.
 - (2) The application of open- and closed-loop drivers should be as follows.
 - (i) For open-loop drivers, the following apply.
 - (A) An open-loop driver should show an essential match to the shape, timing and magnitude of the pilot control input compared with the corresponding validation data parameter.
 - (B) Small, constant offsets throughout the time history may be acceptable to take account of differences between the FSTD pilot control input

and corresponding validation data parameter resulting from the initial trim.

- (ii) For closed-loop drivers, the following apply.
 - (A) Application of a closed-loop driver should result in the target parameter having a close match to the validation data, with any error represented in the pilot control input. Therefore, the flight control match becomes an important indicator of the FSTD quality and should only exhibit small differences from the validation data.
 - (B) In instances in which the flight control (CLH or CLO) and performance and handling (GND, IGE or OGE) features both contain tolerance parameters for the test, any error in terms of differences between the FSTD result and the validation data may be spread across both parameters.
 - (C) The overall shape and trend of a flight control parameter that has a closed loop applied to it should be reflective of the inputs that a pilot would physically provide if they were flying the manoeuvre in the real aircraft. For example, the input should not exhibit rapid oscillations that might be observed with over-corrections.
- (3) Test condition verification should be as follows.
 - (i) When assessing the parameters listed in the initial conditions or snapshots and comparing time history plots against those of the aeroplane, sufficient parameters should be assessed to verify the correct test condition. For example:
 - (A) To show that the control force is within ± 2.2 daN (5 lbf) in a static stability test, data to show correct airspeed, power, thrust or torque, aeroplane configuration, altitude and other appropriate datum identification parameters should also be assessed.
 - (B) To assess short-period dynamics in an FSTD, pitch angle, pitch rate and normal acceleration may be used to establish a match to the aeroplane, but airspeed, altitude, control input, aeroplane configuration and other appropriate data should also be assessed.
 - (C) All airspeed values should be assumed to be calibrated, unless otherwise annotated, and like values should be used for assessment.
- (4) General principles are as follows.
 - (i) Test result assessment for initial conditions and any subsequent snapshots should review all parameters presented to ensure that they show a good match to the validation data, taking into consideration the FCS. The same should be applied to all test results presented as snapshots.

- (ii) Test result assessment for time history plots should review all parameters plotted to ensure that they show a good match to the validation data, taking into consideration the FCS. This should include assessing the FSTD result for all supporting parameters in the primary axis, and also off-axis parameters, to confirm the overall replication of the manoeuvre and implementation of end-to-end testing.
- (c) For recurrent test assessment, the following apply.
 - (1) An essential match of pilot inputs is of paramount importance for a recurrent test in ensuring that an essential match with the MQTG test result can be achieved. If the recurrent test pilot inputs do not match the MQTG test result pilot inputs, the test should not be considered to be compliant with the technical requirements. The reasons for any differences should be investigated and resolved, with the test revised in the MQTG if necessary.
 - (2) If the tolerance specified for recurrent test assessment is 'same as Init', all elements of this subpart that refer to the initial qualification are applicable to recurrent tests.
- (d) For manual test assessment, the following apply.
 - (1) There may be segments of a test result in which a parameter is outside the limits of the prescribed tolerance. Engineering judgement should be applied to the assessment of the test result, taking into consideration aspects such as rapid variations in parameter values and the complexity of the pilot's input that the test is attempting to replicate.
 - (2) A rationale should be provided if a single set of validation data is to be used to validate a test run from either pilot seat.
- (e) The RAeS publication *Aeroplane Flight Simulator Evaluation Handbook – Volume 1*, as amended, is a useful source of guidance on objective tests, the use of open- and closed-loop drivers and flight test validation data, including how and in which cases to apply engineering judgement.

GM1 FSTD.QTG.025(a)(2) Assessing objective tests

ENGINEERING SIMULATOR/SIMULATION VALIDATION DATA TEST TOLERANCES

For the use of engineering simulator/simulation validation data, the following apply.

- (a) There are many reasons why an objective test may not fully match the validation data within the limits of the prescribed tolerances. For example:
 - (1) flight tests are subject to many sources of potential error (e.g. instrumentation errors and atmospheric disturbance during data collection);
 - (2) data that exhibits rapid variation or noise may also be difficult to match;
 - (3) engineering simulator/simulation validation data and other calculated data may exhibit errors due to a variety of potential differences, discussed below.

- (b) If applying tolerances to any test result, good engineering judgement should be applied. If a test result clearly falls outside the prescribed tolerance(s) for no acceptable reason, the test should be judged to have failed.
- (c) If engineering simulator/simulation validation data is used, the basis for its use is that the validation data is produced using the same simulation models as used in the equivalent FSTD; that is, the two sets of results should be 'essentially' similar. If engineering simulator/simulation validation data is used, it is understood that the flight-test-based tolerances should be reduced, since applied tolerances should not include measurement errors inherent to flight test data.
- (d) There are reasons why the FSTD test result would differ from the engineering simulator/simulation validation test data. These reasons include but are not limited to:
 - (1) hardware (avionics units and flight controls);
 - (2) modelling solutions used in the FSTD that are different from those used by the aircraft OEM (e.g. ground-handling models, braking models, engine models);
 - (3) model cascading effects;
 - (4) iteration rates;
 - (5) execution orders;
 - (6) integration methods;
 - (7) processor architecture;
 - (8) digital drift:
 - (i) interpolation methods;
 - (ii) data-handling differences;
 - (iii) auto-test trim tolerances etc.;
 - (iv) open-loop versus closed-loop responses;
 - (v) test duration;
 - (vi) extent of the dependency on contributory aircraft systems, adding to the complexity of the test; and/or
 - (vii) accuracy of the match of the initial conditions.
- (e) Any differences between the FSTD test result and the engineering simulator/simulation validation data should, however, be small. The reasons for any differences, other than those listed above, should be clearly explained.

GM1 FSTD.QTG.025(b) Assessing objective tests

ESSENTIAL MATCH

An 'Essential Match' (EM) is a comparison of two objective test results for which the differences should be negligible. The comparison should be conducted in accordance with the following principles of EM.

- (a) The magnitude of any variation should be extremely small such that an analysis of the test results with a naked eye shows them to be identical in cases both are overlaid. However:
 - (1) analysing results electronically by zooming in to very high levels of magnification and analysing time history plots pixel by pixel is not the intent;
 - (2) analysing initial conditions and snapshot values at a level of significant figure accuracy beyond that which identifies variation in the result is not the intent.
- (b) It is possible that noticeable variations in test results may occur between successive runs of the same test on the same software load, such that an essential match to the MQTG result could be questioned. Examples of reasons for such variation include the following.
 - (1) Issues such as floating-point calculations, differences in execution frame times, etc. that are inherent to real-time simulation software may cause variations.
 - (2) execution of a test with flight control hardware in the loop may cause small variations.
 - (3) Non-deterministic systems such as rotorcraft stabilisation systems may cause small variations.
- (c) It is important that any noticeable variations are thoroughly reviewed to understand the cause.
 - (1) The fundamental principle of the essential match is that an FSTD, with hardware and software that has not changed, should be able to consistently demonstrate the same objective test results as those in the MQTG.
 - (2) If this cannot be achieved, the level of variation of a test result that is acceptable based on the design of the FSTD should be clearly understood and defined. This acceptable level should be described in a rationale to support the test result. Further variations beyond this level should be deemed a test failure.
 - (3) If the variation is due to a known FSTD software and/or hardware change and the MQTG has not been updated to reflect the change, this represents an issue with the configuration control of the MQTG and should be resolved accordingly.

OBJECTIVE TESTS FOR AEROPLANES

CS FSTD.QTG.A.100 Performance taxi – Aeroplane

This CS provides the standards for the objective tests for aeroplane taxi.

1.a: Performance taxi		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
1.a.1	Minimum radius turn	Ground	Aeroplane turn radius	GND		Init: CT&M Rec: EM	Init: ± 0.9 m (3 ft) or ± 20 % Rec: EM
1.a.2	Rate of turn versus nose wheel steering angle	Ground	Turn rate	GND		Init: CT&M Rec: EM	Init: ± 10 % or $\pm 2^\circ/\text{s}$ Rec: EM

Individual test requirements. The individual test requirements defined in the table below are applicable to all fidelity levels unless otherwise stated.

1.a: Performance taxi		Test requirements
1.a.1	Minimum radius turn	- Test should plot both main and nose gear-turning loci and key engine parameter(s). - Data should be gathered for no brake use and the minimum thrust required to maintain a steady turn, except for aeroplanes requiring asymmetric thrust or braking to achieve the minimum radius turn.
1.a.2	Rate of turn versus nose wheel steering angle	Test should record for a minimum of two speeds greater than the minimum turning radius speed, with one at a typical taxi speed, and with a spread of at least 5 kt.

CS FSTD.QTG.A.105 Performance take-off – Aeroplane

This CS provides the standards for the objective tests for aeroplane take-off.

1.b: Performance take-off		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
1.b.1	Ground acceleration time and distance	Take-off	Time	GND		Init: $\pm 5\%$ or $\pm 1.5\text{ s}$ Rec: EM	Init: $\pm 5\%$ or $\pm 1.5\text{ s}$ Rec: EM
			Distance	GND		Init: $\pm 5\%$ or $\pm 61\text{ m}$ (200 ft) Rec: EM	Init: $\pm 5\%$ or $\pm 61\text{ m}$ (200 ft) Rec: EM
1.b.2	Minimum control speed (ground) (V_{MCG})	Take-off	Maximum aeroplane lateral deviation	GND			Init: $\pm 25\%$ of maximum aeroplane lateral deviation or $\pm 1.5\text{ m}$ (5 ft) Rec: EM
			Additional tolerance parameter for aeroplanes with reversible flight control systems				
			Rudder pedal force	GND			Init: $\pm 10\%$ or $\pm 2.2\text{ daN}$ (5 lbf) Rec: EM
1.b.3	Minimum unstick speed (V_{MU}) or equivalent test	Take-off	Airspeed	GND, IGE			Init: $\pm 3\text{ kt}$ Rec: EM
			Pitch angle	GND, IGE			Init: $\pm 1.5^\circ$ Rec: EM
1.b.4	Normal take-off	Take-off	Airspeed	GND, IGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 3\text{ kt}$ Rec: EM

1.b: Performance take-off		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
			Pitch angle	GND, IGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 1.5^\circ$ Rec: EM
			Angle of attack	GND, IGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 1.5^\circ$ Rec: EM
			Height	GND, IGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: ± 6 m (20 ft) Rec: EM
			Additional tolerance parameter for aeroplanes with reversible flight control systems				
			Column force	GND, IGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: ± 10 % or ± 2.2 daN (5 lbf) Rec: EM
1.b.5	Critical engine failure on take-off	Take-off	Airspeed	GND, IGE		Init: CT&M Rec: EM	Init: ± 3 kt Rec: EM
			Pitch angle	GND, IGE		Init: CT&M Rec: EM	Init: $\pm 1.5^\circ$ Rec: EM
			Angle of attack	GND, IGE		Init: CT&M Rec: EM	Init: $\pm 1.5^\circ$ Rec: EM
			Height	GND, IGE		Init: CT&M Rec: EM	Init: ± 6 m (20 ft) Rec: EM
			Roll angle	GND, IGE		Init: CT&M Rec: EM	Init: $\pm 2^\circ$ Rec: EM
			Side-slip angle	GND, IGE		Init: CT&M Rec: EM	Init: $\pm 2^\circ$ Rec: EM

1.b: Performance take-off		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
			Heading angle	GND, IGE		Init: CT&M Rec: EM	Init: $\pm 3^\circ$ Rec: EM
			Additional tolerance parameters for aeroplanes with reversible flight control systems				
			Column force	GND, IGE		Init: CT&M Rec: EM	Init: $\pm 10\%$ or ± 2.2 daN (5 lbf) Rec: EM
			Wheel force	GND, IGE		Init: CT&M Rec: EM	Init: $\pm 10\%$ or ± 1.3 daN (3 lbf) Rec: EM
			Rudder pedal force	GND, IGE		Init: CT&M Rec: EM	Init: $\pm 10\%$ or ± 2.2 daN (5 lbf) Rec: EM
1.b.6	Crosswind take-off	Take-off	Airspeed	GND, IGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: ± 3 kt Rec: EM
			Pitch angle	GND, IGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 1.5^\circ$ Rec: EM
			Angle of attack	GND, IGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 1.5^\circ$ Rec: EM
			Height	GND, IGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: ± 6 m (20 ft) Rec: EM
			Roll angle	GND, IGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 2^\circ$ Rec: EM
			Side-slip angle	GND, IGE		Init: CT&M Rec: EM	Init: $\pm 2^\circ$ Rec: EM

1.b: Performance take-off		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
			Heading angle	GND, IGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 3^\circ$ Rec: EM
			Additional tolerance parameters for aeroplanes with reversible flight control systems				
			Column force	GND, IGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 10\%$ or ± 2.2 daN (5 lbf) Rec: EM
			Wheel force	GND, IGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 10\%$ or ± 1.3 daN (3 lbf) Rec: EM
			Rudder pedal force	GND, IGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 10\%$ or ± 2.2 daN (5 lbf) Rec: EM
1.b.7	Rejected take-off	Take-off	Time	GND	Init: CT&M Rec: EM	Init: $\pm 5\%$ or ± 1.5 s Rec: EM	Init: $\pm 5\%$ or ± 1.5 s Rec: EM
			Distance	GND	Init: CT&M Rec: EM	Init: $\pm 7.5\%$ or ± 76 m (250 ft) Rec: EM	Init: $\pm 7.5\%$ or ± 76 m (250 ft) Rec: EM
1.b.8	Dynamic engine failure after take-off	Take-off CCA: test in normal and non-normal control states	Body angular rates (roll, pitch and yaw)	OGE			Init: $\pm 20\%$ or $\pm 2^\circ/\text{s}$ Rec: EM

Requirements for all the aeroplane take-off tests.

1.b: Performance take-off	Test requirements
All tests	For feature GND or IGE at fidelity level R or S, all aeroplane manufacturer commonly used, certified, take-off flap settings should be demonstrated at least once either in tests for V_{MU} (1.b.3), normal take-off (1.b.4), critical engine failure on take-off (1.b.5) or crosswind take-off (1.b.6).

Individual test requirements. The individual test requirements defined in the table below are applicable to all fidelity levels unless otherwise stated.

1.b: Performance take-off		Test requirements
1.b.1	Ground acceleration time and distance	- Acceleration time and distance should be recorded for a minimum of 80 % of the total time from brake release to rotation speed (V_R). - This may be combined with normal take-off (1.b.4) or rejected take-off (1.b.7). Plotted data should be shown using appropriate scales for each portion of the manoeuvre.
1.b.2	Minimum control speed (ground) (V_{MCG})	- Aerodynamic controls should be only per applicable airworthiness requirement or alternative engine-inoperative test to demonstrate ground control characteristics. - Engine failure speed should be within ± 1 kt of aeroplane engine failure speed. Engine thrust decay should be that resulting from the mathematical model for the engine variant applicable to the FSTD being tested. If the modelled engine is not the same as the aeroplane manufacturer's flight test engine, a further test may be run with the same initial conditions using the thrust from the flight test data as the driving parameter. To ensure only aerodynamic control, nose wheel steering should be disabled (i.e. castored) or the nose wheel held slightly off the ground. - If a V_{MCG} test is not available, an acceptable alternative is a flight test of snap engine deceleration to idle at a speed between the critical decision speed (V_1) and $V_1 - 10$ kt, followed by control of heading using aerodynamic control only, and recovery should be achieved with the main gear on the ground.
1.b.3	Minimum unstick speed (V_{MU}) or equivalent test	- The purpose is to demonstrate early rotation take-off characteristics. V_{MU} is defined as the minimum speed at which the last main landing gear leaves the ground. The main landing gear strut compression or equivalent air/ground signal should be recorded. - If a V_{MU} test is not available, alternative acceptable flight tests are a constant high-attitude take-off run through main gear lift-off or an early rotation take-off. - If either of these alternative solutions is selected, aft body contact / tail strike protection functionality, if present on

1.b: Performance take-off		Test requirements
		the aeroplane, should be active. 5. Test should record time history data from 10 kt before the start of rotation until at least five seconds after the occurrence of main gear lift-off.
1.b.4	Normal take-off	1. For feature GND or IGE at fidelity level R or S, data is required for near maximum certified take-off weight at mid-centre-of-gravity location and light take-off weight at an aft-centre-of-gravity location. 2. For feature GND, IGE or OGE at fidelity level R or S, if the aeroplane has more than one certified take-off configuration, a different configuration should be used for each weight. 3. Test should record take-off profile from brake release to at least 61 m (200 ft) AGL. 4. The test may be used for ground acceleration time and distance (1.b.1). 5. Plotted data should be shown using appropriate scales for each portion of the manoeuvre.
1.b.5	Critical engine failure on take-off	1. Test should record take-off profile to at least 61 m (200 ft) AGL. 2. Engine failure speed should be within ± 3 kt of the validation data. 3. Test should take place at near maximum take-off weight.
1.b.6	Crosswind take-off	1. Test should record take-off profile from brake release to at least 61 m (200 ft) AGL. 2. This test requires test data, including wind profile, for a crosswind component of at least 60 % of the aeroplane performance data value measured at 10 m (33 ft) above the runway. 3. Wind components should be provided as headwind and crosswind values with respect to the runway.
1.b.7	Rejected take-off	1. Test should record at near maximum take-off weight. 2. Speed for reject should be at least 80 % of V_1. 3. Auto-brakes will be used as applicable. 4. Test should use maximum braking effort, auto or manual. 5. If a maximum braking demonstration is not available, an acceptable alternative is a test using approximately 80 % braking and full reverse, if applicable. 6. Time and distance should be recorded from brake release to full stop. 7. For feature GND at fidelity level G or R, test should record time for at least 80 % of the time segment from initiation of the rejected take-off to full stop.
1.b.8	Dynamic engine failure after take-	1. Engine failure speed should be within ± 3 kt of the validation data. 2. Engine failure may be a snap deceleration to idle.

1.b: Performance take-off		Test requirements
	off	3. Test should record hands off from five seconds before engine failure to five seconds after engine failure or 30° bank, whichever occurs first. 4. Note: For safety considerations, aeroplane flight test may be performed out-of-ground effect at a safe altitude, but with correct aeroplane configuration and airspeed. 5. For CCA, test should be in normal and non-normal control states.

CS FSTD.QTG.A.110 Performance climb – Aeroplane

This CS provides the standards for the objective tests for aeroplane climb performance.

1.c: Performance climb		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
1.c.1	Normal climb, all engines operating	Clean	Airspeed	OGE	Init: CT&M Rec: EM	Init: ± 3 kt Rec: EM	Init: ± 3 kt Rec: EM
			Rate of climb	OGE	Init: CT&M Rec: EM	Init: ± 5 % or ± 0.5 m/s (100 ft/min) Rec: EM	Init: ± 5 % or ± 0.5 m/s (100 ft/min) Rec: EM
1.c.2	One-engine-inoperative second-segment climb	Second-segment climb	Airspeed	OGE	Init: CT&M Rec: EM	Init: ± 3 kt Rec: EM	Init: ± 3 kt Rec: EM
			Rate of climb	OGE	Init: CT&M Rec: EM	Init: ± 5 % or ± 0.5 m/s (100 ft/min), but not less than aeroplane performance data requirements Rec: EM	Init: ± 5 % or ± 0.5 m/s (100 ft/min), but not less than aeroplane performance data requirements Rec: EM
1.c.3	One-engine-inoperative en route climb	Clean	Time	OGE		Init: ± 10 % Rec: EM	Init: ± 10 % Rec: EM
			Distance	OGE		Init: ± 10 % Rec: EM	Init: ± 10 % Rec: EM

1.c: Performance climb		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
			Fuel used	OGE		Init: $\pm 10\%$ Rec: EM	Init: $\pm 10\%$ Rec: EM
1.c.4	One-engine-inoperative approach climb (for aeroplanes with icing accountability) (if provided in the aeroplane performance data for this phase of flight)	Approach	Airspeed	OGE		Init: ± 3 kt Rec: EM	Init: ± 3 kt Rec: EM
			Rate of climb	OGE		Init: $\pm 5\%$ or ± 0.5 m/s (100 ft/min), but not less than aeroplane performance data requirements Rec: EM	Init: $\pm 5\%$ or ± 0.5 m/s (100 ft/min), but not less than aeroplane performance data requirements Rec: EM

Individual test requirements. The individual test requirements defined in the table below are applicable to all fidelity levels unless otherwise stated.

1.c: Performance climb		Test requirements
1.c.1	Normal climb, all engines operating	- Flight test data is preferred; however, aeroplane performance manual data is an acceptable alternative. - Test should record at nominal climb speed and mid-initial-climb altitude. - FSTD performance is to be recorded over an interval of at least 300 m (1 000 ft). - For feature OGE at fidelity level G, this test may be a snapshot test.
1.c.2	One-engine-inoperative second-segment climb	- Flight test data is preferred; however, aeroplane performance manual data is an acceptable alternative. - Test should record at nominal climb speed. - FSTD performance is to be recorded over an interval of at least 300 m (1 000 ft). - Test should be at weight, altitude or temperature (WAT) limiting condition. - Test should be performed in a manner similar to that for which a pilot is trained to trim an engine failure condition.

1.c: Performance climb		Test requirements
		6. For feature OGE at fidelity level G, this test may be a snapshot test.
1.c.3	One-engine-inoperative en route climb	1. Flight test data or aeroplane performance manual data may be used. 2. Test should be for at least a 1 550 m (5 000 ft) segment.
1.c.4	One-engine-inoperative approach climb (for aeroplanes with icing accountability) (if provided in the aeroplane performance data for this phase of flight)	1. Flight test data or aeroplane performance manual data may be used. 2. FSTD performance is to be recorded over an interval of at least 300 m (1 000 ft). 3. Test should be at near maximum certified landing weight, as may be applicable to an approach in icing conditions. 4. Aeroplane should be configured with all anti-icing and de-icing systems operating normally, gear up and go-around flap. 5. All icing accountability considerations, in accordance with the aeroplane performance data for an approach in icing conditions, should be applied.

CS FSTD.QTG.A.115 Performance cruise and descent – Aeroplane

This CS provides the standards for the objective tests for aeroplane cruise and descent performance.

1.d: Performance cruise and descent		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
1.d.1	Level flight acceleration	Cruise	Time	OGE	Init: CT&M Rec: EM	Init: $\pm 5\%$ Rec: EM	Init: $\pm 5\%$ Rec: EM
1.d.2	Level flight deceleration	Cruise	Time	OGE	Init: CT&M Rec: EM	Init: $\pm 5\%$ Rec: EM	Init: $\pm 5\%$ Rec: EM
1.d.3	Cruise performance	Cruise	Critical engine parameter	OGE		Init: $\pm 3\%$ Rec: EM	Init: $\pm 3\%$ Rec: EM
			Fuel flow	OGE		Init: $\pm 5\%$ Rec: EM	Init: $\pm 5\%$ Rec: EM
1.d.4	Idle descent	Clean	Airspeed	OGE	Init: CT&M Rec: EM	Init: ± 3 kt Rec: EM	Init: ± 3 kt Rec: EM
			Rate of descent	OGE	Init: CT&M Rec: EM	Init: $\pm 5\%$ or ± 1.0 m/s (200 ft/min) Rec: EM	Init: $\pm 5\%$ or ± 1.0 m/s (200 ft/min) Rec: EM
1.d.5	Emergency descent	As per aeroplane performance data	Airspeed	OGE		Init: ± 5 kt Rec: EM	Init: ± 5 kt Rec: EM
			Rate of descent	OGE		Init: $\pm 5\%$ or ± 1.5 m/s (300 ft/min) Rec: EM	Init: $\pm 5\%$ or ± 1.5 m/s (300 ft/min) Rec: EM

Individual test requirements. The individual test requirements defined in the table below are applicable to all fidelity levels unless otherwise stated.

1.d: Performance cruise and descent		Test requirements
1.d.1	Level flight acceleration	1. Time required to increase airspeed by a minimum of 50 kt, using maximum continuous thrust rating or equivalent, should be tested. 2. For aeroplanes with a small operating speed range, speed change may be reduced to 80 % of operational speed range.
1.d.2	Level flight deceleration	1. Time required to decrease speed a minimum of 50 kt, using idle power, should be tested. 2. For aeroplanes with a small operating speed range, speed change may be reduced to 80 % of operational speed range.
1.d.3	Cruise performance	1. The test may be a single snapshot showing instantaneous fuel flow or a minimum of two consecutive snapshots with a spread of at least three minutes in steady flight. 2. In the case of propeller-driven aircraft, the 'critical engine parameter' tolerance applies to both the engine parameter and propeller RPM.
1.d.4	Idle descent	1. Idle power stabilised descent at normal descent speed at mid altitude should be tested. 2. FSTD performance is to be recorded over an interval of at least 300 m (1 000 ft).
1.d.5	Emergency descent	1. Stabilised descent is to be conducted with speed brakes extended, if applicable, at mid altitude and near M_{MO}/V_{MO} or in accordance with the emergency descent procedure. 2. FSTD performance is to be recorded over an interval of at least 900 m (3 000 ft).

CS FSTD.QTG.A.120 Performance stopping – Aeroplane

This CS provides the standards for the objective tests for aeroplane stopping performance.

1.e: Performance stopping		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
1.e.1	Deceleration time and distance, manual wheel brakes, dry runway, no reverse thrust	Landing	Time	GND	Init: CT&M Rec: EM	Init: $\pm 5\%$ or ± 1.5 s time Rec: EM	Init: $\pm 5\%$ or ± 1.5 s time Rec: EM
			Distance	GND	Init: CT&M Rec: EM	Init: For distances up to 1 220 m (4 000 ft), ± 61 m (200 ft) or $\pm 10\%$, whichever is the smaller For distances greater than 1 220 m (4 000 ft), $\pm 5\%$ distance Rec: EM	Init: For distances up to 1 220 m (4 000 ft), ± 61 m (200 ft) or $\pm 10\%$, whichever is the smaller For distances greater than 1 220 m (4 000 ft), $\pm 5\%$ distance Rec: EM
1.e.2	Deceleration time and distance, reverse thrust, no	Landing	Time	GND		Init: $\pm 5\%$ or ± 1.5 s time Rec: EM	Init: $\pm 5\%$ or ± 1.5 s time Rec: EM

1.e: Performance stopping		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
	wheel brakes, dry runway		Distance	GND		Init: smaller of $\pm 10\%$ or ± 61 m (200 ft) Rec: EM	Init: smaller of $\pm 10\%$ or ± 61 m (200 ft) Rec: EM
1.e.3	Stopping distance, wheel brakes, wet runway	Landing	Distance	GND			Init: $\pm 10\%$ or ± 61 m (200 ft) Rec: EM
1.e.4	Stopping distance, wheel brakes, icy runway	Landing	Distance	GND			Init: $\pm 10\%$ or ± 61 m (200 ft) Rec: EM

Individual test requirements. The individual test requirements defined in the table below are applicable to all fidelity levels unless otherwise stated.

1.e: Performance stopping		Test requirements
1.e.1	Deceleration time and distance, manual wheel brakes, dry runway, no reverse thrust	- Time and distance should be recorded for at least 80 % of the total time from touchdown to full stop. - Position of ground spoilers and brake system pressure should be plotted, if applicable. - Tests are required for medium and near maximum certified landing weight. - Engineering validation data may be used for the medium weight condition.
1.e.2	Deceleration time and distance, reverse thrust, no wheel brakes, dry runway	- Time and distance should be recorded for at least 80 % of the total time from initiation of reverse thrust to full thrust reverser minimum operating speed. - Position of ground spoilers should be plotted, if applicable. - Tests are required for medium and near maximum certified landing weight. - Engineering validation data may be used for the medium weight condition.
1.e.3	Stopping distance,	Either flight test or manufacturer's performance manual validation data should be used, if available.

1.e: Performance stopping		Test requirements
	wheel brakes, wet runway	2. Engineering validation data, based on dry runway flight test stopping distance and the effects of contaminated runway braking coefficients, is an acceptable alternative.
1.e.4	Stopping distance, wheel brakes, icy runway	1. Either flight test or manufacturer's performance manual validation data should be used, if available. 2. Engineering validation data, based on dry runway flight test stopping distance and the effects of contaminated runway braking coefficients, is an acceptable alternative.

CS FSTD.QTG.A.125 Performance engines – Aeroplane

This CS provides the standards for the objective tests for aeroplane engine performance.

1.f: Performance engines		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
1.f.1	Engine acceleration	Either approach or landing	T _i and T _t	GND, IGE, OGE	Init: CT&M Rec: EM	Init: ±10 % T _i or ±0.25 s and ±10 % T _t or ±0.25 s Rec: EM	Init: ±10 % T _i or ±0.25 s and ±10 % T _t or ±0.25 s Rec: EM
1.f.2	Engine deceleration	Ground	T _i and T _t	GND, IGE, OGE	Init: CT&M Rec: EM	Init: ±10 % T _i or ±0.25 s and ±10 % T _t or ±0.25 s Rec: EM	Init: ±10 % T _i or ±0.25 s and ±10 % T _t or ±0.25 s Rec: EM

Individual test requirements. The individual test requirements defined in the table below are applicable to all fidelity levels unless otherwise stated.

1.f: Performance engines		Test requirements
1.f.1	Engine acceleration	1. Test should plot from flight idle to go-around power for a rapid throttle movement.

1.f: Performance engines		Test requirements
		2. T_i = Total time from initial throttle movement until a critical engine parameter reaches 10 % of its total response above idle power. 3. T_t = Total time from initial throttle movement until a critical engine parameter reaches 90 % of its total response above idle power. 4. Total response is the incremental change in the critical engine parameter from idle power to go-around power. 5. Critical engine parameter should be a measure of power (N1, N2, EPR, etc.).
1.f.2	Engine deceleration	1. Test should plot from maximum take-off power to idle for a rapid throttle movement. 2. T_i = Total time from initial throttle movement until a critical engine parameter reaches 10 % of its total response below maximum take-off power. 3. T_t = Total time from initial throttle movement until a critical engine parameter reaches 90 % of its total response below maximum take-off power. 4. Total response is the incremental change in the critical engine parameter from maximum take-off power to idle power. 5. Critical engine parameter should be a measure of power (N1, N2, EPR, etc.).

GM1 FSTD.QTG.A.125 Performance engines – Aeroplane

This GM provides guidance on the assessment of engine performance parameters.

Tests are required to show the response of the critical engine parameter to a rapid throttle movement for an engine acceleration and an engine deceleration. The procedure for evaluating the response is shown in Figures 1 and 2.

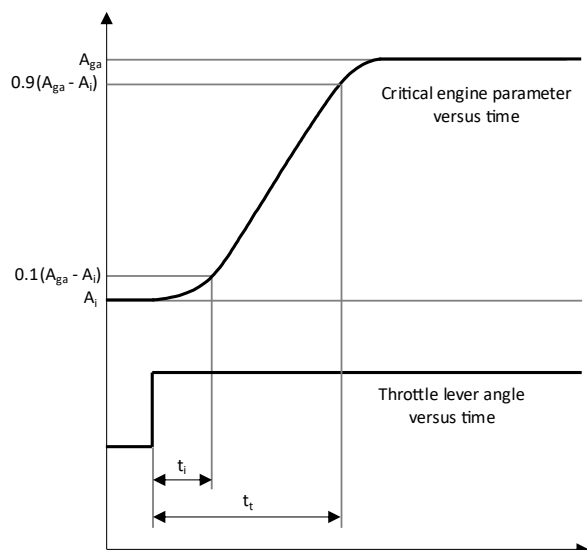


Figure 1. Engine acceleration

Legend to Figure 1:

A_{ga} = critical engine parameter at go-around power

A_i = critical engine parameter at idle power

t_i = total time from initial throttle movement until a critical engine parameter reaches 10 % of its total response above idle power

t_t = total time from initial throttle movement until a critical engine parameter reaches 90 % of its total response above idle power

Total response is the incremental change in the critical engine parameter from idle power to go-around power.

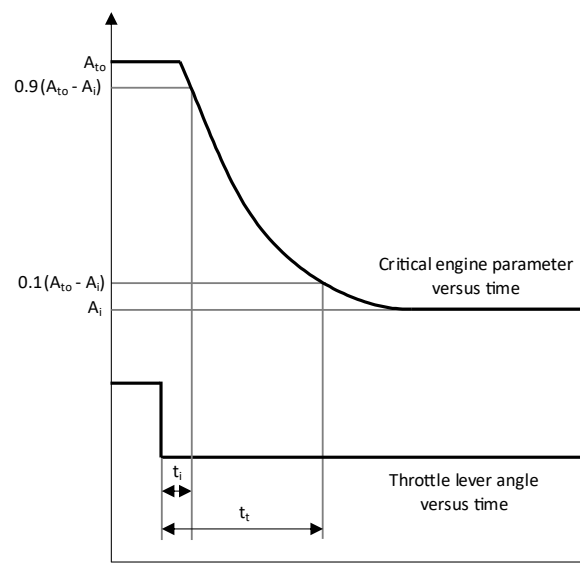


Figure 2. Engine deceleration

Legend to Figure 2:

A_{to} = critical engine parameter at maximum take-off power

A_i = critical engine parameter at idle power

t_i = total time from initial throttle movement until a critical engine parameter reaches 10 % of its total response below maximum take-off power

t_t = total time from initial throttle movement until a critical engine parameter reaches 90 % of its total response below maximum take-off power

Total response is the incremental change in the critical engine parameter from maximum take-off power to idle power.

CS FSTD.QTG.A.200 Static control checks – Aeroplane

This CS provides the standards for the objective tests for aeroplane static control checks.

2.a: Handling qualities static control checks		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
2.a.1	Pitch controller position versus force and surface position calibration	Ground	Breakout force	CLH	Init: CT&M Rec: ± 0.9 daN (2 lbf)	Init: CT&M Rec: ± 0.9 daN (2 lbf)	Init: ± 0.9 daN (2 lbf) Rec: Same as Init
		Ground For aeroplanes with reversible flight controls: ground and approach	Force	CLH	Init: CT&M Rec: ± 2.2 daN (5 lbf) or ± 10 %	Init: CT&M Rec: ± 2.2 daN (5 lbf) or ± 10 %	Init: ± 2.2 daN (5 lbf) or ± 10 % Rec: Same as Init
			Elevator angle	CLO		Init: CT&M Rec: EM	Init: $\pm 2^\circ$ Rec: EM
2.a.2	Roll controller position versus force and surface position calibration	Ground	Breakout force	CLH	Init: CT&M Rec: ± 0.9 daN (2 lbf)	Init: CT&M Rec: ± 0.9 daN (2 lbf)	Init: ± 0.9 daN (2 lbf) Rec: Same as Init
		Ground For aeroplanes with reversible flight controls: ground and approach	Force	CLH	Init: CT&M Rec: ± 1.3 daN (3 lbf) or ± 10 %	Init: CT&M Rec: ± 1.3 daN (3 lbf) or ± 10 %	Init: ± 1.3 daN (3 lbf) or ± 10 % Rec: Same as Init
			Aileron angle	CLO		Init: CT&M Rec: EM	Init: $\pm 2^\circ$ Rec: EM
			Spoiler angle	CLO		Init: CT&M Rec: EM	Init: $\pm 3^\circ$ Rec: EM

2.a: Handling qualities static control checks		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
2.a.3	Rudder pedal position versus force and surface position calibration	Ground	Breakout force	CLH	Init: CT&M Rec: ± 0.9 daN (2 lbf)	Init: CT&M Rec: ± 0.9 daN (2 lbf)	Init: ± 2.2 daN (5 lbf) Rec: Same as Init
		Ground For aeroplanes with reversible flight controls: ground and approach	Force	CLH	Init: CT&M Rec: ± 2.2 daN (5 lbf) or ± 10 %	Init: CT&M Rec: ± 2.2 daN (5 lbf) or ± 10 %	Init: ± 2.2 daN (5 lbf) or ± 10 % Rec: Same as Init
			Rudder angle	CLO		Init: CT&M Rec: EM	Init: $\pm 2^\circ$ Rec: EM
2.a.4	Nose wheel steering controller force and position calibration	Ground	Breakout force	CLH		Init: CT&M Rec: ± 0.9 daN (2 lbf)	Init: ± 0.9 daN (2 lbf) Rec: Same as Init
			Force	CLH		Init: CT&M Rec: ± 1.3 daN (3 lbf) or ± 10 %	Init: ± 1.3 daN (3 lbf) or ± 10 % Rec: Same as Init
			Nose wheel angle	CLO		Init: CT&M Rec: EM	Init: $\pm 2^\circ$ Rec: EM
2.a.5	Rudder pedal steering calibration	Ground	Nose wheel angle	CLO		Init: CT&M Rec: EM	Init: $\pm 2^\circ$ Rec: EM
2.a.6	Pitch trim versus surface position calibration	Ground	Trim angle	CLH	Init: CT&M Rec: $\pm 1^\circ$	Init: CT&M Rec: $\pm 1^\circ$	Init: $\pm 0.5^\circ$ Rec: Same as Init
2.a.7	Pitch trim rate	Ground and approach	Trim rate	CLO	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: ± 10 % or $\pm 0.1^\circ/\text{s}$ Rec: EM

2.a: Handling qualities static control checks		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
2.a.8	Alignment of flight deck throttle lever versus selected engine parameter	Ground	When matching engine parameters				
			TLA	CLH	Init: CT&M Rec: ±5 %	Init: CT&M Rec: ±5 %	Init: ±5° or ±5 % of travel Rec: Same as Init
			When matching detents / lever positions				
			Critical engine parameter	GND, IGE, OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: ±3 % Rec: EM
2.a.9	Brake pedal position versus force and brake system pressure calibration	Ground	Force	CLH		Init: CT&M Rec: ±2.2 daN (5 lbf) or ±10 %	Init: ±2.2 daN (5 lbf) or ±10 % Rec: Same as Init
			Brake system pressure	CLO		Init: CT&M Rec: ±1.0 MPa (150 psi) or ±10 %	Init: ±1.0 MPa (150 psi) or ±10 % Rec: EM
2.a.10	Stick pusher system force calibration (if applicable)	Either ground or flight	Stick/column transient force	CLH			Init: ±2.2 daN (5 lbf) or ±10 % Rec: Same as Init

Requirements for all the aeroplane static control checks.

2.a: Handling qualities static control checks	Test requirements
All static control checks	1. Plots of pitch, roll and yaw controller position versus force should be measured at the control. a. An alternative method in lieu of external test fixtures at the flight controls is to have recording and measuring instrumentation built into the FSTD. The force and position data from this instrumentation should be recorded and matched to the validation data.

2.a: Handling qualities static control checks	Test requirements
	b. Provided the instrumentation is verified by using external measuring equipment while conducting the static control checks, or verified using equivalent means, and that evidence of the satisfactory comparison is included in the QTG, the instrumentation can be used for both initial and recurrent testing for the measurement of all required control checks. c. Verification of the instrumentation using external measuring equipment should be repeated if major modifications or repairs are made to the control loading system. Such a permanent installation can be used without any time being lost for the installation of external devices. 2. Static and dynamic flight control checks should be accomplished at the same feel or impact pressures, as applicable. 3. Pilot control positions should be cross-plotted against control force and surface positions (e.g. elevator, nose wheel angle) or system parameters (e.g. brake pressure). 4. Static control checks from the second set of pilot controls are only required if both sets of controls are not mechanically interconnected in the FSTD. A rationale is required from the validation data provider if a single set of data is applicable to both sets of controls. If the controls are mechanically interconnected in the FSTD, a single set of checks are sufficient. 5. The static control checks must be conducted manually if the FSTD control positions are not backdriven by the control loading. 6. For irreversible flight control systems, if forces are generated solely by use of aircraft hardware in the FSTD, then testing of position versus force is not applicable.

Individual test requirements. The individual test requirements defined in the table below are applicable to all fidelity levels unless otherwise stated.

2.a: Handling qualities static control checks	Test requirements
2.a.1 Pitch controller position versus force and surface position calibration	1. The ground test should be an uninterrupted control sweep to stops over approximately 100 seconds. Test results should be validated (if possible) using in-flight data from tests such as longitudinal static stability and stalls. 2. For aeroplanes with reversible flight controls: a. an additional test in the approach test condition should be provided to validate aerodynamic effects on flight controls;

2.a: Handling qualities static control checks		Test requirements
		b. for feature CLH at fidelity level G: i. the approach test should be an uninterrupted control sweep to stops; ii. the test may be a footprint test run with some FSTD freeze functions active to keep the airspeed and altitude constant; iii. the test should be run in a suitable airspeed condition; c. for feature CLH at fidelity level R or S, a partial control sweep is acceptable and may be validated with controller position, controller force and surface position from the longitudinal static stability test (CS FSTD.QTG.A.210, test 2.c.7).
2.a.2	Roll controller position versus force and surface position calibration	1. The ground test should be an uninterrupted control sweep to stops over approximately 100 seconds. Test results should be validated (if possible) using in-flight data from tests such as engine-out trims and steady-state side-slips. 2. For aeroplanes with reversible flight controls: a. an additional test in the approach test condition should be provided to validate aerodynamic effects on flight controls; b. for feature CLH at fidelity level G: i. the approach test should be an uninterrupted control sweep to stops; ii. the test may be a footprint test run with some FSTD freeze functions active to keep the airspeed and altitude constant; iii. the test should be run in a suitable airspeed condition; c. for feature CLH at fidelity level R or S, a partial control sweep is acceptable and may be validated with controller position, controller force and surface position from the steady-state side-slip test (CS FSTD.QTG.A.215, test 2.d.8) and controller position and surface position from the engine-inoperative trim test (CS FSTD.QTG.A.215, test 2.d.5).
2.a.3	Rudder pedal position versus force and surface position calibration	1. The ground test should be an uninterrupted control sweep to stops over approximately 100 seconds. Test results should be validated (if possible) using in-flight data from tests such as engine-out trims and steady-state side-slips. 2. For aeroplanes with reversible flight controls: a. an additional test in the approach test condition should be provided to validate aerodynamic effects on flight controls;

2.a: Handling qualities static control checks		Test requirements
		b. for feature CLH at fidelity level G: i. the approach test should be an uninterrupted control sweep to stops; ii. the test may be a footprint test run with some FSTD freeze functions active to keep the airspeed and altitude constant; iii. the test should be run in a suitable airspeed condition; c. for feature CLH at fidelity level R or S, a partial control sweep is acceptable and may be validated with controller position, controller force and surface position from the steady-state side-slip test (CS FSTD.QTG.A.215, test 2.d.8) and controller position and surface position from the engine-inoperative trim test (CS FSTD.QTG.A.215, test 2.d.5).
2.a.4	Nose wheel steering controller force and position calibration	1. Test should be an uninterrupted control sweep to stops.
2.a.5	Rudder pedal steering calibration	1. Test should be an uninterrupted control sweep to stops.
2.a.6	Pitch trim versus surface position calibration	1. The purpose of this test is to compare the FSTD surface position indicator and the FSTD flight control model's computed value against the validation data (e.g. design data or equivalent).
2.a.7	Pitch trim rate	1. Trim rate is to be checked at pilot primary induced trim rate and autopilot or pilot primary trim rate in flight at go-around flight conditions. 2. Test is applicable to simulated aeroplanes with electric trim systems.
2.a.8	Alignment of flight deck throttle lever versus selected engine parameter	1. Simultaneous recording for all engines should take place. 2. The test procedure should be either to carefully match the lever positions and check the resulting engine parameters or to match the engine parameters and check the resulting lever positions. 3. For aeroplanes with throttle lever detents, all detents are to be presented and at least one position between detents / end points (if practical) given. For aeroplanes without detents, end points and at least three other positions are to be presented. For aeroplanes with engine reversers, only forward throttle lever positions should be considered. 4. Validation data from a test aeroplane or engineering test bench is acceptable, provided that the same engine

2.a: Handling qualities static control checks		Test requirements
		controller (both hardware and software) is used. - In the case of propeller-driven aircraft, the 'critical engine parameter' tolerance applies to both the engine parameter and propeller RPM. - In the case of propeller-driven aeroplanes, if an additional lever, usually referred to as the propeller lever, is present, it should also be checked to ensure that it is within the same tolerances as the TLA. - This test may be a series of snapshot tests.
2.a.9	Brake pedal position versus force and brake system pressure calibration	- Both left and right pedals from the same pilot seat should be validated. - Test should be an uninterrupted control sweep to stops.
2.a.10	Stick pusher system force calibration (if applicable)	- This test is applicable only to aeroplane FSTDs that are to be qualified for full-stall training tasks. - This test is intended to validate the stick/column transient force resulting from a stick pusher system activation. - This test may be conducted in an on-ground condition through stimulation of the stall protection system in a manner that generates a stick pusher response characteristic of an in-flight condition. - Aeroplane manufacturer design data may be utilised as validation data, if acceptable to the competent authority. - The test requirements may be met through column force validation in conjunction with the full-stall characteristics test (CS FSTD.QTG.A.210, test 2.c.8.a).

GM1 FSTD.QTG.A.200 Static control checks – Aeroplane

All tests related to control system mechanical characteristics should be performed in an integrated manner, so that:

- the measured controls positions and forces match what the crew feels when hands on the control;
- significant changes in the control mechanical characteristics (e.g. increased mechanical friction) or to the control software are reflected in the results of the tests when run in manual and automatic modes.

Test fixture at the control (similar test installation as the one used to measure data on the reference aircraft) is generally the most comprehensive way for measuring flight control mechanical characteristics. However, it may not be efficient or practical to install such an external test bench every time that running a test is required. Therefore, a permanent instrumentation of the FSTD that provides equivalent results to an external test fixture may be suitable.

It should be justified that the FSTD instrumentation is comparable to an external test fixture. This can be achieved by providing a calibration test for each axis that was run concurrently with the external and internal instrumentation; the calibration test of the internal instrumentation should be performed in both automatic and manual modes. This comparison should be performed prior to the initial evaluation and if impacted by a modification of the system.

If the test is run in automatic mode, the system should take into account both the calculated forces and the measured mechanical forces when moving the controls. Any change in friction, inertia or the presence of an obstacle should be visible in the test results. It should be possible to demonstrate this safely by holding the controls during an automatic test: the test results should show an increase in the forces if the controls are held.

In manual mode, the FSTD instrumentation is generally not capable of measuring the mechanical friction introduced between the control grip and the force sensor. Therefore, the manual test mode is useful for assessing the validity of the internal instrumentation but may not be sufficient to detect all changes in control loads.

If there are control boots, in order to measure consistent results, it is acceptable to either remove them or loosen them.

CS FSTD.QTG.A.205 Dynamic control checks – Aeroplane

This CS provides the standards for the objective tests for aeroplane dynamic control checks.

2.b: Handling qualities dynamic control checks		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances			
					G	R	S	
2.b.1	Control dynamic pitch – dynamic releases	Take-off, cruise and landing	For underdamped systems					
			P ₀ : time from 90 % of total initial displacement of pilot controller (A _d) to first zero crossing	CLH			Init: ±10 % of P ₀ or ±0.05 s Rec: Same as Init	
			P ₁ : period of the first oscillation	CLH			Init: ±20 % of P ₁ or ±0.05 s Rec: Same as Init	
			P ₂ : period of the second oscillation	CLH			Init: ±30 % of P ₂ or ±0.05 s Rec: Same as Init	
			P _n : period of the <i>n</i> th oscillation	CLH			Init: ±10(<i>n</i> + 1) % of P _n or ±0.05 s Rec: Same as Init	

2.b: Handling qualities dynamic control checks		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
			Sequential amplitude of overshoot after initial x-axis crossing (A_n): amplitude of the n th significant overshoot (outside the residual band)	CLH			Init: $\pm 10\%$ of $A_{\max}$, where $A_{\max}$ is the largest amplitude or $\pm 0.5\%$ of the total control travel (stop to stop) Rec: Same as Init
			Number of significant overshoots	CLH			Init: ± 1 significant overshoots (a minimum of one significant overshoot) Rec: Same as Init
			For overdamped and critically damped systems only				
			P_0 : time from 90 % of A_d to 10 % of A_d	CLH			Init: $\pm 10\%$ of P_0 or ± 0.05 s Rec: Same as Init

2.b: Handling qualities dynamic control checks		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
	Control dynamic pitch – sweeps	Ground and flight Slow sweep and fast sweep	Force	CLH		Init: ± 0.9 daN (2 lbf) or ± 10 % on the dynamic increment above static test Rec: Same as Init	
2.b.2	Control dynamic roll – dynamic releases	Take-off, cruise and landing	Same as 2.b.1	CLH			Same as 2.b.1
	Control dynamic roll – sweeps	Ground and flight Slow sweep and fast sweep	Same as 2.b.1	CLH		Same as 2.b.1	
2.b.3	Control dynamic yaw – dynamic releases	Take-off, cruise and landing	Same as 2.b.1	CLH			Same as 2.b.1
	Control dynamic yaw – sweeps	Ground and flight Slow sweep and fast sweep	Same as 2.b.1	CLH		Same as 2.b.1	
2.b.4	Small control inputs – pitch	Either approach or	Body pitch rate	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 0.15^\circ/\text{s}$ Rec: EM

2.b: Handling qualities dynamic control checks		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
		landing CCA: test in normal and non-normal control states	Peak body pitch rate applied throughout the time history	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 20\%$ Rec: EM
2.b.5	Small control inputs – roll	Either approach or landing	Body roll rate	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 0.15^\circ/\text{s}$ Rec: EM
		CCA: test in normal and non-normal control states	Peak body roll rate applied throughout the time history	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 20\%$ Rec: EM
2.b.6	Small control inputs – yaw	Either approach or landing	Body yaw rate	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 0.15^\circ/\text{s}$ Rec: EM
		CCA: test in normal and non-normal control states	Peak body yaw rate applied throughout the time history	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 20\%$ Rec: EM

Individual test requirements. The individual test requirements defined in the table below are applicable to all fidelity levels unless otherwise stated.

2.b: Handling qualities dynamic control checks		Test requirements
2.b.1	Control dynamic pitch – dynamic releases	- Test is not applicable to FSTDs if the control forces are completely generated within the aeroplane controller unit installed in the FSTD. - Engine power setting may be that required for level flight unless otherwise specified. - Data should be for normal control displacements in both directions (approximately 25–50 % of full travel or approximately 25–50 % of maximum allowable pitch controller deflection for flight conditions limited by the manoeuvring load envelope). - Tolerances apply against the absolute values of each period (considered independently).

2.b: Handling qualities dynamic control checks		Test requirements
		- For test, n = the sequential period of a full oscillation. - Static and dynamic flight control tests should be accomplished at the same feel or impact pressures as in the real aircraft, as applicable.
	Control dynamic pitch – sweeps	- The method applies to the CLH feature at fidelity level R. Instead of free response measurements, the system is validated by measurements of control force and rate of movement. - The control should be forced to its maximum extreme position (for tests conducted on the ground) or to the maximum safe displacement (for tests conducted in flight) for the following distinct rates. The intent is to validate the flight control force damping characteristics: - for static test, refer to CS FSTD.QTG.A.200, test 2.a.1 ('Pitch controller position versus force and surface position calibration'); - for slow dynamic sweep, achieve a full sweep in approximately 10 seconds or the longest period that can be achieved safely for tests conducted in flight; - for fast dynamic sweep, achieve a full sweep in approximately 4 seconds. - Dynamic sweeps may be limited to forces not exceeding 44.5 daN (100 lbf).
2.b.2	Control dynamic roll – dynamic releases	- Test is not applicable to FSTDs if the control forces are completely generated within the aeroplane controller unit installed in the FSTD. - Engine power setting may be that required for level flight unless otherwise specified. - Data should be for normal control displacement (approximately 25–50 % of full travel or approximately 25–50 % of maximum allowable roll controller deflection for flight conditions limited by the manoeuvring load envelope). - Tolerances apply against the absolute values of each period (considered independently). - For test, n = the sequential period of a full oscillation. - Static and dynamic flight control tests should be accomplished at the same feel or impact pressures as in the real aircraft, as applicable

2.b: Handling qualities dynamic control checks		Test requirements
	Control dynamic roll – sweeps	1. The method applies to the CLH feature at fidelity level R. Instead of free response measurements, the system is validated by measurements of control force and rate of movement. 2. The control should be forced to its maximum extreme position (for tests conducted on the ground) or to the maximum safe displacement (for tests conducted in flight) for the following distinct rates. The intent is to validate the flight control force damping characteristics: a. for static test, refer to CS FSTD.QTG.A.200, test 2.a.2 ('Roll controller position versus force and surface position calibration'); b. for slow dynamic sweep, achieve a full sweep in approximately 10 seconds or the longest period that can be achieved safely for tests conducted in flight; c. for fast dynamic sweep, achieve a full sweep in approximately 4 seconds. 3. Dynamic sweeps may be limited to forces not exceeding 44.5 daN (100 lbf).
2.b.3	Control dynamic yaw – dynamic releases	1. Test is not applicable to FSTDs if the control forces are completely generated within the aeroplane controller unit installed in the FSTD. 2. Engine power setting may be that required for level flight unless otherwise specified. 3. Data should be for normal displacement (approximately 25–50 % of full travel). 4. Tolerances apply against the absolute values of each period (considered independently). 5. For test, n = the sequential period of a full oscillation. 6. Static and dynamic flight control tests should be accomplished at the same feel or impact pressures as in the real aircraft, as applicable.
	Control dynamic yaw – sweeps	1. The method applies to the CLH feature at fidelity level R. Instead of free response measurements, the system is validated by measurements of control force and rate of movement. 2. The control should be forced to its maximum extreme position (for tests conducted on the ground) or to the maximum safe displacement (for tests conducted in flight) for the following distinct rates. The intent is to validate the flight control force damping characteristics: a. for static test, refer to CS FSTD.QTG.A.200, test 2.a.3 ('Rudder pedal position versus force and surface position calibration');

2.b: Handling qualities dynamic control checks		Test requirements
		b. for slow dynamic sweep, achieve a full sweep in approximately 10 seconds or the longest period that can be achieved safely for tests conducted in flight; c. for fast dynamic sweep, achieve a full sweep in approximately 4 seconds. 3. Dynamic sweeps may be limited to forces not exceeding 44.5 daN (100 lbf).
2.b.4	Small control inputs – pitch	1. Control inputs should be typical of minor corrections made while established on an ILS approach (approximately 0.5°/s to 2°/s pitch rate). 2. Test should be in both directions. 3. Test should show time history data from five seconds before until at least five seconds after initiation of control input. 4. If a single test is used to demonstrate both directions, there should be a minimum of five seconds before control reversal to the opposite direction. 5. For CCA, test should be in normal and non-normal control states.
2.b.5	Small control inputs – roll	1. Control inputs should be typical of minor corrections made while established on an ILS approach (approximately 0.5°/s to 2°/s roll rate). 2. Test should be in one direction. 3. For aeroplanes that exhibit non-symmetrical behaviour, test should be in both directions. 4. Test should show time history data from five seconds before until at least five seconds after initiation of control input. 5. If a single test is used to demonstrate both directions, there should be a minimum of five seconds before control reversal to the opposite direction. 6. For CCA, test should be in normal and non-normal control states.
2.b.6	Small control inputs – yaw	1. Control inputs should be typical of minor corrections made while established on an ILS approach (approximately 0.5°/s to 2°/s yaw rate). 2. Test should be in both directions. 3. Test should show time history data from five seconds before until at least five seconds after initiation of control input. 4. If a single test is used to demonstrate both directions, there should be a minimum of five seconds before control reversal to the opposite direction. 5. For CCA, test should be in normal and non-normal control states.

GM1 FSTD.QTG.A.205 Dynamic control checks – Aeroplane

This GM provides guidance on the assessment of flight controls dynamic test parameters.

(a) General guidance is as follows.

The characteristics of an aircraft flight control system have a major effect on handling qualities. A significant consideration in the acceptability of an aircraft to a pilot is the ‘feel’ provided through the flight controls. Considerable effort is expended on aircraft feel system design so that pilots will be comfortable and will consider the aircraft desirable to fly. In order for an FSTD to be representative, it too should present the pilot with the proper feel – that is, that of the aircraft being simulated. Compliance with this requirement should be determined by comparing a recording of the control feel dynamics of the FSTD with actual aircraft measurements in the take-off, cruise and landing configurations.

- (1) Recordings such as free response to a pulse or step function are classically used to estimate the dynamic properties of electromechanical systems. In any case, the dynamic properties can only be estimated, since the true inputs and responses are also only estimated. Therefore, it is imperative that the best possible data be collected, since close matching of the FSTD control loading system to the aircraft systems is essential.
- (2) Control dynamic characteristics should be measured at and recorded directly from the flight controls. This procedure is usually accomplished by measuring the free response of the controls using a step input or pulse input to excite the system. The procedure should be accomplished in the take-off, cruise and landing flight conditions and configurations.
- (3) For aircraft with irreversible control systems, measurements may be obtained on the ground if the proper pitot–static inputs, if applicable, are provided to represent airspeeds typical of those encountered in flight. Likewise, it may be shown that, for some aircraft, take-off, cruise and landing configurations have similar effects. Thus, one configuration may suffice. If either or both of the considerations apply, a rationale should be submitted as a justification for ground tests or for the elimination of a configuration. For FSTDs requiring static and dynamic tests at the controls, special test fixtures should not be required during evaluations if the QTG shows both test fixture results and the results of an alternate approach, such as computer plots that were produced concurrently and show satisfactory agreement. Repeat of the alternate method during an evaluation would then satisfy this test requirement.

(b) Control dynamic assessment guidance is as follows.

The dynamic properties of control systems are often stated in terms of frequency, damping and a number of other classic measurements that can be found in the various documents available on control systems. In order to establish a consistent means of validating test results for FSTD control

loading, criteria are needed that clearly define the interpretation of the measurements and the tolerances to be applied. Criteria are needed for underdamped, critically damped and overdamped systems. In the case of an underdamped system with very light damping, the system may be quantified in terms of frequency and damping. In critically damped or overdamped systems, the frequency and damping are not readily measured from a response time history. Therefore, some other measurement should be used.

Tests to verify that control feel dynamics represent those of the aircraft should show that the dynamic damping cycles (free response of the controls) match those of the aircraft within specified tolerances. The method of evaluating the response and the tolerance to be applied is described for the underdamped and critically damped cases. The response is as follows.

- (1) For underdamped response, the following apply.
 - (i) Two measurements are required for the period: the time to first zero crossing (if a rate limit is present) and the subsequent frequency of oscillation. It is necessary to measure cycles on an individual basis in case there are non-uniform periods in the response. Each period should be independently compared with the relevant period of the aircraft control system and consequently should enjoy the full tolerance specified for that period.
 - (ii) The damping tolerance should be applied to overshoots on an individual basis. Care should be taken when applying the tolerance to small overshoots since the significance of such overshoots becomes questionable. Only those overshoots larger than 5 % of the A_d should be considered. The residual band, labelled ' $T(A_d)$ ' (tolerance applied to residual amplitude) in Figure 1, is ± 5 % of the initial displacement amplitude, A_d , from the steady-state value of the oscillation. Only oscillations outside the residual band are considered significant. If comparing FSTD data with aircraft data, the process should begin by overlaying or aligning the FSTD and aircraft steady-state values and then comparing amplitudes of oscillation peaks, times of the first zero crossings and individual periods of oscillation. The FSTD should show the same number of significant overshoots as in the aircraft data. This procedure for evaluating the response is shown in Figure 1.
- (2) For critically damped and overdamped response, the following applies.

Due to the nature of critically damped and overdamped responses (no overshoots), the time to travel from 90 % of the initial displacement to 10 % of the steady-state (neutral-point) value should be the same as that of the aircraft within ± 10 % or ± 0.05 s. Figure 2 shows the procedure.

(3) Special considerations are as follows.

Control systems that exhibit characteristics other than classical overdamped or underdamped responses should meet specified tolerances. In addition, special consideration should be given to ensuring that significant trends are maintained.

(c) Tolerance guidance is as follows.

The following summarises the values for tolerance applied to period ($T(P)$) (refer to Figures 1 and 2 for the referenced measurements):

$T(P_0) = \pm 10\% \text{ of } P_0 \text{ or } \pm 0.05 \text{ s};$

$T(P_1) = \pm 20\% \text{ of } P_1 \text{ or } \pm 0.05 \text{ s};$

$T(P_2) = \pm 30\% \text{ of } P_2 \text{ or } \pm 0.05 \text{ s};$

$T(P_n) = \pm 10(n + 1)\% \text{ of } P_n \text{ or } \pm 0.05 \text{ s};$

$T(A_n) = \pm 10\% \text{ of } A_{\max}$, where $A_{\max}$ is the largest amplitude or $\pm 0.5\%$ of the total control travel (stop to stop).

Where:

residual band $T(A_d) = \text{either } \pm 5\% \text{ of } A_d \text{ or } \pm 0.5\% \text{ of the maximum control travel};$

significant overshoots = ± 1 significant overshoots (a minimum of one significant overshoot);

steady-state position = within the residual band.

Note: Tolerances should not be applied to period or amplitude after the last significant overshoot.

Note: Oscillations within the residual band are not considered significant and are not subject to tolerances.

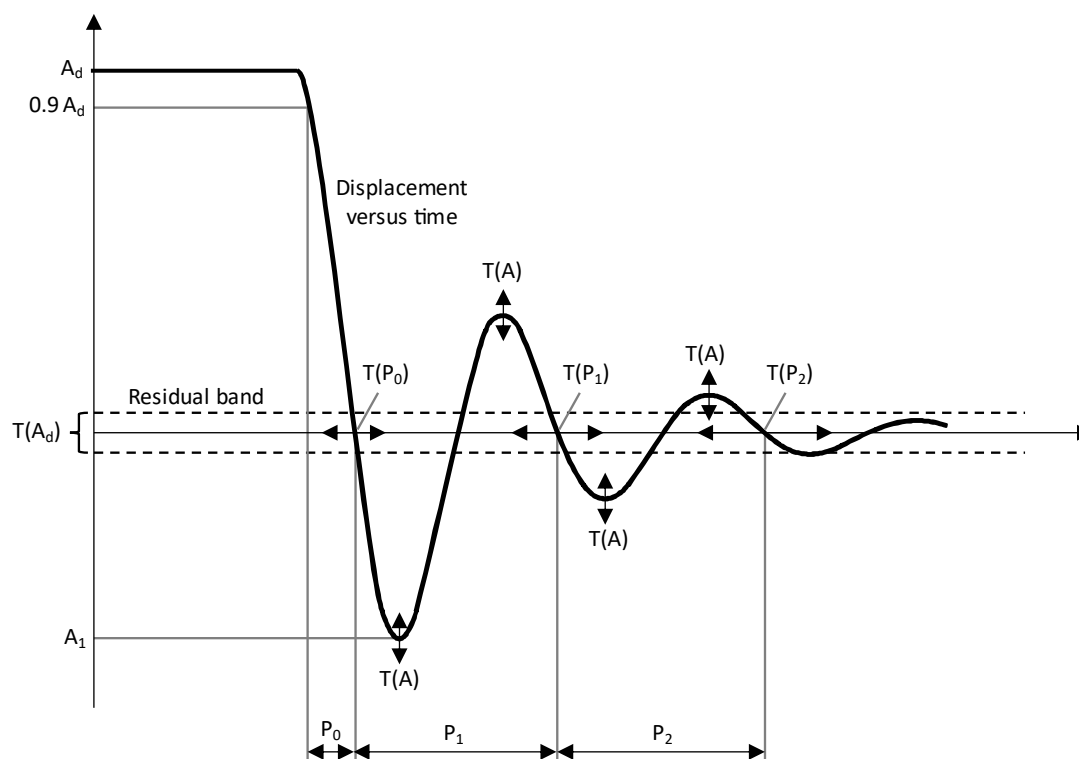


Figure 1. Underdamped step response

Legend to Figure 1:

A = amplitude

P = period

$T(A)$ = tolerance applied to amplitude ($0.1 A_1$)

$T(P)$ = tolerance applied to period (10 % of P_0 , $10(n + 1)$ % of P_n).

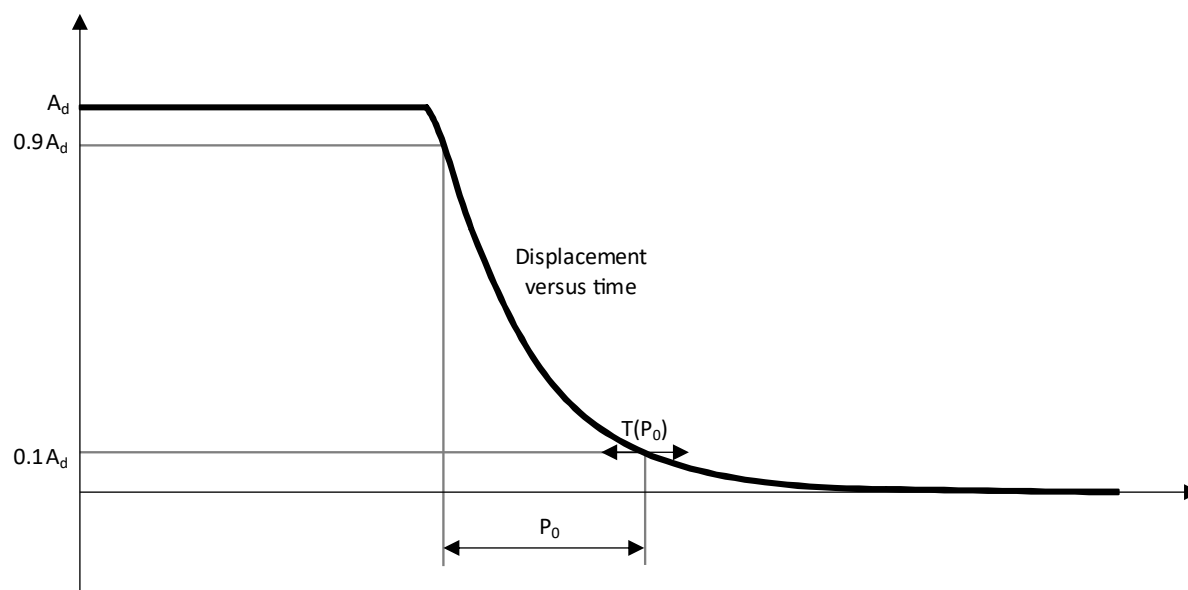


Figure 2. Critically damped step response

The following tolerance applies only to overdamped and critically damped systems (refer to Figure 2 above for the reference measurement):

$$T(P_0) = \pm 10 \% \text{ of } P_0 \text{ or } \pm 0.05 \text{ s.}$$

- (d) Alternate control dynamic test method guidance is as follows.

The method applies to aircraft with hydraulically powered flight controls and artificial feel systems or to FSTDs with the CLH feature at fidelity level R. Instead of free response measurements, the system would be validated by measurements of control force and rate of movement.

These tests should be conducted under typical flight and ground conditions. For each axis of pitch, roll and yaw, the control should be forced to its maximum extreme position for the following distinct rates:

- (1) for static test, slowly move the control such that approximately 100 seconds are required to achieve a full sweep (a full sweep is defined as the movement of the controller from neutral to the stop, usually aft or right stop, then to the opposite stop, then to the neutral position);

- (2) for slow dynamic test, achieve a full sweep in approximately 10 seconds (tolerance: ± 0.9 daN (2 lbf) or ± 10 % on the dynamic increment above static test);
- (3) for fast dynamic test, achieve a full sweep in approximately four seconds (tolerance: ± 0.9 daN (2 lbf) or ± 10 % on the dynamic increment above static test).

Note: Dynamic sweeps may be limited to forces not exceeding 44.5 daN (100 lbf).

The competent authority should consider alternative test methods such as the ones described above. Such alternatives should, however, be justified and appropriate to the application. For example, the method described here may not apply to all manufacturers' systems and certainly not to aircraft with reversible control systems. Hence, each case should be considered on its own merit on an ad hoc basis. Should the competent authority find that alternative methods do not result in satisfactory performance, then more conventionally accepted methods should be used.

(e) Alternate reference line assessment method guidance is as follows.

- (1) Background information on the method is provided below.
 - (i) If evaluating a flight control dynamic response, the periods, amplitudes and residual band are defined with respect to a reference line, which is the steady-state value of the control. This selection is made since it is assumed that the steady-state value is representative of the control's rest position throughout the test. For standard irreversible control systems, this is very often a valid assumption. However, in reversible control systems, for example, aerodynamic forces on the control surfaces influence the instantaneous rest position of the control. During the dynamic test, the control's rest position will vary in response to the variance of the flight conditions. In such a case, the instantaneous rest position and steady-state value at the end of the test are not equivalent. If the tolerances are applied to the entire dynamic response based on the steady-state value, they may become incorrect and lead to problems evaluating the cases.
 - (ii) In such cases, an alternate reference line that attempts to better approximate the true rest position of the control throughout a step response may be used. That reference line is obtained as described in paragraph (2) (on the alternate reference line) below.
 - (iii) The rest position is defined as the position in which the control would eventually settle if no pilot forces were applied to it (i.e. if it were left free). This position may or may not be affected by the aerodynamic conditions, the aircraft configuration and the acceleration to which it is subjected. It will depend on the type of flight control system in the aircraft. Typically, reversible control

systems will be affected while irreversible systems will not. The instantaneous rest position is defined as the theoretical rest position at a particular point in time and under the same conditions as at that moment.

- (2) The following method can be used to identify the alternate reference line.
 - (i) On the control position curve, identify the median points, defined as points on the control position curve located equidistantly between two consecutive peaks, measured vertically (refer to Figure 3). The last median point is the first point at which the dynamic portion of the response has ended, rather than the midpoint between the last peak and the end of the dynamic portion.
 - (ii) Join the median points to produce the 'line of medians'. Then, identify reference points, defined as the intersection of a vertical line passing through a position peak and the line of medians (refer to Figure 4).
 - (iii) The first reference point is the last control position before the start of the excitation. If data on this part is not available, project the first available reference point horizontally to time 0. The last reference point is simply the last median point.
 - (iv) Link all the reference points to obtain the alternate reference line (refer to Figure 4) and append the final non-dynamic portion to it.

- (3) The following applies to tolerances.

The final alternate reference line (refer to Figure 5) may be used to calculate the conventional tolerances. Note that the residual band $T(A_d)$ should be at a distance of $\pm 5\%$ of A_d or $\pm 0.5\%$ of the total control travel (stop to stop) from the alternate reference line. Its shape will therefore follow the alternate reference line.

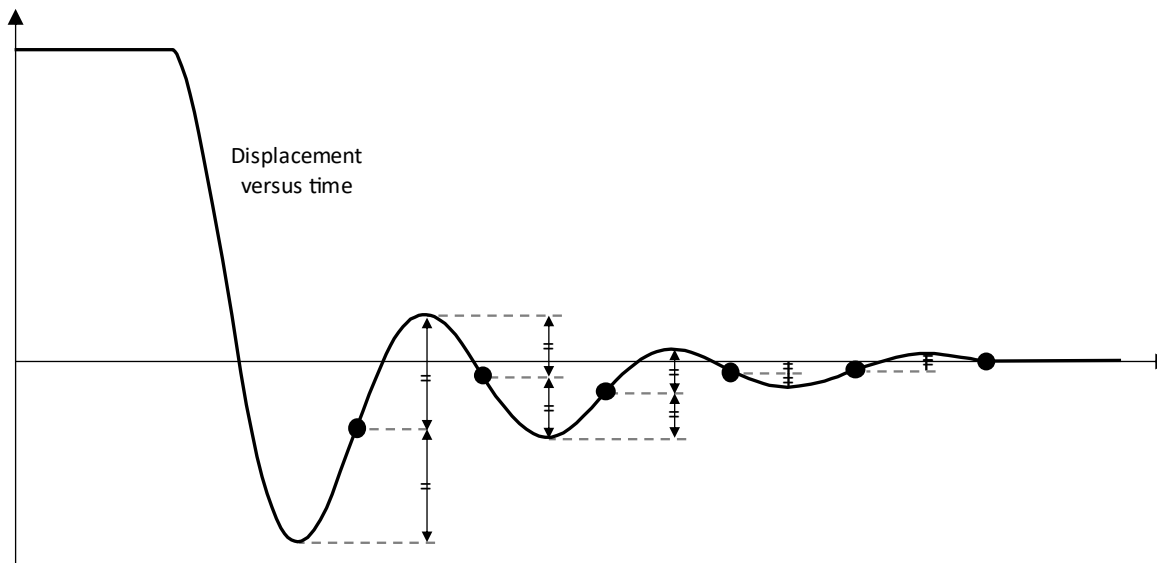


Figure 3. Locating median points

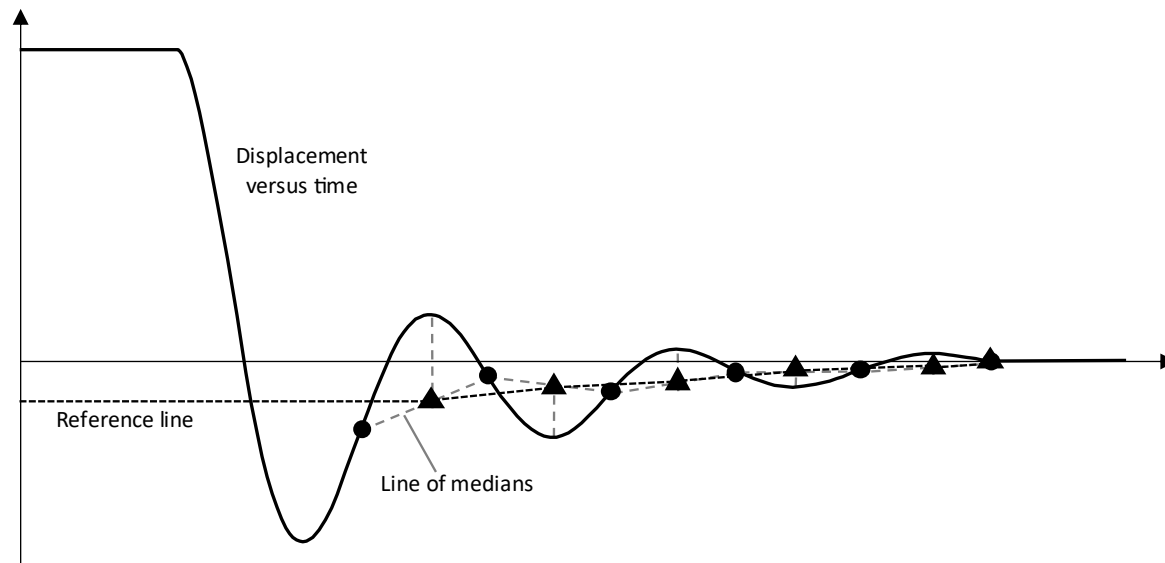


Figure 4. Producing the alternate reference line

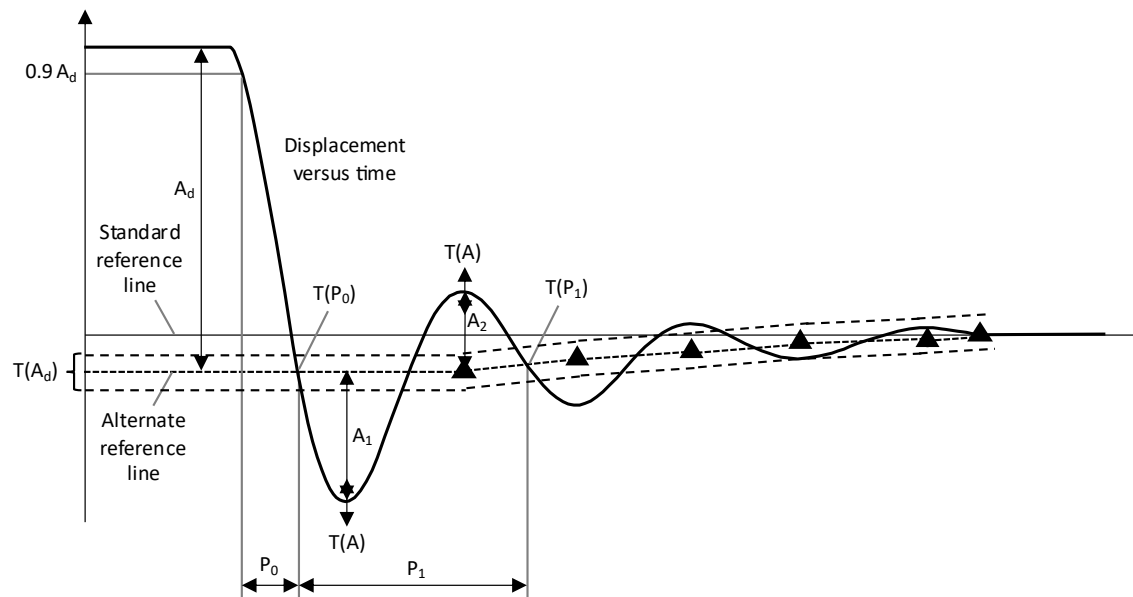


Figure 5. Tolerances applied using the alternate reference line

CS FSTD.QTG.A.210 Handling qualities longitudinal – Aeroplane

This CS provides the standards for the objective tests for aeroplane longitudinal handling qualities.

2.c: Handling qualities longitudinal		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
2.c.1	Power change dynamics	Approach CCA: test in normal and non-normal control states	Airspeed	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: ± 3 kt Rec: EM
			Altitude	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: ± 30 m (100 ft) Rec: EM
			Pitch angle	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 1.5^\circ$ or ± 20 % Rec: EM
			Alternative method for aeroplanes with the OGE feature at fidelity level G				
			Pitch controller force	OGE	Init: CT&M Rec: EM		
2.c.2	Flap change dynamics	Take-off through initial flap retraction, and approach to landing CCA: test in normal and non-normal control states	Airspeed	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: ± 3 kt Rec: EM
			Altitude	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: ± 30 m (100 ft) Rec: EM
			Pitch angle	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 1.5^\circ$ or ± 20 % Rec: EM
			Alternative method for aeroplanes with the OGE feature at fidelity level G				
			Pitch controller force	OGE	Init: CT&M Rec: EM		

2.c: Handling qualities longitudinal		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
2.c.3	Spoiler / speed brake change dynamics	Cruise CCA: test in normal and non-normal control states	Airspeed	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: ± 3 kt Rec: EM
			Altitude	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: ± 30 m (100 ft) Rec: EM
			Pitch angle	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 1.5^\circ$ or ± 20 % Rec: EM
2.c.4	Gear change dynamics	Take-off (retraction) and approach (extension) CCA: test in normal and non-normal control states	Airspeed	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: ± 3 kt Rec: EM
			Altitude	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: ± 30 m (100 ft) Rec: EM
			Pitch angle	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 1.5^\circ$ or ± 20 % Rec: EM
			Alternative method for aeroplanes with the OGE feature at fidelity level G				
			Pitch controller force	OGE	Init: CT&M Rec: EM		
2.c.5	Longitudinal trim	Cruise, approach and landing CCA: test in normal or non-normal control state, as applicable	Elevator angle	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 1^\circ$ Rec: EM
			Pitch trim angle	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 0.5^\circ$ Rec: EM
			Pitch angle	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 1^\circ$ Rec: EM

2.c: Handling qualities longitudinal		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
			Net thrust or equivalent	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 5\%$ Rec: EM
2.c.6	Longitudinal manoeuvring stability (stick force/g)	Cruise, approach and landing CCA: test in normal and non-normal control states, as applicable	Pitch controller force	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: ± 2.2 daN (5 lbf) or $\pm 10\%$ Rec: EM
			Alternative tolerances for aeroplanes that do not exhibit stick-force-per-g characteristics				
			Change of elevator angle	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 1^\circ$ or $\pm 10\%$ Rec: EM
2.c.7	Longitudinal static stability	Approach CCA: test in normal or non-normal control state	Pitch controller force	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: ± 2.2 daN (5 lbf) or $\pm 10\%$ Rec: EM
			Alternative tolerances for aeroplanes that do not exhibit speed stability characteristics				
			Change of elevator angle	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 1^\circ$ or $\pm 10\%$ Rec: EM
2.c.8.a	Full-stall characteristics	Second-segment climb, high-altitude cruise (near	Airspeed for stall warning and stall speed values	OGE			Init: ± 3 kt Rec: EM

2.c: Handling qualities longitudinal		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
		performance-limited condition), and either approach or landing Wings level, turning flight, and power-on stall entry methods CCA: for aeroplanes with stall envelope protection systems, test in normal and non-normal control states	Angle of attack for the buffet threshold of perception and for the initial buffet based on the component of the load factor in the aircraft z-axis direction	VIB			Init: $\pm 2^\circ$ Rec: EM
			Primary flight control inputs	CLO			Init: CT&M Rec: EM
			Approach to stall				
			Pitch angle	OGE			Init: $\pm 2.0^\circ$ Rec: EM
			Angle of attack	OGE			Init: $\pm 2.0^\circ$ Rec: EM
			Bank angle	OGE			Init: $\pm 2.0^\circ$ Rec: EM
			Stall warning up to stall				
			Pitch angle	OGE			Init: $\pm 2.0^\circ$ Rec: EM

2.c: Handling qualities longitudinal		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
			Angle of attack	OGE			Init: $\pm 2.0^\circ$ Rec: EM
			Roll rate	OGE			Init: CT&M Rec: EM
			Yaw rate	OGE			Init: CT&M Rec: EM
			Stall break and recovery				
			Pitch angle	OGE			Init: CT&M Rec: same as Init
			Angle of attack	OGE			Init: CT&M Rec: same as Init
			Roll rate	OGE			Init: CT&M Rec: same as Init
			Yaw rate	OGE			Init: CT&M Rec: same as Init
			Additionally, for those FSTDs with reversible flight control systems or equipped with stick pusher systems				
			Stick/column force (prior to the stall angle of attack)	OGE			Init: $\pm 10\%$ or ± 2.2 daN (5 lbf) Rec: EM
2.c.8.b	Approach-to-stall characteristics	Second-segment climb, high-altitude	Airspeed for stall warning speeds	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: ± 3 kt Rec: EM

2.c: Handling qualities longitudinal		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
		cruise (near performance-limited condition), and either approach or landing CCA: test in normal and non-normal control states	Angle of attack: initial buffet	VIB	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 2.0^\circ$ Rec: EM
			Pitch angle	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 2.0^\circ$ Rec: EM
			Angle of attack	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 2.0^\circ$ Rec: EM
			Bank angle	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 2.0^\circ$ Rec: EM
			Control inputs	CLO	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: CT&M Rec: EM
			Additionally, for those aeroplanes with reversible flight control systems				
			Stick/column force	OGE		Init: CT&M Rec: EM	Init: $\pm 10\%$ or ± 2.2 daN (5 lbf) Rec: EM
2.c.9	Phugoid dynamics	Cruise CCA: test in non-normal control state	Period	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 10\%$ Rec: EM
			Time to one-half or double amplitude, or damping ratio	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 10\%$ time or ± 0.02 of damping Rec: EM
2.c.10	Short-period dynamics	Cruise	Pitch angle or pitch rate	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 1.5^\circ$ or $\pm 2^\circ/s$ Rec: EM

2.c: Handling qualities longitudinal		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
		CCA: test in normal and non-normal control states	Normal acceleration	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: ± 0.1 g Rec: EM

Requirements for all the aeroplane longitudinal tests.

2.c: Handling qualities longitudinal	Test requirements
All tests	1. Power setting may be that required for level flight unless otherwise specified.

Individual test requirements. The individual test requirements defined in the table below are applicable to all fidelity levels unless otherwise stated.

2.c: Handling qualities longitudinal		Test requirements
2.c.1	Power change dynamics	- Test should cover power change from thrust for approach or level flight to maximum continuous or go-around power. - Test should cover time history of uncontrolled free response for a time increment equal to at least five seconds before initiation of the power change to completion of the power change followed by 15 seconds. - For CCA, test should be in normal and non-normal control states. - Alternative method test for aeroplanes with the OGE feature at fidelity level G should provide the force required to maintain constant altitude or vertical speed to complete the configuration change.
2.c.2	Flap change dynamics	- Test should cover time history of uncontrolled free response for a time increment equal to at least five seconds before initiation of the reconfiguration change to completion of the reconfiguration change followed by 15 seconds. - For CCA, test should be in normal and non-normal control states. - Alternative method test for aeroplanes with the OGE feature at fidelity level G should provide the force required to maintain constant altitude or vertical speed to complete the configuration change.
2.c.3	Spoiler / speed	Test should cover time history of uncontrolled free response for a time increment equal to at least five seconds before

2.c: Handling qualities longitudinal		Test requirements
	brake change dynamics	- initiation of the reconfiguration change to completion of the reconfiguration change followed by 15 seconds. 2. Results should be required for both extension and retraction. 3. For CCA, test should be in normal and non-normal control states.
2.c.4	Gear change dynamics	1. Test should cover time history of uncontrolled free response for a time increment equal to at least five seconds before initiation of the configuration change to completion of the reconfiguration change followed by 15 seconds. 2. For CCA, test should be in normal and non-normal control states. 3. Alternative method test for aeroplanes with the OGE feature at fidelity level G should provide the force required to maintain constant altitude or vertical speed to complete the configuration change.
2.c.5	Longitudinal trim	1. Test should be for steady-state wings-level trim with thrust for level flight. 2. This test may be a series of snapshot tests. 3. For CCA, test should be in normal or non-normal control state, as applicable. 4. For feature OGE at fidelity level G, pitch controller position tolerance parameter may be used instead of elevator angle.
2.c.6	Longitudinal manoeuvring stability (stick force/g)	1. Continuous time history data or a series of snapshot tests may be used. 2. Test should cover up to approximately 30° of roll angle for approach and landing configurations. 3. Test should cover up to approximately 45° of roll angle for the cruise configuration. 4. The test is initiated from a steady-state wings-level trim condition with thrust set for level flight. The longitudinal control is used to maintain the trimmed airspeed for the specified roll angles. The trim, engine power or configuration should not be changed during the test. 5. Force tolerance is not applicable if forces are generated solely by the use of aeroplane hardware in the FSTD. 6. For feature OGE at fidelity level S, for CCA, test should be in normal and non-normal control state. 7. For feature OGE at fidelity levels G and R for CCA, test should be in normal mode only. 8. For feature OGE at fidelity levels G and R, tests should be in cruise, approach or landing, if appropriate. 9. For the alternative method, for feature OGE at fidelity level G, pitch controller position tolerance parameter may be used instead of elevator angle.
2.c.7	Longitudinal static stability	1. Data for at least two speeds above and two speeds below trim speed should be used. The speed range should be sufficient to demonstrate stick force versus speed characteristics. 2. This test may be a series of snapshot tests. 3. Force tolerance is not applicable if forces are generated solely by the use of aeroplane hardware in the FSTD.

2.c: Handling qualities longitudinal		Test requirements
		4. For CCA, test should be in normal or non-normal control state, as applicable. 5. For the alternative method, for feature OGE at fidelity level G, pitch controller position tolerance parameter may be used instead of elevator angle.
2.c.8.a	Full-stall characteristics	1. Test is applicable only to aeroplane FSTDs that are to be qualified for full-stall training tasks. 2. Flight phase/test conditions are as follows. a. The test should be performed from approach to stall through to full stall and recovery to normal flight. b. For safety-of-flight considerations, flight test validation data may be limited to the stall angle of attack. Engineering simulator/simulation validation data may be used in lieu of flight test validation data for angles of attack that exceed the activation of a stall protection system or stick pusher system. c. The following stall entry methods should be demonstrated: i. stall entry at wings level (1 g); ii. stall entry in turning flight of at least 25° bank angle (accelerated stall); iii. stall entry in a power-on condition (required only for propeller-driven aeroplanes); d. The entry methods should be demonstrated at least once in the entirety of the tests provided for the 2.c.8.a requirement; demonstrating each entry method in each flight phase is not required. e. The cruise flight phase should be conducted in a flaps-up (clean) configuration. f. The second-segment climb flight phase should use a different flap setting from that for the approach or landing flight condition. g. Tests may be conducted at centre of gravity and weights typically required for aeroplane certification stall testing. 3. For computer-controlled aeroplanes: a. For aeroplanes with stall envelope protection systems, test should be in normal and non-normal control states. b. For the normal control state: i. It is expected that envelope protections will take effect, and it may not be possible to reach the aerodynamic stall condition for some aeroplanes. ii. The test is only required for an angle-of-attack range necessary to demonstrate the correct operation of the envelope protection system. 4. Objective test result requirements are as follows. a. For feature VIB at fidelity level S, the test should be run with the VIB feature active. b. The stall warning signal should occur in the proper relation to buffet/stall. FSTDs of aeroplanes exhibiting a

2.c: Handling qualities longitudinal		Test requirements
		sudden pitch attitude change or ‘g-break’ should demonstrate this characteristic. FSTDs of aeroplanes exhibiting a roll-off or loss-of-roll control authority should demonstrate this characteristic. c. The stall warning signal and initial buffet signal measured by the FSTD accelerometers, if applicable, should be recorded. d. Buffet threshold of perception should be based on 0.03 g peak-to-peak normal acceleration above the background noise at the pilot seat. Initial buffet is to be based on normal acceleration at the pilot seat with a larger peak-to-peak value relative to buffet threshold of perception (some airframe manufacturers have used 0.1 g peak to peak). The test should demonstrate the correct trend in growth of buffet amplitude from initial buffet to V_S for normal and lateral acceleration. e. The maximum buffet may be limited based on motion platform capability/limitations or other FSTD system limitations. If the maximum buffet is limited, the limit should be sufficient to allow proper use in training (e.g. not less than 0.5 g peak to peak) and, in any case, the instructor should be informed of the limitations. f. Numerical tolerances are not applicable past the stall angle of attack, but should demonstrate CT&M through recovery. (Refer to CS FSTD.QB.107.2.STALL for requirements regarding the aerodynamic model SOJ.) g. If engineering simulator/simulation validation data is used, the reduced engineering tolerances (as defined in CS FSTD.QTG.025, paragraph (a)(2)) do not apply.
2.c.8.b	Approach-to-stall characteristics	1. Test is applicable to those FSTDs that are to be qualified for approach-to-stall training tasks. 2. For feature OGE at fidelity levels G and R, test should be in normal mode only. 3. Flight phase/test conditions (for feature OGE at fidelity level S only) are as follows. a. The following stall entry methods should be demonstrated: i. stall entry at wings level (1 g); ii. stall entry in turning flight of at least 25° bank angle (accelerated stall); iii. stall entry in a power-on condition (required only for propeller-driven aeroplanes); b. The entry methods should be demonstrated at least once in the entirety of the tests provided for the 2.c.8.b requirement; demonstrating each entry method in each flight phase is not required. c. The cruise flight phase should be conducted in a flaps-up (clean) configuration. d. The second-segment climb flight phase should use a different flap setting from that for the approach or landing flight condition. e. Tests may be conducted at centre of gravity and weights typically required for aeroplane certification stall testing.

2.c: Handling qualities longitudinal		Test requirements
		4. For computer-controlled aeroplanes, the following applies. For aeroplanes with stall envelope protection systems, test should be in normal and non-normal control states. For the normal control state, the test is only required for an angle-of-attack range necessary to demonstrate the correct operation of the envelope protection system. The tests may be used to satisfy the angle-of-attack test flight manoeuvre and envelope protection function test requirement (CS FSTD.QTG.A.235, test 2.h.6).
2.c.9	Phugoid dynamics	1. The test should include three full cycles of free response or those necessary to determine time to one-half or double amplitude, whichever is less. 2. The test should demonstrate an uncontrolled free response after excitation of the phugoid via a longitudinal control input. 3. For CCA, test should be in non-normal control state.
2.c.10	Short-period dynamics	1. For CCA, test should be in normal and non-normal control states. 2. The test should demonstrate an uncontrolled free response after excitation of the short period via a momentary longitudinal control input.

CS FSTD.QTG.A.215 Handling qualities lateral–directional – Aeroplane

This CS provides the standards for the objective tests for aeroplane lateral–directional handling qualities.

2.d: Handling qualities lateral–directional		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
2.d.1	Minimum control speed (air) (V_{MCA}) or minimum control speed (landing) (V_{MCL})	Either take-off or landing (whichever is most critical in the aeroplane) CCA: test in normal or non-normal control state	Airspeed	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: ± 3 kt Rec: EM
2.d.2	Roll response (rate)	Cruise and either approach or landing	Roll rate	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: ± 10 % or $\pm 2^\circ/\text{s}$ Rec: EM
			For aeroplanes with reversible flight control systems				
			Roll controller force	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: ± 1.3 daN (3 lbf) or ± 10 % Rec: EM

2.d: Handling qualities lateral-directional		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
2.d.3	Step input of flight deck roll controller (or roll overshoot)	Either approach or landing CCA: test in normal and non-normal control states For feature OGE at fidelity level G, test in normal mode only	Bank angle	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 10\%$ or $\pm 2^\circ$ Rec: EM
2.d.4	Spiral stability	Cruise and either approach or landing CCA: test in non-normal control state	Bank angle after 20 s	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: Correct trend and $\pm 2^\circ$ or $\pm 10\%$ Rec: EM
			If alternate test is used				
			Aileron angle	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 2^\circ$ Rec: EM
2.d.5	Engine-inoperative trim	Second-segment climb and either approach or landing	Rudder angle or tab angle or equivalent rudder pedal	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 1^\circ$ Rec: EM
			Side-slip angle	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 2^\circ$ Rec: EM
2.d.6	Rudder response	Either approach or	Yaw rate	OGE		Init: CT&M Rec: EM	Init: $\pm 2^\circ/\text{s}$ or $\pm 10\%$ Rec: EM

2.d: Handling qualities lateral-directional		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
		landing For feature OGE at fidelity levels S and R: test with stability augmentation on and off For feature OGE at fidelity level G: test with stability augmentation off CCA: test in normal and non-normal control states For feature OGE at fidelity level G: test in normal mode only	Yaw rate or heading change	OGE	Init: CT&M Rec: EM		
2.d.7	Dutch roll	Cruise and either approach or	Period	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: ± 0.5 s or ± 10 % Rec: EM

2.d: Handling qualities lateral-directional		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
		landing Yaw damper off CCA: test in non-normal control state	Time to one-half or double amplitude or damping ratio	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: ± 10 % of time to one-half or double amplitude or ± 0.02 of damping ratio Rec: EM
			Time difference between peaks of roll angle and side-slip	OGE			Init: ± 1 s or ± 20 % Rec: EM
2.d.8	Steady-state side-slip	Either approach or landing	For a given rudder position				
			Roll angle	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 2^\circ$ Rec: EM
			Side-slip angle	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 1^\circ$ Rec: EM
			Aileron angle	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 2^\circ$ or ± 10 % Rec: EM
			Spoiler or equivalent roll controller position or force	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 5^\circ$ or ± 10 % Rec: EM
			For aeroplanes with reversible flight control systems				

2.d: Handling qualities lateral-directional		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
			Wheel force	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: ± 1.3 daN (3 lbf) or ± 10 % Rec: EM
			Rudder pedal force	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: ± 2.2 daN (5 lbf) or ± 10 % Rec: EM

Requirements for all the aeroplane lateral-directional tests.

2.d: Handling qualities lateral-directional	Test requirements
All tests	1. Power setting may be that required for level flight unless otherwise specified.

Individual test requirements. The individual test requirements defined in the table below are applicable to all fidelity levels unless otherwise stated.

2.d: Handling qualities lateral-directional		Test requirements
2.d.1	Minimum control speed (air) (V_{MCA}) or minimum control speed (landing) (V_{MCL})	1. Test should be performed per applicable airworthiness standard or for low-speed engine-inoperative handling characteristics in the air. 2. Minimum speed may be defined by a performance or control limit that prevents demonstration of V_{MCA} or V_{MCL} in the conventional manner. 3. Take-off thrust should be set on the operating engine(s). 4. Time history or snapshot data may be used. 5. For CCA, test should be in normal or non-normal control state. 6. In the case of propeller-driven aircraft, the propellers of the inoperative engine should be feathered. 7. For feature OGE at fidelity levels G and R, it is important that there is a realistic speed relationship between V_{MCA} (or V_{MCL}) and V_S for all configurations and, in particular, the most critical full-power engine-out configuration.

2.d: Handling qualities lateral-directional		Test requirements
2.d.2	Roll response (rate)	1. Test should be with normal roll control displacement of about one third of maximum roll controller travel, holding this at least until a steady roll rate is established. 2. This test may be combined with the step input of flight deck roll controller (or roll overshoot) test (test 2.d.3).
2.d.3	Step input of flight deck roll controller (or roll overshoot)	1. With wings level, test should apply a step roll control input using approximately one third of maximum roll controller travel. At an approximately 20°–30° roll angle, test should abruptly return the roll controller to neutral and allow at least 10 seconds of aeroplane free response. 2. This test may be combined with roll response (rate) test (test 2.d.2). 3. For CCA, test should be in normal and non-normal control states. 4. For feature OGE at fidelity level G, test should be in normal mode only.
2.d.4	Spiral stability	1. Validation data averaged from multiple tests may be used. 2. Test should be for both directions. 3. As an alternative test, the lateral control required to maintain a steady turn with a roll angle of approximately 30° can be shown. 4. For CCA, test should be in non-normal control state.
2.d.5	Engine-inoperative trim	1. Test should be performed in a manner similar to that for which a pilot is trained to trim an engine failure condition. 2. Second-segment climb test should be at take-off thrust. 3. Approach or landing test should be at thrust for level flight. 4. This test may be a series of snapshot tests.
2.d.6	Rudder response	1. For feature OGE at fidelity levels S and R, test should be with stability augmentation on and off. 2. For feature OGE at fidelity level G, test should be with stability augmentation off. 3. Test should be with a rapid step input at approximately 25 % of full rudder pedal travel. 4. For CCA, test should be in normal and non-normal control states. 5. For feature OGE at fidelity level G, test should be in normal mode only.
2.d.7	Dutch roll	1. Test for at least six cycles of free response with stability augmentation off. 2. For CCA, test should be in non-normal control state. 3. The test should demonstrate an uncontrolled free response after excitation of the Dutch roll via momentary yaw control inputs.

2.d: Handling qualities lateral-directional		Test requirements
2.d.8	Steady-state side-slip	1. This test may be a series of snapshot tests using at least two rudder positions (in each direction for propeller-driven aeroplanes), one of which should be near the maximum allowable rudder while using sufficient lateral control to maintain a constant heading. 2. For feature OGE at fidelity level G, roll controller position tolerance parameter may be used instead of aileron angle.

CS FSTD.QTG.A.220 Handling qualities landing – Aeroplane

This CS provides the standards for the objective tests for aeroplane landing.

2.e: Handling qualities landing		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
2.e.1	Normal landing	Landing CCA: test in normal and non-normal control states, if applicable	Airspeed	GND, IGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: ± 3 kt Rec: EM
			Pitch angle	GND, IGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 1.5^\circ$ Rec: EM
			Angle of attack	GND, IGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 1.5^\circ$ Rec: EM
			Height	GND, IGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: ± 3 m (10 ft) or ± 10 % Rec: EM
			For aeroplanes with reversible flight control systems				
			Column force	GND, IGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: ± 10 % or ± 2.2 daN (5 lbf) Rec: EM
2.e.2	Minimum flap landing	Minimum certified landing flap configuration	Airspeed	GND, IGE			Init: ± 3 kt Rec: EM
			Pitch angle	GND, IGE			Init: $\pm 1.5^\circ$ Rec: EM
			Angle of attack	GND, IGE			Init: $\pm 1.5^\circ$ Rec: EM

2.e: Handling qualities landing		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
			Height	GND, IGE			Init: ± 3 m (10 ft) or ± 10 % Rec: EM
			For aeroplanes with reversible flight control systems				
			Column force	GND, IGE			Init: ± 10 % or ± 2.2 daN (5 lbf) Rec: EM
2.e.3	Crosswind landing	Landing	Airspeed	GND, IGE		Init: CT&M Rec: EM	Init: ± 3 kt Rec: EM
			Pitch angle	GND, IGE		Init: CT&M Rec: EM	Init: $\pm 1.5^\circ$ Rec: EM
			Angle of attack	GND, IGE		Init: CT&M Rec: EM	Init: $\pm 1.5^\circ$ Rec: EM
			Height	GND, IGE		Init: CT&M Rec: EM	Init: ± 3 m (10 ft) or ± 10 % Rec: EM
			Roll angle	GND, IGE		Init: CT&M Rec: EM	Init: $\pm 2^\circ$ Rec: EM
			Side-slip angle	GND, IGE		Init: CT&M Rec: EM	Init: $\pm 2^\circ$ Rec: EM
			Heading angle	GND, IGE		Init: CT&M Rec: EM	Init: $\pm 3^\circ$ Rec: EM

2.e: Handling qualities landing		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
			For aeroplanes with reversible flight control systems				
			Column force	GND, IGE		Init: CT&M Rec: EM	Init: ±10 % or ±2.2 daN (5 lbf) Rec: EM
			Wheel force	GND, IGE		Init: CT&M Rec: EM	Init: ±10 % or ±1.3 daN (3 lbf) Rec: EM
			Rudder pedal force	GND, IGE		Init: CT&M Rec: EM	Init: ±10 % or ±2.2 daN (5 lbf) Rec: EM
2.e.4	One-engine-inoperative landing	Landing	Airspeed	GND, IGE		Init: CT&M Rec: EM	Init: ±3 kt Rec: EM
			Pitch angle	GND, IGE		Init: CT&M Rec: EM	Init: ±1.5° Rec: EM
			Angle of attack	GND, IGE		Init: CT&M Rec: EM	Init: ±1.5° Rec: EM
			Height	GND, IGE		Init: CT&M Rec: EM	Init: ±3 m (10 ft) or ±10 % Rec: EM
			Roll angle	GND, IGE		Init: CT&M Rec: EM	Init: ±2° Rec: EM
			Side-slip angle	GND, IGE		Init: CT&M Rec: EM	Init: ±2° Rec: EM

2.e: Handling qualities landing		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
			Heading angle	GND, IGE		Init: CT&M Rec: EM	Init: $\pm 3^\circ$ Rec: EM
2.e.5	Autopilot landing	Landing	Flare height	GND, IGE		Init: CT&M Rec: EM	Init: ± 1.5 m (5 ft) Rec: EM
			Duration of flare	GND, IGE		Init: CT&M Rec: EM	Init: ± 0.5 s or ± 10 % Rec: EM
			Rate of descent at touchdown	GND, IGE		Init: CT&M Rec: EM	Init: ± 0.7 m/s (140 ft/min) Rec: EM
			Lateral position on runway during roll-out	GND, IGE		Init: CT&M Rec: EM	Init: ± 3 m (10 ft) Rec: EM
2.e.6	All-engine autopilot go-around	As per aeroplane performance data	Airspeed	IGE		Init: CT&M Rec: EM	Init: ± 3 kt Rec: EM
		CCA: test in normal and non-normal control states, if applicable	Pitch angle	IGE		Init: CT&M Rec: EM	Init: $\pm 1.5^\circ$ Rec: EM
			Angle of attack	IGE		Init: CT&M Rec: EM	Init: $\pm 1.5^\circ$ Rec: EM
2.e.7	One-engine-inoperative go-around	With autopilot and without autopilot	Airspeed	IGE		Init: CT&M Rec: EM	Init: ± 3 kt Rec: EM
		As per aeroplane	Pitch angle	IGE		Init: CT&M Rec: EM	Init: $\pm 1.5^\circ$ Rec: EM

2.e: Handling qualities landing		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
		performance data CCA: non-autopilot test in non-normal control state	Angle of attack	IGE		Init: CT&M Rec: EM	Init: $\pm 1.5^\circ$ Rec: EM
			Roll angle	IGE		Init: CT&M Rec: EM	Init: $\pm 2^\circ$ Rec: EM
			Side-slip angle	IGE		Init: CT&M Rec: EM	Init: $\pm 2^\circ$ Rec: EM
2.e.8	Directional control (rudder effectiveness) with symmetric reverse thrust	Landing	Airspeed	GND		Init: CT&M Rec: EM	Init: ± 5 kt Rec: EM
			Yaw rate	GND		Init: CT&M Rec: EM	Init: $\pm 2^\circ/\text{s}$ Rec: EM
2.e.9	Directional control (rudder effectiveness) with asymmetric reverse thrust	Landing	Airspeed	GND		Init: CT&M Rec: EM	Init: ± 5 kt Rec: EM
			Heading angle	GND		Init: CT&M Rec: EM	Init: $\pm 3^\circ$ Rec: EM

Individual test requirements. The individual test requirements defined in the table below are applicable to all fidelity levels unless otherwise stated.

2.e: Handling qualities landing		Test requirements
2.e.1	Normal landing	- Test should be from a minimum of 61 m (200 ft) AGL to nose wheel touchdown. - Two tests should be shown, including two normal landing flaps, if applicable, one of which should be near maximum certified landing weight and one of which should be at light or medium weight. - For CCA, test should be in normal and non-normal control states, if applicable.
2.e.2	Minimum flap	Test should be from a minimum of 61 m (200 ft) AGL to nose wheel touchdown.

2.e: Handling qualities landing		Test requirements
	landing	2. Test should be at near maximum certified landing weight.
2.e.3	Crosswind landing	1. Test should be from a minimum of 61 m (200 ft) AGL to a 50 % decrease in main landing gear touchdown speed. 2. Test requires validation data, including wind profile, for a crosswind component of at least 60 % of aeroplane performance data value measured at 10 m (33 ft) above the runway. 3. Wind components should be provided as headwind and crosswind values with respect to the runway.
2.e.4	One-engine-inoperative landing	1. Test should be from a minimum of 61 m (200 ft) AGL to a 50 % decrease in main landing gear touchdown speed.
2.e.5	Autopilot landing	1. If autopilot provides roll-out guidance, test should record lateral position on runway from touchdown to a 50 % decrease in main landing gear touchdown speed. 2. Time of autopilot flare mode engagement and main gear touchdown should be noted.
2.e.6	All-engine autopilot go-around	1. Normal all-engine autopilot go-around should be demonstrated at medium weight. 2. For CCA, test should be in normal and non-normal control states, if applicable.
2.e.7	One-engine-inoperative go-around	1. Test of the engine-inoperative go-around procedure is required for near maximum certified landing weight with critical engine(s) inoperative. 2. For CCA, non-autopilot test is to be conducted in non-normal mode.
2.e.8	Directional control (rudder effectiveness) with symmetric reverse thrust	1. Test should apply rudder pedal input in both directions using full reverse thrust until reaching full thrust reverser minimum operating speed.
2.e.9	Directional control (rudder effectiveness) with asymmetric reverse thrust	1. With full reverse thrust on the operating engine(s), test should maintain heading with rudder pedal input until maximum rudder pedal input or thrust reverser minimum operating speed is reached.

CS FSTD.QTG.A.225 Handling qualities ground effect – Aeroplane

This CS provides the standards for the objective tests for aeroplane ground effect handling qualities.

2.f: Handling qualities ground effect		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
2.f.1	Ground effect	Landing CCA: test in normal or non-normal control state	Elevator	IGE		Init: CT&M Rec: EM	Init: $\pm 1^\circ$ Rec: EM
			Pitch trim angle	IGE		Init: CT&M Rec: EM	Init: $\pm 0.5^\circ$ Rec: EM
			Net thrust or equivalent	IGE		Init: CT&M Rec: EM	Init: $\pm 5\%$ Rec: EM
			Angle of attack	IGE		Init: CT&M Rec: EM	Init: $\pm 1^\circ$ Rec: EM
			Height	IGE		Init: CT&M Rec: EM	Init: ± 1.5 m (5 ft) or $\pm 10\%$ Rec: EM
			Airspeed	IGE		Init: CT&M Rec: EM	Init: ± 3 kt Rec: EM
			Pitch angle	IGE		Init: CT&M Rec: EM	Init: $\pm 1^\circ$ Rec: EM

Individual test requirements. The individual test requirements defined in the table below are applicable to all fidelity levels unless otherwise stated.

2.f: Handling qualities ground effect		Test requirements
2.f.1	Ground effect	1. For CCA, test should be in normal or non-normal control state.

GM1 FSTD.QTG.A.225 Handling qualities ground effect – Aeroplane

This GM provides guidance on the definition of aeroplane ground effect test parameters.

- (a) The selection of the test method and procedures to validate ground effect is up to the data provider; however, the flight test should be performed with enough duration near the ground to validate the ground-effect model sufficiently.

Acceptable tests for the validation of ground effect include the following.

- (1) **Level fly-bys.** These should be conducted at a minimum of three altitudes within the ground effect, including one at no more than 10 % of the wingspan above the ground and one at each of approximately 30 % and 50 % of the wingspan, where height refers to the height of the main gear tyre above the ground. In addition, one in level flight trim condition should be conducted out-of-ground effect (e.g. at 150 % of wingspan).
- (2) **Shallow approach landing.** This should be performed at a glide slope of approximately 1°, with negligible pilot activity until flare.

If other methods are proposed, a rationale should be provided to show that the tests performed validate the ground-effect model.

- (b) The lateral-directional characteristics are also altered by ground effect. For example, because of changes in lift, roll damping is affected. The change in roll damping will affect other dynamic modes usually evaluated in FSTD validation. Dutch roll dynamics, spiral stability and the roll rate for a given lateral control input are altered by ground effect. Steady heading side-slips will also be affected. These effects should be accounted for in the FSTD modelling. Several tests – such as 2.e.3 ‘Crosswind landing’, 2.e.4 ‘One-engine-inoperative landing’ and 1.b.5 ‘Engine failure on take-off’ – serve to validate lateral-directional ground effect, since portions of them are accomplished while transiting heights at which ground effect is an important factor.

CS FSTD.QTG.A.230 Wind shear demonstration – Aeroplane

This CS provides the standards for the objective tests for aeroplane wind shear demonstration.

2.g: Wind shear demonstration		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
2.g.1	Wind shear demonstration	Take-off and landing	All plotted parameters	GND, IGE, OGE		Init: n/a Rec: EM	Init: n/a Rec: EM

Individual test requirements. The individual test requirements defined in the table below are applicable to all fidelity levels unless otherwise stated.

2.g: Wind shear demonstration		Test requirements
2.g.1	Wind shear demonstration	- Four tests are required, two in the take-off condition and two in the landing condition. For the two tests in each condition, one should be conducted to establish a reference for FSTD performance in still air and the other should be conducted to demonstrate the effects of the active wind shear model relative to the reference test. - The test should utilise the wind shear model(s) as described in the SOJ required in CS FSTD.QB.113, section 'For aeroplanes: wind shear and microburst'. The test should include a rationale that describes the wind shear effects being demonstrated. - The tests with wind shear active should demonstrate the entry into and exit from the active wind shear.

CS FSTD.QTG.A.235 Flight and manoeuvre envelope protection functions – Aeroplane

This CS provides the standards for the objective tests for aeroplane flight and manoeuvre envelope protection functions.

2.h: Handling qualities flight and manoeuvre envelope protection functions		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
2.h.1	Overspeed	Cruise CCA: test in normal and non-normal control states, if function is different	Airspeed	OGE		Init: CT&M Rec: EM	Init: ± 5 kt Rec: EM
2.h.2	Minimum speed	Take-off, cruise and either approach or landing CCA: test in normal and non-normal control states, if function is different	Airspeed	OGE		Init: CT&M Rec: EM	Init: ± 3 kt Rec: EM
2.h.3	Load factor	Take-off and cruise CCA: test in normal and non-normal control states, if function is different	Normal acceleration	OGE		Init: CT&M Rec: EM	Init: ± 0.1 g Rec: EM

2.h: Handling qualities flight and manoeuvre envelope protection functions		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
2.h.4	Pitch angle	Cruise and approach CCA: test in normal and non-normal control states, if function is different	Pitch angle	OGE		Init: CT&M Rec: EM	Init: $\pm 1.5^\circ$ Rec: EM
2.h.5	Bank angle	Approach CCA: test in normal and non-normal control states, if function is different	Roll angle	OGE		Init: CT&M Rec: EM	Init: $\pm 2^\circ$ or $\pm 10\%$ Rec: EM
2.h.6	Angle of attack	Second-segment climb and either approach or landing CCA: test in normal and non-normal control states, if function is different	Angle of attack	OGE		Init: CT&M Rec: EM	Init: $\pm 1.5^\circ$ Rec: EM

Requirements for all the aeroplane flight and manoeuvre envelope protection functions.

2.h: Handling qualities flight and manoeuvre envelope protection functions	Test requirements
All tests	1. These tests are only applicable to computer-controlled aeroplanes. 2. Time history results of response to control inputs during entry into each envelope protection function (i.e. in normal and degraded control states, if function is different) are required. 3. Test should set thrust as required to reach the envelope protection function.

CS FSTD.QTG.A.240 Engine and airframe icing effects demonstration – Aeroplane

This CS provides the standards for the objective tests for aeroplane engine and airframe icing effects demonstration.

2.i: Handling qualities engine and airframe icing effects		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
2.i.1	Engine and airframe icing effects demonstration (high angle of attack)	Either take-off or approach or landing	All plotted parameters	OGE			Init: n/a Rec: EM

Individual test requirements. The individual test requirements defined in the table below are applicable to all fidelity levels unless otherwise stated.

2.i: Handling qualities engine and airframe icing effects		Test requirements
2.i.1	Engine and airframe icing effects demonstration (high angle of attack)	- Tests are applicable only to FSTDs that simulate an aircraft type and variant certified for flight into icing conditions. - The tests should be time histories starting from a trimmed condition through approach to stall, full stall and initiation of the recovery. The full-stall portion of the manoeuvre is applicable only to FSTDs that are to be qualified for full-stall training tasks. - Two tests are required, both in the same test condition. One test is to establish a reference for FSTD performance without ice accretion and the second test is to demonstrate the characteristic engine and aerodynamic airframe icing effects caused by in-flight ice accretion relative to the reference test. The test with ice accretion should be conducted by initialising and maintaining a fixed amount of ice accretion throughout the manoeuvre in order to consistently evaluate the icing effects. - The tests should utilise the icing model(s) described in the SOJ required in CS FSTD.QB.107 section 'Icing effects'. The tests should include a description of the icing effects being demonstrated. Icing effects may include but are not limited to the following effects, as applicable to the particular aeroplane type: - decreases in the stall angle of attack

2.i: Handling qualities engine and airframe icing effects	Test requirements
	b. changes in the pitching moment c. decreases in control effectiveness d. changes in control forces e. increases in drag f. changes in stall buffet characteristics and threshold of perception and g. engine effects (power reduction/variation, vibration, etc., if expected to be present in the aeroplane in the ice accretion scenario being tested). 5. Plotted parameters should include the following as a minimum: a. altitude b. airspeed c. normal acceleration d. engine power e. angle of attack f. pitch attitude g. bank angle h. flight control inputs and i. stall warning and stall buffet onset. 6. Flight test validation data is not required.

CS FSTD.QTG.A.300 Motion system – Aeroplane

This CS provides the standards for the objective tests for aeroplane motion systems.

3.a–f: Motion system		Test conditions		Tolerance parameter	Primary feature(s) validated	Tolerances		
						G	R	S
3.a	Frequency response	All linear axes		Phase	MTN		Init: 0° to -20° between 0.1 Hz and 1.0 Hz; 0° to -40° between 1.0 Hz and 3.0 Hz; tolerance to be specified by operator above 3.0 Hz and at 15.0 Hz Rec: Same as Init	Init: 0° to -20° between 0.1 Hz and 1.0 Hz; 0° to -40° between 1.0 Hz and 3.0 Hz; tolerance to be specified by operator above 3.0 Hz and at 15.0 Hz Rec: Same as Init
				Gain	MTN		Init: ±2 dB between 0.1 Hz and 1.0 Hz; ±4 dB between 1.0 Hz and 3.0 Hz; tolerance to be specified by operator above 3.0 Hz and at 15.0 Hz Rec: Same as Init	Init: ±2 dB between 0.1 Hz and 1.0 Hz; ±4 dB between 1.0 Hz and 3.0 Hz; tolerance to be specified by operator above 3.0 Hz and at 15.0 Hz Rec: Same as Init

3.a–f: Motion system		Test conditions		Tolerance parameter	Primary feature(s) validated	Tolerances		
						G	R	S
3.b	Leg balance	n/a		Phase shift (each motion jack compared with a datum jack)	MTN		Init: 0° to 1.5° Rec: Same as Init	Init: 0° to 1.5° Rec: Same as Init
3.c	Turnaround	n/a		Deviation from the desired sinusoidal acceleration	MTN		Init: ±0.05 g peak Rec: Same as Init	Init: ±0.05 g peak Rec: Same as Init
3.d		Reserved						
3.e	Motion system repeatability	On ground and in flight		Actual motion platform linear accelerations	MTN		Init: n/a Rec: ±0.05 g actual motion platform linear accelerations compared with the MQTG value	Init: n/a Rec: ±0.05 g actual motion platform linear accelerations compared with the MQTG value

3.a–f: Motion system		Test conditions		Tolerance parameter	Primary feature(s) validated	Tolerances		
						G	R	S
3.f	Motion cueing performance signature	Refer to the test requirement table below for test 3.f		Actual motion platform linear acceleration cues	MTN			Init: Subjective motion assessment to establish MQTG footprint demonstrating high correlation between motion cueing and aircraft simulation model Rec: ± 0.05 g compared to the time histories of the MQTG

3.a–f: Motion system		Test conditions		Tolerance parameter	Primary feature(s) validated	Tolerances		
						G	R	S
				Actual motion platform angular cues	MTN			Init: Subjective motion assessment to establish MQTG footprint demonstrating high correlation between motion cueing and aircraft simulation model Rec: $\pm 0.5^\circ$ in the pitch, roll and yaw attitude of the actual motion platform and $\pm 0.5^\circ/\text{s}$ in the corresponding angular rates when compared with the MQTG time histories
				Flight parameters	MTN			Init: n/a Rec: EM

Requirements for all the aeroplane motion system tests.

3.a–f: Motion system	Test requirements
All tests	1. The intent of tests 3.a ('Frequency response'), 3.b ('Leg balance') and 3.c ('Turnaround') is to demonstrate the performance of the motion system hardware and to check the integrity of the motion system with regard to calibration and wear. These tests are considered as robotic tests and do not need to be run as integrated tests as required by CS FSTD.QTG.020, paragraph (d), as they are independent of the motion cueing software. 2. Manual tests are not required.

Individual test requirements. The individual test requirements defined in the table below are applicable to all fidelity levels unless otherwise stated.

3.a–f: Motion system		Test requirements
3.a	Frequency response	1. The test should demonstrate the motion frequency response for an oscillatory input, for each linear axis. 2. The test should cover at least the frequencies between 0.1 Hz and 15.0 Hz. 3. Plots of phase and attenuation should be plotted against the frequency of the motion movement. 4. If motion accelerations are computed from the leg positions instead of being directly measured (e.g. by using accelerometers), this should be justified in the MQTG. 5. The tolerances for above 3.0 Hz but below 15.0 Hz have been left to the discretion of the applicant for the FSTD qualification. The selected tolerances should be set to demonstrate the performance of the motion system hardware and to check the integrity of the motion system with regard to calibration and wear. The SOJ should explain how the tolerances have been determined.
3.b	Leg balance	1. The phase shift between a datum jack and any other jack should be measured using a heave (vertical) signal of 0.5 Hz at ± 0.25 g.
3.c	Turnaround	1. The test should demonstrate that the motion response during oscillatory manoeuvres does not exhibit excessive noise by driving each motion actuator with a sinusoidal signal and recording the motion movement. 2. The motion base should be driven sinusoidally in the heave axis through a displacement of 150 mm peak to peak at a frequency of 0.5 Hz. 3. The motion movement should be measured (e.g. using an accelerometer) and any deviation compared with the command signal.

3.a–f: Motion system		Test requirements
3.e	Motion system repeatability	1. The test should demonstrate that the motion system hardware and software, in normal FSTD operating mode, continue to perform as initially qualified (i.e. to demonstrate that the motion system software and motion system hardware have not degraded or changed over time). 2. The test inputs should be such that rotational accelerations, rotational rates and linear accelerations are inserted before the transfer from aeroplane centre of gravity to pilot reference point (i.e. pilot seat), with minimum amplitudes of 5°/s/s, 10°/s and 0.3 g, respectively, to provide adequate analysis of the output. 3. Two tests should be performed, one in an on-ground state and the other in an in-air state. This is to cater for possible differences in motion system gains for the on-ground and in-air conditions. The exact test conditions are to be determined by the applicant for FSTD qualification. 4. The parameters to be plotted are: a. the acceleration commands; b. actual linear acceleration measured from accelerometers at the pilot position (the output will comprise accelerations due to both linear and rotational motion accelerations); c. motion actuator positions; and d. the motion system attitude and position (i.e. pitch, roll, yaw, surge, sway, heave).
3.f	Motion cueing performance signature	1. The tests establish a footprint for the motion movement and cueing for the test cases given in the table below. 2. For the initial qualification, the following apply. a. An SOJ should explain the motion movements and give the basis for their magnitude and correlation for each axis for each test case. b. In addition, the FSTD motion cueing should be compared with the aircraft's key acceleration and angular cues as provided by the simulation model, using overplotting of time histories. The motion cueing should maintain a high correlation with the simulation model's key acceleration and angular cues, both during significant changes and in steady-state conditions, while accounting for any washout effects. c. Time histories of the following parameters should be plotted for initial qualification (i.e. MQTG): i. same flight parameters as for the referenced objective tests (refer to the table below); ii. validation data for the flight parameters of the referenced objective tests (if data is available); iii. motion actuator positions for the whole time history of the test;

3.a–f: Motion system		Test requirements									
		iv. motion system attitude and position on all six axes (i.e. pitch, roll, yaw, heave, sway and surge) for the whole time history of the test; and v. correlation of motion cueing with respect to the aircraft simulation model in terms of load factor and angular cues by overplotting. The actual linear acceleration should be measured from accelerometers at the pilot reference point (e.g. under the pilot seat). 3. For recurrent testing, the following applies. Time histories of the following parameters should be overplotted with the MQTG results: a. same flight parameters of the simulation model as for the referenced objective tests (refer to the table below); b. motion actuator positions for the whole time history of the test; c. motion system attitude and position on all six axes (i.e. pitch, roll, yaw, heave, sway and surge) for the whole time history of the test; and d. the actual linear acceleration measured from accelerometers at the pilot reference point (e.g. under the pilot seat). 4. The table below specifies inputs and flight parameters from applicable objective tests. If a referenced test applies to this FSTD, its footprint should be used. If not, the associated motion cueing performance signature test case should be conducted using a manoeuvre similar to that in the referenced test. The table below defines the required test cases for an FSTD incorporating the aircraft simulation features at fidelity level G, R or S, whether individually or in combination. The table applies irrespective of the distribution of fidelity levels across the features. If any feature is at fidelity level N (not simulated), the applicable test cases should be determined in agreement with the competent authority. <table> <tr> <th>No</th><th>Test</th><th>Demonstration method</th><th>Key acceleration and angular cues</th></tr> <tr> <td>1</td><td>Taxiing turn</td><td>Test is to demonstrate motion cueing for taxiing turn. Inputs and flight parameters are as in test 1.a.1 'Minimum radius turn'.</td><td> - Short-term lateral acceleration cue for entry into turn - Long-term sustained centripetal acceleration due to turning </td></tr> </table>		No	Test	Demonstration method	Key acceleration and angular cues	1	Taxiing turn	Test is to demonstrate motion cueing for taxiing turn. Inputs and flight parameters are as in test 1.a.1 'Minimum radius turn'.	- Short-term lateral acceleration cue for entry into turn - Long-term sustained centripetal acceleration due to turning
No	Test	Demonstration method	Key acceleration and angular cues								
1	Taxiing turn	Test is to demonstrate motion cueing for taxiing turn. Inputs and flight parameters are as in test 1.a.1 'Minimum radius turn'.	- Short-term lateral acceleration cue for entry into turn - Long-term sustained centripetal acceleration due to turning								

3.a–f: Motion system		Test requirements			
		2	Take-off to take-off safety speed (V_2)	Test is to demonstrate motion cueing for a normal take-off. Inputs and flight parameters are as in test 1.b.4 'Normal take-off'.	- Long-term x-axis acceleration due to airspeed increase - Pitch rate for rotation - Vertical acceleration (g) due to lift-off trajectory
		3	Brake release and initial acceleration	Test is to demonstrate motion cueing for a take-off with thrust against brake followed by a sudden brake release. This test may be combined with the take-off test.	- Initial x-axis acceleration due to airspeed increase - Long-term x-axis acceleration due to airspeed increase
		4	Rejected take-off (engine failure prior to V_1)	Test is to demonstrate motion cueing for a rejected take-off. Inputs and flight parameters are as in test 1.b.7 'Rejected take-off'.	- Initial x-axis deceleration due to airspeed decrease - Long-term x-axis deceleration due to airspeed decrease
		5	Engine failure in the air	Test is to demonstrate motion cueing for an engine failure. Inputs and flight parameters are as in test 1.b.8 'Dynamic engine failure after take-off'. For CCA, test is in non-normal control state.	All the angular rates due to yaw, bank and pitch changes of the aircraft
		6	Steady-state climb	Test is to demonstrate motion cueing for steady-state climb. Inputs and flight parameters are as in test 1.c.1 'Normal climb, all engines operating'.	Pitch attitude of the aircraft
		7	Level flight acceleration	Test is to demonstrate motion cueing for level flight acceleration. Inputs and flight parameters are as in test 1.d.1 'Level flight acceleration'.	- Initial x-axis acceleration due to airspeed increase - Long-term x-axis acceleration due to airspeed increase

3.a–f: Motion system		Test requirements			
		8	Level flight deceleration	Test is to demonstrate motion cueing for level flight acceleration. Inputs and flight parameters are as in test 1.d.2 ‘Level flight deceleration’.	- Initial x-axis deceleration due to airspeed decrease - Long-term x-axis deceleration due to airspeed decrease
		9	Power change dynamics	Tests are to demonstrate motion cueing for power changes. Inputs and flight parameters are as in test 2.c.1 ‘Power change dynamics’. For CCA, tests are in non-normal control state.	- Pitch rate - Long-term x-axis acceleration/deceleration due to airspeed changes - If applicable, yaw rate of the aircraft due to asymmetric thrust (e.g. single-engine aircraft)
		10	Configuration changes, flaps	Test is to demonstrate motion cueing for configuration changes. Inputs and flight parameters are as in test 2.c.2 ‘Flap change dynamics’. For CCA, test is in non-normal control state.	- Pitch rate - Long-term x-axis acceleration/deceleration due to airspeed changes
		11	Configuration changes, landing gear	Tests are to demonstrate motion cueing for configuration changes. Inputs and flight parameters are as in test 2.c.4 ‘Gear change dynamics’. For CCA, tests are in non-normal control state.	- Pitch rate - Long-term x-axis acceleration/deceleration due to airspeed changes
		12	Turns	Test is to demonstrate motion cueing for turns in the air. Inputs and flight parameters are as in test 2.d.3 ‘Step input of flight deck roll controller (or roll overshoot)’.	- Short-term roll rate cue for roll initiation - Long-term roll cue due to sustained bank

3.a–f: Motion system		Test requirements			
		13	Rudder response	Test is to demonstrate motion cueing for rudder response in the air. Inputs and flight parameters are as in test 2.d.6 ‘Rudder response’.	Yaw rate of the aircraft
		14	Landing	Test is to demonstrate motion cueing for normal landing. Inputs and flight parameters are as in test 2.e.1 ‘Normal landing’.	- Pitch attitude of the aircraft during final approach - Pitch for flare
		15	Go-around	Test is to demonstrate motion cueing for a go-around. Inputs and flight parameters are as in test 2.e.6 ‘All-engine autopilot go-around’. If the FSTD is not equipped with an autopilot system, then the test inputs and flight parameters are to be flown as for a normal go-around profile.	- Pitch rate for rotation - Vertical acceleration (g) due to trajectory

GM1 FSTD.QTG.A.300 Motion system – Aeroplane

MOTION CUEING PERFORMANCE SIGNATURE TESTS

This GM gives guidance on the motion cueing performance signature tests for the motion cueing FSTD feature at the specific fidelity level.

Pilots use continuous information signals to regulate the state of the aircraft. In concert with the instruments and outside-world visual information, full-body motion feedback – including vestibular and proprioceptive cues (e.g. colloquially referred to as ‘seat-of-the-pants’ sensations) – is essential in helping the pilot to perceive and control the aircraft’s dynamics, particularly in the presence of external disturbances. The motion system should therefore meet basic objective performance criteria and be subjectively tuned at the pilot’s seat position to represent the linear and angular accelerations of the aircraft during a prescribed minimum set of manoeuvres and conditions. Moreover, the response of the motion cueing system should be repeatable.

The motion cueing performance signature tests provide objective evidence of the motion movements for different manoeuvres. The purpose of these tests is to demonstrate that the motion is giving appropriate cueing and that the motion cueing remains the same as time passes. For the motion cueing FSTD feature at fidelity level S, the motion system should provide high tilt coordination gain, high rotational gain and, in general, high correlation with respect to

the aircraft simulation model for the key acceleration and angular cues and coordination of acceleration and rotation cues. For a typical motion set-up, it is not possible to sustain acceleration cues for an extended duration. Therefore, the assessment should focus on the key acceleration and angular cues, specifically the moments when changes in acceleration and/or aircraft attitude occur. The motion system should provide a strong initial cue for movement. Additionally, the rotational motion used for angular cueing should exhibit high gain and correlation, while the motion platform tilting used to simulate linear acceleration may be slower to prevent the introduction of incorrect cues. These aspects are further detailed later in this GM.

Realistic motion cueing requires, among other things, an adequate motion movement envelope, the capability to perform quick motion movements and good motion cueing algorithms. The movement envelope is limited by the physical dimensions (e.g. stroke length), geometry (e.g. attachment points of the legs), cueing algorithms and platform safety buffers. A Stewart platform is a widely used six degrees of freedom motion system platform. Such motion systems are capable of tilting (pitch, roll) the flight deck to approximately the same tilt angle regardless of the leg stroke length. However, the stroke length has a clear impact on the translation (surge, sway, heave) and yaw envelope: the shorter the stroke, the shorter the achievable translation and yaw excursion. A Stewart platform is able to sustain acceleration cues on the longitudinal and lateral axes for extended periods by tilting the flight deck. In contrast, vertical axis acceleration cueing relies on the actual up-and-down movement of the flight deck. Due to the limited stroke length, a Stewart platform can provide only momentary vertical acceleration cues and is therefore unable to maintain long-term cues along this axis.

The pilot seats are typically located well above the motion centroid, which means that the pilots are situated on top of a 'lever arm'. If the flight deck is tilted backwards, the pilots also move backwards. In other words, the motion movement has cross-coupling: movement on one axis generates movement on another axis. Humans are able to register all these movements, so the unwanted cross-coupling needs to be compensated for by translating the motion system together with the tilting. This is done by the motion cueing algorithms. It is laborious to tune the motion algorithms to provide only correct cues, while still maintaining the maximum level of cueing. The motion cueing algorithms may use only a portion of the motion's physical envelope but may use the whole envelope, especially for certain manoeuvres (e.g. helicopter autorotation). The motion cueing performance signature tests should demonstrate that the motion provides adequate cueing and that the motion cueing uses the motion envelope to the maximum extent possible, as required by the general requirements in CS FSTD.QB.110.

The motion gives short-term cues (e.g. due to a change in aircraft attitude) and long-term cues (e.g. due to airspeed acceleration). The long-term cue for airspeed acceleration for an aeroplane FSTD is to tilt the motion platform and flight deck backwards to use gravity to create the sensation of forward acceleration (refer to Figure 1). The motion system cannot maintain large excursions for a prolonged time due to the limitations of the motion platform (depending on actuator stroke lengths) and because it has to be ready for the next movement, which could require more displacement. If an aeroplane is stationary on the runway with take-off thrust applied against the brakes and the brakes are suddenly released, the motion system typically does the following. The motion platform moves forwards to give an initial cue for a forward acceleration. Simultaneously, the motion system tilts backwards to give a sustained

long-term cue for the forward acceleration. Soon after the tilting is performed, the motion system may begin a washout. This means that the motion system returns slowly towards a neutral attitude. This is done at such a low rate that it remains below the perception threshold of the semicircular canals in the human vestibular system, making it unnoticeable, especially if other cues (e.g. visual, auditory or instrument indications) support the sensation of continued acceleration. For the sway and surge axes, depending on the chosen compromise within the cueing algorithm and on the amplitude of the aircraft acceleration, the cueing algorithm can either favour a high initial acceleration that is close to the aircraft initial acceleration, at the expense of reducing the maintained acceleration cues when washout activates, or it can favour the same acceleration trend as the aircraft without reducing acceleration cues during washout, at the expense of a reduced initial acceleration. The short- and long-term cues and washout can be seen in the graphs of the motion cueing performance signature tests. The motion system cannot tilt too quickly, since the human vestibular system is able to feel angular rates above the perception threshold and users may feel sick or disoriented if the motion platform tilting rate is above this threshold, with it not being consistent with the visual cues. The motion platform pitch and roll angles should also be limited to appropriate values, since humans are able to feel not only significant angular rates but also significant attitudes.

Additional information can be found in, among other sources, Telban, R. J., *A nonlinear motion cueing algorithm with a human perception model*, PhD report, State University of New York, Department of Mechanical Engineering, Binghamton, 2001.

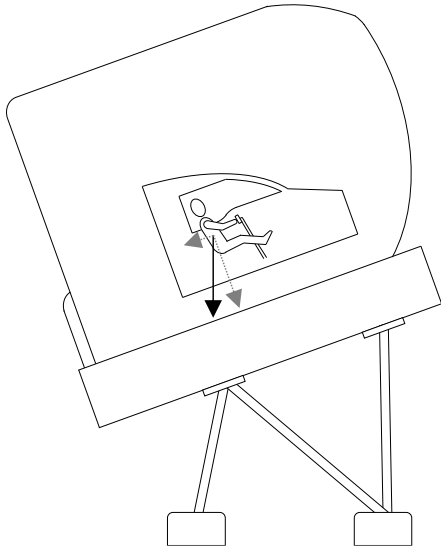


Figure 1. Motion system tilted backwards to use the component of the gravity vector in forward acceleration cueing

EXAMPLE OF A MOTION CUEING PERFORMANCE SIGNATURE TEST

Figures 2–5 below show the results for a simplified example of a motion cueing performance signature test for an aeroplane FSTD take-off. This example shows some of the FSTD result graphs from the QTG, which would be delivered to the competent authority for approval as the MQTG. The future recurrent test results would then be compared with the MQTG result to ensure that no change in the motion cueing had happened.

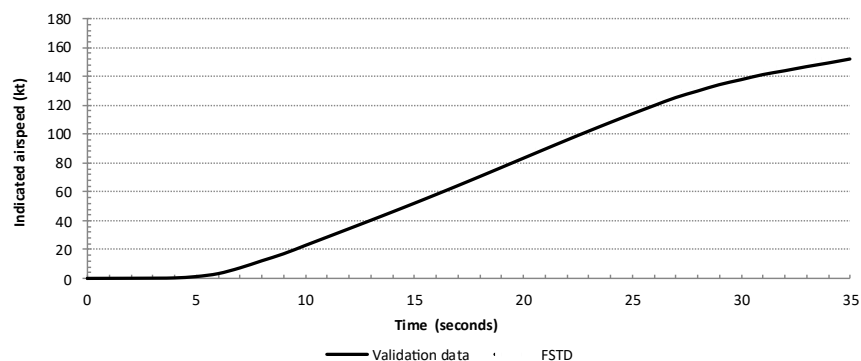


Figure 2. Aeroplane airspeed during take-off

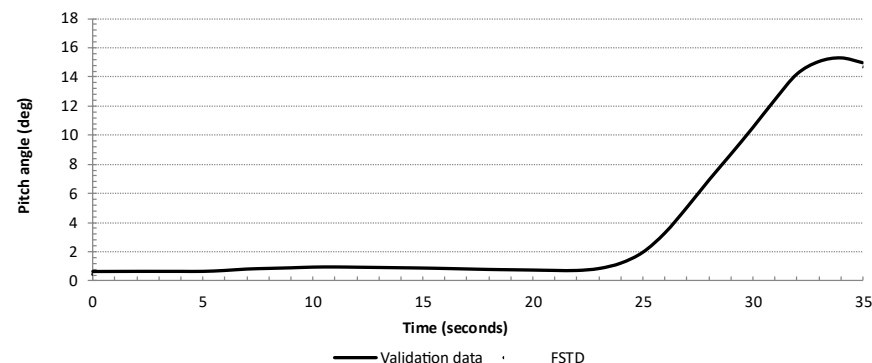


Figure 3. Aeroplane pitch angle during take-off

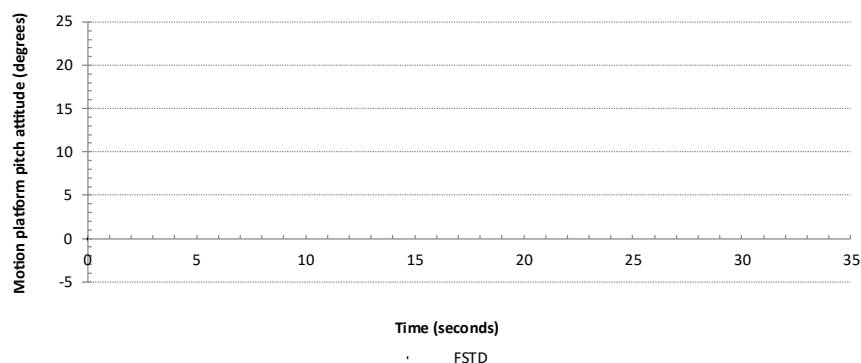


Figure 4. Motion system platform tilting on the pitch axis for take-off

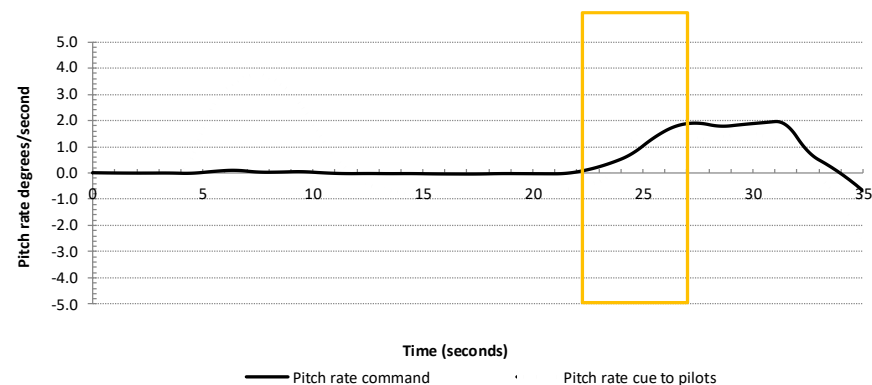


Figure 5. Correlation of the motion system platform tilting on the pitch axis versus the aeroplane, with the period of key angular cue highlighted in the amber box

The inputs in this test are the aircraft flight parameters, and the output is the motion movement. The test results should show graphs of all the applicable flight parameters (e.g. airspeed, pitch attitude, bank angle, heading, flight control positions, altitude, thrust), the motion movement (e.g. motion translations and tilt angles) and a comparison between the commanded and actual motion cues in terms of key acceleration and angular rates.

Figure 2 shows the airspeed. Figure 3, for pitch attitude, shows that the rotation is initiated at about 23 seconds. These graphs show that the FSTD performance matches the validation data.

Figure 4, for the motion platform, shows a footprint of the motion tilt angle. When forward acceleration of the ownship begins, the motion system combines linear acceleration with a backward tilt to use gravity as a cue for forward acceleration. Figure 4 shows the latter effect (i.e. tilting the motion platform to give a sustained cue of forward acceleration by using gravity). The tilt angle depends on the motion cueing algorithms and motion performances (e.g. actuator stroke) but also on the conditions (e.g. weight of aircraft, density altitude) and configuration (e.g. thrust, flaps). If the load factor would be too low, the pilots would not sense an adequate feeling of acceleration. The washout begins at about 12 seconds. The platform is slowly returned towards the neutral position to be ready to give a short-term linear or angular cue (e.g. pitch angle change at about 24 seconds).

Figure 5 shows the correlation between the motion system's angular pitch rate cue and the ownship during take-off. The key angular cue in this test is the rotation, which occurs after 23 seconds and is highlighted with amber colour. Figure 5 shows that, during rotation, the motion system's angular cue in this example corresponds to that of the simulation model. Similar assessments should be performed for all key acceleration and angular cues required for all the tests. The assessment of correlation should be focused on the key acceleration and angular cues for the periods of their duration.

It is important to note that the assessment of the graph in Figure 5 should focus solely on the key angular cue (after approximately 23 seconds), which represents the motion system's response to the aircraft's rotation. Other parts of the graph reflect motion responses to x-axis acceleration and long-term cueing effects resulting from platform tilting. The period between 4 and 11 seconds in Figure 5 may appear to present a false cue, but it is actually caused by the tilting of the motion platform, which generates a sustained forward acceleration cue (refer to Figures 4 and 2). This tilting of the motion platform (between 4 and 11 seconds) produces a pitch rate that is also perceivable by the pilot. The pilots are seated above the motion centroid, effectively positioned on top of a 'lever arm'. A tilt of the motion platform would feel like a false motion cue unless it is compensated by translating the motion system in coordination with the tilt, in order to maintain the position of the pilots' vestibular sensory systems. Therefore, the other graphs in this test should indicate that a translation occurs simultaneously with the backward tilt of the motion platform. The evaluation and statement of compliance for these tests should focus on the integrated cueing as a whole. The example in Figure 5 highlights the importance of recognising and distinguishing key cues in motion cueing (refer to FSTD.QTG.A.300 above) and of providing appropriate information in the SOJ.

For a six-DOF Stewart platform, ensuring that the perceived angular rates align with those of the simulated vehicle is essential for realistic motion cueing. The human vestibular system detects angular velocity and acceleration, so discrepancies between the platform's rotational motion and the visual simulation can lead to cue conflicts, reducing immersion and potentially causing motion sickness. However, due to the platform's limited range of motion, exact replication of the vehicle's angular rates is not always feasible. To address this, motion cueing algorithms apply filtering techniques, such as washout filters and high-

pass/low-pass separation, to maintain realism while considering platform constraints. These algorithms must provide motion cues that feel natural while preventing excessive platform displacement that could lead to mechanical limitations or unrealistic sensations.

Figures 2–5 demonstrate that the motion cueing has the expected elements for short- and long-term cueing (i.e. high-pass and low-pass filters of the motion cueing algorithms) and washout. The motion cueing must have a high correlation with respect to the aircraft simulation model. This can be demonstrated by overlaying the plots of the aircraft acceleration and angular rates with the cues given to the pilots, as shown in Figure 5. The aircraft acceleration (i.e. acceleration command) is given directly by the simulation model. The acceleration cue given to the pilots can be measured using the accelerometers in the FSTD, or it can be derived from the attitude and position of the motion system (see the two components of the gravity vector in Figure 1).

GUIDANCE ON THE CORRELATION OF MOTION CUEING WITH THE AIRCRAFT SIMULATION MODEL

Figure 5 and associated guidance give an example of the correspondence of a key angular cue. Figures 6 and 7 show two examples of the plot for the cueing correlation from another take-off motion cueing signature test. Note that these figures do not correspond to the test used in Figures 2–5 and show a different aeroplane take-off case. The solid line shows the forward acceleration (x-axis) as calculated by the simulation model. The dotted line shows the forward acceleration cue provided by actual linear acceleration combined with tilting the motion system backwards.

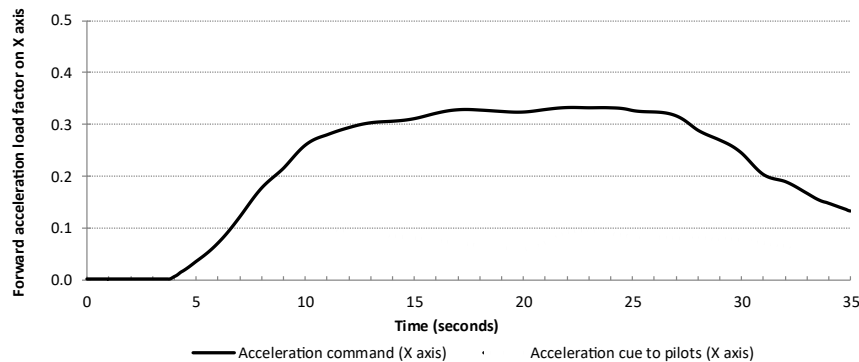


Figure 6. Example of a quick initial movement but limited sustained cue

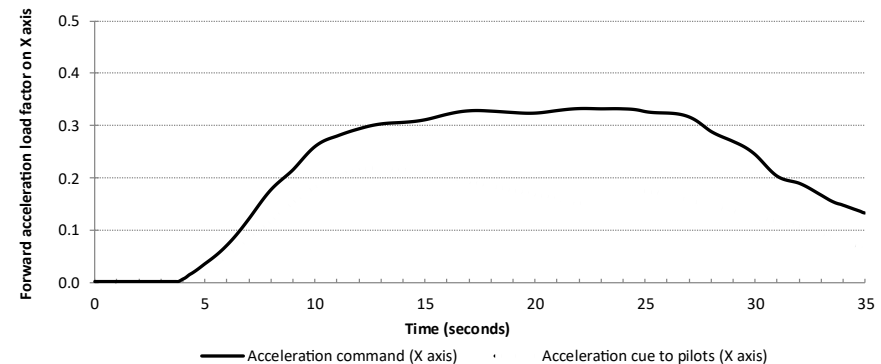


Figure 7. Example of a slower initial movement but stronger sustained cue

Figure 6 demonstrates a cueing algorithm that provides a quick initial reaction (see the period between four and seven seconds) but has limited sustained cues due to platform excursion limits. The washout activates after the initial acceleration to provide the motion platform with additional margins for

subsequent cueing; if properly justified (e.g. at a sufficiently slow rate), the washout should not be considered as failing to meet the test tolerances specified in CS FSTD.QTG.A.300.

Figure 7 demonstrates a cueing algorithm that is less responsive (see the period between 4 and 12 seconds) but is capable of greater amplitudes.

While this correlation has no numerical tolerances, the cueing should have a high correlation with respect to the aircraft simulation model. The cueing depends on the platform excursion limits, and the cueing algorithms may keep some margin for further acceleration changes. In addition, the motion cueing algorithms may not use a washout for all the axes but use a deliberately lower but sustained cue. These factors affect the cueing, and thus the statement of justification should give the basis for the magnitude and correlation.

Figures 6 and 7 show how the correlation graphs between the commanded acceleration are plotted. The results in Figures 6 and 7 demonstrate differences between the cueing and the commanded acceleration. The SOJ should address these differences, explain the motion movements and give the basis for their magnitude and correlation for each axis for each test case. Similar correlation graphs for the translations and short-term cueing (rotation, in this example) should also demonstrate a high correlation with other key acceleration and angular cues (refer to Figure 5 as an example).

The FSTDs and motion systems are different. The shape of the motion movement time histories is different for each device, but their movements should demonstrate the aspects presented above and provide effective cueing. Similar aspects have to be considered in the assessment of all the motion cueing performance signature tests. These aspects should also be noted while doing functions and subjective testing.

Instead of plotting the three components of the measured load factor vector at the pilot position, plotting the norm of the load factor vector and its direction in pitch and in roll may allow for an easier interpretation of the results.

The tests should be performed with the vibration module off, for better readability of the acceleration and angular cues and to ensure the repeatability of the results.

GM2 FSTD.QTG.A.300 Motion system – Aeroplane

ASSESSMENT OF THE MOTION CUEING PERFORMANCE SIGNATURE TESTS

This GM gives guidance on the assessment of the motion cueing performance signature tests for the motion cueing FSTD feature at the specific fidelity level.

(a) For initial qualification, the following apply.

(1) Development and tuning of the motion system

The development and tuning of the motion system is an iterative process that requires subjective assessment. This process should take into account the aspects outlined in GM1 FSTD.QTG.A.300. Once the motion cueing is considered ready and finalised, the required footprints are recorded and the corresponding time history plots are generated.

(2) Statement of justification

The SOJ should be prepared so that it provides separate justifications for the objective test results of all required 3.f objective test cases. It should include a written explanation describing and justifying why each motion cueing performance signature test case demonstrates a high correlation between the motion cueing and the aircraft simulation model for the key acceleration and angular cues. Note that the key cues may be either long in duration or momentary in their nature.

The SOJ should clearly identify the time periods during which the key acceleration or angular cues are present in the time histories. In addition, the SOJ should explain the motion system movements (e.g. use of washout, cross-coupling effects) and provide a rationale for the magnitude of movement in each axis for each test case.

The SOJ should also include a description of the overall motion cueing design principle or concept. This description should explain the intended balance between sustained and transient cueing, the emphasis on specific types of motion cues (e.g. long-duration accelerations or short, sharp onset cues) and how these choices reflect the handling characteristics and operational environment of the simulated aircraft.

The SOJ should take into account the aspects outlined in GM1 FSTD.QTG.A.300. The design of the motion system may, for example, be intended to provide fast-response cueing with lower cueing strength or slower-response cueing with higher cueing strength. Such design choices dictate certain expected cueing characteristics and should be described in the SOJ.

(3) Required plots

Plots showing the correlation between motion cueing and the aircraft simulation model, specifically in terms of load factor and angular cues, are required only for the initial qualification. These plots should be included in the MQTG. Together with the footprints and the SOJ, they are intended to introduce objectivity into the evaluation of the motion cueing performance signature tests and to ensure that the motion movements and cueing strength are clearly understood and provide appropriate cues.

(4) Cue consistency and washout

It is important that the motion cues remain consistent. Cross-coupling effects should be well coordinated and applied in a timely manner. The motion system should not produce false cues, such as uncommanded movements, incorrect timing, a transient loss of cue fidelity or a lack of coordination in cross-coupling responses.

Washout is not considered a false cue if it remains below the perception threshold and is properly justified. If washout is applied, the motion cueing deviates from the acceleration commanded by the simulation model, and this deviation should be visible in the time history plots. The SOJ should explain the use of washout for each test case individually.

(5) The focus of the assessment

The test assessment in the initial qualification should focus on the correlation between motion cueing and the aircraft simulation model for the key acceleration and angular cues (as specified in the motion cueing performance signature test requirements in FSTD.QTG.A.300). Simplified examples can be seen in Figures 5, 6 and 7 of GM1 FSTD.QTG.A.300. In particular, the focus of Figure 5 is the rotation in the period after about 23 seconds. This is the moment of the significant changes in the key angular cue. The correlation between motion cueing and the aircraft simulation model should be high for the rotation pitch rate. Similar applies to other key acceleration and angular cues. The correlation between the motion cueing and the given acceleration or angular cue input should be high. This should be interpreted as correspondence between the two graphs in terms of CT&M (refer to GM1 FSTD.ENG.020(c)(5)), while also accounting for washout and motion system limitations.

(b) For recurrent testing, the following apply.

(1) The purpose of the recurrent tests

The purpose of the recurrent tests is to demonstrate that there has been no perceivable change in the motion cueing. The tests are performed with the motion system operating.

(2) The content of the tests

The tests should show full time histories of the test cases and overplot the recurrent result with the MQTG results for all the required parameters. The comparison graphs for the motion cueing and the aircraft's key acceleration and attitude, as provided by the simulation model, are not required for recurrent tests. In addition, the SOJ need not be updated unless the motion cueing and footprints are modified.

(3) Evaluation criteria

The tolerances for recurrent tests are specified in the 3.f objective test requirements and apply between the objective test results and the MQTG.

CS FSTD.QTG.A.305 Characteristic vibrations/buffet – Aeroplane

This CS provides the standards for the objective tests for aeroplane characteristic vibrations that can be sensed in the flight deck, as applicable, by aeroplane type.

3.g: Characteristic vibrations		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
3.g.1	Thrust effects with brakes set	Ground	Frequency of vibration	VIB		Init: Subjective vibration assessment to establish the MQTG amplitude versus frequency footprint Rec: The FSTD should exhibit the same overall appearance and trends as in the MQTG	Init: The FSTD should exhibit the overall appearance and trends of the validation data, with at least three (3) of the predominant frequency 'spikes' being present within ± 2 Hz of the validation data Rec: Same as Init
3.g.2	Landing-gear-extended buffet	Flight	Frequency of vibration	VIB		Same as 3.g.1	Same as 3.g.1
3.g.3	Flaps-extended buffet	Flight	Frequency of vibration	VIB		Same as 3.g.1	Same as 3.g.1
3.g.4	Speed-brake-deployed buffet	Flight	Frequency of vibration	VIB		Same as 3.g.1	Same as 3.g.1

3.g: Characteristic vibrations		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
3.g.5.a	Full-stall buffet	Cruise (high altitude), second-segment climb and either approach or landing	Frequency of vibration	VIB			Same as 3.g.1
3.g.5.b	Approach-to-stall buffet	Refer to the test requirement table below for test 3.g.5.b	Frequency of vibration	VIB		Same as 3.g.1	Same as 3.g.1
3.g.6	High-speed or Mach buffet	Flight	Frequency of vibration	VIB		Same as 3.g.1	Same as 3.g.1
3.g.7	In-flight vibrations	Flight in clean configuration	Frequency of vibration	VIB		Same as 3.g.1	Same as 3.g.1

Requirements for all the aeroplane characteristic vibration tests.

3.g: Characteristic vibrations	Test requirements
All tests	- The tests should validate amplitude versus frequency. The tests should cover a frequency range of at least 1–20 Hz. - Acceleration power spectral density (PSD) against frequency (in g's root mean squared (grms²)/Hz) for the x-, y- and z-axes should be plotted. - If the amplitude on any axis is negligible, its plots may be ignored, provided that the SOJ explains this. - For feature VIB at fidelity level S, the following apply. - The validation data and FSTD results should be produced using comparable data analysis techniques. The recorded test results for characteristic buffets should allow for the comparison of relative amplitude versus frequency. - The tolerances 'at least three of the predominant frequency "spikes" being present within ± 2 Hz of the validation data' means that at least one spike at low-frequency range (1–4 Hz), mid-frequency range (4–10 Hz) and HF range (10–25 Hz) should be present within ± 2 Hz of the validation data. However, if the predominant frequencies of the

3.g: Characteristic vibrations	Test requirements
	aeroplane are close to each other (e.g. all the spikes are in the same mid-frequency range), then the three dominant spikes should be present within ± 2 Hz of the validation data. 5. For feature VIB at fidelity level R, the following apply. a. Only footprint tests are required. b. The recurrent tests are required to exhibit the same overall appearance and trends as in the MQTG. It is acknowledged that the vibration test results may have some variation when compared with those in the MQTG due to the random nature of the vibration/buffet and due to the mathematical process used to plot the amplitude against frequency (i.e. Fourier transformation). 6. The tests may give the test conditions and flight parameters as numerical snapshot values, provided that flight phases are stable.

Individual test requirements. The individual test requirements defined in the table below are applicable to all fidelity levels unless otherwise stated.

3.g: Characteristic vibrations		Test requirements
3.g.1	Thrust effects with brakes set	1. Test should be conducted at the maximum possible thrust with brakes set.
3.g.2	Landing-gear-extended buffet	1. Test condition should be a normal operational speed and not the gear-limiting speed.
3.g.3	Flaps-extended buffet	1. Test condition should be a normal operational speed and not the flap-limiting speed.
3.g.4	Speed-brake-deployed buffet	1. Test condition should be a typical speed for speed brake extension.
3.g.5.a	Full-stall buffet	1. Tests are applicable only to aeroplane FSTDs that are to be qualified for full-stall training tasks. 2. Tests should be conducted for an angle-of-attack range between the buffet threshold of perception to the pilot and the stall angle of attack. Post-stall characteristics are not required. 3. Test results may be displayed as snapshots for a sample of relevant angle-of-attack conditions (buffet threshold of perception, initial buffet, buffet onset, stall angle of attack). 4. If stabilised flight data between the buffet threshold of perception and stall angle of attack is not available, PSD

3.g: Characteristic vibrations		Test requirements
		analysis should be conducted for a time span between the initial buffet and stall angle of attack.
3.g.5.b	Approach-to-stall buffet	1. Test is applicable to those FSTDs that are to be qualified for approach-to-stall training tasks. 2. Test is required only for those aeroplanes for which approach-to-stall buffet is the first indication of an impending stall event (e.g. approach-to-stall buffet occurs before the activation of the stall warning system). 3. Test conditions should be where approach-to-stall buffet is the first indication of an impending stall event. 4. Test should be conducted for an angle-of-attack range between the buffet threshold of perception to the pilot and the initial buffet. 5. Test results may be displayed as snapshots for a sample of relevant angle-of-attack conditions (buffet threshold of perception, initial buffet). 6. If stabilised flight data between the buffet threshold of perception and initial buffet is not available, PSD analysis should be conducted for a time span around the occurrence of the initial buffet.
3.g.6	High-speed or Mach buffet	1. Test condition should be high-speed manoeuvre buffet / wind-up turn or, alternatively, Mach buffet.
3.g.7	In-flight vibrations	1. Test should be conducted to be characteristic of in-flight vibrations for propeller-driven aeroplanes.

GM1 FSTD.QTG.A.305 Characteristic vibrations/buffet – Aeroplane

(a) The presentation of results is as follows.

The characteristic vibrations/buffets are a means to verify that the FSTD can reproduce the frequency content of the aircraft when flown in particular conditions. The test results should be presented as a PSD plot with frequencies on the horizontal axis and amplitude on the vertical axis. The aircraft data and FSTD data should be presented in the same format with the same scaling. The algorithms used for generating the FSTD data should be the same as those used for the aircraft data. If they are not the same, then the algorithms used for the FSTD data should be proved to be sufficiently comparable. As a minimum, the results along the dominant axes should be presented and a rationale for not presenting the other axes should be provided.

(b) The interpretation of results is as follows.

- (1) For feature VIB at fidelity level S, the following applies.

The overall trend of the PSD plot should be considered while focusing on the dominant frequencies. Humans are particularly sensitive to vibratory accelerations occurring in the frequency range from 1 Hz to 10 Hz, so the primary emphasis of test assessment should be on this range. Less emphasis should be placed on the differences at the HF and low-amplitude portions of the PSD plot.

During the analysis, it should be taken into consideration that certain structural components of the FSTD have resonant frequencies that are filtered and thus may not appear in the PSD plot. If such filtering is required, the notch filter bandwidth should be limited to 1 Hz to ensure that the buffet feel is not adversely affected. In addition, the SOJ should explain this and confirm that the characteristic vibration is not being adversely affected by the filtering.

The amplitude should match the aircraft data as per the description below. However, if for subjective reasons the PSD plot is altered, the SOJ should note this. The human pilot can perceive vibrations over an energy range of several orders of magnitude, and so the power spectrum plots are normally rectilinear displays of log magnitude versus frequency. A buffet of 1×10^{-3} grms²/Hz would be a heavy buffet. On the other hand, a 1×10^{-6} grms²/Hz buffet is barely perceivable but may represent a buffet at low speed. The previous two examples could differ in magnitude by a factor of 1 000. On a PSD plot, this represents three decades (one decade is a change in order of magnitude of 10, two decades is a change in order of magnitude of 100, etc.).

- (2) For feature VIB at fidelity level R, the following applies.

The tests may be footprint tests. While the vibrations amplitudes and frequencies may be derived from relevant sources of design and/or validation data, the vibration performance of the FSTD should be subjectively assessed to establish the MQTG footprint test results. Representation of some dominant frequencies is expected to be evident in the MQTG test results; however, the correlation of these frequencies with the validation data is not required and is not a tolerance parameter for fidelity level R.

CS FSTD.QTG.A.400 Visual display system – Aeroplane

This CS provides the standards for the objective tests for aeroplane FSTD visual display systems.

4.a: Visual display system		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
4.a.1	Visual field of view	n/a	Horizontal and vertical continuous FOV	VIS	Init: Minimum of 45° horizontal and 30° vertical field of view Rec: EM	Init: Minimum of 200° horizontal and 40° vertical field of view Rec: EM	Init: Minimum of 200° horizontal and 40° vertical field of view Rec: EM
4.a.2.a	System geometry – image position – collimated display	n/a	Centre of the image Difference between the left and right horizontal angles	VIS		Init: Between 0° and 2° in the horizontal plane and within ±0.25° in the vertical plane. Difference between the left and right horizontal angles of ±1° Rec: EM	Init: Between 0° and 2° in the horizontal plane and within ±0.25° in the vertical plane. Difference between the left and right horizontal angles of ±1° Rec: EM

4.a: Visual display system		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
4.a.2.b	System geometry – image position – non-collimated display	n/a	Centre of the image	VIS		Init: Less than or equal to 10° in the horizontal plane and within ±0.25° in the vertical plane Rec: EM	
4.a.2.c	System geometry – absolute geometry	n/a	All points	VIS		Init: Within the central 200° × 40°, all points on a 5° grid should fall within 3° of the design position as measured from each pilot eyepoint Rec: EM	Init: Within the central 200° × 40°, all points on a 5° grid should fall within 3° of the design position as measured from each pilot eyepoint Rec: EM

4.a.2.d	System geometry – relative geometry	n/a	Relative position from one point to the next	VIS		Init: Measurements of relative dot positions should be made every 5°. In the area from -10° to the lowest visible point at 15° azimuth inboard, 0°, 30°, 60° and 85° outboard for each pilot position, vertical measurements should be made every 1° to the edge of the visible image. The relative position from one point to the next should not exceed: zone 1, 0.075°/deg; zone 2, 0.15°/deg; and zone 3, 0.2°/deg Rec: EM	Init: Measurements of relative dot positions should be made every 5°. In the area from -10° to the lowest visible point at 15° azimuth inboard, 0°, 30°, 60° and 85° outboard for each pilot position, vertical measurements should be made every 1° to the edge of the visible image. The relative position from one point to the next should not exceed: zone 1, 0.075°/deg; zone 2, 0.15°/deg; and zone 3, 0.2°/deg Rec: EM
4.a.3	Surface resolution	n/a	Surface resolution	VIS	Init: Not greater	Init: Not greater	Init: Not greater

4.a: Visual display system		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
	(object detection)				than 4 arc minutes Rec: Same as Init	than 3 arc minutes Rec: Same as Init	than 2 arc minutes Rec: Same as Init
4.a.4	Light point size	n/a	Light point size	VIS	Init: Not greater than 8 arc minutes Rec: Same as Init	Init: Not greater than 8 arc minutes Rec: Same as Init	Init: Not greater than 5 arc minutes Rec: Same as Init
4.a.5.a	Surface contrast ratio – collimated display	n/a	Surface contrast ratio	VIS		Init: Not less than 5:1 Rec: Same as Init	Init: Not less than 8:1 Rec: Same as Init
4.a.5.b	Surface contrast ratio – non-collimated display	n/a	Surface contrast ratio	VIS		Init: Not less than 5:1 Rec: Same as Init	Init: Not less than 5:1 Rec: Same as Init
4.a.6	Light point contrast ratio	n/a	Light point contrast ratio	VIS		Init: Not less than 25:1 Rec: Same as Init	Init: Not less than 25:1 Rec: Same as Init
4.a.7	Light point brightness	n/a	Light point brightness	VIS		Init: Not less than 17 cd/m ² (5.0 ft-lamberts) from the display measured at the design eye point Rec: Same as Init	Init: Not less than 20 cd/m ² (5.8 ft-lamberts) from the display measured at the design eye point Rec: Same as Init

4.a: Visual display system		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
4.a.8	Surface brightness	n/a	Surface brightness	VIS		Init: Not less than 14 cd/m ² (4.1 ft-lamberts) from the display measured at the design eyepoint Rec: Same as Init	Init: Not less than 20 cd/m ² (5.8 ft-lamberts) from the display measured at the design eyepoint Rec: Same as Init
4.a.9.a	Black level	n/a	Black intensity	VIS		Init: Tolerance criterion is (background brightness minus black polygon brightness) < 0.015 cd/m ² (0.004 ft-lamberts) Rec: Same as Init	Init: Tolerance criterion is (background brightness minus black polygon brightness) < 0.015 cd/m ² (0.004 ft-lamberts) Rec: Same as Init
4.a.9.b	Luminance levels	n/a	Luminance levels	VIS		Init: Minimum of 11 shades of grey Rec: Same as Init	Init: Minimum of 11 shades of grey Rec: Same as Init

4.a: Visual display system		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
4.a.10	Motion blur	n/a	Pattern gap	VIS		Init: If a pattern is rotated around the eyepoint at 10°/s, the smallest detectable gap should be 6 arc minutes or less Rec: Same as Init	Init: If a pattern is rotated around the eyepoint at 10°/s, the smallest detectable gap should be 4 arc minutes or less Rec: Same as Init
4.a.11	Speckle test	n/a	Speckle contrast	VIS		Init: Speckle contrast should be < 10 % Rec: Same as Init	Init: Speckle contrast should be < 10 % Rec: Same as Init

Individual test requirements. The individual test requirements defined in the table below are applicable to all fidelity levels unless otherwise stated.

4.a: Visual display system		Test requirements
4.a.1	Visual field of view	- FOV should be measured using a visual test pattern filling the entire visual scene (all channels), consisting of a matrix of black and white 5° squares. - Installed alignment should be confirmed in an SOJ (this would generally consist of results from acceptance testing). - For feature VIS at fidelity level G, 30° vertical FOV may be insufficient to meet the requirements of the VGS (if required). This needs to be considered in the FOV calculation.
4.a.2.a	System geometry – image position – collimated display	- The image position should be checked relative to the FSTD centre line. - If there is a design offset in the vertical display centre, this should be stated.
4.a.2.b	System geometry –	The image position should be checked relative to the FSTD centre line.

4.a: Visual display system		Test requirements
	image position – non-collimated display	If there is a design offset in the vertical display centre, this should be stated.
4.a.2.c	System geometry – absolute geometry	- The procedure for this test is to display a 5° grid to show the angular spacing. The initial assessment of the visual system should be performed by using a theodolite. The recurrent testing method may use a fixed grid projected from an independent device, calibrated during the initial acceptance of the visual system and confirmed in an SOJ. If the visual system image can be shown to align with the tolerances specified, it may be assumed that the requirements are met based on the initial acceptance of the visual system. If there is any doubt about the test results or the visual system hardware parts have been modified, replaced or detached, then a theodolite measurement should be used to demonstrate compliance. - If a system of more than 200° × 40° is supplied, the geometry outside the central area should not have any distracting discontinuities.
4.a.2.d	System geometry – relative geometry	- Azimuth and elevation are relative to the aircraft centre line and to the horizon. They are positive to the right and above the horizon. - Zone 1 covers +5° to -15° vertically by ±80° horizontally. - Zone 2 covers +10° to -25° vertically by ±90° horizontally, excluding the area covered by zone 1. - Zone 3 covers +20° to -40° vertically by ±105° horizontally, excluding the areas covered by zones 1 and 2. - A light point is set every 5° in azimuth and in elevation in the whole FOV (and also outside zones 1–3 if the FOV is larger). - A light point is set every 1° at elevation -10° and downwards (up to the limit of the FOV), at azimuth -85°, -60°, -30°, -15°, 15°, 30°, 60° and 85°. - From the pilot's eye position, test should verify every visible 5° point on the vertical lines and horizontal lines. In addition, at -15°, 0°, +30°, +60° and +85° in azimuth, test should verify all visible 1° points from the -10° point to the lowest visible point. - From the copilot's eye position, test should verify every visible 5° point on the vertical lines and horizontal lines. In addition, at 15°, 0°, -30°, -60° and -85° in azimuth, test should verify all visible 1° points from the -10° point to the lowest visible point. - Tolerances apply to zones 1–3. For larger FOVs, there should be no distracting discontinuity outside zone 3.

4.a: Visual display system		Test requirements
4.a.3	Surface resolution (object detection)	- Resolution should be demonstrated by a test of objects shown to occupy the required visual angle in each visual display used on a scene from the pilot's eyepoint. - The object will subtend an angle no greater than the tolerance angle to the eye. - This may be demonstrated using threshold bars for a horizontal test. - A vertical test should also be demonstrated. - The subtended angles should be confirmed by calculations in an SOJ.
4.a.4	Light point size	- Light point size should be measured using a test pattern consisting of a centrally located single row of white light points displayed as both horizontal and vertical rows in each visual display. - It should be possible to move the light points closer to and further away from the eyepoint. - At a point at which modulation is just discernible in each visual channel, a calculation should be made to determine the light spacing. - An SOJ is required to demonstrate the test method and calculations.
4.a.5.a and 4.a.5.b	Surface contrast ratio	- Surface contrast ratio should be measured using a test pattern filling the entire visual scene (all channels). - The test pattern should consist of black and white squares, 5° per square, with a white square in the centre of each channel. - Measurement should be made on the centre bright square for each channel using a 1° luminance meter. This value should have a minimum brightness of 7 cd/m² (2 ft-lamberts). Measurements should also be taken for any adjacent dark square. - The contrast ratio is the bright square value divided by the dark square value. - During contrast ratio testing, FSTD aft-cabin and flight deck ambient light levels should be as low as possible. - Measurements should be taken at the centre of squares to avoid light spill into the measurement device.
4.a.6	Light point contrast ratio	- Light point contrast ratio should be measured using a test pattern demonstrating an area of greater than 1° filled with white light points (i.e. light point modulation just discernible) and should be compared with the adjacent background. - Measurements of the background should be taken such that the bright square is just out of the light meter FOV. - During contrast ratio testing, FSTD aft-cabin and flight deck ambient light levels should be as low as possible.
4.a.7	Light point brightness	- Light points should be displayed as a matrix creating a square. - Light points should overlap such that the square is continuous (individual light points will not be visible).
4.a.8	Surface brightness	Surface brightness should be measured on a white polygon, measuring the brightness using the 1° luminance meter.

4.a: Visual display system		Test requirements
4.a.9.a	Black level	1. The light meter should be mounted in a fixed position viewing the forward centre area of each display. 2. All projectors should be turned off and the flight deck environment made as dark as possible. A background reading should be taken of the remaining ambient light on the screen. 3. The projectors should then be turned on and a black polygon displayed. A second reading should then be taken and the difference between this and the ambient level recorded. 4. This test is generally only required for light valve projectors.
4.a.9.b	Luminance levels	1. A test pattern consists of a pair of grey fields with opposite-direction luminance staircases between 0 % and 100 % luminance, with steps of 10 %. The pattern is displayed in each channel. 2. All grey steps should be discernible and without noticeable chromatic aberration. 3. FSTD aft-cabin and flight deck ambient light levels should be as low as possible.
4.a.10	Motion blur	1. A test pattern consists of an array of five peak white squares with black gaps of decreasing width between them. 2. The range of black gap widths should at least extend above and below the required detectable gap and should be in steps of 1 arc minute. 3. The pattern is to be rotated at the required rate. 4. Two arrays of squares should be provided, one rotating in heading and the other in pitch, to provide testing in both axes. 5. A series of stationary numbers identifies the gap number. 6. Note: This test can be limited by the display technology. If this is the case, the competent authority should be consulted on the limitations. 7. This test is generally only required for light valve projectors.
4.a.11	Speckle test	1. An SOJ describing the test method is required. 2. This test is generally only required for laser projectors.

GM1 FSTD.QTG.A.400 Visual display system – Aeroplane

This GM provides guidance on the assessment of visual display systems.

- (a) For the visual display system, the following applies.

For large FOVs, the distortion of the projection of the pattern on a sphere becomes noticeable closer to the poles. The matrix should have actual squares on the horizon (ignoring parallax error between theoretical eye position and actual pilot or copilot position) and polygon height should always be constant, but polygon width will be smaller the further away (up or down) from the horizon (packing the same number of polygons on a parallel that is smaller in diameter).

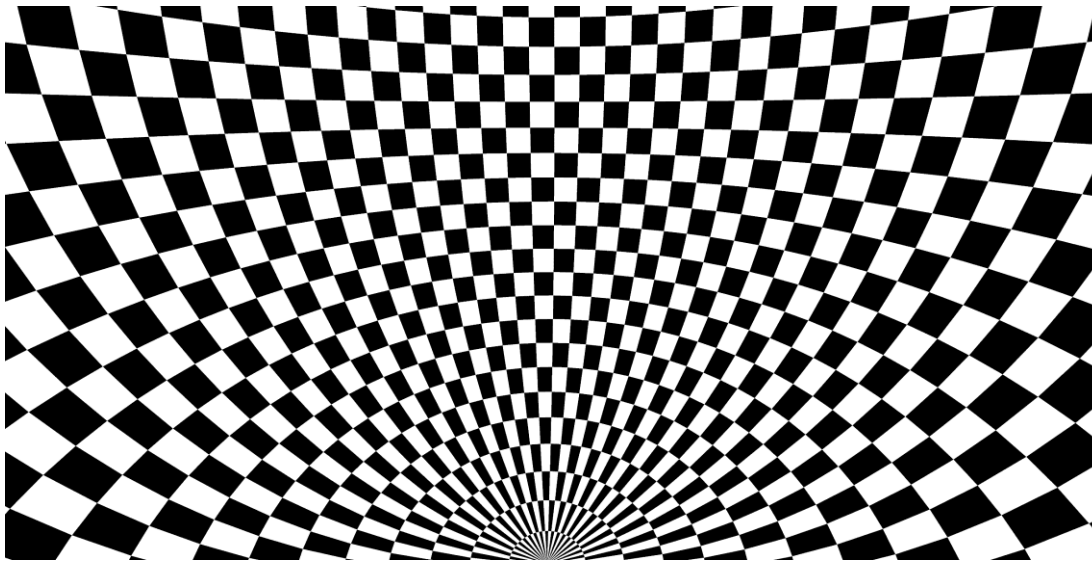


Figure 1. Matrix distortion for large FOV

- (b) For image geometry, the following applies.

The geometry of the final image as displayed to each pilot should meet the criteria defined in the tests. This assumes that the individual optical components have been tested to demonstrate a performance that is adequate to achieve this end result.

- (1) CS FSTD.QTG.A.400, test 4.a.2 (System geometry – image position), sets out the following.
 - (i) If measured from the pilot's and co-pilot's eyepoint, the centre of the image should be positioned horizontally between 0° and 2° inboard and within $\pm 0.25^\circ$ vertically for a collimated display and 0° and 10° inboard and within $\pm 0.25^\circ$ vertically for a non-collimated display relative to the FSTD centre line, taking into account any designed vertical offset.
 - (ii) The differential between the measurements of horizontal position between each eyepoint should not exceed 1°.

Note: The tolerances are based on eye spacings of up to ± 53.3 cm (± 21 in). Greater eye spacings should be accompanied by an explanation of any additional tolerance required. For non-collimated systems (test 4.a.2.b), the image is positioned at the FSTD centre line for these checks.
- (2) CS FSTD.QTG.A.400, test 4.a.2.c (Surface geometry – absolute geometry), sets out the following.

The absolute geometry of any point on the image should not exceed 3° from the theoretical position. This tolerance applies to the central minimal FOV required by test 4.a.1. For larger FOVs, there should be no distracting discontinuities outside this area.

If the calibration device is not able to produce enough light points to fully cover all intersections, then, at a minimum, the horizontal and both diagonal rows of lights should be displayed.
- (3) CS FSTD.QTG.A.400, test 4.a.2.d ('Surface geometry – relative geometry'), sets out the following.
 - (i) The relative geometry check is intended to test the displayed image to demonstrate that there are no significant changes in image size over a small angle of view. With high-detail visual systems, the eye can be a very powerful comparator to discern changes in geometric size. If there are large changes in image magnification over a small area of the picture, the image can appear to 'swim' as it moves across the mirror.
 - (ii) The typical Mylar-based mirror system will naturally tend to form a 'bathtub' shape. This can cause magnification or 'rush' effects at the bottom and top of the image. These can be particularly distracting in the lower half of the mirror when in the final approach phase and hence should be minimised. The tolerances are designed to try to keep these effects to an acceptable level while accepting that the technology is limited in its ability to produce a perfect spherical shape.
 - (iii) The FOV is divided up into three zones to set tolerances for relative geometry, as shown in Figure 2.

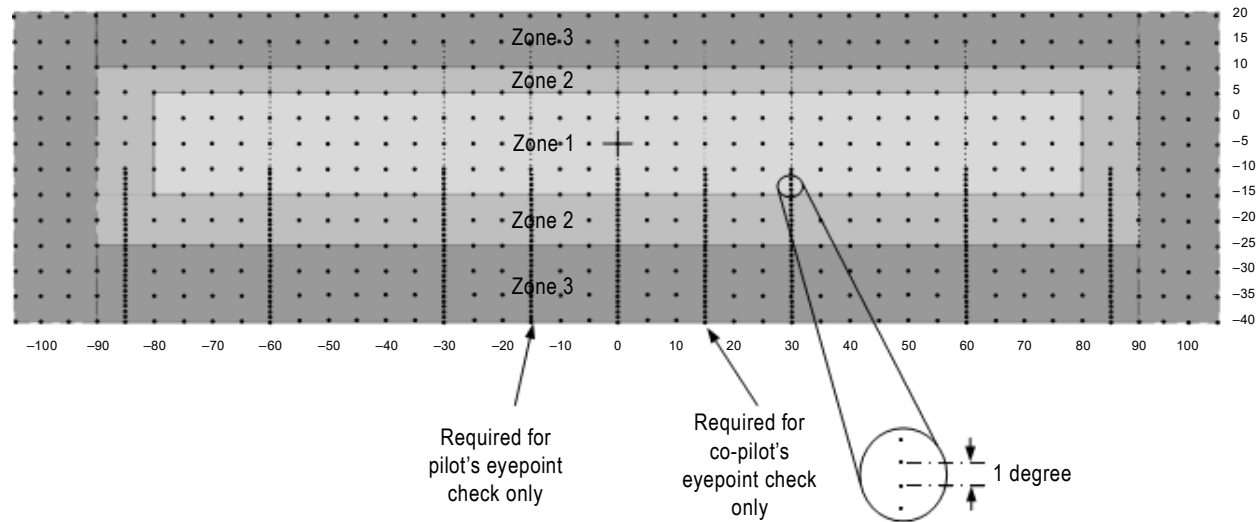


Figure 2. Relative geometry test pattern showing zones

- (iv) Testing of the relative geometry should proceed as follows. Note that not all points depicted on the pattern are measured, but they may be measured if observation suggests a problem.
- (A) The relative spacing of points should not exceed the following tolerances when comparing the gap between one pair of dots with the gap between an adjacent pair. This means:
- Zone 1 < 0.075 deg/deg
- Zone 2 < 0.15 deg/deg
- Zone 3 < 0.2 deg/deg
- (B) Here, as 5° gaps are being measured, the tolerances should be multiplied by five (e.g. one 5° gap should not be more than $(5 \times 0.075) = 0.375^\circ$ more or less than the adjacent gap when in zone 1).
- (C) For larger fields of view, there should be no distracting discontinuities outside this area.

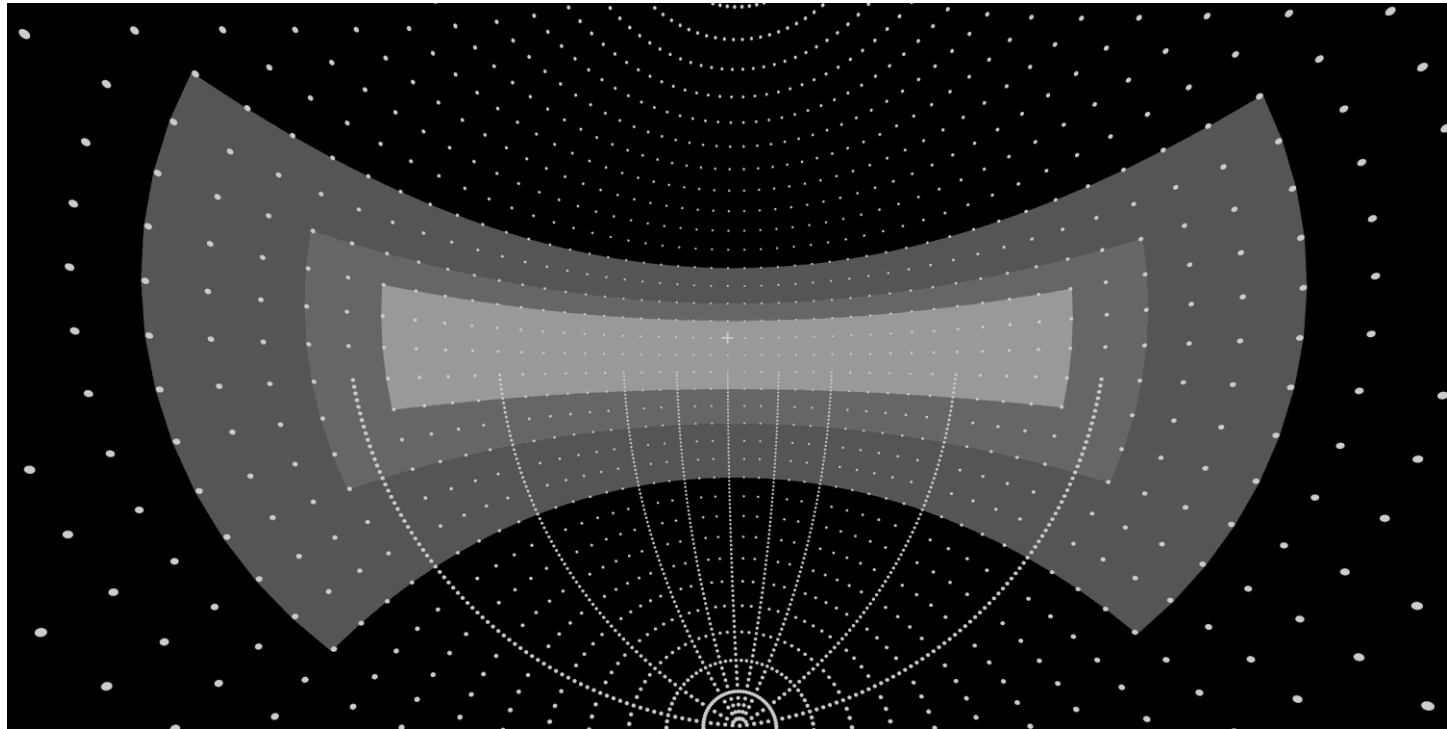


Figure 3. Relative geometry test pattern for large field of views

- (v) For recurrent testing, the use of an optical checking device is encouraged. This device should typically consist of a handheld go/no-go gauge to check that the relative positioning is maintained.

- (c) For luminance levels, the following applies.

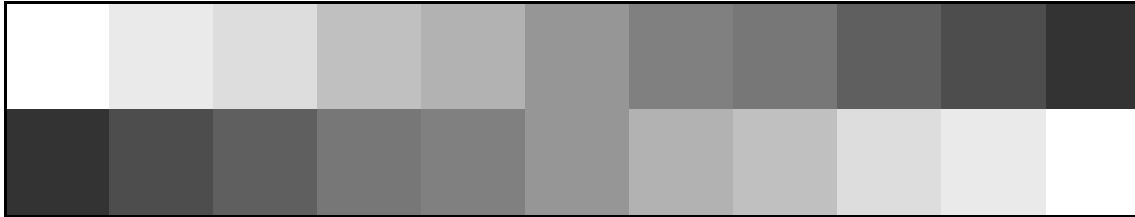


Figure 4. Luminance level pattern

As this is an integrated test, test patterns internal to the projection system should not be used.

- (d) For the laser speckle contrast ratio (laser projection system), the following applies.

The objective measure of speckle contrast that is described in the following subparagraphs considers the grainy structure of speckle and concentrates on the variations of brightness inherently introduced by speckle. Speckle contrast is quite commonly measured in many applications. However, speckle contrast does not take into account the size of the grains (i.e. the spatial wavelength of the speckle pattern).

- (1) The definition of speckle contrast ratio is as follows.

- (i) Due to its noisy character, one adequate measure used to quantify speckle is the root mean square (RMS) deviation derived from statistical theory: in a random distribution, the RMS deviation quantifies the amount of variation from the mean value.
- (ii) If applied to the intensity profile of an illuminated surface, the speckle contrast C is the RMS deviation normalised to the mean value.
- (iii) Given the intensity profile $I(x, y)$ in the considered field of view, the speckle contrast C can be defined as:

$$C = \frac{\sqrt{\langle I^2 \rangle - \langle I \rangle^2}}{\langle I \rangle},$$

where the average operator $\langle \rangle$ operating on a profile $I(x, y)$ is defined as:

$$\langle I \rangle = \frac{1}{A} \cdot \int_{\text{FOV}} I(x, y) dA$$

Hence:

$$C = \frac{\sqrt{A \cdot \int_{\text{FOV}} (I(x, y))^2 dA - \left(\int_{\text{FOV}} I(x, y) dA \right)^2}}{\int_{\text{FOV}} I(x, y) dA}$$

(2) Speckle measurement is as follows.

- (i) The intensity profile $I(x, y)$ can be measured with a charge-coupled device (CCD) camera. The set-up of the measurement (selection of lenses and CCD array) ensures that the granularity of the speckle can easily be resolved; the granularity on the CCD chip should therefore be larger than the pixel size.
- (ii) With the discrete nature of the CCD chip, $I(x, y)$ translates into an array $I_{m,n}$, while $\frac{1}{A} \cdot \int_{\text{FOV}} I(x, y) dA$ translates into $\frac{1}{m \cdot n} \cdot \sum_{\text{FOV}} I_{m,n}$.
- (iii) Therefore:

$$C = \frac{\sqrt{m \cdot n \cdot \sum_{\text{FOV}} I_{m,n}^2 - \left(\sum_{\text{FOV}} I_{m,n} \right)^2}}{\sum_{\text{FOV}} I_{m,n}},$$

where the following definitions apply.

Symbol or notation	Description	Units
Σ	Summation operator	n/a
A	Area	Arbitrary units
C	Speckle contrast	Per cent
FOV	Field of view	Degrees
I	Intensity	Arbitrary units
m	Number of pixel rows within FOV	n/a

n Number of pixel columns within FOV n/a

- (iv) Since the definition of C is also sensitive to the profile's low-frequency variations across the FOV, either the illumination together with the reflectivity of the screen should be homogeneous or the measured intensity profile should be corrected for these variations. This can be accomplished by applying a suitable high-pass filter; for example, by evaluating using sufficiently small FOVs in which low-frequency variations are negligible.
 - (v) To take into account the subjective nature of speckle, the f-number (or $f\#$, which is sometimes called the focal ratio, expressing the diameter of the entrance pupil D divided by the focal length f (i.e. D/f)) of the lens that should be used is the one as close as possible to that of the human eye. The recommended f-number is $1/16$.
- (3) For speckle test (test 4.a.11) tolerance, the following applies.
- If the speckle contrast is more than 10 %, the image begins to appear disturbed. The distractive modulation as an overlay of the image reduces the perceptibility of the projected image and then degrades the perceived resolution. With a speckle contrast below 10 %, the resolution and focus are not affected.

CS FSTD.QTG.A.405 Head-up display – Aeroplane

This CS provides the standards for the objective tests for aeroplane HUDs.

4.b: Head-up display (HUD)		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
4.b.1	Static alignment	n/a	Static alignment with displayed image	VIS		Init: ± 6 arc minutes Rec: Same as Init	Init: ± 6 arc minutes Rec: Same as Init
4.b.2	HUD attitude versus FSTD attitude indicator (pitch and roll of horizon)	Flight	Pitch and roll	VIS		Init: Pitch and roll align with aeroplane instruments Rec: EM	Init: Pitch and roll align with aeroplane instruments Rec: EM

Individual test requirements. The individual test requirements defined in the table below are applicable to all fidelity levels unless otherwise stated.

4.b: Head-up display (HUD)		Test requirements
4.b.1	Static alignment	- HUD boresight should align with the centre of the displayed image spherical pattern. - For feature VIS at fidelity level S, the alignment requirement applies to any HUD system in use or both HUDs simultaneously if they can be used simultaneously for training. - For feature VIS at fidelity level R, the alignment requirement only applies to the pilot flying.
4.b.2	HUD attitude versus FSTD attitude indicator (pitch and roll of horizon)	For feature VIS at fidelity level R, the alignment requirement only applies to the pilot flying.

CS FSTD.QTG.A.410 Enhanced flight vision system – Aeroplane

This CS provides the standards for the objective tests for aeroplane EFVSs.

4.c: Enhanced flight vision system (EFVS)		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
4.c.1	Registration test	Take-off point and on approach at 61 m (200 ft)	Alignment between EFVS display and the window image	VIS		Init: Alignment should represent the alignment typical of the aeroplane and system type Rec: Same as Init	Init: Alignment should represent the alignment typical of the aeroplane and system type Rec: Same as Init
4.c.2	EFVS RVR and visibility calibration	Flight at 350 m (1 200 ft) RVR and 1 609 m (1 sm) RVR	EFVS view including correct light intensity	VIS		Init: The scene should represent the EFVS view, including correct light intensity Rec: Same as Init	Init: The scene should represent the EFVS view, including correct light intensity Rec: Same as Init
4.c.3	Thermal crossover	Day and night	Thermal crossover effects during day-to-night transition	VIS		Init: Demonstration of thermal crossover effects during day-to-night transition Rec: Same as Init	Init: Demonstration of thermal crossover effects during day-to-night transition Rec: Same as Init

Individual test requirements. The individual test requirements defined in the table below are applicable to all fidelity levels unless otherwise stated.

4.c: Enhanced flight vision system (EFVS)		Test requirements
4.c.1	Registration test	1. For the S level, the effects of the alignment tolerance in CS FSTD.QTG.A.405, test 4.b.1 ('Static alignment') should be taken into account 2. For feature VIS at fidelity level R, the alignment requirement only applies to the pilot flying.
4.c.2	EFVS RVR and visibility calibration	1. Test should use infrared scene representative of both 350 m (1 200 ft) and 1 609 m (1 sm) RVR. 2. The visual scene may be removed.
4.c.3	Thermal crossover	1. The scene will correctly represent the thermal characteristics of the scene during a day-to-night transition.

CS FSTD.QTG.A.415 Night-vision imaging system – Aeroplane

This CS provides the standards for the objective tests for aeroplane NVISs.

4.d: Night vision imaging system (NVIS)		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
4.d.1	NVIS visible black level	Night, no moon, no stars, and visual display configured for NVIS operation	Display brightness	VIS			Init: Maximum of 0.01 cd/m ² (0.003 ft-lamberts) Rec: Same as Init
4.d.2	NVIS surface brightness	Visual display configured for NVIS operation	Display brightness	VIS			Init: Minimum of 3.4 cd/m ² (1 ft-lambert) Rec: Same as Init
4.d.3	NVIS surface contrast ratio	Visual display configured for NVIS operation	Contrast ratio through NVIS	VIS			Init: Minimum of 4:1 Rec: Same as Init
4.d.4	Lunar illumination	Night, full moon, and visual display configured for NVIS operation	Display brightness	VIS			Init: 1 cd/m ² ±0.025 cd/m ² (0.3 ft-lamberts ±0.007 ft-lamberts) Rec: EM
4.d.5	NVIS surface resolution	Night, full moon, and visual display configured for NVIS operation	Surface resolution through NVIS	VIS			Init: Not greater than 5 arc minutes Rec: EM

Requirements for all the aeroplane NVIS tests.

4.d: Night vision imaging system (NVIS)	Test requirements
All tests	1. Applicable to actual NVISs only. 2. The visual/display system should be configured for NVIS operation. 3. If measures or observations through the NVIS are required, the automatic gain function of the imaging system should be disabled (as applicable). 4. Flight deck lights should be turned off.

Individual test requirements. The individual test requirements defined in the table below are applicable to all fidelity levels unless otherwise stated.

4.d: Night vision imaging system (NVIS)	Test requirements
4.d.1 NVIS visible black level	1. The scene should be fully dark. The use of a black visual pattern on all channels is acceptable. 2. The measure is performed directly, without using the NVIS.
4.d.2 NVIS surface brightness	1. Test should use a similar pattern as for CS FSTD.QTG.A.400, test 4.a.8 ('Surface brightness'). 2. The measure is performed directly, without using the NVIS.
4.d.3 NVIS surface contrast ratio	1. Test should use a similar pattern as for CS FSTD.QTG.A.400, test 4.a.5 ('Surface contrast ratio'). 2. The measure is performed through the NVIS. 3. The gain of the NVIS should be the same when measuring the white square and the black square.
4.d.4 Lunar illumination	1. Test should use a pattern that displays a unique white 5° square in the centre of the display, with a black background. 2. The luminosity of the white square should be the same as the luminosity of a fully white Lambertian surface under a full moon at 90° elevation, with a clear atmosphere. 3. The measure is performed directly, without using the NVIS.
4.d.5 NVIS surface resolution	1. Test should use a similar pattern as for CS FSTD.QTG.A.400, test 4.a.3 ('Surface resolution (object detection)'). 2. The assessment is performed through the NVIS.

CS FSTD.QTG.A.420 Visual ground segment – Aeroplane

This CS provides the standards for the objective tests for the aeroplane VGS.

4.e: Visual ground segment		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
4.e.1	Visual ground segment (VGS)	Trimmed in the landing configuration at 30 m (100 ft) wheel height above runway threshold on glide slope at a RVR setting of 300 m (1 000 ft)	Near end: lights before threshold	VIS	Init: The correct number of approach lights within the computed VGS should be visible Rec: Same as Init	Init: The correct number of approach lights within the computed VGS should be visible Rec: Same as Init	Init: The correct number of approach lights within the computed VGS should be visible Rec: Same as Init
			Far end: lights after the threshold	VIS	Init: $\pm 20\%$ of the computed VGS Rec: Same as Init	Init: $\pm 20\%$ of the computed VGS Rec: Same as Init	Init: $\pm 20\%$ of the computed VGS Rec: Same as Init
			Threshold lights	VIS	Init: The threshold lights computed to be visible should be visible Rec: Same as Init	Init: The threshold lights computed to be visible should be visible Rec: Same as Init	Init: The threshold lights computed to be visible should be visible Rec: Same as Init

Individual test requirements. The individual test requirements defined in the table below are applicable to all fidelity levels unless otherwise stated.

4.e: Visual ground segment		Test requirements
4.e.1	Visual ground	1. If non-homogeneous fog is used, the vertical variation in horizontal visibility should be described and be included in

4.e: Visual ground segment		Test requirements
	segment (VGS)	the slant range visibility calculation used in VGS computation.

GM1 FSTD.QTG.A.420 Visual ground segment – Aeroplane

This GM provides guidance on the assessment of the VGS test.

(a) The following apply to the VGS.

- (1) This test is designed to assess items impacting the accuracy of the visual scene presented to a pilot at DH on an ILS approach. Those items include:
 - (i) RVR/visibility;
 - (ii) glideslope and localiser modelling accuracy (location and slope) for an ILS;
 - (iii) aircraft and flight deck geometry;
 - (iv) for a given weight, configuration and speed representative of a point within the aircraft operational envelope for a normal approach and landing; and
 - (v) radio altimeter.
- (2) Altitude and RVR for the assessment have been selected in order to produce a visual scene that can be readily assessed for accuracy (RVR calibration) and in which the spatial accuracy (centre line and glide slope) of the aircraft can be readily determined using approach/runway lighting and flight deck instruments.
- (3) The MQTG should indicate the source of data (airport and runway used, ILS glideslope antenna location (airport and aircraft), pilot eye reference point, flight deck cut-off angle, threshold crossing height, etc.) used to make accurate VGS scene content calculations. Refer to Figure 1.
- (4)** Automatic positioning of the ownship on the ILS is encouraged. If such positioning is accomplished, diligent care should be taken to ensure that the correct spatial position and aircraft attitude are achieved. Flying the approach manually or auto-coupled should also produce acceptable results.

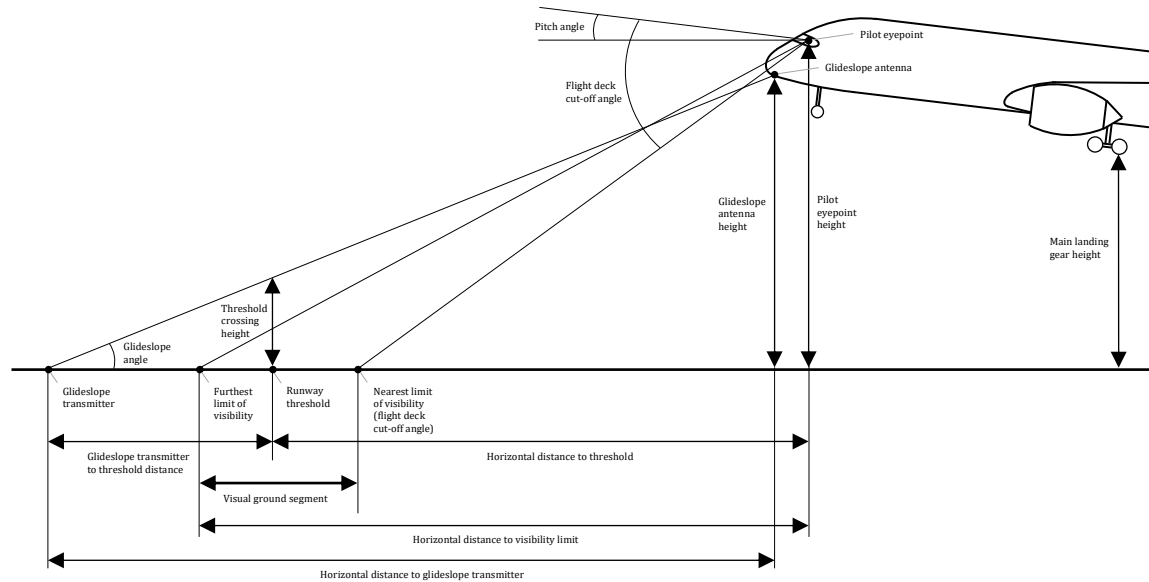


Figure 1. Visual ground segment (VGS) scene content calculations

- (b) As stated above, the visual ground segment should be conducted with the aircraft stabilised and frozen on an ILS approach. If the simulated visual scenes (refer to CS FSTD.QB.114) do not include an ILS approach, the use of a localiser performance with vertical guidance (LPV) approach may be justified. However, it should be noted that ILS and LPV approaches have different decision altitudes and system minima. If the LPV approach is flown manually or auto-coupled, the height for the visual ground testing should not be lower than the approach system minima. This also affects the required RVR setting for the test. For example, if the aircraft is trimmed in the landing configuration at 60 m (200 ft) wheel height above the touchdown zone on glidepath, the corresponding RVR is 500 m (1 700 ft).

CS FSTD.QTG.A.500 Sound systems – Aeroplane

This CS provides the standards for the objective tests for aeroplane sound systems.

5: Sound systems		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
5.a: Turbojet/turbofan aeroplanes							
5.a.1	Ready for engine start	Ground	Sound amplitude	SND	Init: Subjective sound assessment to establish MQTG footprint of overall measured sound pressure level (SPL) Rec: ±3 dB SPL RMS compared with the MQTG	Init: Subjective sound assessment to establish MQTG footprint of one-third octave bands Rec: Should not exceed ±5 dB difference per one-third octave band compared with the MQTG	Init: Within ±5 dB per one-third octave band of the validation data Rec: Same as Init
5.a.2	All engines at idle	Ground	Sound amplitude	SND	Same as 5.a.1	Same as 5.a.1	Same as 5.a.1
5.a.3	All engines at maximum allowable thrust with brakes set	Ground	Sound amplitude	SND	Same as 5.a.1	Same as 5.a.1	Same as 5.a.1
5.a.4	Climb	En route climb	Sound amplitude	SND	Same as 5.a.1	Same as 5.a.1	Same as 5.a.1
5.a.5	Cruise	Cruise	Sound amplitude	SND	Same as 5.a.1	Same as 5.a.1	Same as 5.a.1

5: Sound systems		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
5.a.6	Speed brake / spoilers extended (as appropriate)	Cruise	Sound amplitude	SND	Same as 5.a.1	Same as 5.a.1	Same as 5.a.1
5.a.7	Initial approach	Approach	Sound amplitude	SND	Same as 5.a.1	Same as 5.a.1	Same as 5.a.1
5.a.8	Final approach	Landing	Sound amplitude	SND	Same as 5.a.1	Same as 5.a.1	Same as 5.a.1
5.b: Propeller aeroplanes							
5.b.1	Ready for engine start	Ground	Sound amplitude	SND	Same as 5.a.1	Same as 5.a.1	Same as 5.a.1
5.b.2	All propellers feathered, if applicable	Ground	Sound amplitude	SND	Same as 5.a.1	Same as 5.a.1	Same as 5.a.1
5.b.3	Ground idle or equivalent	Ground	Sound amplitude	SND	Same as 5.a.1	Same as 5.a.1	Same as 5.a.1
5.b.4	Flight idle or equivalent	Ground	Sound amplitude	SND	Same as 5.a.1	Same as 5.a.1	Same as 5.a.1
5.b.5	All engines at maximum allowable power with brakes set	Ground	Sound amplitude	SND	Same as 5.a.1	Same as 5.a.1	Same as 5.a.1
5.b.6	Climb	En route climb	Sound amplitude	SND	Same as 5.a.1	Same as 5.a.1	Same as 5.a.1
5.b.7	Cruise	Cruise	Sound amplitude	SND	Same as 5.a.1	Same as 5.a.1	Same as 5.a.1

5: Sound systems		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
5.b.8	Initial approach	Approach	Sound amplitude	SND	Same as 5.a.1	Same as 5.a.1	Same as 5.a.1
5.b.9	Final approach	Landing	Sound amplitude	SND	Same as 5.a.1	Same as 5.a.1	Same as 5.a.1
5.c–e: All aeroplanes							
5.c	Special cases	As defined by applicant for FSTD qualification	Sound amplitude	SND	Same as 5.a.1	Same as 5.a.1	Same as 5.a.1
5.d	FSTD background noise	n/a	Sound amplitude	SND	Init: Subjective assessment of measured overall SPL Rec: ± 3 dB SPL RMS compared with the MQTG	Init: Each one-third octave band dB level should be lower than a reference line linearly interpolated between 70 dB at 50 Hz, 55 dB at 1 kHz and 30 dB at 16 kHz Rec: Same as Init, and ± 3 dB per one-third octave band compared with the MQTG	Init: Each one-third octave band dB level should be lower than a reference line linearly interpolated between 70 dB at 50 Hz, 55 dB at 1 kHz and 30 dB at 16 kHz Rec: Same as Init, and ± 3 dB per one-third octave band compared with the MQTG

5: Sound systems		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
5.e	Frequency response	n/a	Sound amplitude	SND		Init: n/a Rec: Should not exceed ± 5 dB on three consecutive bands when compared with the MQTG, and the average sum of the absolute differences of all bands between the MQTG and recurrent results should not exceed 2 dB	Init: n/a Rec: Should not exceed ± 5 dB on three consecutive bands when compared with the MQTG, and the average sum of the absolute differences of all bands between the MQTG and recurrent results should not exceed 2 dB

Requirements for all the aeroplane sound system tests.

5: Sound systems	Test requirements
All tests	- For feature SND at fidelity levels G and R, the following applies. The sound performance of the FSTD should be based on subjective tuning from pilot assessment. After this tuning has been completed, the applicable sound tests in Sections 5.a, 5.b and 5.c should be baselined to create footprint test results that will become the MQTG results after the initial evaluation. - For feature SND at fidelity level G, the following applies. All required tests in this section may be presented as a single overall SPL level and are only required if the flight deck is fully or partially enclosed (e.g. installed in a dedicated room).

5: Sound systems	Test requirements
	3. For feature SND at fidelity levels R and S, the following applies. All required tests in this section should be presented using an unweighted one-third octave band format from bands 17–42 (50–16 kHz). A minimum of 20 seconds average should be taken at the location corresponding to the validation dataset. 4. The tests may give the test conditions and flight parameters as numerical snapshot values.

Individual test requirements. The individual test requirements defined in the table below are applicable to all fidelity levels unless otherwise stated.

5: Sound systems		Test requirements
5.a.1	Ready for engine start	1. Normal conditions prior to engine start should be used. The APU should be on if appropriate.
5.a.2	All engines at idle	1. Normal conditions prior to take-off should be used.
5.a.3	All engines at maximum allowable thrust with brakes set	1. Normal conditions prior to take-off should be used. 2. The test is intended to check the maximum stabilised allowable thrust with brakes set, without jeopardising the aeroplane and safety.
5.a.4	Climb	1. Medium altitude should be used.
5.a.5	Cruise	1. Normal cruise configuration should be used.
5.a.6	Speed brake / spoilers extended (as appropriate)	1. Normal and constant speed brake deflection for descent at a constant airspeed and power setting should be used.
5.a.7	Initial approach	1. Constant airspeed should be used, with gear up and flaps/slats as appropriate.
5.a.8	Final approach	1. Constant airspeed should be used, with gear down and landing configuration flaps.
5.b.1	Ready for engine start	1. Normal conditions prior to engine start should be used. The APU should be on if appropriate.

5: Sound systems		Test requirements
5.b.2	All propellers feathered, if applicable	1. Normal conditions prior to take-off should be used.
5.b.3	Ground idle or equivalent	1. Normal conditions prior to take-off should be used.
5.b.4	Flight idle or equivalent	1. Normal conditions prior to take-off should be used.
5.b.5	All engines at maximum allowable power with brakes set	1. Normal conditions prior to take-off should be used.
5.b.6	Climb	1. Medium altitude should be used.
5.b.7	Cruise	1. Normal cruise configuration should be used.
5.b.8	Initial approach	1. Constant airspeed should be used, with gear up, flaps extended as appropriate and RPM as per the operations manual.
5.b.9	Final approach	1. Constant airspeed should be used, with gear down, landing configuration flaps and RPM as per the operations manual.
5.c	Special cases	1. Test applies to special steady-state cases identified as particularly significant to the pilot, important in training or unique to a particular aeroplane type or model.
5.d	FSTD background noise	1. The measurements should be made with the simulation running, the sound muted and a dead flight deck.
5.e	Frequency response	1. White noise should be used to ensure that the full frequency spectrum is demonstrated for each speaker.

GM1 FSTD.QTG.A.500 Sound systems – Aeroplane

This GM provides guidance on the assessment of sound systems.

(a) General guidance is as follows.

The total sound environment in the aircraft is very complex and changes with atmospheric conditions, aircraft configuration, airspeed, altitude, power settings, etc. Flight deck sounds are an important component of the flight deck operational environment and as such provide valuable information to the flight crew. These aural cues can either assist the crew, as an indication of an abnormal situation, or hinder the crew, as a distraction or nuisance. For effective training, the FSTD should provide flight deck sounds that are perceptible to the pilot during normal and abnormal operations and that are comparable to those of the aircraft. Accordingly, the organisation requesting the qualification of an FSTD should carefully evaluate background noises in the location being considered. To demonstrate compliance with the sound requirements, the objective or validation tests have been selected to provide a representative sample of normal static conditions typical of those experienced by a pilot.

(b) Alternate engine fit guidance is as follows.

For FSTDs with multiple propulsion configurations, any condition listed in the tables of FSTD validation tests and feature fidelity levels in CS FSTD.QTG.A (any section) or CS FSTD.QTG.H (any section) that is identified by the data provider as significantly different due to a change in engine model should be presented for initial qualification as test(s) under Section 5.c of the QTG.

(c) Data and data collection system guidance is as follows.

- (1) Information provided to the FSTD manufacturer should comply with ARINC Specification 450-1, *Flight Simulator Design and Performance Data Requirements*, as amended. This information should contain calibration and frequency response data.
- (2) The system used to perform the tests listed in CS FSTD.QTG.A.500 should comply with the following standards.
 - (i) For feature SND at fidelity levels R and S (one-third octave band measurements), the standards that apply are:
 - (A) American National Standards Institute (ANSI) S1.11 (1986), *Specification for octave, half octave and third octave band filter sets*; and
 - (B) International Electrotechnical Commission (IEC) 1094-4 (1995), *Measurement microphones*, type WS2 or better.

- (ii) For feature SND at fidelity level G (overall SPL measurements), IEC 61672, IEC 60651 or ANSI S1.4, *Specifications for sound level meters*, time-weighting, class/type 2 or better, applies. Equipment not meeting the IEC or ANSI standard, such as a smartphone with a 'decibel meter' application or a computer with an external microphone, may be acceptable provided they can achieve repeatable measurements within the validation test tolerances.

(d) Background noise guidance is as follows.

- (1) Background noise is the noise present in the FSTD due to the FSTD's cooling and hydraulic systems that is not associated with the aircraft, and the extraneous noise from other aspects of the FSTD installation. Background noise can seriously impact the correct simulation of aircraft sounds, so the goal should be to keep the background noise below the level of aircraft sounds. In some cases, the sound level of the simulation can be increased to compensate for the background noise. However, this approach is limited by the specified tolerances and by the subjective acceptability of the sound environment to the evaluation pilot. Aeroplane sounds that fall below this limit require careful review and may require a justification to specify lower limits for the background noise.
- (2) The acceptability of the background noise levels is dependent on the normal sound levels in the aircraft or group of aircraft being represented. The background noise levels should fall below the limits defined by the following points (refer to Figure 1 below):
 - (i) 70 dB at 50 Hz
 - (ii) 55 dB at 1 000 Hz
 - (iii) 30 dB at 16 kHz.

These limits are for unweighted one-third octave band sound levels.

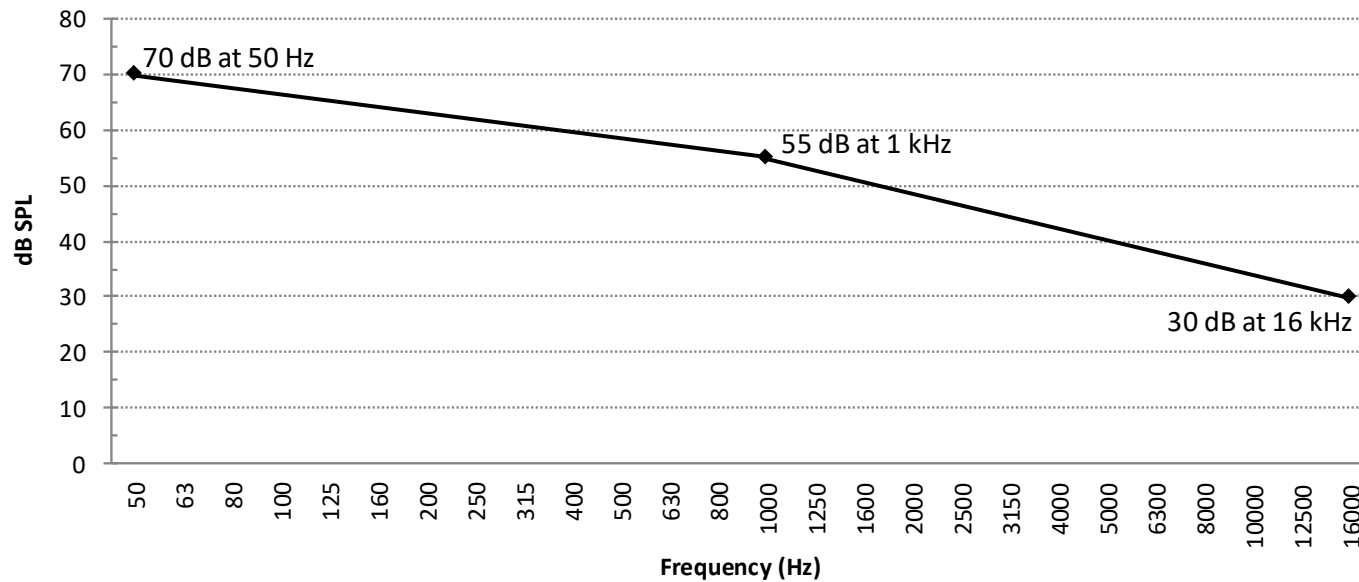


Figure 1. Background noise – maximum levels

(e) Validation testing guidance is as follows.

If aircraft recordings are used as validation data for a test, access to those recordings should be ensured. Deficiencies in aircraft recordings should be considered when applying the specified tolerances to ensure that the simulation is representative of the aircraft. Examples of typical deficiencies are related to the:

- (1) variation of data between tail numbers;
- (2) frequency response of microphones;
- (3) repeatability of the measurements; and
- (4) extraneous sounds during recordings.

-
- (f) The one-third octave band centre frequencies ranging from 50 Hz to 16 kHz cover the following values: 50 Hz, 63 Hz, 80 Hz, 100 Hz, 125 Hz, 160 Hz, 200 Hz, 250 Hz, 315 Hz, 400 Hz, 500 Hz, 630 Hz, 800 Hz, 1 000 Hz, 1 250 Hz, 1 600 Hz, 2 000 Hz, 2 500 Hz, 3 150 Hz, 4 000 Hz, 5 000 Hz, 6 300 Hz, 8 000 Hz, 10 000 Hz, 12 500 Hz and 16 000 Hz.

CS FSTD.QTG.A.600 Systems integration – Aeroplane

This CS provides the standards for the objective tests for aeroplane systems integration.

6.a: Systems integration – system response time		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
6.a.1	System response time – transport delay	Pitch, roll and yaw	Motion system response time	MTN	Init: 200 ms or less after controller movement Rec: Same as Init	Init: 150 ms or less after controller movement Rec: Same as Init	Init: 100 ms or less after controller movement Rec: Same as Init
			Instrument system response time	SYS	Init: 200 ms or less after controller movement Rec: Same as Init	Init: 150 ms or less after controller movement Rec: Same as Init	Init: 100 ms or less after controller movement Rec: Same as Init
			Visual system response time	VIS	Init: 200 ms or less after controller movement Rec: Same as Init	Init: 150 ms or less after controller movement Rec: Same as Init	Init: 100 ms or less after controller movement Rec: Same as Init
			HUD response time	SYS		Init: 150 ms or less after controller movement Rec: Same as Init	Init: 100 ms or less after controller movement Rec: Same as Init

6.a: Systems integration – system response time		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
			EFVS response time	SYS		Init: ± 30 ms from visual system response, and not before motion system response Rec: Same as Init	Init: ± 30 ms from visual system response, and not before motion system response Rec: Same as Init

Individual test requirements. The individual test requirements defined in the table below are applicable to all fidelity levels unless otherwise stated.

6.a: Systems integration – system response time		Test requirements
6.a.1	System response time – transport delay	1. A separate test is required for each axis. 2. The tests to demonstrate compliance with these requirements should simultaneously record: a. the output from the pilot's controller; b. the output from the accelerometer attached to the motion system platform, located as near the pilots' seats as is practicable; c. the output signal to the visual system display; d. the output signal to the HUD/EFVS, if installed; e. the output signal to the pilot's attitude indicator. 3. For feature MTN at fidelity levels R and S, the onset of motion response may occur before the start of the scan of the first visual system video field containing different information and must occur before the end of the scan of the same video field. 4. Visual system response may be measured from the start of the scan of the first video field containing different information. 5. The delay from the aeroplane EFVS electronic elements should be added to the 30 ms tolerance before comparison with visual system response. 6. If the primary features validated by this test are present in the FSTD at differing fidelity levels, the tolerance for the

6.a: Systems integration – system response time		Test requirements
		highest fidelity level should be applied to all the features present.

GM1 FSTD.QTG.A.600 Systems integration – Aeroplane

This GM provides guidance on the assessment of the transport delay tests.

(a) Background information is provided below.

- (1) The purpose of this CS is to provide the methods for conducting transport delay tests.
- (2) The transport delay test is the method for determining the delay introduced into the FSTD due to the time taken for the computations through the FSTD controls, host, motion and visual computer modules. The transport delay test is not dependent on flight test data but may require avionics computer and instrument data from the data provider for some of the cases described below. Figure 1 presents the principles of transport delay testing.

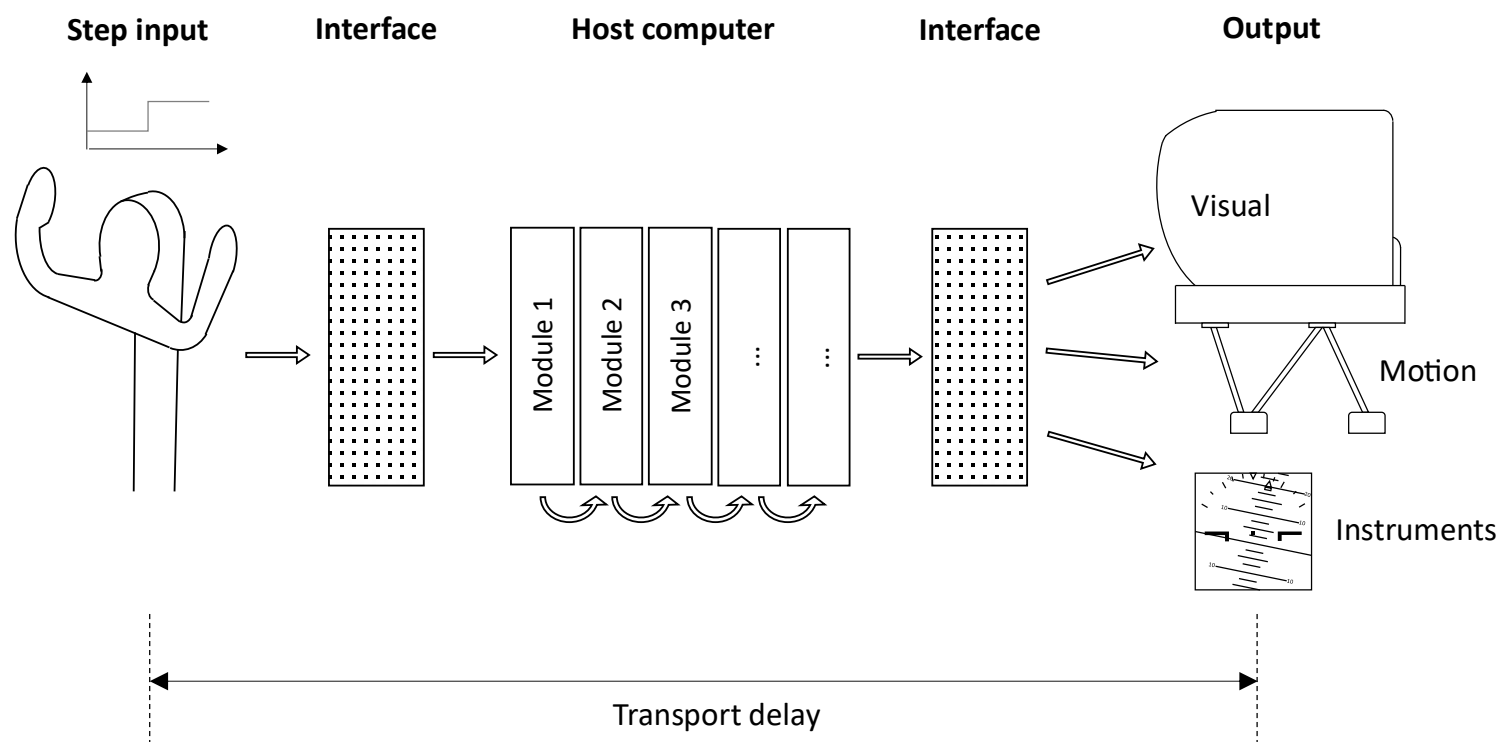


Figure 1. Transport delay testing

(b) Transport delay guidance is as follows.

- (1) The purpose of paragraph (b) is to demonstrate how to determine the introduced transport delay through the FSTD system such that it does not exceed a certain duration. It is not the intention of the transport delay test to arrive at a comparison with the aircraft but rather to demonstrate acceptable performance of the simulation at the point of initial qualification and then to be used as a non-regression test for the software architecture at each recurrent evaluation. The transport delay needs to be measured from control inputs through the interface, through each of the host computer modules and back through the interface to motion, flight instrument and visual systems. The delay needs to be shown to be no more than the required tolerances set out in the validation test tables.

- (2) In all cases, the simulation will have been demonstrated to be dynamically equivalent to the aircraft or group of aircraft in terms of response by the many dynamic tests in the QTG and the subjective handling tests, for both short-term and long-term modes. It is, therefore, only necessary to measure the maximum increased time added by the various interfaces and computing elements in the FSTD that are not present in the aircraft. To do this, a signal is processed through the entire system from the input to the first interface from the control column or stick, through each subsequent computing element or interface and back out to the physical feedback to the pilot, via the motion system, visual system or flight deck instruments. To make this signal more traceable, a handshaking method may be used from element to element such that a clear leading edge is visible at any point through the system. However, it should be noted that the signal needs to be passed through each element of the software and hardware architecture and that the simulation should be running in its normal mode with all software elements active. This is to ensure that recurrent testing can verify that software modifications have not affected the overall path length. A full description of the method chosen, the path of the signal and the input and recording points should be provided.
- (3) The test result analysis requires only that the input and output signals be measured to ensure that they are separated by no more than the limits of the tolerance parameters, according to the FCS of the FSTD. The point of movement will be very simple to determine, since both input and output signals will have clear leading edges.
- (4) For a non-CCA, the total transport delay is equivalent to the delay introduced by the simulation. Refer to Figure 1.
- (5) For CCA, the following points apply.

For FSTDs of aircraft with electronic elements in the path between the input from the pilot and the resulting output, the measured transport delay will include elements of the aircraft itself. These may include flight control systems, avionics or display systems. Since the intention of the transport delay test is to measure only the time specific to the FSTD and not that of the aircraft, the test result time should be offset by the throughput time of the avionics elements. This throughput time should be based on data from the manufacturer of the aircraft or avionics. Alternatively, the aircraft equipment may be bypassed, provided that the signal path is maintained in terms of FSTD interfaces. A schematic diagram should be provided to present the part of the aircraft equipment being considered in this manner and the way in which the signal path has been treated to be representative of all the simulation elements (refer to Figure 2).

- (i) For FSTDs in which the avionics elements in question are replaced by re-hosted, re-targeted or other similar solutions, it is still necessary to offset the test result by the equivalent throughput time of the aircraft elements. However, the schematic diagram should, in this case, demonstrate the equivalence of the simulated avionics to the real avionics in terms of architecture. It is the responsibility

of the developer of the re-hosted, re-targeted or other similar solution to establish the equivalence of the simulated element to the aircraft element being replaced.

- (ii) For CCA, if it can be established that the data path to the instrumentation in the aircraft is subject to computer and data bus asynchronism, uncertainty or 'jitter' of a similar order of magnitude to the transport delay allowance, an SOJ will suffice in place of an actual test. This SOJ should establish the equivalence of the simulated solution to that of the aircraft and provide a rationale regarding the statistical uncertainty. In this situation, the requirement for objective test 6.a.1 for pitch, roll and yaw may be waived.

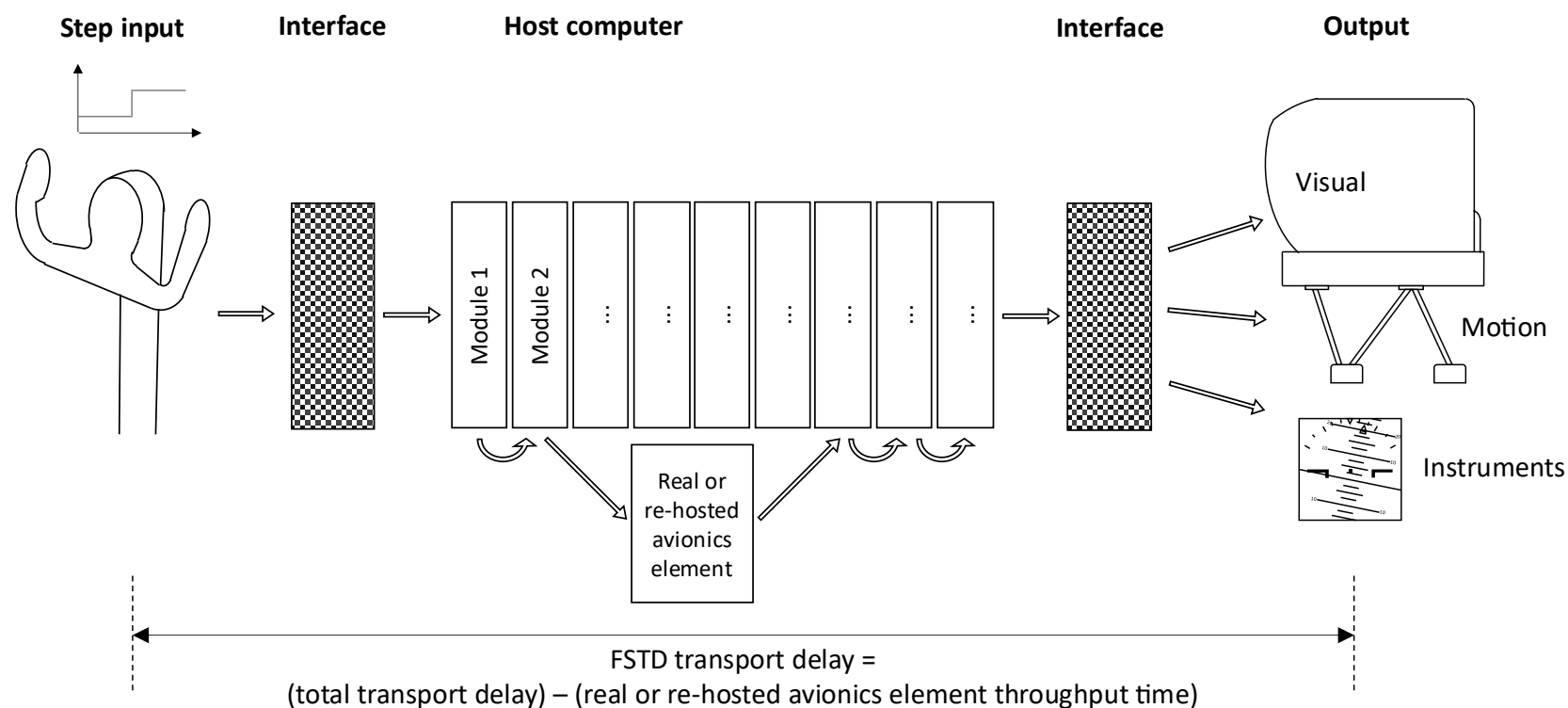


Figure 2. Transport delay with avionics elements

- (6) For recorded signals, the following applies.

The signals recorded to conduct the transport delay calculations should be explained on a schematic block diagram. The FSTD manufacturer should also provide an explanation of why each signal was selected and how they relate to the above descriptions.

- (7) For the interpretation of results, the following applies.

It is normal for FSTD results to vary over time from test to test. This can easily be explained by a simple factor called 'sampling uncertainty'. FSTDs may run at a specific rate at which all modules are executed sequentially in the host computer. The flight control input can occur at any time in the iteration, but this data will not be processed before the start of the new iteration. For an FSTD running at 60 Hz, a worst-case difference of 16.67 ms can be expected. If multiple parallel processors or priority-based execution systems are used, the scatter may be greater. Moreover, in some conditions, the host computer and the visual system do not run at the same iteration rate; therefore, the output of the host computer and the display on the visual system will not always be synchronised.

- (8) If offsetting the measured results by the throughput time of the avionics elements, it is also necessary to recognise that digital equipment will normally give a range of response times dependent on the synchronisation of the control input with the internal equipment frame time. The aircraft or avionics manufacturer should quantify the range of results that should be expected by providing minimum and maximum response times and an indication of the statistical spread in this range. It may be necessary to run the test several times in the FSTD to demonstrate the correctness of the avionics simulation in these conditions.

- (9) The transport delay test should account for daylight, twilight and night modes (as applicable) of operation of the visual system. The required tolerance is as set out in the validation test tables, and motion response should occur before the end of the first video scan containing new information. If it can be demonstrated that the visual system operates at the same execution rate in both day and night modes, a single test in each axis is sufficient.

OBJECTIVE TESTS FOR ROTORCRAFT

CS FSTD.QTG.H.100 Performance – Engine assessment – Rotorcraft

This CS provides the standards for the objective tests for rotorcraft performance for engine assessment.

1.a: Performance – engine assessment		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
1.a.1.a	Start operations – engine start and acceleration (transient)	Ground Rotor brake used and not used	Light-off time	SYS	Init: CT&M Rec: EM	Init: $\pm 10\%$ or ± 1 s Rec: EM	Init: $\pm 10\%$ or ± 1 s Rec: EM
			Torque	GND		Init: CT&M Rec: EM	Init: $\pm 5\%$ Rec: EM
			Rotor speed	GND	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 3\%$ Rec: EM
			Gas generator speed	GND	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 5\%$ Rec: EM
			Power turbine speed	GND		Init: CT&M Rec: EM	Init: $\pm 5\%$ Rec: EM
			Turbine gas temperature	GND	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 30^\circ\text{C}$ Rec: EM
1.a.1.b	Steady-state idle and operating RPM conditions	Ground	Torque	GND	Init: CT&M Rec: EM	Init: $\pm 3\%$ Rec: EM	Init: $\pm 3\%$ Rec: EM
			Rotor speed	GND	Init: CT&M Rec: EM	Init: $\pm 1.5\%$ Rec: EM	Init: $\pm 1.5\%$ Rec: EM

1.a: Performance – engine assessment		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
			Fuel flow	GND			Init: ± 5 % Rec: EM
			Ng	GND			Init: ± 2 % Rec: EM
			Power turbine speed	GND	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: ± 2 % Rec: EM
			Turbine gas temperature	GND			Init: ± 20 °C Rec: EM
1.a.2	Power turbine speed trim	Ground Actuation of trim system in both directions	Total change of power turbine speed	GND	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: ± 10 % Rec: EM
			Or				
			Rotor speed	GND	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: ± 0.5 % Rec: EM
1.a.3	Engine and rotor speed governing	Climb and descent	Torque	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: ± 5 % Rec: EM
			Rotor speed	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: ± 1.5 % Rec: EM

Individual test requirements. The individual test requirements defined in the table below are applicable to all fidelity levels unless otherwise stated.

1.a: Performance – engine assessment		Test requirements
1.a.1.a	Start operations –	1. Test covers time histories of each engine from initiation of start sequence to steady-state idle and from steady-state

1.a: Performance – engine assessment		Test requirements
	engine start and acceleration (transient)	idle to operating RPM in accordance with the recommended rotorcraft flight manual (RFM) procedure (i.e. through steady-state idle or directly to operating RPM in accordance with the RFM). 2. Tolerance is to be applied only in the validity domain of the engine parameter sensors. 3. Light-off time should be defined in accordance with required checks in approved RFM normal engine start procedure. This means, for example: a. starter is disengaged before an Ng value stated in the RFM; or b. main rotor starts spinning before an Ng value stated in the RFM; or c. time is defined in accordance with a particular RFM procedure.
1.a.1.b	Steady-state idle and operating RPM conditions	1. Test may be a series of snapshot tests. 2. Test should present data for both steady-state idle and operating RPM conditions.
1.a.2	Power turbine speed trim	1. Test should cover time history of engine response to trim system actuation. 2. Both directions should have to be demonstrated.
1.a.3	Engine and rotor speed governing	1. Collective step inputs are required. 2. Test can be conducted with climb and descent performance tests. 3. Collective up input should be performed while rotorcraft is initially in descent. A large input is recommended, so that the rotorcraft is no longer in descent after the control input and significant torque variation is obtained. 4. Collective down input should be performed while rotorcraft is initially in climb. A large input is recommended, so that the rotorcraft is no longer in climb after the control input and significant torque variation is obtained.

CS FSTD.QTG.H.105 Ground operations – Rotorcraft

This CS provides the standards for the objective tests for rotorcraft performance for ground operations.

1.b: Ground operations		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
1.b.1	Reserved						
1.b.2	Rate of turn versus pedal deflection or steering wheel angle	Ground Left and right turns	Turn rate	GND	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 10\%$ or $\pm 2^\circ/\text{s}$ Rec: EM
1.b.3	Taxi	Ground	Pitch angle	GND	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 1.5^\circ$ Rec: EM
			Torque	GND	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 3\%$ Rec: EM
			Longitudinal control position	GND	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 5\%$ Rec: EM
			Lateral control position	GND	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 5\%$ Rec: EM
			Directional control position	GND	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 5\%$ Rec: EM
			Collective control position	GND	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 5\%$ Rec: EM
1.b.4	Brake effectiveness	Ground	Time	GND			Init: CT&M Rec: $\pm 10\%$ or $\pm 1\text{ s}$

1.b: Ground operations		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
			Distance	GND			Init: CT&M Rec: $\pm 10\%$ or ± 5 m (15 ft)

Individual test requirements. The individual test requirements defined in the table below are applicable to all fidelity levels unless otherwise stated.

1.b: Ground operations		Test requirements
1.b.2	Rate of turn versus pedal deflection or steering wheel angle	- Test should be without usage of wheel brakes. - Left and right turns must be demonstrated.
1.b.4	Brake effectiveness	Test should record data until full stop.

CS FSTD.QTG.H.110 Take-off – Rotorcraft

This CS provides the standards for the objective tests for rotorcraft performance for take-off.

1.c: Take-off		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
1.c.1	All engines	Take-off from in-ground-effect hover and initial climb	Airspeed	IGE	Init: CT&M Rec: EM	Init: ± 3 kt Rec: EM	Init: ± 3 kt Rec: EM
			Altitude	IGE	Init: CT&M Rec: EM	Init: ± 20 ft (6.1 m) Rec: EM	Init: ± 20 ft (6.1 m) Rec: EM
			Torque	IGE	Init: CT&M Rec: EM	Init: ± 3 % Rec: EM	Init: ± 3 % Rec: EM
			Rotor speed	IGE	Init: CT&M Rec: EM	Init: ± 1.5 % Rec: EM	Init: ± 1.5 % Rec: EM
			Pitch angle	IGE	Init: CT&M Rec: EM	Init: $\pm 2.5^\circ$ Rec: EM	Init: $\pm 1.5^\circ$ Rec: EM
			Bank angle	IGE	Init: CT&M Rec: EM	Init: $\pm 2^\circ$ Rec: EM	Init: $\pm 2^\circ$ Rec: EM
			Heading	IGE	Init: CT&M Rec: EM	Init: $\pm 2^\circ$ Rec: EM	Init: $\pm 2^\circ$ Rec: EM
			Longitudinal control position	IGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: ± 10 % Rec: EM
			Lateral control position	IGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: ± 10 % Rec: EM
			Directional control position	IGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: ± 10 % Rec: EM

1.c: Take-off		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
			Collective control position	IGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 10\%$ Rec: EM
1.c.2	One-engine-inoperative continued take-off	Take-off from in-ground-effect hover and initial climb	Airspeed	IGE	Init: CT&M Rec: EM	Init: ± 3 kt Rec: EM	Init: ± 3 kt Rec: EM
			Altitude	IGE	Init: CT&M Rec: EM	Init: ± 20 ft (6.1 m) Rec: EM	Init: ± 20 ft (6.1 m) Rec: EM
			Torque	IGE	Init: CT&M Rec: EM	Init: $\pm 3\%$ Rec: EM	Init: $\pm 3\%$ Rec: EM
			Rotor speed	IGE	Init: CT&M Rec: EM	Init: $\pm 1.5\%$ Rec: EM	Init: $\pm 1.5\%$ Rec: EM
			Pitch angle	IGE	Init: CT&M Rec: EM	Init: $\pm 2.5^\circ$ Rec: EM	Init: $\pm 1.5^\circ$ Rec: EM
			Bank angle	IGE	Init: CT&M Rec: EM	Init: $\pm 2^\circ$ Rec: EM	Init: $\pm 2^\circ$ Rec: EM
			Heading	IGE	Init: CT&M Rec: EM	Init: $\pm 2^\circ$ Rec: EM	Init: $\pm 2^\circ$ Rec: EM
			Longitudinal control position	IGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 10\%$ Rec: EM
			Lateral control position	IGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 10\%$ Rec: EM
			Directional control position	IGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 10\%$ Rec: EM

1.c: Take-off		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
			Collective control position	IGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 10\%$ Rec: EM
1.c.3	One-engine-inoperative rejected take-off	Take-off from in-ground-effect hover	Airspeed	IGE	Init: CT&M Rec: EM	Init: ± 3 kt Rec: EM	Init: ± 3 kt Rec: EM
			Altitude	IGE	Init: CT&M Rec: EM	Init: ± 20 ft (6.1 m) Rec: EM	Init: ± 20 ft (6.1 m) Rec: EM
			Torque	IGE	Init: CT&M Rec: EM	Init: $\pm 3\%$ Rec: EM	Init: $\pm 3\%$ Rec: EM
			Rotor speed	IGE	Init: CT&M Rec: EM	Init: $\pm 1.5\%$ Rec: EM	Init: $\pm 1.5\%$ Rec: EM
			Pitch angle	IGE	Init: CT&M Rec: EM	Init: $\pm 2.5^\circ$ Rec: EM	Init: $\pm 1.5^\circ$ Rec: EM
			Bank angle	IGE	Init: CT&M Rec: EM	Init: $\pm 1.5^\circ$ Rec: EM	Init: $\pm 1.5^\circ$ Rec: EM
			Heading	IGE	Init: CT&M Rec: EM	Init: $\pm 2^\circ$ Rec: EM	Init: $\pm 2^\circ$ Rec: EM
			Longitudinal control position	IGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 10\%$ Rec: EM
			Lateral control position	IGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 10\%$ Rec: EM
			Directional control position	IGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 10\%$ Rec: EM

1.c: Take-off		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
			Collective control position	IGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 10\%$ Rec: EM
			Distance	IGE	Init: CT&M Rec: EM	Init: $\pm 7.5\%$ or 30 m (100 ft) Rec: EM	Init: $\pm 7.5\%$ or 30 m (100 ft) Rec: EM
1.c.4	Lift-off to hover	Ground	Torque	IGE	Init: CT&M Rec: EM	Init: $\pm 5\%$ Rec: EM	Init: $\pm 5\%$ Rec: EM
			Pitch angle	IGE	Init: CT&M Rec: EM	Init: $\pm 2^\circ$ Rec: EM	Init: $\pm 2^\circ$ Rec: EM
			Bank angle	IGE	Init: CT&M Rec: EM	Init: $\pm 2^\circ$ Rec: EM	Init: $\pm 2^\circ$ Rec: EM
			Heading	IGE	Init: CT&M Rec: EM	Init: $\pm 3^\circ$ Rec: EM	Init: $\pm 3^\circ$ Rec: EM
			Longitudinal control position	IGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 10\%$ Rec: EM
			Lateral control position	IGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 10\%$ Rec: EM
			Directional control position	IGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 10\%$ Rec: EM
			Collective control position	IGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 10\%$ Rec: EM

Requirements for all the rotorcraft take-off tests.

1.c: Take-off	Test requirements
All tests	1. Wind conditions should be justified. In addition, if wind is not constant during the test, it should be plotted. 2. For fidelity level G only, a FSTD footprint test is an acceptable alternative to CT&M for initial evaluation.

Individual test requirements. The individual test requirements defined in the table below are applicable to all fidelity levels unless otherwise stated.

1.c: Take-off		Test requirements
1.c.1	All engines	1. Test should cover time history of take-off from hover. 2. If airspeed is not valid or not reliable, tolerance should be applied to ground speed instead. 3. Test should record data to at least 200 ft (61 m) above terrain elevation or V_Y, whichever comes later. 4. Initial pressure altitude may not be reliable due to rotor downwash; if so, height above ground should be used instead for defining initial conditions and pressure altitude should be assessed once it is reliable. As an alternative, GNSS altitude above ellipsoid can be used (with initial offset compensated) instead of pressure altitude for the whole test.
1.c.2	OEI continued take-off	1. Test should cover time history of open-field category A take-off from hover, with one-engine failure after take-off decision point (TDP). 2. If airspeed is not valid or not reliable, tolerance should be applied to ground speed instead. 3. Test should record data to at least 200 ft (61 m) above terrain elevation or V_Y, whichever comes later. 4. Initial pressure altitude may not be reliable due to rotor downwash; if so, height above ground should be used instead for defining initial conditions and pressure altitude should be assessed once it is reliable. As an alternative, GNSS altitude above ellipsoid can be used (with initial offset compensated) instead of pressure altitude for the whole test. 5. Use of an OEI training mode is acceptable.
1.c.3	OEI rejected take-off	1. Test should cover time history of open-field take-off from hover, with one-engine failure before TDP, until touchdown. 2. Test conditions should be near those limiting performance, as per the aircraft manual. 3. If airspeed is not valid or not reliable, tolerance should be applied to ground speed instead. 4. Pressure altitude may not be reliable due to rotor downwash; if so, either height above ground or GNSS altitude should be used instead. 5. Use of an OEI training mode is acceptable.

1.c: Take-off		Test requirements
1.c.4	Lift-off to hover	1. Test should record a manoeuvre starting on the ground with collective full down until stabilised in hover in-ground effect.

GM1 FSTD.QTG.H.110 Take-off – Rotorcraft

The wind conditions (speed and direction) are generally available from the windsock, control tower or ATIS. However, wind can sometimes change rapidly – for instance, it is common to encounter a wind gradient function of the height above ground due to the friction forces close to the surface of the Earth. In such cases, replaying the reference flight data with the static wind conditions kept constant during the test may not be suitable and the wind gradient should be estimated and set during the test.

The actual wind at the ownship position can often be estimated by comparing airspeed components with ground speed or inertial speed components. This computation is usually performed using a Kalman estimation filter (or a similar method), so that the noise, sampling or inaccuracy of measured data does not result in a fictitious wind being computed. This computation is only possible if airspeed and ground speed / inertial data is valid. Unfortunately, that is not the case at low speed, when slight changes in wind conditions have a significant impact on helicopter behaviour. If this data is not valid, the wind can be extrapolated between the latest and the next valid estimation. It can also be inferred from global helicopter behaviour during the flight test.

If this computation results in strong wind gradients (quick and large changes in wind direction or speed), engineering judgement should be applied in order to evaluate if the estimated wind is judged to be physical and realistic. If not, the computed wind should not be applied in the simulation. Unless the weather is very rough during the flight (in which case the validity and representativity of the validation data should be justified), helicopter main rotor filters wind shear and turbulence and strong wind gradients should not be expected as a result of such computation and may be due to an incorrect estimation.

CS FSTD.QTG.H.115 Hover performance – Rotorcraft

This CS provides the standards for the objective tests for rotorcraft hover performance.

1.d: Hover performance		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
1.d	Hover performance	In-ground effect	Torque	IGE (points in-ground effect)	Init: CT&M Rec: EM	Init: $\pm 3\%$ Rec: EM	Init: $\pm 3\%$ Rec: EM
		Out-of-ground effect		OGE (points out-of-ground effect)			
		Light and heavy gross weights	Pitch angle	IGE (points in-ground effect)		Init: $\pm 1.5^\circ$ Rec: EM	Init: $\pm 1.5^\circ$ Rec: EM
		For height, refer to requirements in the test requirement table below		OGE (points out-of-ground effect)			
			Bank angle	IGE (points in-ground effect)		Init: $\pm 1.5^\circ$ Rec: EM	Init: $\pm 1.5^\circ$ Rec: EM
				OGE (points out-of-ground effect)			
			Longitudinal control position or blade angle	IGE (points in-ground effect)		Init: CT&M Rec: EM	Init: $\pm 5\%$ Rec: EM
				OGE (points out-of-ground effect)			

1.d: Hover performance		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
			Lateral control position or blade angle	IGE (points in-ground effect) OGE (points out-of-ground effect)		Init: CT&M Rec: EM	Init: $\pm 5\%$ Rec: EM
			Directional control position or blade angle	IGE (points in-ground effect) OGE (points out-of-ground effect)		Init: CT&M Rec: EM	Init: $\pm 5\%$ Rec: EM
			Collective control position or blade angle	IGE (points in-ground effect) OGE (points out-of-ground effect)	Init: CT&M Rec: EM	Init: $\pm 5\%$ Rec: EM	Init: $\pm 5\%$ Rec: EM

Individual test requirements. The individual test requirements defined in the table below are applicable to all fidelity levels unless otherwise stated.

1.d: Hover performance		Test requirements
1.d	Hover performance	- Tests may be snapshot tests. - Light gross weight is a weight that is not more than 120 % of the basic operating weight of the reference rotorcraft or as limited by the minimum practical operating weight of the rotorcraft. The basic operating weight is the empty weight of the rotorcraft plus the weight of the following: normal oil quantity, minimum fuel, required crew members and their baggage, standard equipment and servicing fluids. - Heavy gross weight is a weight that is not less than 90 % of the maximum certificated weight of the reference rotorcraft. - Tests should be conducted at a minimum of three heights with in-ground effect, including one at no more than 10 %

1.d: Hover performance		Test requirements
		of the rotor diameter above the ground and one at each of approximately 30 % and 70 % of the rotor diameter, where height refers to the height of the main gear above the ground. In addition, one in level flight trim condition should be conducted out-of-ground effect (e.g. at 150 % of rotor diameter). 5. If the stability augmentation system has no influence on the stable conditions, tests can be performed with augmentation either on or off. Otherwise, both augmentation on and off tests are required. 6. Torque and collective blade angle (or control position) should show correct trend versus height above ground in-ground effect, and correct trend out-of-ground effect compared with in-ground effect. For the initial qualification, this should be demonstrated by cross-plots against height. 7. If tolerances are applied to rotor blade angles, it should be justified that the flight control positions are consistent with the authority range of the autopilot serial actuators.

CS FSTD.QTG.H.120 Vertical climb performance – Rotorcraft

This CS provides the standards for the objective tests for rotorcraft vertical climb performance.

1.e: Vertical climb performance		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
1.e	Vertical climb performance	From out-of-ground-effect hover	Vertical velocity	OGE		Init: CT&M Rec: EM	Init: $\pm 10\%$ or ± 100 fpm (0.5 m/s) Rec: EM
		Light and heavy gross weights	Directional control position or blade angle	OGE			Init: $\pm 5\%$ Rec: EM
			Collective control position or blade angle	OGE		Init: CT&M Rec: EM	Init: $\pm 5\%$ Rec: EM

Individual test requirements. The individual test requirements defined in the table below are applicable to all fidelity levels unless otherwise stated.

1.e: Vertical climb performance		Test requirements
1.e	Vertical climb performance	- Tests may be snapshot tests. - Light gross weight is a weight that is not more than 120 % of the basic operating weight of the reference rotorcraft or as limited by the minimum practical operating weight of the rotorcraft. The basic operating weight is the empty weight of the rotorcraft plus the weight of the following: normal oil quantity, minimum fuel, required crew members and their baggage, standard equipment and servicing fluids. - Heavy gross weight is a weight that is not less than 90 % of the maximum certificated weight of the reference rotorcraft. - If tolerances are applied to rotor blade angles, it should be justified that the flight control positions are consistent with the authority range of the autopilot serial actuators.

1.e: Vertical climb performance		Test requirements
		5. If the stability augmentation system has no influence on the stable conditions, tests can be performed with augmentation either on or off. Otherwise, both augmentation on and off tests are required.

CS FSTD.QTG.H.125 Level flight performance and trimmed flight control position – Rotorcraft

This CS provides the standards for the objective tests for rotorcraft level flight performance and trimmed flight control position.

1.f: Level flight performance and trimmed flight control position		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
1.f	Level flight performance and trimmed flight control position	Straight cruise and level flight	Torque	OGE	Init: CT&M Rec: EM	Init: $\pm 3\%$ Rec: EM	Init: $\pm 3\%$ Rec: EM
		Light and heavy gross weights	Pitch angle	OGE	Init: CT&M Rec: EM	Init: $\pm 1.5^\circ$ Rec: EM	Init: $\pm 1.5^\circ$ Rec: EM
		For airspeed, refer to requirements in the test requirement table below	Side-slip angle	OGE	Init: CT&M Rec: EM	Init: $\pm 2^\circ$ Rec: EM	Init: $\pm 2^\circ$ Rec: EM
			Longitudinal control position or blade angle	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 5\%$ Rec: EM
			Lateral control position or blade angle	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 5\%$ Rec: EM
			Directional control position or blade angle	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 5\%$ Rec: EM

1.f: Level flight performance and trimmed flight control position		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
			Collective control position or blade angle	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 5\%$ Rec: EM

Individual test requirements. The individual test requirements defined in the table below are applicable to all fidelity levels unless otherwise stated.

1.f: Level flight performance and trimmed flight control position		Test requirements
1.f	Level flight performance and trimmed flight control position	- Tests may be snapshot tests. - Light gross weight is a weight that is not more than 120 % of the basic operating weight of the reference rotorcraft or as limited by the minimum practical operating weight of the rotorcraft. The basic operating weight is the empty weight of the rotorcraft plus the weight of the following: normal oil quantity, minimum fuel, required crew members and their baggage, standard equipment and servicing fluids. - Heavy gross weight is a weight that is not less than 90 % of the maximum certificated weight of the reference rotorcraft. - Tests should cover at least three airspeeds (including V_Y, maximum cruise speed and at least one intermediate speed) within the flight envelope, for each weight configuration (a minimum of six tests are required). - If the stability augmentation system has no influence on the stable conditions, tests can be performed with augmentation either on or off. Otherwise, both augmentation on and off tests are required. - For a given weight and centre-of-gravity configuration, torque should show correct trend versus airspeed. For the initial qualification, this should be demonstrated by a cross-plot against airspeed. - If tolerances are applied to rotor blade angles, it should be justified that the flight control positions are consistent with the authority range of the autopilot serial actuators.

CS FSTD.QTG.H.130 Climb performance and trimmed flight control position – Rotorcraft

This CS provides the standards for the objective tests for rotorcraft climb performance and trimmed flight control position.

1.g: Climb performance and trimmed flight control position		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
1.g	Climb performance and trimmed flight control position	All engines operating OEI Light and heavy gross weights	Vertical velocity	OGE	Init: CT&M Rec: EM	Init: ± 100 fpm (0.5 m/s) or 10 % Rec: EM	Init: ± 100 fpm (0.5 m/s) or 10 % Rec: EM
			Pitch angle	OGE	Init: CT&M Rec: EM	Init: $\pm 1.5^\circ$ Rec: EM	Init: $\pm 1.5^\circ$ Rec: EM
			Side-slip angle	OGE	Init: CT&M Rec: EM	Init: $\pm 2^\circ$ Rec: EM	Init: $\pm 2^\circ$ Rec: EM
			Longitudinal control position or blade angle	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: ± 5 % Rec: EM
			Lateral control position or blade angle	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: ± 5 % Rec: EM
			Directional control position or blade angle	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: ± 5 % Rec: EM
			Collective control position or blade angle	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: ± 5 % Rec: EM
			Speed	OGE	Init: CT&M Rec: EM	Init: ± 3 kt Rec: EM	Init: ± 3 kt Rec: EM

Individual test requirements. The individual test requirements defined in the table below are applicable to all fidelity levels unless otherwise stated.

1.g: Climb performance and trimmed flight control position		Test requirements
1.g	Climb performance and trimmed flight control position	1. Tests may be snapshot tests. 2. Light gross weight is a weight that is not more than 120 % of the basic operating weight of the reference rotorcraft or as limited by the minimum practical operating weight of the rotorcraft. The basic operating weight is the empty weight of the rotorcraft plus the weight of the following: normal oil quantity, minimum fuel, required crew members and their baggage, standard equipment and servicing fluids. 3. Heavy gross weight is a weight that is not less than 90 % of the maximum certificated weight of the reference rotorcraft. 4. Data should be presented at relevant climb power conditions, for each weight/centre-of-gravity configuration: a. in all engines operating, close to maximum continuous power; b. in OEI, close to maximum power in line with the selected power rating. 5. The achieved measured vertical velocity of the FSTD and of the validation data should not be less than the appropriate approved RFM values. 6. The test may employ an OEI training mode, provided it is installed and approved for use in the rotorcraft. 7. If the stability augmentation system has no influence on the stable conditions, tests can be performed with augmentation either on or off. Otherwise, both augmentation on and off tests are required. 8. If tolerances are applied to rotor blade angles, it should be justified that the flight control positions are consistent with the authority range of the autopilot serial actuators.

CS FSTD.QTG.H.135 Performance – Descent – Rotorcraft

This CS provides the standards for the objective tests for rotorcraft performance in descent.

1.h: Descent		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
1.h.1	Descent performance and trimmed flight control position	Light and heavy gross weights At or near 1 000 fpm (5 m/s) rate of descent at normal approach speed	Torque	OGE	Init: CT&M Rec: EM	Init: $\pm 3 \%$ Rec: EM	Init: $\pm 3 \%$ Rec: EM
			Pitch angle	OGE	Init: CT&M Rec: EM	Init: $\pm 1.5^\circ$ Rec: EM	Init: $\pm 1.5^\circ$ Rec: EM
			Side-slip angle	OGE	Init: CT&M Rec: EM	Init: $\pm 2^\circ$ Rec: EM	Init: $\pm 2^\circ$ Rec: EM
			Longitudinal control position or blade angle	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 5 \%$ Rec: EM
			Lateral control position or blade angle	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 5 \%$ Rec: EM
			Directional control position or blade angle	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 5 \%$ Rec: EM
			Collective control position or blade angle	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 5 \%$ Rec: EM
1.h.2	Autorotation performance and trimmed flight	Steady descents Light and heavy	Vertical velocity	OGE	Init: CT&M Rec: EM	Init: ± 100 fpm (0.5 m/s) or 10 % Rec: EM	Init: ± 100 fpm (0.5 m/s) or 10 % Rec: EM

1.h: Descent		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
	control position	gross weights	Rotor speed	OGE	Init: CT&M Rec: EM	Init: $\pm 1.5\%$ Rec: EM	Init: $\pm 1.5\%$ Rec: EM
			Pitch angle	OGE	Init: CT&M Rec: EM	Init: $\pm 1.5^\circ$ Rec: EM	Init: $\pm 1.5^\circ$ Rec: EM
			Side-slip angle	OGE	Init: CT&M Rec: EM	Init: $\pm 2^\circ$ Rec: EM	Init: $\pm 2^\circ$ Rec: EM
			Longitudinal control position or blade angle	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 5\%$ Rec: EM
			Lateral control position or blade angle	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 5\%$ Rec: EM
			Directional control position or blade angle	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 5\%$ Rec: EM
			Collective control position or blade angle	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 5\%$ Rec: EM

Requirements for all the rotorcraft descent tests.

1.h: Descent	Test requirements
All tests	1. Light gross weight is a weight that is not more than 120 % of the basic operating weight of the reference rotorcraft or as limited by the minimum practical operating weight of the rotorcraft. The basic operating weight is the empty weight

1.h: Descent	Test requirements
	of the rotorcraft plus the weight of the following: normal oil quantity, minimum fuel, required crew members and their baggage, standard equipment and servicing fluids. - Heavy gross weight is a weight that is not less than 90 % of the maximum certificated weight of the reference rotorcraft. - If tolerances are applied to rotor blade angles, it should be justified that the flight control positions are consistent with the authority range of the autopilot serial actuators. - If the stability augmentation system has no influence on the stable conditions, tests can be performed with augmentation either on or off. Otherwise, both augmentation on and off tests are required.

Individual test requirements. The individual test requirements defined in the table below are applicable to all fidelity levels unless otherwise stated.

1.h: Descent	Test requirements
1.h.1 Descent performance and trimmed flight control position	- Tests may be snapshot tests. - Tests should be at or near a 1 000 fpm (5 m/s) rate of descent at normal approach speed.
1.h.2 Autorotation performance and trimmed flight control position	- Test may be a series of snapshot tests. - Rotor speed tolerance only applies if collective control position is full down. - Test should be a speed sweep from approximately 50 kt to at least maximum glide distance airspeed or maximum allowable power-off airspeed, whichever is slower, for each weight configuration (a minimum of four tests are required).

CS FSTD.QTG.H.140 Performance – Autorotational entry – Rotorcraft

This CS provides the standards for the objective tests for rotorcraft performance for autorotational entry.

1.i: Autorotational entry		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
1.i	Autorotational entry	Either cruise or climb	Torque	OGE	Init: CT&M Rec: EM	Init: $\pm 3\%$ Rec: EM	Init: $\pm 3\%$ Rec: EM
			Rotor speed	OGE	Init: CT&M Rec: EM	Init: $\pm 3\%$ Rec: EM	Init: $\pm 3\%$ Rec: EM
			Pitch angle	OGE	Init: CT&M Rec: EM	Init: $\pm 2^\circ$ Rec: EM	Init: $\pm 2^\circ$ Rec: EM
			Bank angle	OGE	Init: CT&M Rec: EM	Init: $\pm 3^\circ$ Rec: EM	Init: $\pm 3^\circ$ Rec: EM
			Heading	OGE	Init: CT&M Rec: EM	Init: $\pm 5^\circ$ Rec: EM	Init: $\pm 5^\circ$ Rec: EM
			Airspeed	OGE	Init: CT&M Rec: EM	Init: ± 5 kt Rec: EM	Init: ± 5 kt Rec: EM
			Altitude	OGE	Init: CT&M Rec: EM	Init: ± 20 ft (6.1 m) Rec: EM	Init: ± 20 ft (6.1 m) Rec: EM

Individual test requirements. The individual test requirements defined in the table below are applicable to all fidelity levels unless otherwise stated.

1.i: Autorotational entry		Test requirements
1.i	Autorotational entry	1. Test should cover time history of vehicle response to a rapid power reduction to idle. If the approved RFM procedure for intentional entry into autorotation does not allow for a rapid power reduction to idle, supplementary data (e.g. engineering validation data and/or pilot manoeuvre footprint with a statement from a suitably qualified pilot

1.i: Autorotational entry		Test requirements
		acceptable) should be provided to validate quick entry into autorotation, in addition to entry into autorotation in accordance with the RFM. 2. If testing cruise and level flight, data should be presented for the maximum range airspeed or maximum allowable power-off airspeed, whichever is slower. 3. If testing climb, data should be presented for the maximum rate of climb airspeed at or near maximum continuous power. 4. For fidelity level G only, a FSTD footprint test is an acceptable alternative to CT&M for initial evaluation.

CS FSTD.QTG.H.145 Performance – Landing – Rotorcraft

This CS provides the standards for the objective tests for rotorcraft performance for landing.

1.j: Landing		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
1.j.1	All engines landing	Approach and landing	Airspeed	IGE	Init: CT&M Rec: EM	Init: ± 3 kt Rec: EM	Init: ± 3 kt Rec: EM
			Altitude	IGE	Init: CT&M Rec: EM	Init: ± 20 ft (6.1 m) Rec: EM	Init: ± 20 ft (6.1 m) Rec: EM
			Torque	IGE	Init: CT&M Rec: EM	Init: ± 3 % Rec: EM	Init: ± 3 % Rec: EM
			Rotor speed	IGE	Init: CT&M Rec: EM	Init: ± 1.5 % Rec: EM	Init: ± 1.5 % Rec: EM
			Pitch angle	IGE	Init: CT&M Rec: EM	Init: $\pm 2.5^\circ$ Rec: EM	Init: $\pm 1.5^\circ$ Rec: EM
			Bank angle	IGE	Init: CT&M Rec: EM	Init: $\pm 2^\circ$ Rec: EM	Init: $\pm 1.5^\circ$ Rec: EM
			Heading	IGE	Init: CT&M Rec: EM	Init: $\pm 2^\circ$ Rec: EM	Init: $\pm 2^\circ$ Rec: EM
			Longitudinal control position	IGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: ± 10 % Rec: EM
			Lateral control position	IGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: ± 10 % Rec: EM
			Directional control position	IGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: ± 10 % Rec: EM

1.j: Landing		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
			Collective control position	IGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 10\%$ Rec: EM
1.j.2	OEI landing	Approach and landing	Airspeed	IGE	Init: CT&M Rec: EM	Init: ± 3 kt Rec: EM	Init: ± 3 kt Rec: EM
			Altitude	IGE	Init: CT&M Rec: EM	Init: ± 20 ft (6.1 m) Rec: EM	Init: ± 20 ft (6.1 m) Rec: EM
			Torque	IGE	Init: CT&M Rec: EM	Init: $\pm 3\%$ Rec: EM	Init: $\pm 3\%$ Rec: EM
			Rotor speed	IGE	Init: CT&M Rec: EM	Init: $\pm 1.5\%$ Rec: EM	Init: $\pm 1.5\%$ Rec: EM
			Pitch angle	IGE	Init: CT&M Rec: EM	Init: $\pm 2.5^\circ$ Rec: EM	Init: $\pm 1.5^\circ$ Rec: EM
			Bank angle	IGE	Init: CT&M Rec: EM	Init: $\pm 2^\circ$ Rec: EM	Init: $\pm 1.5^\circ$ Rec: EM
			Heading	IGE	Init: CT&M Rec: EM	Init: $\pm 2^\circ$ Rec: EM	Init: $\pm 2^\circ$ Rec: EM
			Longitudinal control position	IGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 10\%$ Rec: EM
			Lateral control position	IGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 10\%$ Rec: EM
			Directional control position	IGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 10\%$ Rec: EM

1.j: Landing		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
			Collective control position	IGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 10\%$ Rec: EM
1.j.3	Balked landing / missed approach	OEI approach	Airspeed	IGE	Init: CT&M Rec: EM	Init: ± 3 kt Rec: EM	Init: ± 3 kt Rec: EM
			Altitude	IGE	Init: CT&M Rec: EM	Init: ± 20 ft (6.1 m) Rec: EM	Init: ± 20 ft (6.1 m) Rec: EM
			Torque	IGE	Init: CT&M Rec: EM	Init: $\pm 3\%$ Rec: EM	Init: $\pm 3\%$ Rec: EM
			Rotor speed	IGE	Init: CT&M Rec: EM	Init: $\pm 1.5\%$ Rec: EM	Init: $\pm 1.5\%$ Rec: EM
			Pitch angle	IGE	Init: CT&M Rec: EM	Init: $\pm 2.5^\circ$ Rec: EM	Init: $\pm 1.5^\circ$ Rec: EM
			Bank angle	IGE	Init: CT&M Rec: EM	Init: $\pm 2^\circ$ Rec: EM	Init: $\pm 1.5^\circ$ Rec: EM
			Heading	IGE	Init: CT&M Rec: EM	Init: $\pm 2^\circ$ Rec: EM	Init: $\pm 2^\circ$ Rec: EM
			Longitudinal control position	IGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 10\%$ Rec: EM
			Lateral control position	IGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 10\%$ Rec: EM
			Directional control position	IGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 10\%$ Rec: EM

1.j: Landing		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
			Collective control position	IGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 10\%$ Rec: EM
1.j.4	Autorotational landing with touchdown	Approach and touchdown	Airspeed	IGE		Init: CT&M Rec: EM	Init: ± 3 kt Rec: EM
			Torque	IGE		Init: CT&M Rec: EM	Init: $\pm 3\%$ Rec: EM
			Rotor speed	IGE		Init: CT&M Rec: EM	Init: $\pm 3\%$ Rec: EM
			Altitude	IGE		Init: CT&M Rec: EM	Init: ± 20 ft (6.1 m) Rec: EM
			Pitch angle	IGE		Init: CT&M Rec: EM	Init: $\pm 2^\circ$ Rec: EM
			Bank angle	IGE		Init: CT&M Rec: EM	Init: $\pm 2^\circ$ Rec: EM
			Heading	IGE		Init: CT&M Rec: EM	Init: $\pm 5^\circ$ Rec: EM
			Longitudinal control position	IGE		Init: CT&M Rec: EM	Init: $\pm 10\%$ Rec: EM
			Lateral control position	IGE		Init: CT&M Rec: EM	Init: $\pm 10\%$ Rec: EM
			Directional control position	IGE		Init: CT&M Rec: EM	Init: $\pm 10\%$ Rec: EM

1.j: Landing		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
			Collective control position	IGE		Init: CT&M Rec: EM	Init: $\pm 10\%$ Rec: EM
1.j.5	Hover to touchdown	In-ground-effect hover	Torque	IGE	Init: CT&M Rec: EM	Init: $\pm 5\%$ Rec: EM	Init: $\pm 5\%$ Rec: EM
			Pitch angle	IGE	Init: CT&M Rec: EM	Init: $\pm 2^\circ$ Rec: EM	Init: $\pm 2^\circ$ Rec: EM
			Bank angle	IGE	Init: CT&M Rec: EM	Init: $\pm 2^\circ$ Rec: EM	Init: $\pm 2^\circ$ Rec: EM
			Heading	IGE	Init: CT&M Rec: EM	Init: $\pm 3^\circ$ Rec: EM	Init: $\pm 3^\circ$ Rec: EM
			Longitudinal control position	IGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 10\%$ Rec: EM
			Lateral control position	IGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 10\%$ Rec: EM
			Directional control position	IGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 10\%$ Rec: EM
			Collective control position	IGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 10\%$ Rec: EM
1.j.6	Level flight acceleration	From hover to V_Y	Airspeed	IGE	Init: CT&M Rec: EM	Init: ± 3 kt Rec: EM	Init: ± 3 kt Rec: EM
			Torque	IGE	Init: CT&M Rec: EM	Init: $\pm 3\%$ Rec: EM	Init: $\pm 3\%$ Rec: EM

1.j: Landing		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
			Pitch angle	IGE	Init: CT&M Rec: EM	Init: $\pm 2^\circ$ Rec: EM	Init: $\pm 2^\circ$ Rec: EM
			Bank angle	IGE	Init: CT&M Rec: EM	Init: $\pm 2^\circ$ Rec: EM	Init: $\pm 2^\circ$ Rec: EM
			Heading	IGE	Init: CT&M Rec: EM	Init: $\pm 2^\circ$ Rec: EM	Init: $\pm 2^\circ$ Rec: EM
			Longitudinal control position	IGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 10\%$ Rec: EM
			Lateral control position	IGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 10\%$ Rec: EM
			Directional control position	IGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 10\%$ Rec: EM
			Collective control position	IGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 10\%$ Rec: EM

Requirements for all the rotorcraft landing tests.

1.j: Landing	Test requirements
All tests	1. Wind conditions should be justified. In addition, if wind is not constant during the test, it should be plotted. 2. For fidelity level G only, a FSTD footprint test is an acceptable alternative to CT&M for initial evaluation.

Individual test requirements. The individual test requirements defined in the table below are applicable to all fidelity levels unless otherwise stated.

1.j: Landing		Test requirements
1.j.1	All engines landing	1. Test should cover time history of approach and landing to hover. 2. If airspeed is not valid or not reliable, tolerance should be applied to ground speed instead. 3. If pressure altitude becomes unreliable due to rotor downwash, height above ground should be used instead. As an alternative, GNSS altitude above ellipsoid can be used (with initial offset compensated) instead of pressure altitude for the whole test.
1.j.2	OEI landing	1. Test should cover time history of approach and landing to touchdown. 2. If airspeed is not valid or not reliable, tolerance should be applied to ground speed instead. 3. If pressure altitude becomes unreliable due to rotor downwash, height above ground should be used instead. As an alternative, GNSS altitude above ellipsoid can be used (with initial offset compensated) instead of pressure altitude for the whole test.
1.j.3	Balked landing / missed approach	1. Test should be from a stabilised approach at the landing decision point (LDP). 2. If airspeed is not valid or not reliable, tolerance should be applied to ground speed instead.
1.j.4	Autorotational landing with touchdown	1. Test should cover time history of autorotational deceleration and touchdown from a stabilised autorotational descent. 2. If airspeed is not valid or not reliable, tolerance should be applied to ground speed instead. 3. If pressure altitude becomes unreliable due to rotor downwash, height above ground should be used instead. As an alternative, GNSS altitude above ellipsoid can be used (with initial offset compensated) instead of pressure altitude for the whole test. 4. If full autorotation landing cannot be safely performed or is not allowed by the RFM, alternate manoeuvres can be used, namely: a. a simulated autorotational flare and reduction of rate of descent at altitude; or b. a power-on termination following an autorotational approach and flare. In such cases, a footprint test for full autorotation landing should be provided, along with a statement from a suitably qualified pilot to validate the autorotation landing.
1.j.5	Hover to touchdown	1. Test should record a manoeuvre starting from in-ground-effect hover and lasting until touchdown, ending with collective control full down.

1.j: Landing		Test requirements
1.j.6	Level flight acceleration	1. Test should record a coordinated level flight acceleration with a minimum approximate speed range increase from hover to V_Y, initiated with a single power increase at the beginning of the manoeuvre (not a continuous power increase). 2. If airspeed is not valid or not reliable, tolerance should be applied to ground speed instead.

GM1 FSTD.QTG.H.145 Performance – Landing – Rotorcraft

Refer to GM1 FSTD.QTG.H.110.

CS FSTD.QTG.H.200 Handling qualities – Control system mechanical characteristics – Rotorcraft

This CS provides the standards for the objective tests for rotorcraft control system mechanical characteristics.

2.a: Control system mechanical characteristics		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
2.a.1	Cyclic force versus position	Static on ground Trim on and off Friction off Longitudinal and lateral	Breakout force	CLH	Init: CT&M Rec: ± 0.111 daN (0.25 lbf) or ± 25 %	Init: ± 0.111 daN (0.25 lbf) or ± 25 % Rec: Same as Init	Init: ± 0.111 daN (0.25 lbf) or ± 25 % Rec: Same as Init
			Force	CLH	Init: CT&M Rec: ± 0.222 daN (0.5 lbf) or ± 10 %	Init: ± 0.222 daN (0.5 lbf) or ± 10 % Rec: Same as Init	Init: ± 0.222 daN (0.5 lbf) or ± 10 % Rec: Same as Init
			Control range	CLH	Init: CT&M Rec: ± 1.5 % of total travel or ± 3 mm	Init: ± 1.5 % of total travel or ± 3 mm Rec: Same as Init	Init: ± 1.5 % of total travel or ± 3 mm Rec: Same as Init
2.a.2	Collective/pedal force versus position	Static on ground Trim on and off Friction off Collective and pedals	Breakout force	CLH	Init: CT&M Rec: ± 0.222 daN (0.5 lbf) or ± 25 %	Init: ± 0.222 daN (0.5 lbf) or ± 25 % Rec: Same as Init	Init: ± 0.222 daN (0.5 lbf) or ± 25 % Rec: Same as Init
			Force	CLH	Init: CT&M Rec: ± 0.445 daN (1 lbf) or ± 10 %	Init: ± 0.445 daN (1 lbf) or ± 10 % Rec: Same as Init	Init: ± 0.445 daN (1 lbf) or ± 10 % Rec: Same as Init
			Control range	CLH	Init: CT&M Rec: ± 1.5 % of total travel or ± 3 mm	Init: ± 1.5 % of total travel or ± 3 mm Rec: Same as Init	Init: ± 1.5 % of total travel or ± 3 mm Rec: Same as Init
2.a.3	Brake pedal force versus position	Static on ground	Force	CLH			Init: ± 2.224 daN (5 lbf) or ± 10 % Rec: Same as Init

2.a: Control system mechanical characteristics		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
			Control range	CLH			Init: ± 1.5 % of total travel or ± 3 mm Rec: Same as Init
2.a.4	Trim system rate	Static on ground Trim on Friction off All applicable axes, in both directions for each axis	Trim rate	CLO, CLH	Init: CT&M Rec: ± 10 %	Init: ± 10 % Rec: Same as Init	Init: ± 10 % Rec: Same as Init
2.a.5	Control dynamics – dynamic releases	For flight conditions, refer to requirements in the test requirement table below All applicable axes, in both directions for each axis Trim on Friction off	For underdamped systems, all the following tolerance parameters				
			P_0 : time from 90 % of A_d to first zero crossing)	CLH			Init: ± 10 % or 0.05 s Rec: Same as Init
			P_1 : period of the first oscillation	CLH			Init: ± 20 % or 0.05 s Rec: Same as Init
			P_2 : period of the second oscillation	CLH			Init: ± 30 % or 0.05 s Rec: Same as Init
			P_n : period of the n th oscillation	CLH			Init: $\pm 10(n + 1)$ % or ± 0.05 s Rec: Same as Init

2.a: Control system mechanical characteristics		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
		Stability augmentation on and off	A_n : amplitude of the n th significant overshoot (outside the residual band)	CLH			Init: $\pm 10\%$ of the amplitude of the largest overshoot ($A_{\max}$) or $\pm 0.5\%$ of the total control travel (stop to stop) Rec: Same as Init
			Number of significant overshoots	CLH			Init: ± 1 (minimum of one significant overshoot) Rec: Same as Init
			For overdamped and critically damped systems				
			P_0 : time from 90 % of A_d to 10 % of A_d	CLH			Init: $\pm 10\%$ or 0.05 s Rec: Same as Init
	Control dynamics – sweeps	Static on ground All applicable axes, slow and fast sweeps Trim on Friction off	Force	CLH		Init: ± 0.9 daN (2 lbf) or $\pm 10\%$ on the dynamic increment above static test Rec: Same as Init	

2.a: Control system mechanical characteristics		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
2.a.6	Free play	Static on ground Friction off All axes	Freeplay	CLH	Init: CT&M Rec: ± 0.1 in (2.5 mm)	Init: CT&M Rec: ± 0.1 in (2.5 mm)	Init: From 0 to validation data value + 0.1 in (2.5 mm) Rec: ± 0.1 in (2.5 mm)
2.a.7	Rotor blade angle versus control position	Static on ground, rotors not turning Stability augmentation off All axes	Blade angle or equivalent	CLO	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: CT&M Rec: EM

Requirements for all the rotorcraft control system mechanical characteristic tests.

2.a: Control system mechanical characteristics	Test requirements
All tests	1. Plots of pitch, roll, yaw and collective controller position versus force should be measured at the control. a. An alternative method in lieu of external test fixtures at the flight controls is to have recording and measuring instrumentation built into the FSTD. The force and position data from this instrumentation should be directly recorded and matched to the validation data. b. Provided that the instrumentation is verified by using external measuring equipment while conducting the static control checks, or verified using equivalent means, and that evidence of the satisfactory comparison is included in the QTG, the instrumentation can be used for both initial and recurrent testing for the measurement of all required control checks. c. Verification of the instrumentation using external measuring equipment should be repeated if major modifications or repairs are made to the control loading system. Such a permanent installation can be used without any time being lost for the installation of external devices. 2. Static control checks from the second set of pilot controls are only required if both sets of controls are not mechanically interconnected in the FSTD. A rationale is required from the validation data provider if a single set of data is applicable to both sets of controls. If the controls are mechanically interconnected in the FSTD, a single set of checks are sufficient. 3. For the control sweep tests (tests 2.a.1, 2.a.2 and 2.a.3), if there is a steep force gradient near control stops, the force tolerance does not apply. 4. Test should be with the hydraulic system, if applicable, pressurised. A supplemental hydraulic pressurisation system may be used. 5. The static control checks must be conducted manually if the FSTD control positions are not backdriven by the control loading. 6. For irreversible flight control systems, if forces are generated solely by use of aircraft hardware in the FSTD, then testing of position versus force is not applicable.

Individual test requirements. The individual test requirements defined in the table below are applicable to all fidelity levels unless otherwise stated.

2.a: Control system mechanical characteristics		Test requirements
2.a.1	Cyclic force versus position	1. Test should be uninterrupted control sweeps. 2. This test is not required for aircraft hardware modular controllers.
2.a.2	Collective/pedal force versus position	1. Test should be uninterrupted control sweeps. 2. This test is not required for aircraft hardware modular controllers.
2.a.4	Trim system rate	1. Feature CLO is applicable if autopilot, a stability augmentation system or similar equipment is involved in trim actuation.
2.a.5	Control dynamics – dynamic releases	1. Control dynamic tests should be performed in hover and in cruise. a. Control dynamics for reversible control systems should be evaluated in flight, considering safety issues or using an alternate method. The alternative method should be proposed to and agreed with the competent authority on a case-by-case basis. b. For irreversible control systems, control dynamic tests may be evaluated in a ground/static condition. This applies if proper pitot-static inputs, if applicable, are provided to represent airspeeds typical of those encountered in flight. 2. Data should be for a normal control displacement in both directions in each axis, approximately 25–50 % of full travel (rest position to stop) or that necessary for proper excitation. 3. For test, n = the sequential period of a full cycle of oscillation. 4. The amplitude of the residual band is ± 5 % of the A_d or ± 0.5 % of the total control travel (stop to stop), whichever is greater. Oscillations within the residual band are not considered significant and are not subject to tolerances. 5. Test should measure the free response of the controls using a step input or pulse input to excite the system.
	Control dynamics – sweeps	1. The method applies to rotorcraft with hydraulically powered flight controls and artificial feel systems. Instead of free response measurements, the system is validated by measurements of control force and rate of movement. 2. The control should be forced to its maximum extreme position for distinct rates, namely: a. for static tests, refer to tests 2.a.1 and 2.a.2; b. for slow dynamic sweeps, achieve a full sweep in approximately 10 seconds;

2.a: Control system mechanical characteristics		Test requirements
		c. for fast dynamic sweeps, achieve a full sweep in approximately 4 seconds. 3. Dynamic sweeps may be limited to forces not exceeding 44.5 daN (100 lbf).
2.a.7	Rotor blade angle versus control position	1. Acceptable alternatives for blade angles are actuator extension or swashplate attitudes. For fidelity level G, rotor disc attitudes are acceptable. 2. Test can be combined with tests 2.a.1 and 2.a.2.

GM1 FSTD.QTG.H.200 Handling qualities – Control system mechanical characteristics – Rotorcraft

Refer to GM1 FSTD.QTG.A.200 'Static control checks – Aeroplane' and GM1 FSTD.QTG.A.205 'Dynamic control checks – Aeroplane', which are also applicable to rotorcraft.

CS FSTD.QTG.H.205 Low-airspeed handling qualities – Rotorcraft

This CS provides the standards for the objective tests for rotorcraft low-airspeed handling qualities.

2.b: Handling – low airspeed		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
2.b.1	Trimmed flight control positions	Translational flight in-ground effect Sideways, rearwards and forwards For ground speed, refer to requirements in the test requirement table below	Torque	IGE		Init: $\pm 3\%$ Rec: EM	Init: $\pm 3\%$ Rec: EM
			Pitch angle	IGE		Init: $\pm 1.5^\circ$ Rec: EM	Init: $\pm 1.5^\circ$ Rec: EM
			Bank angle	IGE		Init: $\pm 2^\circ$ Rec: EM	Init: $\pm 2^\circ$ Rec: EM
			Longitudinal control position or blade angle	IGE, CLO		Init: $\pm 5\%$ Rec: EM	Init: $\pm 5\%$ Rec: EM
			Lateral control position or blade angle	IGE, CLO		Init: $\pm 5\%$ Rec: EM	Init: $\pm 5\%$ Rec: EM
			Directional control position or blade angle	IGE, CLO		Init: $\pm 5\%$ Rec: EM	Init: $\pm 5\%$ Rec: EM
			Collective control position or blade angle	IGE, CLO		Init: $\pm 5\%$ Rec: EM	Init: $\pm 5\%$ Rec: EM
2.b.2	Critical azimuth	Stationary hover in-	Torque	IGE		Init: $\pm 3\%$ Rec: EM	Init: $\pm 3\%$ Rec: EM

2.b: Handling – low airspeed		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
		ground effect For relative wind direction, refer to requirements in the test requirement table below	Pitch angle	IGE		Init: $\pm 1.5^\circ$ Rec: EM	Init: $\pm 1.5^\circ$ Rec: EM
			Bank angle	IGE		Init: $\pm 2^\circ$ Rec: EM	Init: $\pm 2^\circ$ Rec: EM
			Longitudinal control position or blade angle	IGE, CLO		Init: $\pm 5\%$ Rec: EM	Init: $\pm 5\%$ Rec: EM
			Lateral control position or blade angle	IGE, CLO		Init: $\pm 5\%$ Rec: EM	Init: $\pm 5\%$ Rec: EM
			Directional control position or blade angle	IGE, CLO		Init: $\pm 5\%$ Rec: EM	Init: $\pm 5\%$ Rec: EM
			Collective control position or blade angle	IGE, CLO		Init: $\pm 5\%$ Rec: EM	Init: $\pm 5\%$ Rec: EM
2.b.3.a	Longitudinal control response	Hover Stability augmentation on and off	Pitch rate	IGE, OGE	Init: CT&M Rec: EM	Init: $\pm 10\%$ or $\pm 2^\circ/s$ Rec: EM	Init: $\pm 10\%$ or $\pm 2^\circ/s$ Rec: EM
			Pitch angle change	IGE, OGE	Init: CT&M Rec: EM	Init: $\pm 10\%$ or $\pm 1.5^\circ$ Rec: EM	Init: $\pm 10\%$ or $\pm 1.5^\circ$ Rec: EM
2.b.3.b	Lateral control response	Hover Stability augmentation on and off	Roll rate	IGE, OGE	Init: CT&M Rec: EM	Init: $\pm 10\%$ or $\pm 3^\circ/s$ Rec: EM	Init: $\pm 10\%$ or $\pm 3^\circ/s$ Rec: EM
			Bank angle change	IGE, OGE	Init: CT&M Rec: EM	Init: $\pm 10\%$ or $\pm 3^\circ$ Rec: EM	Init: $\pm 10\%$ or $\pm 3^\circ$ Rec: EM

2.b: Handling – low airspeed		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
2.b.3.c	Directional control response	Hover	Yaw rate	IGE, OGE	Init: CT&M Rec: EM	Init: $\pm 10\%$ or $\pm 2^\circ/\text{s}$ Rec: EM	Init: $\pm 10\%$ or $\pm 2^\circ/\text{s}$ Rec: EM
		Both directions					
		Stability augmentation on and off	Heading change	IGE, OGE	Init: CT&M Rec: EM	Init: $\pm 10\%$ or $\pm 2^\circ$ Rec: EM	Init: $\pm 10\%$ or $\pm 2^\circ$ Rec: EM
			Altitude or height or vertical speed	IGE, OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: CT&M Rec: EM
2.b.3.d	Vertical control response	Hover	Normal acceleration	IGE, OGE	Init: CT&M Rec: EM	Init: $\pm 0.1\text{ g}$ Rec: EM	Init: $\pm 0.1\text{ g}$ Rec: EM
		Stability augmentation on and off	Vertical speed	IGE, OGE	Init: CT&M Rec: EM	Init: $\pm 10\%$ or $\pm 100\text{ fpm (0.5 m/s)}$ Rec: EM	Init: $\pm 10\%$ or $\pm 100\text{ fpm (0.5 m/s)}$ Rec: EM

Individual test requirements. The individual test requirements defined in the table below are applicable to all fidelity levels unless otherwise stated.

2.b: Handling – low airspeed		Test requirements
2.b.1	Trimmed flight control positions	- Test should cover several ground speed increments to translational airspeed limits and 45 kt forwards. - Test may be a series of snapshot tests. Each series of snapshots should be obtained during a continuous uninterrupted flight. - Stable flight conditions should be reached at each airspeed increment. - If tolerances are applied to rotor blade angles, it should be justified that the flight control positions are consistent with the authority range of the autopilot serial actuators.
2.b.2	Critical azimuth	- Test should present data for three relative wind directions (including the most critical case) in the critical quadrant. - Test may be a series of snapshot tests. The data provider must ensure that a steady-state condition exists at the instant of time captured by the snapshot. The steady-state condition must exist from four seconds prior to the instant of time captured by the snapshot through to one second following it.

2.b: Handling – low airspeed		Test requirements
		- Precise wind measurement is very difficult and use of simulated wind obtained by translational flight in calm weather condition (no wind) is preferred in order to precisely control flight conditions by using ground speed measurement (usually GNSS). In this condition, it is more practical to realise this test with test 2.b.1 in order to ensure consistency between critical azimuth and other directions (forwards, sideways and rearwards). If tests 2.b.1 and 2.b.2 are performed concurrently in calm weather condition, then only one azimuth (the most critical one) is required for test 2.b.2. - If not defined in the RFM, the critical azimuth should be defined either as the azimuth with minimum directional control margin or as the azimuth with maximum instability. - If tolerances are applied to rotor blade angles, it should be justified that the flight control positions are consistent with the authority range of the autopilot serial actuators.
2.b.3.a	Longitudinal control response	- Test should use step control input. - Attitude change is defined as the difference between the current attitude and the stable attitude prior to the step input. - The tolerance for attitude change is applied continuously from the time of the step input. - This is a short-term response test. - Validation data in-ground effect is preferred, since the pilot has a better reference point. However, for safety reasons, validation data out-of-ground effect is also acceptable. Only one test (either in-ground effect or out-of ground effect) is required.
2.b.3.b	Lateral control response	- Test should use step control input. - Attitude change is defined as the difference between the current attitude and the stable attitude prior to the step input. - The tolerance for attitude change is applied continuously from the time of the step input. - This is a short-term response test. - Validation data in-ground effect is preferred, since the pilot has a better reference point. However, for safety reasons, validation data out-of-ground effect is also acceptable. Only one test (either in-ground effect or out-of-ground effect) is required.
2.b.3.c	Directional control response	- Test should use step control input. - Heading change is defined as the difference between the current heading and the stable heading prior to the step

2.b: Handling – low airspeed		Test requirements
		input. 3. The tolerance for heading change is applied continuously from the time of the step input. 4. This is a short-term response test. 5. Validation data in-ground effect is preferred, since the pilot has a better reference point. However, for safety reasons, validation data out-of-ground effect is also acceptable. Only one test (either in-ground effect or out-of-ground effect) is required for each direction (total of two tests).
2.b.3.d	Vertical control response	1. Test should use step control input. 2. This is a short-term response test. 3. Validation data in-ground effect is preferred, since the pilot has a better reference point. However, for safety reasons, validation data out-of-ground effect is also acceptable. Only one test (either in-ground effect or out-of-ground effect) is required.

GM1 FSTD.QTG.H.205 Low-airspeed handling qualities – Rotorcraft

To better demonstrate the influence of airspeed on trimmed flight control positions, test 2.b.1 should include snapshots of a continuous, uninterrupted flight in stable conditions approximately every 10 kt, with other flight conditions unchanged.

In order to better demonstrate the critical azimuth, it is recommended that the chosen airspeed is close to translational limits in the critical quadrant, within the flight envelope.

CS FSTD.QTG.H.210 Longitudinal handling qualities – Rotorcraft

This CS provides the standards for the objective tests for rotorcraft longitudinal handling qualities.

2.c: Handling – longitudinal		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
2.c.1	Control response	Straight cruise and level flight	Pitch rate	OGE	Init: CT&M Rec: EM	Init: $\pm 10\%$ or $\pm 2^\circ/\text{s}$ Rec: EM	Init: $\pm 10\%$ or $\pm 2^\circ/\text{s}$ Rec: EM
		Stability augmentation on and off For airspeed, refer to requirements in the test requirement table below	Pitch angle change	OGE	Init: CT&M Rec: EM	Init: $\pm 10\%$ or $\pm 1.5^\circ$ Rec: EM	Init: $\pm 10\%$ or $\pm 1.5^\circ$ Rec: EM
2.c.2	Static stability	Either straight cruise and level flight, or straight cruise climb	Longitudinal control position change from trim	OGE, CLO	Init: CT&M Rec: EM	Init: $\pm 10\%$ or $\pm 6.3\text{ mm}$ (0.25 in) Rec: EM	Init: $\pm 10\%$ or $\pm 6.3\text{ mm}$ (0.25 in) Rec: EM
		Autorotation For airspeed, refer to requirements in the test requirement table below	Longitudinal control force change from trim	OGE, CLO	Init: CT&M Rec: EM	Init: $\pm 10\%$ or $\pm 0.2224\text{ daN}$ (0.5 lbf) Rec: EM	Init: $\pm 10\%$ or $\pm 0.2224\text{ daN}$ (0.5 lbf) Rec: EM
2.c.3.a	Dynamic stability –	Straight cruise and	If behaviour is periodic				

2.c: Handling – longitudinal		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
	long-term response (phugoid dynamics)	level flight Stability augmentation off	Period	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 10\%$ or $\pm 0.5\text{ s}$ Rec: EM
			Additionally, one of the following tolerance parameters				
			Time to one-half or double amplitude	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 10\%$ Rec: EM
			Damping ratio	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: ± 0.02 Rec: EM
			If behaviour is non-periodic				
			Pitch angle	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 3^\circ$ Rec: EM
			Airspeed	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 5\text{ kt}$ Rec: EM
2.c.3.b	Dynamic stability – short-term response	Straight cruise flight, either level or climb Stability augmentation on and off For airspeed, refer to requirements in the test requirement table below	Normal acceleration	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 0.1\text{ g}$ Rec: EM
			Additionally, one of the following tolerance parameters				
			Pitch angle	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 1.5^\circ$ Rec: EM
			Pitch rate	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: $\pm 2^\circ/\text{s}$ Rec: EM

2.c: Handling – longitudinal		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
2.c.4	Manoeuvring stability	Either cruise or climb	Longitudinal control position change from trim	OGE, CLO		Init: CT&M Rec: EM	Init: $\pm 10\%$ or $\pm 6.3\text{ mm}$ (0.25 in) Rec: EM
		Stability augmentation on and off Left and right turns For airspeed, refer to requirements in the test requirement table below	Or Longitudinal control force change from trim	OGE, CLO		Init: CT&M Rec: EM	Init: $\pm 10\%$ or $\pm 0.2224\text{ daN}$ (0.5 lbf) Rec: EM

Individual test requirements. The individual test requirements defined in the table below are applicable to all fidelity levels unless otherwise stated.

2.c: Handling – longitudinal		Test requirements
2.c.1	Control response	- Test should cover two cruise airspeeds, to include minimum power required speed. - Test should use step control input. - Attitude change is defined as the difference between the current attitude and the stable attitude prior to the step input. - The tolerance for attitude change is applied continuously from the time of the step input.
2.c.2	Static stability	- Test may be a series of snapshot tests. - Test should cover a minimum of two speeds on each side of the trim speed, with a minimum gap of 5 kt between two consecutive speeds. - If presented as a series of snapshot tests, tolerance for control position change from trim does not apply to the reference trim speed snapshot but does apply to the four subsequent snapshots. A minimum of 10 snapshot tests are required (5 for each speed), with tolerances applied to 8 of them.

2.c: Handling – longitudinal		Test requirements
		4. The rotorcraft is first stabilised in straight cruise and level flight at an initial airspeed; this condition is identified as the reference trim speed. From this reference trim condition, four additional stable straight cruise conditions are reached, two with different airspeeds greater than the reference trim speed and two with different airspeeds lower than the reference trim speed; they can be reached in any order. 5. For the demonstration of rotorcraft static stability, it is recommended that collective position/power remain unchanged when stabilising the four additional cruise conditions with increased or decreased airspeed; therefore, only the reference trim condition will be in level flight, while the additional stable cruise conditions will be in climb or descent. 6. Stability augmentation on and off conditions are required if the augmentation system includes airspeed regulation. 7. Control position change from trim is defined as the difference between the current control position and the position at the reference trim speed. 8. If presented as time histories, the tolerance for control position change from trim is applied continuously from the time the reference trim condition is reached. 9. In order to demonstrate static stability, the sign of the control position change from trim should match: with other flight conditions being unchanged, in order to increase airspeed, cyclic control should be moved forwards; to decrease airspeed, cyclic control should be moved aft. 10. If the rotorcraft has force trim release, follow-up trim or any function that affects the control force and/or the trim position, this function should be disengaged to be able to apply tolerance to control force. Otherwise, tolerance should be applied to flight control position change from trim.
2.c.3.a	Dynamic stability – long-term response (phugoid dynamics)	1. Test should cover time history to validate long-term rotorcraft response (phugoid). Generally, this can be demonstrated with a control doublet input. This test should be performed even if the phugoid cannot be made apparent in the reference rotorcraft. 2. Test should include three full cycles (six overshoots after input completed) or the number sufficient to determine time to one-half or double amplitude, whichever is less. If rotorcraft behaviour diverges, the test should end before unsafe attitudes are reached. If rotorcraft longitudinal long-term response gets masked by off-axis behaviour (e.g. if/when phugoid is masked by Dutch roll), the test should end when longitudinal proper mode is no longer apparent. 3. For non-periodic response, or if the period and damping ratio cannot be determined due to rotorcraft behaviour, the time history of the FSTD should be matched to the validation data. Test should last 20 seconds after control release.

2.c: Handling – longitudinal		Test requirements
2.c.3.b	Dynamic stability – short-term response	1. Test should cover two cruise airspeeds. 2. Test should cover time history to validate short-term rotorcraft response due to control pulse input. 3. Test should check to stop four seconds after completion of input. If rotorcraft behaviour diverges, the test should end before unsafe attitudes are reached.
2.c.4	Manoeuvring stability	1. Test may be a series of snapshot tests. 2. Force may be a cross-plot for irreversible systems. 3. If presented as a series of snapshot tests, tolerance for control position change from trim does not apply to the reference trim snapshot in straight flight but does apply to the subsequent snapshots in left and right turns. A total of 10 snapshot tests are required (5 for each speed), with tolerances applied to 8 of them. 4. Test should cover two airspeeds. 5. Approximately 30° and 45° bank angle data should be presented. 6. The rotorcraft is first stabilised in straight cruise and level flight at a first airspeed. Then, the rotorcraft is stabilised at approximately 30° and 45°, left and right (in any order), while keeping the airspeed constant. It is recommended that collective position/power remain unchanged when stabilising these four turn conditions, which may no longer be in level flight. The above test conditions are repeated at a second airspeed. Each series of snapshots should be obtained during a continuous, uninterrupted flight. 7. For each speed condition, the control position change from trim is defined as the difference between the current control position and the initial stable control position in straight and level flight. 8. If the rotorcraft has force trim release, follow-up trim or any function that affects the control force and/or the trim position, this function should be disengaged to be able to apply tolerance to control force. Otherwise, tolerance should be applied to flight control position change from trim.

CS FSTD.QTG.H.215 Lateral and directional handling qualities – Rotorcraft

This CS provides the standards for the objective tests for rotorcraft lateral and directional handling qualities.

2.d: Handling – lateral and directional		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
2.d.1.a	Control response – lateral	Straight cruise and level flight	Roll rate	OGE	Init: CT&M Rec: EM	Init: $\pm 10\%$ or $\pm 3^\circ/\text{s}$ Rec: EM	Init: $\pm 10\%$ or $\pm 3^\circ/\text{s}$ Rec: EM
		Stability augmentation on and off For airspeed, refer to requirements in the test requirement table below	Bank angle change	OGE	Init: CT&M Rec: EM	Init: $\pm 10\%$ or $\pm 3^\circ$ Rec: EM	Init: $\pm 10\%$ or $\pm 3^\circ$ Rec: EM
2.d.1.b	Control response – directional	Straight cruise and level flight	Yaw rate	OGE	Init: CT&M Rec: EM	Init: $\pm 10\%$ or $\pm 2^\circ/\text{s}$ Rec: EM	Init: $\pm 10\%$ or $\pm 2^\circ/\text{s}$ Rec: EM
		Stability augmentation on and off For airspeed, refer to requirements in the test requirement table below	Heading change	OGE	Init: CT&M Rec: EM	Init: $\pm 10\%$ or $\pm 2^\circ$ Rec: EM	Init: $\pm 10\%$ or $\pm 2^\circ$ Rec: EM
2.d.2	Static stability		One of the following tolerance parameters				

2.d: Handling – lateral and directional		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
		Straight cruise flight, either level or both climb and descent	Bank angle	OGE	Init: CT&M Rec: EM	Init: $\pm 1.5^\circ$ Rec: EM	Init: $\pm 1.5^\circ$ Rec: EM
			Lateral load factor	OGE	Init: CT&M Rec: EM	Init: 0.026 g Rec: EM	Init: ± 0.026 g Rec: EM
		For side-slip, refer to requirements in the test requirement table below	Additionally, one of the following tolerance parameters				
			Longitudinal control position change from trim	OGE, CLO	Init: CT&M Rec: EM	Init: $\pm 10\%$ or ± 6.3 mm (0.25 in) Rec: EM	Init: $\pm 10\%$ or ± 6.3 mm (0.25 in) Rec: EM
			Longitudinal control force change from trim	OGE, CLO	Init: CT&M Rec: EM	Init: $\pm 10\%$ or ± 0.2224 daN (0.5 lbf) Rec: EM	Init: $\pm 10\%$ or ± 0.2224 daN (0.5 lbf) Rec: EM
			Additionally, one of the following tolerance parameters				
			Lateral control position change from trim	OGE, CLO	Init: CT&M Rec: EM	Init: $\pm 10\%$ or ± 6.3 mm (0.25 in) Rec: EM	Init: $\pm 10\%$ or ± 6.3 mm (0.25 in) Rec: EM
			Lateral control force change from trim	OGE, CLO	Init: CT&M Rec: EM	Init: $\pm 10\%$ or ± 0.2224 daN (0.5 lbf) Rec: EM	Init: $\pm 10\%$ or ± 0.2224 daN (0.5 lbf) Rec: EM
			Additionally, one of the following tolerance parameters				
			Directional control position change from trim	OGE, CLO	Init: CT&M Rec: EM	Init: $\pm 10\%$ or ± 6.3 mm (0.25 in) Rec: EM	Init: $\pm 10\%$ or ± 6.3 mm (0.25 in) Rec: EM

2.d: Handling – lateral and directional		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
			Directional control force change from trim	OGE, CLO	Init: CT&M Rec: EM	Init: ±10 % or ±0.4448 daN (1 lbf) Rec: EM	Init: ±10 % or ±0.4448 daN (1 lbf) Rec: EM
2.d.3.a	Dynamic stability – lateral and directional oscillations (Dutch roll)	Straight cruise flight, either level or climb Stability augmentation on and off	For periodic response, the following tolerance parameters				
			Period	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: ±10 % or ±0.5 s Rec: EM
			Time difference between peaks of bank and side-slip	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: ±20 % or ±1 s Rec: EM
			Additionally, one of the following tolerance parameters				
			Time to one-half or double amplitude	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: ±10 % Rec: EM
			Damping ratio	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: ±0.02 Rec: EM
			For non-periodic response, all the following tolerance parameters				
			Airspeed	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: ±10 kt Rec: EM
			Bank angle	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: ±5° Rec: EM
			Heading	OGE	Init: CT&M Rec: EM	Init: CT&M Rec: EM	Init: ±4° Rec: EM
2.d.3.b	Reserved						

2.d: Handling – lateral and directional		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
2.d.3.c	Dynamic stability – adverse/proverse yaw	Straight cruise flight, either level or climb	Side-slip	OGE			Init: $\pm 2^\circ$ Rec: EM
			Or all the following tolerance parameters				
		Stability augmentation on and off	Bank angle	OGE			Init: $\pm 2^\circ$ Rec: EM
			Heading	OGE			Init: $\pm 3^\circ$ Rec: EM
		Left and right turns	Lateral load factor	OGE			Init: ± 0.052 g Rec: EM

Individual test requirements. The individual test requirements defined in the table below are applicable to all fidelity levels unless otherwise stated.

2.d: Handling – lateral and directional		Test requirements
2.d.1.a	Control response – lateral	- Test should cover two airspeeds, to include one at or near the minimum power required speed. - Test should use step control input. - Attitude change is defined as the difference between the current attitude and the stable attitude prior to the step input. - The tolerance for attitude change is applied continuously from the time of the step input.
2.d.1.b	Control response – directional	- Test should cover two airspeeds, to include one at or near the minimum power required speed. - Test should use step control input. - Heading change is defined as the difference between the current heading and the stable heading prior to the step input. - The tolerance for heading change is applied continuously from the time of the step input.
2.d.2	Static stability	- Test should cover steady heading side-slip. - Test should cover a minimum of two side-slip angles on either side of the trim point. The rotorcraft is first stabilised

2.d: Handling – lateral and directional	Test requirements
	in straight flight; this condition is identified as the reference trim. From this reference trim condition, four additional stable straight cruise conditions are reached, two with different left side-slip and two with different right side-slip; they can be reached in any order. For the demonstration of rotorcraft static stability, it is recommended that collective position/power remain unchanged when stabilising the four additional side-slip conditions. Therefore, if initially in level flight, rotorcraft may no more be in level flight when stabilised with side-slip; likewise, if initially in climb/descent, rotorcraft may no longer be in climb/descent when stabilised with side-slip. 3. If initial conditions are not level flight, then both climb and descent conditions should be presented. 4. Force may be a cross-plot for irreversible control systems. 5. Test may be a series of snapshot tests. 6. Control position change from trim is defined as the difference between the current control position and the position at the reference trim. 7. If presented as a series of snapshot tests, tolerance for control position change from trim does not apply to reference trim speed snapshot but does apply to the four subsequent snapshots. A minimum of five snapshot tests are required if cruise and level flight conditions are chosen. A minimum of 10 snapshot tests (5 in climb and 5 in descent) are required if climb and descent flight conditions are chosen. Each series of snapshots should be obtained during a continuous uninterrupted flight. 8. If presented as time histories, the tolerance for control position change from trim is applied continuously from the time the reference trim condition is reached. 9. If the rotorcraft has force trim release, follow-up trim or any function that affects the control force and/or the trim position, this function should be disengaged to be able to apply tolerances to control forces. Otherwise, tolerances should be applied to flight controls positions change from trim.
2.d.3.a	Dynamic stability – lateral and directional oscillations (Dutch roll) 1. Test should cover two cruise airspeeds. 2. Test should cover time history to validate long-term rotorcraft response (Dutch roll). Test should excite with cyclic or pedal doublet. This test should be performed even if the Dutch roll cannot be made apparent on the reference rotorcraft. 3. Test should include six full cycles (12 overshoots after input completed) or the number sufficient to determine time to one-half or double amplitude, whichever is less. If rotorcraft behaviour diverges, test should end before unsafe attitudes are reached. If rotorcraft longitudinal long-term response gets masked by off-axis behaviour, the test

2.d: Handling – lateral and directional		Test requirements
		should end when lateral/directional proper mode is no longer apparent. 4. For non-periodic response or if the period and damping ratio cannot be determined due to rotorcraft behaviour, the test should last 20 seconds after control release.
2.d.3.c	Dynamic stability – adverse/proverse yaw	1. Test should cover time history of initial entry into uncoordinated turns in both directions. Test should use moderate control input rate. From a stable straight cruise and level flight condition, rotorcraft enters an uncoordinated turn in one direction, using either pedal-only or cyclic-only input. The whole manoeuvre is repeated for the opposite turn direction. 2. Adverse yaw is the tendency of the rotorcraft to yaw in the opposite direction to the roll. Proverse yaw is the tendency of the rotorcraft to yaw in the same direction as the roll.

CS FSTD.QTG.H.300 Motion system – Rotorcraft

This CS provides the standards for the objective tests for rotorcraft motion systems.

3: Motion system		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
3.a	Frequency response	All linear and rotational axes	Phase	MTN		Init: 0° to -20° between 0.1 Hz and 1.0 Hz; 0° to -40° between 1.0 Hz and 3.0 Hz Rec: Same as Init	Init: 0° to -20° between 0.1 Hz and 1.0 Hz; 0° to -40° between 1.0 Hz and 3.0 Hz Rec: Same as Init
			Gain	MTN		Init: ±2 dB between 0.1 Hz and 1.0 Hz; ±4 dB between 1.0 Hz and 3.0 Hz Rec: Same as Init	Init: ±2 dB between 0.1 Hz and 1.0 Hz; ±4 dB between 1.0 Hz and 3.0 Hz Rec: Same as Init
3.b	Leg balance	n/a	Phase shift (each motion jack compared with a datum jack)	MTN		Init: 0° to 1.5° Rec: Same as Init	Init: 0° to 1.5° Rec: Same as Init
3.c	Turnaround	n/a	Deviation from the desired sinusoidal acceleration	MTN		Init: ±0.05 g peak Rec: Same as Init	Init: ±0.05 g peak Rec: Same as Init
3.d.1	Motion envelope – pitch	n/a	Displacement	MTN		Init: ±20° Rec: Same as Init	Init: ±25° Rec: Same as Init

3: Motion system		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
			Velocity	MTN		Init: $\pm 15^\circ/\text{s}$ Rec: Same as Init	Init: $\pm 20^\circ/\text{s}$ Rec: Same as Init
			Acceleration	MTN		Init: $\pm 75^\circ/\text{s}^2$ Rec: Same as Init	Init: $\pm 100^\circ/\text{s}^2$ Rec: Same as Init
3.d.2	Motion envelope – roll	n/a	Displacement	MTN		Init: $\pm 20^\circ$ Rec: Same as Init	Init: $\pm 25^\circ$ Rec: Same as Init
			Velocity	MTN		Init: $\pm 15^\circ/\text{s}$ Rec: Same as Init	Init: $\pm 20^\circ/\text{s}$ Rec: Same as Init
			Acceleration	MTN		Init: $\pm 75^\circ/\text{s}^2$ Rec: Same as Init	Init: $\pm 100^\circ/\text{s}^2$ Rec: Same as Init
3.d.3	Motion envelope – yaw	n/a	Displacement	MTN		Init: $\pm 25^\circ$ Rec: Same as Init	Init: $\pm 25^\circ$ Rec: Same as Init
			Velocity	MTN		Init: $\pm 15^\circ/\text{s}$ Rec: Same as Init	Init: $\pm 20^\circ/\text{s}$ Rec: Same as Init
			Acceleration	MTN		Init: $\pm 75^\circ/\text{s}^2$ Rec: Same as Init	Init: $\pm 100^\circ/\text{s}^2$ Rec: Same as Init
3.d.4	Motion envelope – vertical	n/a	Displacement	MTN		Init: ± 22 in Rec: Same as Init	Init: ± 34 in Rec: Same as Init
			Velocity	MTN		Init: ± 16 in/s Rec: Same as Init	Init: ± 24 in/s Rec: Same as Init
			Acceleration	MTN		Init: ± 0.6 g Rec: Same as Init	Init: ± 0.8 g Rec: Same as Init

3: Motion system		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
3.d.5	Motion envelope – lateral	n/a	Displacement	MTN		Init: ± 26 in Rec: Same as Init	Init: ± 45 in Rec: Same as Init
			Velocity	MTN		Init: ± 20 in/s Rec: Same as Init	Init: ± 28 in/s Rec: Same as Init
			Acceleration	MTN		Init: ± 0.4 g Rec: Same as Init	Init: ± 0.6 g Rec: Same as Init
3.d.6	Motion envelope – longitudinal	n/a	Displacement	MTN		Init: ± 27 in Rec: Same as Init	Init: ± 34 in Rec: Same as Init
			Velocity	MTN		Init: ± 20 in/s Rec: Same as Init	Init: ± 28 in/s Rec: Same as Init
			Acceleration	MTN		Init: ± 0.4 g Rec: Same as Init	Init: ± 0.6 g Rec: Same as Init
3.d.7	Motion envelope – initial rotational acceleration rate	n/a	All axes	MTN		Init: $\pm 225^\circ/\text{s}^2/\text{s}$ Rec: Same as Init	Init: $\pm 300^\circ/\text{s}^2/\text{s}$ Rec: Same as Init
3.d.8	Motion envelope – initial linear acceleration rate	n/a	Vertical	MTN		Init: ± 4 g/s Rec: Same as Init	Init: ± 6 g/s Rec: Same as Init
			Lateral	MTN		Init: ± 2 g/s Rec: Same as Init	Init: ± 3 g/s Rec: Same as Init
			Longitudinal	MTN		Init: ± 2 g/s Rec: Same as Init	Init: ± 3 g/s Rec: Same as Init

3: Motion system		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
3.e	Motion cue repeatability	On ground and in flight	Actual motion platform linear accelerations	MTN		Init: n/a Rec: ± 0.05 g actual motion platform linear accelerations compared with the MQTG value	Init: n/a Rec: ± 0.05 g actual motion platform linear accelerations compared with the MQTG value
3.f	Motion cueing performance signature	Refer to the test requirement table below for test 3.f	Actual motion platform linear acceleration cues	MTN			Init: Subjective motion assessment to establish MQTG footprint demonstrating high correlation between motion cueing and aircraft simulation model Rec: ± 0.05 g compared to the time histories of the MQTG

3: Motion system		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
			Actual motion platform angular cues	MTN			Init: Subjective motion assessment to establish MQTG footprint demonstrating high correlation between motion cueing and aircraft simulation model Rec: $\pm 0.5^\circ$ in the pitch, roll and yaw attitude of the actual motion platform and $\pm 0.5^\circ/\text{s}$ in the corresponding angular rates when compared with the MQTG time histories
			Flight parameters	MTN			Init: n/a Rec: EM

Requirements for all the rotorcraft motion system tests.

3.a–f: Motion system	Test requirements
All tests	- The intent of tests 3.a ('Frequency response'), 3.b ('Leg balance'), 3.c ('Turnaround') and 3.d ('Motion envelope') is to demonstrate the performance of the motion system hardware and to check the integrity of the motion system with regard to calibration and wear. These tests are considered as robotic tests and do not need to be run as integrated tests as required by CS FSTD.QTG.020, paragraph (d), as they are independent of the motion cueing software. - Manual tests are not required.

Individual test requirements. The individual test requirements defined in the table below are applicable to all fidelity levels unless otherwise stated.

3: Motion system		Test requirements
3.a	Frequency response	- The test should demonstrate the motion frequency response for an oscillatory input, for each linear and rotational axis. - The test should cover at least the frequencies between 0.1 Hz and 3.0 Hz. - Plots of phase and attenuation should be plotted against the frequency of the motion movement. - If motion accelerations are computed from the leg positions instead of being directly measured (e.g. by using accelerometers), this should be justified.
3.b	Leg balance	The phase shift between a datum jack and any other jack should be measured using a heave (vertical) signal of 0.5 Hz at ± 0.25 g.
3.c	Turnaround	- The test should demonstrate that the motion response during oscillatory manoeuvres does not exhibit excessive noise by driving each motion actuator with a sinusoidal signal and recording the motion movement. - The motion base should be driven sinusoidally in the heave axis through a displacement of 150 mm peak to peak at a frequency of 0.5 Hz. - The motion movement should be measured (e.g. using an accelerometer) and any deviation compared with the command signal.
3.d.7	Motion envelope – initial rotational acceleration rate	Test should cover all rotational axes.

3: Motion system		Test requirements
3.e	Motion cue repeatability	1. The test should demonstrate that the motion system hardware and software, in normal FSTD operating mode, continue to perform as initially qualified (i.e. to demonstrate that the motion system software and motion system hardware have not degraded or changed over time). 2. The characteristics of the test signal should be set so that the acceleration command reaches two thirds of the motion system acceleration envelope for the linear axes and two thirds of the angular velocity envelope for angular axes, as defined in test 3.d. These demands are to be applied at the point at which the aircraft centre-of-gravity accelerations and velocities are transformed into the pilot reference point accelerations and velocities prior to entering the motion cueing software. The test signal should be applied when the system is steady. The tests should clearly demonstrate the initial motion response to the demands and associated washouts. 3. Two tests should be performed, one in an on-ground state and the other in an in-air state. This is to cater for possible differences in motion system gains for the on-ground and in-air conditions. The exact test conditions are to be determined by the applicant for FSTD qualification. 4. The parameters to be plotted are: a. the acceleration/rate commands; b. actual linear acceleration measured from accelerometers close to the pilot position (the output will comprise accelerations due to both linear and rotational motion accelerations); c. motion actuator positions; and d. the motion system attitude and position (i.e. pitch, roll, yaw, surge, sway, heave).
3.f	Motion cueing performance signature	1. The tests establish a footprint for the motion movement and cueing for the test cases given in the table below. 2. For the initial qualification, the following apply. a. An SOJ should explain the motion movements and give the basis for their magnitude and correlation for each axis for each test case. b. In addition, the FSTD motion cueing should be compared with the aircraft's key acceleration and angular cues as provided by the simulation model, using overplotting of time histories. The motion cueing should maintain a high correlation with the simulation model's key acceleration and angular cues, both during significant changes and in steady-state conditions, while accounting for any washout effects.

		c. Time histories of the following parameters should be plotted for initial qualification (i.e. MQTG): i. same flight parameters as for the referenced objective tests (refer to the table below); ii. validation data for the flight parameters of the referenced objective tests (if data is available); iii. motion actuator positions for the whole time history of the test; iv. motion system attitude and position on all six axes (i.e. pitch, roll, yaw, heave, sway and surge) for the whole time history of the test; and v. correlation of motion cueing with respect to the aircraft simulation model in terms of load factor and angular cues by overplotting. The actual linear acceleration should be measured from accelerometers at the pilot reference point (e.g. under the pilot seat). 3. For recurrent testing, the following applies. Time histories of the following parameters should be overplotted with the MQTG results: a. same flight parameters of the simulation model as for the referenced objective tests (refer to the table below); b. motion actuator positions for the whole time history of the test; c. motion system attitude and position on all six axes (i.e. pitch, roll, yaw, heave, sway and surge) for the whole time history of the test; and d. the actual linear acceleration measured from accelerometers at the pilot reference point (e.g. under the pilot seat). 4. The table below specifies inputs and flight parameters from applicable objective tests. If a referenced test applies to this FSTD, its footprint should be used. If not, the associated motion cueing performance signature test case should be conducted using a manoeuvre similar to that in the referenced test. The table below defines the required test cases for an FSTD incorporating the aircraft simulation features at fidelity level G, R or S, whether individually or in combination. The table applies irrespective of the distribution of fidelity levels across the features. If any feature is at fidelity level N (not simulated), the applicable test cases should be determined in agreement with the competent authority.	
No	Test	Demonstration method	Key acceleration and angular cues

3: Motion system		Test requirements			
		1	Take-off (all engines)	Test is to demonstrate motion cueing for a normal take-off. Inputs and flight parameters are as in test 1.c.1 'Take-off – All engines'.	- Long-term x-axis acceleration due to airspeed increase - Pitch rate for rotorcraft pitch attitude changes - Vertical acceleration (G) due to lift-off trajectory
		2	Hover performance (in-ground effect or out-of-ground effect)	Test is to demonstrate motion cueing for hover in either in-ground effect or out-of-ground effect. Inputs and flight parameters are as in test 1.d 'Hover performance'.	Pitch and roll attitude
		3	Control response – longitudinal	Test is to demonstrate motion cueing for a small control input. Inputs and flight parameters are as in test 2.c.1 'Control response'.	Pitch rate for rotorcraft pitch attitude changes
		4	Control response – lateral	Test is to demonstrate motion cueing for a small control input. Inputs and flight parameters are as in test 2.d.1.a 'Control response – lateral'.	Roll rate for rotorcraft roll attitude changes
		5	Control response – directional	Test is to demonstrate motion cueing for a small control input. Inputs and flight parameters as in test 2.d.1.b 'Control response – directional'.	Yaw rate for rotorcraft yaw attitude changes

3: Motion system		Test requirements			
		6	Autorotational entry	Test is to demonstrate motion cueing for autorotational entry. The flight control response to autorotation should have a two-second delay to demonstrate the motion cueing for the angular movements. The test case may be a footprint test or may use the inputs and flight parameters of test 1.i 'Autorotational entry' if they fulfil the above requirement.	• Yaw rate for rotorcraft yaw attitude changes • Roll rate for rotorcraft roll attitude changes • Pitch rate for rotorcraft pitch attitude changes
		7	Go-around	Test is to demonstrate motion cueing for a go-around. Inputs and flight parameters are as in test 1.j.3 'Balked landing / missed approach'.	• Vertical acceleration (G) due to trajectory
		8	Landing (all engines)	Test is to demonstrate motion cueing for normal landing. Inputs and flight parameters are as in test 1.j.1 'All engines landing'.	• Pitch attitude of the aircraft during final approach • Pitch for flare
		9	Autorotational landing	Test is to demonstrate motion cueing for autorotation flare. Inputs and flight parameters are as in test 1.j.4 'Autorotational landing with touchdown'.	• Pitch attitude of the aircraft during final approach • Pitch for flare • Vertical acceleration (G) due to trajectory

GM1 FSTD.QTG.H.300 Motion system – Rotorcraft

Refer to GM1 FSTD.QTG.A.300 and GM2 FSTD.QTG.A.300 on motion cueing performance signature tests, which are also applicable to rotorcraft.

CS FSTD.QTG.H.305 Characteristic vibrations/buffet – Rotorcraft

This CS provides the standards for the objective tests for rotorcraft vibration systems.

3.g: Characteristic vibrations		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
3.g.1	Characteristic vibrations – on ground at idle	On ground at idle	Vibration amplitude at frequency 1/rev	VIB			Init: +3 dB / -6 dB, or ± 10 % of nominal vibration level in cruise flight Rec: Same as Init
			Vibration amplitude at frequency n /rev	VIB			Init: +3 dB / -6 dB, or ± 10 % of nominal vibration level in cruise flight Rec: Same as Init
3.g.2	Characteristic vibrations – on ground at operating RPM	On ground at operating RPM	Vibration amplitude at frequency 1/rev	VIB		Init: CT&M Rec: The FSTD should exhibit the overall appearance and trends as given in the MQTG	Same as 3.g.1

3.g: Characteristic vibrations		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
			Vibration amplitude at frequency n/rev	VIB		Init: CT&M Rec: The FSTD should exhibit the overall appearance and trends as given in the MQTG	Same as 3.g.1
3.g.3	Characteristic vibrations – hover (in-ground effect)	Hover (in-ground effect)	Vibration amplitude at frequency $1/\text{rev}$	VIB		Same as 3.g.2	Same as 3.g.1
			Vibration amplitude at frequency n/rev	VIB		Same as 3.g.2	Same as 3.g.1
3.g.4	Characteristic vibrations – hover (out-of-ground effect)	Hover (out-of-ground effect)	Vibration amplitude at frequency $1/\text{rev}$	VIB		Same as 3.g.2	Same as 3.g.1
			Vibration amplitude at frequency n/rev	VIB		Same as 3.g.2	Same as 3.g.1
3.g.5	Characteristic vibrations – normal climb	Normal climb	Vibration amplitude at frequency $1/\text{rev}$	VIB		Same as 3.g.2	Same as 3.g.1

3.g: Characteristic vibrations		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
			Vibration amplitude at frequency n/rev	VIB		Same as 3.g.2	Same as 3.g.1
3.g.6	Characteristic vibrations – vertical climb	Normal climb	Vibration amplitude at frequency $1/\text{rev}$	VIB			Same as 3.g.1
			Vibration amplitude at frequency n/rev	VIB			Same as 3.g.1
3.g.7	Characteristic vibrations – translational lift	Level flight at effective translational lift airspeed	Vibration amplitude at frequency $1/\text{rev}$	VIB		Same as 3.g.2	Same as 3.g.1
			Vibration amplitude at frequency n/rev	VIB		Same as 3.g.2	Same as 3.g.1
3.g.8	Characteristic vibrations – cruise speed	Level flight at cruise airspeed	Vibration amplitude at frequency $1/\text{rev}$	VIB		Same as 3.g.2	Same as 3.g.1
			Vibration amplitude at frequency n/rev	VIB		Same as 3.g.2	Same as 3.g.1
3.g.9	Characteristic vibrations – high speed	High speed	Vibration amplitude at frequency $1/\text{rev}$	VIB		Same as 3.g.2	Same as 3.g.1

3.g: Characteristic vibrations		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
			Vibration amplitude at frequency n/rev	VIB		Same as 3.g.2	Same as 3.g.1
3.g.10	Characteristic vibrations – descent	Descent	Vibration amplitude at frequency $1/\text{rev}$	VIB		Same as 3.g.2	Same as 3.g.1
			Vibration amplitude at frequency n/rev	VIB		Same as 3.g.2	Same as 3.g.1
3.g.11	Characteristic vibrations – autorotation	Autorotation	Vibration amplitude at frequency $1/\text{rev}$	VIB		Same as 3.g.2	Same as 3.g.1
			Vibration amplitude at frequency n/rev	VIB		Same as 3.g.2	Same as 3.g.1
3.g.12	Characteristic vibrations – left steady turn	Left steady turn, cruise	Vibration amplitude at frequency $1/\text{rev}$	VIB		Same as 3.g.2	Same as 3.g.1
			Vibration amplitude at frequency n/rev	VIB		Same as 3.g.2	Same as 3.g.1
3.g.13	Characteristic vibrations – right steady turn	Right steady turn, cruise	Vibration amplitude at frequency $1/\text{rev}$	VIB		Same as 3.g.2	Same as 3.g.1

3.g: Characteristic vibrations		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
			Vibration amplitude at frequency n/rev	VIB		Same as 3.g.2	Same as 3.g.1
3.g.14	Characteristic vibrations – special conditions	Refer to the test requirement table below for test 3.g.14	Vibration amplitude at frequency $1/\text{rev}$	VIB			Same as 3.g.1
			Vibration amplitude at frequency n/rev	VIB			Same as 3.g.1
3.g.15	Comparison of vibration amplitudes	On ground and in flight	Overall g's root mean (grms) at $1/\text{rev}$	VIB		Init: Correct trend Rec: Same as Init	Init: Correct trend Rec: Same as Init
			Overall grms at n/rev	VIB		Init: Correct trend Rec: Same as Init	Init: Correct trend Rec: Same as Init

Requirements for all the rotorcraft characteristic vibration tests.

3.g: Characteristic vibrations	Test requirements
All tests	- Definitions of $1/\text{rev}$ and n/rev are as follows. $1/\text{rev}$ refers to once-per-revolution oscillations caused primarily by the aerodynamic forces acting on the main rotor blades as they rotate. n/rev refers to the basic $1/\text{rev}$ vibration multiplied by the number of main rotor blades, n. If some harmonics (multiple of $1/\text{rev}$ lower than n) are significant, they should be evaluated with the same tolerances applied as for vibrations at n/rev. - The comparison between FSTD and validation data can be demonstrated by either of the following tolerances. - Tolerance can be +3 dB / -6 dB between either the fast Fourier transform (FFT) or the PSDs of the FSTD result and

3.g: Characteristic vibrations	Test requirements
	the validation data for the frequency spectrum. The relationship (L) in decibels between these two FFTs or PSDs is calculated at each frequency and is evaluated as: $L = 20\log_{10}\left(\frac{\text{FFT}_{\text{FSTD}}}{\text{FFT}_{\text{data}}}\right) \text{ or } L = 10\log_{10}\left(\frac{\text{PSD}_{\text{FSTD}}}{\text{PSD}_{\text{data}}}\right)$ Note: The FFT focuses on the amplitude spectrum (e.g. displacement, velocity or acceleration versus frequency), using a factor of 20 for decibel conversion. The PSD measures power per frequency (e.g. energy per Hz), using a factor of 10, since power is proportional to the square of the amplitude. The frequency resolution of the FFT depends on the number of frequency lines or bins. This affects the comparison of two different vibrating systems. When comparing the FFT results of the validation data and FSTD results, both should be processed to the same bandwidth. The PSD is more useful here because the amplitude value is normalised to the frequency bin width. By normalising the result, a more accurate comparison of vibration levels in signals is possible. Because of this, the use of PSD for the relationship between validation data and FSTD results is encouraged. b. Tolerance can be 10 % of the nominal vibration level for cruise flight at each frequency that is evaluated. For all tests conditions this is, for 1/rev, ± 10 % of the amplitude of the 1/rev vibration in cruise and level flight and, for n/rev, ± 10 % of the amplitude of the n/rev vibration in cruise and level flight. 3. The parameters to be plotted are acceleration FFT or PSD against frequency (FFT in g and PSD in grms^2/Hz) for the x-, y- and z-axes. 4. The tolerances are applied only at the 1/rev and n/rev frequencies (and at the significant harmonics, if applicable). The vibration amplitude ‘spikes’ should be present within ± 0.5 Hz of the validation data. 5. If the amplitude on any axis is negligible, its plot may be ignored, provided that the statement of justification explains this. 6. The recurrent tests have the same tolerances as the initial evaluation. The recurrent tests are also expected to have very similar results as those in the MQTG. It is acknowledged that the vibration test results may have some variation when compared with those in the MQTG due to the random nature of the vibration/buffet and due to the mathematical process used to plot the amplitude against frequency (i.e. FFT). 7. The tests may give the test conditions and flight parameters as numerical snapshot values, ensuring that flight phases are stable.

Individual test requirements. The individual test requirements defined in the table below are applicable to all fidelity levels unless otherwise stated.

3.g: Characteristic vibrations		Test requirements
3.g.1	Characteristic vibrations – on ground at idle	1. Test is to demonstrate the normal vibration level with rotorcraft on the ground and all engines operating at normal idle power settings.
3.g.2	Characteristic vibrations – on ground at operating RPM	1. Test is to demonstrate the normal vibration level with rotorcraft on the ground and all engines operating at normal flight power settings.
3.g.3	Characteristic vibrations – hover (in-ground effect)	1. Test is to demonstrate the normal vibration level with rotorcraft in hover condition in-ground effect.
3.g.4	Characteristic vibrations – hover (out-of-ground effect)	1. Test is to demonstrate the normal vibration level with rotorcraft in hover condition out-of-ground effect.
3.g.5	Characteristic vibrations – normal climb	1. Test is to demonstrate the normal vibration level with rotorcraft in normal climb at normal climb speed and all engines operating.
3.g.6	Characteristic vibrations – vertical climb	1. Test is to demonstrate the normal vibration level with rotorcraft in vertical climb from hover condition.
3.g.7	Characteristic vibrations – translational lift	1. Test is to demonstrate the normal vibration level with rotorcraft in forward translational level flight.
3.g.8	Characteristic vibrations – cruise	1. Test is to demonstrate the normal vibration level with rotorcraft in flight at normal cruise speed.

3.g: Characteristic vibrations		Test requirements
	speed	
3.g.9	Characteristic vibrations – high speed	1. Test is to demonstrate the normal vibration level with rotorcraft in flight at high speed (near or at the never exceed speed).
3.g.10	Characteristic vibrations – descent	1. Test is to demonstrate the normal vibration level with rotorcraft in normal powered descent at normal speed and all engines operating.
3.g.11	Characteristic vibrations – autorotation	1. Test is to demonstrate the normal vibration level with helicopter in autorotational descent, all engines inoperative (or at least at idle), at nominal main rotor RPM and recommended autorotation speed.
3.g.12	Characteristic vibrations – left steady turn	1. Test is to demonstrate the normal vibration level with rotorcraft in stabilised turn at various bank angles; at least two conditions are to be demonstrated (e.g. for a standard rate of turn and a higher bank angle of around 45° in order to demonstrate the effect of rotor disk load on vibration level, if any).
3.g.13	Characteristic vibrations – right steady turn	1. Test is to demonstrate the normal vibration level with rotorcraft in stabilised turn at various bank angles; at least two conditions are to be demonstrated (e.g. for a standard rate of turn and a higher bank angle of around 45° in order to demonstrate the effect of rotor disk load on vibration level, if any).
3.g.14	Characteristic vibrations – special conditions	1. This applies to special steady-state cases identified as particularly significant to the pilot, important in training or unique to a particular helicopter type or model. These may include the effects of a tandem-rotor helicopter (e.g. translational lift for yawing), icing effect and all relevant vibration cues due to normal and abnormal operations of the rotor and transmission system. 2. The SOJ should describe the test cases. 3. Steady-state tests are acceptable. 4. Refer to CS FSTD.ENG.015 for the validation data requirements applicable to test cases in which flight test validation data may not be available, such as operations in icing conditions or flights outside the certified flight domain or envelope.
3.g.15	Comparison of	1. Correct trend refers to a comparison of vibration/buffet amplitudes between different manoeuvres. For example, if

3.g: Characteristic vibrations		Test requirements
	vibration amplitudes	the 1/rev vibration amplitude in the rotorcraft is higher during steady-state turns than in level flight, this increasing trend should be demonstrated in the FSTD tests. 2. Plots of 1/rev and n/rev overall grms for all the test conditions listed above should be presented to support the evaluation of the correct vibration trend for different test conditions.

GM1 FSTD.QTG.H.305 Characteristic vibrations/buffet – Rotorcraft

VIBRATION HARMONICS – ROTORCRAFT

The higher-order vibration harmonics can be caused by a variety of factors, including the interaction of rotor blades with the air and mechanical imbalances. The n value represents an integer multiple of the rotational frequency of the rotor. If $n = 2$, the vibration occurs twice per revolution of the rotor, and so on. For example, if the rotor RPM is 300 and the number of blades is four, the vibration peaks are expected to occur 300, 600, 900 and 1 200 times per minute or 5, 10, 15 and 20 times per second (i.e. at 5 Hz, 10 Hz, 15 Hz and 20 Hz).

CS FSTD.QTG.H.400 Visual display system – Rotorcraft

This CS provides the standards for the objective tests for rotorcraft FSTD visual display systems.

4.a: Visual display system		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
4.a.1	Visual field of view	n/a	Horizontal and vertical continuous field of view	VIS	Init: Continuous visual FOV providing each pilot with 150° horizontal and 40° vertical FOV Rec: EM	Init: Continuous visual FOV providing each pilot with 200° horizontal and 60° vertical FOV Rec: EM	Init: Continuous visual FOV providing each pilot with 200° horizontal and 60° vertical FOV Rec: EM
4.a.2.a	System geometry – image position – collimated display	n/a	Centre of the image Difference between the left and right horizontal angles	VIS		Init: Between 0° and 2° in the horizontal plane and within $\pm 0.25^\circ$ in the vertical plane; Difference between the left and right horizontal angles of $\pm 1^\circ$ Rec: EM	Init: Between 0° and 2° in the horizontal plane and within $\pm 0.25^\circ$ in the vertical plane; Difference between the left and right horizontal angles of $\pm 1^\circ$ Rec: EM

4.a: Visual display system		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
4.a.2.b	System geometry – image position – non-collimated display	n/a	Centre of the image	VIS		Init: Less than or equal to 10° in the horizontal plane and within ±0.25° in the vertical plane Rec: EM	Init: Less than or equal to 10° in the horizontal plane and within ±0.25° in the vertical plane Rec: EM
4.a.2.c	System geometry – absolute geometry	n/a	All points	VIS		Init: Within the central 200° × 60°, all points on a 5° grid should fall within 3° of the design position as measured from each pilot eyepoint Rec: EM	Init: Within the central 200° × 60°, all points on a 5° grid should fall within 3° of the design position as measured from each pilot eyepoint Rec: EM

4.a: Visual display system		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
4.a.2.d	System geometry – relative geometry	n/a	Relative position from one point to the next	VIS		Init: Measurements of relative dot positions should be made every 5°. In the area from -10° to the lowest visible point at 15° azimuth inboard, 0°, 30°, 60° and 85° outboard for each pilot position, vertical measurements should be made every 1° to the edge of the visible image. The relative position from one point to the next should not exceed: zone 1, 0.075°/deg; zone 2, 0.15°/deg; and zone 3, 0.2°/deg Rec: EM	Init: Measurements of relative dot positions should be made every 5°. In the area from -10° to the lowest visible point at 15° azimuth inboard, 0°, 30°, 60° and 85° outboard for each pilot position, vertical measurements should be made every 1° to the edge of the visible image. The relative position from one point to the next should not exceed: zone 1, 0.075°/deg; zone 2, 0.15°/deg; and zone 3, 0.2°/deg Rec: EM

4.a: Visual display system		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
4.a.3	Surface resolution (object detection)	n/a	Surface resolution	VIS	Init: Not greater than 3 arc minutes Rec: Same as Init	Init: Not greater than 3 arc minutes Rec: Same as Init	Init: Not greater than 3 arc minutes Rec: Same as Init
4.a.4	Light point size	n/a	Light point size	VIS	Init: Not greater than 8 arc minutes Rec: Same as Init	Init: Not greater than 8 arc minutes Rec: Same as Init	Init: Not greater than 5 arc minutes Rec: Same as Init
4.a.5.a	Surface contrast ratio – collimated display	n/a	Surface contrast ratio	VIS		Init: Not less than 5:1 Rec: Same as Init	Init: Not less than 8:1 Rec: Same as Init
4.a.5.b	Surface contrast ratio – non-collimated display	n/a	Surface contrast ratio	VIS		Init: Not less than 5:1 Rec: Same as Init	Init: Not less than 5:1 Rec: Same as Init
4.a.6	Light point contrast ratio	n/a	Light point contrast ratio	VIS		Init: Not less than 25:1 Rec: Same as Init	Init: Not less than 25:1 Rec: Same as Init
4.a.7	Light point brightness	n/a	Light point brightness	VIS		Init: Not less than 17 cd/m ² (5.0 ft-lamberts) from the display measured at the design eyepoint Rec: Same as Init	Init: Not less than 20 cd/m ² (5.8 ft-lamberts) from the display measured at the design eyepoint Rec: Same as Init

4.a: Visual display system		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
4.a.8	Surface brightness	n/a	Surface brightness	VIS		Init: Not less than 14 cd/m ² (4.1 ft-lamberts) from the display measured at the design eyepoint Rec: Same as Init	Init: Not less than 20 cd/m ² (5.8 ft-lamberts) from the display measured at the design eyepoint Rec: Same as Init
4.a.9.a	Black level	n/a	Black intensity	VIS		Init: Tolerance criterion is (background brightness minus black polygon brightness) < 0.015 cd/m ² (0.004 ft-lamberts) Rec: Same as Init	Init: Tolerance criterion is (background brightness minus black polygon brightness) < 0.015 cd/m ² (0.004 ft-lamberts) Rec: Same as Init
4.a.9.b	Luminance levels	n/a	Luminance levels	VIS		Init: Minimum 11 shades of grey Rec: Same as Init	Init: Minimum 11 shades of grey Rec: Same as Init

4.a: Visual display system		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
4.a.10	Motion blur	n/a	Pattern gap	VIS		Init: When a pattern is rotated around the eyepoint at 10°/s, the smallest detectable gap should be 6 arc minutes or less Rec: Same as Init	Init: When a pattern is rotated around the eyepoint at 10°/s, the smallest detectable gap should be 4 arc minutes or less Rec: Same as Init
4.a.11	Speckle test	n/a	Speckle contrast	VIS		Init: Speckle contrast should be < 10 % Rec: Same as Init	Init: Speckle contrast should be < 10 % Rec: Same as Init

Individual test requirements. The individual test requirements defined in the table below are applicable to all fidelity levels unless otherwise stated.

4.a: Visual display system		Test requirements
4.a.1	Visual field of view	- FOV should be measured using a visual test pattern filling the entire visual scene (all channels), consisting of a matrix of black and white 5° squares. - Installed alignment should be confirmed in an SOJ (this would generally consist of results from acceptance testing).
4.a.2.a	System geometry – image position – collimated display	- The image position should be checked relative to the FSTD centre line. - If there is a design offset in the vertical display centre, this should be stated.
4.a.2.b	System geometry –	The image position should be checked relative to the FSTD centre line.

4.a: Visual display system		Test requirements
	image position – non-collimated display	4. If there is a design offset in the vertical display centre, this should be stated.
4.a.2.c	System geometry – absolute geometry	1. The procedure for this test is to display a 5° grid to show the angular spacing. The initial assessment of the visual system should be performed by using a theodolite. The recurrent testing method may use a fixed grid projected from an independent device, calibrated during the initial acceptance of the visual system and confirmed in an SOJ. If the visual system image can be shown to align with the tolerances specified, it may be assumed that the requirements are met based on the initial acceptance of the visual system. If there is any doubt about the test results or the visual system hardware parts have been modified, replaced or detached, then a theodolite measurement should be used to demonstrate compliance. 2. If a system of more than 200° × 60° is supplied, the geometry outside the central area should not have any distracting discontinuities.
4.a.2.d	System geometry – relative geometry	1. Azimuth and elevation are relative to the aircraft centre line and to the horizon. They are positive to the right and above the horizon. 2. Zone 1 covers +5° to -15° vertically by ±80° horizontally. 3. Zone 2 covers +10° to -25° vertically by ±90° horizontally, excluding the area covered by zone 1. 4. Zone 3 covers +20° to -40° vertically by ±105° horizontally, excluding the areas covered by zones 1 and 2. 5. A light point is set every 5° in azimuth and in elevation in the whole FOV (and also outside zones 1–3 if the FOV is larger). 6. A light point is set every 1° at elevation -10° and downwards (up to the limit of the FOV), at azimuth -85°, -60°, -30°, -15°, 15°, 30°, 60° and 85°. 7. From the pilot's eye position, test should verify every visible 5° point on the vertical lines and horizontal lines. In addition, at -15°, 0°, +30°, +60° and +85° in azimuth, test should verify all visible 1° points from the -10° point to the lowest visible point. 8. From the copilot's eye position, test should verify every visible 5° point on the vertical lines and horizontal lines. In addition, at 15°, 0°, -30°, -60° and -85° in azimuth, test should verify all visible 1° points from the -10° point to the lowest visible point. 9. Tolerances apply to zones 1–3. For larger FOVs, there should be no distracting discontinuity outside zone 3.

4.a: Visual display system		Test requirements
4.a.3	Surface resolution (object detection)	- Resolution should be demonstrated by a test of objects shown to occupy the required visual angle in each visual display used on a scene from the pilot's eyepoint. - The object will subtend an angle no greater than the tolerance angle to the eye. - This may be demonstrated using threshold bars for a horizontal test. - A vertical test should also be demonstrated. - The subtended angles should be confirmed by calculations in an SOJ.
4.a.4	Light point size	- Light point size should be measured using a test pattern consisting of a centrally located single row of white light points displayed as both horizontal and vertical rows in each visual display. - It should be possible to move the light points closer to and further away from the eyepoint. - At a point at which modulation is just discernible in each visual channel, a calculation should be made to determine the light spacing. - An SOJ is required to demonstrate the test method and calculations.
4.a.5.a and 4.a.5.b	Surface contrast ratio	- Surface contrast ratio should be measured using a test pattern filling the entire visual scene (all channels). - The test pattern should consist of black and white squares, 5° per square, with a white square in the centre of each channel. - Measurement should be made on the centre bright square for each channel using a 1° luminance meter. This value should have a minimum brightness of 7 cd/m² (2 ft-lamberts). Measurements should also be taken for any adjacent dark square. - The contrast ratio is the bright square value divided by the dark square value. - During contrast ratio testing, FSTD aft-cabin and flight deck ambient light levels should be as low as possible. - Measurements should be taken at the centre of squares to avoid light spill into the measurement device.
4.a.6	Light point contrast ratio	- Light point contrast ratio should be measured using a test pattern demonstrating an area of greater than 1° filled with white light points (i.e. light point modulation just discernible) and should be compared with the adjacent background. - Measurements of the background should be taken such that the bright square is just out of the light meter FOV. - During contrast ratio testing, FSTD aft-cabin and flight deck ambient light levels should be as low as possible.
4.a.7	Light point	Light points should be displayed as a matrix creating a square.

4.a: Visual display system		Test requirements
	brightness	2. Light points should overlap such that the square is continuous (individual light points will not be visible).
4.a.8	Surface brightness	1. Surface brightness should be measured on a white polygon, measuring the brightness using the 1° luminance meter.
4.a.9.a	Black level	1. The light meter should be mounted in a fixed position, viewing the forward centre area of each display. 2. All projectors should be turned off and the flight deck environment made as dark as possible. A background reading should be taken of the remaining ambient light on the screen. 3. The projectors should then be turned on and a black polygon displayed. A second reading should then be taken and the difference between this and the ambient level recorded. 4. This test is generally only required for light valve projectors.
4.a.9.b	Luminance levels	1. A test pattern consists of a pair of grey fields with opposite-direction luminance staircases between 0 % and 100 % luminance, with steps of 10 %. The pattern is displayed in each channel. 2. All grey steps should be discernible and without noticeable chromatic aberration. 3. FSTD aft-cabin and flight deck ambient light levels should be as low as possible.
4.a.10	Motion blur	1. A test pattern consists of an array of five peak white squares with black gaps of decreasing width between them. 2. The range of black gap widths should at least extend above and below the required detectable gap and should be in steps of 1 arc minute. 3. The pattern is to be rotated at the required rate. 4. Two arrays of squares should be provided, one rotating in heading and the other in pitch, to provide testing in both axes. 5. A series of stationary numbers identifies the gap number. 6. Note: This test can be limited by the display technology. If this is the case, the competent authority should be consulted on the limitations. 7. This test is generally only required for light valve projectors.
4.a.11	Speckle test	1. An SOJ describing the test method is required. 2. This test is generally only required for laser projectors.

GM1 FSTD.QTG.H.400 Visual display system – Rotorcraft

Refer to GM1 FSTD.QTG.A.400 'Visual display system – Aeroplane', which is also applicable to rotorcraft.

CS FSTD.QTG.H.405 Head-up display – Rotorcraft

Refer to CS FSTD.QTG.A.405 'Head-up display – Aeroplane', which is also applicable to rotorcraft. This CS is applicable to installed airborne HUDs but not to helmet-mounted HUDs.

CS FSTD.QTG.H.410 Enhanced flight vision system – Rotorcraft

Refer to CS FSTD.QTG.A.410 'Enhanced flight vision system – Aeroplane', which is also applicable to rotorcraft. This CS is applicable only to EFVSs that fit the following definition: an installed airborne system that provides the flight crew with a real-time display of the external scene topography through the use of imaging sensors, that is integrated with a flight guidance system and that is implemented on a HUD or equivalent display system.

CS FSTD.QTG.H.415 Night-vision imaging system – Rotorcraft

Refer to CS FSTD.QTG.A.415 'Night-vision imaging system – Aeroplane', which is also applicable to rotorcraft.

CS FSTD.QTG.H.420 Visual ground segment – Rotorcraft

This CS provides the standards for the objective tests for the rotorcraft VGS.

4.e: Visual ground segment		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
4.e.1	Visual ground segment (VGS)	Trimmed in the landing configuration at 30 m (100 ft) landing gear height above runway threshold on glide slope at a RVR setting of 300 m (1 000 ft)	Near end: lights before threshold	VIS	Init: The correct number of approach lights within the computed VGS should be visible Rec: Same as Init	Init: The correct number of approach lights within the computed VGS should be visible Rec: Same as Init	Init: The correct number of approach lights within the computed VGS should be visible Rec: Same as Init
			Far end: lights after the threshold	VIS	Init: $\pm 20\%$ of the computed VGS Rec: Same as Init	Init: $\pm 20\%$ of the computed VGS Rec: Same as Init	Init: $\pm 20\%$ of the computed VGS Rec: Same as Init
			Threshold lights	VIS	Init: The threshold lights computed to be visible should be visible Rec: Same as Init	Init: The threshold lights computed to be visible should be visible Rec: Same as Init	Init: The threshold lights computed to be visible should be visible Rec: Same as Init

Individual test requirements. The individual test requirements defined in the table below are applicable to all fidelity levels unless otherwise stated.

4.e: Visual ground segment		Test requirements
4.e.1	Visual ground	1. If non-homogeneous fog is used, the vertical variation in horizontal visibility should be described and be included in

4.e: Visual ground segment		Test requirements
	segment (VGS)	the slant range visibility calculation used in the VGS computation.

GM1 FSTD.QTG.H.420 Visual ground segment – Rotorcraft

Refer to GM1 FSTD.QTG.A.420 ‘Visual ground segment – Aeroplane’, which is also applicable to rotorcraft.

CS FSTD.QTG.H.500 Sound systems – Rotorcraft

This CS provides the standards for the objective tests for rotorcraft sound systems.

5: Sound systems		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
5.a.1	Ready for engine start	Ground, ready for engine start	Sound amplitude	SND	Init: Subjective sound assessment to establish MQTG footprint of overall measured SPL Rec: ± 3 dB SPL RMS compared with the MQTG	Init: Subjective sound assessment to establish MQTG footprint of one-third octave bands Rec: Should not exceed ± 5 dB difference per one-third octave band compared with the MQTG	Init: within ± 5 dB per one-third octave band of the validation data Rec: Same as Init
5.a.2	All engines at idle	Ground, all engines at idle, rotor not turning (if applicable)	Sound amplitude	SND		Same as 5.a.1	Same as 5.a.1
5.a.3	All engines at operating RPM	Ground, all engines at operating RPM, rotor turning	Sound amplitude	SND	Same as 5.a.1	Same as 5.a.1	Same as 5.a.1
5.a.4	Hover	Hover	Sound amplitude	SND	Same as 5.a.1	Same as 5.a.1	Same as 5.a.1
5.a.5	Climb	En route climb	Sound amplitude	SND		Same as 5.a.1	Same as 5.a.1

5: Sound systems		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
5.a.6	Cruise	Straight cruise and level flight	Sound amplitude	SND	Same as 5.a.1	Same as 5.a.1	Same as 5.a.1
5.a.7	Final approach	Final approach to landing	Sound amplitude	SND		Same as 5.a.1	Same as 5.a.1
5.b	Reserved						
5.c	Special cases	As defined by applicant for FSTD qualification	Sound amplitude	SND		Same as 5.a.1	Same as 5.a.1
5.d	FSTD background noise	n/a	Sound amplitude	SND	Init: Subjective assessment of measured overall SPL Rec: ± 3 dB SPL RMS compared with the MQTG	Init: Each one-third octave band level should be lower than a reference line linearly interpolated between 70 dB at 50 Hz, 55 dB at 1 kHz and 30 dB at 16 kHz Rec: Same as Init, and ± 3 dB per one-third octave band compared with the MQTG	Init: Each one-third octave band level should be lower than a reference line linearly interpolated between 70 dB at 50 Hz, 55 dB at 1 kHz and 30 dB at 16 kHz Rec: Same as Init, and ± 3 dB per one-third octave band compared with the MQTG

5: Sound systems		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
5.e	Frequency response	n/a	Sound amplitude	SND		Init: n/a Rec: should not exceed ± 5 dB on three consecutive bands when compared with the MQTG, and the average of the absolute differences between the MQTG and recurrent results should not exceed 2 dB	Init: n/a Rec: should not exceed ± 5 dB on three consecutive bands when compared with the MQTG, and the average of the absolute differences between the MQTG and recurrent results should not exceed 2 dB

Requirements for all the rotorcraft sound system tests.

5: Sound systems	Test requirements
All tests	- For feature SND at fidelity levels G and R, the following applies. The sound performance of the FSTD should be based on subjective tuning from pilot assessment. After this tuning has been completed, the applicable sound tests in Sections 5.a and 5.c should be baselined to create footprint test results that will become the MQTG results after the initial evaluation. - For feature SND at fidelity level G, the following applies. All required tests in this section may be presented as a single overall SPL level. They are only required if the flight deck is fully or partially enclosed (e.g. installed in a dedicated room). - For feature SND at fidelity levels R and S, the following applies.

5: Sound systems	Test requirements
	All required tests in this section should be presented using an unweighted one-third octave band format from band 17–42 (50–16 kHz). A minimum of 20 seconds average should be taken at the location corresponding to the validation dataset. 4. The tests may give the test conditions and flight parameters as numerical snapshot values.

Individual test requirements. The individual test requirements defined in the table below are applicable to all fidelity levels unless otherwise stated.

5: Sound systems		Test requirements
5.a.1	Ready for engine start	1. Normal conditions prior to engine start should be used. The APU should be on if appropriate.
5.a.2	All engines at idle	1. Rotor should not be turning, if applicable. Otherwise, rotor should be turning.
5.a.3	All engines at operating RPM	1. Normal conditions prior to lift-off should be used.
5.a.5	Climb	1. Medium altitude should be used.
5.a.6	Cruise	1. Normal cruise configuration should be used.
5.c	Special cases	1. Test applies to special steady-state cases identified as particularly significant to the pilot, important in training or unique to a particular aeroplane type or model.
5.d	FSTD background noise	1. The measurements should be made with the simulation running, the sound muted and a dead flight deck.
5.e	Frequency response	1. White noise should be used to ensure that the full frequency spectrum is demonstrated for each speaker.

GM1 FSTD.QTG.H.500 Sound systems – Rotorcraft

Refer to GM1 FSTD.QTG.A.500 ‘Sound systems – Aeroplane’, which is also applicable to rotorcraft.

CS FSTD.QTG.H.600 Systems integration – Rotorcraft

This CS provides the standards for the objective tests for rotorcraft system integration.

6.a: Systems integration – system response time		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
6.a.1	System response time – transport delay	Pitch, roll and yaw	Motion system response time	MTN	Init: 200 ms or less after controller movement Rec: Same as Init	Init: 150 ms or less after controller movement Rec: Same as Init	Init: 100 ms or less after controller movement Rec: Same as Init
			Instruments system response time	SYS	Init: 200 ms or less after controller movement Rec: Same as Init	Init: 150 ms or less after controller movement Rec: Same as Init	Init: 100 ms or less after controller movement Rec: Same as Init
			Visual system response time	VIS	Init: 200 ms or less after controller movement Rec: Same as Init	Init: 150 ms or less after controller movement Rec: Same as Init	Init: 100 ms or less after controller movement Rec: Same as Init
			HUD response time	SYS		Init: 150 ms or less after controller movement Rec: Same as Init	Init: 100 ms or less after controller movement Rec: Same as Init

6.a: Systems integration – system response time		Test conditions	Tolerance parameter	Primary feature(s) validated	Tolerances		
					G	R	S
			EFVS response time	SYS		Init: ±30 ms from visual system response, and not before motion system response Rec: Same as Init	Init: ±30 ms from visual system response, and not before motion system response Rec: Same as Init

Individual test requirements. The individual test requirements defined in the table below are applicable to all fidelity levels unless otherwise stated.

6.a: Systems integration – system response time		Test requirements
6.a.1	System response time – transport delay	1. A separate test is required for each axis. 2. The tests to demonstrate compliance with these requirements should simultaneously record: a. the output from the pilot's controller; b. the output from the accelerometer attached to the motion system platform, located as near the pilots' seats as is practicable; c. the output signal to the visual system display; d. the output signal to the HUD/EFVS, installed; e. the output signal to the pilot's attitude indicator. 3. For feature MTN at fidelity levels R and S, the onset of motion response may occur before the start of the scan of the first visual system video field containing different information and must occur before the end of the scan of the same video field. 4. Visual system response may be measured from the start of the scan of the first video field containing different information. 5. HUD requirements apply in the same conditions as stated in CS FSTD.QTG.H.405 and EFVS requirements apply as stated in CS FSTD.QTG.H.410. 6. The delay from the aeroplane EFVS electronic elements should be added to the 30 ms tolerance before comparison

6.a: Systems integration – system response time		Test requirements
		with visual system response. 7. If the primary features validated by this test are present in the FSTD at differing fidelity levels, the tolerance for the highest fidelity level should be applied to all the features present.

GM1 FSTD.QTG.H.600 Systems integration – Rotorcraft

Refer to GM1 FSTD.QTG.A.600 'System integration – Aeroplane', which is also applicable to rotorcraft.

SUBPART E – FUNCTIONS AND SUBJECTIVE TESTS

GENERAL

CS FSTD.FST.001 General

- (a) The basis and criteria for the functions and subjective testing (F&S) are the general requirements presented in Subpart B. The general requirement tables in Subpart B indicate if verification and validation through F&S is required.
- (b) Functions tests are objective tests to verify the FSTD's aircraft system functionalities and operations in accordance with aircraft documentation such as flight manuals and CSs. Testing should ensure that the systems function as expected and that particular aircraft aspects (e.g. automation systems unique to the aircraft) operate as expected.
- (c) Subjective tests provide a basis for evaluating the FSTD's capability to perform over a typical training period and to verify that the correct operation, cueing and handling characteristics of the FSTD are in place. While objective tests (refer to Subpart D) provide validation only for the conditions and configurations for which they are performed and focus on particular features of the FCS, the subjective tests are used to test the device in more conditions and configurations to subjectively evaluate whether the performance and handling and flight controls match expectations throughout the flight envelope.
- (d) The list of applicable functions and subjective tests, along with the testing results, should be documented in the functions and subjective test table in the MQTG. The 'Result' column should indicate that the test has been assessed with satisfactory results. Any relevant details regarding the tested item (e.g. the runway used for testing an approach, the configuration used) should be added in the dedicated 'Remark' column.

GM1 FSTD.FST.001 General

F&S is a process that requires pilot expertise to assess the FSTD. The necessity of F&S arises from the need to confirm that the simulation has produced a fully integrated and compliant simulation of the aircraft type, variant or group, as applicable. This testing has multiple objectives, such as the evaluation of:

- (a) the compliance of the FSTD with the requested FCS;
- (b) the FSTD's capability to perform over a typical training session;
- (c) the functionalities and systems of the simulated aircraft;
- (d) the performance and handling during manoeuvres and training tasks in various conditions and configurations;
- (e) the cueing;

- (f) the simulated malfunctions or failures;
- (g) the environmental simulation, conditions and settings;
- (h) the IOS functionalities, such as reposition, freeze and slew functions;
- (i) the integration of the whole FSTD;
- (j) the entries declared in the ESL.

GM2 FSTD.FST.001 General

The efficient use of the FSTD in a training environment requires the minimisation of training interruptions and the use of different kinds of freezes, repositions, snapshot recalls, flight plan loads, resets, etc. These capabilities can be divided into five functional categories, namely:

- (a) simulation control (e.g. flight freeze, fuel freeze, position freeze);
- (b) scenario set-up (e.g. weight change, reposition, environment, internal system status reset);
- (c) optimisation (e.g. simulation rate factor, snapshot);
- (d) maintenance set-up (e.g. built-in test equipment, fault memory clear);
- (e) other functions (e.g. head-up displays, flight management data export).

Testing these capabilities, if implemented, should be part of the F&S.

These capabilities should be designed to minimise training disruptions, including snapshot loading time, repositioning and system resets after malfunctions.

ARINC Report 610C, *Guidance for design of aircraft equipment and software for use in training devices*, as amended, gives guidance to support the compliance and building of these capabilities.

CS FSTD.FST.005 Testing methods

This CS defines the testing methods for functions and subjective tests.

- (a) The organisation requesting the qualification of an FSTD should perform functions and subjective tests to ensure that the device complies with the general requirements in Subpart B for each feature and its associated fidelity level for the requested FCS.
- (b) The testing should ensure that the different features are all properly integrated for a complete flight simulation.
- (c) The simulation of aircraft systems with respect to the requested fidelity level should be tested at each flight crew member position. All system functions should be assessed for normal and, as appropriate, alternate or backup operations.
- (d) Handling qualities, performance and FSTD systems operation as they pertain to the simulated aircraft, and also FSTD cueing (e.g. visual and motion cueing) and other supporting systems (e.g. IOS), should be subjectively assessed.
- (e) Normal, abnormal and emergency procedures associated with a flight phase should be assessed during the evaluation of manoeuvres or events within that flight phase. The effects

of the malfunctions should be sufficient to allow pilots to follow the aircraft-related procedures, normally contained in the pilot checklists or quick reference handbook. Aircraft systems are listed separately in CS FSTD.FST.020 and CS FSTD.FST.030 under heading 'Any appropriate flight phase' to ensure that appropriate attention is paid to system checks in all the appropriate flight phases.

- (f) Thorough and integrated malfunction testing should be conducted to verify the alignment of the described effects with the expected aircraft behaviour. The simulated malfunctions should be verified before the initial qualification process.
- (g) Functions and subjective tests should be run in a logical flight sequence.
- (h) During the functions and subjective tests, the organisation requesting qualification of an FSTD should ensure that the FSTD can be used for a typical training session duration without repositioning, freezes or resets, thereby demonstrating proof of reliability.
- (i) For aircraft simulation FSTD features at the generic fidelity level, the FSTD may simulate a group of aircraft, and the simulation may be based on characteristics of multiple aircraft. The F&S should evaluate the performance, handling qualities, individual aircraft systems and integrated flight deck operation and demonstrate that they are acceptable to the pilot. The results of the simulation should be characteristic of the simulated group and should correspond to what would be acceptable in a real aircraft of that group.
- (j) The system operation should be tested in terms of pilot actions and configurations that may reasonably be used by the pilot, even if they would not be in accordance with normal procedures. The simulation should not contribute to negative transfer of training.
- (k) F&S should also be performed for any additional or alternate engine types, thrust ratings or avionics configurations applicable.

GM1 FSTD.FST.005 Testing methods

- (a) It is important that the personnel testing the FSTD understand the criteria and expectations for each test (refer to Subpart B). This is of even greater importance if the FCS has features at mixed fidelity levels (i.e. not all at the same level but a mixture of N, G, R and S).
 - (1) The general requirements in Subpart B specify the criteria and expectations for the simulation of each feature.
 - (2) Each feature should be assessed and tested to ensure that the device complies with the general requirements for that feature. In addition, the interaction and integration of the features should be assessed for every functions and subjective test item.
- (b) During F&S testing, all features and their associated fidelity levels should be considered simultaneously in order to assess the integrated performance of the FSTD, including the combined effect of all systems and sensory cues, for each test item. Refer to GM1 FSTD.QB.010(e);(f) for an example of the integration of the features.
- (c) The personnel conducting the testing should apply all relevant human senses to assess whether the integrated FSTD simulation is compliant and appropriate for each test item. To

enable comprehensive sensory assessment, it may be necessary to repeat certain manoeuvres to ensure that all relevant sensory cues are perceivable and can be evaluated adequately. Examples of the use of human sensory and physiological systems in testing are:

- (1) visual sense – perception of visual cues and image elements, flight instruments, indicators and flight deck components;
 - (2) haptic sense – perception of tactile feedback, control forces and vibration or buffet cues through body contact with the hardware;
 - (3) vestibular sense (balance) – perception of motion cues and accelerations;
 - (4) auditory sense – perception of sound cues.
- (d) F&S should cover each test item in detail, under different conditions and during appropriate flight phases. While it is not possible to cover an infinite number of conditions and configurations, the evaluation should systematically and adequately sample them to demonstrate that the simulation produces appropriate and expected results. The evaluated conditions and configurations should be recorded in the test documentation. The following are examples of test conditions, configurations and flight phases for test items.
- (1) Evaluation of an aircraft system malfunction should begin by evaluating whether the indications (e.g. master caution and a notification) are as expected (e.g. as presented in the AFM or pilot operating handbook). Then, it should be evaluated whether the malfunction leads to the expected consequences (e.g. a generator failure may lead to the redistribution of electrical power, change in the load of operating generators, etc.). The malfunction may have different implications due to inhibit functions and therefore the different flight phases should be evaluated (e.g. cruise versus final approach for auto-land).
 - (2) Subjective evaluation of flight characteristics should cover a range of environmental conditions (e.g. altitude, wind, outside air temperature, QNH, visibility) and aircraft configurations (e.g. clean, flap settings, landing gear extended, varying weight and centre of gravity, one or more engines inoperative) across the flight envelope. The purpose is to ensure that the simulation supports positive transfer of training and that relevant aerodynamic and handling characteristics are appropriately simulated and consistent with the conditions (e.g. primary flight control forces relative to airspeed, performance with altitude, aircraft stability with altitude).
 - (3) Evaluation of autopilot operations should cover a variety of conditions and configurations, such as different airspeeds, altitudes and joining angles; for example, for NAV or localiser capture.
 - (4) For CCA, the primary flight control authority may change as the aircraft configuration changes. Therefore, testing should cover all configurations and flight phases. Similarly, any reversion mode of a CCA may affect the stability of the aircraft; consequently, all reversion modes should be tested across all configurations and flight phases.
- (e) It is advisable to record appropriate parameters (e.g. by writing them down in a table) during testing to allow for consistency checks. For example, noting the vertical speed, airspeed, pitch

attitude and main engine parameters during stabilised AEO and OEI climbs – at every 1 000 ft increment from initial climb to ceiling – can effectively identify inconsistencies or discontinuities in the simulation model and support a comparison with AFM performance data.

- (f) Refer to GM1 FSTD.FST.020 and GM1 FSTD.FST.030 for guidance on the application of test procedures for various functions and subjective tests related to vibration, buffet and motion cueing.
- (g) The RAeS publication *Aeroplane Flight Simulator Evaluation Handbook – Volume 2*, as amended, is a useful source of guidance on conducting F&S.

CS FSTD.FST.010 Table of applicable functions and subjective tests

This CS defines the procedure to be used for the determination of the functions and subjective tests applicable to an FSTD.

- (a) The organisation requesting qualification of an FSTD should prepare a F&S table applicable to the FSTD. The table should include all applicable test items to demonstrate compliance with the FSTD general requirements (refer to Subpart B).
- (b) For any FSTD, the organisation requesting the qualification of an FSTD should prepare a table of functions and subjective tests by undertaking the following steps:
 - (1) Start with the tables in CS FSTD.FST.020 and CS FSTD.FST.030.
 - (2) Remove the tests that are not applicable for the simulated aircraft or for the requested FCS.
 - (3) Expand the list of tests to cover all the applicable general requirements in Subpart B for the requested FCS.
 - (4) Expand the list of tests to cover the aircraft systems operation.
 - (5) If applicable, expand the list to cover a HUD (refer to CS FSTD.QB.104.HUD), EFVS (refer to CS FSTD.QB.104.EFVS), NVIS (refer to CS FSTD.QB.104.NVIS) and/or SATCE (refer to CS FSTD.FST.040).
 - (6) If applicable, expand the list for additional capabilities (refer to CS FSTD.QB.017) of the FSTD.
 - (7) If applicable, expand the list of tests to cover the training areas of specific emphasis (TASE), as applicable to the simulated aircraft and presented in the OSD report.
 - (8) If applicable, add any items covered by special conditions.
- (c) If the FSTD is missing a function or capability compared to the simulated aircraft and FCS, a rationale should be provided.

GM1 FSTD.FST.010 Table of applicable functions and subjective tests

- (a) The organisation requesting qualification of an FSTD should use the tables in CS FSTD.FST.020 and CS FSTD.FST.030 only as a starting point when preparing a list of applicable functions and subjective tests. The customised list should reflect the capabilities of the device and address the applicable general requirements for the requested FCS.
- (b) The customised list of applicable functions and subjective tests should provide sufficient detail for the test personnel to understand what is to be tested.
 - (1) The general requirements (refer to Subpart B) are not fully repeated in the tables of CS FSTD.FST.020 and CS FSTD.FST.030. Instead, the tables specify associated test items.
 - (2) The aircraft system tests listed under heading 'Any appropriate flight phase' in CS FSTD.FST.020 and CS FSTD.FST.030 should be expanded and detailed to reflect the simulated aircraft and then allocated to the appropriate flight phases.
 - (3) For the aircraft features (e.g. performance-and-handling-related tests), the tables in CS FSTD.FST.020 and CS FSTD.FST.030 do not separate the test items for fidelity levels G, R and S. This is because the features should be evaluated simultaneously (see GM1 FSTD.FST.005), and the tables in this CS-FSTD cannot cover all combinations of fidelity levels and their associated fidelity requirements. As appropriate, the customised list should provide sufficient detail to clarify the expectations for each feature and its fidelity level.
 - (4) For the cueing features – sound, vibration, motion and visual – the tables in CS FSTD.FST.020 and CS FSTD.FST.030 provide separate test items for qualification levels G, R and S. Due to the nature of these features, the test items can be separated for different levels. The separation of cueing test items by fidelity level supports the development of an effective and structured checklist for test planning and execution. Nevertheless, cueing should still be evaluated in conjunction with each test item to ensure the integration and consistency of the overall simulation.
- (c) In accordance with CS FSTD.FST.010, any additional capabilities should be covered by the F&S. For example, if a rotorcraft FSTD has a hoist system, it should be covered by the testing (refer to CS FSTD.QB.017). Any additional systems or capabilities should be tested to ensure that they do not contribute to negative transfer of training. The customised list of functions and subjective tests should contain appropriate tests.
- (d) The functions and subjective test list should not include tests that do not apply to the FSTD. The following are examples of cases in which functions and subjective test items of CS FSTD.FST.020 and CS FSTD.FST.030 are not applicable:
 - (1) If the motion cueing FSTD feature is at level N, then the motion cueing testing is not applicable.

- (2) If the FSTD does not have primary flight control hardware (i.e. the flight control forces and hardware FSTD feature is at fidelity level N), then the items requiring manual flying are not applicable.
- (3) If the FSTD simulates a single-engine aircraft, then the items related to multi-engine aircraft are not applicable.
- (4) If the FSTD is not equipped with an autopilot, then the items related to the use of autopilot are not applicable.
- (5) If the FSTD is not equipped with an ACAS system, then ACAS tests are not applicable.
- (6) If the installed avionics do not support 3D required navigation performance (RNP) approaches, then tests involving this type of approach are not applicable.
- (7) If the FSTD is not equipped with the applicable equipment for category III operations with a DH below 100 ft or no DH and RVR in accordance with the operational weather minima (CAT III) approaches, then tests involving CAT III approaches are not applicable.
- (8) If the FSTD has the atmosphere and weather FSTD feature at a level that does not require wind shear simulation, then wind shear tests are not applicable.

GM2 FSTD.FST.010 Table of applicable functions and subjective tests

This GM gives guidance on the use of the functions and subjective test table.

- (a) General guidance is as follows.

Table 1 below provides a fictitious, partially completed example that includes sample entries showing how information could be recorded during F&S. The organisation requesting qualification of an FSTD should record details at a level appropriate to the requested FCS, ensuring that test procedures and results can be demonstrated to the competent authority.

- (b) CS FSTD.FST.010 paragraph (b)(4) requires the expansion of the lists of tests in CS FSTD.FST.020 and CS FSTD.FST.030 to cover the aircraft system operation. Some examples of this are presented in italics in Table 1 and demonstrate how such an expansion may be represented.

- (c) Result column guidance is as follows.

The 'Result' column should indicate whether or not a test's result has been assessed as satisfactory. If a result has been declared unsatisfactory, it is useful to annotate a reference to any relevant declared discrepancy to ensure the traceability of corrective actions.

- (1) 'OK' in Table 1 indicates that the item was tested and was found to be satisfactory.
- (2) 'Fail' indicates that the item was tested with unsatisfactory results.
- (3) The abbreviation 'DR' stands for 'discrepancy report'. The following number refers to the report number used for tracking purposes.

(d) Remark column guidance is as follows.

The 'Remark' column should contain any relevant details for the corresponding item to support the traceability of what has been tested and under what conditions. This information may also assist the organisation in completing or verifying the related sections of the ESL. The following are examples of uses of the 'Remark' column:

- (1) assessed configurations and conditions;
- (2) ATA chapter malfunctions assessed;
- (3) visual scenes assessed (e.g. EDDF stands for an airport code);
- (4) specific performance-based navigation approaches conducted;
- (5) low-visibility taxi routes assessed;
- (6) low-visibility minima used.

Table 1. Fictional example of a partially completed functions and subjective test list

Section	Table of functions and subjective tests	Result	Remark
B	Preparation for flight		
	Functionality of all switches, indicators, systems and equipment at all crew members stations and IOS - Flight deck when electrical power is off - Electrical system with battery power on (synoptics and functionalities) - Electrical system with external power connected 	OK OK OK OK	The flight deck design and functions are identical to those of the real aircraft
	The flight deck design and functions are in accordance with the fidelity level of the FDK feature	OK	
	The ambient lighting provides an even level of illumination and is not distracting to the flight crew members	OK	
	Automated weather reporting - COM1, COM2 - Headsets and loudspeaker	OK OK OK	ATIS at EDDF matches the IOS settings
C	Surface operations		
C.1	APU/engine start and run-up		
	APU start - Synoptics for start sequence - Electrical and air conditioning systems with APU running - APU fire and automatic shutdown 	OK OK OK OK	
	Normal engine start - Auto-start sequence (FF at 18 % N2, starter OFF at 55 % N2) - Automatic ignition system selection logic - Engine primary and secondary parameter rates 	OK OK OK OK	APU assisted; OAT at 15 °C, 35 °C and -25 °C. Oil temperature and pressure versus OAT is realistic
	Alternate engine start procedures - External air for start - Manual start sequence - Cross-bleed start (requires 35 psi and correct valve positions) 	OK OK OK OK	
	Abnormal engine starts and shutdowns - FADEC protection for hot start triggers before EGT red line - Hung start indications and lower-than-normal fuel flow - No protections for manual start (i.e. hot start would exceed EGT limit) - Caution and warning system indications	OK OK OK OK	

	o		
C.2	Taxi		
	Friction of the engine control levers	Fail DR #13	Left throttle lever is too stiff to operate.
	Pushback	OK	EDDF A25, left, right and straight
	Thrust required to taxi	OK	MTOW, medium and light weight
	Thrust response	OK	MTOW, medium and light weight
	Ground handling	OK	
	o Tiller deflection	OK	
	o Pedal connection to nose wheel	OK	
	o Brake pressures	OK	
	o Heading hold function (calm wind and 20 kt crosswind)	OK	
	o		
			
G	Descent and arrival		
	Normal rate	OK	
	o Automatic pressurisation (vertical speed, pressure difference)	OK	
	o Manual pressurisation (use of outflow valve versus parameters)	OK	
	o FMS estimates (time at waypoints, time of arrival, fuel)	OK	
	o		
	Maximum rate/emergency (clean and with speed brake etc.)	OK	Emergency descent from FL330 to FL80
	o Spoiler indications and buffet	OK	
	o Overspeed warning (master warning, aural, speed tape)	OK	
	o		
	With autopilot	OK	
	o Vertical speed mode, with altitude preselect capture	OK	
	o Managed mode	OK	
	o		
H	Instrument approach operations		
H.1	Precision approach procedures		
	ILS CAT I approach: manual approach with/without FD, including landing	OK	EDDF 25C, CAT I day and night, both with and without FD
	o Managed mode	OK	
	o FD sensitivity	OK	
	o Aural call-out for minima	OK	
	o		
	ILS CAT I approach: autopilot and auto-throttle coupled approach and manual landing	OK	EDDF 25L, 20 kt crosswind with gusts
	o Managed mode	OK	
	o FD/autopilot and auto-thrust sensitivity	OK	
	o Glideslope angle	OK	
	o		
			

GM3 FSTD.FST.010 Table of applicable functions and subjective tests

FUNCTION AND SUBJECTIVE TESTING OF NON-PRECISION APPROACHES

The test sections in CS FSTD.FST.020 and CS FSTD.FST.030 for non-precision approaches do not explicitly define all required test cases. The required test cases depend on the simulated aircraft and the capabilities of the FSTD. For example, an RNP approach in accordance with the navigation specification 'RNP APCH' of International Civil Aviation Organization Doc 9613 for performance-based navigation (RNP APCH) may be flown using various FD/autopilot modes depending on the aircraft (e.g. heading hold mode, navigation tracking mode or aircraft-specific modes).

The organisation requesting qualification should include appropriate non-precision approach tests in the customised list of applicable functions and subjective tests. The emphasis is on sufficient coverage rather than on repeating every possible combination of FD/autopilot modes, approach types, engine configurations (all engines operating / OEI) and approaches with and without continuous descent final approach (CDFA). For example, FD modes that are independent of the navigation source (e.g. heading hold mode, vertical speed mode) do not need to be tested repeatedly across approach types if their function is not affected. In addition, if the aircraft does not provide CDFA guidance (e.g. a computed descent path), it is not necessary to repeat each type of non-precision approach with and without CDFA, as a single manually flown CDFA approach is sufficient.

Test cases should be selected to provide confidence that the simulated aircraft behaviour and all relevant guidance functions operate correctly in the context of non-precision approaches, without unnecessary repetition or excessive test volume.

FUNCTION AND SUBJECTIVE TESTS FOR AEROPLANES

CS FSTD.FST.020 Table of functions and subjective tests for aeroplane flight simulation training devices

Table 1 below presents a table of functions and subjective tests for aeroplane FSTDs, to be further modified in accordance with CS FSTD.FST.010.

Table 1. Function and subjective tests for aeroplane FSTDs

Section	Functions and subjective tests (aeroplanes)
A	Flight deck layout and structure and aircraft system operation Refer to CS FSTD.QB.101 for specified requirements.
	● Enclosure of the flight deck layout ● Flight deck windows ● Flight crew member seating ● Basic flight instruments ● Appearance of the flight deck panels, instruments and switches ● Haptics of the flight deck panels, instruments and switches ● Circuit breakers ● Flight deck equipment (e.g. headsets, oxygen masks, EFBs) ● Instrument lighting ● Flight deck lighting ● Ambient lighting
B	Preparation for flight
	● Functionality of all switches, indicators, systems and equipment at all crew members stations and IOS ● The flight deck design and functions are in accordance with the fidelity level of the FDK feature ● The ambient lighting provides an even level of illumination and is not distracting to the flight crew members ● Automated weather reporting
C	Surface operations
C.1	APU/engine start and run-up
	● APU start ● Normal engine start ● Alternate engine start procedures ● Abnormal engine starts and shutdowns (hot start, hung start, tailpipe fire, etc.) ● After-start system checks (e.g. electrical, hydraulic, flight control, autopilot, radio and lighting systems)
C.2	Taxi

Section	Functions and subjective tests (aeroplanes)
	● Friction of the engine control levers ● Pushback/powerback ● Thrust required to taxi ● Thrust response ● Ground handling ● Minimum radius turn ● Nose wheel scuffing ● Taxi aids (e.g. taxi camera, moving map) ● Low visibility ● Surface contaminants (wet, icy, patchy wet, patchy ice, wet or rubber residue) ● Surface roughness
C.3	Brake operation
	● Brake operation (normal, automatic and alternate/emergency) ● Brake fade ● Brake failure
D	Take-off and departure
D.1	Normal take-off
	● Aeroplane/engine parameter relationships, including run-up ● Acceleration characteristics ● Nose wheel and rudder steering ● Crosswind ○ Crosswind (maximum demonstrated) ○ Gusting crosswind ● Performance configurations and conditions ○ All flap/slat configurations certified for the simulated aircraft ○ Reduced V_1 ○ Maximum engine derating ○ Soft surface ○ Short-field / short take-off and landing operations ○ Obstacle (performance over visual obstacle) ● Low-visibility take-off ● Visual traffic circuit ● Landing gear and wing flap leading-edge device operation ● Contaminated runway operation
D.2	Abnormal/emergency procedure during take-off and departure
	● Rejected take-off ● Rejected take-off on contaminated runway ● Rejected take-off special performance (e.g. reduced V_1, maximum engine derating, soft field, short take-off and landing operations) ● Take-off with a propulsion system malfunction of the most critical engine (allowing for an analysis of causes, symptoms, recognition and the effects on aeroplane performance and handling) at the following points: ○ prior to V_1 ○ between V_1 and V_R ○ between V_R and 500 ft AGL ○ after gear-up during climb out ● Other simulated engine failures (e.g. different engines and simulated failures)

Section	Functions and subjective tests (aeroplanes)
	Flight control system failures, reconfiguration modes, manual reversion and associated handling
E	Climb
	- Normal climb - Climb performance (all engines operating and OEI) compared with AFM performance data - Best rate - Best angle - One or more engines inoperative - Level-off
F	Cruise
	Note: Tests in this section should be conducted at various altitudes, not only at typical cruising levels.
F.1	Performance characteristics
	- Straight and level flight performance (speed versus power, configuration and attitude) - Accelerations and decelerations / changes of airspeed - High-speed buffet - Overspeed warning (in excess of V_{MO} or M_{MO})
F.2	Manoeuvres and flying qualities
	- Turns - Normal - Standard rates (rate $\frac{1}{2}$, 1 and 2) - Steep (45° of bank) - Turns with/without speed brake / spoilers deployed - Low-air-speed handling (e.g. turns, airspeed stability) - High-air-speed handling (e.g. turns) - High-altitude handling - High-Mach-number handling (Mach tuck, Mach buffet) and recovery (trim change) - Gliding to a forced landing; terrain accuracy for forced landing area selection
F.3	Characteristics of the particular aircraft
	- Flight characteristics of the particular aircraft (e.g. CCA, direct lift control) - Flight envelope protection functions (high angle of attack, bank limit, overspeed, etc.) - Flight control system failures, reconfiguration modes, manual reversion and associated handling
F.4	High angle of attack and FSTD validation envelope
	- High angle of attack - High angle of attack, approach to stalls, stall warning and stall buffet (take-off, cruise, approach and landing configuration), including reaction of the auto-flight system and stall protection system - High angle of attack, approach to stalls, stall warning, stall buffet and stall (and g-break, if applicable) (take-off, cruise, approach and landing configuration), including reaction of the auto-flight system and stall protection system - Upset prevention and recovery manoeuvre within the FSTD validation envelope
F.5	One engine inoperative

Section	Functions and subjective tests (aeroplanes)
	• In-flight engine shutdown • Manoeuvring with one or more engines inoperative, as appropriate • In-flight engine restart (assisted and windmill)
F.6	Enroute navigation
	• Terrain accuracy for visual navigation • Radionavigation • GNSS navigation
G	Descent and arrival
	• Normal descent • Maximum rate descent / emergency (clean and with speed brake etc.) • With autopilot • Flight control system failures, reconfiguration modes, manual reversion and associated handling
H	Instrument approach operations The FSTD should be tested for all relevant types of instrument approach, using appropriate aircraft configurations, flight conditions and system states (e.g. calm winds, limiting wind velocities, wind shear, relevant aircraft system failures). Testing should cover all applicable aircraft systems, equipment (e.g. HUD, EFVS) and procedures. The customised list of F&S should contain all the types of approaches defined in the ESL and applicable to the simulated aircraft.
H.1	Precision approach procedures
	Testing should cover all applicable aircraft systems, equipment (e.g. HUD, EFVS) and procedures.
	• CAT I approach (ILS, GBAS landing system, microwave landing system) ○ Manual approach and landing with/without FD ○ Manual approach, one or more engines inoperative ○ Autopilot and auto-throttle coupled approach and manual landing ○ Autopilot and auto-throttle coupled approach, one or more engines inoperative • category II operations with a DH below 200 ft and down to 100 ft and RVR in accordance with the operational weather minima (CAT II) approach ○ Autopilot and auto-throttle coupled approach to DH and landing ○ Autopilot and auto-throttle coupled approach with OEI approach to DH and go-around ○ Manual approach and landing • CAT III approach ○ Autopilot and auto-throttle coupled approach to landing and roll-out ○ Autopilot and auto-throttle coupled approach to DH and go-around ○ Autopilot and auto-throttle coupled approach to landing and roll-out with OEI ○ Autopilot and auto-throttle coupled approach to DH and go-around with OEI ○ Manual approach and landing (if applicable, such as with HUD) • Autopilot/auto-throttle coupled approach (to a landing or to a go-around) ○ With generator failure ○ With maximum tailwind component certified or authorised

Section	Functions and subjective tests (aeroplanes)
	○ With maximum crosswind component certified or authorised
H.2	Non-precision approach procedures Testing should cover manual flight and the use of all applicable FD/autopilot modes across non-precision approach types. Modes intended for OEI operations should also be tested in the OEI configuration. Testing should cover approaches flown with and without CDFA if such guidance is supported (e.g. a computed descent path). If CDFA guidance is not available, one manually flown CDFA approach is sufficient.
	● RNP APCH approach to localiser performance (LP) and lateral navigation (LNAV) minima, tested under varying temperature conditions, including down to the minimum authorised temperature ● VOR and VOR/DME approach ● ILS-localiser-only (LOC) approach ● ILS localiser back course (LOC-BC) approach ● ILS offset localiser approach ● NDB approach ● Circling approach (approach prior to visual circling manoeuvre)
H.3	Approach procedures with vertical guidance (APV – 3D approaches)
	● RNP APCH barometric-VNAV approach to LNAV/VNAV minima, with all engines operating and with one or more engines inoperative ● RNP APCH approach based on SBAS (LPV minima), with all engines operating and with one or more engines inoperative ● Loss of SBAS signal during approach
H.4	RNP approach in accordance with the navigation specification ‘RNP AR APCH’ of International Civil Aviation Organization Doc 9613 for performance-based navigation (RNP AR APCH) Note: Equipment requirements for RNP AR APCH differ if the required navigation accuracy is 0.3 NM or lower. Refer to airworthiness and operational regulations for criteria applicable to real aircraft.
	● RNP AR APCH approach procedures, with all engines operating and with one or more engines inoperative ● Go-around with one engine inoperative
I	Visual approaches and instrument approach visual segments
	● Manoeuvring, normal approach and landing with all engines operating, with and without visual and navigational approach aid guidance ● Approach and landing with one or more engines inoperative ● Operation of landing gear, flap/slats and speed brakes (normal and abnormal) ● Approach and landing with crosswind ○ Maximum demonstrated ○ Gusting ● Approach and landing with flight control system failures, reconfiguration modes, manual reversion and associated handling ● Approach and landing with trim malfunctions ○ Longitudinal trim malfunction ○ Lateral-directional trim malfunction ● Landing from a visual traffic pattern

Section	Functions and subjective tests (aeroplanes)
	- Approach and landing with standby (minimum) electrical/hydraulic power - Approach and landing from circling conditions (circling approach) Note: For this test, the visual field of view (refer to CS FSTD.QB.111) should be at least 200° horizontally to allow visual contact to be maintained with the airport during the whole circling procedure after descending below the minimum descent altitude (MDA). In addition, the operating sites and terrain FSTD feature should be, as a minimum, at the representative fidelity level (refer to CS FSTD.QB.114) to enable adequate visual content in the visual image. Any associated hazard lights or any other visual aids for use as part of the published circling procedure should be included in the correct positions and be of the appropriate colours, directionality and behaviour.
J	Missed approach
	- All engines operating, manual and autopilot - One or more engines inoperative, manual and autopilot - Rejected landing - With auto-flight - With flight control system failures, reconfiguration modes and manual reversion - Approach climb in icing (for aeroplanes with icing accountability)
K	Surface operations (landing, after landing and post-flight)
K.1	Landing roll and taxi
	- Spoiler operation - Reverse thrust operation - Directional control and ground handling, both with and without reverse thrust - Reduction of rudder effectiveness with increased reverse thrust (rear pod-mounted engines) - Brake and anti-skid operation - Brake and anti-skid operation in dry, wet, icy, patchy wet, patchy ice, wet or rubber residue in touchdown zone conditions - Brake and anti-skid operation in dry and wet conditions - Brake operation in dry conditions - Auto-braking system operation, as applicable - HUD/EFVS
L	Engine shutdown and parking
	- Engine and system operation - Parking brake operation
M	Any appropriate flight phase
M.1	Flight control forces and hardware in all flight phases
	Refer to CS FSTD.QB.102 for specified requirements.
	- Primary flight control forces - Integration of the primary flight control hardware with the aircraft systems - Hardware, forces and feel of other flight controls - Hardware, forces and feel of controls that affect aircraft configuration
M.2	Flight control systems operation in all flight phases
	Refer to CS FSTD.QB.103 for specified requirements.

Section	Functions and subjective tests (aeroplanes)
	• Primary flight control systems operation (normal and non-normal modes, backup systems) • Primary flight control systems indications • Operation and indications of other flight controls • Operation and indications of controls that affect aircraft configuration • Flight and manoeuvre envelope protection functions • Flight control integration with aircraft systems
M.3	Aeroplane and power plant system operation (if fitted), including associated abnormal and emergency procedures in all appropriate flight phases Refer to CS FSTD.QB.104 for specified requirements.
	• Instrument indications • Flight display systems • Navigation systems • Autopilot and flight director • Flight management computers and systems • Flight control computers, including stability and control augmentation • Caution and warning systems • Communications systems • Surveillance systems • Electrical systems • Air conditioning and pressurisation (environmental control) system • De-icing/anti-icing systems • Auxiliary power plant / APU • Fire and smoke detection and suppression system • Fuel and oil systems • Hydraulic systems • Pneumatic systems • Landing gear • Lighting (internal and external) • Oxygen systems • Power plant systems • Airborne radar • Terrain awareness warning systems (e.g. TAWS) • Collision avoidance systems (e.g. ACAS) • Surveillance and data link communications systems (e.g. ADS-B, CPDLC, ACARS, FANS) • Head-up guidance and HUDs (including EFVS if applicable) • Stall warning/avoidance • Wind shear avoidance/recovery guidance equipment • Flight envelope protections • EFB • Automatic checklists (normal, abnormal, emergency and deferred procedures) • Runway alerting and advisory systems
M.4	Airborne procedures

Section	Functions and subjective tests (aeroplanes)			
	• Holding • Air hazard avoidance (traffic and weather, including visual correlation) • Effects of airframe and engine icing (refer to CS FSTD.QB.107.1.ICING) Note: Preferably performed in level flight with autopilot on and anti-icing/de-icing off, followed by icing activation and monitoring of effects (e.g. power, pitch, vibration, performance), then manual flight to assess handling changes and finally activation of anti-icing/de-icing systems to evaluate their function and effects. • Airborne weather radar • Wind shear ○ Prior to take-off rotation ○ At lift-off ○ During initial climb ○ On final approach, below 500 ft AGL • Microburst			
M.5	Non-normal flight conditions			
	• Tests of simulated malfunctions (refer to CS FSTD.FST.005; CS FSTD.QB.115, section 'Simulated malfunctions'; and CS FSTD.QB.104) • Non-normal flight conditions, for example (as applicable): ○ recovery from unusual attitudes ○ trim runaway ○ flight control surface failure			
N	Visual cueing The test items in this section supplement the requirements specified in CS FSTD.QB.111 and should be performed to ensure satisfactory results for the applicable fidelity level (G, R or S). Note: The visual cueing should also be subjectively evaluated as part of the integrated simulation simultaneously with other cueing features and the complete simulation in all flight phases.			
		Feature fidelity level		
		G	R	S
N.1	Display			
	Image is free from optical discontinuities and artefacts	✓	✓	✓
	The visual channels have the same image quality for day, twilight and night	✓	✓	✓
	Image system light points are free from distracting jitter, smearing or streaking	✓	✓	✓
	Surfaces and textural cues are free from apparent quantisation (aliasing)		✓	✓
	Surfaces and textural cues do not create distracting quantisation (aliasing)	✓		
	Image is stable, seamless and realistic	✓	✓	✓
	Visual display geometry is correct		✓	✓
	Visual image geometry has no distracting discontinuities	✓	✓	✓
	The FOV replicates the aircraft FOV without obstruction			✓

Section	Functions and subjective tests (aeroplanes)			
	System is capable of portraying full-colour realistic textural cues	✓	✓	✓
	The flight deck cut-off angle matches the simulated aircraft	✓	✓	✓
N.2	Image qualities			
	For collimated displays: - the parallax effects are minimised - the centre of the image is aligned with the flight deck centre line and the horizon		✓	✓
	For non-collimated displays: - the parallax effects are minimised - the visual system can be aligned at any time to each of the pilot eye positions and to the flight deck centre line - when the image is aligned to the pilot flying, the misalignment observed by the non-flying pilot does not exceed 20°	✓	✓	
	Image is not distorted	✓	✓	✓
	Display contrast ratio is sufficient to clearly differentiate the brightest and darkest areas of the display	✓	✓	✓
	Light point contrast ratio is sufficient to be able to clearly differentiate light points	✓	✓	✓
	Surface brightness is sufficient to be able to clearly identify visual elements in the visual scene	✓	✓	✓
	Image's black level is sufficient to support operations in daylight, twilight and night conditions	✓	✓	✓
	Image's luminance levels are sufficient to support operations in daylight, twilight and night conditions	✓	✓	✓
	Motion blur is below a distracting level	✓	✓	✓
	Speckle is below a distracting level	✓	✓	✓
	The ground segment visible to the pilot when conducting a landing manoeuvre in day, twilight and night low-visibility conditions conforms to that of the aircraft	✓	✓	✓
	The visual system has an effect for a crash	✓	✓	✓
	All moving objects in the visual image are free from jitter	✓	✓	✓
	The level of detail in the visual image supports accurate perception and recognition of elements, objects and terrain Curved and complex shapes appear smooth and realistic The scene content is realistically affected by the viewing distance	✓	✓	✓
	Visual image gives cues to assess sink rate and enable depth perception during take-off and landing		✓	✓
	System is capable of providing focus effects that simulate rain		✓	✓
	System is capable of providing light point perspective growth		✓	✓
N.3	Visual scene content			
	Continuous and sufficient horizontal and vertical FOVs specific to the aircraft being simulated are provided			✓

Section	Functions and subjective tests (aeroplanes)			
	Visual cues to assess the rate of change in height and attitude and the translational displacements are provided		✓	✓
	An identifiable horizon is displayed under day, twilight and night conditions with unlimited visibility If the visibility is reduced, the horizon becomes indistinct or obscured	✓	✓	✓
	Terrain contours are displayed at the resolution demanded by the environment			✓
	Landing areas are visible from a sufficient distance	✓	✓	✓
	Focus and clarity of terrain and ground objects are sharp		✓	✓
	Surface shading effects are consistent with simulated sun position Shadows of 3D objects and lighting effects are appropriate for the directionality of light (e.g. sunshine)			✓
	Images for daylight, twilight and night visuals are supported	✓	✓	✓
	Image shows operational aircraft landing lights for twilight and night scenes	✓	✓	✓
	The visual image corresponds to other traffic or terrain		✓	✓
	For aircraft capable of landing on water, the sea states and effects of wind lanes on water surface are shown		✓	✓
	Image displays the ownship parts that are visible within the FOV (e.g. wings, nacelles, propellers, spinners) Any movement (e.g. ailerons, propellers) is displayed	✓	✓	✓
	Image displays the icing effects on the aircraft parts that are visible within the FOV (e.g. wing leading edges, spinners)	✓	✓	✓
	For NVIS, as applicable: - the visual image supports NVIS for the black level, surface brightness, surface contrast ratio, lunar illumination, image colour and surface resolution - surface shading effects are consistent with the simulated moon position			✓
N.4	Visual element recognition Note: The following are the minimum distances at which runway elements should be visible. Distances are measured from the runway threshold to an aircraft aligned with the runway on an extended 3° glide slope in suitable simulated meteorological conditions. For circling approaches, all tests below apply to both the runway used for the initial approach and the runway of intended landing.			
	Runway definition, strobe lights, approach lights and runway edge white lights from 8 km (5 sm) from the runway threshold	✓	✓	✓
	Visual approach aid lights from 8 km (5 sm) from the runway threshold			✓
	Visual approach aids lights from 4.8 km (3 sm) from the runway threshold	✓	✓	
	Runway centre line lights and taxiway definition from 4.8 km (3 sm)	✓	✓	✓

Section	Functions and subjective tests (aeroplanes)			
	Threshold lights and touchdown zone lights from 3.2 km (2 sm)	✓	✓	✓
	Runway markings within range of landing lights for night scenes; as required by the surface resolution test for day scenes			✓
	For circling approaches, runway of intended landing and associated lighting fade into view in a non-distracting manner		✓	✓
O	Operating sites and terrain – visual representation The test items in this section supplement the requirements specified in CS FSTD.QB.114 and should be performed to ensure satisfactory results for the applicable fidelity level (G, R or S). For the operating sites and terrain FSTD feature at the specific fidelity level, refer to CS FSTD.QB.114.1 for requirements for tier I operating site models.			
		Feature fidelity level		
		G	R	S
O.1	Runways and taxiways			
	Realistic runways and taxiways			✓
	Representative runways and taxiways		✓	
	Generic runways and taxiways	✓		
	If appropriate for the airport, two parallel runways and one crossing runway can be displayed simultaneously; at least two runways can be lit simultaneously			✓
	Runway threshold elevations and locations are modelled to provide correlation with aeroplane systems (e.g. compass, altimeter, GNSS, HUD)	✓	✓	✓
	Slopes in runways, taxiways and ramp areas may not cause distracting or unrealistic effects, including pilot eyepoint height variation			✓
	The displayed scene corresponds to the appropriate surface contaminants and includes runway lighting reflections for wet, partially obscured lights for snow or suitable alternative effects			✓
	An airport with a snow scene is to include terrain snow and snow-covered taxiways and runways			✓
O.2	Runway surface, markings and elements for each ‘in-use’ runway			
	Threshold markings	✓	✓	✓
	Runway numbers	✓	✓	✓
	Touchdown zone markings	✓	✓	✓
	Fixed distance markings		✓	✓
	Edge markings	✓	✓	✓
	Centre line markings	✓	✓	✓
	Distance remaining signs			✓
	Signs at intersecting runways and taxiways			✓
	Windsock that gives appropriate wind cues		✓	✓

Section	Functions and subjective tests (aeroplanes)			
	Windssock lighting		✓	✓
O.3	Runway lighting of appropriate colours, directionality, behaviour and spacing for each 'in-use' runway, including the following, if appropriate			
	Threshold lights	✓	✓	✓
	Edge lights	✓	✓	✓
	End lights	✓	✓	✓
	Centre line lights	✓	✓	✓
	Touchdown zone lights	✓	✓	✓
	Lead-on/lead-off lights			✓
	Visual landing aid(s)	✓	✓	✓
	Appropriate approach lighting system for the runway		✓	✓
	Approach lighting system for the runway	✓		
O.4	Taxiway surface and markings (associated with each 'in-use' runway), including the following, if appropriate			
	Edge markings	✓	✓	✓
	Centre line markings	✓	✓	✓
	Runway holding position markings	✓	✓	✓
	ILS critical area markings			✓
	All taxiway markings, lighting and signage to taxi, as a minimum, from a designated parking position to a designated runway and return, after landing on the designated runway, to a designated parking position. A low-visibility taxi route (e.g. surface movement guidance control system, follow-me truck, daylight taxi lights) must also be demonstrated for operations authorised in low-visibility conditions. The designated runway and taxi routing must be consistent with that airport for operations in low-visibility conditions.			✓
O.5	Taxiway lighting of appropriate colours, directionality, behaviour and spacing (associated with each 'in-use' runway), including the following, if appropriate			
	Edge lights	✓	✓	✓
	Centre line lights	✓	✓	✓
	Runway holding position and ILS critical area lights	✓	✓	✓
O.6	Visual model correlation			
	Airport model aligned with the navigational aids that are associated with operations at the 'in-use' runway	✓	✓	✓
	Simulation of runway contaminants correlates with the displayed runway surface and lighting			✓
O.7	Airport buildings, structures, objects and lighting			
	Airport buildings, structures, objects and lighting			✓
	Representative airport buildings, structures, objects and lighting		✓	
	Generic airport buildings, structures, objects and lighting	✓		
	At least one usable gate, set at the appropriate height (required only for aeroplanes that typically operate from terminal gates)		✓	✓

Section	Functions and subjective tests (aeroplanes)			
	Moving and static operating site clutter (e.g. other aeroplanes, power carts, tugs, fuel trucks, additional gates)		✓	✓
	Gate/apron markings (e.g. hazard markings, lead-in lines, gate numbering), lighting and gate docking aids or a marshaller		✓	✓
O.8	Terrain and obstacles			
	Terrain and obstacles within 46 km (25 NM) of the reference airport with appropriate colours and textures for the simulated area (including ground objects of sufficient number and appropriate size and perspective)			✓
	Representative depiction of terrain and obstacles within 46 km (25 NM) of the reference airport		✓	
	Visual scene with sufficient scene content to allow a pilot to successfully accomplish a visual landing	✓		
	Visual scene with adequate terrain elements to permit VFR navigation by sole reference to visual landmarks in accordance with appropriate charts (typically 1:500 000, 1:250 000 or 1:100 000 scale mapping) and with suitable terrain contouring represented		✓	✓
	Buildings, trees or other vertical obstructions in the immediate vicinity of the landing area		✓	✓
O.9	Significant, identifiable natural and cultural elements			
	Significant, identifiable natural and cultural elements within 46 km (25 NM) of the reference airport Note: This refers to natural and cultural elements that are typically used for pilot orientation in flight. Outlying airports not intended for landing need only provide a reasonable facsimile of runway orientation.			✓
	Representative depiction of significant and identifiable natural and cultural elements within 46 km (25 NM) of the reference airport Note: This refers to natural and cultural elements that are typically used for pilot orientation in flight. Outlying airports not intended for landing need only provide a reasonable facsimile of runway orientation.		✓	
O.10	Visual scene management			
	All airport runway, approach and taxiway lighting and cultural element lighting intensity for any approach are capable of being set to five different intensities and an off setting; all visual scene light points fade into view appropriately in accordance with the conditions set on the IOS			✓
	Airport runway, approach and taxiway lighting and cultural feature lighting intensity for any approach can be set at an intensity representative of that used in training for the visibility level set; all visual scene light points fade into view appropriately	✓	✓	

Section	Functions and subjective tests (aeroplanes)			
	The directionality of strobe lights, approach lights, runway edge lights, visual landing aids, runway centre line lights, threshold lights and touchdown zone lights on the runway of intended landing is appropriate	✓	✓	✓
O.11	Selectable airport visual scene capability for the following			
	Night	✓	✓	✓
	Twilight (i.e. dusk and dawn)	✓	✓	✓
	Day	✓	✓	✓
	Moving ground vehicles undertaking predefined manoeuvres (conflicting and not conflicting with the ownship), for example another aeroplane crossing the active runway or converging airborne traffic		✓	✓
	Aircraft traffic undertaking predefined manoeuvres (conflicting and not conflicting with the ownship), for example another aircraft crossing the active runway or converging airborne traffic		✓	✓
	Operational visual scenes that portray physical relationships known to cause landing illusions, for example short runways, landing approaches over water, uphill or downhill runways, rising terrain on the approach path and unique topographical elements		✓	✓
O.12	Correlation with aeroplane and associated equipment			
	Visual cues to relate to simulated aeroplane attitudes and responses	✓	✓	✓
	Visual image sufficient to support changes in approach path by using runway perspective; changes in visual cues during take-off, approach and landing cannot distract the pilot		✓	✓
	Visual scene correlates with aeroplane systems, if fitted, for example: • TAWS indications versus terrain in the visual image • ACAS versus traffic in the visual image • EFVS versus visual image • reduction of visual image defocus due to rain when wipers are used		✓	✓
	Visual effects for each visible, ownship, external light (e.g. taxi and landing light lobes)	✓	✓	✓
P	Atmosphere and weather The test items in this section supplement the requirements specified in CS FSTD.QB.113 and should be performed to ensure satisfactory results for the applicable fidelity level (G, R or S).			
		Feature fidelity level		
		G	R	S
P.1	Environment simulation			
	Dynamic environment simulation with weather effects (on ground / in air) integrated with the entire simulation		✓	✓

Section	Functions and subjective tests (aeroplanes)			
	A basic atmospheric model, including pressure, temperature, visibility, cloud base and winds	✓		
P.2	Atmosphere simulation			
	The atmosphere simulation includes: - wind speed, direction and gusts - air temperature and its lapse rate - air pressure - turbulence levels	✓	✓	✓
	Turbulence effects by using applicable cues on all axes, including: - effects on instruments - effects on aerodynamics and handling as a result of the atmospheric disturbances - effects on the controls, as applicable (e.g. reversible flight controls) - motion and/or vibration effects, if installed	✓	✓	✓
P.3	Visual weather representation			
	The visual scene includes the following: - various visibility and RVR conditions, including fog and patchy fog effects - effects on operating site lighting (including variable intensity and fog effects)		✓	✓
	Patchy fog giving the effect of variable RVR. The lowest RVR must be that selected on the IOS (i.e. variability is only greater than the RVR set on the IOS).		✓	✓
	Effects of fog on airport lighting, such as halos and defocus		✓	✓
	The visual scene includes various visibility and RVR conditions	✓		
	Visibility and RVR measured in terms of distance. Visibility/RVR must be checked at and below a height of 2 000 ft above the operating site and within a radius of 16 km (8.6 nm; 10 sm) of the operating site.	✓	✓	✓
	Visual image for: - various visibility and RVR conditions - a realistic transition to IMCs - a realistic transition to visual contact from IMCs	✓	✓	✓
	Gradual breakout to ambient visibility/RVR, defined as up to 10 % of the relevant cloud base or top. This transition layer should be no less than 20 ft and no more than 200 ft in vertical extent. Transition effects should be complete when the IOS cloud base or top is reached when exiting and start when entering the cloud (i.e. transition effects must occur within the IOS defined cloud layer). Cloud effects must be checked at and below a height of 2 000 ft above the operating site and within a radius of 16 km (8.6 nm; 10 sm) of the operating site.	✓	✓	✓

Section	Functions and subjective tests (aeroplanes)			
	Visual image has cloud layers representing few, scattered, broken and overcast cloud conditions, giving partial or complete obstruction of the ground scene	✓	✓	✓
	3D clouds			✓
	Visual image provides an in-cloud airspeed effect (variable cloud density and ambient changes)			✓
	The clouds, fog and weather effects affect the ownship external lights (e.g. landing lights, strobes and beacons) in the visual scene, such as reflected glare		✓	✓
	The clouds, fog and weather effects affect the other aircraft external lights within visual range in the visual scene		✓	✓
P.4	Weather effects			
	Weather effects, including visual representations as applicable, for: - rain - hail - storm cells (i.e. localised storm area, rain cell) - thunderstorms (i.e. including thunder and lightning effects) - falling snow - blowing snow and - blowing sand			✓
	The precipitation and blowing snow or sand are affected by the wind direction and velocity			✓
	Sound, motion and visual effects of light, medium and heavy precipitation near a thunderstorm on take-off, approach and landing at and below an altitude of 600 m (2 000 ft) above the airport surface and within a radius of 16 km (10 sm) of the operating site			✓
Q	Vibration cueing The test items in this section supplement the requirements specified in CS FSTD.QB.109 and should be performed to ensure satisfactory results for the applicable fidelity level (G, R or S). Note: The vibration and buffet cueing should also be subjectively evaluated as part of the integrated simulation simultaneously with other cueing features and the complete simulation in all flight phases.			
		Feature fidelity level		
		G	R	S
	Cues on the ground due to thrust effect with brakes set		✓	✓
	Cues on the ground due to spoiler / speed brake extension and thrust reversal		✓	✓
	Cues due to the extension and retraction of landing gear		✓	✓
	Cues in the air due to flap and spoiler / speed brake extension		✓	✓
	Cues due to atmospheric disturbances		✓	✓
	Cues due to Mach and manoeuvre buffet		✓	✓

Section	Functions and subjective tests (aeroplanes)			
	Cues due to approach-to-stall buffet and stall buffet		✓	✓
	Cues due to configuration changes (e.g. flaps, landing gear, spoilers, speed brakes)	✓		
	Cues due to thrust	✓		
	Cues due to aerodynamic regimes (e.g. overspeed, approach to stall)	✓		
R	Motion cueing The test items in this section supplement the requirements specified in CS FSTD.QB.110 and should be performed to ensure satisfactory results for the applicable fidelity level (G, R or S). Motion effects should give cues realistic to the simulated aircraft dynamics to indicate the threshold at which a flight crew member recognises an event or situation. Note: The motion cueing should also be subjectively evaluated as part of the integrated simulation simultaneously with other cueing features and the complete simulation in all flight phases.			
		Feature fidelity level		
		G	R	S
R.1	Motion cues – general			
	The pilot receives an effective motion cue and stimulus, which provide the appropriate sensations of the acceleration of the aircraft's six degrees of freedom. These sensations reflect the aircraft's translational accelerations (longitudinal, lateral, vertical) and angular motion (pitch, roll, yaw), including their onset, rate and intensity, and also include quasi-static platform attitudes.			✓
	The motion system is capable of generating sustained acceleration cues and generating the key acceleration and angular cues that are characteristic of the aircraft dynamics in all axes for: - all flight phases - manoeuvres (e.g. rotation, turns, acceleration, deceleration, side-slips, flare, landing, go-around) - failures (e.g. engine failures, engine-out operations, abnormal operations, emergencies)			✓
	The FSTD motion cueing system has a high tilt coordination gain, high rotational gain and high correlation with respect to the aircraft simulation model for all the manoeuvres			✓
	The pilot receives a motion cue and stimulus, which provide the appropriate sensations of acceleration. These sensations reflect the aircraft's translational accelerations (longitudinal, lateral, vertical) and angular motion (pitch, roll, yaw), including their onset, rate and intensity, and also include quasi-static platform attitudes, within its available degrees of freedom.		✓	

Section	Functions and subjective tests (aeroplanes)			
	The motion system is capable of generating sustained acceleration cues and generating the key acceleration and angular cues that are characteristic of the aircraft dynamics in the appropriate axes. The motion cueing in each case should be assessed with consideration given to the system's degrees of freedom and physical limitations for: - all flight phases; - manoeuvres (e.g. rotation, turns, acceleration, deceleration, side-slips, flare, landing, go-around); - failures (e.g. engine failures, engine-out operations, abnormal operations, emergencies).		✓	
	The cross-coupling effects of the motion movements are compensated for so that they are not discernible by a human during flight		✓	✓
	The washout is below the human-perceivable threshold during flight and does not exhibit any adverse motion cueing effects		✓	✓
R.2	Motion effects			
	Taxiway surface roughness			✓
	Effects of runway and taxiway rumble, oleo deflections, ground speed, uneven runway, runway contamination with the associated anti-skid characteristics, and running over runway centre line lights		✓	✓
	Effects of runway and taxiway rumble, oleo deflections, uneven runway, runway contamination with the associated anti-skid characteristics, and running over runway centre line light characteristics (with such effects as functions of ground speed)	✓		
	Bumps associated with landing gear mechanism movement		✓	✓
	Touchdown cues for main and nose gear	✓	✓	✓
	Nose wheel scuffing	✓	✓	✓
	Tyre failure		✓	✓
	Engine damage	✓	✓	✓
	Tail, pod and wing strike		✓	✓
	Airframe ground strike (e.g. tail strike)	✓		

Section	Functions and subjective tests (aeroplanes)			
S	Sound cueing			
	The requirements specified in CS FSTD.QB.108 for the applicable fidelity level of the sound cueing FSTD feature should be included in the list of functions and subjective tests. The following supplements those requirements. The following checks should be performed during a normal flight profile with motion and/or vibration system(s) on, as applicable. If headsets are used during normal aircraft operation, headsets with similar characteristics and performance should also be used during FSTD sound system testing. Note: The sound cueing should also be subjectively evaluated as part of the integrated simulation simultaneously with other cueing features and the complete simulation in all flight phases.			
		Feature fidelity level		
		G	R	S
	Significant sounds perceptible to the flight crew during flight operations replicate those of the simulated aircraft type and variant in all flight phases			✓
	Significant sounds perceptible to the flight crew during flight operations are characteristic of the simulated aircraft type in all flight phases		✓	
	Significant sounds perceptible to the flight crew during flight operations are characteristic of the simulated group of aircraft, including basic engine, propeller, airframe and aircraft system sounds, in all flight phases	✓		
	Precipitation			✓
T	Rain removal equipment (e.g. wipers)			✓
	Significant aeroplane noises perceptible to the pilot during normal operations, such as noises from engines, propellers, flaps, gears, anti-skid systems, spoiler extension/retraction and thrust reverser, to be of a comparable level to that found in the aeroplane	✓	✓	✓
	Abnormal operations for which there are associated sound cues including but not limited to engine malfunctions, landing gear / tyre malfunctions, tail and engine pod/propeller strike and pressurisation malfunctions	✓	✓	✓
	Sound of a crash when the FSTD is landed in excess of limitations	✓	✓	✓
T	Special effects			

Section	Functions and subjective tests (aeroplanes)
	● Braking dynamics ○ Normal braking dynamics ○ Anti-skid braking dynamics ○ Brake failure dynamics (including anti-skid) and decreased brake efficiency due to high brake temperatures based on aeroplane-related data. These representations should be realistic enough to enable the pilot's identification of the problem and the implementation of appropriate procedures. FSTD pitch, sideloading and directional control characteristics are representative of the aeroplane. ● Applicable only to aeroplane FSTDs that are to be qualified for UPRT. Refer to CS FSTD.QB.107.1.STALL and CS FSTD.QB.107.2.STALL.
U	Instructor operating station The requirements specified in CS FSTD.QB.115 should be included in the list of functions and subjective tests. The following supplements those requirements. Note: The list below is not exhaustive but is intended to cover the sorts of functionalities that should be available to support the intended use, depending on the FSTD and FCS. Refer to all the general requirements for IOS requirements.
	● Repositions ○ Parking spot ○ Take-off position ○ Approach position – at least three positions at approximately 1.8 km, 5.5 km and 9.3 km (1 NM, 3 NM and 5 NM, respectively) from the runway threshold Note: Repositions are in trim at the appropriate speed and configuration for the point. ● Resets ○ System ○ Temperature ○ Fluids and agents ● Weather presets (as applicable) ○ Unlimited, CAVOK, VFR, non-precision, precision (CAT I, CAT II, CAT III), EFVS ● Visual effects ○ Time of day (day, twilight or night) ○ Clouds (bases, tops, layers, types, density) ○ Visibility, with the units corresponding to those in use at the operating site ○ RVR, with the units corresponding to those in use at the operating site ○ Special effects (precipitation, thunderstorms, blowing snow, blowing sand, etc.) ○ Sea state conditions ● Wind speed and direction ○ Surface ○ Intermediate levels ○ Typical gradient ○ Gusts with the associated heading and speed variance ○ Turbulence ● Temperature – surface ● Atmospheric pressure (QNH, QFE) ● Airport visual effects to include

Section	Functions and subjective tests (aeroplanes)
	○ Active runway or landing area selection ○ Airport lighting controls ○ Airport preset positions (take-off, approach, etc.) ○ Landing surface conditions (rough, smooth, icy, wet, etc.) ○ Dynamic effects, including ground and airborne traffic ● Aircraft configuration in units appropriate to the aircraft type ○ Gross weight ○ Fuel loading ○ Payload ○ Centre of gravity ○ Aircraft system status/control (e.g. rapid alignment of navigation systems, such as inertial navigation systems, GNSS or attitude and heading reference systems; others) ○ Ground crew functions (ground power etc.) ● Storing and recalling the FMS data, unless precluded by installed equipment ● Ground crew functions (e.g. external power, air, pushback) ● Plotting and recording (take-off and approach) ● Aircraft malfunctions (inserting and removing, trigger based on speed, altitude, etc.) ● Sound controls (on, off and adjustment; indication of when the sound level is set to a value other than the approved level) ● Motion system (on, off and emergency stop) ● Control loading system (on, off and emergency stop) ● Vibration system (on, off and emergency stop) ● FSTD master/emergency power switch off ● Observer seat position/adjustment system and positive restraint system for FSTD with motion (e.g. safety belt, shoulder harness) ● Communication between the instructor/observer(s) and the flight crew ● Freezes/resets ○ Complete simulation freeze ○ Flight/problem freeze ○ Position freeze ○ Fuel freeze ○ Ground speed control ○ Standard atmosphere reset ○ Aircraft system resets (e.g. resetting engine fire bottles after their use) ● Applicable only to aeroplane FSTDs that are to be qualified for UPRT, UPRT instructor feedback mechanism as described in CS FSTD.QB.115 and CS FSTD.QB.125: ○ FSTD validation envelope ○ Flight control inputs ○ Aeroplane operational limits ○ Activation of upset scenarios ○ Operation of the recording mechanism

GM1 FSTD.FST.020 Table of functions and subjective tests for aeroplane flight simulation training devices

Table 1 below gives guidance on the application of test procedures for various aeroplane functions and subjective tests related to vibration, buffet and motion cueing. Refer also to GM2 FSTD.QB.110.

Table 1. Guidance on test procedures for aeroplane vibration, buffet and motion cueing test items

Section	Test item	Test procedure
Q	Cues on the ground due to thrust effect with brakes set	With the simulated aeroplane set with the brakes on at the take-off point, increase the engine power until buffet is experienced and evaluate its characteristics. Confirm that the buffet increases appropriately with increasing engine thrust.
Q	Cues on the ground due to spoiler / speed brake extension and thrust reversal	Perform a normal landing and use ground spoilers and reverse thrust – either individually or in combination with each other – to decelerate the simulated aeroplane. Do not use wheel braking, so that only the buffet due to the ground spoilers and thrust reversers is felt.
Q	Cues due to the extension and retraction of landing gear	Operate the landing gear. Check that the motion cues of the buffet experienced are characteristic of the simulated aeroplane.
Q	Cues in the air due to flap and spoiler / speed brake extension	First, perform an approach and extend the flaps and slats, specifically with airspeeds deliberately in excess of the normal approach speeds, then repeat at normal speeds. In cruise configuration, verify the buffets associated with the spoiler / speed brake extension. The above effects should also be verified with different combinations of speed brake / flap / gear settings to assess the interaction effects.
Q	Cues due to Mach and manoeuvre buffet	With the simulated aeroplane trimmed in 1 g flight while at high altitude, increase the engine power such that the Mach number exceeds the reference value (e.g. refer to AFM) at which Mach buffet is experienced. Check that the buffet begins at the same Mach number as it does in the aeroplane for the same configuration and that buffet levels are a reasonable representation of those in the actual aeroplane. For some aeroplanes, a manoeuvre buffet could also be verified for the same effects. Manoeuvre buffet can occur during turning flight at conditions greater than 1 g, particularly at higher altitudes.
Q	Cues due to approach-to-stall buffet and stall buffet	Conduct an approach to stall with engines at idle and a deceleration of 1 kt/s. Check that the motion cues of the buffet, including the level of buffet increase with decreasing speed, are reasonably representative of those of the simulated aeroplane. For FSTDs that are to be qualified for full-stall training tasks, modelling is to account for any increase in buffet amplitude from the initial buffet threshold of perception to the critical angle of attack or deterrent buffet as a function of the angle of attack. The

		stall buffet modelling should include the effects of N_z , as well as N_x and N_y , if relevant.
R	Effects of runway and taxiway rumble, oleo deflections, ground speed, uneven runway, runway contamination with the associated anti-skid characteristics, running over runway centre line lights and taxiway characteristics	After the aeroplane has been preset to the take-off position and then released, perform taxi at various speeds, first with a smooth runway, and note the general characteristics of the simulated runway rumble effects of oleo deflections. Next, repeat the manoeuvre with a runway roughness of 50 % and then, finally, with maximum roughness. The associated motion effects for the above tests should also include an assessment of the effects of rolling over the centre line lights, surface discontinuities of uneven runways and various taxiway characteristics. The associated effects should be affected by ground speed and runway roughness. Different gross weights may also affect the associated effects, depending on the aeroplane type.
R	Bumps associated with the landing gear mechanism movement	Perform a normal take-off, paying special attention to the bumps that could be perceptible due to maximum oleo extension after lift-off. If the landing gear is extended or retracted, motion bumps could be felt when the gear locks into position.
R	Touchdown cues for main and nose gear	Fly several normal approaches with various rates of descent. Check that the motion cues of the touchdown bump for each descent rate reasonably reflect those of the simulated aeroplane.
R	Nose wheel scuffing	Taxi the simulated aeroplane at various ground speeds and manipulate the nose wheel steering to cause yaw rates to develop that cause the nose wheel to vibrate against the ground ('scuffing'). Evaluate the speed / nose wheel combination needed to produce scuffing and check that the resultant vibrations reasonably reflect those of the simulated aeroplane.
R	Tyre failure	Depending on the aeroplane type, a single tyre failure may not necessarily be noticed by the pilot and therefore there may not be any noticeable motion effect. There may possibly be some sound and/or vibration associated with the tyre losing pressure. With multiple tyre failure selected on the same side, the pilot should notice yawing that requires controlling of the aeroplane.
R	Engine damage	The characteristics of an engine malfunction, as defined in the malfunction definition document for the specific FSTD, should describe the associated motion effects perceived by the pilot. These depend on the simulated failure, aircraft type and engine model. In addition, the corresponding engine instrument indications vary depending on the nature of the malfunction.
R	Tail, pod and wing strike	Tail strikes can be checked by the over-rotation of the aeroplane at a speed below V_R while performing a take-off. The effects can also be verified during a landing. The motion effect should be felt as a noticeable bump. If the tail strike affects the aeroplane's angular rates, the cueing provided by the motion system should have an associated effect. Excessive banking of the aeroplane during its

		take-off / landing roll can cause a pod strike. The motion effects should be felt as a noticeable bump. If the pod strike affects the aeroplane's angular rates, the cueing provided by the motion system should have an associated effect.
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FUNCTION AND SUBJECTIVE TESTS FOR ROTORCRAFT

CS FSTD.FST.030 Table of functions and subjective tests for rotorcraft flight simulation training devices

Table 1 below presents a table of functions and subjective tests for rotorcraft FSTDs, to be further modified in accordance with CS FSTD.FST.010.

Table 1. Function and subjective tests for rotorcraft FSTDs

Section	Functions and subjective tests (rotorcraft)
A	Flight deck layout and structure and aircraft system operation
	Refer to CS FSTD.QB.101 for specified requirements.
	• Enclosure of the flight deck layout • Flight deck windows • Flight crew member seating • Basic flight instruments • Appearance of the flight deck panels, instruments and switches • Haptics of the flight deck panels, instruments and switches • Circuit breakers • Flight deck equipment (e.g. headsets, oxygen masks, EFBs) • Instrument lighting • Flight deck lighting • Ambient lighting
B	Preparation for flight
	• Functionality of all switches, indicators, systems and equipment at all crew members stations and IOS • The flight deck design and functions are in accordance with the fidelity level of the FDK feature • The ambient lighting provides an even level of illumination and is not distracting to the flight crew members • Automated weather reporting
C	Surface operations
C.1	APU/engine start and run-up
	• APU start • Normal engine start • Alternate engine start procedures • Abnormal engine starts and shutdowns (hot start, hung start, fire, etc.) • Rotor start/engagement and acceleration, disengagement and deceleration ◦ Rotor start/engagement and acceleration ◦ Ground resonance (if applicable to type) ◦ Icy/slippery surface • After-start system checks (e.g. electrical, hydraulic, flight control, autopilot, radio and lighting systems)
C.2	Taxi
	• Collective lever / cyclic friction setting • Power required to taxi / cyclic input response

Section	Functions and subjective tests (rotorcraft)
	● Ground handling ● Minimum radius turn ● Taxi aids (e.g. taxi camera, moving map) ● Brake operation (effectiveness/failure) ● Water taxi / handling / floats ● Tail / nose wheel lock operation ● Surface contaminants (wet, icy, patchy wet, patchy ice, wet or rubber residue) ● Surface roughness ● Surface type (hard, soft, etc.) ● Brake and tyre failures
D	Take-off and departure
D.1	Hover and translational flight
	● Lift-off characteristics with and without wind ● Hover characteristics and engine and flight instrument response (in-ground effect and out-of-ground effect) Note: Hovering tests include stability augmentation system on and off, height stability on and off, cyclic trim on and off, follow-up trims and automation and protections as applicable to the simulated aircraft. ○ In-ground effect ○ Out-of-ground effect ○ With and without wind ● Hover power check ○ In-ground effect ○ Out-of-ground effect ● Hover turns (around a spot, about the nose/tail) ● Anti-torque/directional control effect ● Translating tendency ● No wind / headwind / crosswind / tailwind hover ● Critical azimuth ● Air taxi / transit / translational flight (forwards, sideways, rearwards) ● Low visibility ● Abnormal/emergency procedures ○ Engine failure(s) ○ Fuel governing system failure ○ Hydraulic system failure ○ Stability system failure ○ Directional control malfunctions
D.2	Normal take-off
	● Engine and transmission parameters ● Normal take-off ○ From ground ● From take-off hover ○ Category A and/or performance class 1 / performance class 2 for all certified profiles ○ Category B or performance class 3 ● Running ● Crosswind/tailwind

Section	Functions and subjective tests (rotorcraft)
	• Maximum performance • Maximum certificated take-off mass • Instrument (in IMC) • Confined area • Slope • Obstacle clearance • Elevated heliport/helideck/pinnacle/platform • Vertical • High altitude • Take-off in snow, sand and dust • Transition into forward flight • Instrument departure
D.3	Abnormal/emergency procedure during take-off and departure
	• Engine failure (different simulated failures) • Rejected take-off/forced landing: ○ Over land ○ Over water • Category A and/or performance class 1 / performance class 2 ○ Engine failure prior to TDP ○ Engine failure at or after TDP • With autopilot / stability augmentation system failure
E	Climb
	• Normal climb • Climb performance (all engines operating and OEI) compared with AFM performance data • Obstacle clearance • Best rate • Best angle • Vertical climb • One or more engines inoperative • Level-off • Category A and/or performance class 1 / performance class 2 operation for all certified profiles with engine failure up to 300 m (1 000 ft) above heliport elevation
F	Cruise Note: Tests in this section should be conducted at various altitudes, not only at typical cruising levels.
F.1	Performance characteristics
	• Straight and level flight performance • Accelerations and decelerations / changes of airspeed • High-speed vibrations • Overspeed warning
F.2	Manoeuvres and flying qualities
	• Turns ○ Normal ○ Standard rates (rates of ½, 1 and 2) ○ Steep (45° of bank) ○ Flight control servo actuator transparency effects

Section	Functions and subjective tests (rotorcraft)
	• Low-air-speed handling (not below effective translational lift speed) and turns • High-air-speed handling (e.g. turns) • High-altitude handling • Flight and manoeuvre envelope protection functions • Gliding to a forced landing; terrain accuracy for forced landing area selection
F.3	Characteristics of the particular aircraft
	• Flight characteristics of the particular aircraft • With autopilot / stability augmentation system failure
F.4	En route navigation
	• Terrain accuracy for visual navigation • Radionavigation • GNSS navigation
G	Descent and arrival
	• Normal descent • Maximum rate descent / non-emergency autorotation (VFR and IFR) • Autorotative descent ○ Straight in ○ With turn ○ IMCs ○ To landing ○ Power recovery • Instrument arrival • With autopilot / stability augmentation system failure
H	Instrument approach operations The FSTD should be tested for all relevant types of instrument approach, using appropriate aircraft configurations, flight conditions and system states (e.g. calm winds, limiting wind velocities, wind shear, relevant aircraft system failures). The customised list of F&S should contain all the types of approaches defined in the ESL and applicable to the simulated aircraft.
H.1	Precision approach procedures Testing should cover all applicable aircraft systems, equipment (e.g. HUD, EFVS) and procedures.
	• CAT I approach (ILS, GBAS landing system, microwave landing system) ○ Manual approach and landing with/without FD ○ Manual approach, one or more engines inoperative ○ 3- or 4-axis autopilot coupled approach and manual landing ○ 3- or 4-axis autopilot coupled approach and manual landing, one or more engines inoperative • CAT II approach ○ Autopilot coupled approach to DH and landing ○ Autopilot coupled approach with OEI approach to DH and go-around (manual and autopilot)
H.2	Non-precision approach procedures

Section	Functions and subjective tests (rotorcraft)
	Testing should cover manual flight and the use of all applicable FD/autopilot modes across non-precision approach types. Modes intended for OEI operations should also be tested in the OEI configuration. Testing should cover approaches flown with and without CDFA if such guidance is supported (e.g. a computed descent path). If CDFA guidance is not available, one manually flown CDFA approach is sufficient.
	• RNP APCH approach to localiser performance (LP) and lateral navigation (LNAV) minima, tested under varying temperature conditions, including down to the minimum authorised temperature • VOR and VOR/DME approach • ILS-localiser-only (LOC) approach • ILS localiser back course (LOC-BC) approach • ILS offset localiser approach • NDB approach • Circling approach (approach prior to visual circling manoeuvre) • Airborne radar approach • Point-in-space approach
H.3	Approach procedures with vertical guidance (APV – 3D approaches)
	• RNP APCH barometric-VNAV approach to LNAV/VNAV minima, with all engines operating and with one or more engines inoperative • RNP APCH approach based on SBAS (LPV minima), with all engines operating and with one or more engines inoperative • Loss of SBAS signal during approach
I	Visual approaches and instrument approach visual segments
	• Normal • Steep • Shallow • Vertical • Elevated landing sites / pinnacle / ridgelines / slope • Helideck (ship) / rig • Confined area • Crosswind/tailwind • Landing from a visual traffic pattern • Visual circling to land manoeuvre after an instrument approach Note: For this test, the visual FOV (refer to CS FSTD.QB.111) should be at least 200° horizontally to allow visual contact to be maintained with the airport during the whole circling procedure after descending below the minimum descent altitude. In addition, the operating sites and terrain FSTD feature should be, as a minimum, at the representative fidelity level (refer to CS FSTD.QB.114) to enable adequate visual contents in the visual image. Any associated hazard lights or any other visual aids for use as part of the published circling procedure should be included in the correct positions and be of the appropriate colours, directionality and behaviour. • Quick stop • Transition to hover • With NVGs, HUDs and/or EFVS

Section	Functions and subjective tests (rotorcraft)
	• Category A or performance class 1 / performance class 2 operation for all certified profiles from 1 000 ft above heliport elevation to or beyond LDP • Abnormal/emergency procedures: ○ Engine failure(s) ○ Fuel governing system failure ○ Hydraulic system failure ○ With autopilot / stability augmentation system failure ○ Directional control malfunctions
J	Missed approach
	• Manual and with 3- or 4-axis autopilot • All engines operating • One or more engines inoperative • Balked rig/deck landing • With autopilot / stability augmentation system failure
K	Landing transition / touchdown
	• From hover • Running • Slope • Surface (hard, soft, water) • Crosswind/tailwind • High altitude • Elevated landing sites / pinnacle / ridgelines • Helideck (ship) / rig • From a visual approach • From an instrument approach to minima and visual final approach thereafter • From autorotation • With anti-torque/directional control malfunction • Category A operation for all certified profiles for performance class 1 / performance class 2 ○ Landing with engine failure prior to LDP ○ Landing with engine failure at or after LDP
L	Engine shutdown and parking
	• Engine and system operation • Parking brake operation • Rotor disengagement and deceleration • Rotor brake operation • Emergency evacuation
M	Any appropriate flight phase
M.1	Flight control forces and hardware in all flight phases Refer to CS FSTD.QB.102 for specified requirements.
	• Primary flight control forces • Integration of the primary flight control hardware with the aircraft systems • Hardware, forces and feel of other flight controls • Hardware, forces and feel of controls that affect aircraft configuration
M.2	Flight control systems operation in all flight phases Refer to CS FSTD.QB.103 for specified requirements.

Section	Functions and subjective tests (rotorcraft)
	• Primary flight control systems operation in all flight phases (normal and non-normal modes, backup systems) • Primary flight control systems indications • Operation and indications of other flight controls • Operation and indications of controls that affect aircraft configuration • Flight and manoeuvre envelope protection functions • Flight control integration with aircraft systems • Stability augmentation system failures (all applicable modes)
M.3	Rotorcraft and power plant system operation (if fitted), including associated abnormal and emergency procedures in all appropriate flight phases Refer to CS FSTD.QB.104 for specified requirements.
	• Instrument indications • Flight display systems • Navigation systems • Autopilot and FD • Flight management and control computers and systems • Stability and control augmentation system • Caution and warning systems • Communication systems • Surveillance systems • Electrical system • Air conditioning and ventilation system • De-icing/anti-icing systems • Auxiliary power plant / APU • Fire and smoke detection and suppression system • Fuel and oil systems • Stabiliser/stabilator • Hydraulic systems • Pneumatic systems • Landing gear • Floats • Lighting (internal and external) • Oxygen systems • Power plant systems • Transmission systems • Rotor systems • Airborne radar used for weather avoidance, offshore operations and approaches • Terrain awareness warning systems (e.g. HTAWS) • Airborne collision avoidance systems (e.g. ACAS) • Surveillance and data link communication systems (e.g. ADS-B, CPDLC, ACARS, FANS) • Head-up guidance and HUDs (including EFVS if applicable) • HUDs (including EFVS, if appropriate) • EFB • Automatic checklists (normal, abnormal and emergency procedures)
M.4	Airborne and other miscellaneous procedures
	• Holding

Section	Functions and subjective tests (rotorcraft)		
	• Air hazard avoidance (traffic and weather, including visual correlation) • Effects of airframe and engine icing (refer to CS FSTD.QB.107.1.ICING) Note: Preferably performed in level flight with autopilot on and anti-icing/de-icing off, followed by icing activation and monitoring of effects (e.g. power, pitch, vibration, performance), then manual flight to assess handling changes and finally activation of anti-icing/de-icing systems to evaluate their function and effects. • Mast bumping • Inadvertent entry into IMCs • Recovery from unusual attitudes • Engine shutdown and in-flight restart • Wind shear models that provide training for recognition and necessary corrective pilot actions • Microburst models		
M.5	Non-normal flight conditions		
	• Tests of simulated malfunctions (refer to CS FSTD.FST.005; CS FSTD.QB.115, section ‘Simulated malfunctions’; and CS FSTD.QB.104) • Non-normal flight conditions, for example (as applicable): ○ engine fire on ground or in hover ○ engine fire in forward flight ○ autorotation to engine-off landing ○ incipient vortex ring / power settling at altitude ○ incipient vortex ring / power settling on approach ○ recovery from unusual attitudes ○ hydraulic failure in hover ○ hydraulic failure in forward flight ○ hydraulic jack stall (servo transparency) ○ unanticipated yaw ○ tail rotor drive failure in hover ○ tail rotor drive failure in forward flight ○ tail rotor control failure in hover ○ tail rotor control failure in forward flight ○ dynamic rollover ○ severe vibration ○ ground resonance ○ retreating blade stall ○ ditching		
N	Visual cueing The test items in this section supplement the requirements specified in CS FSTD.QB.111 and should be performed to ensure satisfactory results for the applicable fidelity level (G, R or S). Note: The visual cueing should also be subjectively evaluated as part of the integrated simulation simultaneously with other cueing features and the complete simulation in all flight phases.		
		Feature fidelity level	
		G	R
			S

Section	Functions and subjective tests (rotorcraft)			
N.1	Display			
	Image is free from optical discontinuities and artefacts	✓	✓	✓
	The visual channels have the same image quality for day, twilight and night	✓	✓	✓
	Image system light points are free from distracting jitter, smearing or streaking	✓	✓	✓
	Surfaces and textural cues are free from apparent quantisation (aliasing)		✓	✓
	Surfaces and textural cues do not create distracting quantisation (aliasing)	✓		
	Image is stable, seamless and realistic	✓	✓	✓
	Visual display geometry is correct		✓	✓
	Visual image geometry has no distracting discontinuities	✓	✓	✓
	The FOV replicates the aircraft FOV without obstruction			✓
	System is capable of portraying full-colour realistic textural cues	✓	✓	✓
	The flight deck cut-off angle matches the simulated aircraft	✓	✓	✓
N.2	Image qualities			
	For collimated displays: - the parallax effects are minimised - the centre of the image is aligned with the flight deck centre line and the horizon		✓	✓
	For non-collimated displays: - the parallax effects are minimised - the visual system can be aligned at any time to each of the pilot eye positions and to the flight deck centre line. - when the image is aligned to the pilot flying, the misalignment observed by the non-flying pilot does not exceed 20°	✓	✓	✓
	Image is not distorted	✓	✓	✓
	Display contrast ratio is sufficient to clearly differentiate the brightest and darkest areas of the display	✓	✓	✓
	Light point contrast ratio is sufficient to be able to clearly differentiate light points	✓	✓	✓
	Surface brightness is sufficient to be able to clearly identify visual elements in the visual scene.	✓	✓	✓
	Image's black level is sufficient to support operations in daylight, twilight and night conditions	✓	✓	✓
	Image's luminance levels are sufficient to support operations in daylight, twilight and night conditions	✓	✓	✓
	Motion blur is below a distracting level	✓	✓	✓
	Speckle is below a distracting level	✓	✓	✓
	The ground segment visible to the pilot when conducting a landing manoeuvre in day, twilight and night low-visibility conditions conforms to that of the aircraft	✓	✓	✓
	The visual system has an effect for a crash	✓	✓	✓

Section	Functions and subjective tests (rotorcraft)			
	All moving objects in the visual image are free from jitter	✓	✓	✓
	The level of detail in the visual image supports accurate perception and recognition of elements, objects and terrain	✓	✓	✓
	Curved and complex shapes appear smooth and realistic			
	The scene content is realistically affected by the viewing distance			
	Visual image gives cues to assess sink rate and enable depth perception during take-off and landing		✓	✓
	System is capable of providing focus effects that simulate rain		✓	✓
	System is capable of providing light point perspective growth		✓	✓
N.3	Visual scene content			
	Continuous and sufficient horizontal and vertical FOVs specific to the aircraft being simulated are provided			✓
	Visual cues to assess the rate of change in height and attitude and the translational displacements are provided		✓	✓
	An identifiable horizon is displayed under day, twilight and night conditions with unlimited visibility	✓	✓	✓
	If the visibility is reduced, the horizon becomes indistinct or obscured			
	Terrain contours are displayed at the resolution demanded by the environment			✓
	Landing areas are visible from a sufficient distance	✓	✓	✓
	Focus and clarity of terrain and ground objects are sharp		✓	✓
	Surface shading effects are consistent with simulated sun position			✓
	Shadows of 3D objects and lighting effects are appropriate for the directionality of light (e.g. sunshine)			
	Images for daylight, twilight and night visuals are supported	✓	✓	✓
	Image shows operational aircraft landing lights for twilight and night scenes	✓	✓	✓
	The visual image corresponds to other traffic or terrain		✓	✓
	For aircraft capable of landing on water, the sea states and effects of wind lanes on water surface are shown		✓	✓
	Image displays the ownship parts that are visible within the FOV (e.g. rotor blades)	✓	✓	✓
	Any movement (e.g. rotor start-up and stop, orientation of the rotor disc due to pilot control input) is displayed			
	Image displays the icing effects on the aircraft parts that are visible within the FOV	✓	✓	✓
	For NVIS, as applicable: - the visual image supports NVIS for the black level, surface brightness, surface contrast ratio, lunar illumination, image colour and surface resolution - surface shading effects are consistent with the simulated moon position			✓
N.4	Visual element recognition for runways			

Section	Functions and subjective tests (rotorcraft)			
	Note: The following are the minimum distances at which runway elements should be visible. Distances are measured from the runway threshold to an aircraft aligned with the runway on an extended 3° glide slope in suitable simulated meteorological conditions. For circling approaches, all tests below apply to both the runway used for the initial approach and the runway of intended landing.			
	Runway definition, strobe lights, approach lights and runway edge white lights from 8 km (5 sm) from the runway threshold	✓	✓	✓
	Visual approach aids lights from 8 km (5 sm) from the runway threshold			✓
	Visual approach aids lights from 4.8 km (3 sm) from the runway threshold	✓	✓	
	Runway centre line lights and taxiway definition from 4.8 km (3 sm)	✓	✓	✓
	Threshold lights and touchdown zone lights from 3.2 km (2 sm)	✓	✓	✓
	Runway markings within range of landing lights for night scenes; as required by the surface resolution test for day scenes			✓
	For circling approaches, runway of intended landing and associated lighting fades into view in a non-distracting manner		✓	✓
N.5	Visual element recognition for rotorcraft landing areas Note: The following are the minimum distances at which landing area elements should be visible. Distances are measured from the landing area to a rotorcraft aligned with the landing area on an extended 3° glide slope in suitable simulated meteorological conditions.			
	Heliport definition, strobe lights and approach lights from 4.8 km (3 sm)	✓	✓	✓
	Visual approach aid lights (e.g. helicopter approach path indicator lights) through approach angles up to 12°	✓	✓	✓
	Taxiway definition from 3.2 km (2 sm)	✓	✓	✓
	Markings within range of landing lights for night or twilight scenes	✓	✓	✓
	Markings as required by the surface resolution test for day scenes		✓	✓
	Landing direction lights and raised FATO lights from 1.6 km (1 sm)		✓	✓
	Flush mounted FATO lights, touchdown and lift-off (TLOF) lights and the lighted windsock from 800 m (0.5 sm)	✓	✓	✓
	Hover taxiway lighting (yellow/blue/yellow cylinders) from TLOF	✓	✓	✓
O	Operating sites and terrain – visual representation The test items in this section supplement the requirements specified in CS FSTD.QB.114 and should be performed to ensure satisfactory results for the applicable fidelity level (G, R or S). For the operating sites and terrain FSTD feature at the S fidelity level, refer to CS FSTD.QB.114.1 for requirements for tier I operating site models.			

Section	Functions and subjective tests (rotorcraft)	Feature fidelity level		
		G	R	S
O.1	Landing areas, runways and taxiways			
	Realistic landing areas, airport runways and taxiways			✓
	Representative landing areas, runways and taxiways		✓	
	Generic landing areas, runways and taxiways	✓		
	If appropriate for the airport, two parallel runways and one crossing runway can be displayed simultaneously; at least two runways can be lit simultaneously			✓
	Rotorcraft landing areas and runway threshold elevations and locations are modelled to provide correlation with rotorcraft systems (e.g. compass, altimeter, GNSS, HUD)	✓	✓	✓
	Slopes in landing areas, runways, taxiways and ramp areas may not cause distracting or unrealistic effects, including pilot eyepoint height variation			✓
	The displayed scene corresponds to the appropriate surface contaminants and includes runway lighting reflections for wet, partially obscured lights for snow or suitable alternative effects			✓
	An airport with a snow scene is to include terrain snow and snow-covered taxiways and runways			✓
	The effects of ownship downwash on various surfaces (e.g. snow, sand, water, grass, trees and vegetation) are simulated		✓	✓
	The white-out or brownout recirculation effects of ownship's rotor downwash on various surfaces (e.g. snow, sand, water, grass) are simulated, including the effects of reduced visibility beginning at a height above the ground equal to approximately one half of the rotor diameter		✓	✓
O.2	Rotorcraft landing area surface, markings and elements			
	Markings for standard heliport identification ('H' and location) and other specific marking such as a hospital pad, aiming point and aiming circle, as appropriate, properly sized, coloured and oriented		✓	✓
	TLOF area markings		✓	✓
	FATO designation markings as appropriate (including design size, perimeter, maximum rotorcraft dimensions and weight limitations, as appropriate)		✓	✓
	Safety areas, obstacle-free sector and limited obstacle sector as appropriate		✓	✓
	Taxiways / taxi routes and their markings	✓	✓	✓
	Aprons and their markings		✓	✓
	Signs as appropriate for the model used		✓	✓
	Signs at intersecting runways and taxiways			✓
	Windsock that gives appropriate wind cues		✓	✓
	Windsock lighting		✓	✓

Section	Functions and subjective tests (rotorcraft)			
O.3	Lighting for the rotorcraft landing area of appropriate colours, directionality, behaviour and spacing, associated with each pad			
	Landing direction	✓	✓	✓
	Raised and flush FATO, TLOF perimeter and flood lighting		✓	✓
	Visual approach aids	✓	✓	✓
	Approach lighting of appropriate colours	✓	✓	✓
O.4	Taxiway lighting of appropriate colours, directionality, behaviour and spacing, associated with each pad			
	Lighting for taxiways / taxi routes	✓	✓	✓
	Lighting for aprons		✓	✓
O.5	Airport surface, markings and elements			
	Threshold markings	✓	✓	✓
	Runway numbers	✓	✓	✓
	Touchdown zone markings	✓	✓	✓
	Fixed distance markings		✓	✓
	Edge markings	✓	✓	✓
	Centre line markings	✓	✓	✓
	Distance remaining signs		✓	✓
	Signs at intersecting runways and taxiways		✓	✓
	Windsock that gives appropriate wind cues		✓	✓
	Windsock lighting		✓	✓
O.6	Airport runway lighting of appropriate colours, directionality, behaviour and spacing for each 'in-use' runway, including the following, if appropriate			
	Threshold lights	✓	✓	✓
	Edge lights	✓	✓	✓
	End lights	✓	✓	✓
	Centre line lights	✓	✓	✓
	Touchdown zone lights	✓	✓	✓
	Lead-off lights		✓	✓
	Visual landing aid(s)	✓	✓	✓
	Appropriate approach lighting system for the runway		✓	✓
	Approach lighting system for the runway	✓		
O.7	Airport taxiway surface and markings (associated with each 'in-use' runway), including the following, if appropriate			
	Edge markings	✓	✓	✓
	Centre line markings	✓	✓	✓
	Runway holding position markings	✓	✓	✓
	ILS critical area markings			✓
	All taxiway markings, lighting and signage to taxi, as a minimum, from a designated parking position to a designated runway and return, after landing on the designated runway, to a designated parking position. A low-visibility taxi route (e.g. surface movement guidance control system, follow-me truck, daylight			✓

Section	Functions and subjective tests (rotorcraft)			
	taxi lights) must also be demonstrated for operations authorised in low-visibility conditions. The designated runway and taxi routing must be consistent with that airport for operations in low-visibility conditions.			
O.8	Taxiway lighting of appropriate colours, directionality, behaviour and spacing (associated with each 'in-use' runway), including the following, if appropriate			
	Edge lights	✓	✓	✓
	Centre line lights	✓	✓	✓
	Runway holding position and ILS critical area lights	✓	✓	✓
O.9	Visual model correlation			
	Airport or rotorcraft landing area model aligned with the navigational aids that are associated with operations at the 'in-use' runway or rotorcraft landing area	✓	✓	✓
	Simulation of runway or rotorcraft landing area contaminants correlates with the displayed runway surface and lighting			✓
O.10	Airport and rotorcraft landing area buildings, structures, objects and lighting			
	Airport or rotorcraft landing area buildings, structures, objects and lighting			✓
	Representative airport or rotorcraft landing area buildings, structures, objects and lighting		✓	
	Generic airport or rotorcraft landing area buildings, structures, objects and lighting	✓		
	Moving and static operating site clutter (e.g. other rotorcraft and aeroplanes, power carts, tugs, fuel trucks)		✓	✓
	Apron markings (e.g. hazard markings, lead-in lines, stand numbering), lighting and a marshaller		✓	✓
O.11	Terrain and obstacles			
	Terrain and obstacles within 46 km (25 NM) of the reference airport or rotorcraft landing area with appropriate colours and textures for the simulated area (including ground objects of sufficient number and appropriate size and perspective)			✓
	Representative depiction of terrain and obstacles within 46 km (25 NM) of the reference airport		✓	
	Visual scene with sufficient scene content to allow a pilot to successfully accomplish a visual landing	✓		
	Visual scene with adequate terrain elements to permit VFR navigation by sole reference to visual landmarks in accordance with appropriate charts (typically 1:500 000, 1:250 000 or 1:100 000 scale mapping) and with suitable terrain contouring represented		✓	✓
	Buildings, trees or other vertical obstructions in the immediate vicinity of the landing area		✓	✓
	Suspended wires in the immediate vicinity of the landing area		✓	✓
O.12	Significant, identifiable natural and cultural elements			
	Significant, identifiable natural and cultural elements within 46 km (25 NM) of the reference airport or rotorcraft landing area			✓

Section	Functions and subjective tests (rotorcraft)			
	Note: This refers to natural and cultural elements that are typically used for pilot orientation in flight. Outlying airports not intended for landing need only provide a reasonable facsimile of runway orientation.			
	Representative depiction of significant and identifiable natural and cultural elements within 46 km (25 NM) of the reference airport or rotorcraft landing area Note: This refers to natural and cultural elements that are typically used for pilot orientation in flight. Outlying airports not intended for landing need only provide a reasonable facsimile of runway orientation.		✓	
O.13	Visual scene management			
	All landing area and airport runway, approach and taxiway lighting and cultural element lighting intensity for any approach are capable of being set to five different intensities and an off setting; all visual scene light points fade into view appropriately in accordance with the conditions set on the IOS			✓
	Airport runway, approach and taxiway lighting, rotorcraft landing area approach lighting and cultural feature lighting intensity for any approach can be set at an intensity representative of that used in training for the visibility level set; all visual scene light points fade into view appropriately	✓	✓	
	The directionality of strobe lights, approach lights, runway edge lights, visual landing aids, runway centre line lights, threshold lights, touchdown zone lights on the runway of intended landing and TLOF or FATO lights is appropriate	✓	✓	✓
O.14	Selectable airport visual scene capability for the following			
	Night	✓	✓	✓
	Twilight (i.e. dusk and dawn)	✓	✓	✓
	Day	✓	✓	✓
	Moving ground vehicles undertaking predefined manoeuvres (conflicting and not conflicting with the ownship), for example an aeroplane or another rotorcraft crossing the active runway or converging airborne traffic, or a supply ship in the vicinity of the oil rig during rotorcraft's final approach for landing		✓	✓
	Aircraft traffic undertaking predefined manoeuvres (conflicting and not conflicting with the ownship), for example another aircraft crossing the active runway or converging airborne traffic		✓	✓
	Operational visual scenes that portray physical relationships known to cause landing illusions, for example short runways, landing approaches over water, uphill or downhill runways, rising terrain on the approach path and unique topographical elements		✓	✓
O.15	Correlation with rotorcraft and associated equipment			
	Visual cues relate to simulated rotorcraft attitudes and responses	✓	✓	✓

Section	Functions and subjective tests (rotorcraft)			
	Visual terrain representation gives sufficient cues to determine speed, altitude, attitude and position	✓	✓	✓
	Visual scene correlates with aircraft systems, if fitted, for example: ● TAWS indications versus terrain in the visual image ● ACAS versus traffic in the visual image ● EFVS versus visual image ● reduction of visual image defocus due to rain when wipers are used		✓	✓
	Visual effects for each visible, ownship, external light (e.g. taxi, landing and search light lobes)	✓	✓	✓
P	Atmosphere and weather The test items in this section supplement the requirements specified in CS FSTD.QB.113 and should be performed to ensure satisfactory results for the applicable fidelity level (G, R or S).			
		Feature fidelity level		
		G	R	S
P.1	Environment simulation			
	Dynamic environment simulation with weather effects (on ground / in air) integrated with the entire simulation		✓	✓
	A basic atmospheric model, including pressure, temperature, visibility, cloud base and winds	✓		
P.2	Atmosphere simulation			
	The atmosphere simulation includes: ● wind speed, direction and gusts ● air temperature and its lapse rate ● air pressure ● turbulence levels	✓	✓	✓
	Turbulence effects by using applicable cues on all axes, including: ● effects on instruments ● effects on aerodynamics and handling as a result of the atmospheric disturbances ● effects on the controls, as applicable (e.g. reversible flight controls) ● motion and/or vibration effects, if installed	✓	✓	✓
P.3	Visual weather representation			
	The visual scene includes the following: ● various visibility and RVR conditions, including fog and patchy fog effects ● effects on operating site lighting (including variable intensity and fog effects)		✓	✓
	Patchy fog giving the effect of variable RVR. The lowest RVR must be that selected on the IOS (i.e. variability is only greater than the RVR set on the IOS).		✓	✓
	Effects of fog on airport lighting, such as halos and defocus		✓	✓

Section	Functions and subjective tests (rotorcraft)			
	The visual scene includes various visibility and RVR conditions	✓		
	Visibility and RVR measured in terms of distance. Visibility/RVR must be checked at and below a height of 2 000 ft above the operating site and within a radius of 16 km (8.6 nm; 10 sm) of the operating site.	✓	✓	✓
	Visual image for: - various visibility and RVR conditions - a realistic transition to IMCs - a realistic transition to visual contact from IMCs	✓	✓	✓
	Gradual breakout to ambient visibility/RVR, defined as up to 10 % of the relevant cloud base or top. This transition layer should be no less than 20 ft and no more than 200 ft in vertical extent. Transition effects should be complete when the IOS cloud base or top is reached when exiting and start when entering the cloud (i.e. transition effects must occur within the IOS defined cloud layer). Cloud effects must be checked at and below a height of 2 000 ft above the operating site and within a radius of 16 km (8.6 nm; 10 sm) of the operating site.	✓	✓	✓
	Visual image has cloud layers representing few, scattered, broken and overcast cloud conditions, giving partial or complete obstruction of the ground scene	✓	✓	✓
	3D clouds			✓
	Visual image provides an in-cloud airspeed effect (variable cloud density and ambient changes)			✓
	The clouds, fog and weather effects affect the ownship external lights (e.g. landing lights, strobes and beacons) in the visual scene, such as reflected glare		✓	✓
	The clouds, fog and weather effects affect the other aircraft external lights within visual range in the visual scene		✓	✓
P.4	Weather effects			
	Weather effects, including visual representations as applicable, for: - rain - hail - storm cells (i.e. localised storm area, rain cell) - thunderstorms (i.e. including thunder and lightning effects) - falling snow - blowing snow and - blowing sand			✓
	The precipitation and blowing snow or sand are affected by the wind direction and velocity			✓

Section	Functions and subjective tests (rotorcraft)			
	Sound, motion and visual effects of light, medium and heavy precipitation near a thunderstorm on take-off, approach and landing at and below an altitude of 600 m (2 000 ft) above the airport surface and within a radius of 16 km (10 sm) of the operating site			✓
Q	Vibration cueing The test items in this section supplement the requirements specified in CS FSTD.QB.109 and should be performed to ensure satisfactory results for the applicable fidelity level (G, R or S). Note: The vibration and buffet cueing should also be subjectively evaluated as part of the integrated simulation simultaneously with other cueing features and the complete simulation in all flight phases.			
		Feature fidelity level		
		G	R	S
	Cues due to lift/thrust system effects, including main and tail rotor speed and disk loading vibration effects and cues due to main rotor and anti-torque system blade-tracking and balance effects		✓	✓
	Cues due to translational lift and the effect of speed changes (acceleration and deceleration)	✓	✓	✓
	Cues due to primary anti-torque system (e.g. tail rotor, fenestron, NOTAR system)		✓	✓
	Cues due to transverse flow effects		✓	✓
	Cues due to the extension and retraction of landing gear (if applicable to the simulated aircraft)		✓	✓
	Cues due to autorotation		✓	✓
	Cues due to retreating blade stall Note: Retreating blade stall should result in both vibration/buffet cues (refer to CS FSTD.QB.109) and motion cues.		✓	✓
	Cues due to vortex ring (setting with power) Note: Entering the vortex ring should result in both vibration/buffet cues (refer to CS FSTD.QB.109) and motion cues.		✓	✓
	Cues due to high-speed rotor vibration		✓	✓
	Cues due to basic anti-torque system failure		✓	✓
	Cues due to operation in the low-relative-speed environment (e.g. relative wind from all directions)		✓	✓
	Cues due to icing effects		✓	✓
	Cues due to atmospheric disturbances		✓	✓
	Cues due to hydraulic system failures		✓	✓
	Cues due to detailed malfunction effects (to include those associated with the main rotor damper, vibration suppression systems, and anti-torque bearing, coupling, gearbox or driveshaft)			✓

Section	Functions and subjective tests (rotorcraft)			
	If the simulated rotorcraft type is equipped with a vibration reduction system, vibration cues simulating the situation if the vibration reduction system is inoperative		✓	✓
	Primary main rotor speed and disk loading vibration effects	✓		
	Buffet and vibration due to translational lift and the effect of speed changes	✓		
R	Motion cueing The test items in this section supplement the requirements specified in CS FSTD.QB.110 and should be performed to ensure satisfactory results for the applicable fidelity level (G, R or S). Motion effects should give cues realistic to the simulated aircraft dynamics to indicate the threshold at which a flight crew member recognises an event or situation. Note: The motion cueing should also be subjectively evaluated as part of the integrated simulation simultaneously with other cueing features and the complete simulation in all flight phases.			
		Feature fidelity level		
		G	R	S
R.1	Motion cues – general			
	The pilot receives an effective motion cue and stimulus, which provides the appropriate sensations of the acceleration of the aircraft’s six degrees of freedom. These sensations reflect the aircraft’s translational accelerations (longitudinal, lateral, vertical) and angular motion (pitch, roll, yaw), including their onset, rate and intensity, and also include quasi-static platform attitudes.			✓
	The motion system is capable of generating sustained acceleration cues and generating the key acceleration and angular cues that are characteristic of the aircraft dynamics in all axes for: - all flight phases - manoeuvres (e.g. vertical take-off to hover, pedal turns, translations, banked turns, quick stop, flare, landing, go-around, vortex ring state, retreating blade stall) - failures (e.g. engine failures, engine-out operations, abnormal operations, emergencies)			✓
	The FSTD motion cueing system has a high tilt coordination gain, high rotational gain and high correlation with respect to the aircraft simulation model for all the manoeuvres			✓
	The pilot receives a motion cue and stimulus, which provides the appropriate sensations of acceleration. These sensations reflect the aircraft’s translational accelerations (longitudinal, lateral, vertical) and angular motion (pitch, roll, yaw), including their onset, rate and intensity, and also include quasi-static platform attitudes, within its available degrees of freedom.		✓	

Section	Functions and subjective tests (rotorcraft)			
	The motion system is capable of generating sustained acceleration cues and generating the key acceleration and angular cues that are characteristic of the aircraft dynamics in the appropriate axes. The motion cueing in each case should be assessed with consideration given to the system's degrees of freedom and physical limitations for: - all flight phases; - manoeuvres (e.g. vertical take-off to hover, pedal turns, translations, banked turns, quick stop, flare, landing, go-around, vortex ring state, retreating blade stall); - failures (e.g. engine failures, engine-out operations, abnormal operations, emergencies).		✓	
	The cross-coupling effects of the motion movements are compensated for so that they are not discernible by a human during flight		✓	✓
	The washout is below the human-perceivable threshold during flight and does not exhibit any adverse motion cueing effects		✓	✓
R.2	Motion effects			
	Effects of runway and taxiway rumble, oleo deflections, ground speed, uneven runway, runway contamination, running over runway centre line lights and taxiway characteristics			✓
	Bumps associated with landing gear mechanism movement		✓	✓
	Touchdown cues for main and nose gear or for skids, based on their geometry		✓	✓
	Touchdown cues (relative to each landing gear) that reflect the effects of lateral and directional cues resulting from asymmetry or side-slip at touchdown and from the rate of descent		✓	✓
	Touchdown cues for main and nose gear or for skids	✓		
	Friction drag from skid-type landing gear	✓	✓	✓
	Ground resonance		✓	✓
	Tyre failure	✓	✓	✓
	Airframe ground strike (e.g. tail strike)	✓	✓	✓
	Engine, transmission or rotor damage	✓	✓	✓
S	Sound cueing The requirements specified in CS FSTD.QB.108 for the applicable fidelity level of the sound cueing FSTD feature should be included in the list of functions and subjective tests. The following supplements those requirements. The following checks should be performed with the motion and vibration systems on. If headsets are used during normal aircraft operation, headsets with similar characteristics and performance should also be used during FSTD sound system testing. Note: The sound cueing should also be subjectively evaluated as part of the integrated simulation simultaneously with other cueing features and the complete simulation in all flight phases.			

Section	Functions and subjective tests (rotorcraft)	Feature fidelity level		
		G	R	S
	Significant sounds perceptible to the flight crew during flight operations should replicate those of the simulated aircraft type and variant in all flight phases			✓
	Significant sounds perceptible to the flight crew during flight operations are characteristic of the simulated aircraft type in all flight phases		✓	
	Significant sounds perceptible to the flight crew during flight operations are characteristic of the simulated group of aircraft, including basic engine, rotor, airframe and aircraft system sounds, in all flight phases	✓		
	Precipitation			✓
	Rain removal equipment (e.g. wipers)			✓
	Significant rotorcraft noises perceptible to the pilot during normal operations, such as noises from engines, transmissions, rotors or other sources, to be of a comparable level to that found in the rotorcraft	✓	✓	✓
	Significant rotorcraft flight deck sounds and those that result from an action by the pilot	✓	✓	✓
	Abnormal operations for which there are associated sound cues including but not limited to engine, rotors, transmission malfunctions, landing gear / tyre malfunctions and tail or landing gear / skid strikes	✓	✓	✓
	Sound of a crash when the rotorcraft is landed in excess of limitations or in unusual attitudes	✓	✓	✓
T	Special effects			
	Airflow patterns and relevant effects associated with large structures, such as buildings and oil rigs, confined areas and mountain peaks, including demarcation lines			
	Special atmospheric effects, as may be required for a particular training programme, such as arctic sea smoke, katabatic winds, mountain waves and effects (rotors, demarcation lines, etc.), oil rig exhaust turbulence, and wake vortices and downwash effects from other aircraft			
U	Instructor operating station			
	The requirements specified in CS FSTD.QB.115 should be included in the list of functions and subjective tests. The following supplements those requirements. Note: The list below is not exhaustive but is intended to cover the sorts of functionalities that should be available to support the intended use, depending on the FSTD and FCS. Refer to all the general requirements for IOS requirements.			
	● Repositions ○ Parking spot ○ Take-off position ○ Approach position – at least three positions at approximately 1.8 km, 5.5 km and 9.3 km (1 NM, 3 NM and 5 NM, respectively) from the runway threshold			

Section	Functions and subjective tests (rotorcraft)
	○ Elevated surface position (building top, offshore oil rig, etc.) ○ Confined landing area ○ On a slope - Note: Repositions are in trim at the appropriate speed and configuration for the point. ● Resets ○ System ○ Temperature ○ Fluids and agents ● Weather presets (as applicable) ○ Unlimited, cloud ceiling and visibility OK, VFRs, non-precision, precision (CAT I, CAT II, CAT III), EFVS ● Visual effects ○ Time of day (day, twilight, night) ○ Clouds (bases, tops, layers, types, density) ○ Visibility, with the units corresponding to those in use at the operating site ○ RVR, with the units corresponding to those in use at the operating site ○ Special effects (precipitation, thunderstorms, blowing snow, blowing sand, etc.) ○ Sand/dust/snow/water downwash/recirculation effect on/off ○ Sea state conditions ● Wind speed and direction ○ Surface ○ Intermediate levels ○ Typical gradient ○ Gusts with the associated heading and speed variance ○ Turbulence ● Temperature – surface ● Atmospheric pressure (QNH, altimeter pressure setting related to a particular reference datum point) ● Airport/heliport visual effects: ○ To include active runway or landing area selection ○ Airport/heliport lighting controls ○ Airport/heliport preset positions (take-off, approach, oil rig, etc.) ○ Landing surface conditions (rough, smooth, icy, wet, etc.) ○ Dynamic effects, including ground and airborne traffic ● Aircraft configuration in units appropriate to the aircraft type ○ Gross weight ○ Fuel loading ○ Payload ○ Centre of gravity ○ Aircraft system status/control (e.g. rapid alignment of navigation systems, such as inertial navigation systems, GNSS or attitude and heading reference systems; others) ○ Ground crew functions (ground power etc.) ● Storing and recalling the FMS data unless precluded by installed equipment ● Ground crew functions (e.g. external power, air, pushback) ● Plotting and recording (take-off and approach)

Section	Functions and subjective tests (rotorcraft)
	• Aircraft malfunctions (inserting and removing, trigger based on speed, altitude, etc.) • Sound controls (on, off and adjustment; indication of when the sound level is set to a value other than the approved level) • Motion system (on, off and emergency stop) • Control loading system (on, off and emergency stop) • Vibration system (on, off and emergency stop) • FSTD master/emergency power switch off • Observer seats position/adjustment system and positive restraint system for FSTD with motion (e.g. safety belt, shoulder harness) • Communication between the instructor/observer(s) and the flight crew • Freezes/resets ○ Complete simulation freeze ○ Flight/problem freeze ○ Position freeze ○ Fuel freeze ○ Ground speed control ○ Standard atmosphere reset ○ Aircraft system resets (e.g. resetting engine fire bottles after their use)

GM1 FSTD.FST.030 Table of functions and subjective tests for rotorcraft flight simulation training devices

Table 1 below gives guidance on the application of test procedures for various rotorcraft functions and subjective tests related to vibration, buffet and motion cueing. Refer also to GM2 FSTD.QB.110.

Table 1. Guidance on test procedures for rotorcraft vibration, buffet and motion cueing test items

Section	Test item	Test procedure
Q	Cues due to translational lift and the effect of speed changes (acceleration and deceleration)	From a stabilised in-ground-effect hover, begin a slow forward acceleration in level flight. If passing through the effective translational lift range, the noticeable effect may be a possible nose pitch-up (in some rotorcraft), an increase in the rate of climb and a temporary increase in vibration level (in some cases, this vibration may be pronounced). This effect is experienced again upon deceleration through the appropriate speed range. During deceleration, the pitch and rate of climb will have the reverse effect, but there will be a similar, temporary increase in vibration level.
R	Effects of runway and taxiway rumble, oleo deflections, ground speed, uneven runway, runway contamination, running over runway centre line lights and	After the rotorcraft has been preset to the take-off position and then released, perform taxi at various speeds, first with a smooth runway, and note the general characteristics of the simulated runway rumble effects. Next, repeat the manoeuvre with a runway roughness of 50 % and then, finally, with maximum roughness. The associated motion vibrations should be affected by ground speed and runway roughness. Similar tests should be conducted on taxiways at various taxi speeds.

	taxiway characteristics	The associated motion effects for the above tests should also include an assessment of the effects of rolling over the centre line lights, surface discontinuities of uneven runways and various taxiway characteristics. The associated effects should be affected by ground speed and runway roughness. Different gross weights may also affect the associated effects, depending on the aeroplane type.
R	Bumps associated with landing gear	Perform a normal take-off, paying special attention to the bumps that could be perceptible due to maximum oleo extension after lift-off. If the landing gear is extended or retracted, motion bumps may be felt when the gear locks into position.
R	Touchdown cues for main and nose gear or for skids, based on their geometry	Conduct several touchdowns with various rates of descent, from hover and a run-on. Check that the motion cues of the touchdown bumps for each descent rate and speed reflect those of the simulated rotorcraft.
R	Friction drag from skid-type landing gear	Perform a running take-off or a running landing and note any change (increase or decrease) with speed in fuselage vibrations (as opposed to rotor vibrations) due to the friction of dragging the skid along the surface. This vibration should lessen as the ground speed decreases.
R	Tyre failure	Simulate tyre failures and note the effects of yaw, motion, vibration and sound effects. The pilot should notice yawing with the failure of multiple tyres selected on the same side. This should require controlling of the rotorcraft. Depending on the rotorcraft type, a single tyre failure may not necessarily be noticed by the pilot and therefore there may not be any noticeable motion effect. There may possibly be some sound and/or vibration associated with the tyre losing pressure.
R	Airframe ground strike (e.g. tail strike)	Tail strikes can be checked by the over-rotation of the rotorcraft at a quick stop or during autorotation to the ground. They can also be checked in hover by a rapid aft cyclic input. The motion effect should be felt initially as a motion bump as the tail guard hits the surface. A nose-down pitching moment may possibly be noticeable after the strike.
R	Engine, transmission or rotor damage	The characteristics of an engine malfunction, as defined in the malfunction definition document for the specific FSTD, should describe the associated motion effects perceived by the pilot. These depend on the simulated failure and aircraft type. In addition, the corresponding engine instrument indications vary depending on the nature of the malfunction.
R	Vortex ring state	Procedures may differ between rotorcraft and may be prescribed by the rotorcraft manufacturer. The following information is provided for illustrative purposes. To enter the manoeuvre, reduce power to below hover power, descending vertically or near vertically and allowing sink rate to increase to 300 ft/min or more (actual sink rate value will depend on the rotorcraft type simulated).

		Adjust attitude to obtain airspeeds of less than 10 kt. The aircraft should shudder when entering the vortex ring state. During the initial stage (if a large amount of excess power is available), a large application of collective pitch may arrest rapid descent. If done carelessly or too late, a collective increase can aggravate the situation, resulting in more turbulence, an increased rate of descent and an increase in the vibration level. In single-rotor rotorcraft, the recovery can be accomplished by applying cyclic input to gain airspeed and arrest the upward induced flow of air and/or by lowering the collective input (altitude permitting). Normally, gaining airspeed is the preferred method, as less altitude is lost. In tandem-rotor rotorcraft, fore and aft cyclic inputs aggravate the situation. By lowering thrust (altitude permitting) and applying lateral cyclic input or pedal input to arrest the upward induced flow of air, the pilot can accomplish recovery.
R	Retreating blade stall	Procedures may differ between rotorcraft and may be prescribed by the rotorcraft manufacturer. The following information is provided for illustrative purposes. To enter the manoeuvre, increase forward airspeed; the effect should be recognised through the development of a low frequency vibration, the pitching-up of the nose and a roll in the direction of the retreating blade. High-weight, low-rotor RPM, high-density altitude, turbulence or steep abrupt turns are all conducive to retreating blade stall at high forward airspeeds. Correct recovery from retreating blade stall requires the collective to be lowered first, which reduces the disk loading. Aft cyclic input can then be used to slow the rotorcraft.

FUNCTIONS AND SUBJECTIVE TESTS FOR SIMULATED AIR TRAFFIC CONTROL ENVIRONMENT

CS FSTD.FST.040 Table of functions and subjective tests for simulated air traffic control environment

If the FSTD to be qualified includes a SATCE system, the functions and subjective testing of the FSTD should be performed to ensure that the SATCE system complies with the requirements specified in CS FSTD.QB.160. The focus of the functions and subjective testing of SATCE should be on those functions that support key training objectives rather than those that provide a complete, highly detailed, synthetic representation of real-world operations. The list of functions and subjective tests should meet the applicable requirements specified in CS FSTD.QB.160, corresponding to the fidelity level of the FSTD. Table 1 below supplements those requirements.

Table 1. Function and subjective tests for Simulated Air Traffic Control Environment (SATCE)

Section	SATCE
SATCE.A	Automated weather reporting
	● Message format (ICAO or regional) ● Reported weather conditions (multiple) ● Reported weather conditions (assignment) ● Message format (ICAO) ● Reported weather conditions (single) ● Instructor control (override) ● Instructor control ● ATC correlation with reported weather ● Message content ● Flight crew access ● Flight crew access (information)
SATCE.B	Other traffic
	● Ground vehicles ● Ground vehicle performance ● Ground vehicle behaviour ● Ground vehicle behaviour (dynamic) ● Other aircraft ● Other aircraft performance ● Other aircraft behaviour (dynamic) ● Other aircraft behaviour (routing) ● Other aircraft behaviour (movements) ● Other aircraft behaviour (taxiing) ● Other aircraft behaviour (parking) ● Other traffic transponder state ● Other traffic transponder mode of operation ● Other aircraft types and liveries ● Other aircraft visual effects ● Other traffic automation

	• Other traffic correlation with ATC • Other traffic separation • Other aircraft call signs and liveries • Ground vehicle conflicts • Other aircraft ground conflicts
SATCE.C	Background radio communications
	• Presence • ATC services and other traffic operations • Correctness • Radio discipline (overstepping) • Radio discipline (interrupting) • Radio discipline (ownship – overstepping) • Radio discipline (ownship – interrupting) • Ownship radio support
SATCE.D	ATC services
	• ATC procedures (location-specific) • Radio ranging • ATC service provision • Roles and frequency allocation • ATC procedures • ATC procedures (correlation) • ATC service continuity • Human-provided ATC services to ownship
SATCE.E	Language and phraseology
	• Standard phraseology (regional) • Standard phraseology • Language
SATCE.F	Voice characteristics
	• ATC voices (multiple) • Other traffic voices (multiple) • ATC voices • ATC voice assignment • Dedicated ATC voices • Other traffic voices
SATCE.G	Operating site and airspace modelling
	• Operating sites (location-specific) • Controlled airspace (location-specific) • Multiple connected ground movement areas • Dual-direction runway movements • Multiple take-off and landing areas • Runway operation modes • Take-off and landing area lighting • Holding point lighting • Taxi route lighting • Operating sites • Controlled airspace • Minimum connected ground movement areas

	• Single-direction runway movements
SATCE.H	Weather
	• Site operations and reported weather • ATC procedures and reported weather
SATCE.I	Voice communications
	• Ownship non-normal conditions • Ownship emergency conditions • Ownship routing (taxi routing and parking) • Voice continuity • Time of day • Communication initiation • ATC services and ownship operations • ATC service continuity • Standby • Say again • Content errors and omissions • Incorrect frequency transmissions • Clearance deviations • Ownship routing (flight plan) • Ownship routing (published routes) • Ownship routing (take-off and landing areas)
SATCE.J	ATC data link communications
	• Message sequence • Message indications • Message presentation • Timing delays • Message content • Air traffic service clearances • Data link weather • DLIC • Connection management • CPDLC • CPDLC (regional protocols) • ADS-C • FIS-B • Service failures
SATCE.K	System correlation
	• ACAS (standard procedures) • Traffic on the visual system • Traffic on the visual system (alignment) • Visual system clutter • Navigation data alignment (other aircraft) • Navigation data alignment (airspace) • Traffic on flight deck displays • ADS-B traffic • ACAS

SATCE.L	IOS interfaces and controls
	● Access to data link communications ● Other traffic context and information ● Access to radio communications ● Simulator functions (common) ● Disable SATCE ● Mute and restore (background radio communications) ● Other traffic (configurable presence) ● Other traffic (controllable presence) ● Other traffic (configurable flow)